

# The FEMU Manual

## Reference Manual and User Guide for the FEMU NVMe and CXL SSD Emulator

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<https://github.com/MoatLab/FEMU>

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### Abstract

FEMU is an NVMe and CXL SSD emulator built on QEMU/KVM. A guest operating system sees an emulated SSD as a real PCIe device and drives it with its own NVMe or CXL drivers, while FEMU charges NAND, channel and garbage collection time from a configurable timing model. FEMU emulates several kinds of SSD: a fast DRAM-backed NVMe device with no media timing, a conventional SSD with a device FTL, Zoned Namespace, Open-Channel, key-value and computational storage SSDs, and a CXL Type-3 memory device backed by flash.

This manual describes how to build and run FEMU, how it is built inside, each mode and feature with its parameters, the counters it reports and how to measure with them, and how to test and change it. The chapters are generated from the Markdown documentation in the FEMU repository, which remains the reference when the two differ.

**Keywords:** SSD emulation, NVMe, flash translation layer, zoned storage, CXL, QEMU

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# 1 Introduction

FEMU is an SSD emulator built into QEMU. It adds three devices to QEMU: `femu`, an NVMe controller; `femu-subsys`, an NVMe subsystem that groups controllers and namespaces; and `femu-cxl-ssd`, a CXL Type-3 memory device. A guest operating system finds the controllers and the CXL device on its PCIe bus and drives them with its own NVMe and CXL drivers; the subsystem is visible only through the controllers it groups. In the block modes, applications, file systems and the block layer in the guest run unchanged, so a whole storage stack can be studied on an SSD whose internals you choose (Figure 1).

This manual is modelled on the DiskSim reference manual: it explains how to run FEMU, then describes each component with its parameters, the statistics it produces and how far it has been validated, and closes with how to extend it. Its chapters are generated from the Markdown pages under `hw/femu/docs/` in the FEMU repository. Those pages are the reference; when this manual and the repository differ, the repository is right.

## 1.1 What FEMU does

- **Emulates several kinds of SSD.** The `femu_mode` property of `-device femu` selects one: 0 Open-Channel SSD (OCSSD), 1 BlackBox SSD with a device FTL (BBSSD), 2 NoSSD, a DRAM device with no media timing and the default, 3 Zoned Namespace (ZNS), 4 computational storage (CSD), and 5 key-value (KV). Flexible Data Placement, several namespaces per controller, namespace management, and per-block metadata with protection information are features layered on these modes. `femu-cxl-ssd` puts a DRAM page cache in front of the BBSSD FTL and exposes the result as CXL memory.
- **Charges flash time.** The modes with flash compute how long each command would take on the emulated device: NAND reads, programs and erases, channel transfers, garbage collection, and optional host-link and controller-firmware costs. For NVMe commands the data itself is copied at once; FEMU holds the completion until the host clock reaches the computed time, so the guest sees the modelled latency on its own clock. On the CXL SSD, the vCPU that made an access waits out the media time before the access completes.
- **Reports what the device did.** For BBSSD, CSD and KV, a vendor log page (C0h) gives the write amplification factor and media counters. The SMART log gives host read and write totals, wear and media errors, and the CXL SSD exposes cache counters as QOM properties readable over QMP.
- **Is configured from the command line.** Every geometry, timing and FTL parameter is a device property, documented in Appendix A. Launcher scripts in `hw/femu/scripts/` start a guest with a working configuration for most modes; KV mode has none.

## 1.2 What FEMU does not do

- **It does not keep data across runs.** The emulated medium lives in host memory. A guest reboot or a controller reset keeps it (unless you trigger a simulated power loss on a device with `power_loss=on`, which drops the write buffer); QEMU exiting loses it. FEMU never writes an NVMe namespace to a file. A `femu-cxl-ssd` given a file-backed memory backend leaves its bytes in that file.
- **It does not support migration or snapshots.** The devices are marked `unmigratable`, so QEMU refuses `migrate` and `savevm`.

- **It does not slow the host down to the model.** Latencies are lower bounds: a completion is posted on the first poller sweep after it is due, so a poller thread that does not get a host CPU adds delay on top. Section 6 explains how to get repeatable numbers.
- **It does not reproduce a particular commercial SSD.** Its timing comes from the parameters you set. Calibrating them against a real drive is your part of the experiment.
- **It does not run everywhere.** FEMU needs an x86\_64 Linux host with hardware virtualization and KVM. Open-Channel mode needs a guest kernel from 4.16 to 5.14 (4.17 for Open-Channel 2.0, the default), because Linux removed LightNVM in 5.15.

## 1.3 Organization of this manual

### Section 2, Getting started

Host requirements, building FEMU, preparing a guest image, and a first run that measures write amplification.

### Section 3, Architecture and design

The layers of FEMU from the guest interface to the memory backend, its threads, and the design of the NVMe front end, the FTL and the NAND timing model.

### Section 4, Modes and features

How to choose a mode, then one part per mode or feature: what it emulates, how to turn it on, its parameters and its limits.

### Section 5, Running and configuring

The parameter manual, tutorials, performance tuning, and the security model and host sizing.

### Section 6, Measuring and validation

The counters FEMU reports, recipes for repeatable measurements, and the timing model in detail.

### Section 7, Troubleshooting

Answers to common questions and how to debug FEMU.

### Section 8, Contributing

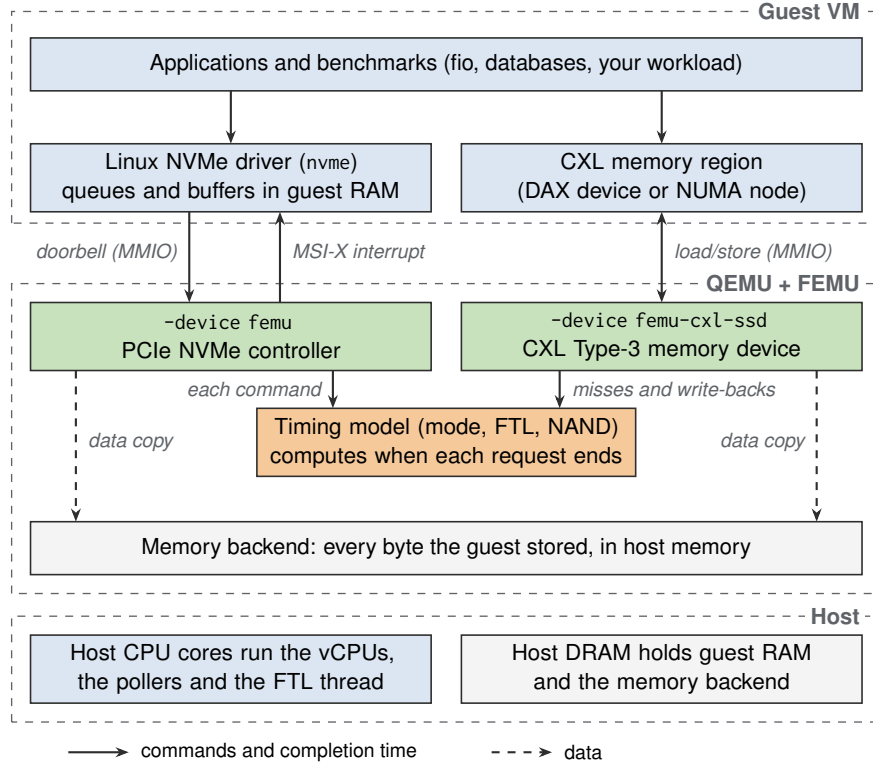
The source tree, the tests, and the checks that keep this documentation in step with the code.

### Appendices

The device property reference generated from the binary, log pages and counters, the scripts shipped with FEMU, and the changelog.

## 1.4 Conventions

Commands, file names, properties and values are set in monospace. A reference such as (§4.2) points to another part of this manual. A footnote gives the GitHub address of a file in the FEMU repository that the manual does not include, or of an outside page. Command blocks are taken from the Markdown pages, where continuous integration runs most `-device femu` examples to check that they still work. Diagrams drawn in text keep their original layout and may be set in smaller type to fit the page.



**Figure 1:** Where FEMU sits: the guest drives FEMU’s NVMe or CXL device, the timing model decides when each request completes, and the data goes straight to the memory backend in host memory.

## 1.5 How to cite FEMU

If you use FEMU in your research, cite the FAST ’18 paper. The same reference is in CITATION.cff at the top of the repository.

```
@inproceedings{Li+18-FEMU,
  author    = {Huaicheng Li and Mingzhe Hao and Michael Hao Tong and
               Swaminathan Sundararaman and Matias Björling and Haryadi S. Gunawi},
  title     = {{The CASE of FEMU: Cheap, Accurate, Scalable and Extensible Flash Emulator}},
  booktitle = {16th USENIX Conference on File and Storage Technologies (FAST 18)},
  year      = {2018},
}
```

If you use one of the following modes, also cite the paper it comes from: FDP comes from WARP (FAST ’26, Section 4.5.8), the CXL SSD from Cylon (FAST ’26, Section 4.18.11) and CSD from CEMU (ASPLOS ’26, Section 4.13.9).



## 2 Getting started

### 2.1 Requirements

What the host and the guest need before you build and run FEMU. The quick start (§2.4) assumes everything on this page.

#### 2.1.1 Host

##### Operating system and CPU

FEMU runs on an x86\_64 Linux host with hardware virtualization (Intel VT-x or AMD-V). Run it on a physical machine. Nested virtualization works but distorts the emulated latency, and WSL is not supported.

FEMU is based on QEMU 10.1, which needs Python 3.9 or newer and GLib 2.66 or newer. Check yours:

```
python3 --version
pkg-config --modversion glib-2.0
```

| Distribution           | Python | GLib | Status                           |
|------------------------|--------|------|----------------------------------|
| Ubuntu 24.04 LTS       | 3.12   | 2.80 | Built in CI                      |
| Ubuntu 22.04 LTS       | 3.10   | 2.72 | Built in CI                      |
| Ubuntu 20.04 and older | 3.8    | 2.64 | Cannot build with stock packages |

Other distributions with new enough Python and GLib should build FEMU, but CI does not test them. `pkgdep.sh` installs dependencies on Debian and Ubuntu only; on other distributions install the equivalents of the packages listed in `build.md` (§2.2.2).

CI builds FEMU with `femu-compile.sh` on Ubuntu 22.04 and 24.04 (`.github/workflows/ci.yml`). It runs the unit tests and `qtests` and starts every mode without a guest. It does not boot a guest. The guest-level quick start (§2.4) was last run end to end on a Pop!\_OS 24.04 host (Ubuntu 24.04 base) with Linux 6.18.

##### KVM access

The run scripts start QEMU with `-enable-kvm`. Check that KVM is loaded and that you can open it:

```
ls -l /dev/kvm
id -nG | grep -w kvm
```

If `/dev/kvm` is missing, enable virtualization in the BIOS and load the module (`sudo modprobe kvm_intel` or `sudo modprobe kvm_amd`). If you are not in the `kvm` group, add yourself and log in again:

```
sudo usermod -aG kvm "$USER"
```

The `run-*.sh` launchers run QEMU with `sudo`, so they work without the group. `make-guest-image.sh` runs without root and needs the group.

##### Memory

The emulated SSD lives in host DRAM. Nothing is stored in a file, and nothing written to it survives a shutdown. Plan host memory as:

```
host RAM needed = devsz_mb (the emulated SSD) + guest RAM (-m) + about 1 GiB for QEMU
```

| Launcher                                                            | devsz_mb | Guest -m | Free host RAM needed |
|---------------------------------------------------------------------|----------|----------|----------------------|
| run-blackbox.sh (BBSSD)                                             | 12288    | 4G       | about 17 GiB         |
| run-nossd.sh, run-zns.sh, run-whitebox.sh, run-csd.sh, run-kvssd.sh | 4096     | 4G       | about 9 GiB          |

To fit a smaller host, lower `ssd_size` in `run-blackbox.sh` and the geometry with it (see the property reference (§A.1)).

FEMU locks the device memory (`mlock`) so page faults do not distort latency. Under `sudo` the limit does not apply. If you run QEMU as a normal user, raise the limit with `ulimit -l unlimited` (or `/etc/security/limits.conf`). Without it the device still starts and prints a warning, and latency is less precise.

## Hugepages

The standard `run-*.sh` launchers do not need hugepages. `femu-cxl-ssd` with `der=cylon` needs a shared, preallocated `hugetlb` memory backend and a modified host kernel (see `cxlssd.md`<sup>1</sup>).

## Disk

The source tree and one build take about 3 GiB. The guest image built by `make-guest-image.sh` uses about 2.5 GiB on disk (32 GiB virtual), plus a 0.6 GiB download cache.

## CPU cores

The launchers give the guest 4 vCPUs (`-smp 4`). FEMU adds its own polling threads. 8 host cores is a comfortable minimum.

### 2.1.2 Guest

#### Kernel per mode

| Mode                          | femu_mode                                | Guest kernel                           | Notes                                                                                                                                                  |
|-------------------------------|------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| NoSSD                         | 2 (default)                              | any with NVMe                          |                                                                                                                                                        |
| BlackBox SSD (BBSSD)          | 1                                        | any with NVMe                          |                                                                                                                                                        |
| OpenChannel SSD 1.2 / 2.0     | 0                                        | 4.16 to 5.14 (2.0 needs 4.17 or newer) | LightNVM was removed in Linux 5.15. On newer kernels use SPDK.                                                                                         |
| Zoned Namespace (ZNS)         | 3                                        | 5.9 or newer, CONFIG_BLK_DEV_ZONED=y   |                                                                                                                                                        |
| Key-value (KV)                | 5                                        | 6.0 or newer                           | No block device. The namespace appears as the generic character device <code>/dev/ngXnY</code> . Linux 5.9 to 5.19 do not attach the namespace at all. |
| Computational storage (CSD)   | 4                                        | any with NVMe                          | Guest tools in <code>hw/femu/tests/csd/</code> .                                                                                                       |
| Flexible Data Placement (FDP) | 1, with <code>femu-subsys, fdp=on</code> | any with NVMe                          | Use passthrough commands or <code>io_uring</code> to send placement hints.                                                                             |

<sup>1</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md>

| Mode                   | femu_mode       | Guest kernel                                                                                                | Notes                                                            |
|------------------------|-----------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| CXL SSD (femu-cxl-ssd) | not a femu_mode | CXL region support: CONFIG_CXL_BUS, CXL_PCI, CXL_ACPI, CXL_MEM, CXL_PORT, CXL_REGION, DEV_DAX, DEV_DAX_KMEM | Needs <code>cxl-cli</code> and <code>daxctl</code> in the guest. |

The image from `make-guest-image.sh` runs Ubuntu 24.04 with Linux 6.8. It covers every mode except OCSSD.

### Guest tools

- `nvme-cli` and `fio` for every NVMe mode. `make-guest-image.sh` installs both.
- ZNS needs `nvme-cli` 1.12 or newer for `nvme zns`.
- `nvme-cli` 2.8, the version in Ubuntu 24.04, double-frees and crashes after every successful Persistent Event Log read (`nvme persistent-event-log`). Build `nvme-cli` 3.x from source in the guest if you need that log. Other commands work.
- CXL: `cxl-cli`, `daxctl` and `ndctl` (`make-guest-image.sh --cxl`).

## 2.2 Build FEMU

How to get the source, install dependencies and build the FEMU binary. Check `requirements.md` (§2.1) first.

### 2.2.1 Clone

```
git clone https://github.com/MoatLab/FEMU.git
cd FEMU
```

The repository has no git submodules. QEMU's build downloads a few meson subprojects (for example `keycodemapdb` and `berkeley-softfloat-3`) during configure, so the first build needs network access.

### 2.2.2 Dependencies

On Debian and Ubuntu, `pkgdep.sh` installs what the build needs. It must run as root:

```
mkdir build-femu
cd build-femu
cp ../femu-scripts/femu-copy-scripts.sh .
./femu-copy-scripts.sh
sudo ./pkgdep.sh
```

`femu-scripts` is a link to `hw/femu/scripts`. `femu-copy-scripts.sh` copies the build helpers, the image builder and the run scripts into `build-femu/`.

`pkgdep.sh` installs `gcc` `pkg-config` `git` `libglib2.0-dev` `libfdt-dev` `libpixman-1-dev` `zlib1g-dev` `libdw-dev` `libaio-dev` `libslirp-dev` `libnuma-dev` `ninja-build`. QEMU's configure also needs `python3-venv`, `flex` and `bison`; install them if configure asks:

```
sudo apt install python3-venv flex bison
```

CI installs a longer list, which is a known-good superset (`.github/workflows/ci.yml`, step "Install dependencies"):

```
sudo apt install -y build-essential pkg-config libglib2.0-dev \
    libpixman-1-dev libfdt-dev zlib1g-dev libaio-dev \
    libcap-ng-dev libattr1-dev ninja-build python3-pip \
    libslirp-dev libseccomp-dev libcurl4-gnutls-dev \
    libiscsi-dev libnfs-dev librbd-dev librados-dev \
    libssh-dev liblz2-dev libsnappy-dev libbz2-dev \
    liblzma-dev libzstd-dev libgcrypt20-dev libgnutls28-dev \
    uuid-dev libcap-dev libxml2-dev libmount-dev \
    liburing-dev flex bison
```

### 2.2.3 Compile

From build-femu/:

```
| ./femu-compile.sh
```

The script runs `make clean`, then

```
| ../configure --enable-kvm --target-list=x86_64-sofmmu --enable-slirp \
    --disable-libnfs --disable-libiscsi --disable-curl
```

and `make` with one job per CPU. A first build takes 3 to 15 minutes, depending on the number of cores.

The binary is `build-femu/qemu-system-x86_64`. The run scripts expect it in the current directory, so run them from `build-femu/`. The build also produces `build-femu/qemu-img`, which `make-guest-image.sh` uses.

Check that the FEMU devices are registered:

```
| ./qemu-system-x86_64 -device help | grep femu
```

You should see the NVMe controller, the CXL SSD and the NVMe subsystem:

```
| name "femu", bus PCI, desc "FEMU Non-Volatile Memory Express"
| name "femu-cxl-ssd", bus PCI, desc "FEMU CXL SSD"
| name "femu-subsys", desc "FEMU NVMe Subsystem (FDP)"
```

`./qemu-system-x86_64 -device femu,help` lists every property with its default; the property reference (§A.1) explains them.

### 2.2.4 Optional features

#### CSD with uBPF programs

Computational storage mode (`femu_mode=4`) loads shared-library programs with no extra build option. To also run uBPF programs, build with uBPF:

```
| ./femu-compile.sh --enable-csd-ubpf # libubpf found by pkg-config
| ./femu-compile.sh --enable-csd-ubpf=/path/to/ubpf-cemu # a ubpf-cemu source tree
```

With a path, the build links `<path>/build/lib/libubpf.a`. These are the only options `femu-compile.sh` accepts.

#### CXL SSD

`femu-cxl-ssd` is built by default for `x86_64-sofmmu`. It needs no option. Check it with:

```
| ./qemu-system-x86_64 -device help | grep femu-cxl-ssd
```

## Debug build

femu-compile.sh has no debug switch. Run configure yourself from build-femu/:

```
./configure --enable-kvm --target-list=x86_64-softmmu --enable-slirp \  
--disable-libnfs --disable-libiscsi --disable-curl \  
--enable-debug --enable-debug-info \  
make -j"$(nproc)"
```

Add `--extra-cflags=-DFEMU_FTL_ASSERT` to turn on the FTL consistency checks that are compiled out by default.

### 2.2.5 Rebuild

After you change the source, rerun make from build-femu/. It rebuilds only what changed:

```
make -j"$(nproc)"
```

`./femu-compile.sh` always starts with `make clean` and reruns `configure`, so use it only for a full rebuild.

After you pull new scripts, run `./femu-copy-scripts.sh` again. It overwrites the copies in build-femu/, including any run script you edited there.

### 2.2.6 Common build errors

| Message                                                 | Cause                                                                                                                | Fix                                                                                            |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| ERROR: Cannot find Ninja                                | ninja-build is missing                                                                                               | <code>sudo apt install ninja-build</code>                                                      |
| ERROR: python venv creation failed                      | Debian and Ubuntu ship venv in a separate package                                                                    | <code>sudo apt install python3-venv</code>                                                     |
| ERROR: Cannot use 'python3', Python >= 3.9 is required. | Python is older than 3.9 (Ubuntu 20.04)                                                                              | Use Ubuntu 22.04 or newer                                                                      |
| Dependency "glib-2.0" not found or a version below 2.66 | GLib is missing or too old                                                                                           | <code>sudo apt install libglib2.0-dev</code> ; on Ubuntu 20.04, upgrade the distribution       |
| Errors around <code>nfs_pread_async</code>              | Older FEMU trees used <code>libnfs 6</code> , whose API changed; the current QEMU base ignores <code>libnfs 6</code> | Build current master. <code>femu-compile.sh</code> also passes <code>--disable-libnfs</code> . |
| A meson subproject fails to download                    | No network during configure                                                                                          | Build once with network access                                                                 |
| <code>pkgdep: unsupported system type</code>            | Not Debian or Ubuntu                                                                                                 | Install the equivalent packages by hand                                                        |
| A warning stops the build ( <code>-Werror</code> )      | A newer compiler warns where QEMU 10.1's did not                                                                     | Add <code>--disable-werror</code> to the configure line, and please report the warning         |

## 2.3 Guest image

FEMU emulates an SSD inside a virtual machine, so you need a guest disk image. The run scripts boot `~/images/u20s.qcow2` by default. The name is historical; the image can hold any Linux release.

### 2.3.1 Build one with `make-guest-image.sh`

From build-femu/ (after the build (§2.2)):

```
./make-guest-image.sh
```

The script:

1. downloads the Ubuntu 24.04 cloud image and checks it against Ubuntu's SHA256 sums (and their GPG signature when gpgv and the Ubuntu cloud-image keyring are installed);
2. boots it once without a display to create user femu with passwordless sudo, install nvme-cli and fio, and turn on the serial console;
3. powers the guest off and writes the result.

It writes these files to ~/images/:

| File                               | Purpose                                               |
|------------------------------------|-------------------------------------------------------|
| u20s.qcow2                         | The guest disk the run scripts boot                   |
| femu-guest-key, femu-guest-key.pub | SSH key for user femu                                 |
| cache/                             | The downloaded cloud image, reused on the next run    |
| seed.iso, provision.log            | Provisioning input and serial log, kept for debugging |

It takes a few minutes and needs no root access. The host needs curl, sha256sum, ssh-keygen, timeout, qemu-img, and one of cloud-localds, genisoimage, xorriso or mkisofs. It uses the qemu-system-x86\_64 and qemu-img next to it in build-femu/ when they exist. If tools are missing, the script names them. On Ubuntu:

```
| sudo apt install curl cloud-image-utils
```

It also needs read and write access to /dev/kvm (see requirements.md (§2.1.1)).

Options:

```
| ./make-guest-image.sh --size 64G                # disk size (default 32G)
| ./make-guest-image.sh --ssh-key ~/.ssh/id_ed25519.pub
| ./make-guest-image.sh --password femu           # also allow console login
| ./make-guest-image.sh --cxl                     # also install ndctl, daxctl, cxl-cli
| ./make-guest-image.sh --packages build-essential,git # extra Ubuntu packages
| ./make-guest-image.sh -o /data/images           # another output directory
| ./make-guest-image.sh --force                   # replace an existing image
```

./make-guest-image.sh --help lists every option. The script refuses to overwrite an existing image unless you pass --force. If provisioning fails, read provision.log in the output directory.

Without --password, user femu has no password, so you can log in only over SSH. The serial console in the run script's terminal shows the boot and a login prompt you cannot use. Pass --password if you want console login too.

## 2.3.2 Use an image in another place

Every run-\*.sh script reads two variables:

- IMGDIR: the image directory (default ~/images);
- OSIMGF: the image file (default \$IMGDIR/u20s.qcow2).

```
| OSIMGF=/data/images/u20s.qcow2 ./run-blackbox.sh
```

make-guest-image.sh and run-guest-ssh.sh read IMGDIR too, so one export IMGDIR=/data/images covers all three.

### 2.3.3 Kernel per mode

The cloud image runs Linux 6.8. That covers NoSSD, BBSSD, ZNS, KV, CSD, FDP and CXL. OCSSD needs a guest kernel older than 5.15, because Linux removed LightNVM in 5.15. See the kernel table (§2.1.2) for every mode.

To run OCSSD with LightNVM, install an older release (Ubuntu 20.04 ships 5.4) by hand as below, or use SPDK in the guest.

### 2.3.4 Log in with SSH

The run scripts forward host port 8080 to the guest's SSH port. From build-femu/, while a run script is running:

```
| ./run-guest-ssh.sh          # interactive shell  
| ./run-guest-ssh.sh sudo nvme list  # one command
```

run-guest-ssh.sh uses the key \$IMGDIR/femu-guest-key and user femu. Set SSH\_PORT, SSH\_KEY or GUEST\_USER to change them. It skips host key checks, because the guest is rebuilt often and listens only on localhost. The plain equivalent is:

```
| ssh -i ~/images/femu-guest-key -p 8080 femu@localhost
```

Only one guest can use a port at a time. To run a second guest, or when another program holds 8080, set SSH\_PORT for both the run script and run-guest-ssh.sh:

```
| SSH_PORT=8081 ./run-blackbox.sh      # terminal 1  
| SSH_PORT=8081 ./run-guest-ssh.sh     # terminal 2
```

### 2.3.5 Alternatives

#### Prebuilt image

The FEMU team shares a prebuilt Ubuntu 20.04 image on request through the FEMU VM image form<sup>2</sup>. Extract it to ~/images/ and rename it to u20s.qcow2. It holds Ubuntu 20.04; issue #57 reports Linux 5.4, which runs OCSSD but not ZNS. Log in with ssh -p 8080 <user>@localhost, using the account described with the image.

#### Install from an ISO

This needs a display for the installer.

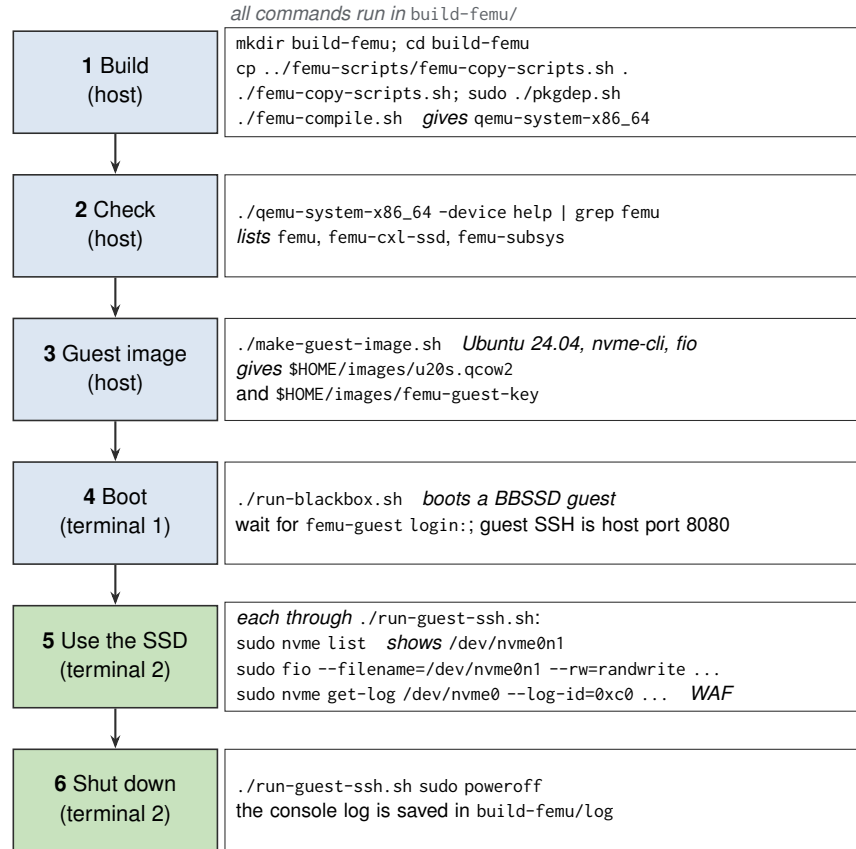
```
| mkdir -p ~/images  
| cd ~/images  
| wget https://releases.ubuntu.com/24.04/ubuntu-24.04.3-live-server-amd64.iso  
| qemu-img create -f qcow2 u20s.qcow2 80G  
| qemu-system-x86_64 -enable-kvm -cpu host -smp 8 -m 8192 \  
|     -cdrom ubuntu-24.04.3-live-server-amd64.iso -boot d \  
|     -hda u20s.qcow2 -net nic -net user
```

If the ISO link has moved, pick the current one from <https://releases.ubuntu.com>. Install openssh-server, nvme-cli and fio.

The run scripts use -nographic, so the guest must use the serial console. Inside the installed guest, edit /etc/default/grub:

---

<sup>2</sup><https://forms.gle/nEZaEe2fkj5B1bxt9>



**Figure 2:** Quick start: build FEMU, make the guest image, boot a BBSSD guest, then use the SSD over SSH. Host steps are blue, steps inside the guest green.

```
GRUB_CMDLINE_LINUX="console=tty0 console=ttyS0,115200n8"
GRUB_TERMINAL="console serial"
GRUB_SERIAL_COMMAND="serial --unit=0 --speed=115200"
```

Then run `sudo update-grub` and power the guest off. Log in later with `ssh -p 8080 <user>@localhost`.

## 2.4 Quick start

From a fresh clone to a running BlackBox SSD (BBSSD). On a 20-core host the whole run took 5 minutes, most of it compiling. You build FEMU, build a guest image, boot the guest with an emulated SSD, write to the SSD with fio and read its write amplification factor (WAF).

You need an x86\_64 Ubuntu or Debian host with KVM, sudo, network access, and about 17 GiB of free RAM. requirements.md (§2.1) has the details.

You use two terminals. Terminal 1 runs the virtual machine. Terminal 2 talks to it over SSH.

### 2.4.1 Build FEMU (terminal 2)

```
git clone https://github.com/MoatLab/FEMU.git
cd FEMU
mkdir build-femu
```



```
cd build-femu
cp ../femu-scripts/femu-copy-scripts.sh .
./femu-copy-scripts.sh
sudo ./pkgdep.sh
./femu-compile.sh
```

femu-compile.sh takes 3 to 15 minutes, depending on the number of cores, and ends with ==> FEMU compilation done .... Check that the build has the FEMU device:

```
./qemu-system-x86_64 -device help | grep femu

name "femu", bus PCI, desc "FEMU Non-Volatile Memory Express"
name "femu-cxl-ssd", bus PCI, desc "FEMU CXL SSD"
name "femu-subsys", desc "FEMU NVMe Subsystem (FDP)"
```

If the build fails, see build.md (§2.2.6).

## 2.4.2 Build the guest image (terminal 2)

Still in build-femu/:

```
sudo apt install curl cloud-image-utils
./make-guest-image.sh
```

This downloads the Ubuntu 24.04 cloud image (about 600 MB) and provisions it, which takes one to a few minutes. It ends with:

```
Image ready: /home/<you>/images/u20s.qcow2
```

If it fails, read ~/images/provision.log. guest-image.md (§2.3) explains the options and alternatives.

## 2.4.3 Boot the guest with a BBSSD (terminal 1)

Open a second terminal, go to the same build-femu/ directory, and run:

```
./run-blackbox.sh
```

The script prints the -device femu, ... line it uses, asks for your sudo password, and boots the guest on this terminal. Wait for the login prompt, about 30 seconds:

```
femu-guest login:
```

You cannot log in here (user femu has no password). Leave the terminal running.

The guest's SSH port is forwarded to host port 8080. If another program already uses 8080, QEMU stops with Could not set up host forwarding rule. Pick a free port and set it for both scripts, in both terminals:

```
export SSH_PORT=8081
```

## 2.4.4 Look at the SSD (terminal 2)

Back in terminal 2, in build-femu/:

```
./run-guest-ssh.sh sudo nvme list
```

| Node         | Generic              | SN     | Model                        | Namespace |
|--------------|----------------------|--------|------------------------------|-----------|
| Usage        | Format               | FW Rev |                              |           |
| -----        |                      |        |                              |           |
| /dev/nvme0n1 | /dev/ng0n1           | vSSD0  | FEMU BlackBox-SSD Controller | 0x1       |
| 12.88 GB /   | 12.88 GB 512 B + 0 B | 1.0    |                              |           |

/dev/nvme0n1 is the emulated SSD. The guest's own disk is /dev/sda. If SSH says Connection refused, the guest is still booting; wait and retry.

## 2.4.5 Write to it with fio (terminal 2)

```
./run-guest-ssh.sh sudo fio --name=qs --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio
--rw=randwrite --bs=4k --iodepth=16 --size=1G
```

fio writes 1 GiB of random 4 KiB blocks and prints the bandwidth and latency it saw. The latency includes the emulated NAND program time.

## 2.4.6 Read the write amplification factor (terminal 2)

FEMU reports its media counters in the vendor log page C0h. The first 4 bytes are the WAF times 1000:

```
./run-guest-ssh.sh "sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b | od -An -t u4 -N 4"
1000
```

1000 means a WAF of 1.000: every page the host wrote was programmed once, and garbage collection has not run yet. The next three 8-byte counters are the pages the host wrote, the pages garbage collection moved, and the pages programmed in total:

```
./run-guest-ssh.sh "sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b | od -An -t u8 -j 8
-N 24 -w24"
262144 0 262144
```

Write more than the drive's capacity, and the second counter and the WAF grow. log-pages-and-counters.md (§B) lists every counter.

## 2.4.7 Shut down (terminal 2)

```
./run-guest-ssh.sh sudo poweroff
```

The guest powers off and run-blackbox.sh returns in terminal 1. The data on the emulated SSD is gone: FEMU keeps it only in host memory.

If the guest hangs, stop QEMU from terminal 1 with Ctrl-a then x.

run-blackbox.sh saves the console output of the run in build-femu/log and overwrites it on the next run.

## 2.4.8 Next steps

- Change the geometry and latency at the top of run-blackbox.sh, or pick another mode with another run-\*.sh script. The doc map<sup>3</sup> lists the guides.
- Every property: reference/properties.md (§A.1).

<sup>3</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/README.md>

## 3 Architecture and design

### 3.1 FEMU architecture

FEMU is QEMU with a set of emulated storage devices under `hw/femu/`. This page describes those devices as a stack of layers, from what the guest sees down to the host memory that holds the data. Each section says what the layer does, where its code lives, which threads run it, and where it charges time.

Two ideas explain most of the design:

- **Data and time are separate.** The guest's data is copied into a host memory buffer as soon as a command is parsed. The FTL and NAND model never touch the data; they only compute how long the command would take on a real device, and the completion is held back until that time has passed. (The power-loss model is the exception: with `power_loss=on` the FTL thread applies the data.)
- **NVMe I/O commands run on FEMU's own threads, not on the vCPU.** A doorbell write only records the new queue tail. Polling threads fetch and execute the commands, and an FTL thread computes their timing.

The CXL SSD (`femu-cxl-ssd`) is the exception to the second rule: its loads and stores are served on the vCPU thread that issued them. The CXL SSD design note<sup>4</sup> covers it in depth.

#### 3.1.1 The layers

*Figure 3 draws this diagram.*

Solid arrows carry commands and timing. Dotted arrows carry the data, which goes straight to the memory backend.

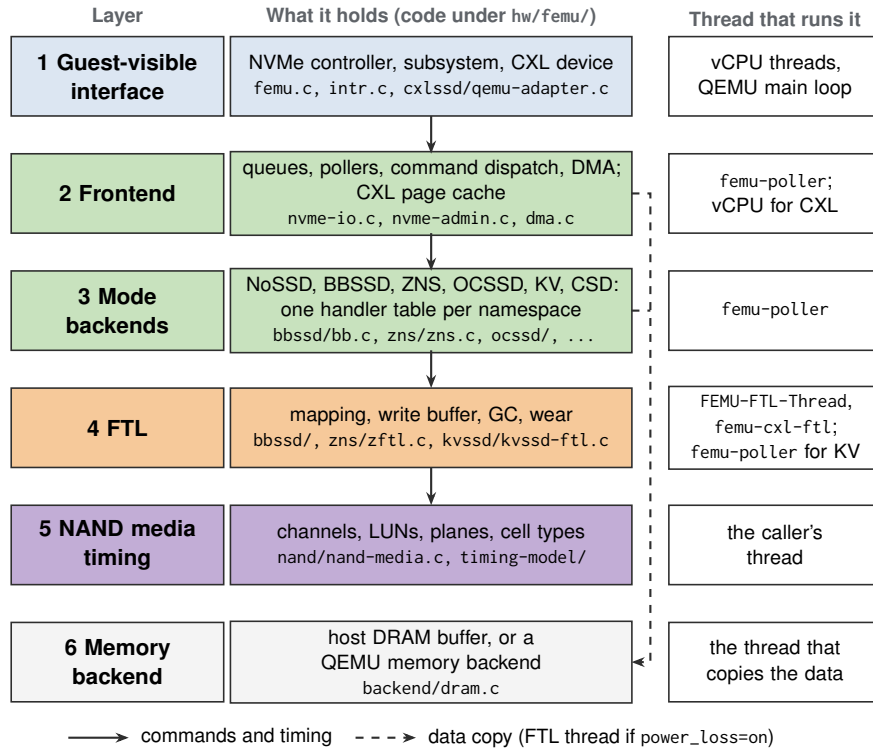
| Layer                      | Main source files                                                                                                                                                                                        | Threads                                         |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 1. Guest-visible interface | <code>hw/femu/femu.c</code> , <code>hw/femu/femu-props.c</code> , <code>hw/femu/intr.c</code> , <code>hw/femu/cxlssd/qemu-adapter.c</code>                                                               | vCPU threads, QEMU main loop                    |
| 2. Frontend                | <code>hw/femu/nvme-io.c</code> , <code>hw/femu/nvme-admin.c</code> , <code>hw/femu/dma.c</code> , <code>hw/femu/lib/</code> , <code>hw/femu/cxlssd/cxlssd.c</code> , <code>hw/femu/cxlssd/cache.c</code> | femu-poller, vCPU threads                       |
| 3. Mode backends           | <code>hw/femu/nossd/</code> , <code>hw/femu/bbssd/bb.c</code> , <code>hw/femu/zns/zns.c</code> , <code>hw/femu/ocssd/</code> , <code>hw/femu/kvssd/</code> , <code>hw/femu/csd/</code>                   | femu-poller                                     |
| 4. FTL                     | <code>hw/femu/bbssd/</code> , <code>hw/femu/zns/zftl.c</code> , <code>hw/femu/kvssd/kvssd-ftl.c</code>                                                                                                   | FEMU-FTL-Thread, femu-poller (KV), femu-cxl-ftl |
| 5. NAND media timing       | <code>hw/femu/nand/</code> , <code>hw/femu/ocssd/oc-timing.c</code>                                                                                                                                      | the caller's thread                             |
| 6. Memory backend          | <code>hw/femu/backend/dram.c</code> ; for CXL, a QEMU memory backend object                                                                                                                              | the thread that copies the data                 |

#### 3.1.2 Guest-visible interface

FEMU registers three device types.

**-device femu** is a PCIe NVMe controller. It reports NVMe version 1.4 and PCI ID `1d1d:1f1f` by default (vid, did). BAR0 holds the controller registers and the doorbells, MSI-X has one vector per queue plus the admin

<sup>4</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md>



**Figure 3:** FEMU's six layers, the code that implements each one, and the thread that runs it; data is copied from the frontend straight to the memory backend, past the FTL and NAND model.

queue, and an optional Controller Memory Buffer (cmb) must be placed on BAR2 with cmbloc. femu\_mode selects which kind of SSD the controller emulates (see layer 3 (§3.1.4) and choosing a mode (§4.1)).

Namespaces are not separate devices. There is no FEMU namespace device to put on the command line; the controller builds its namespaces at realize from namespaces, namespace\_sizes and namespace\_modes, and Namespace Management can add more at run time when ns\_mgmt=on. Migration and snapshots are blocked for this device.

**-device femu-subsys** is an NVMe subsystem. It is not on any bus. A controller joins it with subsys=<id>, and the subsystem must appear first on the command line. It is needed for Flexible Data Placement (fdp=on) and for namespaces shared between controllers (ns\_mgmt=on on the subsystem).

**-device femu-cxl-ssd** is a CXL Type-3 volatile memory device, a subclass of QEMU's cxl-type3. Its data lives in the memory backend named by volatile-memdev. FEMU installs an I/O region named femu-cxl-media over every CXL fixed memory window that can reach the device, so guest loads and stores to the device's address range reach FEMU. Optional extras are a cache control interface on BAR5 (cca=on) and an experiment control channel through the label storage area (lsa-control=on). A femu controller can serve the same medium as an NVMe namespace with cxl-ssd=<id>.

Where it lives:

- hw/femu/femu.c: QOM types femu and femu-subsys, femu\_realize(), nvme\_check\_constraints(), PCI and BAR setup, MMIO and doorbell handlers (nvme\_mmio\_write()).
- hw/femu/femu-props.c: the help text of the femu and femu-subsys properties. The properties and their defaults are defined in femu.c (femu\_props[]). The generated property reference (§A.1) is built from

the binary.

- hw/femu/intr.c: MSI-X, MSI and pin interrupts (nvme\_isr\_notify\_io()).
- hw/femu/cxlssd/qemu-adapter.c: QOM type femu-cxl-ssd, realize checks, window overlay and address decoding, and its properties (cxl\_props[]). hw/femu/cxlssd/props.c: their help text.

Threads: register and doorbell writes are MMIO exits handled on the vCPU thread that made them, under QEMU's big lock (BQL).

### 3.1.3 Frontend

#### NVMe queues and the admin path

The admin queue is handled synchronously. A write to the admin submission queue doorbell calls nvme\_process\_sq\_admin() on the vCPU thread, which runs nvme\_admin\_cmd() in hw/femu/nvme-admin.c for each new entry. Commands the generic code does not know go to the mode's admin hook, for example the BBSSD vendor command 0xEF.

I/O queues are handled asynchronously. A write to an I/O submission queue doorbell only stores the new tail (nvme\_process\_db\_io()). The host can also set up shadow doorbells with the standard Doorbell Buffer Config command (nvme\_set\_db\_memory()); a queue with a shadow doorbell is then read from that buffer, and FEMU publishes EventIdx values so the host can skip doorbell writes it does not need.

#### Pollers

When the host enables the controller, nvme\_start\_dataplane() creates the poller threads (femu-poller) once. Each poller loops over the I/O queues it owns (nvme\_poller() in hw/femu/nvme-io.c). How many pollers there are is set by two properties:

| multipoller_enabled | Pollers                     | Queues per poller                              |
|---------------------|-----------------------------|------------------------------------------------|
| 0 (default)         | 1                           | all I/O queues                                 |
| 1                   | ceil(queues / poller_ratio) | queues i, i + N, i + 2N, ... for poller i of N |

Any other value fails realize. A poller fetches commands from a queue without a lock, so every queue must have exactly one owner.

Each poller owns three structures, all indexed by poller number:

- to\_ftl[i]: a ring of requests waiting for timing.
- to\_poller[i]: a ring of timed requests coming back from the FTL thread.
- pq[i]: a priority queue of timed requests, ordered by completion time.

The rings are in hw/femu/lib/rte\_ring.c and the priority queue in hw/femu/lib/pqueue.c.

#### I/O dispatch

For each new submission queue entry, nvme\_process\_sq\_io():

1. copies the command and stamps the request with the current host time (QEMU\_CLOCK\_REALTIME) as both its start time and its completion time;
2. calls nvme\_io\_cmd(), which handles the commands every block mode shares (Flush, Dataset Management, Compare, Write Zeroes, Copy, Verify, Write Uncorrectable, I/O Management) and passes the rest to the namespace's mode (ns->ext\_ops.io\_cmd);

3. puts the request on `to_ftl[i]`.

For Read and Write on NoSSD, BBSSD and CSD, the mode handler calls `nvme_rw()`, which checks the command, maps the guest's PRP or SGL list (`hw/femu/dma.c`) and copies the data between guest memory and the memory backend with `backend_rw()`. The data transfer is finished at this point, on the poller thread, before any time has been charged. ZNS and OCSSD have their own read and write paths that copy to the same backend, and KV copies values into a store of its own.

On a NoSSD controller, NoSSD namespaces skip the rings: the poller posts the completion inside the same sweep, unless the host-link or firmware-CPU model is enabled.

## Completion

`nvme_process_cq_cpl()` runs at the end of every sweep. It drains the timed requests (from `to_poller[i]` when an FTL thread exists, otherwise straight from `to_ftl[i]`), adds the optional host-link and firmware-CPU time, and inserts them into `pq[i]`. Then it posts every request whose completion time has passed and raises one interrupt per completion queue it posted to. A request that is not due yet stays in the queue for the next sweep. This is where FEMU enforces latency: nothing sleeps, the completion is simply not posted before its time.

## CXL accesses

The CXL device has no queues. A guest load or store to its range is an MMIO access on the vCPU thread, routed through the host bridge and endpoint decoders to `femu_cxl_access()` in `hw/femu/cxlssd/cxlssd.c`. A page-granular cache (`hw/femu/cxlssd/cache.c`, policies `fifo`, `lifo`, `clock`, `s3-fifo`) decides whether the access needs the FTL. The cache keeps only page numbers and dirty bits; the data is always in the memory backend. With `der=memslot` or `der=cylon`, cached pages can be mapped into the guest so that later accesses do not reach QEMU at all (see the CXL load walkthrough (§3.1.11)).

### 3.1.4 Mode backends

A mode is a table of handlers (`FemuExtCtrlOps` in `hw/femu/nvme.h`). `nvme_register_extensions()` in `hw/femu/femu.c` installs the table for the controller's `femu_mode`, and `nvme_register_extensions_ns()` gives each namespace the table for its own mode, so one controller can mix modes with `namespace_modes`. I/O commands are dispatched on the namespace's mode, not the controller's.

| Mode  | femu_<br>mode | Code                                                                                                                 | What the I/O handler does                           | Where time is computed                                                                           |
|-------|---------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------|
| OCSSD | 0             | <code>hw/femu/ocssd/oc12.c</code> ( <code>lver=1</code> ), <code>hw/femu/ocssd/oc20.c</code> ( <code>lver=2</code> ) | Open-Channel vector commands; the host runs the FTL | in the poller, from chip and channel busy-until times ( <code>hw/femu/nand/nand-media.c</code> ) |
| BBSSD | 1             | <code>hw/femu/bbssd/bb.c</code>                                                                                      | Read and Write through <code>nvme_rw()</code>       | FTL thread, <code>bb_ftl_process_req()</code>                                                    |
| NoSSD | 2             | <code>hw/femu/nosssd/nop.c</code>                                                                                    | Read and Write through <code>nvme_rw()</code>       | none                                                                                             |
| ZNS   | 3             | <code>hw/femu/zns/zns.c</code>                                                                                       | zoned command set, zone state machine               | FTL thread, <code>zns_ftl_process_req()</code>                                                   |
| CSD   | 4             | <code>hw/femu/csd/csd.c</code>                                                                                       | computational storage commands plus block I/O       | FTL thread for NAND; programs and compute-unit time on the <code>femu-csd-cu</code> threads      |

| Mode | femu_<br>mode | Code                  | What the I/O handler does                                   | Where time is computed                      |
|------|---------------|-----------------------|-------------------------------------------------------------|---------------------------------------------|
| KV   | 5             | hw/femu/kvssd/kvssd.c | key-value command set: Store, Retrieve, Delete, Exist, List | in the poller, through the BBSSD NAND model |

Features that are not modes:

- **Flexible Data Placement:** enabled on femu-subsys; only BBSSD places data by reclaim unit (hw/femu/bbssd/ftl-fdp.c).
- **Several namespaces and per-namespace modes:** namespaces, namespace\_sizes, namespace\_modes (nvme\_init\_namespaces() in hw/femu/femu.c).
- **Namespace Management:** ns\_mgmt=on on a NoSSD or BBSSD controller.
- **Metadata and protection information:** meta, mc, pi (hw/femu/nvme-pi.c), NoSSD and BBSSD only.
- **Streams:** streams=on (hw/femu/nvme-streams.c).
- **Persistent Event Log:** always on, kept in pel\_file if set (hw/femu/nvme-pel.c).

Choosing a mode (§4.1) lists which of these combine.

Threads: one FEMU-FTL-Thread per controller, started at realize only when a namespace is BBSSD, ZNS or CSD, or the controller shares a BBSSD subsystem (femu\_needs\_ftl\_thread()). It reads every poller's to\_ftl[i] ring, calls femu\_ftl\_process\_req(), adds the returned latency to the request's completion time and puts it on to\_poller[i]. A controller whose namespaces are all NoSSD, OCSSD or KV has no FTL thread.

### 3.1.5 FTL

#### BBSSD

The BBSSD FTL in hw/femu/bbssd/ is a page-level device FTL. CSD uses it unchanged, and the CXL SSD runs a private instance of it. Its state is struct ssd (hw/femu/bbssd/ftl.h).

- **Geometry.** Channels (nchs) hold LUNs (luns\_per\_ch), LUNs hold planes (pls\_per\_lun), planes hold blocks (blks\_per\_pl), blocks hold pages (pgs\_per\_blk) of secs\_per\_pg sectors of secsz bytes. With the defaults a page is 4 KiB. A line (superblock) is the same block index on every plane of every LUN, so there are blks\_per\_pl lines.
- **Write pointer.** New pages are allocated from the current line in the order channel, then LUN, then plane, then page, so consecutive pages land on different channels and LUNs and can be programmed in parallel.
- **Mapping** (mapping): page (default, a full table in memory), dftl (a cached mapping table of mapping\_cache\_mb whose misses cost NAND reads), hybrid and fast (log-block schemes). Code: hw/femu/bbssd/ftl-map\*.c.
- **Write buffer** (buffer\_size, in pages): a DRAM buffer that accepts writes. Once it reaches buffer\_thres\_pcent, the next write that needs a slot programs a batch of the least recently written pages. Without power\_loss it holds page numbers only, since the data is already in the memory backend. FUA writes and stream writes go straight to NAND, and Flush drains it. Code: hw/femu/bbssd/ftl-datapath.c.
- **Read cache** (read\_cache\_mb, cache\_evict): a DRAM cache of recently read pages (hw/femu/bbssd/ftl-cache.c).
- **Garbage collection.** After every request the FTL thread runs one background GC pass if the share of lines in use has reached gc\_thres\_pcent. A write that finds it at gc\_thres\_pcent\_high first runs GC in

the foreground until it drops below. The victim line is chosen by `gc_policy` (greedy by default). GC reads, programs and erases occupy the same LUNs as host I/O. Code: `hw/femu/bbssd/ftl-line-gc.c`.

- **Placement extras:** `hot_cold_sep` writes overwritten pages to separate lines, `read_reclaim_limit` and `retention_limit_sec` rewrite lines that were read too often or hold old data, and FDP maps reclaim units to lines.
- **Wear:** each block counts its erases. SMART Percentage Used compares the average erase count with `pe_cycles_rated`, `nand_bad_blocks` lowers Available Spare, and `ecc_step_ns` makes reads of worn or old blocks slower.

Entry point: `bb_ftl_process_req()` in `hw/femu/bbssd/ftl.c`, which dispatches to `ssd_read()`, `ssd_write()`, `ssd_trim()` and the others in `hw/femu/bbssd/ftl-datapath.c`.

### Other FTLs

- **ZNS** (`hw/femu/zns/zftl.c`): zones map onto blocks striped over `zns_chnls_per_zone` channels. There is no device GC; the host resets zones. Writes collect in per-zone SRAM write caches (`zns_num_wc`) and are programmed when a cache fills or is evicted. Zone Reset erases the zone's blocks.
- **KV** (`hw/femu/kvssd/kvssd-ftl.c`): a hash index of keys, with values stored on NAND lines laid out with the BBSSD geometry and timing.
- **OCSSD** has no device FTL. The host addresses channels, LUNs, blocks and pages directly.

Threads: FEMU-FTL-Thread runs the BBSSD, CSD and ZNS FTLs. The CXL SSD's FTL runs on its own `femu-cxl-ftl` worker; when an NVMe controller is linked with `cxl_ssd=`, both threads use the same FTL and take turns on one mutex.

### 3.1.6 NAND media timing

`hw/femu/nand/nand-media.c` turns one NAND operation into a completion time. `nand_media_op()` takes a location (channel, LUN, plane, block, page), an operation (read, program, erase) and a start time. It keeps a busy-until time per LUN (BBSSD, CSD, KV) or per plane (ZNS): the operation starts when both the request and that unit are ready, and the unit stays busy until it ends. Units that differ run in parallel. A channel bus with command, data and status phases is modelled only when one of those phases has a non-zero time.

Read, program and erase times come either from flat properties or from built-in per-cell-type tables (`hw/femu/nand/nand.h` for BBSSD, CSD and KV; `hw/femu/zns/zns.h` for ZNS). OCSSD also uses the media layer, with its `flash_type` times set in `hw/femu/ocssd/oc-timing.c`.

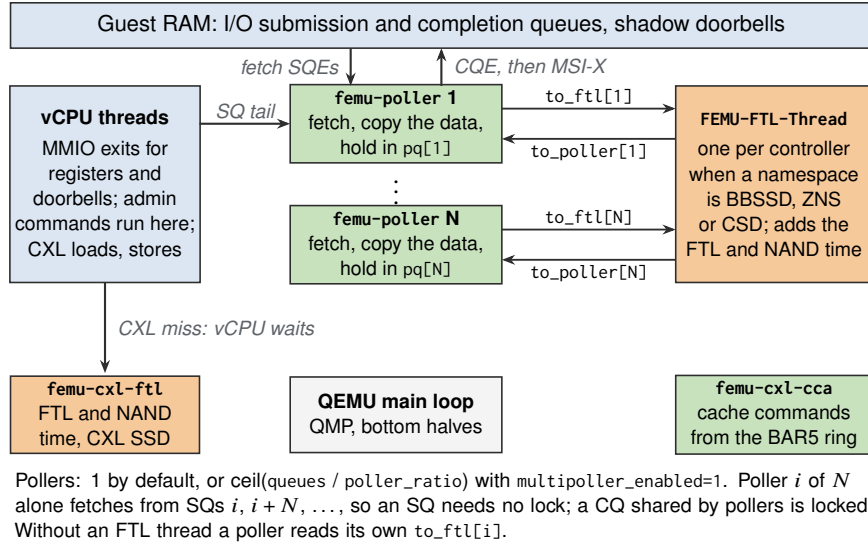
Timing model (§6.2) explains the rules and the properties that control them.

Threads: none of its own. It runs on whichever thread asks: the FTL thread for BBSSD, CSD and ZNS, the poller for KV and OCSSD, and the `femu-cxl-ftl` worker for the CXL SSD.

### 3.1.7 Memory backend

For femu, the data lives in one host buffer of `devsz_mb` MiB (the full NAND capacity for BBSSD with `op_pcent`), allocated zero-filled at realize and locked into RAM when `RLIMIT_MEMLOCK` allows (`init_dram_backend()` in `hw/femu/backend/dram.c`). Namespaces are slices of it, packed back to back. Controllers that share namespaces through a subsystem share one buffer. The block modes read and write it with `backend_rw()`; KV keeps its values in a separate buffer of its own. Nothing is written to a file, so the contents are lost when QEMU exits. `FEMU_MBE_INTERLEAVE` controls its NUMA placement (see the environment variables (§A.1.5)).





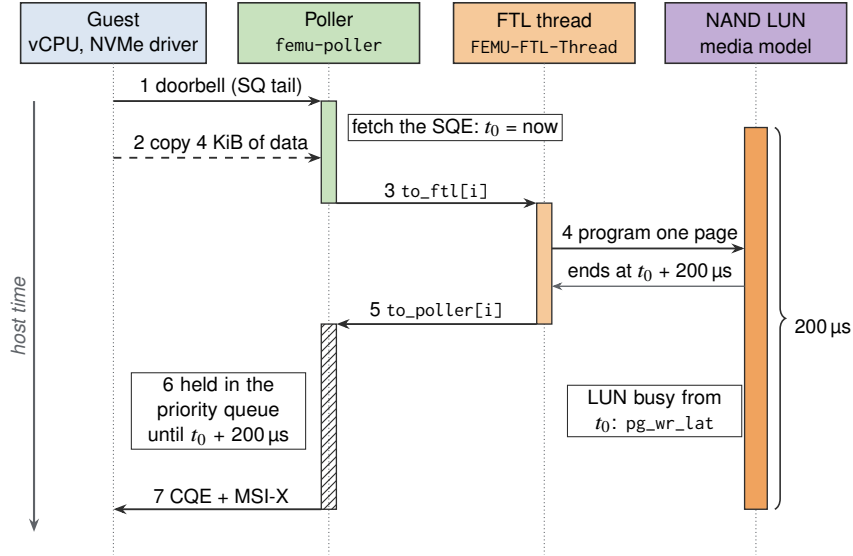
**Figure 4:** FEMU's threads and the queues between them: vCPUs store doorbells, each poller owns its submission queues and its `to_ftl/to_poller` rings, and one FTL thread per controller times BBSSD, ZNS and CSD requests.

For `femu-cxl-ssd`, the data lives in the QEMU memory backend object named by `volatile-memdev`, for example `-object memory-backend-ram,id=cxlmem,size=4G`. It must be a non-zero multiple of 256 MiB and at most 120 GiB. `der=cylon` needs a shared, preallocated `hugetlbfs` backend. The vCPU thread copies data to and from it directly. A femu controller linked with `cxl_ssd=` uses this backend instead of allocating its own.

### 3.1.8 Threads

| Thread name                         | Created by                                                               | How many                                                  | Runs                                                                                                               |
|-------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| vCPU threads and the QEMU main loop | QEMU                                                                     | one per vCPU, plus one main loop                          | MMIO: controller registers, doorbells, admin commands; CXL accesses and their media wait; QMP and monitor commands |
| femu-poller                         | <code>nvme_start_dataplane()</code> when the host enables the controller | 1, or $\text{ceil}(\text{queues} / \text{poller\_ratio})$ | I/O command fetch and execution, data copy, completion and interrupts                                              |
| FEMU-FTL-Thread                     | <code>femu_realize()</code>                                              | 0 or 1 per controller                                     | BBSSD, CSD and ZNS FTL and NAND timing                                                                             |
| femu-cxl-ftl                        | <code>femu_cxl_start()</code>                                            | 1 per femu-cxl-ssd with <code>ftl=on</code>               | FTL and NAND timing for cache misses and write-backs                                                               |
| femu-cxl-cca                        | <code>femu_cxl_cca_start()</code>                                        | 1 per femu-cxl-ssd with <code>cca=on</code>               | cache control commands from BAR5                                                                                   |
| femu-csd-cu                         | <code>csd_init()</code>                                                  | <code>nr_cu</code> per controller with a CSD namespace    | CSD programs (Execute) and their compute-unit time                                                                 |

### 3.1.9 Where latency is charged and enforced



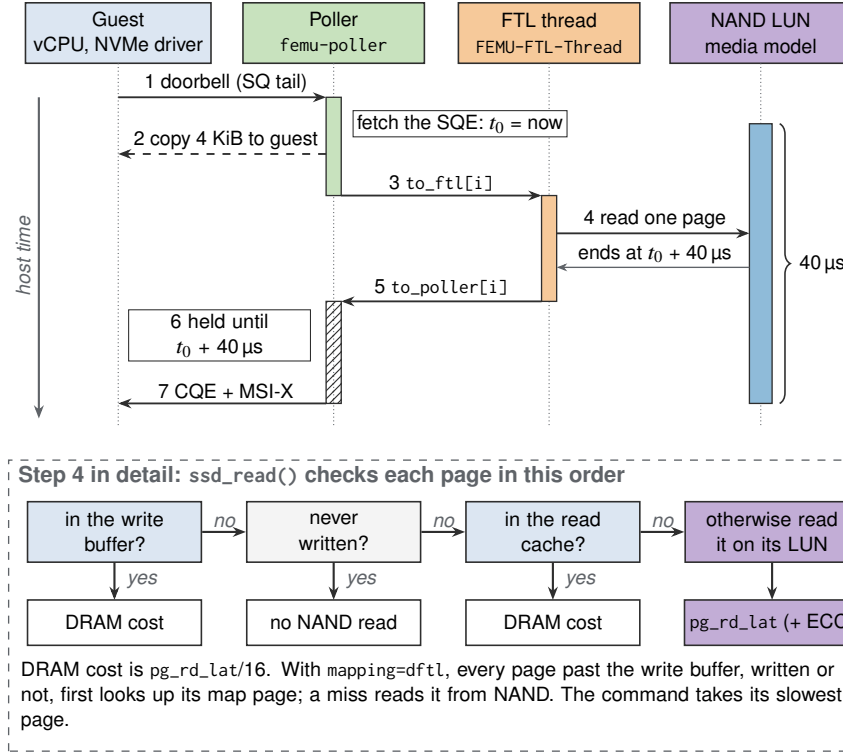
**Figure 5:** One 4 KiB write on BBSSD over host time: the poller copies the data at fetch, the FTL thread books a 200  $\mu$ s program on the LUN from that moment, and the poller holds the completion until it is due.

| Device or mode                                                               | Computed in                                                                                                                                                      | Enforced by                                              |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| NoSSD                                                                        | nothing to compute                                                                                                                                               | completion posted in the same sweep                      |
| BBSSD, CSD, ZNS                                                              | FEMU-FTL-Thread adds the FTL's latency to the completion time; for a CSD Execute, the femu-csd-cu thread that ran the program adds its compute-unit time instead | the poller posts once the time has passed                |
| OCSSD, KV                                                                    | the poller, while executing the command                                                                                                                          | the poller posts once the time has passed                |
| Host link, firmware CPU (pcie_bandwidth_mbps, pcie_prop_delay_ns, fw_cpu_ns) | the poller, before queueing the completion                                                                                                                       | same                                                     |
| femu-cxl-ssd                                                                 | femu-cxl-ftl returns the media time of a miss or write-back                                                                                                      | the vCPU waits it out before the load or store completes |

### 3.1.10 I/O walkthrough: one 4 KiB write on BBSSD

Assume the defaults: femu\_mode=1, one poller, buffer\_size=0, 4 KiB pages, pg\_wr\_lat=200000 (200 us), no channel bus model.

1. **Submit.** The guest NVMe driver writes the command into its submission queue in guest memory and writes the new tail to the queue's doorbell. The vCPU exits to QEMU, and nvme\_process\_db\_io() stores the tail. Nothing else happens on the vCPU.
2. **Fetch and copy.** The poller sees the new tail on its next sweep. nvme\_process\_sq\_io() copies the command and stamps it with the current time  $t_0$ . nvme\_io\_cmd() passes it to bb\_io\_cmd(), which calls nvme\_rw(): the LBA range and size are checked, the PRPs are mapped, and backend\_rw() copies the 4 KiB from guest memory into the memory backend at the namespace's offset. The data is now where later reads will find it.
3. **Hand to the FTL.** The poller puts the request on to\_ftl[1] and goes on with other queues.



**Figure 6:** One 4 KiB read on BBSSD: the guest buffer is filled at fetch and only the completion waits; below, the checks `ssd_read()` makes for each page and what each one costs.

4. **Time it.** The FTL thread takes the request and calls `bb_ftl_process_req()`, which calls `ssd_write()`. If too few lines are free, foreground GC runs first. The one logical page is mapped to the next physical page at the write pointer, the old copy (if any) is marked invalid, and `ssd_advance_status()` asks the media layer for a program on that page's LUN. If the LUN is idle the program takes 200 us from  $t_0$ ; if it is busy until  $t_1$ , it ends at  $t_1 + 200$  us. The LUN stays busy until then. If enough lines are now in use, one background GC pass follows; its NAND operations make the LUNs they use busy for later requests.
5. **Return.** The FTL thread adds the latency to the request's completion time and puts it on `to_poller[1]`.
6. **Hold.** On a following sweep, `nvme_process_cq_cpl()` moves the request into the priority queue. It stays there until the host clock passes its completion time.
7. **Complete.** The poller writes the completion entry into the guest's completion queue and, at the end of the sweep, raises the queue's interrupt. The guest sees the write complete about 200 us after the poller fetched it on an idle device.

With `buffer_size` set, step 4 puts the page in the write buffer instead and charges a DRAM access (`pg_rd_lat / 16`, 2.5 us with the defaults). The program happens on a later write that pushes the buffer past its threshold, and that write pays for it.

### 3.1.11 CXL load walkthrough

The guest loads 8 bytes from a `femu-cxl-ssd` region at host physical address `hpa`. With the cache on (`cache-pages > 0`):

1. **Trap.** Unless the page is mapped directly (below), the load exits to QEMU on the vCPU thread. The `femu-cxl-media` overlay routes it through the window, host bridge and endpoint decoders to a device

address, and calls `femu_cxl_access()`.

2. **Hit.** If the page is in the cache, no media time is charged. The vCPU copies the 8 bytes from the memory backend and returns. The cost is the MMIO exit itself, paid on every access.
3. **Miss.** The page must be brought into the cache:
  - The page is read from NAND: the vCPU drops the BQL, queues a read for the `femu-cxl-ftl` worker, and waits for the latency it returns. A page the FTL has never mapped costs nothing to read.
  - If the page's set is full, the policy evicts a victim. A dirty victim is written back, which costs a NAND program.
  - `femu_cxl_delay()` waits out the rest of the media time, sleeping until 100 us before the deadline and spinning for the rest, with the BQL dropped so other vCPUs keep running.
  - The vCPU copies the data from the memory backend. With `prefetch-degree` set, the next pages are inserted into the cache with no NAND read.

A store follows the same path and marks the page dirty. A store miss reads the page first, as a write-allocate cache does.

What happens to the next access to the same page depends on `der`:

| <code>der</code>           | After a miss or hit                                                                   | Next access to the page                                                                                                  |
|----------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <code>off</code> (default) | nothing                                                                               | traps again; a hit with no media time                                                                                    |
| <code>memslot</code>       | the page is mapped as a one-page KVM memory slot alias (at most 1024 for all devices) | goes straight to host memory with no exit and no time charged; the page counts as dirty, so its eviction costs a program |
| <code>cylon</code>         | the page is mapped by writing the guest's EPT entry through a Cylon host kernel       | goes straight to host memory; dirty state comes from EPT dirty bits                                                      |

A direct mapping is removed when its page is evicted, flushed or invalidated, and the next access traps again. Both direct modes map pages only in the single-endpoint topology the design note describes; elsewhere accesses stay on MMIO. `der=memslot` fails realize under TCG. `der=cylon` fails realize without `cylon-kernel-ack=on`, and falls back to MMIO with a warning when the Cylon host kernel or the `hugetlbfs` backend is missing. The CXL SSD design note<sup>5</sup> gives the details.

### 3.1.12 Related pages

- Choosing a mode (§4.1)
- Timing model (§6.2)
- CXL SSD design note<sup>6</sup>
- Device property reference (§A.1)

## 3.2 FEMU design

This is the design document for FEMU's emulated storage devices. It explains how FEMU is built, one component per chapter, so you can predict what a configuration will do, read the numbers it reports, and

<sup>5</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md>

<sup>6</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md>

change the code.

Read this page first. It says what FEMU models and what it leaves out, shows how the components fit together, follows one request from the guest to the memory backend and back, and gives the reasons for the main design choices. Each component then has its own chapter (see Design pages (§3.2.6)). Every chapter has the same parts: purpose, place in the hierarchy, data structures, algorithms, threads and where time is charged, parameters, counters, validation, limits, how to extend it, and a source map.

For a shorter tour, read Architecture (§3.1). For the property list, read the generated property reference (§A.1). Where this document and the code disagree, the code is right; please report it.

### 3.2.1 What FEMU models

FEMU is QEMU with extra device models under `hw/femu/`. A guest operating system runs unmodified on KVM and sees a storage device built from these models. FEMU models:

- **The host interface.** A PCIe NVMe controller (`-device femu`) with admin and I/O queues, doorbells, shadow doorbells, MSI-X, MSI and pin interrupts, Identify, features, log pages and asynchronous events. A CXL Type-3 memory device (`-device femu-cxl-ssd`) whose loads and stores reach an SSD model.
- **Several kinds of SSD.** `femu_mode` selects one: a conventional SSD with a device FTL (BBSSD), a Zoned Namespace SSD (ZNS), an Open-Channel SSD (OCSSD), a key-value SSD (KV), computational storage (CSD), or a plain memory-backed drive with no media timing (NoSSD). Namespaces of one controller can run different modes.
- **The FTL.** Address mapping (page-level, DFTL, two log-block schemes), a write buffer, a read cache, garbage collection with several victim policies, wear, read and retention reclaim, Flexible Data Placement.
- **NAND timing.** Channels, LUNs, planes, blocks and pages; per-cell-type read, program and erase times; a channel bus; program and erase suspend; ECC read retries as extra time.
- **Optional host-side costs.** A host link with a bandwidth and a propagation delay, and a single controller CPU with a fixed cost per command.

### 3.2.2 What FEMU does not model

- **Cycles.** Nothing is simulated instruction by instruction or clock by clock. Each operation has a cost in nanoseconds, and FEMU holds the completion until that much host time has passed (see Design choices (§3.2.7)).
- **Persistent data.** Device data lives in host memory. A guest reboot or a controller reset keeps it; QEMU exiting loses it. Migration and snapshots are refused (Security and limits (§5.13.2)).
- **Data errors from the medium.** The media model never changes a byte. ECC shows up as extra read time (`ecc_step_ns`). The medium never fails a command on its own: failures come from the injection properties (`err_read_unc_ppm`, `err_write_fail_ppm`) or from rules the host sees (Write Uncorrectable, protection information, Open-Channel chunk state).
- **The PCIe protocol.** There are no TLPs, credits or link training. The optional host-link model is a bandwidth plus a fixed delay.
- **Controller firmware internals.** The optional firmware-CPU model is one fixed cost per Read, Write and Zone Append on one modelled core.
- **Power and heat.** The controller reports one power state, and the temperature is the value of the temperature property. Interrupt coalescing settings are reported but not applied.

- **Firmware update and security commands.** Firmware Download and Commit, Security Send and Security Receive are refused.
- **Exact guest-visible latency.** FEMU enforces a lower bound. A completion is posted on the first poller sweep after it is due, so a poller that does not get a host core adds time (Timing model (§6.2)).

### 3.2.3 Component hierarchy

Everything FEMU adds sits inside the QEMU process. The guest reaches it in two ways: NVMe through PCIe registers and DMA, and CXL through loads and stores to host-managed device memory. Below the interface the two paths share the same FTL and NAND timing code, each device with its own instance.

## GUEST VM

|                                                                              |                                     |
|------------------------------------------------------------------------------|-------------------------------------|
| +-----+ +-----+                                                              |                                     |
| fio, databases, nvme-cli, your program                                       | programs using CXL memory           |
| Linux NVMe driver (or SPDK, xNVMe, ...)                                      | Linux CXL driver: region, dax, NUMA |
| +-----+ +-----+                                                              |                                     |
| MMIO: registers, doorbells                                                   | loads and stores to                 |
| DMA: queue entries, PRP/SGL data                                             | the device's range                  |
| ===== KVM exit or guest memory =====                                         |                                     |
| QEMU PROCESS v                                                               | v                                   |
| +-----+ +-----+                                                              |                                     |
| 1. INTERFACE -device femu                                                    | 1. INTERFACE -device femu-cxl-ssd   |
| PCIe NVMe controller, NVMe 2.1                                               | CXL Type-3 volatile memory,         |
| BAR0 registers + doorbells                                                   | subclass of cxl-type3               |
| BAR4 MSI-X, MSI, pin; BAR2 CMB                                               | I/O overlay "femu-cxl-media" on     |
| -device femu-subsys (optional)                                               | each reachable CXL window           |
| CNTLID space, FDP endurance group,                                           | BAR5 cache control (cca=on)         |
| namespaces shared between controllers                                        | label area channel (lsa-control)    |
| +-----+ +-----+                                                              |                                     |
|                                                                              |                                     |
| +-----+ +-----+                                                              |                                     |
| 2. NVMe FRONTEND                                                             | 2. CXL ACCESS PATH                  |
| admin queue: vCPU thread                                                     | HDM decode, page cache model        |
| I/O queues: femu-poller threads                                              | (fifo, lifo, clock, s3-fifo),       |
| rings to_ftl[i], to_poller[i]                                                | direct mapping (der=memslot,        |
| completion heap pq[i] per poller                                             | der=cylon); vCPU waits media time   |
| +-----+ +-----+                                                              |                                     |
| ns->ext_ops.io_cmd (per namespace)                                           | cache misses,                       |
| +-----+ +-----+                                                              |                                     |
| 3. MODE EXTENSIONS (FemuExtCtrlOps)                                          | write-backs                         |
| NoSSD BBSSD ZNS OCSSD KV CSD                                                 |                                     |
| +-----+ +-----+                                                              |                                     |
|                                                                              |                                     |
| +-----+ +-----+                                                              |                                     |
| 4. FTL BBSSD: page   dftl   hybrid   fast mapping, write buffer, read cache, |                                     |
| GC policies, wear, reclaim, FDP placement (also used by CSD, CXL SSD)        |                                     |
| ZNS: zone FTL and write caches KV: hash index over NAND lines                |                                     |
| OCSSD: none, the host is the FTL                                             |                                     |
| +-----+ +-----+                                                              |                                     |
|                                                                              |                                     |
| +-----+ +-----+                                                              |                                     |
| 5. NAND TIMING nand-media.c: busy-until time per LUN or plane, channel bus,  |                                     |
| cell-type tables, suspend, ECC oc-timing.c: OCSSD 0xEE times                 |                                     |
| +-----+ +-----+                                                              |                                     |

## 6. MEMORY BACKEND (data only; layers 4 and 5 never touch it)

femu: one host buffer of devsz\_mb MiB (backend/dram.c), namespaces packed in it;  
BBSSD with op\_pcent: the NAND capacity; shared subsystem or cxl\_ssd:  
borrowed from the subsystem or the medium  
femu-cxl-ssd: the memory-backend object named by volatile-memdev  
KV: a value store of its own  
NVMe: layers 2 and 3 copy data before any time is charged (power\_loss=on:  
the FTL thread does). CXL: the vCPU copies after the media wait.

| Layer        | What it does                                             | Chapter                                                               |
|--------------|----------------------------------------------------------|-----------------------------------------------------------------------|
| 1. Interface | PCI function, registers, CXL decode, subsystem           | nvme-frontend.md (§3.3),<br>cxl-ssd.md (§4.19), namespaces.md (§4.16) |
| 2. Frontend  | queues, pollers, dispatch, completion timing, interrupts | nvme-frontend.md (§3.3)                                               |

| Layer              | What it does                                      | Chapter                                                                                           |
|--------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 3. Mode extensions | the command set and behaviour of each kind of SSD | ocssd.md (§4.10), zns.md (§4.8), kvssd.md (§4.12), csd.md (§4.14), nossd.md (§4.3), fdp.md (§4.6) |
| 4. FTL             | mapping, buffering, GC, wear                      | ftl.md (§3.4)                                                                                     |
| 5. NAND timing     | how long each NAND operation takes                | nand-timing.md (§3.5)                                                                             |
| 6. Memory backend  | where the bytes live                              | this page, namespaces.md (§4.16)                                                                  |

### Device types

| QOM type     | Parent                         | On the command line                                         | Role                                                           |
|--------------|--------------------------------|-------------------------------------------------------------|----------------------------------------------------------------|
| femu         | TYPE_PCI_DEVICE, PCIe endpoint | -device femu,...                                            | the NVMe controller and all its namespaces                     |
| femu-subsys  | TYPE_DEVICE, on no bus         | -device femu-subsys,id=s0,... before the controllers        | joins controllers (subsys=s0), holds FDP and shared namespaces |
| femu-cxl-ssd | cxl-type3                      | -device femu-cxl-ssd,volatile-memdev=... on a CXL root port | a CXL memory device backed by an SSD model                     |

There is no namespace device: a controller builds its namespaces from namespaces, namespace\_sizes and namespace\_modes, and Namespace Management adds more at run time. femu-subsys creates an nvme-bus (a type FEMU borrows from QEMU's own NVMe controller) but no FEMU device plugs into it. A femu controller can also serve a femu-cxl-ssd medium as its one namespace with cxl\_ssd=<id>; it then uses the medium's memory and FTL instead of its own.

### 3.2.4 How a request flows

#### One NVMe write on a BBSSD namespace

The data moves first; the time is computed afterwards and enforced at completion. Time runs downward.

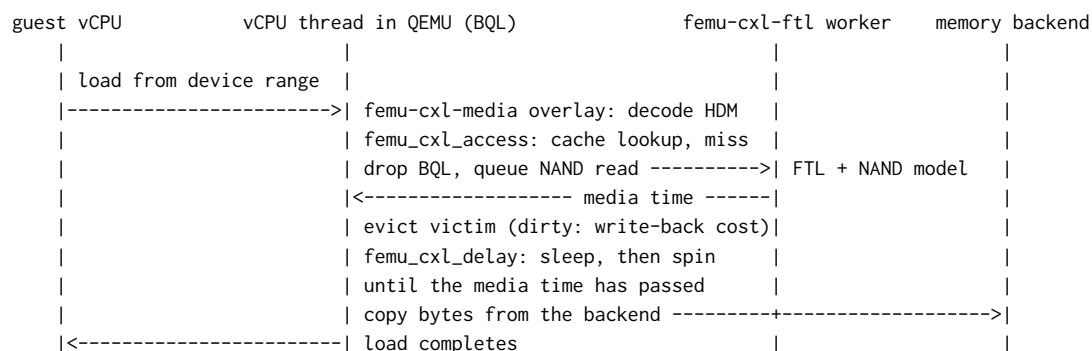


| guest driver | vCPU thread             | femu-poller i                     | FEMU-FTL-Thread      | backend |
|--------------|-------------------------|-----------------------------------|----------------------|---------|
| 1            | SQE into SQ ring        |                                   |                      |         |
|              | tail doorbell --->      | nvme_process_db_io                |                      |         |
|              | (or shadow entry)       | stores sq->tail                   |                      |         |
|              |                         | (nothing else)                    |                      |         |
| 2            |                         | sweep: nvme_process_sq_io         |                      |         |
|              |                         | copy SQE, stime = now,            |                      |         |
|              |                         | expire_time = now                 |                      |         |
| 3            |                         | nvme_io_cmd -> ns ext_ops         |                      |         |
|              |                         | -> nvme_rw: check, map PRP        |                      |         |
|              |                         | backend_rw copies data ---+-----> |                      |         |
| 4            |                         | to_ftl[i] enqueue ----->          |                      |         |
| 5            |                         |                                   | femu_ftl_process_req |         |
|              |                         |                                   | FTL: map, GC, ...    |         |
|              |                         |                                   | NAND: program time   |         |
|              |                         |                                   | expire_time += lat   |         |
|              |                         | <----- to_poller[i] ---           |                      |         |
| 6            |                         | nvme_process_cq_cpl:              |                      |         |
|              |                         | + host link, + firmware CPU,      |                      |         |
|              |                         | insert into heap pq[i]            |                      |         |
|              |                         | next sweeps: is now >=            |                      |         |
|              |                         | expire_time? not yet: keep        |                      |         |
| 7            | <----- CQE written ---- | yes: post CQE, then one           |                      |         |
|              | <----- interrupt -----  | interrupt per CQ per sweep        |                      |         |
| 8            | CQ head doorbell->      | head stored                       |                      |         |

1. The guest writes a submission queue entry and rings the tail doorbell. The vCPU exits to QEMU, which stores the new tail and returns. With shadow doorbells the guest also writes the value to its own memory and rings the register only when the EventIdx protocol asks for it.
2. The poller that owns the queue finds the new entry on its next sweep and stamps the request with the host time.
3. The command is checked and dispatched to the namespace's mode. For a block Read or Write the data is copied between guest memory and the backend now, on the poller thread. (With power\_loss=on the FTL thread executes the command, data copy included, in step 5.)
4. The request goes to the FTL thread through a lock-free ring.
5. The FTL decides where the page goes, runs GC if needed, and asks the NAND model when the program would end. The difference is added to the request's completion time.
6. Back on the poller, optional host-link and firmware-CPU time is added, and the request waits in a min-heap ordered by completion time.
7. Once the host clock passes that time, the poller writes the completion entry and raises the queue's interrupt.
8. The guest consumes the entry and moves the completion queue head.

Modes differ only in steps 3 to 5. NoSSD charges nothing and, on a NoSSD controller with the default settings, posts the completion inside step 3's sweep. OCSSD and KV compute their time on the poller in step 3 and skip the FTL thread. ZNS and CSD use the FTL thread like BBSSD. nvme-frontend.md (§3.3) has the full command lifecycle.

## One CXL load that misses



A hit skips the worker and the wait. With direct mapping (der), subsequent accesses to a cached page do not leave the guest at all. cxl-ssd.md (§4.19) covers the cache, the decoders and direct mapping.

## 3.2.5 Threads and where time is charged

vCPU threads ----- MMIO: registers, doorbells, admin queue, AER, CXL accesses (+ wait)  
 main loop ----- QMP, interrupt bottom halves without irqfd, AER posting, CXL BHs  
 femu-poller x N -- I/O fetch, command execution and data copy (not with power\_loss=on),  
 OCSSD/KV/CSD-compute time,  
 host-link and firmware-CPU time, completion posting, interrupts  
 FEMU-FTL-Thread -- BBSSD, CSD and ZNS FTL and NAND time (0 or 1 per controller);  
 with power\_loss=on it also executes each I/O command; for a  
 cxl\_ssd link it calls into the medium's FTL  
 femu-cxl-ftl ----- FTL and NAND time for CXL misses and write-backs (1 per medium)  
 femu-cxl-cca ----- cache control commands from BAR5 (only with cca=on)

| Device or mode                                         | Time computed in                        | Time enforced by                           |
|--------------------------------------------------------|-----------------------------------------|--------------------------------------------|
| NoSSD                                                  | nothing to compute                      | poller heap, usually in the same sweep     |
| BBSSD, ZNS, CSD NAND time                              | FEMU-FTL-Thread                         | poller heap                                |
| OCSSD, KV, CSD compute units and CSD memory-copy reads | the poller, while executing the command | poller heap                                |
| Host link, firmware CPU                                | the poller, before the heap             | poller heap                                |
| femu-cxl-ssd                                           | femu-cxl-ftl                            | the vCPU waits before the access completes |

## 3.2.6 Design pages

| Page                    | Component                                                                                                                                             |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| nvme-frontend.md (§3.3) | The NVMe controller: PCI and BAR layout, doorbells and shadow doorbells, queues, pollers, dispatch, completion timing, interrupts, reset and teardown |
| ftl.md (§3.4)           | The BBSSD FTL: mapping schemes, write buffer, read cache, lines, GC and victim policies, wear and reclaim                                             |
| nand-timing.md (§3.5)   | The NAND media layer: geometry, cell types, per-LUN and per-plane busy time, channel bus, suspend, ECC                                                |
| zns.md (§4.8)           | Zoned Namespaces: zone state machine, zone FTL, write caches, Zone Append, ZRWA                                                                       |
| fdp.md (§4.6)           | Flexible Data Placement: endurance group, reclaim units and handles, placement and GC                                                                 |

| Page                  | Component                                                                                          |
|-----------------------|----------------------------------------------------------------------------------------------------|
| namespaces.md (§4.16) | Namespaces and the subsystem: sizing, per-namespace modes, Namespace Management, shared namespaces |
| ocssd.md (§4.10)      | Open-Channel SSD 1.2 and 2.0: the host-managed interface and its timing                            |
| kvssd.md (§4.12)      | The NVMe Key Value command set and its FTL                                                         |
| csd.md (§4.14)        | Computational storage: compute units, device memory, programs                                      |
| nossd.md (§4.3)       | NoSSD: memory-backed NVMe with no media timing                                                     |
| cxl-ssd.md (§4.19)    | The CXL SSD: address decoding, page cache, direct mapping, cache control, NVMe link                |

### 3.2.7 Design choices

Each choice below says what FEMU does and why.

**Charge latency, then hold the completion.** FEMU does not simulate hardware cycles. A command's data is moved at memory speed, a model computes how long the command would take, and the completion is withheld until the host clock passes that time. The guest therefore runs at native speed with its real driver and file system, and still sees device-like latencies. The price is that FEMU sets a lower bound: if the poller does not run, the completion is posted after it is due. Give pollers and the FTL thread their own cores when you measure (performance tuning (§5.12.1)).

**Host wall-clock time.** Requests are stamped with `QEMU_CLOCK_REALTIME`, the clock the guest also observes. A 200 us program makes the guest wait 200 us. There is no virtual time that could diverge from what the guest measures.

**Data in host memory, separate from timing.** The FTL and NAND model keep metadata only: mappings, valid bits, busy-until times. The bytes go straight to one host buffer at the namespace's offset. This keeps the models simple and fast, makes a read return the last write no matter what the FTL did, and lets a mapping bug change timing but never corrupt data. The cost is that capacity is bounded by host RAM and data does not survive QEMU. The power-loss model (`power_loss=on`) is the one place the FTL thread applies data, because it has to roll back unflushed writes.

**Polling threads instead of trapping each I/O.** A doorbell write only stores a number. Dedicated femu-poller threads fetch commands and post completions. Doing the work on the vCPU would make every command pay a VM exit and stall the guest CPU, and a trap-driven design cannot post a completion at a chosen microsecond. The cost is host cores that spin while the controller is enabled.

**One owner per queue.** A poller fetches from a submission queue without a lock, so each queue has exactly one poller. `multipoller_enabled` accepts only 0 (one poller, all queues) or 1 (queues split round-robin). An older value greater than 1 started several pollers that each swept every queue, and two of them then executed and completed the same command; realize now refuses it.

**A separate FTL thread with single-threaded state.** One FEMU-FTL-Thread per controller owns the FTL and NAND state of every BBSSD, ZNS and CSD namespace, so that state needs no locks. Pollers hand requests over through single-consumer rings and keep fetching while the FTL computes.

**Modes as handler tables.** A mode is a `FemuExtCtrlOps` table: init, exit, start, admin, I/O and log hooks. The frontend handles everything shared (queues, Identify, features, common NVM commands) and passes the rest to the table of the namespace a command addresses. Adding a kind of SSD means adding a table, and one controller can serve namespaces of different modes.

**Refuse, do not guess.** Settings that cannot work together fail realize with a message that names the property,

with a few exceptions the property reference states (for example `ns_mgmt`, which stays off without an error on a controller that cannot support it), and properties that are accepted but have no effect print a warning. A configuration that starts is one FEMU can serve as described.

**CXL accesses wait on the vCPU.** A load or store cannot complete out of order the way an NVMe command can, so the `femu-cxl-ssd` path charges media time on the vCPU thread that made the access, with the BQL released so other vCPUs keep running.

### 3.2.8 Glossary

| Term                                               | Meaning                                                                                                                                         |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| BAR                                                | PCI Base Address Register: a window of device registers or memory. <code>femu</code> uses BAR0 (registers, doorbells), BAR2 (CMB), BAR4 (MSI-X) |
| BBSSD                                              | black-box SSD, <code>femu_mode=1</code> : a conventional SSD with a device FTL                                                                  |
| BQL                                                | QEMU's big lock. vCPU MMIO handlers and the main loop hold it; pollers and FTL threads do not                                                   |
| CC, CSTS                                           | Controller Configuration and Controller Status registers; <code>CC.EN</code> enables the controller, <code>CSTS.RDY</code> says it is ready     |
| CMB                                                | Controller Memory Buffer, device memory the host can place queues or data in ( <code>cmbsz</code> , <code>cmbloc</code> )                       |
| CNTLID                                             | controller identifier, unique within a subsystem                                                                                                |
| CQ, SQ, CQE, SQE                                   | completion and submission queues and their entries                                                                                              |
| CSD                                                | computational storage device, <code>femu_mode=4</code>                                                                                          |
| DER                                                | direct mapping of cached CXL SSD pages into the guest ( <code>der=memslot</code> or <code>der=cylon</code> )                                    |
| DFTL                                               | a demand-loaded page mapping table with a cached mapping table ( <code>mapping=dftl</code> )                                                    |
| Doorbell                                           | a BAR0 register the host writes a queue's new tail (SQ) or head (CQ) to                                                                         |
| EventIdx                                           | in the Doorbell Buffer Config scheme, the value the controller publishes so the host knows when it must still ring the real doorbell            |
| <code>expire_time</code>                           | a request's completion time in host nanoseconds; the poller posts it no earlier                                                                 |
| <code>ext_ops</code>                               | a mode's <code>FemuExtCtrlOps</code> handler table, per controller and per namespace                                                            |
| FDP                                                | Flexible Data Placement: reclaim units and placement handles on <code>femu-subsys</code>                                                        |
| FTL                                                | flash translation layer: maps logical pages to NAND pages and reclaims space                                                                    |
| GC                                                 | garbage collection: copying valid pages out of a line so it can be erased                                                                       |
| HDM                                                | host-managed device memory, the address range a CXL Type-3 device decodes                                                                       |
| <code>irqfd</code>                                 | a KVM mechanism that lets a thread inject an MSI-X interrupt without the BQL                                                                    |
| KV                                                 | key-value SSD, <code>femu_mode=5</code> , the NVMe Key Value command set                                                                        |
| Line                                               | a superbloc: the same block index on every plane of every LUN; the unit of GC                                                                   |
| LUN                                                | a NAND die that runs one operation at a time; channels hold LUNs                                                                                |
| NoSSD                                              | <code>femu_mode=2</code> , the default: NVMe backed by memory with no media timing                                                              |
| OCSSD                                              | Open-Channel SSD, <code>femu_mode=0</code> : the host addresses NAND directly                                                                   |
| <code>pq[i]</code>                                 | poller <code>i</code> 's min-heap of timed requests, ordered by <code>expire_time</code>                                                        |
| PRP, SGL                                           | the two ways an NVMe command describes its data buffers                                                                                         |
| Poller                                             | a <code>femu-poller</code> thread that fetches I/O commands and posts completions                                                               |
| Realize                                            | QEMU's device construction step; property checks happen here                                                                                    |
| Shadow doorbell                                    | a host-memory copy of the doorbells set up with Doorbell Buffer Config; FEMU polls it                                                           |
| Sweep                                              | one pass of a poller over its queues and its completion heap                                                                                    |
| <code>to_ftl[i]</code> , <code>to_poller[i]</code> | the rings between poller <code>i</code> and the FTL thread                                                                                      |

| Term | Meaning                                                                 |
|------|-------------------------------------------------------------------------|
| WAF  | write amplification factor: NAND pages programmed per host page written |
| ZNS  | Zoned Namespace SSD, femu_mode=3                                        |

### 3.2.9 Source map

| Path                                                        | Contents                                                                                                                      |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| hw/femu/femu.c                                              | QOM types femu and femu-subsys, realize and exit, property checks, registers and doorbells, the FTL thread, mode registration |
| hw/femu/femu-props.c                                        | property help text                                                                                                            |
| hw/femu/nvme-admin.c, nvme-io.c, nvme-util.c, dma.c, intr.c | the NVMe frontend                                                                                                             |
| hw/femu/nvme-pel.c, nvme-pi.c, nvme-streams.c               | Persistent Event Log, protection information, Streams                                                                         |
| hw/femu/nossd/, bbssd/, zns/, ocssd/, kvssd/, csd/          | the modes; bbssd/ also holds the FTL                                                                                          |
| hw/femu/nand/, hw/femu/ocssd/oc-timing.c                    | NAND timing                                                                                                                   |
| hw/femu/backend/dram.c                                      | the memory backend                                                                                                            |
| hw/femu/cxlssd/                                             | the CXL SSD                                                                                                                   |
| hw/femu/lib/                                                | rings and the priority queue                                                                                                  |
| hw/femu/nvme.h                                              | shared structures, constants and the mode enum                                                                                |

Code structure (§8.1) lists every file.

### 3.2.10 Related pages

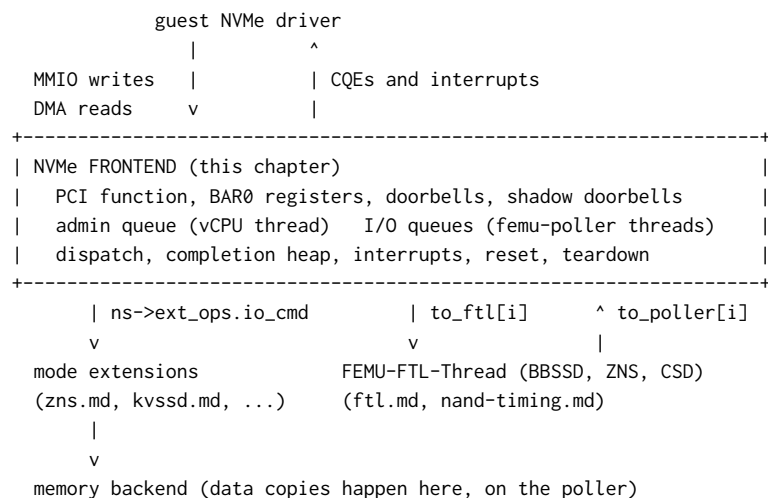
- Architecture (§3.1)
- Timing model (§6.2)
- Choosing a mode (§4.1)
- Security and limits (§5.13)
- Device property reference (§A.1)
- Log pages and counters (§B)

## 3.3 NVMe frontend

The NVMe frontend is the part of `-device femu` that a guest driver talks to: the PCI function, the controller registers, the admin and I/O queues, the poller threads that fetch and complete commands, and the interrupts. It turns NVMe commands into calls to a mode (BBSSD, ZNS, KV and the others) and holds each completion until the time the mode charged has passed.

This chapter describes the frontend as the code in this tree implements it. The design overview (§3.2) shows where it sits among the other components.

### 3.3.1 Place in the hierarchy



### 3.3.2 PCI function and BAR layout

The controller is a PCIe endpoint with class code 01 08 02 (NVMe). Its vendor and device IDs come from vid and did (by default 1d1d:1f1f), and it reports NVMe version 1.4. It has INTA, an MSI capability with up to 32 vectors at config offset 0x50, a PCIe capability at 0x80, and MSI-X with queues + 1 vectors in a BAR of its own. `nvme_init_pci()` sets this up.

| BAR | Contents                                    | When                                     |
|-----|---------------------------------------------|------------------------------------------|
| 0-1 | controller registers and doorbells (64-bit) | always                                   |
| 2-3 | Controller Memory Buffer (64-bit)           | cmbsz non-zero; cmbloc must select BAR 2 |
| 4-5 | MSI-X table and pending bits                | always                                   |

```

BAR0, reg_size bytes
offset  register  FEMU behaviour
0x0000  CAP      MQES = entries, CQR = cqr, AMS = 0, TO = 0xf, DSTRD = stride,
                NSSRS = 0, CSS = NVM + I/O command sets by CSI,
                MPSMIN = mpsmin, MPSMAX = mpsmax,
                CRMS = CRWMS (not with OCSSD)                                read only
0x0008  VS       0x00020100 (2.1); 0x00010400 (1.4) with OCSSD                read only
0x000c  INTMS    set interrupt mask bits (MSI and pin; ignored with MSI-X)
0x0010  INTMC    clear interrupt mask bits; a held MSI is sent when unmasked
0x0014  CC       EN, SHN, MPS, IOSQES, IOCQES, CSS; AMS must be 0 (round
                robin); see the state machine below
0x001c  CSTS     RDY, CFS, SHST                                              read only
0x0020  NSSR     reads 0; writes ignored (no subsystem reset)
0x0024  AQA     admin ASQS and ACQS, 1..4095 (0's based): 2..4096 entries
0x0028  ASQ     admin SQ base, 64 bits, may be written as two dwords
0x0030  ACQ     admin CQ base, 64 bits, may be written as two dwords
0x0038  CMBLOC  cmbloc                                                        read only
0x003c  CMBSZ   cmbsz                                                        read only
0x0068  CRT0    CRWMT = CAP.TO, CRIMT = 0 (reads 0 with OCSSD)              read only
0x1000  doorbells: DB = 4 << stride bytes
                0x1000 + (2 * qid)      * DB    SQ qid tail
                0x1000 + (2 * qid + 1) * DB    CQ qid head
                qid 0 is the admin pair, 1 .. queues are I/O queues

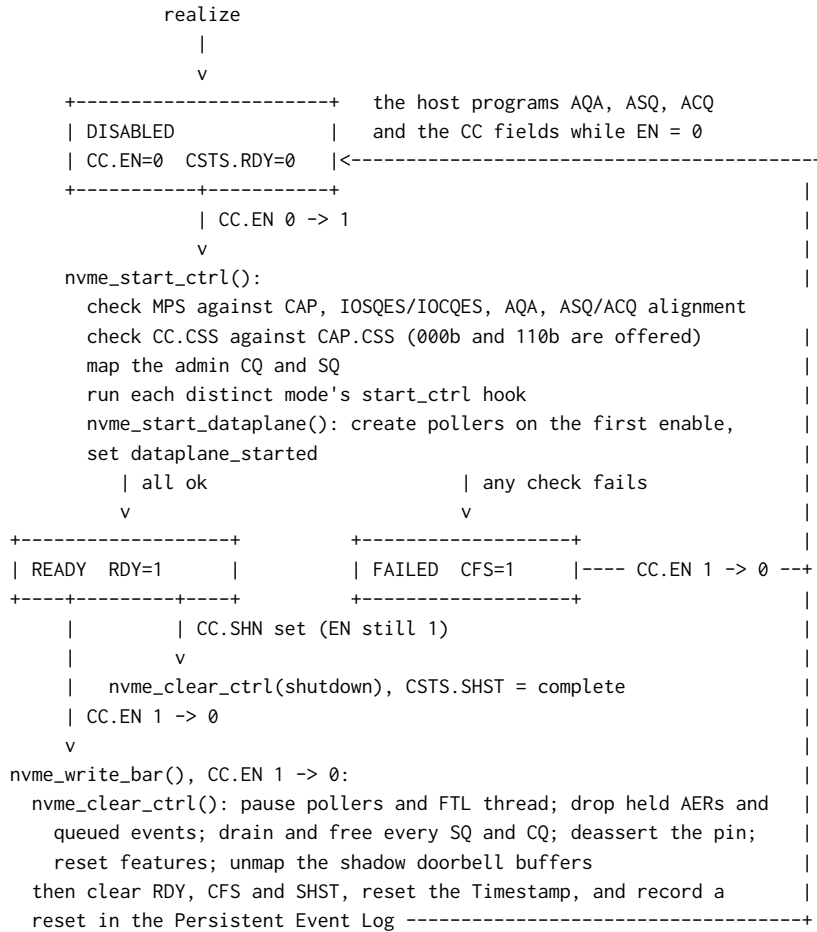
reg_size = max(16 KiB, pow2ceil(0x1000 + 2 * (queues + 1) * DB))

```

BAR0 is at least 16 KiB because the PCIe transport makes bits 13:4 of the BAR read only. Writes go to `nvme_mmio_write()`: offsets inside the register block go to `nvme_write_bar()`, the admin doorbell pair to `nvme_process_db_admin()`, and everything else to `nvme_process_db_io()`. All of them run on the vCPU thread that made the access, holding the BQL.

The CMB is an I/O memory region (`nvme_cmb_ops`) backed by a host buffer of the size `cmbosz` describes. PRP and SGL data pointers, and the entries of PRP-list queues, that fall inside it are served from that buffer (`nvme_addr_is_cmb()` in `hw/femu/dma.c`). A physically contiguous queue is mapped directly and has no CMB case.

### 3.3.3 Controller states



A shutdown with CC.EN still 1 runs `nvme_clear_ctrl()` but leaves RDY set and resets neither the Timestamp nor the Persistent Event Log context.

Pollers are created once and survive a reset; a reset only clears `dataplane_started` until the next enable. With `power_loss=on`, a normal shutdown (CC.SHN = 01b) first flushes the BBSSD write buffer, and a failed flush sets CSTS.CFS.

### 3.3.4 Queues

#### Data structures

| Structure   | Fields that matter                                                                                                                                                                                                                                                 |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NvmeSQueue  | sqid, cqid (the CQ it reports to), head, tail, size; dma_addr and dma_addr_hva (contiguous ring) or prp_list; io_req (one NvmeRequest per slot) and req_list (the free ones); db_addr/db_addr_hva and eventidx_addr/eventidx_addr_hva (shadow doorbell); is_active |
| NvmeCQueue  | cqid, head, tail, size, phase; vector, irq_enabled, virq and guest_notifier (irqfd route), irq_bh (fallback); sq_list; post_lock; the same four shadow doorbell fields; is_active                                                                                  |
| NvmeRequest | cmd, cqe, sq, ns, status; slba, nlb, is_write, xfer_bytes; stime, expire_time, reqlat (ns); cxl_seq                                                                                                                                                                |



| Structure | Fields that matter                                                                                                                                                                                                             |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FemuCtrl  | sq[], cq[] (index 0 is admin); nr_pollers, poller[]; to_ftl[], to_poller[], pq[], cpl_backlog[], should_isr, poller_in_sweep[], poller_ctr[] (all indexed by poller, 1-based); dataplane_started, ftl_in_sweep, use_ftl_thread |

A submission queue owns one request per slot, so it can have at most size commands outstanding; when its free list is empty the poller leaves the rest of the queue for a following sweep.

### Admin queue

The admin queue pair is created from AQA, ASQ and ACQ when the controller is enabled. It is processed synchronously: a write to the admin SQ tail doorbell calls `nvme_process_sq_admin()` on the vCPU thread, which fetches each new entry, runs `nvme_admin_cmd()`, writes the completion and notifies the admin CQ's interrupt. It fetches only while the admin CQ has a free slot; a CQ head doorbell write resumes it. The admin queue always uses its doorbell registers, even after Doorbell Buffer Config.

Admin commands that free or replace state the I/O path reads (Delete SQ and CQ, Format NVM, Namespace Management and Attachment, Sanitize, Streams, some Set Features) first stop the pollers and the FTL thread with `nvme_pause_pollers()` and restart them after (see Quiescing the data plane (§3.3.10)).

### I/O queues

The host creates I/O queues with Create I/O CQ and Create I/O SQ. Queue ids run from 1 to queues, sizes from 2 to entries + 1. With `cqr=1` (the default) queues must be physically contiguous; with `cqr=0` a queue may be a PRP list. A completion queue's interrupt vector must be at most queues, and below the number of MSI vectors the guest enabled when MSI is in use. Several SQs may share one CQ.

A new SQ is published with a write barrier before `is_active` is set, and the poller reads `is_active` with a read barrier before using the queue, so a poller never sees a half-built queue. Delete I/O SQ pauses the data plane, returns the queue's requests to it, and frees it. Delete I/O CQ is refused while an SQ still reports to it.

Set Features Number of Queues reports queues for both SQs and CQs and is accepted only before any I/O queue exists.

## 3.3.5 Doorbells and shadow doorbells

### Doorbell registers

A doorbell write records the new value and returns. For an I/O SQ the vCPU stores `sq->tail`; the poller notices it on its next sweep. For an I/O CQ it stores `cq->head`, notifies the CQ again if entries remain, and on the pin updates the interrupt level. A write to a doorbell that does not exist, or of a value past the end of its queue, is ignored and, while the controller is ready, raises an Error asynchronous event (event information 00h, write to an invalid doorbell register, or 01h, invalid doorbell write value). An I/O doorbell write that is not aligned to DB, and a register write to a queue that has a shadow doorbell, are dropped without an event (for a shadowed CQ the write only updates the pin level).

### Doorbell Buffer Config

The controller always advertises Doorbell Buffer Config (OACS bit 8). The host gives two page-aligned pages of its memory with this admin command: a shadow doorbell page, where it writes tails and heads instead of the registers, and an EventIdx page, where the controller writes back hints. `nvme_set_db_memory()` maps both pages and points every existing I/O queue at its two entries:

| shadow doorbell page (host writes) |                   | EventIdx page (controller writes) |                    |
|------------------------------------|-------------------|-----------------------------------|--------------------|
| entry size DB = 4 << stride bytes  |                   | same layout                       |                    |
| offset                             | +-----+           | +-----+                           |                    |
| 0 * DB                             | SQ0 tail (unused) | SQ0 EventIdx                      | admin pair: kept   |
| 1 * DB                             | CQ0 head (unused) | CQ0 EventIdx                      | equal to the value |
| 2 * DB                             | SQ1 tail          | SQ1 EventIdx                      | just written       |
| 3 * DB                             | CQ1 head          | CQ1 EventIdx                      |                    |
| ...                                | ...               | ...                               |                    |
| 2q * DB                            | SQq tail          | SQq EventIdx                      |                    |
| (2q + 1) * DB                      | CQq head          | CQq EventIdx                      |                    |
|                                    | +-----+           | +-----+                           |                    |

Rules the code enforces:

- Both addresses must be non-zero and page aligned, and both pages must map completely; otherwise the command fails with Invalid Field and nothing is recorded.
- The pages are set once per enable. A second command before a reset fails.
- All entries must fit in one page: (2 x queues + 1) x DB bytes at most. With a 4 KiB page and stride=0 that is 511 I/O queues.
- Each queue that already exists is seeded with its current tail or head before it is switched to its entry, so a poller never reads a stale zero. A queue created afterwards starts at 0 and points at its entry at once, relying on the host's zeroed entry.
- A controller reset unmaps both pages and forgets them.

Once a queue has a shadow doorbell, its register doorbell no longer moves it. The poller reads the SQ tail from the shadow at the start of every sweep (nvme\_update\_sq\_tail()) and the CQ head whenever it checks for space or posts an interrupt (nvme\_update\_cq\_head()). A value past the end of the queue is ignored.

## EventIdx

With shadow doorbells the host rings the real doorbell only when its new value passes the EventIdx the controller published. FEMU's pollers read the shadow values of every active I/O queue on every sweep, so I/O queues do not depend on doorbell writes. Two queues are handled differently:

- **The admin pair** keeps its EventIdx equal to the value just written, so a host that follows the EventIdx protocol still rings it (the admin queue is driven only by its registers).
- **Pin interrupts.** With neither MSI-X nor MSI enabled, the CQ EventIdx is the head itself. The pin is level triggered and can only drop after the host reports its new head, so the host must ring the CQ head doorbell after every batch; the register write then updates the pin level.

## 3.3.6 Pollers

### How many and who owns what

nvme\_init\_poller() creates the pollers the first time the host enables the controller. Their number depends on two properties:

| multipoller_enabled | Pollers                                                 | Poller i of N owns        |
|---------------------|---------------------------------------------------------|---------------------------|
| 0 (default)         | 1                                                       | every I/O SQ              |
| 1                   | ceil(queues / poller_ratio); poller_ratio=0 counts as 1 | SQs i, i + N, i + 2N, ... |
| anything else       | realize fails                                           |                           |

```

multipoller_enabled=0                multipoller_enabled=1, queues=8, poller_ratio=3
                                     N = ceil(8 / 3) = 3

femu-poller 1: SQ1 SQ2 ... SQ8      femu-poller 1: SQ1 SQ4 SQ7
                                     femu-poller 2: SQ2 SQ5 SQ8
                                     femu-poller 3: SQ3 SQ6

per poller i:  to_ftl[i]    ring, poller i -> FTL thread, holds 65535
                to_poller[i] ring, FTL thread -> poller i, holds 65535
                pq[i]       min-heap on expire_time, starts at 65536, grows
                cpl_backlog[i] completions waiting for CQ space
                should_isr row which CQs to interrupt after this sweep
                poller_in_sweep[i], poller_ctr[i]

| -device femu,femu_mode=2,queues=8,multipoller_enabled=1,poller_ratio=3

```

Every SQ has exactly one owner because a poller fetches from it without a lock. Values of `multipoller_enabled` greater than 1 used to start one poller per shard but had each of them sweep every queue; two pollers then fetched, executed and completed the same command. Realize now refuses them.

A CQ is not owned. Its SQs may belong to different pollers, so with more than one poller a completion is posted under the CQ's `post_lock`, and each poller keeps its own row of `should_isr` flags.

### The sweep

```

nvme_poller(i), forever until poller_stopping:
    if !dataplane_started: in_sweep = false, sleep 1 ms, retry
    in_sweep = true; smp_mb; recheck dataplane_started
    for each SQ this poller owns, active and with an active CQ:
        nvme_process_sq_io(sq, i)          fetch and execute new commands
    nvme_process_cq_cpl(i)                  time and post completions
    smp_mb; in_sweep = false

```

A poller never sleeps while the data plane runs: it spins on a host core. Plan one core per poller and one for the FTL thread (performance tuning (§5.12.1)).

### 3.3.7 Command lifecycle

```
SQ slot in guest memory (or the CMB)
| nvme_process_sq_io(): copy the 64-byte SQE, take a free NvmeRequest,
|   stime = expire_time = now (QEMU_CLOCK_REALTIME)
|   v
nvme_io_cmd()
|   nsid -> nvme_ns(): allocated and attached, else Invalid Namespace
|   Flush, Dataset Management, Compare, Write Zeroes, Copy, Verify,
|   Write Uncorrectable, I/O Management Send/Receive: handled here
|   any other opcode -> ns->ext_ops.io_cmd() of the namespace's mode
|   NoSSD, BBSSD, CSD Read/Write: nvme_rw() maps PRP/SGL, copies data
|   ZNS, OCSSD, KV: their own paths; OCSSD and KV add NAND time now
|   status stored; on success the SMART host counters are updated
|   v
to_ftl[i] (every request, failed ones included)
|
+-- controller has an FTL thread: it dequeues, calls
|   femu_ftl_process_req(), adds the latency to expire_time,
|   enqueues on to_poller[i]
+-- no FTL thread: the poller reads to_ftl[i] itself
|   v
nvme_process_cq_cpl(i)
|   add host-link time (Read, Write) and firmware-CPU time
|   (Read, Write, Zone Append), insert into pq[i]
|   retry cpl_backlog[i]
|   while the heap's earliest request is due (now >= expire_time)
|   and, for a CXL-linked request, the medium has dropped stale pages:
|       CQ inactive -> request back to the free list
|       CQ full     -> cpl_backlog[i]
|       otherwise   -> post CQE
|   v
request back on sq->req_list; should_isr[cqid] = true
end of sweep: for each flagged CQ, raise its interrupt
```

Things to note:

- **Data moves before time is charged.** By the time a request reaches `to_ftl[i]`, a Read's data is already in guest memory and a Write's data is already in the backend. The FTL thread only computes time. (With `power_loss=on` the poller skips `nvme_io_cmd()` and the FTL thread executes every I/O command, data copy included, through `nvme_power_io()`.)
- **Failed commands take the same path.** A request with an error status is still enqueued; the mode charges it nothing, and it completes as soon as the poller sees it again (in the same sweep when there is no FTL thread), plus any host-link or firmware-CPU time, which is charged regardless of status. Skipping the ring would leak the request.
- **Order.** Completions are posted in `expire_time` order per poller, not in submission order.
- **Full completion queues.** A due completion whose CQ has no free slot waits in the poller's backlog, and the poller retries it every sweep.
- **Abort** looks for the command in the SQ between head and tail with the pollers paused. If it is there, the controller records it in its own state (the SQ itself is never written) and the Abort completes with `dword 0 bit 0` clear; the command then completes with `Command Abort Requested` when it is fetched, without running. A command already fetched is not aborted, and the Abort completes with `bit 0` set. Admin commands run one at a time, so the Aborts outstanding together are the one running and those still in the admin SQ; with more than `acl` of them waiting, the running one fails with `Abort Command Limit Exceeded`. A command queued behind an Abort on the admin queue can be aborted too.

- **Media errors carry DNR.** Unrecovered Read Error (a block marked by Write Uncorrectable, or an injected read error) and Compare Failure always set Do Not Retry, on every command that can return them: a retry would fail again.
- **Admin commands use PRPs.** An admin command whose PSDT is not 00b fails with Invalid Field in Command.

### 3.3.8 Completion timing

`nvme_process_cq_cpl()` is where FEMU enforces latency. A request's `expire_time` starts at its fetch time and grows as each model adds to it:

| Added by                               | When                                                                          | Property                                                           |
|----------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------|
| OCSSD, KV, CSD compute units           | while the command executes on the poller                                      | mode properties                                                    |
| FTL thread (BBSSD, ZNS, CSD NAND time) | before <code>to_poller[i]</code>                                              | FTL and NAND properties                                            |
| host link                              | in <code>nvme_process_cq_cpl()</code> ; a per-direction next-free time        | <code>pcie_bandwidth_mbps</code> , <code>pcie_prop_delay_ns</code> |
| firmware CPU                           | in <code>nvme_process_cq_cpl()</code> ; one next-free time for the controller | <code>fw_cpu_ns</code>                                             |

The heap `pq[i]` is a binary heap (`hw/femu/lib/pqueue.c`) keyed on `expire_time`. Each sweep pops every request that is due and stops at the first that is not. Nothing sleeps: a request simply stays in the heap until a sweep finds it due. The guest therefore sees at least the charged time, plus however long the poller takes to come round.

The clock is `QEMU_CLOCK_REALTIME`. Only the qtest-only property `x-oc12-clock` switches it to the virtual clock, for deterministic tests.

### 3.3.9 Interrupts

Each I/O CQ created with interrupts enabled gets its vector. After a sweep that posted to a CQ through the heap, the poller calls `nvme_isr_notify_io()` once for that CQ:

```
nvme_isr_notify_io(cq)                                (poller thread, no BQL)
|
+-- cq->virq > 0 (MSI-X under KVM, route added at Create I/O CQ):
|   event_notifier_set() -> KVM irqfd injects the MSI-X vector
|
+-- otherwise: qemu_bh_schedule(cq->irq_bh)           (main loop, BQL)
    nvme_isr_notify_legacy():
        MSI-X enabled -> msix_notify(vector)
        MSI enabled   -> msi_notify(vector), or hold the bit in
                        irq_status while INTMS masks it
        neither (pin) -> nvme_irq_update(): assert INTA while any
                        enabled CQ has head != tail and INTMS allows
```

The admin CQ is notified on the vCPU thread through the same legacy path. MSI-X mask, unmask and pending-bit polling are handled by vector notifiers (`nvme_vector_mask()`, `nvme_vector_unmask()`, `nvme_vector_poll()`) that move the irqfd routes.

There is no interrupt coalescing. `intc`, `intc_thresh` and `intc_time` set the values the Interrupt Coalescing and Interrupt Vector Configuration features report, and nothing else. On the heap path a poller raises at most

one interrupt per CQ per sweep.

### 3.3.10 Quiescing the data plane

Admin commands, controller reset and some features must change state the pollers and the FTL thread read without locks. `nvme_pause_pollers()` waits until neither is inside a sweep:

|                                                  |                                            |                                         |
|--------------------------------------------------|--------------------------------------------|-----------------------------------------|
| caller (vCPU, BQL)                               | poller i                                   | FTL thread                              |
| <code>dataplane_started = false</code>           | <code>in_sweep[i] = true</code>            | <code>ftl_in_sweep = true</code>        |
| <code>smp_mb</code>                              | <code>smp_mb</code>                        | <code>smp_mb</code>                     |
| <code>wait while any in_sweep[i]</code>          | <code>dataplane_started? no:</code>        | <code>dataplane_started? no:</code>     |
| <code>or ftl_in_sweep is true</code>             | <code>in_sweep[i] = false, back off</code> | <code>ftl_in_sweep = false, skip</code> |
| <code>... change queues or namespaces ...</code> | <code>... sweep ...</code>                 | <code>... one request ...</code>        |
| <code>nvme_resume_pollers():</code>              | <code>smp_mb</code>                        | <code>ftl_in_sweep = false</code>       |
| <code>dataplane_started = true</code>            | <code>in_sweep[i] = false</code>           |                                         |

Either the caller sees a sweep in progress and waits for it, or the thread sees the cleared flag and backs off. With shared namespaces (`femu-subsys, ns_mgmt=on`) the pause covers every controller in the subsystem, and a nested pause does not resume its caller's work.

### 3.3.11 Teardown

`femu_exit()` runs when the device is unplugged (`device_del`). QEMU does not call it when the process exits. It stops the threads before freeing anything they read:

1. Set `poller_stopping` and join every poller. Pollers go first because they feed the FTL thread.
2. Set `ftl_stopping` and join the FTL thread.
3. Close the Persistent Event Log, detach a linked CXL medium.
4. Run each distinct mode's exit hook once (not for a controller on a shared subsystem's storage).
5. `nvme_clear_ctrl(shutdown)`, then free the rings, heaps and per-poller arrays, the AER bottom half and the memory backend (unless it is shared or lent by a CXL medium).

### 3.3.12 Commands, namespaces and modes

#### Dispatch into modes

A mode is a `FemuExtCtrlOps` table (`hw/femu/nvme.h`):

| Hook                                                  | Called from                                                                                                                                                                                                                           | For                                                                                     |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <code>init</code>                                     | realize, once per namespace of that mode, and Namespace Management create                                                                                                                                                             | build the mode's state for the namespace                                                |
| <code>init_ctrl_name</code>                           | realize, only with <code>ns_mgmt</code> (otherwise each mode's <code>init</code> sets the name); only a namespace of the controller's <code>femu_mode</code> names it, and the controller's own table names it when no namespace does | Identify model number and serial                                                        |
| <code>start_ctrl</code>                               | controller enable, once per distinct mode                                                                                                                                                                                             | refuse settings the mode cannot serve; derive values from CC                            |
| <code>io_cmd</code>                                   | poller, for opcodes the frontend does not handle                                                                                                                                                                                      | the mode's I/O commands                                                                 |
| <code>admin_cmd,</code><br><code>admin_cmd_cqe</code> | vCPU, for admin opcodes the frontend does not handle                                                                                                                                                                                  | mode admin commands, such as the Open-Channel commands, BBSSD 0xEF and the CSD commands |

| Hook          | Called from                                    | For                                                 |
|---------------|------------------------------------------------|-----------------------------------------------------|
| get_log       | vCPU, for log ids the frontend does not handle | OC 2.0 chunk information, the ZNS Changed Zone List |
| rw_check_req  | not called through the table                   | OC 2.0 sets it and calls its checker directly       |
| exit, ns_exit | teardown, namespace deletion                   | free the mode's state                               |

`nvme_register_extensions()` installs the table for the controller's `femu_mode`, and `nvme_register_extensions_ns()` gives each namespace the table of its own mode, so I/O dispatches on the namespace, not the controller. Admin commands go to the controller's table; Get Log Page tries the named namespace's table first, then the controller's.

### Capability registry

`hw/femu/nvme-caps.c` is the one place that says what the controller handles. `nvme_admin_effects()` and `nvme_io_effects()` return the Commands Supported and Effects entry of an admin opcode, or of an I/O opcode in one command set, and zero for an opcode that is not handled. `nvme_log_support()` returns the Supported Log Pages entry of a log id in one command set.

An I/O opcode's entry for a command set is what the namespaces of that set handle between them (`nvme_ns_io_effects()` for one namespace): Flush, the optional commands and I/O Management, which `nvme_io_cmd()` serves itself, and then the mode's own table, so log 05h lists the Open-Channel vector commands, the CSD commands and the key value commands of the modes present, and leaves Read and Write out for Open-Channel 1.2, which refuses them. A command set with no namespace reports what a namespace of it would handle. The admin entries add the commands of the controller's mode: BBSSD 0xEF, the Open-Channel and CSD admin commands. `nvme_caps_id_ctrl()` fills OACS, ONCS, OCFS, LPA and SANICAP from those three each time Identify Controller is answered. LPA bit 5 stays clear: setting it would oblige a Command Scope in every log 05h entry, and Linux warns about any I/O entry bit beyond CSUPP and LBCC as an unusual effect.

Logs 05h and 00h are built from the same functions, and `nvme_io_cmd()` runs an optional NVM command (Compare, Dataset Management, Write Zeroes, Copy, Verify, Write Uncorrectable) only where `nvme_io_effects()` lists it for the namespace's command set. Features follow the same rule through `nvme_fid_supported()`, which Get Features, Set Features and log 12h share.

The `caps-*` qtests check the registry against dispatch. Four differences are known and left for a decision, and the test holds each to its current answer: I/O Management Send and Receive take their no-operation on every namespace though log 05h lists them only for NVM with FDP on; 0xEE answers Invalid Field outside Open-Channel; the Changed Zone List (BFh) and Chunk Information (CAh) pages answer when the command names another command set; and with a subsystem whose FDP is off the FDP pages answer FDP Disabled without being listed. Outside the test's reach, log 00h takes its command set from CDW14 even when CC.CSS selects NVM only, Get Features serves the Select field while ONCS bit 4 is clear, and CNS 1Ch always lists the NVM, zoned and key value sets.

### Namespace routing

`nvme_ns()` maps an NSID to a namespace that is both allocated and attached to this controller. NSIDs run from 1 to namespaces (or to 256 with Namespace Management). An unknown NSID fails with Invalid Namespace, and with Namespace Management an unallocated or detached NSID within range fails with Invalid Field. A Flush to every namespace (NSID FFFFFFFFh) is refused, which Identify reports as Flush Behavior 10b.

The FTL thread routes each request by its namespace: a namespace borrowed from a linked CXL medium goes to the medium's FTL; with `power_loss=on` the command is first executed; then ZNS namespaces go to `zns_ftl_process_req()`, BBSSD and CSD namespaces to `bb_ftl_process_req()`, and others charge nothing. One FTL thread serves all of them, and it exists only when a namespace needs it. With shared namespaces, the subsystem's `ns_lock` is held around command execution and FTL processing. `namespaces.md` (§4.16) covers sizing, per-namespace modes and Namespace Management.

### Log pages and features at a glance

| Log page                                          | Id       | Notes                                                                                                |
|---------------------------------------------------|----------|------------------------------------------------------------------------------------------------------|
| Supported Log Pages                               | 00h      | lists the ids the controller advertises (04h only with Namespace Management)                         |
| Error Information                                 | 01h      | newest <code>elpe</code> + 1 entries                                                                 |
| SMART / Health                                    | 02h      | host totals from the pollers, media wear, temperature                                                |
| Firmware Slot                                     | 03h      | one slot, read-only; Identify FRMW reports the same                                                  |
| Changed Namespace List                            | 04h      | namespaces whose attributes changed                                                                  |
| Commands Supported and Effects                    | 05h      | per command set (NVM, zoned, KV)                                                                     |
| Device Self-test                                  | 06h      | tests complete at once                                                                               |
| Telemetry Host / Controller                       | 07h, 08h | 07h: header plus the C0h counters captured by the last Create; 08h: header only                      |
| Endurance Group                                   | 09h      | with <code>femu-subsys</code> ; Identify reports the group (CTRATT bit 4, ENDGIDMAX, ENDGID) with it |
| Persistent Event                                  | 0Dh      | kept in <code>pel_file</code> if set                                                                 |
| LBA Status                                        | 0Eh      |                                                                                                      |
| Feature Identifiers Supported and Effects         | 12h      | built from the check Get and Set Features use; per command set (CSI) when CC.CSS is 110b             |
| NVMe-MI Commands Supported and Effects            | 13h      | all zero: no NVMe-MI Send or Receive                                                                 |
| FDP Configurations, RUH Usage, Statistics, Events | 20h-23h  | with FDP                                                                                             |
| Sanitize Status                                   | 81h      |                                                                                                      |
| Changed Zone List                                 | BFh      | ZNS, from the mode                                                                                   |
| Chunk Information                                 | CAh      | Open-Channel 2.0, from the mode                                                                      |
| FEMU media counters                               | C0h      | WAF and FTL counters                                                                                 |

Get Log Page refuses an id that log 00h lists for no command set with Invalid Log Page (`nvme_log_answered()`). Logs 00h, 05h and 12h refuse a CSI other than NVM, KV and zoned with I/O Command Set Not Supported when CC.CSS selects by CSI. The one exception is the FDP pages on a controller with a subsystem: while FDP is off they are not listed but still answer, with FDP Disabled.

Get and Set Features answer Arbitration (01h), Power Management (02h, one power state), LBA Range Type (03h), Temperature Threshold (04h), Error Recovery (05h), Volatile Write Cache (06h, only with `vwec=1`), Number of Queues (07h), Interrupt Coalescing (08h), Interrupt Vector Configuration (09h), Write Atomicity (0Ah), Asynchronous Event Configuration (0Bh), Timestamp (0Eh), Host Behavior Support (16h), Command Set Profile (19h), FDP (1Dh, 1Eh, only with `femu-subsys`), Key Value Configuration (20h, only with a KV namespace) and Software Progress Marker (80h). Any other identifier, or one of these without what it needs, fails with Invalid Field in Command, and log 12h lists exactly the identifiers that answer. Only the FDP features report a saved value: FDP Mode is fixed at realize, FDP Events can be changed at run time. A controller reset (`nvme_reset_features()`) restores Arbitration, Power Management, the temperature



thresholds, Volatile Write Cache, Number of Queues, Interrupt Coalescing, Interrupt Vector Configuration, Write Atomicity, Asynchronous Event Configuration, Host Behavior Support and each namespace's Error Recovery; the Software Progress Marker, LBA Range Type data, Key Value Configuration and FDP event filters keep their values.

### Asynchronous events

The controller holds up to  $aerl + 1$  Asynchronous Event Requests; one more fails with Asynchronous Event Request Limit Exceeded. An event is queued (at most 16; more are dropped) and posted by `nvme_process_aers()` from a main-loop bottom half, because a poller may raise it and must not touch the admin CQ. A request that arrives while events are queued is matched at once on the vCPU thread. The event types and their log pages are listed in log pages and counters (§B.2).

### 3.3.13 Parameters

The frontend reads these `-device femu` properties. Their defaults and ranges are in the generated reference; follow the link of each group.

| Property                                           | Effect on the frontend                                                                                                   | Reference                                             |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| queues                                             | number of I/O queue pairs; sizes BAR0, MSI-X vectors, <code>sq[]</code> , <code>cq[]</code> and the shadow doorbell page | queues, pollers and interrupts (§A.1.1)               |
| entries                                            | CAP.MQES; the largest queue is <code>entries + 1</code>                                                                  | same                                                  |
| max_sqes, max_cqes                                 | Identify SQES and CQES; only 64-byte SQES and 16-byte CQES are accepted                                                  | same                                                  |
| stride                                             | doorbell spacing, $4 \ll \text{stride bytes}$ ; also the shadow doorbell entry size                                      | same                                                  |
| multipoller_enabled, poller_ratio                  | number of pollers and queue ownership                                                                                    | same                                                  |
| hiops_inline                                       | NoSSD only: performance option, on by default; set off only when debugging                                               | same                                                  |
| aerl, elpe                                         | held AER limit, Error log length                                                                                         | same                                                  |
| mdts                                               | largest data transfer per command                                                                                        | same                                                  |
| intc, intc_thresh, intc_time                       | values reported by the coalescing features only                                                                          | same                                                  |
| vid, did                                           | PCI IDs                                                                                                                  | controller identity and capabilities (§A.1.1)         |
| cqr                                                | whether queues must be physically contiguous                                                                             | same                                                  |
| mpsmin, mpsmax                                     | memory page sizes CC.MPS may choose                                                                                      | same                                                  |
| oacs, oncs, vwc                                    | which optional admin and NVM commands are answered                                                                       | same                                                  |
| sgl                                                | accept SGL data pointers                                                                                                 | same                                                  |
| cmbsz, cmbloc                                      | Controller Memory Buffer on BAR 2                                                                                        | same                                                  |
| femu_mode, namespaces, namespace_modes             | which mode tables are installed, NSID range                                                                              | mode, capacity and namespaces (§A.1.1)                |
| ns_mgmt, streams, power_loss                       | admin commands answered; FTL thread applies data with <code>power_loss</code>                                            | namespace management, streams and power loss (§A.1.1) |
| pcie_bandwidth_mbps, pcie_prop_delay_ns, fw_cpu_ns | time added in <code>nvme_process_cq_cpl()</code>                                                                         | host link and controller firmware (§A.1.1)            |
| subsys                                             | CNTLID, FDP, shared namespaces                                                                                           | femu-subsys (§A.1.2)                                  |

Interactions:

- With the defaults (queues=8, stride=0) BAR0 is 16 KiB. Large queues or stride values grow it; queues above 511 with stride=0 and 4 KiB pages leaves Doorbell Buffer Config unusable (the host then rings the registers).
- `multipoller_enabled=1` with `poller_ratio=1` starts one poller per queue, which needs that many spare host cores.
- The host's CC.MPS must fall within `mpsmin..mpsmax`, or enabling the controller sets CSTS.CFS.

### 3.3.14 Counters

| Counter                                                               | Where kept                                                                                                                                                              | How to read it                                                                           |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Host read and write commands and bytes                                | <code>poller_ctr[i]</code> , counted on the poller after execution, successful commands only, for every mode (the FTL thread counts them with <code>power_loss</code> ) | SMART log: Host Read/Write Commands, Data Units Read/Written                             |
| Completions, and completions posted 20 us or more past their due time | <code>poller_ctr[i]</code>                                                                                                                                              | BBSSD vendor command 0xEF, CDW10=5, prints both and resets them (timing model (§6.2.11)) |
| Error log entries                                                     | <code>num_errors</code>                                                                                                                                                 | SMART log, Error Information log                                                         |

The per-poller counters are padded to separate cache lines so pollers never share one; readers sum them. Media counters (WAF, GC pages, buffer hits) come from the FTL, not the frontend; see log pages and counters (§B.1).

### 3.3.15 Validation

The frontend is covered by the `qtest` suite in `hw/femu/tests/qtest/femu-test.c`, which drives the controller through its registers with no guest. Cases that target this chapter include:

| Case                                                                                                | Checks                                                                                |
|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <code>bar0-size</code>                                                                              | BAR0 size for the queue count and stride                                              |
| <code>cc-states</code>                                                                              | enable, disable, shutdown and failure transitions of CC and CSTS                      |
| <code>doorbell-errors</code>                                                                        | invalid doorbell writes raise Error events                                            |
| <code>io-by-doorbell</code> , <code>io-by-shadow-doorbell</code>                                    | I/O through registers and through Doorbell Buffer Config                              |
| <code>dbuf-too-many-queues</code>                                                                   | Doorbell Buffer Config refused when the entries do not fit                            |
| <code>intx-shadow-doorbell</code>                                                                   | the pin drops with shadow doorbells                                                   |
| <code>queue-mapping</code>                                                                          | Create I/O queue refused when the ring cannot be mapped                               |
| <code>shared-cq</code> , <code>shared-cq-pollers</code>                                             | SQs that share one CQ, with one and several pollers                                   |
| <code>poller-burst</code> , <code>poller-burst-per-queue</code> , <code>poller-burst-sharded</code> | full-queue bursts complete exactly once, in their own queue, under each poller layout |
| <code>cq-full</code>                                                                                | completions wait for CQ space                                                         |
| <code>delete-sq-in-flight</code> , <code>ns-retire-pollers</code>                                   | queue and namespace removal with I/O in flight                                        |
| <code>aer-limit</code>                                                                              | the AER limit                                                                         |
| <code>abort</code>                                                                                  | Abort leaves the SQ unwritten, aborts queued admin and I/O commands, and enforces ACL |
| <code>cc-css</code>                                                                                 | an unoffered CC.CSS fails the enable                                                  |

| Case                                                                                                                                              | Checks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| admin-psdt                                                                                                                                        | admin commands naming SGLs are refused                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| media-dnr                                                                                                                                         | Unrecovered Read and Compare Failure set DNR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| fid-effects, fid-effects-kv, fid-effects-fdp                                                                                                      | log 12h agrees with Get Features; log 13h is zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| frmw                                                                                                                                              | FRMW and the firmware slot log agree                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| caps-* (22 configurations: every mode, the optional commands on, off and in pairs, Namespace Management, Streams, PI, FDP, mixed namespace modes) | every admin opcode, every I/O opcode on every namespace and every log id per command set is answered exactly when logs 05h and 00h list it; log 12h agrees with Get Features; OACS, ONCS, OCFS, LPA, SANICAP, VWC, FRMW, SGLS, CNS 1Ch and the Copy limits in Identify Namespace agree with the logs and with what the controller does; ONCS and the Format bit match what the configuration asks for; VS, CAP.CRMS, CRT0 and BPCAP match the reported version (2.1, or 1.4 for Open-Channel), CAP.AMS is 0, and LPA bit 5 and every command scope are clear |
| ns-mgmt-before-identify                                                                                                                           | a managed bbssd namespace is addressed correctly before the host reads Identify Controller                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| v2-refusals                                                                                                                                       | CC.AMS other than round robin fails the enable; Identify CNS 00h on a KV namespace fails with Invalid I/O Command Set while CNS 08h answers; logs 00h and 05h refuse an unknown CSI; CNS 07h for KV is refused under CC.CSS 000b while CNS 1Ah is not; CNS 1Fh answers an allocated NSID and refuses 0 and FFFFFFFFh                                                                                                                                                                                                                                         |
| endgrp-reported                                                                                                                                   | with a subsystem and FDP off, CTRATT bit 4, ENDGIDMAX and the ENDGID of a block and a KV namespace (CNS 08h, and CNS 05h for KV) report the one endurance group                                                                                                                                                                                                                                                                                                                                                                                              |
| features-reset, features-reset-vwc                                                                                                                | features return to defaults on reset                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| admin-fuzz, io-fuzz and its variants                                                                                                              | structured fuzzing of admin and I/O commands                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

Run them all, or one by name, from a build directory (testing (§8.2)):

```
QTEST_QEMU_BINARY=./qemu-system-x86_64 ./tests/qtest/qos-test -m quick \
-p /x86_64/pc/i440FX-pcihost/pci-bus-pc/pci-bus/femu/femu-tests/io-by-shadow-doorbell
```

The documentation example above (frontend-sharded-pollers) is started by check-docs, which enables the controller and moves one block.

### 3.3.16 Limits

- At most 2047 I/O queue pairs (the MSI-X table size), at most 65535 entries per queue.
- SQEs are 64 bytes and CQEs 16 bytes; other sizes are refused at realize.
- One controller per PCI function. No SR-IOV, no NVM Subsystem Reset, no controller memory for PMR.
- No interrupt coalescing; the coalescing features only store values.
- Weighted round robin arbitration is not implemented: CAP.AMS advertises it, but each poller sweeps its queues in id order.
- A queue can have at most size commands outstanding, and a poller's rings hold 65535 requests each.
- Abort only reaches commands not yet fetched.
- Latency is a lower bound that depends on poller scheduling.
- The controller is not migratable.

### 3.3.17 Extension points

- **A new mode:** write a FemuExtCtrlOps table and a nvme\_register\_\*() function, add the mode to the enum in hw/femu/nvme.h, to the femu\_mode bound in nvme\_check\_constraints(), to nvme\_mode\_from\_token() for namespace\_modes, to nvme\_register\_extensions(), and to femu\_needs\_ftl\_thread() and femu\_ftl\_process\_req() if it charges time on the FTL thread. A mode that charges time on the poller adds it to req->expire\_time in its io\_cmd.
- **A new I/O command shared by all block modes:** add it to nvme\_io\_cmd() and to nvme\_io\_effects() in hw/femu/nvme-caps.c.
- **A new admin command, feature or log page:** nvme\_admin\_cmd() and nvme\_admin\_effects(); nvme\_set\_feature(), nvme\_get\_feature() and the support tables nvme\_feature\_support[] and nvme\_feature\_cap[] (with nvme\_fid\_supported() for a feature that needs something else present); nvme\_get\_log() and nvme\_log\_support(). An Identify bit that sums up commands or pages is derived in nvme\_caps\_id\_ctrl(), not set by hand.
- **A new host-side cost:** add it next to the host-link and firmware-CPU models in nvme\_process\_cq\_cpl().
- **Anything that changes state the I/O path reads:** wrap it in nvme\_pause\_pollers() and nvme\_resume\_pollers().
- **A new property:** define it in femu\_props[] in hw/femu/femu.c, describe it in hw/femu/femu-props.c, check it in nvme\_check\_constraints(), and regenerate the property reference (docs maintenance (§8.3)).

### 3.3.18 Source map

| File                                            | Functions                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hw/femu/femu.c                                  | nvme_init_pci(), nvme_init_cmb(), nvme_init_ctrl() (Identify Controller, CAP), nvme_check_constraints(), nvme_mmio_write(), nvme_write_bar(), nvme_process_db_admin(), nvme_process_db_io(), nvme_start_ctrl(), nvme_clear_ctrl(), nvme_reset_features(), femu_ftl_thread(), femu_ftl_process_req(), femu_needs_ftl_thread(), nvme_register_extensions(), nvme_register_extensions_ns(), femu_realize(), femu_exit() |
| hw/femu/nvme-admin.c                            | nvme_create_sq(), nvme_create_cq(), nvme_del_sq(), nvme_del_cq(), nvme_init_poller(), nvme_start_dataplane(), nvme_set_db_memory(), nvme_identify(), nvme_get_feature(), nvme_set_feature(), nvme_get_log(), nvme_abort_req(), nvme_admin_cmd(), nvme_process_aers(), nvme_process_sq_admin()                                                                                                                        |
| hw/femu/nvme-caps.c                             | nvme_admin_effects(), nvme_io_effects(), nvme_ns_io_effects(), nvme_log_support(), nvme_log_answered(), nvme_caps_id_ctrl()                                                                                                                                                                                                                                                                                          |
| hw/femu/nvme-io.c                               | nvme_poller(), nvme_process_sq_io(), nvme_update_sq_eventidx(), nvme_process_cq_cpl(), nvme_post_cqe(), nvme_rw(), nvme_io_cmd()                                                                                                                                                                                                                                                                                     |
| hw/femu/nvme-util.c                             | nvme_pause_pollers(), nvme_resume_pollers(), nvme_update_sq_tail(), nvme_update_cq_head(), nvme_update_cq_eventidx(), nvme_init_sq(), nvme_init_cq()                                                                                                                                                                                                                                                                 |
| hw/femu/intr.c                                  | nvme_isr_notify_io(), nvme_isr_notify_admin(), nvme_irq_update(), nvme_irq_mask_changed(), nvme_setup_virq(), vector notifiers                                                                                                                                                                                                                                                                                       |
| hw/femu/dma.c                                   | PRP and SGL mapping, femu_dma_rw                                                                                                                                                                                                                                                                                                                                                                                     |
| hw/femu/lib/rte_ring.c,<br>hw/femu/lib/pqueue.c | the rings and the completion heap                                                                                                                                                                                                                                                                                                                                                                                    |
| hw/femu/nvme.h                                  | NvmeSQueue, NvmeCQueue, NvmeRequest, FemuCtrl, FemuExtCtrlOps, FemuPollerCtrl, register layout                                                                                                                                                                                                                                                                                                                       |

### 3.3.19 Related pages

- Design overview (§3.2)
- Architecture (§3.1)
- Timing model (§6.2)
- Device property reference (§A.1)
- Log pages and counters (§B)
- Performance tuning (§5.12)

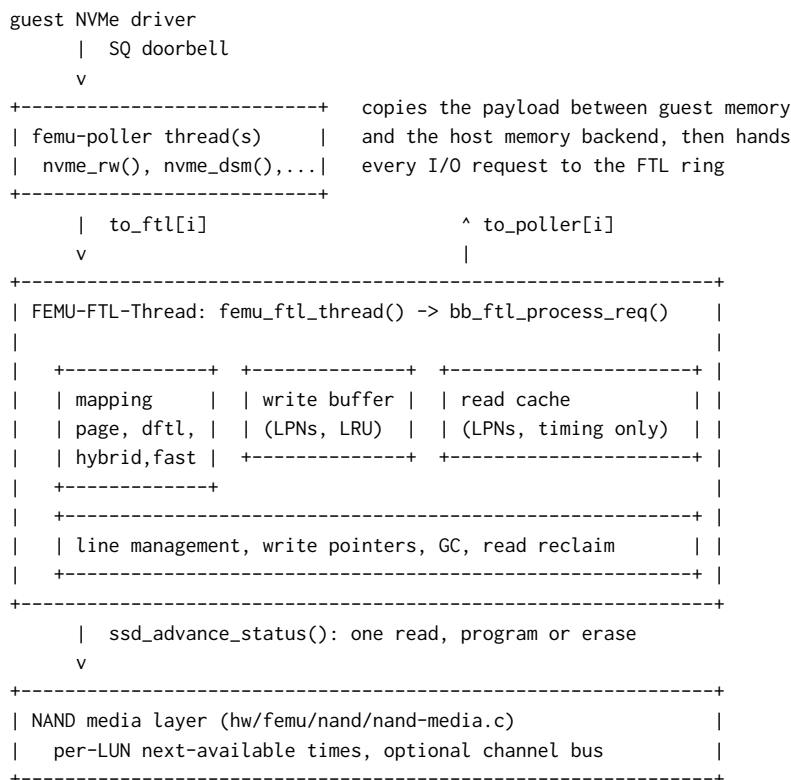
### 3.4 The BlackBox FTL

This chapter describes the flash translation layer (FTL) behind BlackBox mode (`femu_mode=1`): how it maps logical pages to NAND pages, how it organises NAND into lines and write pointers, how it collects garbage, and how its write buffer, read cache, wear and refresh models work. It is written for readers who want to know what the model does, where its time goes, and how to change it. To run the mode, read the BlackBox guide (§4.4) first; this chapter does not repeat its launch and guest steps.

The code lives in `hw/femu/bbssd/`. The same FTL serves CSD namespaces, which build it through the same `ssd_init()`. KV namespaces call `ssd_init()` for the geometry and NAND timing but run their own key-value FTL. A `femu-cxl-ssd` builds a private instance on its own worker thread.

NAND operation timing (per-LUN timelines, the channel bus, suspend, ECC tiers) is the subject of the NAND timing chapter (§3.5) in this directory, and of the timing model (§6.2). This chapter covers what the FTL asks of the NAND, not how long each operation takes.

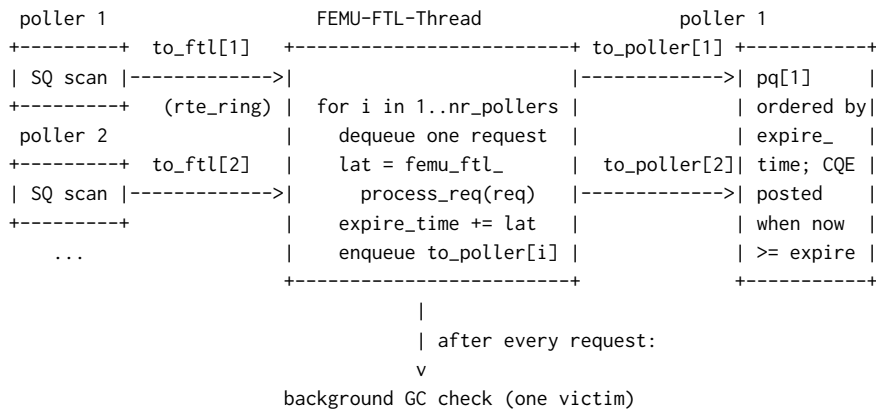
### 3.4.1 Place in the hierarchy



Two things follow from this picture.

- **The FTL models time and placement, not data.** The poller copies the payload to the host memory backend before the request reaches the FTL. The FTL tracks which NAND page each logical page would occupy and how long each operation takes, but it never stores the payload. The one exception is the power-loss model (`power_loss=on`), which runs the data copy on the FTL thread and keeps undo copies of buffered pages; see Power loss (§3.4.7).
- **The FTL returns a duration.** Each handler returns the time the request spends in the device, measured from the request's arrival time `stime`. The FTL thread adds it to `req->expire_time`, and the poller completes the request when the host clock reaches that time.

### 3.4.2 The FTL thread and its queues



- **One thread per controller.** `femu_ftl_thread()` in `hw/femu/femu.c` starts at realize when a namespace is `bbssd`, `ZNS` or `CSD` (or the controller shares a `bbssd` subsystem), after every namespace is built. It is joined at device removal.
- **One pair of rings per poller.** Each poller `i` feeds `to_ftl[i]` and reads `to_poller[i]`. The FTL thread visits the rings in turn and takes one request from each non-empty ring per pass, so no poller can starve another.
- **Every I/O command goes through the ring,** including Flush, Dataset Management and commands that already failed in the I/O layer. A failed request is not processed: `bb_ftl_process_req()` returns 0 for it, and the poller posts the carried error status.
- **No lock protects the FTL state.** The FTL thread is the only writer of the mapping, the lines, the write buffer and the caches. With Streams on, each request runs under `n->streams_lock`; on a shared subsystem namespace it runs under the subsystem's `ns_lock`. An NVMe controller linked to a `femu-cxl-ssd` runs its requests on its own FTL thread against the medium's FTL, under the medium's lock, which also serializes the medium's worker.
- **Admin commands that change FTL state pause the dataplane.** `nvme_pause_pollers()` waits until the FTL thread is outside a request (the `ftl_in_sweep` flag), so the run-time switch (admin opcode `0xEF`), the volatile write cache feature, Format and Sanitize change FTL state with nothing in flight.

`bb_ftl_process_req()` in `hw/femu/bbssd/ftl.c` dispatches on the opcode:

| Opcode | Handler                 | What the FTL does                                            |
|--------|-------------------------|--------------------------------------------------------------|
| Read   | <code>ssd_read()</code> | translate each page; buffer hit, read cache hit or NAND read |

| Opcode              | Handler                                           | What the FTL does                                             |
|---------------------|---------------------------------------------------|---------------------------------------------------------------|
| Write               | <code>ssd_write()</code>                          | foreground GC, read reclaim, then buffer or program each page |
| Write Zeroes        | <code>ssd_write_zeroes()</code>                   | with the Deallocate bit: unmap; without it: program the range |
| Dataset Management  | <code>ssd_trim()</code>                           | deallocate each range                                         |
| Copy                | <code>ssd_copy()</code>                           | read every source range, then write the destination           |
| Flush               | <code>ssd_buffer_destage(ssd, 0, .<br/>..)</code> | program everything the write buffer holds                     |
| I/O Management Send | <code>ssd_fdp_update_ruhs()</code>                | FDP only: update reclaim unit handles                         |

Read and Write latency is the largest per-page latency of the request: pages on different LUNs proceed in parallel, and pages on one LUN queue on its timeline. Copy costs the slowest source read plus the destination write. After the opcode handler, the thread runs one background GC step if the device is past the background watermark (Triggers (§3.4.6)).

### 3.4.3 Data structures

All of the FTL's state hangs off struct `ssd` (`hw/femu/bbssd/ftl.h`), one per `bbssd` or CSD namespace.

#### Physical page address

A struct `ppa` packs a NAND location into 64 bits:

```

63 62          51 50          44 43  40 39          32 31          16 15          0
+-----+-----+-----+-----+-----+-----+
| rsv | ch (12) | lun (7) | pl (4) | sec (8) | pg (16) | blk (16) |
+-----+-----+-----+-----+-----+-----+

```

The field widths bound the geometry: at most 4096 channels, 128 LUNs per channel, 16 planes per LUN, 65536 blocks per plane, 65536 pages per block and 256 sectors per page. `bb_check_geometry()` refuses larger values, and also refuses a geometry whose total sector count exceeds `INT_MAX`, because the derived totals in struct `ssdparams` are `int`. The FTL always addresses whole pages; the `sec` field is unused by the mapping.

#### NAND state

```

ssd->ch[nchs]
    .next_ch_avail_time
    .lun[luns_per_ch]
        .next_lun_avail_time    <- the timeline every NAND op is gated on
        .pl[pls_per_lun]
            .blk[blks_per_pl]
                .vpc, .ipc        valid and invalid page counts
                .erase_cnt        P/E cycles of this block
                .read_cnt         reads since the last erase
                .pg[pgs_per_blk].status FREE, VALID or INVALID

```

A page goes FREE -> VALID when it is programmed (`mark_page_valid()`), VALID -> INVALID when its logical page is overwritten, deallocated or moved by a log-block merge (`mark_page_invalid()`), and back to FREE when its block is erased (`mark_block_free()`). A page that line GC relocates stays VALID until its block is erased. Closing a stream's line marks its unwritten pages INVALID directly.

## Mapping tables

maptbl: logical page -> physical page  
(one struct ppa per logical page)

rmap: physical page -> logical page  
(indexed by ppa2pgidx(), one u64 each)

| LPN | PPA                   |        | pgidx                 |     | LPN |
|-----|-----------------------|--------|-----------------------|-----|-----|
| 0   | ch0 lun0 pl0 blk7 pg3 | -----> | ch0 lun0 pl0 blk7 pg3 | 0   |     |
| 1   | UNMAPPED              |        | ch1 lun0 pl0 blk7 pg3 | 5   |     |
| 2   | ch3 lun1 pl0 blk2 pg0 |        | ...                   | ... |     |
| ... | ...                   |        | (stale copy)          | INV |     |

maptbl is the forward map and rmap the reverse map that GC uses to find which logical page a valid physical page holds. Both have one entry per NAND page (tt\_pgs), 8 bytes each: 32 MiB apiece for the default 16 GiB geometry. Every mapping scheme keeps these two tables as the source of truth.

## Lines

```
typedef struct line {
    int id;                /* the block index this line spans */
    int ipc, vpc;          /* invalid and valid pages over the whole line */
    size_t pos;            /* slot in the victim priority queue, 0 if none */
    uint64_t close_time;   /* when the line filled; age-based policies */
    uint64_t close_seq;    /* order in which lines filled; the FIFO key */
    uint64_t stream_tag;   /* Streams: the stream whose data it holds */
    bool reclaiming;       /* being rewritten by read reclaim */
    ...
} line;
```

struct line\_mgmt keeps a free list (a FIFO), a victim priority queue keyed by valid page count (by close\_seq under gc\_policy=fifo), and a full list.

### 3.4.4 Address mapping

#### From LBA to logical page

ssd\_lpn\_range() turns an LBA range into a range of logical page numbers (LPNs). A page is secsz \* secs\_per\_pg bytes. The byte offset is the LBA times the LBA size, plus the namespace's offset in the backend:

```
LPN = (backend_offset + slba * lba_size) / page_size
```

So a namespace's LPNs start where its data starts in the backend. A controller with Namespace Management on uses offset 0, because each managed namespace has a private FTL. A request whose last LPN is at or beyond tt\_pgs fails with LBA Out of Range.

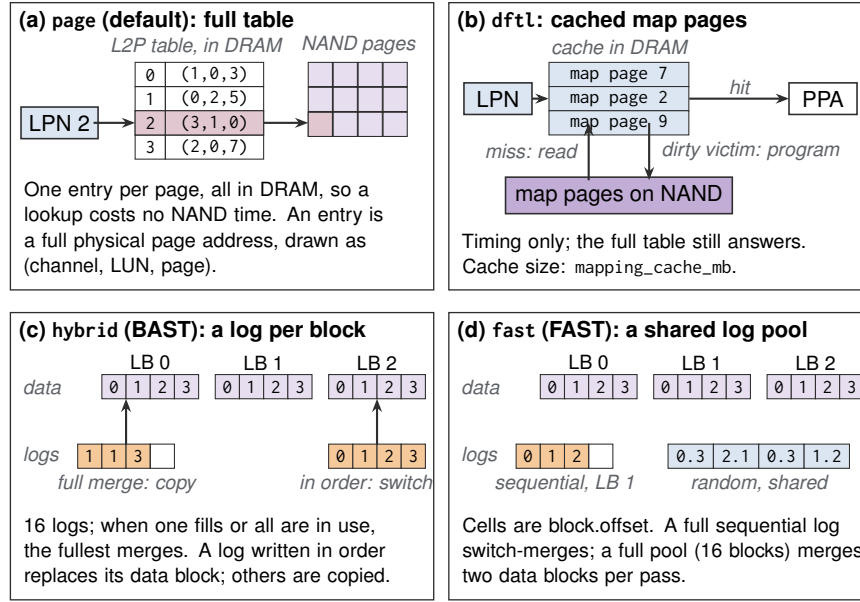
#### Page mapping (mapping=page)

The default. translate() returns maptbl[lpn]. A write invalidates the old physical page, if any, and points the LPN at the new one. A read of an LPN with no mapping costs no NAND time.

#### DFTL (mapping=dftl)

DFTL keeps the page-level mapping on NAND and caches the translation pages that are in use. FEMU models the cost of that cache, not its contents: the flat maptbl stays the source of truth, and cmt\_touch() in hw/femu/bbssd/ftl-map-cmt.c charges NAND time for misses.





**Figure 7:** The four BBSSD mapping schemes (mapping=): the full page table, DFTL’s cache of map pages, BAST’s log block per data block, and FAST’s shared log pool.

```

host read/write of lpn
|
v
tp_id = lpn / lpn_per_tp      lpn_per_tp = page_size / 8
|                             (512 for a 4 KiB page)
v
+-----+
| cached mapping table (CMT) | capacity = mapping_cache_mb / page_size
| CLOCK, one slot per TP    | slots
+-----+
| hit: no cost; a write marks the slot dirty
| miss:
v
evict a slot by CLOCK
| dirty victim: program it on LUN (victim tp_id % tt_luns)
v
read the TP on LUN (tp_id % tt_luns), after the write-back
|
v
latency = write-back + read, charged to this request

```

With mapping\_cache\_mb left at 0, DFTL uses 4 MiB, which holds 1024 translation pages and so covers 2 GiB of logical space on a 4 KiB page. The translation-page traffic occupies the LUN timelines like any other NAND operation but is not counted as host or GC writes. Host reads, host writes and buffer write-backs are charged; GC relocation, deallocate, Write Zeroes and the FDP write path update maptbl without a CMT access.

### Log-block mapping: BAST (mapping=hybrid) and FAST (mapping=fast)

The two log-block schemes write every page through a separate LOG write pointer and model the merges a log-block FTL has to run. Like DFTL, they use maptbl for correctness, so a read always finds the newest copy. A logical block (LBN) is pgs\_per\_blk consecutive LPNs.

fast (FAST): one sequential log  
plus a shared random-write pool  
SW log: one LBN, in-order run

SW log: one LBN, in-order run

```

+-----+
| LBN 7: off 0,1,2,... |
+-----+

```

```

RW pool: 16 blocks of pages,
any LBN, dirty LBN list
+-----+
| LBN 3, LBN 90, LBN 12... |
+-----+

```

```
| pool full: merge two
| dirty LBNs per pass
v
```

random merge: relocate each  
live page of those LBNs

```
SW log full in order:
switch merge, one erase
```

```
SW log full in order:
switch merge, one erase
```

- **hybrid** follows BAST (Kim et al. 2002): one log per logical block, from a pool of 16. Each program consumes a log slot, overwritten or not; neither an overwrite nor a deallocate frees a slot, only a merge does. After every programmed page, if a log is full or every log in the pool is taken, the fullest log is merged. A log written strictly in order switch-merges for the cost of one erase. Any other log full-merges: every live page of its LBN is read and programmed into DATA-class space, counted as GC writes. Hybrid counts its merges for log page C0h. The scheme models merge cost, not BAST's physical block placement: logs of different LBNs share physical lines, and line GC still runs underneath and can add copies.
- **fast** follows FAST (Lee et al. 2007): a single sequential-write log for an in-order run that starts at offset 0 of an LBN, and a shared pool of 16 blocks' worth of pages for everything else. When the pool fills (or the dirty-LBN list reaches its 4095 limit), a reclaim relocates the live pages of up to two dirty LBNs, and repeats on later writes until the pool is no longer full. A full in-order sequential log switch-merges with one erase. The sequential log is released only when it fills in order; a broken run keeps holding it, and later sequential runs go to the shared pool. FAST reclaims once per request, not once per page, and its merge counts are not exported.

Merge reads, programs and erases are charged to the request that triggered the merge, and the relocated pages count in the write amplification factor.

## Choosing a scheme

| mapping | Models                                               | Extra cost charged                                           | Extra write pointer | Counters            |
|---------|------------------------------------------------------|--------------------------------------------------------------|---------------------|---------------------|
| page    | a full page map in DRAM                              | none                                                         | none                | none                |
| dftl    | a page map on NAND with a cache of translation pages | translation page read on a miss, program on a dirty eviction | none                | none exported       |
| hybrid  | BAST log blocks                                      | switch and full merges per log                               | LOG                 | C0h bytes 88 to 111 |
| fast    | FAST log blocks                                      | bounded random merges, sequential switch merges              | LOG                 | none exported       |

Streams need page or dftl. FDP supports only page.

### 3.4.5 Lines, superblocks and write pointers

#### A line is a superblock

A line is the block with the same index on every plane of every LUN of every channel. There are `blks_per_pl` lines, each `nchs * luns_per_ch * pls_per_lun` blocks wide. A write pointer fills a line in this order: channel first, then LUN, then plane, then page.

|      |           |       |       |     |         |                                          |
|------|-----------|-------|-------|-----|---------|------------------------------------------|
|      | ch0       | ch1   | ch2   | ... | ch7     |                                          |
| lun0 | pg0 [ 0 ] | [ 1 ] | [ 2 ] |     | [ 7 ]   | numbers are the order in                 |
| lun1 | pg0 [ 8 ] | [ 9 ] | [10 ] |     | [15 ]   | which the write pointer                  |
| ...  |           |       |       |     |         | hands out pages of line N                |
| lun7 | pg0 [56 ] | [57 ] | [58 ] |     | [63 ]   | (one plane per LUN)                      |
| lun0 | pg1 [64 ] | [65 ] | ...   |     |         |                                          |
| ...  |           |       |       |     |         |                                          |
| lun7 | pg255     |       |       |     | [16383] | -> line full, take the next<br>free line |

Consecutive pages land on different channels, then different LUNs, so a large write or a burst of small ones spreads over the whole device. With several planes, the pointer sweeps every channel and LUN on plane 0, then on plane 1, and so on, before the page index moves on.

#### Why there are several write pointers

Each write pointer holds one line open and appends to it. Data written through one pointer shares lines, so the lines die together or not. Keeping data of different lifetimes apart is what makes GC victims mostly invalid. The FTL has these pointers:

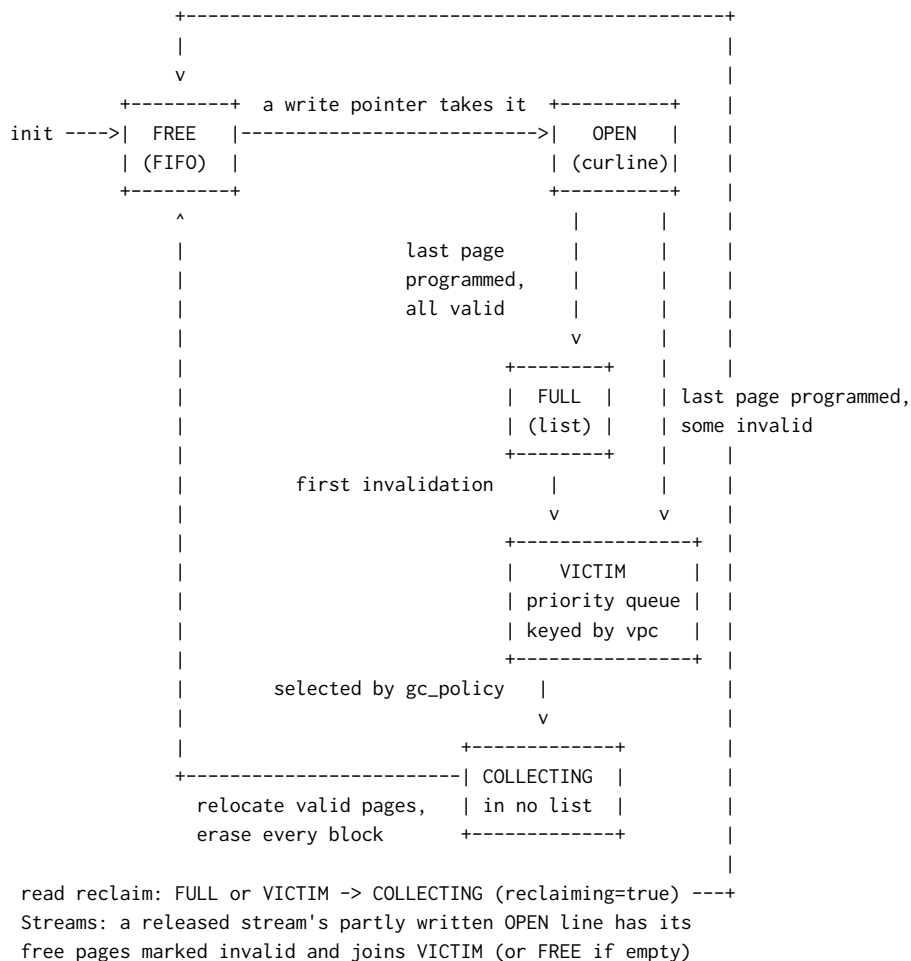
| Pointer      | In struct    | ssd | Takes                                                                                                               | Opened       |
|--------------|--------------|-----|---------------------------------------------------------------------------------------------------------------------|--------------|
| data         | wp           |     | host writes (only first writes with <code>hot_cold_sep=on</code> ), GC relocations, read reclaim, merge relocations | at init      |
| hot          | hot_wp       |     | overwrites of mapped pages, with <code>hot_cold_sep=on</code>                                                       | on first use |
| log          | log_wp       |     | every host write under hybrid or fast                                                                               | on first use |
| stream       | stream_wp[i] |     | writes of open stream i                                                                                             | on first use |
| stream GC    | stream_gc_wp |     | relocated pages of a released stream                                                                                | on first use |
| reclaim unit | per handle   |     | FDP placement; see FDP (§4.5)                                                                                       | FDP init     |

When a hot or log pointer cannot get a line, the write falls back to the data pointer. Every open pointer pins a line, so `bb_check_capacity()` adds one reserved line per pointer the configuration can hold open (Capacity (§3.4.12)).

#### Hot/cold separation

With `hot_cold_sep=on` and page or DFTL mapping, `prepare_write()` sends a write to the hot pointer when its LPN is already mapped. An overwrite is the cheapest predictor of another overwrite, so hot lines fill with pages that are invalidated together. GC relocations go through the data pointer, so pages that survive a collection join the cold data. The log-block schemes ignore the setting, but the extra reserved line is still counted. FDP refuses it.

## Line state machine



An open line belongs to no list; invalidations only lower its valid count. When a line fills, `close_time` and `close_seq` are set. The free list is a FIFO: lines are taken from the head and returned to the tail. The FTL has no explicit wear levelling; the FIFO spreads erases over the free lines.

### 3.4.6 Garbage collection

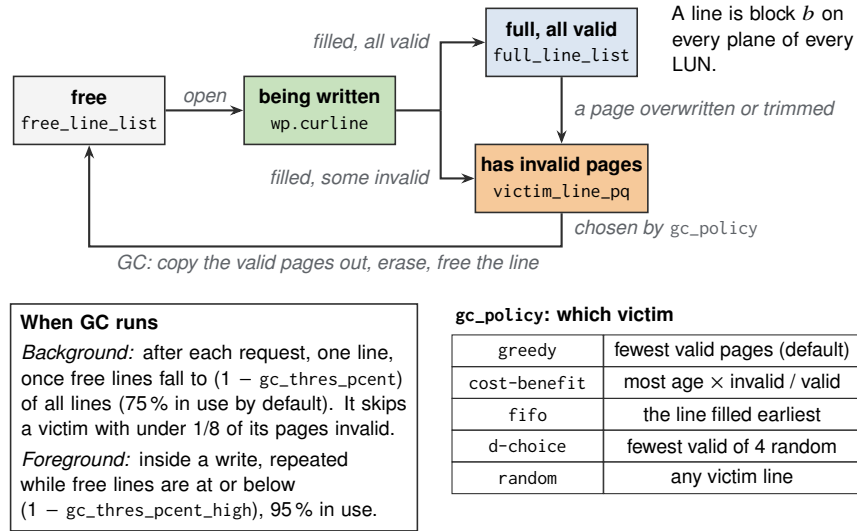
The code is in `hw/femu/bbssd/ftl-line-gc.c`.

## Triggers and watermarks

The two watermarks are stored as free-line counts:

```
gc_thres_lines      = (int)((1 - gc_thres_pcent/100)      * tt_lines)
gc_thres_lines_high = (int)((1 - gc_thres_pcent_high/100) * tt_lines),
    at least 1 with hot_cold_sep, Streams, or a hybrid
    or fast mapping (bb_gc_forced_lines())
```

The floor matters below 20 lines, where the default high watermark rounds to zero. A second write pointer can then take the last free line while the data pointer, which GC writes into, is nearly full, and GC would have nowhere to put a victim's pages (with Streams, GC refuses such a victim outright). One free line always holds a whole victim. The floor costs those small geometries one line of exposable capacity.



**Figure 8:** BBSSD line states and garbage collection: when a line moves between the free, written, full and victim sets, when background and foreground GC run, and how each gc\_policy picks a victim.

| Kind                | When                                  | Where                                                                                                                                | Victim filter                                               | How many                                       |
|---------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------|
| Background          | free lines $\leq$ gc_thres_lines      | after every request that reaches the FTL without an error, any opcode                                                                | the chosen line must have at least 1/8 of its pages invalid | one line                                       |
| Foreground (forced) | free lines $\leq$ gc_thres_lines_high | at the start of a write, and before every page that a write, a write-back of the buffer, or Write Zeroes without Deallocate programs | none                                                        | repeats until above the watermark or no victim |

Forced GC runs per page, not once per command: one command can program more lines than the watermark keeps free, and a collection that starts after the last line is gone has nowhere to move pages to.

A GC step returns -1 when there is no victim, when the background filter rejects it, or when the victim's valid pages cannot all be moved; the forced loop then stops. With Streams on, a step also fails when it frees nothing, because lines of distinct retired streams cannot be combined, or when the victim holds valid pages and no line is free.

### Victim policies

gc\_policy selects one entry of femu\_ftl\_policies[]:

| gc_policy    | Chooses                                                                                                                                                                  | Cost per choice                       |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| greedy       | the line with the fewest valid pages: the top of the priority queue                                                                                                      | $O(\log n)$                           |
| random       | a uniformly random line of the queue, drawn from the seeded generator; a background step puts it back if it fails the 1/8 filter                                         | $O(\log n)$                           |
| cost-benefit | the largest age $\times (1 - u) / 2u$ , with $u = vpc / pgs\_per\_line$ and age = now - close_time, compared in 128-bit integers; a line with no valid pages always wins | $O(n)$ scan                           |
| fifo         | the line that filled first (lowest close_seq), whatever its valid count: the top of the queue, which this policy orders by close_seq                                     | $O(1)$ to find, $O(\log n)$ to remove |

| gc_policy | Chooses                                                                                                | Cost per choice |
|-----------|--------------------------------------------------------------------------------------------------------|-----------------|
| d-choice  | the fewest valid pages among 4 queue slots drawn from the seeded generator; a slot can be picked twice | O(1) sample     |

Only lines in the victim queue are candidates. A line that closed with every page valid stays in the full list until something invalidates one of its pages. Under fifo the queue is keyed by `close_seq`, which does not change while a line is queued, so an invalidation lowers the valid count without moving the line.

random, d-choice and the FDP `gc_strategy=2` (random) draw from a `splitmix64` generator in each namespace's FTL (`ftl_gc_rand()`), seeded from `gc_seed` (default 0). The same configuration, seed and command sequence pick the same victims and give the same write amplification; give each run a different `gc_seed` to vary it. The other policies use no random numbers. `cost-benefit` still reads the host clock for line age, so its picks depend on when commands arrive.

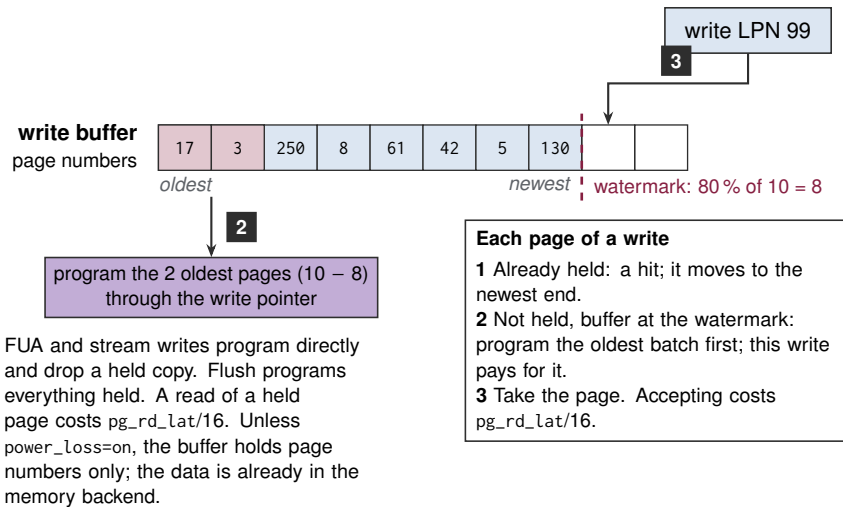
### Collecting a line

```
do_gc(force)
|
v
victim = policy->select_victim_line(force) -- none? return -1
|
v
move: for ch, lun, pl, each page of block (ch, lun, pl, victim->id):
    if VALID:
        new = next page of the data pointer      (stream pointer for
            (takes a free line if it has none)    Streams data)
        none? requeue the victim, return -1
        NAND read at the old page                (GC_IO)
        lpn = rmap[old]
        maptbl[lpn] = new, rmap[new] = lpn
        old page -> INVALID, rmap[old] = none
        NAND program at the new page            (GC_IO)
        gc_write_pages++
    |
    v
erase: for ch in 0..nchs-1, lun in 0..luns_per_ch-1:
    mark each plane's block free: erase_cnt++, read_cnt = 0
    one multi-plane erase of the LUN's blocks    (GC_IO, tplebsy between
                                                planes)
    |
    v
line -> free list tail
```

The relocated page goes wherever the data pointer is, not to the victim's LUN: the model has no copyback. Every valid page is moved before any block is erased. If the data pointer runs out of lines part way, the victim goes back to the victim queue (or the full list, if nothing was moved from a full line) holding the pages it still has, and nothing is erased. The pages that were moved are already invalid in the victim, so its counts stay true and a later step moves only what is left. Writes that then find no line fail with Capacity Exceeded. The watermark floor, per-page forced GC and the capacity reserve keep a valid configuration from getting there.

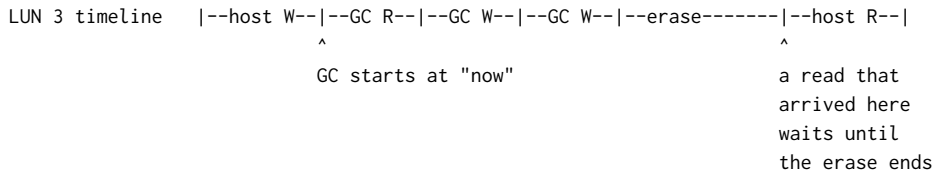
### How GC time is charged

GC operations enter the NAND model with `stime = 0`, which the media bridge replaces with the current host time. Each one moves its LUN's next-available time forward, and the latency it returns is discarded. GC time



**Figure 9:** The BBSSD write buffer with `buffer_size=10` and `buffer_thres_pcent=80`: a write that finds the buffer at its watermark first programs the two oldest pages and pays for them.

therefore reaches the host only through the LUN timelines: a later host read or program on a LUN that GC is using starts when GC is done with it.



- In background GC the request that triggered it has already been timed, so it pays nothing. The requests that follow pay, on the LUNs GC occupied.
- In foreground GC the write itself waits, because its own programs queue behind the GC operations on the same LUNs.
- Admin opcode 0xEF code 2 turns GC timing off (`enable_gc_delay`): GC still moves the pages and erases the blocks, but issues no NAND operations. Code 1 turns it back on.

The LUN field `gc_endtime` is updated as GC programs and erases, and the channel field is never written; nothing reads either.

## FDP reclaim units

With Flexible Data Placement the FTL places data in reclaim units, one line each, through per-handle write pointers, and collects reclaim units instead of lines (`do_gc_fdp_style()` in `hw/femu/bbssd/ftl-fdp.c`). `gc_strategy` picks that victim policy; `gc_policy` and the other FTL knobs listed under Interactions (§3.4.15) are refused. The FDP chapter (§4.6) and the FDP guide (§4.5) describe it.

### 3.4.7 Write buffer

The write buffer models DRAM in front of the NAND. It holds logical page numbers, not data: a write it accepts costs a DRAM access, and the program is charged later to whatever forces the page out. Repeated writes to a buffered page cost one program. The code is in `hw/femu/bbssd/ftl-datapath.c`.

```

host write, pages P..Q
  |
  v
forced GC if free lines <= high watermark; one read-reclaim step
  |
  v
buffer enabled, not FUA, not a stream write? ---- no ----+
  | yes                                                    |
  v                                                    v
for each page:                                           drop buffered copies of
  already held? -> move to tail, write hit                these pages; if the cache
  else if count >= watermark:                             was disabled, write back
    write back `batch` pages from the head                everything
    (oldest first), cost charged to this                    |
    write                                                    v
  insert at tail                                           program each page
    |                                                       (NAND time)
    v
latency = max(DRAM access, write-back cost)

watermark = max(1, (int)(buffer_size * buffer_thres_pcent / 100))
batch      = max(1, buffer_size - watermark)
DRAM access = pg_rd_lat / 16 (1000 ns when pg_rd_lat is 0)

```

- **Structure.** A tail queue in order of last write, LRU (write\_buffer) and a GLib tree keyed by LPN (wb\_tree) for lookup. Capacity is buffer\_size pages.
- **Admission is per page.** A command larger than the buffer cannot push it past its size: it writes back a batch whenever the buffer is at the watermark. If a write-back frees nothing because the device is out of lines, the write fails with Capacity Exceeded.
- **Write-back** (ssd\_buffer\_destage()) takes pages from the head (the least recently written), runs forced GC as needed, charges the DFTL lookup and programs each page exactly as a direct write would. Hybrid merges after each page as usual; FAST merges once per batch. A page costs the same buffered or not; only the request that pays differs.
- **Reads.** A read of a buffered page costs one DRAM access and does not reach NAND, even when the buffer has stopped accepting writes.
- **FUA and stream writes** program directly. A direct write first drops any buffered copy of its own pages, so an older version is never programmed after the newer one.
- **Deallocate** drops buffered copies before unmapping, so discarded data is never programmed later.
- **Flush** reaches the FTL like any other command and writes back the whole buffer, charged to the Flush.
- **Volatile write cache.** With vwc=1 the controller advertises a cache. When the host turns it off with feature 06h, the buffer stops accepting writes and the next write drains it. With vwc=0 the buffer still works, and the host has no way to turn it off; Linux sends no Flush to a controller that advertises no cache.

host\_write\_pages counts pages when the host writes them and nand\_write\_pages when they are programmed, so a buffer that absorbs overwrites makes the write amplification factor drop below 1.

## Power loss

With power\_loss=on, the poller no longer touches payload: the FTL thread runs the whole I/O command (nvme\_power\_io()), so the data copy and the buffer change together. The first time a page enters the buffer, the FTL saves the page's previous bytes and per-LBA state as an undo record. When the page is programmed the record is dropped. Setting the QOM property simulate-power-loss (runtime properties (§A.2.1)) resets



the controller, restores every buffered page from its undo record, empties the buffer and counts an unsafe shutdown.

In this mode a normal shutdown, turning the cache off and Sanitize first write back every namespace's buffer; Flush, Format, Dataset Management, Copy, Write Zeroes (except on a PI-formatted namespace) and Write Uncorrectable write back their own namespace's. If the device has no line to write to, they fail with Capacity Exceeded (a shutdown sets Controller Fatal Status instead) rather than claim the data is durable. An FUA write programs a whole NAND page, so buffered neighbours in that page become durable with it.

### 3.4.8 Read cache

read\_cache\_mb adds a DRAM read cache in front of NAND reads (hw/femu/bbssd/ftl-cache.c). Like the write buffer it holds LPNs only.

```
host read of a mapped lpn (not in the write buffer)
    |
    v
lookup in an open-addressed hash over capacity slots
hit:  cost pg_rd_lat / 16, no NAND read, no block read count
miss: insert (evict by cache_evict if full), then NAND read
```

- Capacity is read\_cache\_mb MiB divided by the page size.
- Only reads fill it. A host overwrite, a deallocate and a power cut remove the page's entry. A GC relocation does not, since the content is the same.
- cache\_evict picks the victim: clock (second chance), random (a fixed-seed generator, so runs repeat), lru (exact, by scan) or arc, a scan-resistant 2Q variant that evicts pages read once before pages read twice. It is not full ARC.
- Hit and miss counts are kept in ssd->rcache but not exported.

### 3.4.9 Deallocate, Write Zeroes, Format and Sanitize

- **Dataset Management deallocate** (ssd\_trim()) unmaps every LPN of every range: it drops the buffered copy, invalidates the physical page, clears maptbl and rmap, and drops the read cache entry. A log-block scheme does this through its trim() hook. A range past the device is skipped. The latency is trim\_lat\_ns times the number of ranges, or 0.
- **Write Zeroes with Deallocate** unmaps the range in the same way and costs no time. **Without Deallocate** it programs the range directly, after forced GC if needed, and counts the pages as host writes.
- **Format** and **Sanitize** call bbssd\_deallocate\_all() on BlackBox namespaces (not CSD), which unmaps every LPN of the namespace's FTL. The lines keep their wear.

ONCS bit 0x8 (Write Zeroes) and 0x100 (Copy) are off unless oncs sets them.

### 3.4.10 Wear, read reclaim and retention refresh

#### Program/erase cycles

Every erase increments the block's erase\_cnt and the FTL-wide total\_erases. SMART Percentage Used is

```
| percentage_used = min(255, total_erases * 100 / (tt_blks * rated_pe_cycles))
```

where rated\_pe\_cycles is pe\_cycles Rated, or the rating of nand\_cell\_type (SLC 100000, MLC 3000, TLC 1000, QLC 300), or 0, in which case the field reads 0. nand\_bad\_blocks marks a number of blocks as

factory bad for SMART Available Spare only:  $100 - \text{bad} * 100 / \text{tt\_blks}$ . Placement ignores them. With several namespaces the controller reports the most worn namespace and the lowest spare.

With `ecc_step_ns` set, a block's erase count (one tier per 750 erases) and the age of its line add time to each read. That model belongs to the NAND timing chapter.

### Read reclaim and retention refresh

```

host NAND read of block B in line L
|
+--- read_reclaim_limit set and B.read_cnt >= limit  ---+
+--- retention_limit_sec set and now - L.close_time >= ----> queue L
    limit                                              (one line at
                                                         a time)

next host write:
|
v
do_read_reclaim(): queued line, high watermark not reached,
at least 2 free lines, L full or in the victim queue?
| yes
v
take L out of its list, reclaiming = true, collect L like GC
read_reclaims++ or retention_refreshes++

```

Every NAND read the FTL issues, host, GC or merge, increments its block's `read_cnt` (GC reads are issued only while GC timing is on); an erase resets it. Reads answered by the write buffer, the read cache or a DFTL translation page do not count. The rewrite happens on a write, where relocation already costs something, so a read never stalls behind a whole line. One line is refreshed per write at most. A line that nothing reads is never refreshed, and a workload that only reads refreshes nothing: FEMU runs no background media scan. The request is dropped if GC collects the line first, or if the line is still open for writing when the next write checks it.

### 3.4.11 Fault insertion

`err_read_unc_ppm` and `err_write_fail_ppm` turn a rate into a period,  $1000000 / \text{ppm}$ , at least 1. Every Nth Read command completes with Unrecovered Read Error (with DNR, as every Unrecovered Read Error in FEMU is) and every Nth Write command with Write Fault. The counters run per namespace FTL, so a run repeats exactly. The command is still timed and, for a write, still programmed and mapped; only its status changes. The injected counts feed SMART Media and Data Integrity Errors. Write Zeroes and Copy are never failed.

### 3.4.12 Over-provisioning and capacity

#### The reserve

`bb_check_capacity()` in `hw/femu/bbssd/bb.c` refuses a namespace that would leave GC no room. The reserve, in lines:

```

reserve = gc_thres_lines_high      forced watermark, with its floor
      + 1                        data pointer
      + 1 if hot_cold_sep          hot pointer
      + 1 if mapping is hybrid or fast  log pointer
      + streams.max + 1 if streams    stream and stream GC pointers

usable = (blks_per_pl - reserve) * pages_per_line * page_size
refused when the namespace is larger than usable,
or when blks_per_pl <= reserve

```

Each namespace has its own FTL built from the whole geometry, so the check applies per namespace.

### Sizing the namespace

Without `op_pcent`, the backend is `devsz_mb` MiB, split across the namespaces, and the spare area is whatever the geometry has beyond that. With `op_pcent`, the backend is the raw NAND capacity and each namespace gets  $\text{raw} \times 100 / (100 + \text{op\_pcent}) / \text{namespaces}$ , rounded down to 512 bytes (an even split unless `namespace_sizes` is given); `devsz_mb` is ignored.

### Worked example: the default geometry

The default properties describe 8 channels x 8 LUNs x 1 plane x 256 blocks x 256 pages x 8 sectors x 512 bytes:

|                 |                            |                            |
|-----------------|----------------------------|----------------------------|
| page            | = 8 x 512                  | = 4 KiB                    |
| NAND pages      | = 8 x 8 x 1 x 256 x 256    | = 4,194,304                |
| raw capacity    | = 4,194,304 x 4 KiB        | = 16 GiB                   |
| lines           | = <code>blks_per_pl</code> | = 256                      |
| blocks per line | = 8 x 8 x 1                | = 64                       |
| pages per line  | = 64 x 256                 | = 16,384 (64 MiB)          |
| background GC   | = (int)(0.25 x 256)        | = 64 free lines (75% used) |
| forced GC       | = (int)(0.05 x 256)        | = 12 free lines (95% used) |
| victim filter   | = 16,384 / 8               | = 2,048 invalid pages      |
| reserve         | = 12 + 1                   | = 13 lines                 |
| usable          | = 243 x 64 MiB             | = 15,552 MiB               |

```

devsz_mb=12288 (the launcher): 12 GiB = 192 lines of data
-> the data pointer holds a line from init, so free lines reach
    64 and background GC starts as the last of the 192 lines is
    opened, just before one full pass completes
op_pcent=25: 16 GiB x 100 / 125 = 13,107.2 MiB exposed
op_pcent=7:  16 GiB x 100 / 107 = 15,312.1 MiB exposed

```

```

host memory for the FTL: maptbl 32 MiB + rmap 32 MiB
                        + page status 16 MiB + block and line arrays

```

With the default `devsz_mb` of 1024 the same geometry exposes 1 GiB, about 6% of the NAND, and GC barely runs. To give GC work on a small device, shrink the geometry instead; this one has 512 MiB of NAND and exposes about 410 MiB:

```

| -device femu,femu_mode=1,nchs=2,luns_per_ch=4,blks_per_pl=64,op_pcent=25

```

### 3.4.13 Run-time controls

Admin opcode 0xEF changes the FTL while the guest runs. Only a BlackBox controller accepts it. Codes 1 to 4 apply to every namespace with an FTL, with the dataplane paused; codes 5 to 7 act on the controller.

| CDW10 | Effect                                                      |
|-------|-------------------------------------------------------------|
| 1     | GC operations take NAND time (the default)                  |
| 2     | GC operations take no NAND time                             |
| 3     | flat read, program, erase times back to 40 us, 200 us, 2 ms |
| 4     | flat read, program and erase times to 0                     |
| 5     | reset the poller completion counters                        |
| 6, 7  | turn the per-request log lines on or off                    |

Codes 3 and 4 set the built-in flat times, not the values given on the command line, and do not affect `nand_cell_type` tables. See changing timing at run time (§6.2.11).

### 3.4.14 Statistics

#### Vendor log page C0h

`nvme_collect_media_stats()` in `hw/femu/nvme-admin.c` sums the FTL counters over the controller's bbssd, CSD and KV namespaces. `log-pages-and-counters.md` (§B.1) lists the offsets.

| C0h field                                | FTL counter                                                   | Moves when                                                                                     |
|------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| WAF x 1000                               | $(\text{nand} + \text{gc}) \times 1000 / \text{host}$         | after the first host write                                                                     |
| host write pages                         | <code>host_write_pages</code>                                 | Write, the destination of Copy, and Write Zeroes without Deallocate, per page, buffered or not |
| GC write pages                           | <code>gc_write_pages</code>                                   | GC, read reclaim and log-block merges relocate a page                                          |
| NAND write pages                         | <code>nand_write_pages</code>                                 | a host page is programmed                                                                      |
| max block reads                          | <code>largest_read_cnt</code>                                 | NAND reads; reset by erase                                                                     |
| read reclaims, retention refreshes       | <code>read_reclaims</code> , <code>retention_refreshes</code> | a queued line is rewritten                                                                     |
| buffer reads and read hits               | <code>sp.read_cnt</code> , <code>sp.read_hit_cnt</code>       | host read pages, and those the buffer held                                                     |
| buffer writes and write hits             | <code>sp.write_cnt</code> , <code>sp.write_hit_cnt</code>     | host write pages, and those already buffered                                                   |
| hybrid switch, full merges, merge erases | <code>struct femu_map_hybrid</code>                           | mapping=hybrid only                                                                            |

The buffer read and write counts move with or without a buffer, so their ratio is the hit rate. Telemetry log 07h captures the same 512 bytes.

#### SMART and Endurance Group

| Field                               | Source                                                                                                                                  |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Percentage Used                     | <code>ssd_percentage_used()</code> , most worn namespace                                                                                |
| Available Spare                     | <code>ssd_available_spare()</code> , lowest namespace; 20 or below sets the spare critical warning                                      |
| Media and Data Integrity Errors     | injected read and write faults (and ZNS write faults)                                                                                   |
| Data units, host commands           | the pollers' host I/O counters, not the FTL                                                                                             |
| Endurance Group Media Units Written | $(\text{NAND} + \text{GC write pages}) \times \text{page size}$ , in units of $10^9$ bytes rounded up; needs a <code>femu-subsys</code> |

### Counters kept but not exported

DFTL hits and misses (ssd->cmt), read cache hits and misses (ssd->rcache), FAST merge counts and the per-block erase\_cnt stay inside the FTL. debug\_ftl=on prints hybrid and FAST merge counts to stdout.

### 3.4.15 Parameters

Every property is listed with its type, default and range in the property reference (§A.1); this table says what each does inside the FTL and what it interacts with.

#### Geometry and capacity

NAND geometry (§A.1.1), mode and capacity (§A.1.1)

| Property                              | Effect in the FTL                                             | Interacts with                                     |
|---------------------------------------|---------------------------------------------------------------|----------------------------------------------------|
| secsz, secs_per_pg                    | page size; LPN = byte offset / page size                      | the LBA format; DFTL entries per translation page  |
| pgs_per_blk                           | pages per block and per line column; the log-block merge unit | at most 512 with nand_cell_type                    |
| blks_per_pl                           | the number of lines                                           | the watermarks and the reserve are fractions of it |
| pls_per_lun, luns_per_ch, nchs        | line width and write striping                                 | parallelism; tplebsy for multi-plane erase         |
| devsz_mb, namespaces, namespace_sizes | the exposed capacity                                          | must fit the usable lines per namespace            |
| op_pcent                              | exposes a fixed fraction of raw NAND                          | overrides devsz_mb; refused with cxl_ssd           |

#### Garbage collection, mapping and caches

Garbage collection, mapping and caches (§A.1.1)

| Property                        | Effect in the FTL                                                | Interacts with                                                    |
|---------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------|
| gc_thres_pcent                  | background GC watermark                                          | must not exceed gc_thres_pcent_high                               |
| gc_thres_pcent_high             | forced GC watermark; sizes the reserve                           | lower values cost exposed capacity                                |
| gc_policy                       | line victim selection                                            | refused with FDP                                                  |
| gc_seed                         | seed for the random and d-choice policies and FDP random reclaim | no effect on the other policies                                   |
| gc_strategy                     | reclaim unit victim selection                                    | FDP only                                                          |
| mapping                         | L2P scheme                                                       | hybrid and fast reserve a line and refuse Streams; FDP needs page |
| mapping_cache_mb                | DFTL cache size                                                  | used only with mapping=dftl                                       |
| read_cache_mb, cache_evict      | read cache size and eviction                                     | hit time is pg_rd_lat / 16 at realize; 0xEF does not change it    |
| hot_cold_sep                    | hot write pointer for overwrites                                 | page or DFTL only; reserves a line; refused with FDP              |
| buffer_size, buffer_thres_pcent | write buffer capacity in pages, and the write-back watermark     | vwc, power_loss; refused with FDP                                 |
| fdp_trim_erase_all              | FDP deallocate resets every reclaim unit                         | FDP only                                                          |
| debug_ftl                       | prints invalid page transitions and merge counts                 | none                                                              |

## Reliability and wear

### Reliability and wear (§A.1.1)

| Property                             | Effect in the FTL                             | Interacts with                             |
|--------------------------------------|-----------------------------------------------|--------------------------------------------|
| pe_cycles Rated                      | Percentage Used denominator                   | overrides the nand_cell_type rating        |
| nand_bad_blocks                      | Available Spare                               | placement ignores it                       |
| ecc_step_ns, ecc_retention_sec       | read time grows with erase count and line age | ecc_retention_sec refused with FDP         |
| err_read_unc_ppm, err_write_fail_ppm | fixed-period command failures                 | counted in SMART media errors              |
| read_reclaim_limit                   | read count that queues a line for rewrite     | needs host writes to act; refused with FDP |
| retention_limit_sec                  | line age that queues a line for rewrite       | same                                       |

## NAND timing the FTL charges

### NAND timing (§A.1.1)

The FTL charges `trim_lat_ns` per deallocate range (refused with FDP) and issues a multi-plane erase per LUN during GC, whose inter-plane time is `tplebsy`. All the other timing properties are applied by the NAND media layer.

## Controller properties the FTL reads

| Property             | Where documented             | Effect in the FTL                                                     |
|----------------------|------------------------------|-----------------------------------------------------------------------|
| vwc                  | controller identity (§A.1.1) | lets the host turn the write buffer off                               |
| oncs                 | same                         | Write Zeroes and Copy reach the FTL only when enabled                 |
| power_loss           | power loss (§A.1.1)          | undo records and simulate-power-loss                                  |
| streams, streams.max | same                         | per-stream write pointers; reserve <code>streams.max + 1</code> lines |

## Interactions and refusals

- **FDP** refuses `buffer_size`, `hot_cold_sep`, `read_reclaim_limit`, `retention_limit_sec`, `ecc_retention_sec`, `trim_lat_ns`, a mapping other than page and a `gc_policy` other than greedy, with FEMU bbsd: `<name>` has no effect under FDP.
- **Unknown names** for mapping, `gc_policy` and `cache_evict` are refused at realize.
- **Watermarks** outside `[1, 100]`, or a high watermark below the low one, are refused.
- **The write buffer** is off when `power_loss=on` and `vwc=0`, and while the host has the advertised cache turned off.
- **DFTL cache size** has no effect under page, hybrid or fast.

Examples, each checked by `check-doc-examples.py`:

```
-device femu,devsz_mb=1024,femu_mode=1,mapping=dftl,mapping_cache_mb=8
-device femu,devsz_mb=1024,femu_mode=1,hot_cold_sep=on,gc_policy=cost-benefit
-device femu,devsz_mb=1024,femu_mode=1,buffer_size=4096,buffer_thres_pcent=75,vwc=1
-device femu,devsz_mb=1024,femu_mode=1,mapping=hybrid
```

### 3.4.16 Validation status

What the automated tests check:

| Area                              | Test                                                                                                                                                         | Checks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BAST merges                       | unit test test-femu-hybrid-oracle; qtests hybrid-oracle-*, hybrid-batch-occupancy, hybrid-destage-occupancy, hybrid-switch-trim-erase, hybrid-trim-occupancy | the unit test checks the reference model; the qtests compare FEMU's switch, full merge and erase counts against it, including deallocate and the write buffer                                                                                                                                                                                                                                                                                                                                                     |
| Victim queue                      | unit test test-femu-pqueue                                                                                                                                   | priority queue operations, including random pop driven by the caller's number                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Reproducible GC                   | qtests gc-seed-d-choice, gc-seed-random, gc-seed-fifo                                                                                                        | two devices with the same configuration and 2048 random page writes report the same relocation count every 32 writes and the same WAF; a third with another gc_seed differs (FIFO: matches)                                                                                                                                                                                                                                                                                                                       |
| NAND timing<br>C0h counters       | unit test test-femu-nand-media<br>qtest media-counters                                                                                                       | the media layer the FTL calls<br>host and NAND page counts and the WAF move with writes                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Write buffer                      | qtest buffer-counters, flush-without-vwc and power-loss-*                                                                                                    | hit counts; Flush (also with vwc=0), FUA, write-back, cache disable, shutdown and power-cut rollback                                                                                                                                                                                                                                                                                                                                                                                                              |
| Streams                           | qtest streams-gc and the other streams-* cases                                                                                                               | stream placement and GC of stream lines                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| GC with no free line              | qtests gc-no-destination, gc-no-destination-hot-cold, streams-gc-floor, gc-no-destination-fdp                                                                | on a geometry whose forced watermark rounds to zero, random single-page and 64-page writes never fail, no mapping names an erased page and no valid page is orphaned (read through the qtest-only x-ftl-check property)                                                                                                                                                                                                                                                                                           |
| Charged work for a fixed workload | qtests ftl-trace-bbssd, ftl-trace-hot-cold, ftl-trace-fdp                                                                                                    | a seeded queue-depth-one write and read workload with collection gives exact command, host, NAND, relocated-page and erase counts and exact read, program and erase commands charged to the media layer (qtest-only x-ftl-trace); a refactor of the allocator, collection or media charge must not move them. A different victim policy or a dropped or duplicated charge fails it. The modelled-latency sum depends on host load and is only held to 25%, so NAND timing values are left to test-femu-nand-media |
| Format, Sanitize                  | qtests format-ftl, sanitize                                                                                                                                  | after Format, GC relocates nothing; Sanitize status and zeroed data (the FTL state is not checked)                                                                                                                                                                                                                                                                                                                                                                                                                |
| Robustness                        | qtests io-fuzz, io-fuzz-fdp, config-refused                                                                                                                  | malformed I/O, refused configurations                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Start-up                          | doc-examples                                                                                                                                                 | every tagged example on this page and the BlackBox guide starts and moves one block                                                                                                                                                                                                                                                                                                                                                                                                                               |

Apart from the reproducibility checks above, no automated test checks the behaviour of the random, cost-benefit, fifo and d-choice policies, dftl and fast mapping, hot/cold separation, the read cache, read reclaim, retention refresh, read and write fault insertion on BlackBox, or the wear and spare figures. They are covered only by the start-up examples and by guest runs during development; femu-test.sh (guest-side

tests (§8.2.5)) checks that the counters move. FEMU's latencies and write amplification have not been calibrated against a specific commercial drive.

### 3.4.17 Limits

- Data placement is modelled per page; the FTL never tracks sectors within a page, so writes smaller than a page cost a whole page program.
- GC relocations go through the data pointer (or a stream pointer for stream data), never back to the victim's LUN (no copyback), and GC runs on the FTL thread, one line at a time.
- There is no explicit wear levelling, and bad blocks do not affect placement.
- Read reclaim and retention refresh act only on reads followed by writes.
- cost-benefit GC uses the host clock for age, so it is not reproducible run to run.
- An injected write fault still programs and maps the data.
- FAST merge counts and DFTL and read cache hit rates are not exported.
- Each namespace's FTL is built from the whole geometry, so host memory for the FTL grows with the namespace count.

### 3.4.18 Extending the FTL

#### Adding a GC victim policy

1. Write static struct line \*select\_victim\_line\_<name>(struct ssd \*ssd, bool force) in hw/femu/bbssd/ftl-line-gc.c. Candidates are the lines in ssd->lm.victim\_line\_pq; slots 1 to size - 1 of pq->d[] hold them.
2. Apply the background filter: when !force and the line has fewer than pgs\_per\_line / 8 invalid pages, return NULL and leave the queue as it was.
3. Remove the chosen line with pqueue\_remove() (or pqueue\_pop() for the top), set line->pos = 0 and decrement lm->victim\_line\_cnt. reclaim\_line() expects a line that is in no list.
4. Add { .name = "<name>", .select\_victim\_line = ... } to femu\_ftl\_policies[]. femu\_ftl\_policy\_known() then accepts the name.
5. Describe it in the gc\_policy description in hw/femu/femu-props.c, regenerate the property reference, and add a qtest that tells the policy apart from greedy.

#### Adding a mapping scheme

1. Implement struct femu\_mapping\_ops in a new ftl-map-<name>.c: at least translate, prepare\_write, commit\_write and gc\_relocate\_commit. Keep maptbl and rmap correct: commit\_write must invalidate the old page and set both tables, because GC and reads depend on them.
2. Keep private state in ssd->map\_priv, allocated in init and freed in exit.
3. Set uses\_log\_class if writes go through the LOG pointer; the reserve then grows by one line and Streams are refused. Set uses\_cmt to get the DFTL cost model.
4. For merges, provide needs\_reclaim and reclaim(ssd, budget). reclaim() charges its own NAND operations and returns the latency to add to the triggering request. reclaim\_per\_page runs it after every page instead of once per request. Allocate a destination before invalidating the source, so a full device never loses the only copy.
5. Provide trim if the scheme keeps per-page state.
6. Add the ops to femu\_mapping\_extra[] in hw/femu/bbssd/ftl-map.c, declare them in ftl-internal.h, and add the file to the FEMU source list in hw/femu/meson.build.

Run the unit tests and the qtests (testing guide (§8.2)), and check that the C0h counters move for the new code



path: a counter that never changes in a mode means the mode is not reached.

### 3.4.19 Source map

| File                           | Contents                                                                                       |
|--------------------------------|------------------------------------------------------------------------------------------------|
| hw/femu/bbssd/bb.c             | mode registration, bb_check_capacity(), FDP refusals, the 0xEF switch, teardown                |
| hw/femu/bbssd/ftl.c            | ssd_init(), bb_ftl_process_req(), Copy, counters, Percentage Used, Available Spare, ssd_free() |
| hw/femu/bbssd/ftl.h            | struct ssd, struct ppa, lines, write pointers, the mapping and policy ops                      |
| hw/femu/bbssd/ftl-internal.h   | address helpers, ssd_lpn_range(), GC watermark tests                                           |
| hw/femu/bbssd/ftl-geom.c       | bb_check_geometry(), ssd_init_params(), NAND array allocation                                  |
| hw/femu/bbssd/ftl-datapath.c   | read, write, write buffer, deallocate, Write Zeroes, power-loss rollback                       |
| hw/femu/bbssd/ftl-line-gc.c    | lines, write pointers, Streams pointers, GC, victim policies, read reclaim                     |
| hw/femu/bbssd/ftl-map.c        | maptbl, rmap, page and DFTL schemes, scheme registry                                           |
| hw/femu/bbssd/ftl-map-cmt.c    | DFTL cached mapping table cost                                                                 |
| hw/femu/bbssd/ftl-map-hybrid.c | BAST log-block scheme and its C0h counters                                                     |
| hw/femu/bbssd/ftl-map-fast.c   | FAST log-block scheme                                                                          |
| hw/femu/bbssd/ftl-cache.c      | read cache and its eviction policies                                                           |
| hw/femu/bbssd/ftl-media.c      | bridge to the NAND media layer, block read counts                                              |
| hw/femu/bbssd/ftl-fdp.c        | FDP reclaim units, handles and their GC                                                        |
| hw/femu/bbssd/ftl-exp.c        | debug tracing of marked pages, off unless FEMU_EXP_LOG or FEMU_DUMP_LPN is set                 |
| hw/femu/femu.c                 | the FTL thread, op_pcent sizing, simulate-power-loss                                           |
| hw/femu/nvme-admin.c           | C0h, SMART and Endurance Group counters, Sanitize, the cache feature                           |

### 3.4.20 Related pages

- BlackBox guide (§4.4): launch, guest use, refusals
- Timing model (§6.2)
- Architecture (§3.1.5)
- Measuring (§6.1.2)
- Log pages and counters (§B)
- Property reference (§A.1)

## 3.5 NAND media and timing model

This chapter describes the component that decides how long a flash operation takes in FEMU. It covers the geometry the model works on, the time it charges for a page read, a page program and a block erase, how operations on the same die, plane or channel wait for each other, and how those times become the completion time of an NVMe command. It also covers the two models that sit after the media: the host link and the controller firmware CPU.

The timing model concept page (§6.2) explains the same model from the user's side. This page is for readers who want the algorithms, the data structures and the source.

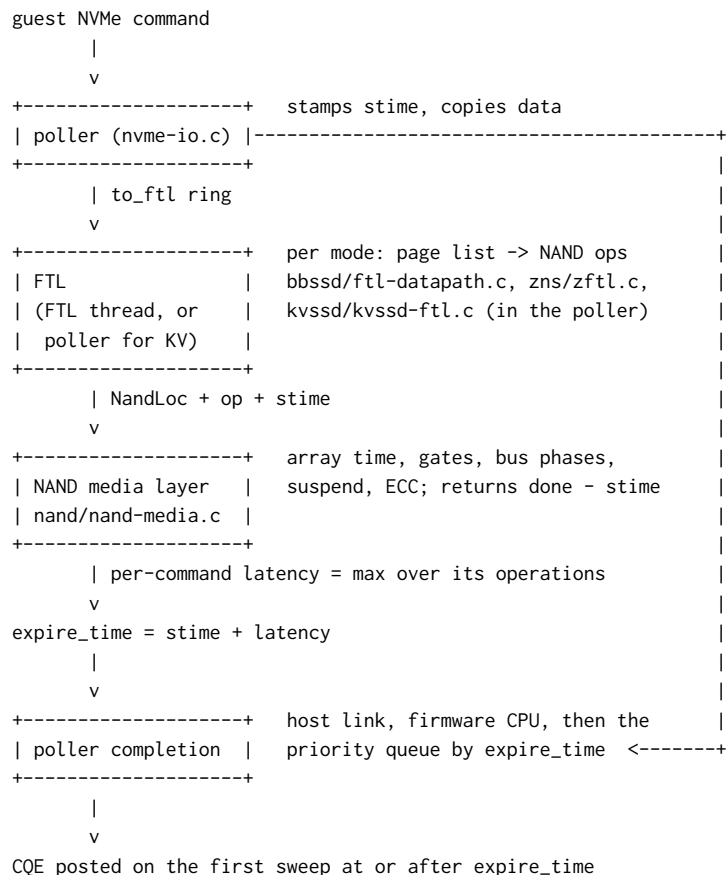
### 3.5.1 Contents

- Purpose and place in FEMU (§3.5.2)
- Geometry (§3.5.3)
- Data structures (§3.5.4)
- Operation timing (§3.5.5)
- Cell types and page types (§3.5.6)
- Channel bus (§3.5.7)
- Plane parallelism and multi-plane operations (§3.5.8)
- Program and erase suspend (§3.5.9)
- ECC read time (§3.5.10)
- From operations to a completion time (§3.5.11)
- How garbage collection is charged (§3.5.12)
- Host link and firmware CPU (§3.5.13)
- OCSSD timing model (§3.5.14)
- Runtime switches (§3.5.15)
- Parameters (§3.5.16)
- Statistics (§3.5.17)
- Calibrating against a real device (§3.5.18)
- Validation status (§3.5.19)
- What is not modelled (§3.5.20)
- Adding a timing feature (§3.5.21)
- Source map (§3.5.22)

### 3.5.2 Purpose and place in FEMU

FEMU copies a command's data as soon as it parses the command. Timing is computed separately: a model works out when the command would have finished on the emulated device, and the poller holds the completion until the host clock reaches that time. The NAND media model is the part of that computation that charges flash operations.

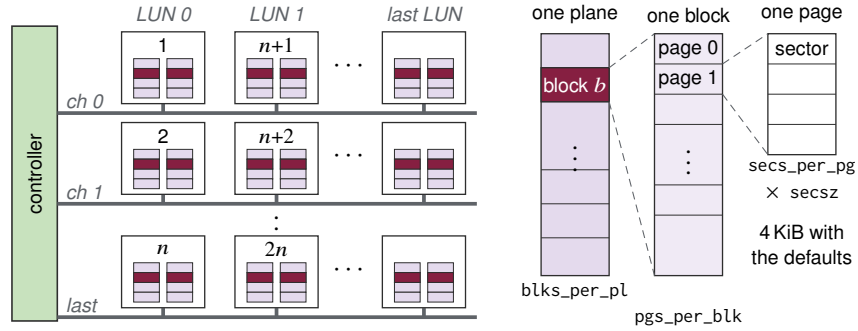
The **media layer**, `nand_media_op()` in `hw/femu/nand/nand-media.c`, is the one media engine. BBSSD, CSD, KV, ZNS, the FTL behind a `femu-cxl-ssd`, and Open-Channel 1.2 and 2.0 use it. It never includes a controller header. Each mode decodes its own address into a `NandLoc`, gives the layer a configuration, and lends it pointers to its busy-until fields. `hw/femu/ocssd/oc-timing.c` keeps the Open-Channel table times, the geometry check and 0xEE (OCSSD timing model (§3.5.14)).



| Mode           | Engine                                 | Built in                                       | Called from         |
|----------------|----------------------------------------|------------------------------------------------|---------------------|
| BBSSD, CSD     | media layer                            | bb_nand_media_init() in bbssd/ftl-media.c      | FTL thread          |
| KV             | media layer, through the BBSSD wrapper | ssd_init() from kvssd/kvssd-ftl.c              | poller              |
| ZNS            | media layer                            | zns_nand_media_init() in zns/zftl.c            | FTL thread          |
| femu-cxl-ssd   | media layer, through the BBSSD wrapper | femu_cxl_start() in cxlssd/cxlssd.c            | femu-cxl-ftl worker |
| OCSSD 1.2, 2.0 | media layer                            | oc12_media_init(), oc20_media_init() in ocssd/ | poller              |
| NoSSD          | none                                   |                                                |                     |

### 3.5.3 Geometry

Every mode that models NAND describes it with the same hierarchy. Only the names of the properties and the field widths differ.



nchs channels, luns\_per\_ch LUNs per channel and pls\_per\_lun planes per LUN (two drawn).  
 ■ A line is block  $b$  on every plane of every LUN, so there are  $\text{blks\_per\_pl}$  lines. The numbers are the order in which the write pointer hands out pages ( $n = \text{nchs}$ ): channel first, then LUN, then plane, then page, so the pages of a large write land on different LUNs.

**Figure 10:** The NAND hierarchy (channel, LUN, plane, block, page, sector), a line as one block index across every plane of every LUN, and the order in which the write pointer hands out pages.

```
device
+-- channel 0 .. nchs-1          one shared bus per channel
+-- LUN 0 .. luns_per_ch-1      a die: runs one array operation at a time
    |                          (one per plane under the ZNS plane gate)
+-- plane 0 .. pls_per_lun-1
+-- block 0 .. blks_per_pl-1    erase unit
+-- page 0 .. pgs_per_blk-1    program and read unit
+-- sector 0 .. secs_per_pg-1  (secsz bytes each)
```

BBSSD line (superblock)  $k = \text{block } k \text{ on every (channel, LUN, plane)}$   
 $\text{lines} = \text{blks\_per\_pl} \quad \text{pages per line} = \text{nchs} * \text{luns\_per\_ch} * \text{pls\_per\_lun} * \text{pgs\_per\_blk}$

## Address formats

| Mode           | Address                                                    | Fields, low bit first                                             | Where                       |
|----------------|------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------|
| BBSSD, CSD, KV | struct ppa                                                 | blk 16, pg 16, sec 8, pl 4, lun 7, ch 12, reserved 1              | bbssd/ftl.h                 |
| ZNS            | struct ppa                                                 | spg 2, pg 16, blk 16, fc (LUN) 8, pl 3, ch 7, valid 1, reserved 8 | zns/zns.h                   |
| OCSSD 1.2      | 64-bit PPA with masks and offsets from the geometry        | ch, lun, pln, blk, pg, sec                                        | PPA_* macros in nand/nand.h |
| OCSSD 2.0      | 64-bit LBA with group, parallel unit, chunk, sector fields | group, PU, chunk, sector                                          | OC20_LBA_GET_* in ocssd/    |

A field width is an upper bound on its axis. `bb_check_geometry()` in `bbssd/ftl-geom.c` refuses a count larger than its field can hold, and refuses a geometry whose total sector count does not fit in a signed 32-bit integer. `zns_check_params()` bounds each ZNS axis by its field. `oc_timing_geometry_ok()` requires every OCSSD axis to be non-zero, `lnum_ch` at most 32 and `lnum_ch * lnum_lun` at most 128, the sizes of the per-channel and per-LUN arrays.

## How pages land on the geometry

The timing a workload sees depends on where the FTL puts its pages, because the media model only charges what it is asked to.

- **BBSSD, CSD, KV** allocate from a write pointer that moves channel first, then LUN, then plane, then page (`ssd_advance_write_pointer_common()` in `bbssd/ftl-line-gc.c`). Consecutive pages of a large write therefore land on different channels, then different LUNs, and are programmed in parallel.
- **ZNS** buffers writes in a per-zone write cache. A flush programs one program unit per plane: `zns_flash_type` pages of 16 KiB (`ZNS_PAGE_SIZE`) per plane, charged as one program operation per plane (`zns_wc_flush()` in `zns/zftl.c`). A 4 KiB logical page is a sub-page (spg) of a 16 KiB page.
- **OCSSD** takes physical addresses from the host, so the host chooses the parallelism.

### 3.5.4 Data structures

The media layer's types are in `hw/femu/nand/nand-media.h`.

| Type             | Fields that matter                                                                                                                                                                                                          | Role                                                                                                                                                                                        |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NandLoc          | ch, lun, pl, blk, pg, flash_type, page_type, pe_cycles, age_sec, xfer_secs                                                                                                                                                  | One operation's position and the facts timing depends on. Filled by the mode's decoder ( <code>bb_decode_loc()</code> , <code>zns_advance_status()</code> , <code>oc12_media_op()</code> ). |
| NandMediaTiming  | rd_ns, wr_ns, er_ns (flat); rd_table_ns, wr_table_ns, er_table_ns (by cell and page type); pgtype_mult; cmd_addr_ns, page_xfer_ns, status_ns; tplebsy_ns and three unused multi-plane and cache-read times; ecc_*; tsusp_ns | Every duration.                                                                                                                                                                             |
| NandMediaPolicy  | array_gate, channel_mode, pe_suspend, ecc_on_read, use_flat_timing, cache_read                                                                                                                                              | Which mechanisms are on.                                                                                                                                                                    |
| NandTimelineOps  | ch_avail, lun_avail, plane_avail, page_reg_ready, lock_lun, unlock_lun                                                                                                                                                      | Accessors that return pointers into the mode's own busy-until fields.                                                                                                                       |
| NandMedia        | cfg, bus_res (per-channel booked windows), susp (per-position suspend state)                                                                                                                                                | One instance per namespace FTL.                                                                                                                                                             |
| NandOpCompletion | done_ns, latency_ns                                                                                                                                                                                                         | Absolute end time, and done_ns - stime.                                                                                                                                                     |

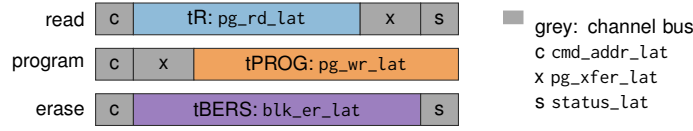
The busy-until times ("availability timestamps") live in the mode's structures, not in the media layer:

| Resource    | BBSSD, CSD, KV                              | ZNS                                                      | OCSSD                                                                 |
|-------------|---------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------|
| Channel bus | <code>ssd_channel.next_ch_avail_time</code> | <code>zns_ch.next_ch_avail_time</code>                   | <code>0c12Ctrl.ch_avail[]</code> (1.2 only)                           |
| LUN (die)   | <code>nand_lun.next_lun_avail_time</code>   | <code>zns_fc.next_fc_avail_time</code> (never consulted) | <code>0c12Ctrl.lun_avail[]</code> , <code>0c20Ctrl.lun_avail[]</code> |
| Plane       | not kept                                    | <code>zns_plane.next_plane_avail_time</code>             | not kept                                                              |

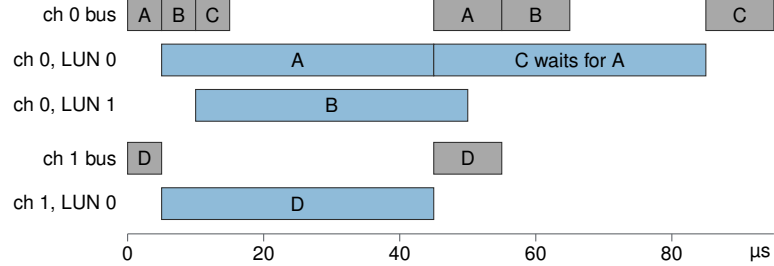
Each mode configures the layer differently:

| Setting      | BBSSD, CSD, KV, CXL                                                                       | ZNS                               |
|--------------|-------------------------------------------------------------------------------------------|-----------------------------------|
| array_gate   | <code>NAND_GATE_LUN_ONLY</code>                                                           | <code>NAND_GATE_PLANE_ONLY</code> |
| channel_mode | <code>NAND_CH_STAGED</code> when any bus phase is non-zero, else <code>NAND_CH_OFF</code> | same rule                         |

**(a) The phases of one operation**



**(b) Four reads issued at  $t = 0$  ( $c = 5$ ,  $tR = 40$ ,  $x = 10$ ,  $s = 0 \mu s$ )**



C waits for LUN 0, which A holds. B's data-out waits for the bus, which A's data-out holds. D is on another channel and overlaps A completely. All planes of a LUN are busy together on BBSSD; ZNS keeps a timeline per plane. With every bus phase at 0 (the default) only the LUN rows exist.

**Figure 11:** How the media layer times an operation: the phases of a read, program and erase, and four reads issued together showing LUN serialization, a shared channel bus and parallel channels.

| Setting         | BBSSD, CSD, KV, CXL                                  | ZNS                                        |
|-----------------|------------------------------------------------------|--------------------------------------------|
| use_flat_timing | true when nand_cell_type=0, false with a cell type   | false (per-type table, page type always 0) |
| pe_suspend      | pe_suspend property                                  | zns_pe_suspend property                    |
| ecc_on_read     | true; the adder is still 0 unless ecc_step_ns is set | false                                      |
| cache_read      | false                                                | false                                      |

The enum also has NAND\_CH\_NOOP and NAND\_GATE\_LUN\_AND\_PLANE. No mode selects either today.

### 3.5.5 Operation timing

#### Array time

array\_lat() picks the time the die spends on the operation:

```
erase : flat ? er_ns : er_table_ns[flash_type]
read : (flat ? rd_ns : rd_table_ns[flash_type][page_type]) + ecc_extra
program : flat ? wr_ns : wr_table_ns[flash_type][page_type]
        if flat and pgtype_lat: * pgtype_mult[page_type] / 1000
```

#### Phase sequence of one operation

With the channel bus off (the default), an operation is one array phase:

```
start = max(stime, gate busy-until)
done = start + array time
gate busy-until = done
```

With the bus on (NAND\_CH\_STAGED), each operation also has bus phases. "Now" phases queue on the bus in arrival order; a read's data-out happens after the array finishes, so it is booked as a future window (Channel bus (§3.5.7)).

```

READ      |cmd/addr|..wait for die..|=== tR array ===|data-out|status|
          bus now                die                bus later  bus later

PROGRAM   |cmd/addr|data-in|..wait for die..|===== tPROG array =====|
          bus now  bus now                die (done = array end)

ERASE      |cmd/addr|..wait for die..|===== tBERS array =====|status|
          bus now                die                bus later

```

A program completes when its array phase ends. A read completes after its status phase, an erase after its status phase. A zero-length phase touches nothing, so a bus with only `pg_xfer_lat` set has no command or status cost.

### Default times

| Mode                                                     | Read                         | Program                        | Erase                        | Source                                                              |
|----------------------------------------------------------|------------------------------|--------------------------------|------------------------------|---------------------------------------------------------------------|
| BBSSD, CSD, KV flat<br>( <code>nand_cell_type=0</code> ) | <code>pg_rd_lat</code> 40 us | <code>pg_wr_lat</code> 200 us  | <code>blk_er_lat</code> 2 ms | property defaults in <code>femu.c</code>                            |
| <code>femu-cxl-ssd</code>                                | <code>read-ns</code> 40 us   | <code>program-ns</code> 200 us | <code>erase-ns</code> 2 ms   | <code>cxlssd/qemu-adapter.c</code>                                  |
| ZNS SLC                                                  | 4 us                         | 75 us                          | 2 ms                         | <code>zns/zns.h</code> , erase from <code>nand/nand.h</code>        |
| ZNS TLC                                                  | 32 us                        | 937.5 us                       | 3 ms                         | same                                                                |
| ZNS QLC (default <code>zns_flash_type</code> )           | 85 us                        | 12.196 ms                      | 3 ms                         | same                                                                |
| ZNS MLC, PLC                                             | none built in                | none built in                  | none built in                | realize refuses them unless all three <code>zns*_lat</code> are set |

The ZNS program time covers a whole program unit (`zns_flash_type` pages of 16 KiB on one plane). The header cites the ISSCC papers the figures come from.

## 3.5.6 Cell types and page types

### BBSSD, CSD, KV: `nand_cell_type`

| <code>nand_cell_type</code> | Cell | Read by page type (us)      | Program by page type (us)                                             | Erase                   |
|-----------------------------|------|-----------------------------|-----------------------------------------------------------------------|-------------------------|
| 0                           | flat | <code>pg_rd_lat</code>      | <code>pg_wr_lat</code> , optionally scaled by <code>pgtype_lat</code> | <code>blk_er_lat</code> |
| 1                           | SLC  | 40                          | 800                                                                   | 2 ms                    |
| 2                           | MLC  | lower 48, upper 64          | lower 850, upper 2300                                                 | 3 ms                    |
| 3                           | TLC  | 56.5, 77.5, 106             | 820.5, 2225, 5734                                                     | 3 ms                    |
| 4                           | QLC  | 59.325, 85.25, 127.2, 169.6 | 861.525, 2447.5, 6880.8, 9174.4                                       | 3 ms                    |

Values come from `nand/nand.h`: MLC from a profile of Micron L95B parts, TLC from SimpleSSD, QLC scaled from the TLC values. Any other value is reset to 0 with a message (`ssd_init()` in `bbssd/ftl.c`); PLC has no table here. With a cell type set, `pg_rd_lat`, `pg_wr_lat` and `blk_er_lat` no longer set NAND times (`pg_rd_lat` still sets the write buffer and read cache hit cost), and `pgs_per_blk` must be at most 512, the size of the page-type tables.

The page type of page `pg` in a block comes from the pairing tables `mlc_tbl`, `tlc_tbl` and `qlc_tbl`, built by `init_nand_flash()` in `nand/nand.c`; `slc_tbl` is all zeros. They follow a shadow programming order: the first

pages of a block are lower pages, then consecutive pairs of pages cycle through the page types from lower to upper.

### BBSSD, CSD, KV: pgtype\_lat and cell\_pages

With `nand_cell_type=0` and `pgtype_lat` non-zero, the flat program time is multiplied by a factor that depends on the page's position in its wordline. The page type is `pg % cell_pages`; `cell_pages 0` is set to 3 when `pgtype_lat` is on (`ssd_init()`). The factors, in thousandths, are in `bb_nand_media_init()`:

| cell_pages | Page type 0 | 1    | 2    | 3    | 4    |
|------------|-------------|------|------|------|------|
| 1          | 1000        |      |      |      |      |
| 2          | 600         | 1400 |      |      |      |
| 3          | 600         | 1000 | 1400 |      |      |
| 4          | 500         | 850  | 1150 | 1500 |      |
| 5          | 400         | 750  | 1000 | 1250 | 1600 |

Each row averages to 1000, so the mean program time stays `pg_wr_lat` when `pgs_per_blk` is a multiple of `cell_pages`. Reads and erases are not scaled.

### ZNS and OCSSD

ZNS keeps one read, program and erase time per cell type and always uses page type 0. Each `zns*_lat` property replaces the built-in value for the configured `zns_flash_type` only. OCSSD uses the `nand_cell_type` tables above, selected by `flash_type`; Open-Channel 1.2 looks up the page type of each page, Open-Channel 2.0 always uses page type 0.

### 3.5.7 Channel bus

The bus is modelled only when a phase has a duration: `cmd_addr_lat`, `pg_xfer_lat` (or `ch_xfer_lat` when `pg_xfer_lat` is 0) or `status_lat` for BBSSD, CSD and KV; `zns_cmd_addr_lat`, `zns_pg_xfer_lat` or `zns_status_lat` for ZNS. Otherwise transfers take no time and channels never contend.

Two kinds of phase share a channel:

- `bus_now()`: command/address, a program's data-in. It starts at `max(earliest, ch_avail)`, moves past any booked window it would overlap, and advances `ch_avail` to its end.
- `bus_later()`: a read's data-out and status, which happen after the array finishes. It starts no earlier than `ch_avail` and is booked as a window `[start, end)` in `bus_res[ch]` and does not advance `ch_avail`, so another die's "now" phase can use the bus before the window opens.

Windows that ended before the current operation's `stime` are pruned. A channel holds at most `NAND_BUS_RES_MAX` (32) windows; when it is full, the data-out is charged FIFO from `ch_avail` instead. With `policy.bus_res_unbounded` (OCSSD 1.2) the list grows and nothing falls back.

A data phase moves one page in `page_xfer_ns`. When the mode sets `secs_per_page` and the operation's `NandLoc.xfer_secs`, it moves only those sectors and takes `DIV_ROUND_UP(page_xfer_ns * xfer_secs, secs_per_page)`, as OCSSD 1.2 charges a partial page.



channel 0 bus, cmd/addr 1 us, data 10 us (times in us)

```

0  1      11 12 13      52      62
|P c|P data-in|R1c|R2c|.... idle .....|R1 out |
                                ^ booked window, R1's array ended at 52
a phase that arrives at 20 fits in the idle gap; one that would overlap
[52,62) moves to 62

```

### 3.5.8 Plane parallelism and multi-plane operations

The array gate decides which operations exclude each other:

|                                 |                           |
|---------------------------------|---------------------------|
| LUN gate (BBSSD, CSD, KV)       | plane gate (ZNS)          |
| LUN 0: [pl0 op][pl1 op][pl0 op] | LUN 0 pl0: [op][op]       |
| one at a time per LUN           | pl1: [op][op] in parallel |

Under the LUN gate, all planes of a LUN are busy together, so `pls_per_lun > 1` adds capacity and line width but no single-operation parallelism, except for the one batched operation below.

`nand_media_multiplane()` runs one operation on several planes of a LUN:

```

per plane i: cmd/addr on bus (+ data-in for a program); tplxbsy between planes
array:      start = max(bus end, gate of every plane); done = start + array time
read:      status, then per plane cmd/addr + data-out (booked windows)
program/erase: status (if set)

```

| Operation                                       | Batched across planes                                                        | Caller                                                                                                                                |
|-------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Erase of a line's block on every plane of a LUN | yes, one erase time plus <code>tplbsy</code> per extra plane                 | BBSSD GC ( <code>bbssd/ftl-line-gc.c</code> ), FDP GC ( <code>bbssd/ftl-fdp.c</code> ), KV reclaim ( <code>kvssd/kvssd-ftl.c</code> ) |
| Host reads and programs                         | no, one operation per page                                                   | <code>bbssd/ftl-datapath.c</code>                                                                                                     |
| GC page moves                                   | no                                                                           | <code>gc_read_page()</code> , <code>gc_write_page()</code>                                                                            |
| ZNS zone reset                                  | no batching needed: each plane has its own gate, so per-plane erases overlap | <code>zns_zone_reset()</code>                                                                                                         |
| Copyback ( <code>nand_media_copyback()</code> ) | not called                                                                   | none                                                                                                                                  |

`tplpbsy`, `tplrbsy` and `trcbysy` are accepted for compatibility and have no effect; setting them warns at realize. With the defaults (no bus, no `tplbsy`) a two-plane erase takes exactly one erase time.

### 3.5.9 Program and erase suspend

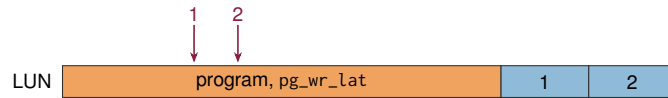
With `pe_suspend` (BBSSD, CSD, KV) or `zns_pe_suspend` (ZNS), a read that finds its LUN (ZNS: plane) running a program or erase goes ahead of it. The layer keeps, per position, `pe_end` (when the newest program or erase there ends) and `rd_end` (when the newest read there ends). `suspend_read_start()` decides:

```

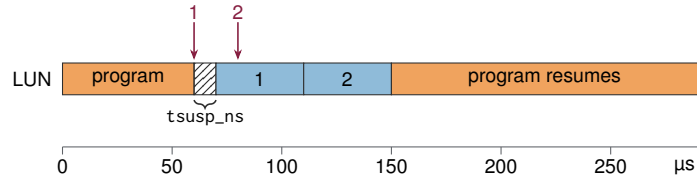
read arrives at t, gate busy until B
if t >= B or t >= pe_end:      ordinary gate (nothing to suspend)
elif rd_end > t:              join the open suspension
    start = rd_end,          shift = tR
else:                         open a suspension
    start = t + tsusp,        shift = tR + tsusp
done = start + tR
pe_end += shift; every gate busy-until > t += shift; rd_end = done

```

(a) `pe_suspend=0`: reads wait for the program



(b) `pe_suspend=1`, `tsusp_ns = 10 μs`: reads go first



Read 1 takes 50  $\mu$ s instead of 180; read 2 takes 70 instead of 200. Only the read that opens a suspension pays `tsusp_ns`; later reads queue behind it. The program ends at 290  $\mu$ s, late by `tsusp_ns` plus every read that went ahead. An erase is suspended the same way. BBSSD suspends per LUN, ZNS per plane.

**Figure 12:** Program suspend on one LUN: with `pe_suspend=1` two reads go ahead of a running program, the first pays `tsusp_ns`, and the program finishes later by that overhead plus both reads.

Only programs and erases are suspended: a read that finds the die busy with another read waits for it. Reads that queue behind an open suspension pay `tsusp_ns` once. A read that arrives after the suspension's reads have ended opens a new one and pays it again.

Worked example, one LUN, `pg_wr_lat` 200  $\mu$ s, `pg_rd_lat` 40  $\mu$ s, `tsusp_ns` 5  $\mu$ s, bus off. Numbers produced by `nand_media_op()`:

```
time (us)  0    50 55    95    135 150 155    195            330            530
LUN busy   |== P ===|ss|  R1  |  R2  |= P =|ss| R3 |===== P =====|===== W2 =====|
```

```
P  program  arrives 0      runs 0-50, 135-150, 195-330 (suspended twice)
R1 read     arrives 50     opens a suspension: starts 50 + 5, done 95
R2 read     arrives 60     joins it behind R1: starts 95, done 135, no tsusp
R3 read     arrives 150    R1 and R2 are over, so it opens a new one: 155-195
W2 program  arrives 160    waits for the LUN until P ends: 330-530
(ss = tsusp_ns)
```

```
P's die time: 200 alone; with the reads 200 + (5+40) + 40 + (5+40) = 330
```

P's own command still completes at 200. Its latency was computed when it was issued and is never revisited. Suspension moves only the die's busy-until time, from 200 to 330, so operations issued afterwards, such as W2, start after 330. A suspended program or erase never makes its own command complete afterwards; it delays the commands that follow.

`nand_media_multiplane()` does not consult the suspend state, so a read issued through it never suspends; only erases use it today. The state is read and written without a lock, so operations on one position must be serialized. BBSSD and ZNS run them on the single FTL thread. A device with suspend on also leaves the lock-free fast path described in Concurrency (§3.5.11).

### 3.5.10 ECC read time

`ecc_step_ns` adds read time for worn or old data (`ecc_extra_ns()`):

```

tiers = pe_cycles / 750 + age_sec / ecc_retention_sec  (second term only if set)
tiers = min(tiers, 4)
extra read time = tiers * ecc_step_ns

```

pe\_cycles is the block's erase count, age\_sec the time since its line was filled (line.close\_time). The constants are FEMU\_ECC\_PE\_PER\_TIER and FEMU\_ECC\_MAX\_TIERS in bbssd/ftl.h. A multi-plane read pays the largest extra of its planes. Read-retry as a sequence of separate array reads is not modelled; this adder stands in for it. ZNS and OCSSD have no ECC time.

### 3.5.11 From operations to a completion time

#### The per-operation rule

For every resource the operation needs, it starts no earlier than that resource's busy-until time, and it moves the resource's busy-until time to when it stops using it. With the bus off and the LUN gate this reduces to:

```

done(op)      = max(stime, lun_avail) + array_time(op)
lun_avail     = done(op)
latency(op)   = done(op) - stime

```

stime is the command's submission time, stamped by the poller from QEMU\_CLOCK\_REALTIME (nvme-io.c; a test-only property switches it to the virtual clock). Background work (GC) passes stime = 0, which the wrappers replace with the current time.

#### The per-command rule

The FTL issues all of a command's NAND operations with the command's stime and returns the largest latency:

```

latency(cmd) = max over the command's operations of latency(op)
               (and of write buffer, read cache and mapping cache costs)
expire_time  = stime + latency(cmd)          femu_ftl_thread(), femu.c

```

Then the poller adds the optional host link and firmware CPU (§3.5.13) models and queues the request in a priority queue ordered by expire\_time. The completion is posted on the first poller sweep at or after it.

#### Worked example: operations overlapped across LUNs and channels

Geometry 2 channels by 2 LUNs, 1 plane, flat 40 us read and 200 us program, staged bus with cmd\_addr\_lat=1000 and pg\_xfer\_lat=10000. Four operations arrive at time 0 in this order. The numbers are what nand\_media\_op() returns.

```

op  target    arrives
P   ch0 LUN0  0      program
R1  ch0 LUN1  0      read
R2  ch0 LUN0  0      read (same LUN as P)
R3  ch1 LUN0  0      read (other channel)

time (us)  0 1  11 12 13      51 52  62      211      251 261
ch0 bus    |c|-d--|c|c|      .  |o1-|      .      |o2-|
           P P   R1 R2      R1      .      R2
ch0 LUN0   .      [===== P tPROG 200 =====)[R2 tR )
ch0 LUN1   [R1 tR 40)
ch1 bus    |c|      |o3-|
ch1 LUN0   [R3 tR 40)

done:  P 211  R1 62  R2 261  R3 51

```

- P: command 0-1, data-in 1-11, array 11-211.
- R1: its command waits for P's data-in (11-12), its array runs on the idle LUN1 (12-52), its data-out is booked 52-62.
- R2: command 12-13, then waits for LUN0 until P ends at 211, array 211-251, data-out 251-261.
- R3: on channel 1, which nothing else uses: 0-1, 1-41, 41-51.

With the bus off, the same four operations end at P 200, R1 40, R2 240, R3 40. With `pe_suspend=1` and `tsusp_ns=5000`, R2 suspends P instead: it starts at  $13 + 5 = 18$ , ends its array at 58, waits for R1's booked window and moves its data at 62-72 (done 72), and the die stays busy until 256. P's own completion stays at 211.

If R1 and R2 were the two pages of one 8 KiB read command, its media latency would be  $\max(62, 261) = 261$  us.

### Worked example: full command completion

A host read of two written 4 KiB pages on idle LUNs of the default BBSSD (bus off), with `pcie_bandwidth_mbps=4000`, `pcie_prop_delay_ns=500` and `fw_cpu_ns=1000`, submitted at time S:

```
S                stime stamped by the poller
S + 40 000        FTL: max(40 000, 40 000) -> expire_time
S + 40 000 .. 42 048 link: 8192 B * 1000 / 4000 MB/s = 2 048 ns on the
                  device-to-host queue (starts at max(queue free, expire))
S + 42 548        + propagation 500 ns
S + 42 548 .. 43 548 firmware core: max(core free, expire) + 1 000 ns
S + 43 548        expire_time; CQE posted on the first sweep at or after it
```

The link and firmware queues are shared by every command of the controller, so under load each command also waits for the ones queued before it on those queues.

### Concurrency

- BBSSD and ZNS call the layer from the single FEMU-FTL-Thread, so their timelines need no lock. KV calls it from the poller and `femu-cxl-ssd` from its `femu-cxl-ftl` worker, each under its own device lock.
- With the LUN gate, no bus and no suspend, `nand_media_op()` updates `next_lun_avail_time` with a compare-and-swap loop that gives the same result as a locked read-max-add-store. Any other configuration takes the general path, which calls `lock_lun/unlock_lun` when the mode provides them; BBSSD and ZNS do not.
- OCSSD 1.2 with channel timing takes the channel's spinlock (`chnl_locks[]`) through `lock_lun/unlock_lun`; without it, and OCSSD 2.0, use the compare-and-swap path.

### 3.5.12 How garbage collection is charged

GC operations go through the same timelines as host operations, with `stime = 0` (the current time) and `type = GC_IO`:

- `gc_read_page()` charges a read of each valid page, `gc_write_page()` a program at its new location, and `reclaim_line()` one multi-plane erase per LUN (`bbssd/ftl-line-gc.c`). All of a victim's pages are moved before any of its blocks is erased, so the erases follow the last relocation. FDP GC does the same in `bbssd/ftl-fdp.c`.
- The returned latency is discarded. GC never adds to a command's latency directly. It makes LUNs and channels busy, and the host operations that need them afterwards wait.

- **Foreground GC** runs when the share of used lines has reached `gc_thres_pcent_high`: at the start of `ssd_write()`, and before each page that a write, a write buffer destage or a Write Zeroes programs (`bbssd/ftl-datapath.c`), until the share drops below it. The command's own programs then queue behind the GC operations on the LUNs they share.
- **Read reclaim** (`read_reclaim_limit`, `retention_limit_sec`) rewrites one queued line through `reclaim_line()` at the start of a write, charged the same way as GC.
- **Background GC** runs one victim after a request finishes in `bb_ftl_process_req()` (`bbssd/ftl.c`), once the latency of that request has been computed. It delays the commands that follow.
- Because GC is stamped with the current time and a host command with its submission time, a command submitted before a GC pass can still wait for it when the FTL thread gets to the command after the pass.
- GC time can be turned off with 0xEF code 2 (Runtime switches (§3.5.15)). GC still moves pages and erases blocks; its operations are then not charged at all.

```
LUN 3 timeline during foreground GC for one write (bus off)
|-- GC read --|-- GC program --| ... |== multi-plane erase ==|-- host program --|
^ write arrives                                     ^ write's done
```

KV is different. Its reclaim erases and compaction moves are stamped with the triggering command's `stime`, the command completes when the last of them and of its own operations does (`kvssd/kvssd-ftl.c`), and 0xEF does not apply to KV, so they cannot be switched off.

ZNS has no device GC. A Zone Reset charges one erase per block of the zone on every plane it spans, with the command's `stime`, and returns the largest (`zns_zone_reset()`).

### Costs that are not NAND operations

The FTL adds these to a command's latency alongside its NAND operations:

- **BBSSD write buffer** (`buffer_size > 0`): a write the buffer accepts costs `pg_rd_lat / 16` (1 us when `pg_rd_lat` is 0), plus the programs of any destage it forces. A Flush pays for programming everything buffered (`bbssd/ftl.c`). A read of a buffered page costs the same DRAM access.
- **BBSSD read cache and DFTL mapping cache**: a hit costs a DRAM access; a mapping miss adds a NAND read of the translation page, after a program if the evicted entry is dirty.
- **ZNS write cache**: a write only enters the zone's write cache and pays `SRAM_WRITE_LATENCY_NS` (1 us) per 4 KiB page, summed over the command (`zns_write()` in `zns/zftl.c`). It pays program time only when the cache fills, or when it has to evict another zone's cache. A cache holds one stripe: `16 KiB * zns_flash_type * zns_num_plane * zns_num_ch * zns_num_lun`. A read of data still in the cache is not mapped yet and costs nothing.

### 3.5.13 Host link and firmware CPU

Both are applied in `nvme_process_cq_cpl()` in `hw/femu/nvme-io.c`, after the media latency is in `expire_time` and before the request enters the priority queue. Both are off by default.

| Model                                              | Applies to                                                                                                                                                                                                                           | Computation                                                                                                                                                                                                                                                                         |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Link transfer ( <code>pcie_bandwidth_mbps</code> ) | NVMe Read and Write (opcodes 01h and 02h) in BBSSD, CSD, KV, ZNS and NoSSD; KV store and retrieve share those opcodes and use the bytes moved. Not Zone Append, and not the Open-Channel vector commands, so OCSSD 1.2 never pays it | $\text{trans} = \text{bytes} * 1000 / \text{Mbps ns}$ . One queue per direction: writes on <code>pcie_rx_next_avail_time</code> , reads on <code>pcie_tx_next_avail_time</code> . $\text{start} = \max(\text{queue}, \text{expire})$ , $\text{queue} = \text{start} + \text{trans}$ |

| Model                            | Applies to                                                                   | Computation                                                                           |
|----------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Propagation (pcie_prop_delay_ns) | same                                                                         | expire = queue + delay; does not occupy the queue                                     |
| Firmware CPU (fw_cpu_ns)         | Read, Write (01h, 02h) and Zone Append; not the Open-Channel vector commands | one modelled core: start = max(core, expire), core = start + fw_cpu_ns, expire = core |

The link is modelled after the media for both directions. A read's transfer to the host never overlaps its NAND time, and a write's transfer from the host is charged after its programs, not before them. The firmware cost is charged at the end of a command, which caps the command rate at about one per fw\_cpu\_ns but does not delay the start of the media operations. The link model is enabled when either link property is non-zero (pcie\_enabled in femu.c).

### 3.5.14 OCSSD timing model

Open-Channel devices run on the media layer, each with its own NandMedia and the LUN gate (oc12\_media\_init(), oc20\_media\_init()):

- Per chip (flat LUN id ch \* num\_lun + lun), an operation ends at max(now, busy) + time. Times come from the flash\_type table: 1.2 as a table by page type, 2.0 as flat lower-page times.
- Open-Channel 1.2 with channel timing uses the staged bus with no command, address or status phase: bus\_now() for a program's data-in and bus\_later() for a read's data-out.
- A write moves its data over the channel, then programs the chip. A read occupies the chip, then moves its data out. Erase charges each chip in the address list.
- Open-Channel 1.2 charges a channel transfer only with oc12\_channel\_timing=on: ch\_xfer\_lat ns per page, or the table value for flash\_type when it is 0, scaled by the sectors used out of lsecs\_per\_pg. Open-Channel 2.0 charges no channel time and uses page type 0.
- Open-Channel 1.2 charges one chip operation per NAND page of the address list. Open-Channel 2.0 groups addresses that differ only in the sector field, which spans the whole chunk, so a run of consecutive sectors in one chunk pays one chip operation whatever its length.
- The latency is max over operations of (done - stime), written straight into expire\_time in the poller (oc12\_advance\_status(), oc20\_advance\_status()).
- Open-Channel 1.2 keeps its data-out windows in an unbounded list per channel (policy.bus\_res\_unbounded), where the other modes cap it at 32.

### 3.5.15 Runtime switches

A BBSSD controller accepts the vendor admin command 0xEF (bb\_flip() in bbssd/bb.c). The action is in CDW10. For codes 1 to 4 it pauses the pollers, applies the change to every namespace of the controller that has a BBSSD FTL, and resumes them. Controllers of every other mode return Invalid Opcode.

| CDW10 | Constant (bbssd/ftl.h) | Effect on timing                                                                                        |
|-------|------------------------|---------------------------------------------------------------------------------------------------------|
| 1     | FEMU_ENABLE_GC_DELAY   | GC operations are charged (default)                                                                     |
| 2     | FEMU_DISABLE_GC_DELAY  | GC operations are not charged                                                                           |
| 3     | FEMU_ENABLE_DELAY_EMU  | Flat read, program and erase set to the built-in 40 us, 200 us and 2 ms, not to the command-line values |
| 4     | FEMU_DISABLE_DELAY_EMU | Flat read, program and erase set to 0                                                                   |

| CDW10 | Constant (bbssd/ftl.h)            | Effect on timing                                                       |
|-------|-----------------------------------|------------------------------------------------------------------------|
| 5     | accounting reset                  | Prints and resets the per-poller completion counters; no timing change |
| 6, 7  | FEMU_ENABLE_LOG, FEMU_DISABLE_LOG | Per-command debug log on or off; no timing change                      |

Codes 3 and 4 refresh only the flat rd\_ns, wr\_ns and er\_ns in the media layer (bb\_nand\_media\_refresh\_timing()). With nand\_cell\_type 1 to 4 the flat times are not used, so the codes have no effect on NAND time. They never change the channel bus phases, pgtype\_lat, ECC time or trim\_lat\_ns. Code 4 makes a write buffer hit cost 1 us; the read cache keeps the hit cost it computed at start. On a controller linked to a femu-cxl-ssd, codes 1 to 4 change the medium's FTL instead, and code 3 restores its read-ns, program-ns and erase-ns.

From the guest:

```
sudo nvme admin-passthru /dev/nvme0 --opcode=0xef --cdw10=2 # stop charging GC
sudo nvme admin-passthru /dev/nvme0 --opcode=0xef --cdw10=4 # zero NAND times
```

The vendor command 0xEE (NVME\_ADM\_CMD\_FEMU\_DEBUG) sets an Open-Channel controller's NAND times at run time, in nanoseconds: CDW10 upper page read, CDW11 lower page read, CDW12 upper page program, CDW13 lower page program, CDW14 block erase and CDW15 channel transfer per page. The lower page is page type 0 and the upper page the highest page type of flash\_type; the centre pages of TLC and QLC keep their table times, and SLC, with one page type, takes the lower page values. The times belong to that controller (FemuCtrl.oc\_pg\_rd\_lat and the fields beside it, filled from the flash\_type table by set\_latency()), so other devices are not affected. Other modes refuse 0xEE with Invalid Field.

```
sudo nvme admin-passthru /dev/nvme0 --opcode=0xee --cdw10=64000 --cdw11=48000 \
--cdw12=2300000 --cdw13=850000 --cdw14=3000000 --cdw15=52433
```

### 3.5.16 Parameters

The generated property reference (§A.1) is the authority for types, defaults and ranges. This table groups the timing properties and states how they interact.

| Group                    | Properties                                                                   | Interactions                                                                                                                     |
|--------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Geometry (§A.1.1)        | secsz, secs_per_pg, pgs_per_blk, blks_per_pl, pls_per_lun, luns_per_ch, nchs | nchs * luns_per_ch is the number of independent dies. pls_per_lun adds no read or program parallelism under the LUN gate.        |
| Array times (§A.1.1)     | pg_rd_lat, pg_wr_lat, blk_er_lat                                             | Ignored when nand_cell_type is 1 to 4. 0xEF codes 3 and 4 overwrite them. pg_rd_lat / 16 is also the cost of a write buffer hit. |
| Cell type (§A.1.1)       | nand_cell_type, cell_pages, pgtype_lat                                       | pgtype_lat applies only with nand_cell_type=0. nand_cell_type limits pgs_per_blk to 512.                                         |
| Bus phases (§A.1.1)      | cmd_addr_lat, pg_xfer_lat, ch_xfer_lat, status_lat                           | Any non-zero value turns the bus on. pg_xfer_lat takes precedence over ch_xfer_lat.                                              |
| Multi-plane (§A.1.1)     | tplbsy (tplpbsy, tplrbsy, trcb sy have no effect)                            | Used only by multi-plane erase, so only with pls_per_lun > 1.                                                                    |
| Suspend (§A.1.1)         | pe_suspend, tsusp_ns                                                         | tsusp_ns matters only with pe_suspend.                                                                                           |
| ECC (§A.1.1)             | ecc_step_ns, ecc_retention_sec                                               | ecc_retention_sec matters only with ecc_step_ns.                                                                                 |
| Other FTL costs (§A.1.1) | trim_lat_ns                                                                  | Charged per Dataset Management range by the FTL, not by the media layer. Refused with FDP.                                       |

| Group                           | Properties                                                                                                                                                                                         | Interactions                                                                                           |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| ZNS (§A.1.1)                    | zns_num_ch, zns_num_lun, zns_num_plane, zns_num_blk, zns_flash_type, zns_pg_rd_lat, zns_pg_wr_lat, zns_blk_er_lat, zns_cmd_addr_lat, zns_pg_xfer_lat, zns_status_lat, zns_pe_suspend, zns_tsusp_ns | The zns_*_lat overrides apply to the configured cell type only. zns_num_plane scales the program unit. |
| OCSSD (§A.1.1)                  | flash_type, oc12_channel_timing, ch_xfer_lat, lnum_ch, lnum_lun, lnum_pln, lsecs_per_pg, lpgs_per_blk                                                                                              | ch_xfer_lat is shared with BBSSD. lnum_ch is at most 32 and lnum_ch * lnum_lun at most 128.            |
| Host link and firmware (§A.1.1) | pcie_bandwidth_mbps, pcie_prop_delay_ns, fw_cpu_ns                                                                                                                                                 | Applied after the media time, to the opcodes listed in Host link and firmware CPU (§3.5.13).           |
| femu-cxl-ssd (§A.1.3)           | channels, luns-per-channel, pages-per-block, blocks-per-plane, read-ns, program-ns, erase-ns, channel-ns                                                                                           | One plane per LUN; feeds the same media layer.                                                         |

A BBSSD configuration with the bus and suspend on:

```
| -device femu,devsz_mb=1024,femu_mode=1,cmd_addr_lat=1000,pg_xfer_lat=10000,pe_suspend=1,tsusp_ns=5000
```

A BBSSD with TLC page-type timing:

```
| -device femu,devsz_mb=1024,femu_mode=1,nand_cell_type=3
```

### 3.5.17 Statistics

The media layer keeps no counters of its own. What it charges shows up in:

- **Latency seen by the guest:** fio completion latency. This is the primary output of the model.
- **Vendor log page C0h** (BBSSD, CSD): host, NAND and GC page counts, write amplification, buffer hits. See log pages and counters (§B).
- **Per-block counters** used by the model: nand\_block.erase\_cnt (ECC tiers, SMART Percentage Used) and nand\_block.read\_cnt (read reclaim), updated in ssd\_advance\_status() and on erase.
- **femu-cxl-ssd:** the QOM counter media-time-ns, the sum of the latencies the FTL returned. See runtime properties (§A.2).
- **Debug:** nand\_lun.gc\_endtime records when the last GC operation on a LUN ends; nothing reports it.

### 3.5.18 Calibrating against a real device

1. **Take array times from the datasheet.** pg\_rd\_lat is tR, pg\_wr\_lat is tPROG, blk\_er\_lat is tBERS. For a part with different lower and upper page times, either use nand\_cell\_type if one of the built-in tables is close, or set the mean tPROG and turn on pgtype\_lat with cell\_pages equal to bits per cell.
2. **Derive the bus phases from the interface rate.** pg\_xfer\_lat is the page size divided by the interface rate: a 16 KiB page on an 8-bit bus at 1600 MT/s takes 16384 / 1.6e9 s, about 10.2 us. cmd\_addr\_lat and status\_lat are the command, address and status cycle times, usually well under 1 us.
3. **Match the parallelism.** Set nchs and luns\_per\_ch to the device's channels and dies per channel, and secs\_per\_pg \* secsz to its page size.
4. **Check the unloaded latency.** Write the range first (a read of an unwritten page costs nothing), then run a queue depth 1 4 KiB random read. The median completion latency should be about tR plus the data-out and status phases plus guest and poller overhead. Measure that overhead once with flat timing, no bus phases, no caches and 0xEF code 4, and subtract it.



5. **Check the saturated throughput.** A random read at high queue depth should approach  $nchs * luns\_per\_ch / tR$  operations per second when the bus is not the limit, or  $nchs / (pg\_xfer\_lat + cmd\_addr\_lat + status\_lat)$  when it is. Compare with the real device at the same queue depth.
6. **Add the host side.** If the real device's peak is below what the media allows, set `pcie_bandwidth_mbps` to its link rate and `fw_cpu_ns` to the inverse of its peak command rate.
7. **Check GC separately.** Precondition the device, run sustained random writes, and compare the write amplification in log page C0h and the steady-state write bandwidth. Toggle 0xEF codes 1 and 2 to see how much of the gap GC time explains.

Give each poller and the FTL thread its own host core while measuring; performance tuning (§5.12) and measuring (§6.1) cover the setup.

### 3.5.19 Validation status

- **Unit tests:** `hw/femu/tests/unit/test-nand-media.c` checks the media layer's arithmetic without QEMU: ECC tiers, the staged bus and its booked windows, the plane gate with the bus, multi-plane erase against serial erases, copyback, and program/erase suspend. Run it with `make -C hw/femu/tests check`.
- **qtests:** `hw/femu/tests/qtest/femu-test.c` covers the OCSSD 1.2 channel model (`oc12-channel-timing`, `oc12-channel-gap`, `oc12-channel-default`, `oc12-channel-off`, `oc12-ppa-timing`, `oc12-flash-type`, `oc12-page-count`, `oc12-transfer-cost`, and the exact per-command times of `oc12-trace-on` and `oc12-trace-off`), the warning for the timing properties that have no effect (`ignored-props`), the 0xEF flips on a linked `femu-cxl-ssd` (`cxl-nvme-flip`), and the two example configurations on this page (`doc-examples`).
- **Not covered by automated tests:** BBSSD and ZNS timing inside QEMU, the host link, `fw_cpu_ns`, the 0xEF flips on a plain BBSSD controller, and end-to-end guest latency against a reference device. The built-in MLC table is a profile of real parts; the other tables are from published papers, as cited in the headers. Check numbers that matter to you with the calibration steps above.

| `make -C hw/femu/tests check`

### 3.5.20 What is not modelled

- Read-retry as repeated array reads, soft-decision decoding, and read disturb errors. The ECC adder is a fixed time per tier, and data is never corrupted.
- Multi-plane reads and programs: host pages are issued one plane at a time. Only line erases are batched.
- Cache read pipelining: the code path exists but no mode enables it. There is no cache program model.
- Copyback for GC. The function exists and has no caller, because GC's destination is almost never on the source LUN.
- Program suspend for programs, or erase suspend by another erase. Only reads suspend.
- Per-die power or thermal limits, and temperature-dependent timing.
- Plane-level parallelism for BBSSD, CSD and KV reads and programs (the gate is per LUN), and the channel bus for OCSSD 2.0.
- PLC outside ZNS. ZNS accepts PLC with times you supply; BBSSD resets `nand_cell_type=5` to 0 and OCSSD refuses `flash_type=5`.
- Queueing inside the controller other than the link and the single firmware core: there is no DRAM bandwidth or NAND controller queue model.
- Timing accuracy below the poller sweep. A completion is posted on the first sweep after it is due, and a poller without its own host core adds delay.

### 3.5.21 Adding a timing feature

1. **Decide the resource.** If the feature makes an existing resource busy for longer, it is an `array_lat()` or bus-phase change. If it adds a resource, add an accessor to `NandTimelineOps` and keep the state in the mode's structures, like the existing `busy-until` fields.
2. **Add the duration to `NandMediaTiming` and a switch to `NandMediaPolicy`** in `nand/nand-media.h`. Default it to 0 or off so that existing configurations keep their exact timing. The layer must not include a controller header or branch on the mode.
3. **Implement it in `nand-media.c`.** Keep the `max()` ordering of the existing paths. A phase that happens at issue goes through `bus_now()`; one that happens after the array goes through `bus_later()`.
4. **Wire the property.** Add it with `DEFINE_PROP_*` in `femu.c` and a description in `femu-props.c`, copy it into `ssdparams` in `ssd_init_params()` (`bbssd/ftl-geom.c`), validate it in `bb_check_geometry()`, and set the media config in `bb_nand_media_init()`. For ZNS, the same in `zns_init_params()` and `zns_nand_media_init()`.
5. **Test the arithmetic** in `tests/unit/test-nand-media.c`. Include a case that fails if the feature is ignored, so the test cannot pass with the feature off.
6. **Update the docs.** Regenerate the property reference with `gen-property-docs.py`, add the parameter to this page and to the timing model concept page (§6.2), and add a `CHANGELOG` entry. Keeping the documentation correct (§8.3) lists the checks.

### 3.5.22 Source map

| File                                                                              | What it holds                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hw/femu/nand/nand-media.h</code> <sup>7</sup>                               | Media layer types and API: <code>NandLoc</code> , <code>NandMediaTiming</code> , <code>NandMediaPolicy</code> , <code>NandTimelineOps</code> , <code>nand_media_op()</code> , <code>nand_media_multiplane()</code> , <code>nand_media_copyback()</code> |
| <code>hw/femu/nand/nand-media.c</code> <sup>8</sup>                               | Array time, gates, bus booking ( <code>bus_now()</code> , <code>bus_later()</code> ), suspend, ECC, multi-plane, copyback                                                                                                                               |
| <code>hw/femu/nand/nand.h</code> <sup>9</sup> , <code>nand.c</code> <sup>10</sup> | Cell-type timing tables, page pairing tables, rated P/E cycles                                                                                                                                                                                          |
| <code>hw/femu/bbssd/ftl-media.c</code> <sup>11</sup>                              | BBSSD adapter: <code>bb_decode_loc()</code> , <code>bb_nand_media_init()</code> , <code>ssd_advance_status()</code> , <code>ssd_advance_status_multiplane()</code> , <code>bb_nand_media_refresh_timing()</code>                                        |
| <code>hw/femu/bbssd/ftl-geom.c</code> <sup>12</sup>                               | Geometry checks and parameter copy ( <code>bb_check_geometry()</code> , <code>ssd_init_params()</code> )                                                                                                                                                |
| <code>hw/femu/bbssd/ftl-datapath.c</code> <sup>13</sup>                           | Host read and write: per-page operations and max latency                                                                                                                                                                                                |
| <code>hw/femu/bbssd/ftl-line-gc.c</code> <sup>14</sup>                            | Write pointer order, GC reads, programs and multi-plane erase                                                                                                                                                                                           |
| <code>hw/femu/bbssd/bb.c</code> <sup>15</sup>                                     | 0xEF handler ( <code>bb_flip()</code> , <code>bb_flip_apply()</code> )                                                                                                                                                                                  |
| <code>hw/femu/zns/zftl.c</code> <sup>16</sup>                                     | ZNS adapter, write cache flush, zone reset erase                                                                                                                                                                                                        |

<sup>7</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nand/nand-media.h>

<sup>8</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nand/nand-media.c>

<sup>9</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nand/nand.h>

<sup>10</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nand/nand.c>

<sup>11</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/bbssd/ftl-media.c>

<sup>12</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/bbssd/ftl-geom.c>

<sup>13</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/bbssd/ftl-datapath.c>

<sup>14</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/bbssd/ftl-line-gc.c>

<sup>15</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/bbssd/bb.c>

<sup>16</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/zns/zftl.c>

| File                                                      | What it holds                                                                 |
|-----------------------------------------------------------|-------------------------------------------------------------------------------|
| hw/femu/zns/zns.c <sup>17</sup> , zns.h <sup>18</sup>     | ZNS timing values and property overrides (zns_init_params())                  |
| hw/femu/ocssd/oc-timing.c <sup>19</sup>                   | OCSSD flash_type times, geometry check, 0xEE                                  |
| hw/femu/ocssd/oc12.c <sup>20</sup> , oc20.c <sup>21</sup> | OCSSD per-command timing (oc12_advance_status(), oc20_advance_status())       |
| hw/femu/femu.c <sup>22</sup>                              | FTL thread: expire_time += latency; timing properties                         |
| hw/femu/nvme-io.c <sup>23</sup>                           | stime stamp, host link and firmware CPU models, priority queue and completion |
| hw/femu/tests/unit/test-nand-media.c <sup>24</sup>        | Media layer unit tests                                                        |

<sup>17</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/zns/zns.c>

<sup>18</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/zns/zns.h>

<sup>19</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/ocssd/oc-timing.c>

<sup>20</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/ocssd/oc12.c>

<sup>21</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/ocssd/oc20.c>

<sup>22</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/femu.c>

<sup>23</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nvme-io.c>

<sup>24</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/tests/unit/test-nand-media.c>

## 4 Modes and features

### 4.1 Choosing a mode

An NVMe femu device emulates one kind of SSD, chosen with `femu_mode`. Features such as Flexible Data Placement or several namespaces are added on top of a mode with more properties. The CXL SSD is a separate device type, `femu-cxl-ssd`, not a `femu_mode`.

If you set nothing, you get NoSSD: `femu_mode` defaults to 2.

#### 4.1.1 `femu_mode` values

| Value | Name  | Emulates                                                                                                                             |
|-------|-------|--------------------------------------------------------------------------------------------------------------------------------------|
| 0     | OCSSD | An Open-Channel SSD. The host runs the FTL. <code>lver=1</code> selects Open-Channel 1.2, <code>lver=2</code> (default) selects 2.0. |
| 1     | BBSSD | A conventional ("black-box") SSD with a device FTL, garbage collection and NAND timing.                                              |
| 2     | NoSSD | An NVMe drive with no media timing. Default.                                                                                         |
| 3     | ZNS   | A Zoned Namespace SSD (NVMe Zoned Namespace command set).                                                                            |
| 4     | CSD   | A computational storage drive that runs programs next to the data, on top of the BBSSD FTL.                                          |
| 5     | KV    | A key-value SSD (NVMe Key Value command set).                                                                                        |

Any other value fails realize with "`femu_mode` must be 0 (OpenChannel), 1 (black-box), 2 (no-SSD), 3 (zoned), 4 (computational storage) or 5 (key-value)". With `femu_mode=0`, an `lver` other than 1 or 2 fails too.

#### 4.1.2 Decision table

Settings are added to `-device femu, ...` unless the row says otherwise. Each launcher is in `hw/femu/scripts/`. Each guide covers launch, configuration, guest-side use, refusals and troubleshooting for its mode or feature.

| Goal                                                                  | Mode or feature | Key settings                                                                                                                                                     | Launcher                     | Guide           |
|-----------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------|
| Fastest emulated NVMe drive, for host software stack work             | NoSSD           | <code>femu_mode=2</code> (or nothing)                                                                                                                            | <code>run-nossd.sh</code>    | NoSSD (§4.2)    |
| A conventional SSD with realistic latency, GC and write amplification | BBSSD           | <code>femu_mode=1</code> ; <code>geometry nchs, luns_per_ch, pls_per_lun, blks_per_pl, pgs_per_blk</code> ; <code>timing pg_rd_lat, pg_wr_lat, blk_er_lat</code> | <code>run-blackbox.sh</code> | BlackBox (§4.4) |
| A zoned drive for ZNS-aware filesystems and databases                 | ZNS             | <code>femu_mode=3</code> ; <code>zns_num_ch, zns_num_lun, zns_num_plane, zns_num_blk, zns_flash_type</code>                                                      | <code>run-zns.sh</code>      | ZNS (§4.7)      |
| Host-managed flash: your own FTL in the host                          | OCSSD           | <code>femu_mode=0</code> , <code>lver=2</code> (or 1); <code>lnum_ch, lnum_lun, lnum_pln, lpgs_per_blk</code>                                                    | <code>run-whitebox.sh</code> | OCSSD (§4.9)    |
| Store and fetch values by key, with no file system                    | KV              | <code>femu_mode=5</code>                                                                                                                                         | none                         | KV (§4.11)      |

| Goal                                                                        | Mode or feature      | Key settings                                                                                      | Launcher            | Guide                                      |
|-----------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------|---------------------|--------------------------------------------|
| Run filters or other programs inside the drive                              | CSD                  | femu_mode=4, fdm_size (required), nr_cu, csd_program_dir                                          | run-csd.sh          | CSD (§4.13)                                |
| Steer writes to reclaim units to reduce write amplification                 | FDP on BBSSD         | -device femu-subsys, id=s0, fdp=on, fdp.nruh=N before the controller, then femu_mode=1, subsys=s0 | run-blackbox-fdp.sh | FDP (§4.5)                                 |
| Several namespaces on one controller, possibly of different modes           | multi-namespaces     | namespaces=N, optional namespace_sizes=..., namespace_modes=...                                   | none                | Several namespaces (§4.15)                 |
| Create and delete namespaces from the guest                                 | Namespace Management | ns_mgmt=on on a NoSSD or BBSSD controller; bbssd_ns_limit for BBSSD                               | none                | Namespace management (§4.17.2)             |
| Metadata and end-to-end protection information                              | metadata, PI         | meta=8 (or more), mc, pi=on                                                                       | none                | Metadata and PI (§4.17.3)                  |
| SSD capacity that the guest uses as memory, with SSD timing on cache misses | CXL SSD              | -device femu-cxl-ssd, volatile-memdev=... on a CXL topology; cache-pages, cache-policy, der       | run-cxlssd.sh       | CXL SSD (§4.18); design note <sup>25</sup> |
| The same CXL medium also as an NVMe block device                            | CXL NVMe front end   | femu-cxl-ssd first, then -device femu, femu_mode=1, cxl_ssd=<id>                                  | none                | CXL NVMe link (§4.21)                      |

Commas inside a property value are doubled on the QEMU command line, for example namespace\_modes=bbssd,,znssd.

### 4.1.3 Which features combine

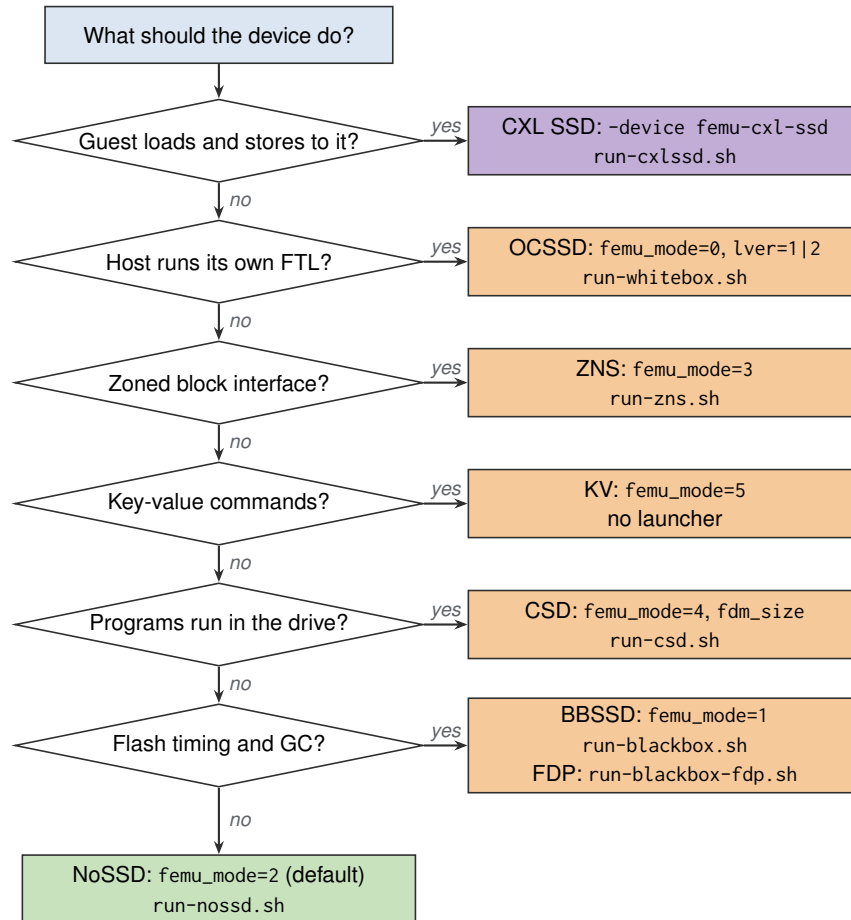
The checks below are made at realize; a combination they refuse stops QEMU with an error naming the rule.

**Flexible Data Placement.** fdp=on is a property of femu-subsys. Only BBSSD has an FDP write path; other modes accept a subsystem with FDP, but placement then has no effect on where data goes. FDP requires a single controller in the subsystem, a single namespace (namespaces=1), fdp.nrg=1 and fdp.nruh from 1 to fdp.nru. It cannot be combined with a KV namespace, meta, Streams or a subsystem with ns\_mgmt=on. A BBSSD controller under FDP also refuses buffer\_size, hot\_cold\_sep, read\_reclaim\_limit, retention\_limit\_sec, ecc\_retention\_sec, trim\_lat\_ns, a mapping other than page and a gc\_policy other than greedy.

**Several namespaces and per-namespaces modes.** namespaces is 1 to 256. The backend of devsz\_mb MiB is split evenly, or by namespace\_sizes. namespace\_modes lists one mode per namespace from nossd, bbssd, znssd, ocssd, csd and kvssd; without it every namespace runs femu\_mode. Limits:

- OCSSD supports one namespace only, and the controller must be OCSSD too.
- At most one CSD namespace per controller.
- ZNS, KV, BBSSD and NoSSD namespaces can be mixed.
- meta and Streams need every namespace to be NoSSD or BBSSD.

<sup>25</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md>



**Figure 13:** Choosing a mode: answer the questions top to bottom; the first “yes” names the mode, its femu\_mode value and its launcher in hw/femu/scripts/.

**Namespace Management.** `ns_mgmt=on` takes effect only when the controller and every namespace are NoSSD, or every one is BBSSD; with other modes, or with `dps`, it is accepted and stays off. On a controller that joins a subsystem without `ns_mgmt`, it fails realize. To share namespaces between controllers, set `ns_mgmt=on` on the `femu-subsys` instead. A shared subsystem takes NoSSD or BBSSD controllers that all have the same mode, `meta`, `mc`, `pi`, `dpc`, `nlbaf`, `vwc` and `oncs`, and no `Streams`, `dps` or `namespace_modes`.

**Metadata and protection information.** `meta` (bytes per block) works on NoSSD and BBSSD namespaces only, not with FDP, and needs a matching `mc` bit. `pi=on` offers protection information types 1 to 3 when `meta` is at least 8; the guest selects one with Format NVM or Namespace Management. `pi` cannot be combined with `power_loss` or `cxl_ssd`.

**CXL NVMe front end.** A `femu` controller with `cxl_ssd=<id>` must be BBSSD with one namespace, and must not use `namespace_modes`, `namespace_sizes`, `ns_mgmt`, `subsys`, `streams`, `power_loss`, `buffer_size`, `op_pcent`, `meta`, `pi` or `dps`. The `femu-cxl-ssd` must come first on the command line and have `ftl=on`. Its geometry and timing apply; the controller's own are ignored, and `devsz_mb` must be left at its default or equal the CXL medium's size.

#### Other combinations.

- `Streams` (`streams=on`): NoSSD or BBSSD, not with FDP or a shared subsystem; BBSSD needs mapping page or `dftl`.
- Power-loss model (`power_loss=on`): BBSSD with `buffer_size > 0`, page-aligned namespaces, and no `meta`, `pi`, `ns_mgmt`, `subsys`, `namespace_modes` or `cxl_ssd`. Without `vwc=1` the write buffer holds nothing, so there is nothing to lose.
- ZNS needs a logical block size of 4 KiB or less (`lba_index`).

#### 4.1.4 Every mode at a glance

Generated from `modes.py`<sup>26</sup>: how to turn each mode or feature on, the guest kernel and tools it needs, what the host needs, and what CI checks. `requirements.md` (§2.1.2) has the guest kernel configuration in more detail.

| Mode or feature              | Use it for                                    | Turn it on with                        | Guest kernel                                                              | Guest tools                                                   | Host needs                                             | Launcher                     | Checked                                                                               |
|------------------------------|-----------------------------------------------|----------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------|------------------------------|---------------------------------------------------------------------------------------|
| NoSSD (§4.2)                 | fast NVMe device in DRAM, no flash timing     | <code>femu_mode=2</code> (the default) | any with the NVMe driver                                                  | <code>nvme-cli</code> , <code>fio</code>                      | none beyond the common ones                            | <code>run-nosssd.sh</code>   | CI: realize, Identify, write and read back                                            |
| BlackBox SSD (BBSSD) (§4.4)  | a commercial SSD: device FTL, GC, NAND timing | <code>femu_mode=1</code>               | any with the NVMe driver                                                  | <code>nvme-cli</code> , <code>fio</code>                      | about 17 GiB free RAM for the launcher's 12 GiB device | <code>run-blackbox.sh</code> | CI: realize, Identify, write and read back; guest: quick start, run end to end (§2.4) |
| Zoned Namespace (ZNS) (§4.7) | zoned storage research                        | <code>femu_mode=3</code>               | 5.9 or newer with <code>CONFIG_BLK_DEV_ZONED=y</code> ; 4 KiB guest pages | <code>nvme-cli</code> 1.12 or newer for <code>nvme zns</code> | none beyond the common ones                            | <code>run-zns.sh</code>      | CI: realize, Identify, write and read back                                            |

<sup>26</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/modes.py>

| Mode or feature                               | Use it for                                       | Turn it on with                                                                        | Guest kernel                                                                    | Guest tools                                      | Host needs                                                                             | Launcher            | Checked                                    |
|-----------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------|---------------------|--------------------------------------------|
| Open-Channel SSD 1.2 (§4.9)                   | host-managed FTL research                        | femu_mode=0, lver=1                                                                    | 4.16 to 5.14 (LightNVM was removed in 5.15)                                     | LightNVM tools, or SPDK on newer kernels         | none beyond the common ones                                                            | run-whitebox.sh     | CI: realize, Identify                      |
| Open-Channel SSD 2.0 (§4.9)                   | host-managed FTL research                        | femu_mode=0 (lver=2 is the default)                                                    | 4.17 to 5.14 (LightNVM was removed in 5.15)                                     | LightNVM tools, or SPDK on newer kernels         | none beyond the common ones                                                            | run-whitebox.sh     | CI: realize, Identify                      |
| Key-value SSD (KV) (§4.11)                    | key-value store research                         | femu_mode=5                                                                            | 6.0 or newer; no block device, the namespace is /dev/ngXnY                      | nvme-cli io-passthru, hw/femu/scripts/kv-probe.c | none beyond the common ones                                                            | run-kvssd.sh        | CI: realize, Identify, store and retrieve  |
| Computational storage (CSD) (§4.13)           | running programs next to the data                | femu_mode=4, fdm_size=<MiB>                                                            | any with the NVMe driver                                                        | hw/femu/tests/csd tools                          | csd_program_dir for shared-library programs; --enable-csd-ubpf build for eBPF programs | run-csd.sh          | CI: realize, Identify, write and read back |
| Flexible Data Placement (FDP) (§4.5)          | placement hints on a BBSSD                       | femu-subsys, fdp=on, fdp.nruh=<n> and femu, femu_mode=1, subsys=<id>                   | any with the NVMe driver; placement hints need passthrough or io_uring commands | nvme-cli with nvme fdp                           | none beyond the common ones                                                            | run-blackbox-fdp.sh | CI: realize, Identify, write and read back |
| Multiple namespaces (§4.15)                   | several namespaces, each with its own mode       | namespaces=<n>, optionally namespace_sizes and namespace_modes                         | any with the NVMe driver (ZNS namespaces need what ZNS needs)                   | nvme-cli                                         | none beyond the common ones                                                            | none                | CI: realize, Identify, write and read back |
| Namespace management (§4.17.2)                | create, delete and attach namespaces at run time | ns_mgmt=on on a NoSSD or BBSSD controller; femu-subsys, ns_mgmt=on to share namespaces | any with the NVMe driver                                                        | nvme-cli create-ns, attach-ns                    | none beyond the common ones                                                            | none                | CI: realize, Identify, write and read back |
| Metadata and protection information (§4.17.3) | per-block metadata, PI types 1 to 3              | meta=<bytes>, mc=<mask>, plus pi=on with meta of 8 or more                             | CONFIG_BLK_DEV_INTEGRITY=y to use metadata formats through the block layer      | nvme-cli format                                  | none beyond the common ones                                                            | none                | CI: realize, Identify, write and read back |



| Mode or feature                     | Use it for                                             | Turn it on with                                                 | Guest kernel                                                                                                                                 | Guest tools                                       | Host needs                                                                                                                                                      | Launcher      | Checked                                    |
|-------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------|
| CXL SSD, der=off (§4.18)            | CXL memory backed by flash, all accesses trapped       | femu-cxl-ssd below pxb-cxl and cxl-rp on - machine q35, cxl=on  | CONFIG_CXL_BUS, CXL_PCI, CXL_ACPI, CXL_MEM, CXL_PORT, CXL_REGION, CXL_REGION_INVALIDATION_TEST (in a VM), DEV_DAX, DEV_DAX_CXL, DEV_DAX_KMEM | cxl-cli, daxctl, ndctl                            | a build with CONFIG_CXL_MEM_DEVICE                                                                                                                              | run-cxlssd.sh | CI: realize                                |
| CXL SSD, der=memslot (§4.18.7)      | cached pages mapped into the guest as KVM memory slots | der=memslot on femu-cxl-ssd                                     | as for der=off                                                                                                                               | as for der=off                                    | KVM (TCG is refused)                                                                                                                                            | run-cxlssd.sh | CI: realize                                |
| CXL SSD, der=cylon (§4.18.7)        | cached pages mapped by a Cylon host kernel             | der=cylon, cylon-kernel-ack=on on femu-cxl-ssd                  | as for der=off                                                                                                                               | as for der=off                                    | Cylon host kernel; KVM with EPT A/D bits and the TDP MMU; 4 KiB host pages; a shared, preallocated hugetlb backend. Without them the device warns and uses MMIO | run-cxlssd.sh | CI: realize                                |
| CXL caching API (CCA) (§4.20)       | guest pins, unpins and invalidates cached pages        | cca=on on femu-cxl-ssd                                          | as for der=off; a devdax region                                                                                                              | hw/femu/tools/cca (ccactl, cca-test), run as root | as for der=off                                                                                                                                                  | run-cxlssd.sh | CI: realize                                |
| NVMe front end on a CXL SSD (§4.21) | the same media as CXL memory and as an NVMe namespace  | femu,bus=pcie.0,femu_mode=1,cxl_ssd=<id> after the femu-cxl-ssd | as for der=off, plus the NVMe driver                                                                                                         | as for der=off, plus nvme-cli                     | as for der=off                                                                                                                                                  | none          | CI: realize, Identify, write and read back |

The image built by `make-guest-image.sh` runs Linux 6.8 and covers every mode except OCSSD.

### 4.1.5 Related pages

- Architecture (§3.1): how the modes fit into FEMU.
- Timing model (§6.2): what each mode charges time for.
- Device property reference (§A.1)
- Mode guides: NoSSD (§4.2), BlackBox (§4.4), ZNS (§4.7), OCSSD (§4.9), KV (§4.11), CSD (§4.13)
- Feature guides: FDP (§4.5), several namespaces (§4.15), namespace management and PI (§4.17)

| Mode<br>femu_mode  | NoSSD<br>2      | BBSSD<br>1      | ZNS<br>3        | OCSSD<br>0        | KV<br>5        | CSD<br>4        | CXL SSD<br>device |
|--------------------|-----------------|-----------------|-----------------|-------------------|----------------|-----------------|-------------------|
| <b>Interface</b>   | NVMe<br>NVM     | NVMe<br>NVM     | NVMe<br>Zoned   | OC<br>vector      | NVMe<br>KV     | NVM and<br>CSD  | CXL.mem<br>loads  |
| <b>FTL</b>         | none            | map<br>and GC   | zone<br>FTL     | in the<br>host    | hash<br>index  | BBSSD<br>FTL    | cache<br>+ FTL    |
| <b>NAND timing</b> | none            | nand/           | nand/           | timing-<br>model/ | BBSSD<br>model | nand/,<br>CUs   | nand/             |
| <b>Data store</b>  | DRAM<br>backend | DRAM<br>backend | DRAM<br>backend | DRAM<br>backend   | value<br>arena | DRAM<br>backend | memory<br>backend |
| <b>Timing on</b>   | poller          | FTL<br>thread   | FTL<br>thread   | poller            | poller         | FTL +<br>poller | vCPU,<br>cxl-ftl  |

#### FEATURES

|              |   |   |   |  |   |   |   |
|--------------|---|---|---|--|---|---|---|
| FDP          |   | ● |   |  |   |   |   |
| Write buffer |   | ● |   |  |   | ● |   |
| Namespaces   | ● | ● | ● |  | ● | ○ |   |
| NS mgmt      | ● | ● |   |  |   |   |   |
| Metadata, PI | ● | ● |   |  |   |   |   |
| Streams      | ● | ● |   |  |   |   |   |
| NVMe link    |   |   |   |  |   |   | ● |
| CCA          |   |   |   |  |   |   | ● |

● supported    ○ one CSD namespace per controller, mixed with others

**Figure 14:** Modes against layers: the interface each mode offers, where its FTL runs, which timing model it uses, where its data lives and which thread charges its time; below, the features each mode supports (CUs are CSD compute units).

## 4.2 NoSSD

NoSSD mode (`femu_mode=2`, the default) emulates an NVMe drive with no media model. Reads and writes copy data between guest memory and a host DRAM buffer and complete as soon as the poller finishes the copy. There is no FTL, no GC and no NAND time.

Use it when the storage device should not be the bottleneck: work on the host I/O stack, the NVMe driver, polling, SPDK or `io_uring`, or as an upper bound to compare a timed mode against. For a device with SSD latency and garbage collection, use `BlackBox` (§4.4).

### 4.2.1 Requirements

- Host and guest: see the mode table (§4.1.4) and `requirements.md` (§2.1.2). Any guest kernel with the NVMe driver works.
- Host memory: `devsz_mb` plus guest RAM plus about 1 GiB for QEMU, about 9 GiB for `run-nossd.sh` (a 4 GiB device and a 4 GiB guest).
- Host cores: the pollers spin on host cores. For throughput work, give each `femu-poller` thread a core of its own.

### 4.2.2 Launch

From `build-femu/`:

```
| ./run-nossd.sh
```

The FEMU device in that script is:

```
| -device femu,devsz_mb=4096,id=nvme0
```

`femu_mode` is not set, so it is 2. Writing `femu_mode=2` has the same effect.

### 4.2.3 Configuration

NoSSD ignores the NAND geometry, timing and FTL properties. These are the ones that change its behaviour.

#### Capacity and namespaces

Properties: `mode`, `capacity` and `namespaces` (§A.1.1).

`devsz_mb` is the namespace size. `namespaces`, `namespace_sizes` and `namespace_modes` split it into several namespaces (multi-namespace guide (§4.15)).

#### Queues and pollers

Properties: `queues`, `pollers` and `interrupts` (§A.1.1).

- `queues` sets the number of I/O queue pairs (default 8).
- By default one poller thread serves every I/O queue. With `multipoller_enabled=1`, FEMU runs  $\text{ceil}(\text{queues} / \text{poller\_ratio})$  pollers, each owning a round-robin share of the queues:

```
| -device femu,devsz_mb=4096,queues=8,multipoller_enabled=1,poller_ratio=2
```

This starts four pollers. Give each one a host core.

## Host link and controller firmware

Properties: host link and controller firmware (§A.1.1).

`pcie_bandwidth_mbps`, `pcie_prop_delay_ns` and `fw_cpu_ns` add link and firmware time to each Read and Write. With any of them set, NoSSD completes commands through the same timed queue as the other modes instead of in the poller sweep, which adds work per command.

```
| -device femu,devsz_mb=4096,pcie_bandwidth_mbps=3500,pcie_prop_delay_ns=1000
```

The preset `hw/femu/scripts/configs/nossd.conf` sets all three to the figures of the NVMeCHA FPGA controller with a null backend (Qiu et al., IEEE TCAD, doi:10.1109/TCAD.2021.3088784): a 7000 MB/s link and about 2.4 us per 4 KiB read. A 4 KiB transfer at 7000 MB/s takes 585 ns, so with 1300 ns of link delay and 500 ns of firmware a 4 KiB Read or Write costs 2385 ns, and the link holds 4 KiB I/O near the controller's 1.7 million IOPS. FEMU charges writes the same as reads, below the controller's 3.2 us, and the guest also sees FEMU's own per-command time on top. `ssd-config.sh` (§5.11) expands it to:

```
| -device femu,id=nvme0,devsz_mb=4096,namespaces=1,pcie_bandwidth_mbps=7000,pcie_prop_delay_ns=1300,fw_cpu_ns=500,femu_mode=2
```

## Optional commands and features

Properties: controller identity and capabilities (§A.1.1), LBA formats, metadata and protection (§A.1.1).

- `oncs` turns on optional NVM commands. The default (0x14) offers Dataset Management and Save/Select. Add 0x8 for Write Zeroes, 0x1 for Compare, 0x100 for Copy.
- NoSSD supports namespace management, metadata and protection information (§4.17) and Streams (`streams=on`, which tracks streams but places nothing).

### 4.2.4 Use it from the guest

Check the device:

```
| sudo nvme list  
| sudo nvme id-ctrl /dev/nvme0 | grep -E '^(mn|sn) '
```

The model is FEMU NoSSD NVMe Controller and the serial number starts with vNoSSD.

Measure latency and throughput with `fio`. Unlike BlackBox, unwritten blocks read at the same speed as written ones:

```
| sudo fio --name=lat --filename=/dev/nvme0n1 --direct=1 --ioengine=io_uring \  
| --rw=randread --bs=4k --iodepth=1 --runtime=30 --time_based  
| sudo fio --name=tput --filename=/dev/nvme0n1 --direct=1 --ioengine=io_uring \  
| --rw=randread --bs=4k --iodepth=64 --numjobs=4 --group_reporting \  
| --runtime=30 --time_based
```

The SMART log counts host reads and writes in every mode, NoSSD included:

```
| sudo nvme smart-log /dev/nvme0
```

The vendor log page C0h stays zero: NoSSD has no media to count.

### 4.2.5 Limits and refusals

| Message                                                                                                         | Cause and fix                  |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------|
| multiplier_enabled must be 0 (one poller for all queues) or 1 (each poller owns poller_ratio queues)            | Any other value.               |
| queues must be in [1, 2047]                                                                                     | queues out of range.           |
| oncs may only set Compare, Write Uncorrectable, DSM, Write Zeroes, Save/Select Feature Support, Verify and Copy | A bit FEMU does not implement. |

If the host cannot allocate devsz\_mb of memory, QEMU aborts in GLib with failed to allocate N bytes.

## 4.2.6 Verify

1. `sudo nvme list` shows the model FEMU NoSSD NVMe Controller.
2. A queue depth 1 random read reports a completion latency of a few microseconds to tens of microseconds, depending on the host, with no difference between written and unwritten blocks.
3. `sudo nvme smart-log /dev/nvme0` shows the read and write command counts moving.

## 4.2.7 Troubleshooting

- **Throughput does not grow with more guest jobs.** A single poller serves every queue by default. Set `multiplier_enabled=1` with a `poller_ratio` that leaves one poller per host core you can spare, and pin the threads.
- **Latency varies from run to run.** The pollers compete with vCPUs and other host work. Pin vCPUs and pollers to separate cores, and keep the host CPU at a fixed frequency (see performance tuning (§5.12)).
- **Data is gone after QEMU exits.** FEMU keeps the device only in host memory. A guest reboot keeps the data.

Related issues: #52, #69.

## 4.2.8 Related pages

- Choosing a mode (§4.1)
- Architecture: pollers (§3.1.3)
- Timing model (§6.2)

# 4.3 NoSSD: the mode without a media model

This chapter describes NoSSD (`femu_mode=2`, the default): what it leaves out compared with the other modes, the path a command takes, and when to use it. To run the mode, see the NoSSD guide (§4.2).

## 4.3.1 Purpose

NoSSD is a plain NVMe namespace stored in host memory. It keeps FEMU's whole NVMe frontend (queues, pollers, admin commands, optional commands, namespaces) and drops everything below it:

| Layer                                | BlackBox ( <code>femu_mode=1</code> ) | NoSSD |
|--------------------------------------|---------------------------------------|-------|
| NVMe frontend, pollers               | yes                                   | yes   |
| FTL: mapping, write buffer, GC, wear | yes, on the FTL thread                | none  |
| NAND timing: chips, channels, planes | yes                                   | none  |

| Layer           | BlackBox (femu_mode=1)      | NoSSD                                         |
|-----------------|-----------------------------|-----------------------------------------------|
| FTL thread      | started                     | not started for NoSSD namespaces              |
| Data            | memory backend              | memory backend                                |
| Completion time | arrival + FTL and NAND time | arrival, plus optional link and firmware time |

A command completes as soon as the poller has copied its data. What the guest then measures is the cost of the emulation itself: the guest driver, the doorbell exit, the poller, the copy and the interrupt.

Use NoSSD when:

- the storage device should not be the bottleneck, for example to work on the guest's I/O stack, the NVMe driver, `io_uring` or `SPDK`;
- you need an upper bound to compare a timed mode against. The difference between a BlackBox run and a NoSSD run with the same queues and pollers is mostly the time FEMU charges for the media and the FTL, but it also includes the cost of BlackBox's longer path through the FTL thread;
- you test NVMe features that do not depend on media, such as namespace management, metadata and protection information, or Streams.

Use BlackBox (§4.4) when you need SSD latency, garbage collection or write amplification.

### 4.3.2 Place in the hierarchy

```

guest NVMe driver
  | submission queue entry, doorbell
+-----v-----+
| poller thread (femu-poller), one sweep |
|                                         |
| nvme_process_sq_io()                   |
|   stamp arrival time (stime = expire_time = now) |
|   nvme_io_cmd()                         |
|   Flush, DSM, Compare, Write Zeroes, Copy, ... (shared)|
|   Read, Write -> nop_io_cmd() -> nvme_rw() |
|               map PRP or SGL, copy to backend |
|   complete: via the poller's priority queue (see below) |
+-----+-----+
|                                         |
+-----v-----+
| memory backend backend/dram.c | devsz_mb of host memory
+-----+-----+

```

not used by NoSSD namespaces: FTL thread, struct `ssd`, NAND media model

`nvme_register_nossd()` (`hw/femu/nossd/nop.c`) installs a handler table with three entries that do work: `init` and `init_ctrl_name`, which set the model and serial strings, and `io_cmd`, which sends Read and Write to the shared `nvme_rw()` and returns Invalid Opcode for anything else that reaches it. The optional commands (Dataset Management, Compare, Write Zeroes, Copy, Verify, Write Uncorrectable) are handled in `nvme_io_cmd()` before the mode is consulted, when `oncs` turns them on. With `streams=on`, a successful write also records its stream, though NoSSD places nothing differently.

### 4.3.3 Data path

`nvme_rw()` checks the LBA range and MDTs, maps the guest's PRP or SGL list, and copies between guest memory and the memory backend, normally with `backend_rw()`. The copy is done on the poller thread before the command completes. The backend is one `devsz_mb` buffer allocated at realize; FEMU tries to pin it with `mlock()` and prints a notice when `RLIMIT_MEMLOCK` is too small, since page faults would then add to the measured latency. Namespaces are slices of that buffer (multi-namespace guide (§4.15)).

The data survives a guest reboot and is lost when QEMU exits.

### 4.3.4 Completion path

Every request is stamped with its arrival time as both its start time and its completion time (`expire_time`). NoSSD adds nothing to it, so a NoSSD request is due the moment it is stamped.

```
fetch -> execute -> copy -> to_ftl ring
      -> nvme_process_cq_cpl(): + link time, firmware time
      -> priority queue -> post when due -> interrupt
```

With no FTL thread, the poller drains its own `to_ftl` ring at the end of the sweep, so the request is posted in the same sweep unless a link or firmware time pushes it into the future or the completion queue is full. A completion that finds the queue full waits in a per-poller backlog. When a black-box, ZNS or CSD namespace on the same controller has started the FTL thread, the request goes through that thread and back.

`hiops_inline` (§A.1.1) applies to NoSSD only: performance option, on by default; set off only when debugging.

### 4.3.5 Threads and pollers

NoSSD runs on the poller threads alone. How many there are, and which queues each owns, is set by `multipoller_enabled` and `poller_ratio` (§A.1.1) as in every mode; the architecture page (§3.1.3) explains the split. Because there is no media time to hide behind, the pollers' CPU is the device's speed limit in this mode. Give each poller a host core of its own, away from the vCPUs, and see performance tuning (§5.12).

### 4.3.6 Timing

None from the media. The optional models in the host link and controller firmware (§A.1.1) group apply to NoSSD as to every mode:

- `pcie_bandwidth_mbps`: transfer time per Read and Write on one device-wide link queue per direction;
- `pcie_prop_delay_ns`: a fixed delay after each transfer;
- `fw_cpu_ns`: a fixed time per Read, Write and Zone Append on one modelled controller core.

### 4.3.7 Parameters

NoSSD ignores the NAND geometry, NAND timing and FTL properties. The ones that matter:

| Group                                      | Properties                                                                                                                         |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Mode, capacity and namespaces (§A.1.1)     | <code>devsz_mb</code> , <code>namespaces</code> , <code>namespace_sizes</code> , <code>namespace_modes</code>                      |
| Queues, pollers and interrupts (§A.1.1)    | <code>queues</code> , <code>multipoller_enabled</code> , <code>poller_ratio</code> , <code>hiops_inline</code> , <code>mdts</code> |
| Host link and controller firmware (§A.1.1) | <code>pcie_bandwidth_mbps</code> , <code>pcie_prop_delay_ns</code> , <code>fw_cpu_ns</code>                                        |

| Group                                                 | Properties                     |
|-------------------------------------------------------|--------------------------------|
| Controller identity and capabilities (§A.1.1)         | oncs, cmbasz, vwc              |
| LBA formats, metadata and protection (§A.1.1)         | meta, mc, pi, nlbaf, lba_index |
| Namespace management, streams and power loss (§A.1.1) | ns_mgmt, streams               |

A NoSSD device with four pollers, each owning two of eight queues:

```
| -device femu,devsz_mb=1024,queues=8,multipoller_enabled=1,poller_ratio=2
```

### 4.3.8 Counters

- The SMART log counts host read and write commands and bytes, kept per poller and summed when the log is read.
- The vendor log page C0h stays zero on a controller whose namespaces are all NoSSD: there is no media to count. Black-box, CSD and KV namespaces on the same controller fill it.

### 4.3.9 Validation status

- NoSSD is the default mode, so most qtest cases in `hw/femu/tests/qtest/femu-test.c` run on it through the default device options. Cases named for it include `io-fuzz-nossd`, a fuzzer over I/O command fields, and the namespace management cases `ns-retire-nossd`, `ns-shared-remove-nossd` and `ns-shared-retire-nossd`. The poller cases `poller-burst`, `poller-burst-per-queue`, `poller-burst-sharded` and `shared-cq-pollers` run on NoSSD because it is the default.
- The documentation check starts each NoSSD example and writes and reads back one block.
- In a guest, `femu-test.sh` runs its block checks on a NoSSD namespace.

### 4.3.10 Limits

- No latency model, so results say nothing about SSD behaviour.
- Throughput is bounded by poller CPU and by how the host schedules the pollers and vCPUs, which varies from run to run unless they are pinned.
- Host memory equal to `devsz_mb` is allocated at realize. If the allocation fails, QEMU aborts. If it succeeds but cannot be pinned, the pages are faulted in as they are used, and a host short of memory may run out then instead.
- Data is not kept across QEMU restarts.

Refusal messages are listed in the NoSSD guide (§4.2.5).

### 4.3.11 Extending the mode

- **A fixed device latency.** Add a constant to `req->expire_time` in `nop_io_cmd()` and run with `hiops_inline=off`; the completion step then holds the request until it is due.
- **Another I/O command.** Add a case to `nop_io_cmd()` for a NoSSD-only command, or to `nvme_io_cmd()` in `hw/femu/nvme-io.c` when every block mode should get it.
- **Starting point for a new mode.** `hw/femu/nossd/nop.c` is the smallest complete mode. Copy it and register it in `nvme_register_extensions()` in `hw/femu/femu.c`. Then add the new `femu_mode` value to the enum in `hw/femu/nvme.h`, raise the bound in the realize check (`nvme_check_constraints()`), which refuses values above the KV mode), teach `backend_rw()` in `hw/femu/backend/dram.c` the mode (it



asserts on an unknown one), add a token to `nvme_mode_from_token()` for `namespace_modes`, and add a `modes.py` entry.

### 4.3.12 Source map

| File                                              | Contents                                                                                                                                                  |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hw/femu/nossd/nop.c</code> <sup>27</sup>    | the mode: handler table, Read and Write dispatch, model string                                                                                            |
| <code>hw/femu/nvme-io.c</code> <sup>28</sup>      | <code>nvme_process_sq_io()</code> , <code>nvme_io_cmd()</code> , <code>nvme_rw()</code> , <code>nvme_process_cq_cpl()</code> , <code>nvme_poller()</code> |
| <code>hw/femu/backend/dram.c</code> <sup>29</sup> | the memory backend and <code>backend_rw()</code>                                                                                                          |
| <code>hw/femu/femu.c</code> <sup>30</sup>         | mode registration, FTL thread decision                                                                                                                    |

### 4.3.13 Related pages

- NoSSD guide (§4.2): launching, fio examples, troubleshooting
- Architecture: pollers (§3.1.3)
- Timing model: host link and controller firmware (§6.2.9)

## 4.4 BlackBox SSD (BBSSD)

BlackBox mode (`femu_mode=1`) emulates a conventional NVMe SSD: the device runs its own flash translation layer (FTL), garbage collection (GC) and NAND timing, and the guest sees an ordinary block device. Use it to study how a host workload behaves on a real-looking SSD: latency under GC, write amplification, the effect of over-provisioning, caches and write buffers.

If you want a drive with no media timing at all, use NoSSD (§4.2). If you want the host to run the FTL, use OCSSD (§4.9) or ZNS (§4.7).

### 4.4.1 Requirements

- Host and guest: see the mode table (§4.1.4) and `requirements.md` (§2.1.2). Any guest kernel with the NVMe driver works.
- Host memory: the whole namespace lives in host DRAM. `run-blackbox.sh` creates a 12 GiB device, so it needs about 17 GiB of free RAM (memory planning (§2.1.1)).
- Guest tools: `nvme-cli` and `fio`, both in the image from `make-guest-image.sh`.

### 4.4.2 Launch

From `build-femu/`, with a guest image in place:

```
| ./run-blackbox.sh
```

The script prints the `-device` line it builds. It is equivalent to:

---

<sup>27</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nossd/nop.c>

<sup>28</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nvme-io.c>

<sup>29</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/backend/dram.c>

<sup>30</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/femu.c>

```
-device femu,devsz_mb=12288,namespaces=1,femu_mode=1,secsz=512,secs_per_pg=8,pgs_per_blk=256,blks_per_pl=256,
pls_per_lun=1,luns_per_ch=8,nchs=8,pg_rd_lat=40000,pg_wr_lat=200000,blk_er_lat=2000000,ch_xfer_lat=0,gc_
thres_pcent=75,gc_thres_pcent_high=95
```

To change the geometry or latency, edit the variables at the top of `run-blackbox.sh`. Its console output goes to `build-femu/log`.

### 4.4.3 Configuration

Every property below is listed with its default and range in the property reference (§A.1). This section says which ones matter for BlackBox and how they interact.

#### Capacity and geometry

Properties: mode and capacity (§A.1.1), NAND geometry (§A.1.1).

The NAND geometry is `nchs` channels, each with `luns_per_ch` LUNs, each with `pls_per_lun` planes of `blks_per_pl` blocks of `pgs_per_blk` pages. A page is `secs_per_pg` sectors of `secsz` bytes. The raw NAND capacity is the product of all seven. With the defaults it is 16 GiB ( $8 \times 8 \times 1 \times 256 \times 256 \times 8 \times 512 = 17,179,869,184$  bytes), and a page is 4 KiB.

The namespace the guest sees is `devsz_mb` MiB, which can be smaller than the raw NAND. The rest is spare area for GC. FEMU refuses a namespace that leaves GC no free lines (see Limits and refusals (§4.4.5)). A line is one block on every plane of every LUN, so there are `blks_per_pl` lines.

To state the over-provisioning directly, set `op_pcent`. FEMU then backs the device with the full raw NAND and exposes `raw / (1 + op_pcent/100)`, and `devsz_mb` is ignored. This example has 512 MiB of NAND and exposes about 410 MiB:

```
-device femu,femu_mode=1,nchs=2,luns_per_ch=4,blks_per_pl=64,op_pcent=25
```

#### NAND timing

Properties: NAND timing (§A.1.1). How the time is computed: timing model (§6.2.3).

- `pg_rd_lat`, `pg_wr_lat` and `blk_er_lat` set flat read, program and erase times in nanoseconds.
- `nand_cell_type` (1 SLC, 2 MLC, 3 TLC, 4 QLC) replaces the flat times with built-in per-page-type timing:

```
-device femu,devsz_mb=1024,femu_mode=1,nand_cell_type=3
```

- `cmd_addr_lat`, `pg_xfer_lat` (or `ch_xfer_lat`) and `status_lat` add a shared channel bus. It is modelled only when one of them is non-zero.
- `pe_suspend` and `tsusp_ns` let a read suspend a program or erase on its LUN.
- `trim_lat_ns` charges each Dataset Management deallocate range.

#### Garbage collection

Properties: garbage collection, mapping and caches (§A.1.1).

Background GC starts when the share of lines in use reaches `gc_thres_pcent` (75). Foreground GC runs inside writes from `gc_thres_pcent_high` (95). `gc_policy` picks the victim line:

- greedy (the default): the line with the fewest valid pages.
- random: a random line among the GC candidates.

- cost-benefit: the line with the largest age  $\times (1 - u) / 2u$ , where  $u$  is the share of valid pages.
- fifo: the oldest closed line.
- d-choice: samples 4 candidate lines at random (a line can be drawn twice) and takes the one with the fewest valid pages.

random and d-choice draw from a generator seeded by `gc_seed` (default 0), so the same configuration and workload pick the same victims and give the same write amplification on every run. Set a different `gc_seed` per run to vary them.

## Mapping and caches

Properties: garbage collection, mapping and caches (§A.1.1).

- mapping: page (default, a full table in DRAM), `dftl` (a cached table of `mapping_cache_mb` MiB whose misses cost NAND reads), `hybrid` or `fast` (log-block schemes).
- `read_cache_mb` and `cache_evict` add a DRAM read cache. It models timing only: a hit costs DRAM time and skips the NAND read, but the cache holds no data, so NAND stays the source of truth. `cache_evict` is `clock` (the default), `random`, `lru` or `arc` (a scan-resistant 2Q variant, not full ARC).

The two log-block schemes follow published designs:

| mapping | Model                                                                                                                      |
|---------|----------------------------------------------------------------------------------------------------------------------------|
| hybrid  | BAST log-block mapping (Kim 2002): one log block per data block, merged when that log block fills or the log pool runs out |
| fast    | FAST log-block mapping (Lee et al. 2007): a sequential log block plus a shared, fully associative pool for random writes   |

Both suit some workloads more than others: sequential overwrites merge cheaply, random overwrites force full merges. A merge is charged to the NAND timeline and counted as relocated pages, so it shows in latency and in the write amplification factor.

- `hot_cold_sep` writes overwrites of mapped pages to separate lines.

## Write buffer and power loss

Properties: garbage collection, mapping and caches (§A.1.1), namespace management, streams and power loss (§A.1.1).

`buffer_size` is the DRAM write buffer capacity in NAND pages, not bytes. Once the buffer is `buffer_thres_pcent` full, the next write programs a batch of the least recently written pages. A write the buffer absorbs costs no NAND time; the cost moves to the write that evicts it. Flush drains the buffer. Set `vwc=1` so that the guest sees a volatile write cache: Linux then sends Flush, and the guest can turn the buffer off with feature 06h. A read of a page the buffer still holds costs no NAND time, and deallocating such a page drops it instead of writing it out later.

```
| -device femu,devsz_mb=1024,femu_mode=1,buffer_size=1024,vwc=1
```

`power_loss=on` makes the buffer hold data, and the QOM property `simulate-power-loss` drops what it holds (runtime properties (§A.2.1)).

## Wear, retention and errors

Properties: reliability and wear (§A.1.1).

All of these are off by default.

- `ecc_step_ns` and `ecc_retention_sec` make reads of worn or old blocks slower.
- `pe_cycles_rated` and `nand_bad_blocks` feed SMART Percentage Used and Available Spare.
- `err_read_unc_ppm` and `err_write_fail_ppm` fail a fixed share of reads or writes. The failures come at a fixed period, so a run repeats exactly.
- `read_reclaim_limit` and `retention_limit_sec` rewrite lines that were read too often or hold old data. The line is picked on a read and rewritten on a following write, so a workload that never writes never triggers it.

Both refresh knobs act only when something reads the line. A region nothing reads is never refreshed: modelling a background media scan would need a timer that FEMU does not run.

`read_reclaim_limit` is the number of reads a block may take before its line is refreshed:

```
| -device femu,devsz_mb=1024,femu_mode=1,read_reclaim_limit=100000
```

The line is chosen on the read but rewritten on the next write, where relocation already costs something, so a read never waits behind a whole line. One line is queued at a time and at most one is refreshed per write, so the rate follows how often the host writes, not how hard it reads. The cost shows up as write amplification. On a 512 MiB region read six times over with a low limit, 5 lines were refreshed and 77824 pages relocated, and the write amplification factor went from 1.000 to 1.542. Lowering the limit from 500 to 10 moved it only from 1.542 to 1.628. A workload that only reads never refreshes anything, where a real drive would do it in the background, so model read-only ageing some other way.

## Host link and controller firmware

Properties: host link and controller firmware (§A.1.1).

`pcie_bandwidth_mbps`, `pcie_prop_delay_ns` and `fw_cpu_ns` add link and firmware time to every Read and Write. See the timing model (§6.2.9).

## Features on top of BlackBox

- Flexible Data Placement (§4.5) (`femu-subsys`, `fdp=on`).
- Several namespaces (§4.15), each with its own FTL.
- Namespace management, metadata and protection information (§4.17).
- Streams (`streams=on`), which needs mapping page or `dftl`.

### 4.4.4 Use it from the guest

These commands run inside the guest. From the host, prefix each with `./run-guest-ssh.sh` as in the quick start (§2.4).

Check the device:

```
| sudo nvme list  
| sudo nvme id-ctrl /dev/nvme0 | grep -E '^(mn|sn) '
```

The model is FEMU BlackBox-SSD Controller and the serial number starts with `vSSD`.

Write before you read. A read of a page that was never written costs no NAND time, so a read benchmark on a fresh device measures nothing. Fill the range first, then read it:

```

sudo fio --name=fill --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
--rw=write --bs=128k --iodepth=16 --size=4G
sudo fio --name=rr --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
--rw=randread --bs=4k --iodepth=1 --size=4G --runtime=30 --time_based

```

With an idle device and queue depth 1, the completion latency (clat) is about `pg_rd_lat` plus guest and poller overhead.

To make GC run, write more than the free lines hold. GC starts when 75% of the lines are in use, so on the default 16 GiB of NAND that takes about 12 GiB of writes, one pass over the 12 GiB namespace. Random writes over the namespace several times keep GC busy:

```

sudo fio --name=age --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
--rw=randwrite --bs=4k --iodepth=32 --loops=5

```

Read the write amplification factor and media counters from the vendor log page C0h. The first four bytes are the WAF times 1000, bytes 4 to 7 are reserved, and the three 8-byte counters from byte 8 are host pages written, pages GC moved, and pages programmed:

```

sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b | od -An -t u4 -N 4
sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b | od -An -t u8 -j 8 -N 24 -w24

```

`log-pages-and-counters.md` (§B.1) lists every field. Wear and host totals are in the SMART log:

```

sudo nvme smart-log /dev/nvme0

```

The vendor admin command 0xEF switches GC time and the NAND times on or off while the guest runs, for example to fill a device quickly:

```

sudo nvme admin-passthru /dev/nvme0 --opcode=0xef --cdw10=4 # NAND times to 0
sudo nvme admin-passthru /dev/nvme0 --opcode=0xef --cdw10=3 # back to 40 us, 200 us, 2 ms

```

Code 3 restores the built-in times, not the ones on your command line. Codes 3 and 4 change only the flat times, so they have no effect with `nand_cell_type` set. The timing model (§6.2.11) lists every code.

## 4.4.5 Limits and refusals

FEMU checks the configuration at realize and stops QEMU with a message. The common ones:

| Message                                                                                                                                                                                                            | Cause and fix                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FEMU bbssd: namespace 1 exposes 1024 MiB of the 1024 MiB this geometry has, leaving garbage collection no room; expose at most 960 MiB per namespace (lower <code>devsz_mb</code> , or set <code>op_pcent</code> ) | <code>devsz_mb</code> (or the namespace size) does not leave the free lines GC needs. Lower <code>devsz_mb</code> , grow the geometry, or set <code>op_pcent</code> . |
| FEMU bbssd: the geometry has only N lines, fewer than the M garbage collection needs free                                                                                                                          | <code>blks_per_pl</code> is too small for the reserve that <code>gc_thres_pcent_high</code> , <code>hot_cold_sep</code> , Streams and log-block mapping add up to.    |
| FEMU bbssd: <code>nchs</code> must be greater than 0, got 0                                                                                                                                                        | A geometry property is 0 or negative. The same message names any of them.                                                                                             |
| FEMU bbssd: the geometry describes N sectors, which exceeds the 2147483647 the FTL can address; ...                                                                                                                | The NAND is too large; reduce one of the geometry properties.                                                                                                         |
| FEMU bbssd: unknown mapping "x", unknown <code>gc_policy</code> "x", unknown <code>cache_evict</code> "x"; ...                                                                                                     | A misspelled policy name.                                                                                                                                             |

If the host cannot allocate `devsz_mb` (or, with `op_pcent`, the raw NAND) of memory, QEMU aborts in GLib with failed to allocate N bytes and a core dump.

Reserve per namespace:  $(1 - \text{gc\_thres\_pcent\_high}/100) * \text{blks\_per\_pl}$  lines, rounded down, plus one for the write pointer, one more with `hot_cold_sep`, one more with `mapping=hybrid` or `fast`, and `streams.max + 1` with Streams. The namespace must fit in the remaining lines.

#### 4.4.6 Verify

1. `sudo nvme list` in the guest shows `/dev/nvme0n1` with model FEMU BlackBox-SSD Controller and the size you set.
2. After a 4 KiB random write run, the C0h WAF reads 1000 or more and the host page counter (byte 8) equals the pages fio wrote.
3. A queue depth 1 random read of written data takes about `pg_rd_lat`.

#### 4.4.7 Troubleshooting

- **Reads are faster than the configured latency.** The range was never written. Unwritten pages have no mapping and cost no NAND time. Fill the range first.
- **The WAF stays at 1.000.** GC has not run, or found only lines with no valid pages, as after a sequential overwrite. GC starts after about 12 GiB of writes on the default geometry. Run random writes over the namespace more than once, or use `op_pcent` to make the namespace a known fraction of the NAND.
- **Property 'femu.cell\_type' not found, or flash\_type has no effect.** The BlackBox cell type is `nand_cell_type`. `flash_type` belongs to OCSSD.
- **QEMU aborts with failed to allocate.** The device is in host DRAM. Lower `devsz_mb` or free host memory.
- **Latency is higher than the model.** Completions are posted by the poller, which needs a host core. Give each `femu-poller` thread and the FEMU-FTL-Thread a core of its own (timing model (§6.2.1)).
- **Data is gone after QEMU exits.** FEMU keeps the device only in host memory and writes nothing to a file. A guest reboot keeps the data.

Related issues: #16, #50, #52, #92, #113, #130, #132, #137.

#### 4.4.8 Related pages

- Timing model (§6.2)
- Architecture: the BBSSD FTL (§3.1.5)
- Quick start (§2.4), which runs this mode end to end
- Device property reference (§A.1)

### 4.5 Flexible Data Placement (FDP)

Flexible Data Placement lets the host tell the SSD which writes belong together. The device groups its NAND into reclaim units and offers a few reclaim unit handles. Each write may carry a placement identifier that names a handle, and the device puts all writes of one handle into the same open reclaim unit. When data with similar lifetimes shares a reclaim unit, GC finds it mostly invalid and copies less, so write amplification drops.

FDP is not a `femu_mode`. It is a property of an NVMe subsystem, `femu-subsys`, which a BlackBox (§4.4) controller joins. In FEMU a reclaim unit is one superblock (line) of the BlackBox FTL. FEMU's FDP support comes from WARP; if you use it, cite WARP as well as FEMU (citation (§4.5.8)).

Use it to measure how placement hints from an application, a file system or fio change write amplification and tail latency.

### 4.5.1 Requirements

- Host and guest: see the mode table (§4.1.4) and requirements.md (§2.1.2). Any guest kernel with the NVMe driver sees the device.
- Placement hints: the guest has to send them. Ordinary writes through the block device carry no placement identifier unless the guest kernel adds one, so use passthrough commands (nvme write with a directive), or fio's `io_uring_cmd` engine on the generic device `/dev/ng0n1`.
- Guest tools: `nvme-cli` with the `fdp` commands (2.x), and `fio` built with `io_uring_cmd` support for placement from fio.

### 4.5.2 Launch

From `build-femu/`:

```
| ./run-blackbox-fdp.sh
```

The script builds two devices. The subsystem must come first on the command line, because the controller names it with `subsys=`:

```
| -device femu-subsys,id=femu-subsys-0,nqn=subsys0,fdp=on,fdp.nruh=4,fdp.nrg=1,fdp.nru=256 -device  
femu,devsz_mb=12288,namespaces=1,femu_mode=1,secsz=512,secs_per_pg=8,pgs_per_blk=256,blks_per_pl=256,  
pls_per_lun=1,luns_per_ch=8,nchs=8,pg_rd_lat=40000,pg_wr_lat=200000,blk_er_lat=2000000,ch_xfer_lat=0,gc_  
thres_pcent=75,gc_thres_pcent_high=95,subsys=femu-subsys-0
```

The shortest form, with the default 64 MiB reclaim units and 128 of them:

```
| -device femu-subsys,id=fdp0,fdp=on,fdp.nruh=4 -device femu,devsz_mb=1024,femu_mode=1,subsys=fdp0
```

`run-blackbox-fdp.sh` writes its console output to `/tmp/femu-fdp.log`.

### 4.5.3 Configuration

#### Subsystem

Properties: Flexible Data Placement (§A.1.2).

- `fdp=on` turns FDP on for the controller that joins the subsystem. An FDP subsystem takes a single controller.
- `fdp.nruh`: number of reclaim unit handles, that is, placement handles the host can name. Required, at least 1.
- `fdp.nru`: reclaim units per reclaim group (default 128). FEMU uses at most one per superblock, so the effective count is the smaller of `fdp.nru` and `blks_per_pl`. It needs at least  $2 * fdp.nruh + 1$  of them: one open unit per handle, one spare per handle for GC, and one more. The namespace gets what is left once each handle has a unit open, each Persistently Isolated handle one to collect into, and forced GC its free units (at least one unless `gc_thres_pcent_high` is 100); a larger `devsz_mb` is refused. See the reserve (§4.6.3).
- `fdp.nrg`: reclaim groups. Must be 1.
- `fdp.runs`: reclaim unit size in bytes. Leave it at 0, or set it to the size of one superblock:  $nchs * luns_per_ch * pls_per_lun * pgs_per_blk * secs_per_pg * secsz$  (64 MiB with the default geometry).
- `fdp.isolation_mode`: 0 makes every handle Persistently Isolated; any other value makes the last handle Initially Isolated.

## Controller

The controller is an ordinary BlackBox controller with `subsys=<id>`, so the BlackBox geometry, timing and GC properties (§4.4.3) apply. Two properties are specific to FDP (garbage collection, mapping and caches (§A.1.1)):

- `gc_strategy`: how GC picks a victim reclaim unit: 0 greedy (default), 1 cost-benefit, 2 random, 4 per-handle. `gc_policy` does not apply under FDP. The random strategy draws from a generator seeded by `gc_seed`, so it repeats run to run.
- `fdp_trim_erase_all`: non-zero makes a deallocate reset every reclaim unit instead of the given ranges.

`FEMU_FDP_DEBUG` in QEMU's environment prints placement and reclaim traces (environment variables (§A.1.5)).

### 4.5.4 Use it from the guest

Check that the controller supports FDP. Bit 19 (0x80000) of `CTRATT` is FDP:

```
| sudo nvme id-ctrl /dev/nvme0 | grep ctratt
```

Read the configuration, handle usage, statistics and events of endurance group 1:

```
| sudo nvme fdp configs /dev/nvme0 -e 1
| sudo nvme fdp usage /dev/nvme0 -e 1
| sudo nvme fdp stats /dev/nvme0 -e 1
| sudo nvme fdp events /dev/nvme0 -e 1 -E
| sudo nvme fdp status /dev/nvme0n1
```

`configs` lists the handles and the reclaim unit size, `usage` gives each handle's attributes, `stats` gives host and media bytes written, and `status` lists the placement identifiers the namespace accepts with the space left in each one's reclaim unit (RUAMW). Every handle backs a placement handle of the namespace, so `usage` reports each one as host specified:

```
| Reclaim Unit Handle 0 Attributes: 0x1 (Host Specified)
| Reclaim Unit Handle 1 Attributes: 0x1 (Host Specified)
| Reclaim Unit Handle 2 Attributes: 0x1 (Host Specified)
| Reclaim Unit Handle 3 Attributes: 0x1 (Host Specified)
```

`events -E` shows host events, such as a write with an invalid placement identifier; every event type is enabled by default.

### Write with a placement identifier

A write with directive type 2 (data placement) uses the directive specific field as its placement identifier. This writes 8 blocks of 512 bytes at LBA 0 through handle 1:

```
| head -c 4096 /dev/urandom > data.bin
| sudo nvme write /dev/nvme0n1 -s 0 -c 7 -z 4096 -d data.bin -T 2 -S 1
```

A write with no directive, or with an identifier the namespace does not have, goes to handle 0. An invalid identifier also logs a host event.

`fio` places data from its `io_uring_cmd` engine. It reads the placement identifier list from the device and cycles through the entries whose indexes are listed in `fdp_pli`:

```
| sudo fio --name=fdp --filename=/dev/ng0n1 --ioengine=io_uring_cmd --cmd_type=nvme \
| --fdp=1 --fdp_pli=0,1,2,3 --rw=randwrite --bs=4k --iodepth=16 --size=4G
```



## Measure the effect

The vendor log page C0h reports the write amplification factor for FDP as for plain BlackBox. Compare a run with placement against the same run without it:

```
| sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b | od -An -t u4 -N 4
```

### 4.5.5 Limits and refusals

| Message                                                                                                                                                                                                 | Cause and fix                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fdp.nruh must be non-zero                                                                                                                                                                               | Set fdp.nruh.                                                                                                                                                                                                                                                  |
| fdp.nruh (200) must not exceed fdp.nru (128)                                                                                                                                                            | More handles than reclaim units.                                                                                                                                                                                                                               |
| fdp.nrg must be 1: placement into a reclaim group other than the first is not implemented                                                                                                               | Leave fdp.nrg at 1.                                                                                                                                                                                                                                            |
| FEMU bbssd: placement needs N reclaim units for M handles across 1 groups and this geometry gives K; raise blks_per_pl or fdp.nru, or lower fdp.nruh                                                    | Too few reclaim units for the handles.                                                                                                                                                                                                                         |
| FEMU bbssd: namespace 1 exposes 1024 KiB, but placement keeps 9 of the 24 reclaim units for 4 handles and collection, leaving 960 KiB; lower devsz_mb or fdp.nruh, raise blks_per_pl, or raise op_pcent | The namespace does not fit in what placement leaves.                                                                                                                                                                                                           |
| FEMU bbssd: fdp.runs must be 67108864, the size of one superblock of this geometry, or unset                                                                                                            | fdp.runs does not match the geometry.                                                                                                                                                                                                                          |
| FDP supports a single namespace; set namespaces=1 or disable FDP on the subsystem                                                                                                                       | namespaces above 1.                                                                                                                                                                                                                                            |
| femu-subsys with fdp=on takes a single controller                                                                                                                                                       | A second controller joins the FDP subsystem.                                                                                                                                                                                                                   |
| FEMU bbssd: buffer_size has no effect under FDP                                                                                                                                                         | FDP has its own write path. The same message names hot_cold_sep, read_reclaim_limit, retention_limit_sec, ecc_retention_sec, trim_lat_ns, a mapping other than page and a gc_policy other than greedy. A CSD controller gets the same messages with FEMU csd:. |
| meta: not supported with placement (fdp)                                                                                                                                                                | meta (with a valid mc) under FDP.                                                                                                                                                                                                                              |
| streams requires bbssd or NoSSD with FDP disabled                                                                                                                                                       | streams=on under FDP.                                                                                                                                                                                                                                          |
| the key-value command set and FDP cannot share a controller: ...                                                                                                                                        | A KV namespace under FDP.                                                                                                                                                                                                                                      |
| namespace management does not support FDP                                                                                                                                                               | ns_mgmt=on together with fdp=on on the subsystem.                                                                                                                                                                                                              |
| Device 'fdp0' not found                                                                                                                                                                                 | The controller comes before its subsystem on the command line.                                                                                                                                                                                                 |

NoSSD, ZNS and OCSSD controllers accept an FDP subsystem, report FDP to the host, and ignore it for placement. KV is refused. CSD goes through the BlackBox FDP write path, with the same refusals and reclaim unit size as BlackBox.

When every reclaim unit is in use and GC cannot free one, a write fails with Capacity Exceeded and QEMU prints `ssd_stream_write: device full, no RU for ruh N`. Deallocating or overwriting data gives GC units to free again, and writes succeed once it has.

### 4.5.6 Verify

1. `ctratt` has bit 19 set, and `nvme fdp configs` shows `fdp.nruh` handles in one reclaim group.

2. After writes with `-T 2 -S 1`, `nvme fdp status /dev/nvme0n1` shows the RUAMW of placement identifier 1 going down.
3. `nvme fdp stats` shows host bytes written growing with your writes.

### 4.5.7 Troubleshooting

- **Property 'femu.fdp' not found, or FDP does not show up.** FDP is a property of `femu-subsys`, not of `femu`. Create the subsystem first and join it with `subsys=`.
- **QEMU crashed in GC under a long write workload (`select_victim_ru`, `victim_ru_get_pri`, No free RUs left).** These were bugs in the reclaim unit victim queue, fixed on master. Update FEMU.
- **The WAF is the same with and without placement.** Check that the writes really carry directive type 2: block-device writes do not unless the guest kernel sets placement. Use `nvme write -T 2 -S N` or `fiio` with `--ioengine=io_uring_cmd --fdp=1`.

Related issues: #153, #186, #189, #191.

### 4.5.8 Citation

FDP support in FEMU comes from WARP, the FDP SSD emulator described in Characterizing and Emulating FDP SSDs with WARP<sup>31</sup> (FAST '26). If you use the FDP mode, please also cite:

```
@inproceedings{Song+26-WARP,
  author    = {Inho Song and Shoaib Asif Qazi and Javier Gonz{\'a}lez and
              Matias Bj{\'o}rling and Sam H. Noh and Huaicheng Li},
  title     = {{Characterizing and Emulating FDP SSDs with WARP}},
  booktitle = {24th USENIX Conference on File and Storage Technologies (FAST 26)},
  pages     = {347--362},
  year      = {2026},
}
```

### 4.5.9 Related pages

- BlackBox SSD (§4.4)
- Choosing a mode: which features combine (§4.1.3)
- Architecture: features that are not modes (§3.1.4)

## 4.6 FDP: Flexible Data Placement

This chapter describes how FEMU implements NVMe Flexible Data Placement (FDP): how the subsystem builds reclaim groups, reclaim units and reclaim unit handles, how a write is placed, how reclaim units map onto the BlackBox FTL's NAND, how garbage collection works under placement, and what the FDP log pages and features report. To launch an FDP device and write with placement identifiers from a guest, read the FDP feature guide (§4.5) first. FEMU's FDP support comes from WARP (FAST '26); the guide has its citation (§4.5.8).

### 4.6.1 Purpose

FDP lets the host group writes it expects to die together. The device offers reclaim unit handles (RUHs); each write may name one through a placement identifier, and the device puts all writes of one handle into the same

<sup>31</sup><https://www.usenix.org/conference/fast26/presentation/song>

open reclaim unit (RU). When an RU holds data of similar lifetime, it is mostly invalid by the time garbage collection picks it, so GC copies less and write amplification drops. FEMU models the host-visible FDP interface and backs it with a placement-aware write and GC path in the BlackBox FTL, so you can measure that effect.

FDP is not a femu\_mode. It is turned on with `fdp=on` on a `femu-subsys` device, and a controller joins the subsystem with `subsys=`. Only a BlackBox (`bbssd`) namespace has the placement data path.

## 4.6.2 Place in the hierarchy

```
femu-subsys device, fdp=on          NVMe subsystem
| endurance group 1 (NvmeEnduranceGroup.fdp)
|   configuration: nrg reclaim groups x nru reclaim units, nruh handles
|   host-visible state: RUH type and attributes, RUAMW per RU,
|   HBMW/MBMW/MBE, host and controller event rings
|
+-- femu device, femu_mode=1, subsys= one controller
    |
    +-- namespace 1                  placement handles PH 0..nruh-1
        |                          (PH i -> RUH i)
        +-- struct ssd (BlackBox FTL)
            FemuRuHandle[nruh]      current RU, GC RU, counters
            FemuReclaimGroup[nrg]   free list, full list,
                                   victim queues
            FemuReclaimUnit[nru]    one line (superblock) each,
                                   write pointer, valid pages
            lines -> blocks on every channel, LUN and plane
```

Two layers of state exist side by side. The subsystem holds what the host sees (`NvmeRuHandle`, `NvmeReclaimUnit` with its `RUAMW`, the statistics and events) in `nvme.h`. The FTL holds what places and collects data (`FemuRuHandle`, `FemuReclaimGroup`, `FemuReclaimUnit` in `bbssd/ftl.h`). Each FTL object points at its host-visible twin, and the write path updates both.

## 4.6.3 Configuration

`nvme_subsys_setup_fdp()` in `femu.c` builds one endurance group (ID 1) with a single FDP configuration when the subsystem is realized:

| Item                                   | Value                                                                                                                                 | Property                        |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Reclaim groups (NRG)                   | 1; any other value is refused                                                                                                         | <code>fdp.nrg</code>            |
| Reclaim units per group                | as set, at most 65536                                                                                                                 | <code>fdp.nru</code>            |
| Reclaim unit handles (NRUH)            | 1 to <code>fdp.nru</code> ; must be set                                                                                               | <code>fdp.nruh</code>           |
| Handle type (RUHT)                     | Persistently Isolated; with <code>fdp.isolation_mode</code> non-zero the last handle is Initially Isolated                            | <code>fdp.isolation_mode</code> |
| Reclaim unit size (RUNS)               | one BlackBox superblock (see below); the subsystem default of 96 MiB applies only when no <code>bbssd</code> controller overwrites it | <code>fdp.runs</code>           |
| Reclaim group identifier format (RGIF) | 0 with one group                                                                                                                      | derived                         |
| MAXPIDS                                | 128 (reported zero-based as 127)                                                                                                      | fixed                           |
| Event filter                           | every supported event type enabled on every handle                                                                                    | Set Features 1Eh                |

When a controller joins, `nvme_ns_init_fdp()` gives namespace 1 every handle in order: placement handle i

is RUH i, the namespace's ENDGID is 1, and each handle's RUAMW starts at a full unit. The controller then sets CTRATT bit 19 (FDP), and the effects log lists I/O Management Send and Receive. CTRATT bit 4 (Endurance Groups), ENDGIDMAX = 1 and each namespace's ENDGID = 1 come with any subsystem, FDP or not, since its one endurance group is what log 09h reports.

The BlackBox side (bb.c, ftl-fdp.c) adds the following, for a CSD namespace as for a bbssd one, since CSD runs the same FTL (bb\_init\_fdp()):

- RUNS is overwritten with the bytes of one superblock:  $nchs * luns\_per\_ch * pls\_per\_lun * pgs\_per\_blk * secs\_per\_pg * secsz$ . fdp.runs must be unset or exactly that.
- The FTL uses  $\min(lines, fdp.nru)$  reclaim units, one line each ( $lines\_per\_ru = 1$ ). Lines left over are not used by the placement write path.
- It needs at least  $fdp.nruh * fdp.nrg + fdp.nruh + 1$  units: one open unit per handle, one collection unit per handle, and one spare.
- gc\_thres\_pcent and gc\_thres\_pcent\_high set the free-unit watermarks instead of free-line watermarks. The forced watermark keeps at least one unit free, even where the percentage rounds to none, unless gc\_thres\_pcent\_high is 100 (bb\_fdp\_forced\_units()): a pass can fill its destination part way through a victim and needs a free unit to go on.
- The namespace must fit in the units left after the reserve (bb\_check\_capacity()), or the controller is refused at realize:

```

reserve = fdp.nruh * fdp.nrg          one open unit per handle and group
          + Persistently Isolated handles one collection unit each
          + forced watermark          free units forced GC keeps
usable  = (units - reserve) * superblock size

```

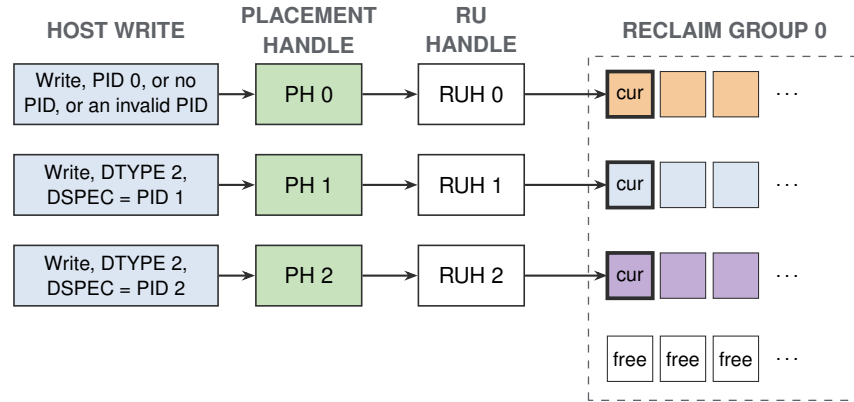
The Initially Isolated handle collects into its open unit and adds nothing. One namespace is checked against the whole pool because nothing else can draw on it: FDP takes a single namespace and a single controller, and namespace management is refused with it. A 1 MiB namespace on 64 KiB superblocks with four Persistently Isolated handles needs  $16 + 4 + 4 + 1 = 25$  units; with 19 or 23 lines, both accepted before, random writes over all four handles failed nearly all of them. A partly exposed last page counts as a whole one. bbssd refuses a namespace that does not fit rather than shrink it, as it does without placement and with op\_pcent.

#### 4.6.4 Placement of writes

```

Write: CDW12 bits 23:20 DTYPE, CDW13 bits 31:16 DSPEC
|
| DTYPE == 2 (data placement) and DSPEC is a valid placement identifier?
| no  -> placement handle 0, reclaim group 0
|      (if DTYPE was 2: host event "Invalid Placement Identifier")
| yes -> PID = (RG << (16 - RGIF)) | PH ; with RGIF 0, PID = PH
v
PH --ns->fdp.phs[]--> RUH --curr_ru--> active RU --wptr--> next page
|
| per FTL page (secsz * secs_per_pg, 4 KiB by default): invalidate the old copy, map LPN -> new page,
| RUAMW == logical blocks written, advance the write pointer,
| charge a NAND program
v
RU full -> retire it (full list or victim queue), take a free RU
          for this handle; none free -> Capacity Exceeded

```



With one reclaim group (`fdp.nrg=1`, the only value) the PID has no group bits: it picks a placement handle, which the namespace maps to a reclaim unit handle. Each handle writes into its current RU and takes the next free RU when that one is full. One RU is one BBSSD line: one block on every channel, LUN and plane. With the default `fdp.isolation_mode=0` every handle is persistently isolated, so garbage collection moves its valid pages into a GC RU of the same handle.

**Figure 15:** FDP placement on a BBSSD: a write's placement identifier picks a placement handle, the namespace maps it to a reclaim unit handle, and the handle fills its current reclaim unit (one FTL line) before taking a free one.

`ssd_stream_write_lpns()` does this on the FTL thread. Write, Write Zeroes without the Deallocate bit, and the destination of Copy all go through it; Copy and Write Zeroes read their placement fields from the same CDW12 and CDW13 bits.

The order inside `ssd_stream_write_lpns()` is:

1. If the handle has no active unit (the last one filled and no free unit was left), take a free unit without running GC; if there is none, fail with Capacity Exceeded. A unit that fills on a command's last page therefore makes the next write to that handle fail when the device is full.
2. Run foreground GC while the free-unit count is at or below the high watermark (see Garbage collection (§4.6.5)).
3. Place the pages, running foreground GC again before each one: a command can take more units than the watermark keeps free. If the device runs out part way, the command completes with Capacity Exceeded and only the pages already placed count.

Writes of different handles never share an RU, but they share the NAND: every RU spans all channels, so concurrent handles compete for the same planes and channel buses.

### Reclaim units over NAND

An RU is one BlackBox line: block *b* on every channel, LUN and plane. Its write pointer moves across channels first, then LUNs, then planes, then down the pages of the block, so consecutive pages of one handle land on different dies:

RU k = line k (block index k), nchs=4, luns\_per\_ch=2, pls\_per\_lun=2

```

page order:  ch0/L0/pl0  ch1/L0/pl0  ch2/L0/pl0  ch3/L0/pl0
              ch0/L1/pl0  ch1/L1/pl0  ch2/L1/pl0  ch3/L1/pl0
              ch0/L0/pl1  ...              ch3/L1/pl1  <- page 0 done
              ch0/L0/pl0 (page 1) ...              <- page 1
              ...              last page  <- RU full

```

RUNS = all of that = one superblock

fdp\_advance\_ru\_pointer() implements the order; fdp\_get\_new\_page() turns the pointer into an address. The NAND timing of each program is the BlackBox timing (see NAND timing (§3.5)).

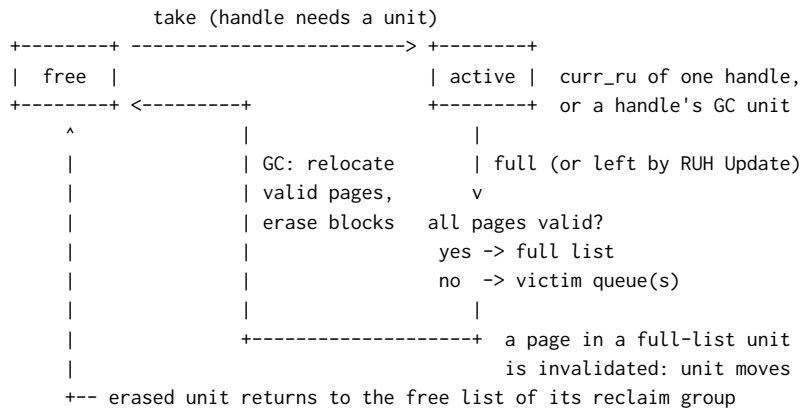
## Reclaim Unit Handle Update

I/O Management Send with operation 1 (RUH Update) lists placement identifiers. The poller checks every identifier first and fails the command with Invalid Field if one is invalid, so nothing changes. On a bbssd namespace `ssd_fdp_update_ruhs()` then, for each handle:

- skips a handle that has no current unit (device full); the command still succeeds;
- keeps the current unit if nothing was written to it yet;
- otherwise takes a fresh unit (running one GC pass if none is free) and retires the old one as if it had filled, or fails the command with Capacity Exceeded if no unit can be freed;
- in both cases records a host event "Reclaim Unit Not Fully Written" when the unit still had room (RUAMW not zero) and that event is enabled on the handle, so an update of a handle whose unit is still empty records one too.

I/O Management Receive with operation 1 (RUH Status) returns one descriptor per placement handle and reclaim group: PID, RUHID, EARUTR (always 0) and the RUAMW of the handle's current unit, or 0 while the handle has none because its last unit filled with no free unit to follow it.

## 4.6.5 Garbage collection



### When it runs

- Background: after every request the FTL thread handles, if a reclaim group's free units are at or below  $(1 - \text{gc\_thres\_pcent} / 100) * \text{units}$ , it runs one pass. A background pass puts its victim back, whatever the policy, unless the victim is empty or at least 1/8 of its pages are invalid.
- Foreground: before a placed write and before each of its pages, while free units are at or below  $(1 - \text{gc\_thres\_pcent\_high} / 100) * \text{units}$  (at least one unless `gc_thres_pcent_high` is 100), it runs

passes until the pressure clears or no victim is left. The host write waits for them. Foreground passes, and the pass RUH Update may run, always collect their victim.

### Victim selection

gc\_strategy selects the policy. The BlackBox gc\_policy property does not apply under FDP and only greedy is accepted with it.

| gc_strategy | Policy                                                                                                                                                                                                                                                                                                                                 |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 (default) | greedy: the unit with the fewest valid pages                                                                                                                                                                                                                                                                                           |
| 1           | cost-benefit: an empty unit first, otherwise the largest $(1 - u) * \text{age} / u$ , with $u$ the valid pages over the pages written (over the unit's pages until a page is invalidated) and age the time since a page in the unit was last invalidated, computed at selection time; a unit never invalidated counts as maximally old |
| 2           | random among the victims                                                                                                                                                                                                                                                                                                               |
| 4           | per handle: the unit with the fewest valid pages among the per-handle queues of Persistently Isolated handles, falling back to greedy                                                                                                                                                                                                  |

### Where relocated data goes

The handle type decides the destination of the valid pages, which is what "isolation" means here:

```
victim from a Persistently Isolated handle h
-> h's own GC unit (gc_ru, taken from the free list on demand)
    data of h stays apart from every other handle, before and after GC
```

```
victim from an Initially Isolated handle
-> the current unit of the last handle (RUH nruh-1)
    data may be mixed with other relocated data after GC
```

With the default `fdp.isolation_mode=0` every handle is Persistently Isolated. With it set, only the last handle is Initially Isolated, so its relocated data goes back into its own current unit.

A pass copies every valid page of the victim, invalidating each old copy as it goes, and only then erases the planes of each LUN as one multi-plane erase when `enable_gc_delay` is on (the default). It adds the copied bytes to MBMW and the erased bytes to MBE, records a controller event for the handle, and returns the unit to the free list. If the destination runs out of space part way, the pass stops before erasing anything: the pages already moved live at their new location, the rest stay in the victim, and the victim goes back on the queue to be finished by a later pass.

Both destinations can be left without a unit: one that fills with no free unit to follow it is dropped. The next pass with a page to move takes a free unit for it from the victim's reclaim group, whatever state the handle's own current unit is in. A victim with no valid page left needs no destination, so it is collected even when no unit is free; that is how a device that ran out gets a unit back once the host deallocates or overwrites data.

### Deallocate

Dataset Management deallocate on an FDP namespace unmaps the given ranges; the pages become invalid and GC reclaims their units later. Write Zeroes with the Deallocate bit does the same. `fdp_trim_erase_all` (test only) instead resets every unit, handle and mapping on any Dataset Management deallocate, ignoring the ranges; Write Zeroes with Deallocate still unmaps only its range.

### 4.6.6 Log pages and features

All four FDP log pages take endurance group 1 in the Log Specific Identifier, and need a subsystem (without one they fail with Invalid Log Page); 21h, 22h and 23h fail with FDP Disabled when FDP is off. The supported log pages list (00h) shows them only while FDP is on.

| Log                           | Content                                                                                                                                                                          | Code                              |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 20h FDP Configurations        | header with NUMFDPC 0 (one configuration), then one descriptor: FDPA (valid, RGIF), NRG, NRUH, MAXPIDS, NNSS 256, RUNS, and NRUH handle descriptors with the RUHT of each handle | <code>nvme_fdp_confs()</code>     |
| 21h Reclaim Unit Handle Usage | header with NRUH, then one descriptor per handle with its RUHA (1 = host-specified for every handle the namespace was given)                                                     | <code>nvme_fdp_ruh_usage()</code> |
| 22h FDP Statistics            | HBMW, MBMW and MBE as 128-bit counters; FEMU fills the low 64 bits                                                                                                               | <code>nvme_fdp_stats()</code>     |
| 23h FDP Events                | number of events, then the events of the ring selected by LSP bit 0 (1 = host events, 0 = controller events)                                                                     | <code>nvme_fdp_events()</code>    |

#### The 21h layout

The log is an 8-byte header followed by one 8-byte Reclaim Unit Handle Usage Descriptor per handle, as the specification defines it. Each descriptor holds the handle's RUHA in byte 0, and bytes 1 to 7 are reserved. `NvmeRuhDescr` in `nvme.h` has that layout, and a `QEMU_BUILD_BUG_ON` keeps it at 8 bytes. The log is  $8 + 8 * \text{NRUH}$  bytes:

| offset      | content                            |
|-------------|------------------------------------|
| 0           | NRUH (2 bytes), reserved (6 bytes) |
| 8           | handle 0: RUHA, reserved           |
| 16          | handle 1: RUHA, reserved           |
| $8 + 8 * i$ | handle i: RUHA, reserved           |

The log carries no per-handle byte counters. The endurance-group totals are in 22h.

#### Statistics

| Counter                                  | Grows by                                                                   | Where                                                           |
|------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------|
| HBMW (host bytes with metadata written)  | bytes a Write, Copy or Write Zeroes without Deallocate actually programmed | <code>fdp_count_write()</code>                                  |
| MBMW (media bytes with metadata written) | the same host bytes, plus bytes relocated by GC                            | <code>fdp_count_write()</code> , <code>do_gc_fdp_style()</code> |
| MBE (media bytes erased)                 | bytes of every block erased by GC                                          | <code>do_gc_fdp_style()</code>                                  |

The ratio MBMW / HBMW is the write amplification of the endurance group. The counters are charged after the write, from the pages it placed, so a write that ran out of space adds only what it wrote. They are reset by a deallocate with `fdp_trim_erase_all` set.

#### Events

There are two rings of 63 events each, one for host events and one for controller events. When a ring is full the oldest event is overwritten. Each event carries the Timestamp feature's current value.



| Type                                              | Ring       | Generated when                                                                                                                                                                                      |
|---------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h Reclaim Unit Not Fully Written                 | host       | RUH Update moves a handle off a unit that still had room                                                                                                                                            |
| 3h Invalid Placement Identifier                   | host       | a placed write (Write, Copy, Write Zeroes) on a bbssd namespace has DTYPE 2 and a DSPEC that is not a valid PID; checked against the event filter of placement handle 0; other modes never raise it |
| 81h (implicit reclaim unit change)<br>1h, 2h, 80h | controller | GC frees a unit of the handle<br>accepted by the event filter, never generated                                                                                                                      |

Get and Set Features 1Eh (FDP Events) read and change the enabled event types of one placement handle, named in CDW11; an unsupported type in a Set fails with Invalid Field. Get Features 1Dh (FDP Mode) for endurance group 1 reports whether FDP is on. Set Features 1Dh fails with Invalid Field for an endurance group other than 1 or without a subsystem, and otherwise with Command Sequence Error, since the mode may only change while the endurance group has no namespaces and FEMU builds the namespace at realize.

#### 4.6.7 Threads, locks and where latency is charged

| Work                                                                               | Thread               | Notes                                          |
|------------------------------------------------------------------------------------|----------------------|------------------------------------------------|
| Parsing DTYPE and DSPEC, validating RUH Update identifiers, I/O Management Receive | poller               | data copy to the DRAM backend happens here too |
| Placement, RU rotation, GC, deallocate, RUH Update on bbssd                        | FTL thread           | sole owner of the FTL structures               |
| Event append                                                                       | poller or FTL thread | events_lock in the endurance group             |
| Log pages 20h to 23h, Features 1Dh and 1Eh                                         | admin path           | 23h reads under events_lock                    |

A placed write is charged the largest NAND program latency of its pages, after any foreground GC passes, whose erases and relocations occupy the same LUNs. GC programs and erases use the BlackBox timing with enable\_gc\_delay (see the FTL chapter (§3.4)).

#### 4.6.8 Parameters

Subsystem properties are in the Flexible Data Placement section (§A.1.2) of the property reference; the controller-side ones are under garbage collection, mapping and caches (§A.1.1).

| Property           | On          | Effect and interactions                                                             |
|--------------------|-------------|-------------------------------------------------------------------------------------|
| fdp                | femu-subsys | Turns FDP on. Refused together with ns_mgmt=on.                                     |
| fdp.nruh           | femu-subsys | Handles and placement handles. Required. Raises the units the geometry must supply. |
| fdp.nru            | femu-subsys | Units per group; the FTL uses at most one per line.                                 |
| fdp.nrg            | femu-subsys | Must be 1.                                                                          |
| fdp.runs           | femu-subsys | Unset, or the superbblock size.                                                     |
| fdp.isolation_mode | femu-subsys | Non-zero makes the last handle Initially Isolated.                                  |
| gc_strategy        | femu        | Victim policy, above. No effect without FDP.                                        |
| fdp_trim_erase_all | femu        | Whole-device reset on deallocate. No effect without FDP.                            |

| Property                                                                                         | On   | Effect and interactions                                          |
|--------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------|
| gc_thres_pcent, gc_thres_pcent_high                                                              | femu | Background and foreground watermarks, as fractions of the units. |
| BlackBox geometry (nchs, luns_per_ch, pls_per_lun, blks_per_pl, pgs_per_blk, secs_per_pg, secsz) | femu | Sets RUNS (one superblock) and the number of lines, hence units. |

Refused under FDP on a bbssd controller: buffer\_size, hot\_cold\_sep, read\_reclaim\_limit, retention\_limit\_sec, ecc\_retention\_sec, trim\_lat\_ns, a mapping other than page, and a gc\_policy other than greedy. Refused on any controller with FDP: more than one namespace, a second controller in the subsystem, meta, streams, a KV namespace. Each message is listed in the feature guide (§4.5.5).

A device with eight handles, every one Persistently Isolated except the last:

```
| -device femu-subsys,id=fdp8,fdp=on,fdp.nruh=8,fdp.isolation_mode=1 -device femu,devsz_mb=1024,femu_mode=1,subsys=fdp8,gc_strategy=1
```

## 4.6.9 Statistics and outputs

| Output                               | What it shows                                                                                                                                             |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Logs 20h to 23h                      | configuration, handle usage, endurance-group byte counters, events                                                                                        |
| Vendor log C0h                       | write amplification and page counts of the FTL; placed writes count as host and programmed pages, relocations as GC pages (log pages and counters (§B.1)) |
| SMART / Health (02h)                 | host data units and commands, as for every mode                                                                                                           |
| FEMU_FDP_DEBUG in QEMU's environment | traces of RU rotation and GC passes on stderr                                                                                                             |

## 4.6.10 Validation

| Check                                          | What it covers                                                                                                                                                                                                                                          |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| qtest cases in hw/femu/tests/qtest/femu-test.c | fdp-events, fdp-features, fdp-report-length, fdp-ruh-usage, fdp-write-zeroes, fdp-write-zeroes-placed, fdp-ruh-update, fdp-ruh-update-full, wide-lba-fdp, io-fuzz-fdp, copy-fdp, log-contents-fdp, ns-mgmt-unavailable-fdp, fdp-csd-knobs, fdp-csd-runs |
| Documentation examples                         | each tagged FDP example starts under qtest and moves one block                                                                                                                                                                                          |
| hw/femu/scripts/fdp-test-nvme-admin.sh         | in-guest nvme-cli checks against the run-blackbox-fdp.sh configuration; manual                                                                                                                                                                          |
| hw/femu/tests/unit/test-pqueue.c               | the priority queue the victim queues are built on                                                                                                                                                                                                       |

## 4.6.11 Limits

- One reclaim group, one FDP configuration, one namespace and one controller per FDP subsystem.
- An RU is exactly one line; RUNS cannot be smaller or larger than a superblock.
- Only bbssd places data. A NoSSD, ZNS or OCSSD controller in an FDP subsystem answers the log pages and features, and RUH Update resets the handle's RUAMW and records event 0h if the unit still had room; no data is placed. CSD uses the bbssd path, with the same refused knobs and RUNS. KV is refused.
- Page mapping only, no write buffer, no read reclaim or retention model under FDP.
- EARUTR is always 0; there is no active reclaim unit time limit, so event 1h is never raised.

- The 128-bit statistics carry only their low 64 bits.
- Metadata, Streams and Namespace Management do not combine with FDP.

#### 4.6.12 How to extend it

- More reclaim groups: the FTL keeps one active unit per handle; a group other than 0 needs a current unit per (handle, group) before `fdp.nrg` can be lifted in `nvme_subsys_setup_fdp()`.
- Larger reclaim units: `lines_per_ru` in `ssd_init_fdp_params()` is the hook, but `fdp_advance_ru_pointer()` retires a unit after its first line; it must walk `ru->lines[]` first.
- A new victim policy: add a value to the GC strategy enum in `bbssd/ftl.h`, a case in `select_victim_ru()`, and accept it in the geometry checks of `bbssd/ftl-geom.c`. Keep the two heap positions (pos for the group queue, `ruh_pos` for a handle queue) separate.
- A new event: generate it with `nvme_fdp_record_event()` after checking the handle's filter, and add the type to `nvme_fdp_events_supported[]` and `nvme_fdp_evf_shifts[]` in `nvme.h`.

#### 4.6.13 Source map

| File                                             | What is there                                                                                                                                                                               |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hw/femu/femu.c</code>                      | femu-subsys properties, <code>nvme_subsys_setup_fdp()</code> , <code>nvme_ns_init_fdp()</code> , <code>nvme_ns_refresh_fdp()</code> , single-controller and single-namespace checks, CTRATT |
| <code>hw/femu/nvme.h</code>                      | FDP wire structures (configuration, usage, statistics, events, RUH status), endurance group and handle state, event tables                                                                  |
| <code>hw/femu/nvme-io.c</code>                   | DTYPE and DSPEC parsing, PID helpers, event recording, I/O Management Send and Receive                                                                                                      |
| <code>hw/femu/nvme-admin.c</code>                | log pages 20h to 23h, Features 1Dh and 1Eh, supported logs, effects log                                                                                                                     |
| <code>hw/femu/bbssd/ftl-fdp.c</code>             | reclaim group, unit and handle setup, placed write, RU rotation, RUH Update, GC, deallocate                                                                                                 |
| <code>hw/femu/bbssd/bb.c</code>                  | FDP checks on the bbssd geometry and refused knobs, RUNS                                                                                                                                    |
| <code>hw/femu/bbssd/ftl.c</code>                 | FTL-thread dispatch to the FDP paths, background GC trigger                                                                                                                                 |
| <code>hw/femu/scripts/run-blackbox-fdp.sh</code> | launcher                                                                                                                                                                                    |

#### 4.6.14 Related pages

- FDP feature guide (§4.5)
- BlackBox mode (§4.4)
- Namespaces and subsystems (§4.16)
- Property reference: Flexible Data Placement (§A.1.2)

### 4.7 Zoned Namespace SSD (ZNS)

ZNS mode (`femu_mode=3`) emulates an NVMe SSD with the Zoned Namespace command set. The namespace is divided into zones. The host writes each zone sequentially at its write pointer (or with Zone Append), and resets a zone before writing it again. The device runs no garbage collection: reclaiming space is the host's job. FEMU models the zone state machine, open and active zone limits, a pool of zone write caches and the NAND time of each read, program and erase.

Use it to develop and measure zoned software: zoned file systems (btrfs, f2fs), zonefs, RocksDB or

other log-structured stores, and zone-aware schedulers. For a conventional SSD with a device FTL, use BlackBox (§4.4).

### 4.7.1 Requirements

- Host and guest: see the mode table (§4.1.4) and requirements.md (§2.1.2). The guest kernel needs zoned block device support (CONFIG\_BLK\_DEV\_ZONED=y) and must be new enough to drive NVMe ZNS. The Linux 6.8 image from make-guest-image.sh works.
- The host driver must set a 4 KiB NVMe memory page size (CC.MPS), as Linux does. The controller refuses to become ready with another page size.
- Guest tools: nvme-cli with the zns commands, blkzone from util-linux, and fio for zoned workloads (--zonemode=zbd).

### 4.7.2 Launch

From build-femu/:

```
| ./run-zns.sh
```

The FEMU device in that script is:

```
| -device femu,devsz_mb=4096,namespaces=1,zns_num_ch=8,zns_num_lun=4,zns_num_plane=2,zns_num_blk=32,zns_flash_type=4,femu_mode=3
```

That is a 4 GiB namespace with 16 zones of 256 MiB and 512-byte logical blocks. The variables at the top of run-zns.sh set the geometry.

### 4.7.3 Configuration

ZNS has its own geometry and timing properties, all prefixed zns\_. The BlackBox geometry and timing properties (nchs, pg\_rd\_lat and so on) do not apply. Every property is in the ZNS section of the property reference (§A.1.1).

#### Zone size and zone count

There is no zone size property. The zone size follows from the namespace size and the geometry:

```
| pages per block = namespace size / 16 KiB / (zns_num_ch * zns_num_lun * zns_num_blk)
| zone width      = zns_chnls_per_zone, or zns_num_ch when it is 0
| zone size       = zone width * zns_num_lun * zns_num_plane * pages per block * 16 KiB
| zone count      = namespace size / zone size
|                  = zns_num_ch * zns_num_blk / (zone width * zns_num_plane)
```

So, when the divisions are exact, the zone count does not depend on the namespace size, and the zone size grows with it. Pages per block are rounded down, so an uneven size gives a smaller zone and a leftover. Some configurations:

| Settings                             | Namespace | Zones | Zone size |
|--------------------------------------|-----------|-------|-----------|
| no zns_ properties, devsz_mb=1024    | 1 GiB     | 16    | 64 MiB    |
| run-zns.sh                           | 4 GiB     | 16    | 256 MiB   |
| run-zns.sh with zns_num_blk=128      | 4 GiB     | 64    | 64 MiB    |
| run-zns.sh with zns_chnls_per_zone=2 | 4 GiB     | 64    | 64 MiB    |

To get more, smaller zones, raise `zns_num_blk` or narrow the zones with `zns_chnls_per_zone`, which must divide `zns_num_ch`:

```
-device femu,devsz_mb=4096,femu_mode=3,zns_num_ch=8,zns_num_lun=4,zns_num_plane=2,zns_num_blk=128
-device femu,devsz_mb=4096,femu_mode=3,zns_num_ch=8,zns_num_lun=4,zns_num_plane=2,zns_num_blk=32,zns_chnls_per_zone=2
```

A narrower zone spans fewer channels, so a single zone has less internal parallelism.

Linux uses a ZNS namespace only when its zone size is a power of two. Keep `devsz_mb` (or each ZNS namespace's size), `zns_num_ch`, `zns_num_lun`, `zns_num_plane` and `zns_num_blk` powers of two, or check the result with the formula above.

`zns_zone_cap` sets a zone capacity below the zone size, in bytes. The host can write only the first `zns_zone_cap` bytes of each zone.

### Logical block size

The namespace starts with 512-byte blocks. `lba_index` picks another of the `nlbaf` formats (512 bytes doubling with each index), so `lba_index=3` gives 4 KiB blocks. ZNS accepts block sizes up to 4 KiB.

```
-device femu,devsz_mb=4096,femu_mode=3,lba_index=3
```

Properties: LBA formats (§A.1.1).

### Open and active zone limits

`zns_max_open` and `zns_max_active` set the Maximum Open and Maximum Active Resources. Both default to 0, which means no limit. Neither may exceed the zone count, and `zns_max_open` may not exceed `zns_max_active`:

```
-device femu,devsz_mb=4096,femu_mode=3,zns_num_ch=8,zns_num_lun=4,zns_num_plane=2,zns_num_blk=128,zns_max_active=16,zns_max_open=8
```

`zns_num_wc` sets the size of the pool of zone write caches. A zone being written takes a cache from the pool; a write goes into it and costs 1 us per 4 KiB page. The cached pages are programmed when the cache fills, or when another zone needs a cache and none is free. The default is `zns_max_open`, or 3 when that is 0.

### NAND timing

`zns_flash_type` picks the cell type: 1 SLC, 2 MLC, 3 TLC, 4 QLC (the default) or 5 PLC. SLC, TLC and QLC have built-in read, program and erase times. MLC and PLC have none, so they need `zns_pg_rd_lat`, `zns_pg_wr_lat` and `zns_blk_er_lat`. Those three also override the built-in times for the other types. `zns_cmd_addr_lat`, `zns_pg_xfer_lat` and `zns_status_lat` add a shared channel bus, and `zns_pe_suspend` with `zns_tsusp_ns` lets reads suspend a program or erase. See the timing model (§6.2.6).

### Optional zone features

All of these are off by default.

- **Zone Random Write Area.** Set `zns_zrwa_size` and `zns_zrwafg_size` (in logical blocks) and `zns_zrwa_num` (zones that may hold one at once). A zone opened with the ZRWA flag accepts writes anywhere in a window above the write pointer. The write pointer advances only in whole flush-granularity units: when a write crosses the end of the window, or when the host flushes explicitly. Finishing or resetting the zone returns its ZRWA resource. With all three properties at 0 the namespace advertises no ZRWA.

```
-device femu,devsz_mb=1024,femu_mode=3,zns_zrwa_size=64,zns_zrwafg_size=8,zns_zrwa_num=2
```

- **Conventional zones.** `zns_num_conv_zones=N` makes the first N zones accept writes anywhere inside the zone. They keep no write pointer (reported as all ones) and reject zone management and Zone Append. Leave it at 0 for a Linux guest: the NVMe ZNS command set defines only the sequential-write-required zone type, so Linux rejects a conventional zone, fails the whole zone report with `EINVAL`, and the namespace reports no zones, which leaves it unusable for zoned btrfs, f2fs, zonefs or dm-zoned. To mix random-write and zoned capacity, give the controller a BlackBox namespace and a ZNS namespace (§4.15) instead.
- **Reads across zone boundaries.** By default a read that runs past the end of its zone fails with a zone boundary error. `zns_cross_zone_read=on` allows it and advertises OZCS bit 0, which tells the host it may issue one. Every zone the read spans must still be in a readable state.
- **Zone Append size limit.** `zns_zasl_bs` (default 128 KiB) caps one Zone Append, and must be a power-of-two multiple of 4 KiB, because Identify reports the limit (ZASL) as a power-of-two count of 4 KiB pages; other values are refused, not rounded down. 0 follows mdt's. The host reads ZASL to size its appends, and a larger append fails with Invalid Field in Command.
- **Zone descriptor extensions.** `zns_zd_ext_size` in bytes, a multiple of 64.
- **Write failures.** `err_write_fail_ppm` fails a fixed share of writes. The zone of a failed write becomes read only and is added to the Changed Zone List log page.

#### 4.7.4 Use it from the guest

Check that Linux sees a zoned device:

```
sudo nvme list
cat /sys/block/nvme0n1/queue/zoned      # host-managed
cat /sys/block/nvme0n1/queue/nr_zones
cat /sys/block/nvme0n1/queue/chunk_sectors # zone size in 512-byte sectors
sudo nvme zns id-ns /dev/nvme0n1
```

The model is FEMU ZMS-SSD Controller [by Misao] and the serial number starts with vZNSSD.

List the zones, with their state and write pointer:

```
sudo nvme zns report-zones /dev/nvme0n1 -d 4
sudo blkzone report /dev/nvme0n1
```

Manage one zone by its start LBA (-s), or every zone (-a):

```
sudo nvme zns open-zone /dev/nvme0n1 -s 0
sudo nvme zns finish-zone /dev/nvme0n1 -s 0
sudo nvme zns reset-zone /dev/nvme0n1 -s 0
sudo nvme zns reset-zone /dev/nvme0n1 -a
```

Append 4 KiB to zone 0 and let the device pick the LBA:

```
head -c 4096 /dev/urandom > data.bin
sudo nvme zns zone-append /dev/nvme0n1 -s 0 -z 4096 -d data.bin
```

Write sequentially with fio, read the written zones back at random, then reset the zones before the next run:

```
sudo fio --name=zns --filename=/dev/nvme0n1 --direct=1 --ioengine=psync \
--zonemode=zbd --rw=write --bs=128k --size=1G
sudo fio --name=zr --filename=/dev/nvme0n1 --direct=1 --ioengine=psync \
--zonemode=zbd --rw=randread --bs=4k --size=1G --runtime=30 --time_based
sudo blkzone reset /dev/nvme0n1
```

With ZRWA configured, open a zone with the ZRWA flag, write inside its window with passthrough write commands (the block layer always writes at the write pointer), then commit the window up to an LBA. With `zns_zrwafg_size=8`, the flush must end on a multiple of 8 blocks:

```
sudo nvme zns open-zone /dev/nvme0n1 -s 0 --zrwaa
sudo nvme zns zrwa-flush-zone /dev/nvme0n1 -l 63
```

Read the Changed Zone List log page (BFh). It needs an explicit namespace:

```
sudo nvme get-log /dev/nvme0 --log-id=0xbf --log-len=4096 --namespace-id=1
```

## Changed Zone List

The page is per namespace, which is why the command above names one: `nvme-cli` otherwise sends the broadcast identifier. It holds an 8-byte count followed by up to 511 zone start LBAs, and lists only zone changes the host did not cause. As the ZNS specification requires, it leaves out changes that follow a Zone Management Send command, writes that open or fill a zone, and the controller closing a zone to free a resource. Reading it without `--rae` clears both the list and the event behind it.

With `err_write_fail_ppm` unset nothing adds to the list, and it stays empty. With it set, one write in every 1,000,000 / `err_write_fail_ppm` fails and makes its zone read only, the way a controller does when it can no longer program the zone. The failing write completes with Write Fault, later writes to that zone are refused as read only, the zone is added to this list. A host that enabled Zone Descriptor Changed notices (bit 27 of Asynchronous Event Configuration, Set Features 0Bh) and has an Asynchronous Event Request outstanding then gets the notice; without that bit the zone is still listed, but no event is raised. The failures come at a fixed count, so a run repeats exactly. `hw/femu/scripts/zone-aen-probe.c` checks this path and sets bit 27 itself, since Linux leaves it clear. Add `,err_write_fail_ppm=10000` to the device in `run-zns.sh`, copy the probe into the guest, and run it on a fresh device; it prints `ZAEN PASS`:

```
gcc -O2 -o zone-aen-probe zone-aen-probe.c # inside the guest
sudo ./zone-aen-probe /dev/nvme0 /dev/nvme0n1
```

Do not run `mkfs.ext4` on the namespace: it needs random writes. Use a file system with zoned support, `zonedfs`, or zone-aware applications.

## 4.7.5 Limits and refusals

| Message                                                                                                                                           | Cause and fix                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <code>max_open_zones</code> value 20 exceeds the number of zones 16                                                                               | <code>zns_max_open</code> above the zone count.               |
| <code>max_active_zones</code> value N exceeds the number of zones M                                                                               | <code>zns_max_active</code> above the zone count.             |
| <code>zns_num_wc</code> value N exceeds the number of zones M, <code>zns_num_wc</code> (N) exceeds the M zones this geometry can have at most     | <code>zns_num_wc</code> above the zone count.                 |
| <code>max_open_zones</code> value 10 exceeds <code>max_active_zones</code> 8                                                                      | <code>zns_max_open</code> above <code>zns_max_active</code> . |
| <code>zns_chnls_per_zone</code> 3 must divide <code>zns_num_ch</code> 2                                                                           | The zone width must divide the channel count.                 |
| <code>zns_flash_type</code> 2 has no built-in timing; set <code>zns_pg_rd_lat</code> , <code>zns_pg_wr_lat</code> and <code>zns_blk_er_lat</code> | MLC or PLC without the three latencies.                       |
| <code>zns_flash_type</code> must be in [1, 5]                                                                                                     | A cell type outside SLC to PLC.                               |
| zoned namespaces need a logical block of 4 KiB or less; <code>lba_index</code> selects 8192 bytes                                                 | <code>lba_index</code> picks a block larger than 4 KiB.       |

| Message                                                                                                              | Cause and fix                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| zns geometry gives N pages per block; must be in [1, 65536]                                                          | The namespace is too small or too large for $\text{zns\_num\_ch} * \text{zns\_num\_lun} * \text{zns\_num\_blk}$ .                                                                                                                                                                                                                 |
| FEMU zns: N zones exceed M blocks per plane ...<br>the device is too small to hold a single zone of N logical blocks | Raise $\text{zns\_num\_blk}$ or the namespace size.<br>The geometry gives less than one zone: $\text{zns\_num\_ch} * \text{zns\_num\_blk}$ is below the zone width times $\text{zns\_num\_plane}$ . Raise $\text{zns\_num\_blk}$ or $\text{zns\_num\_ch}$ , or lower $\text{zns\_num\_plane}$ or $\text{zns\_chnls\_per\_zone}$ . |
| $\text{zns\_zrwafg\_size}$ must be between 1 and 65535 when ZRWA is enabled, got 0                                   | ZRWA needs all three $\text{zns\_zrwa}*$ properties. Other messages check that the window is a multiple of the flush granularity and that the zone capacity is too.                                                                                                                                                               |
| $\text{zns\_zasl\_bs}$ N must be a power-of-two multiple of the 4096 byte controller page size                       | Fix $\text{zns\_zasl\_bs}$ .                                                                                                                                                                                                                                                                                                      |

On a zoned namespace, Dataset Management (deallocate), Write Zeroes, Write Uncorrectable and Copy fail with Invalid Opcode: they would change blocks without going through the zone state machine. Reset the zone instead.

#### 4.7.6 Verify

1. `cat /sys/block/nvme0n1/queue/zoned` prints `host-managed`.
2. `nr_zones` and `chunk_sectors` match the zone count and size from the formula above.
3. `sudo nvme zns report-zones /dev/nvme0n1 -d 1` shows zone 0 empty; after the fio run the first zones are full and their write pointers moved.

#### 4.7.7 Troubleshooting

- **nvme list shows the device but nr\_zones is 0, or blkzone says unable to determine zone size.** The guest kernel cannot drive ZNS. Use a kernel from the requirements table; the image from `make-guest-image.sh` works. Check `dmesg` for the reason. A message with `invalid zone size` means the zone size is not a power of two.
- **No zoned device after setting  $\text{zns\_num\_conv\_zones}$ .** Linux rejects conventional zones and fails the whole zone report. Set it back to 0.
- **The device disappears after a geometry change.** The geometry is checked at `realize`, so a bad one stops QEMU with a message from the table above. Read the console log (`run-zns.sh` writes `build-femu/log`).
- **I want 4 KiB blocks.** Set `lba_index=3`. Older guides patched `zns.c`; that is no longer needed.
- **I want a different zone size or more zones.** Change the geometry, not the code. See Zone size and zone count (§4.7.3).
- **A large ZNS device fails with failed to allocate N bytes.** The namespace lives in host DRAM. A 256 GiB device needs 256 GiB of free host memory.
- **fio fails with an I/O error on the first write.** The zone is full or not at its start. Reset the zones (`blkzone reset /dev/nvme0n1`) and run with `--zonemode=zbd`.

Related issues: #57, #76, #112, #115, #126, #144, #161, #177, #182, #193.

#### 4.7.8 Related pages

- Timing model: ZNS write cache (§6.2.6)



- Architecture: other FTLs (§3.1.5)
- Multiple namespaces (§4.15), to mix zoned and conventional namespaces
- Device property reference: ZNS (§A.1.1)

## 4.8 ZNS: the zoned namespace model

This chapter describes how FEMU implements the NVMe Zoned Namespace (ZNS) command set: the zone state machine, how zones are laid out on the emulated NAND, where each cost is charged, and what every property changes. To run a ZNS device and use it from a guest, read the ZNS mode guide (§4.7) first; this page explains what happens inside.

### 4.8.1 Purpose

A zoned namespace splits its logical blocks into zones. The host writes each zone sequentially at the zone's write pointer, or with Zone Append, and resets a zone before it writes it again. The device keeps no page-level remapping for overwrites and runs no garbage collection; reclaiming space is the host's job. FEMU's ZNS mode gives you:

- the zone state machine of the ZNS specification, with open and active zone limits, Zone Append, Zone Random Write Area (ZRWA), conventional zones, zone descriptor extensions and the Changed Zone List log page;
- a zone-to-NAND layout that decides which channels, LUNs and planes a zone occupies, so zone width changes internal parallelism;
- a per-zone write cache and NAND read, program and erase timing, so a zoned workload sees realistic latency.

ZNS is selected with `femu_mode=3` for the whole controller, or per namespace with `znssd` in `namespace_modes` (see namespaces (§4.16.5)).

### 4.8.2 Place in the hierarchy

```

guest: blkzone, nvme zns, fio --zonemode=zbd, zoned btrfs/f2fs, zonefs
|
| NVMe queues (Read, Write, Zone Append, Zone Mgmt Send/Receive)
v
poller thread -- nvme-io.c: nvme_io_cmd() --> ns->ext_ops.io_cmd = zns_io_cmd()
|
|          zns.c: zone checks, write pointer, state machine,
|                  data copy to the DRAM backend (backend_rw)
| to_ftl ring
v
FTL thread    -- femu.c: femu_ftl_process_req() --> zftl.c: zns_ftl_process_req()
|
|          write cache, L2P table, zone erase, NAND timing
v
NAND media    -- nand/: nand_media_op(), plane-gated timelines,
|                  optional channel bus and program/erase suspend
|
| latency added to req->expire_time; to_poller ring
v
poller completes the command when the clock passes expire_time

```

The data itself always lives in the controller's DRAM backend. The NAND model holds no data: it tracks where each 4 KiB logical page would sit and how long each operation takes. The general request path is described in architecture (§3.1.3).

### 4.8.3 Data structures

| Structure                  | File      | Fields that matter                                                                                                                                                                                                                                                                                                 |
|----------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NvmeZone                   | zns/zns.h | d (the zone descriptor the host reads: type zt, state zs, attributes za, zcap, zslba, wp), w_ptr (the write pointer used to place writes), list entry                                                                                                                                                              |
| NvmeNamespace (zoned part) | nvme.h    | zone_array, num_zones, zone_size and zone_capacity in logical blocks, zone_size_log2, the lists exp_open_zones, imp_open_zones, closed_zones, full_zones, the counters nr_open_zones and nr_active_zones with their limits, the ZRWA fields zrwa_size, zrwa_size, zrwa_num, zrwa_avail, zd_extensions, id_ns_zoned |
| struct zns_ssd             | zns/zns.h | NAND geometry (num_ch, num_lun, num_plane, num_blk, num_page), the ch -> fc -> plane -> blk tree with per-block page_wp, maptbl (L2P per 4 KiB page), cache (write caches), program_unit, stripe_unit, chnls_per_zone, zone_wp_slot[], media, zone_lock, the changed zone list                                     |
| struct zns_write_cache     | zns/zns.h | sblk (the zone it serves), used, cap, lpns[]                                                                                                                                                                                                                                                                       |

There are two write pointers per zone. w\_ptr moves when a write is accepted, before its data is copied, so that two writes on different pollers, or two appends, never get the same LBA. d.wp is the pointer the host sees in a zone report; it moves when the write is finalized after the copy. Without a ZRWA they agree once every accepted write has finished.

### 4.8.4 Zone geometry

#### Size, capacity and count

There is no zone size property. zns\_init\_zone\_cap() derives it from the namespace size and the zns\_geometry, with a fixed 16 KiB NAND page (ZNS\_PAGE\_SIZE):

```

pages per block = namespace bytes / 16 KiB / (zns_num_ch * zns_num_lun * zns_num_blk)
zone width      = zns_chnls_per_zone, or zns_num_ch when it is 0
zone size       = zone width * zns_num_lun * zns_num_plane * pages per block * 16 KiB
zone count      = namespace bytes / zone size (rounded down)
zone capacity    = zns_zone_cap, or the zone size when it is 0

```

The namespace exposes zone\_count \* zone\_size blocks (NSZE). Bytes past the last whole zone are not exposed. Each zone's capacity is the same; a zone accepts writes only below ZSLBA + ZCAP and reads up to ZSLBA + ZSZE. The mode guide (§4.7.3) has a table of zone sizes for common settings.

#### Zone layout over NAND

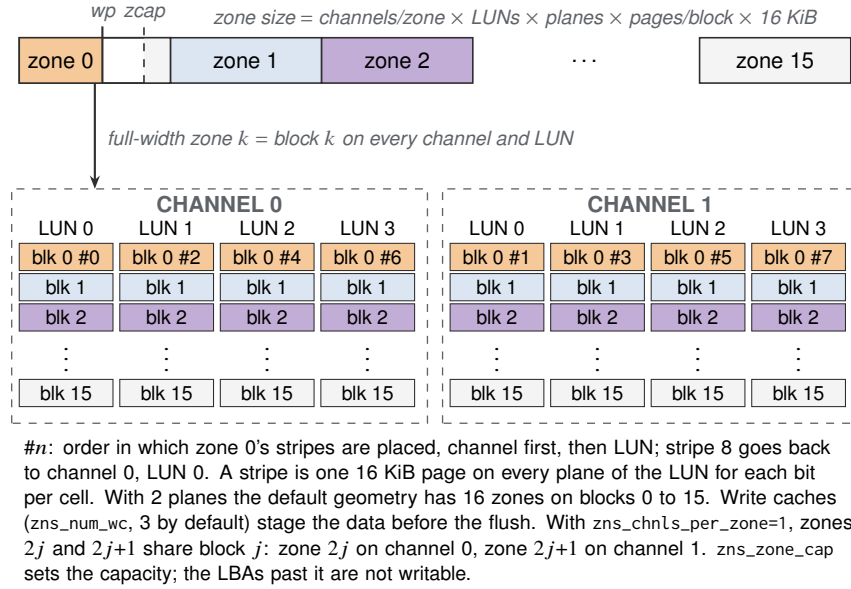
Zone i occupies one block index in a set of channels. With full-width zones (zns\_chnls\_per\_zone unset) the zone owns block i on every channel, LUN and plane:

full width: zns\_num\_ch=4, zns\_num\_lun=2, zns\_num\_plane=2

|     | ch0  |      | ch1  |      | ch2  |      | ch3  |      |                     |
|-----|------|------|------|------|------|------|------|------|---------------------|
|     | LUN0 | LUN1 | LUN0 | LUN1 | LUN0 | LUN1 | LUN0 | LUN1 |                     |
| pl0 | b0   | b0   | b0   | b0   | b0   | b0   | b0   | b0   | <- zone 0 = block 0 |
| pl1 | b0   | b0   | b0   | b0   | b0   | b0   | b0   | b0   | everywhere          |
| pl0 | b1   | b1   | b1   | b1   | b1   | b1   | b1   | b1   | <- zone 1 = block 1 |
| pl1 | b1   | b1   | b1   | b1   | b1   | b1   | b1   | b1   |                     |
| ... |      |      |      |      |      |      |      |      |                     |

A narrower zone spans zns\_chnls\_per\_zone channels. The channels form groups = zns\_num\_ch / zns\_

## HOST VIEW: LBAS OF ONE ZONED NAMESPACE



**Figure 16:** How zones sit on NAND with the default geometry: zone  $k$  is block  $k$  on every channel and LUN, and its pages are placed channel first, then LUN.

chnls\_per\_zone groups, and the zones that share one block index sit in different groups:

zns\_num\_ch=4, zns\_chnls\_per\_zone=2 -> 2 channel groups

|         | group 0 (ch0, ch1) | group 1 (ch2, ch3) |
|---------|--------------------|--------------------|
| block 0 | zone 0             | zone 1             |
| block 1 | zone 2             | zone 3             |
| block 2 | zone 4             | zone 5             |

zone  $i$ : channels ( $i \% \text{groups}$ ) \* width .. + width - 1, block  $i / \text{groups}$

So narrowing the zones doubles the zone count for each halving of the width, and each zone has fewer channels to program in parallel. The geometry check refuses a configuration with more zones than  $\text{zns\_num\_blk} * \text{groups}$ , since every zone needs its own block.

## Placement inside a zone

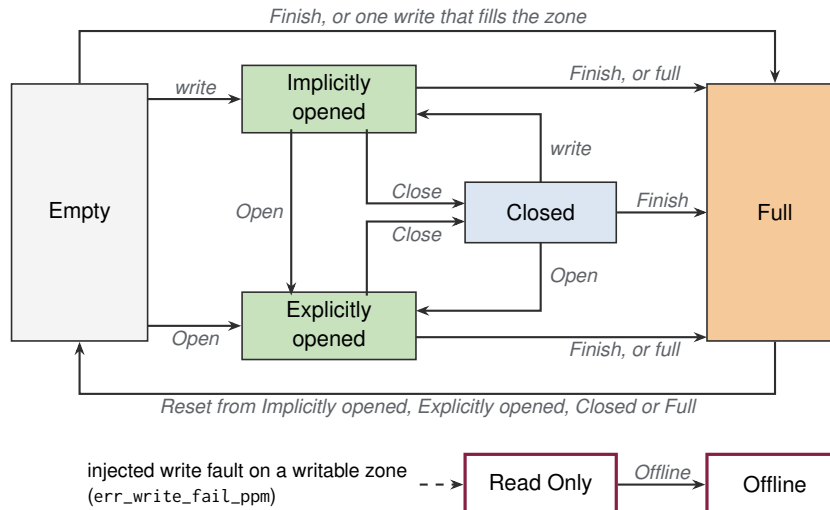
`get_new_page()` in `zftl.c` picks the next location from the zone's slot counter `zone_wp_slot[zone]`. A slot is one (channel, LUN) pair. Slots go channel first, then LUN:

```
slot s (full width):  ch = s % zns_num_ch
                      lun = (s / zns_num_ch) % zns_num_lun
slot s (narrow):      ch = (zone % groups) * width + s % width
                      lun = (s / width) % zns_num_lun
```

one slot programs, on each plane of that LUN, `zns_flash_type` pages of 16 KiB (4 logical 4 KiB pages each):

```
program unit = 16 KiB * zns_flash_type * zns_num_plane
stripe unit  = program unit * zns_num_ch * zns_num_lun
```

With `run-zns.sh` (8 channels, 4 LUNs, 2 planes, QLC) the program unit is 128 KiB and the stripe unit 4 MiB. Each block keeps its own page write pointer `page_wp`; a zone reset sets it, and the zone's slot counter, back to



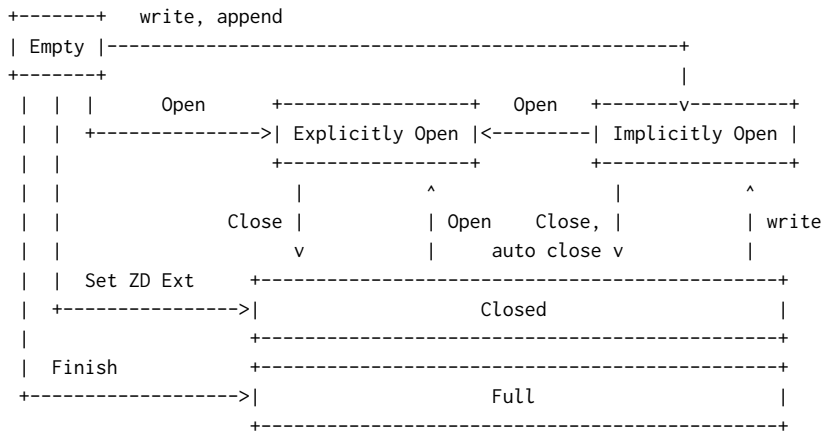
Open limit (zns\_max\_open): opening one more zone first closes the oldest implicitly opened zone, and fails if every open zone is explicit. Active limit (zns\_max\_active) counts open and closed zones; at the limit the request fails. Not drawn: Set Zone Descriptor Extension (Empty to Closed) and ZRWA flush (Closed to Implicitly opened). Conventional zones have no state.

**Figure 17:** The zone state machine in FEMU's ZNS mode: writes open zones implicitly, Open/Close/Finish/Reset move them explicitly, a zone becomes Full when its write pointer reaches the capacity, and only an injected write fault makes it Read Only.

zero.

#### 4.8.5 Zone state machine

The states and transitions below are the ones `zns.c` implements. Every transition the host drives goes through `zns_assign_zone_state()`, which also moves the zone between the per-state lists.



also to Full: a write that reaches ZCAP (from Empty, either Open state or Closed), or Finish from either Open state or Closed  
Reset: Explicitly Open, Implicitly Open, Closed, Full --> Empty  
controller: injected write fault --> Read Only --(Offline action)--> Offline

| From                           | Event                                                 | To                       | Code                                                                   |
|--------------------------------|-------------------------------------------------------|--------------------------|------------------------------------------------------------------------|
| Empty                          | write or Zone Append                                  | Implicitly Open          | <code>zns_auto_open_zone()</code> , <code>zns_advance_zone_wp()</code> |
| Empty, Implicitly Open, Closed | Open Zone                                             | Explicitly Open          | <code>zns_open_zone()</code>                                           |
| Empty                          | Set Zone Descriptor Extension                         | Closed (extension valid) | <code>zns_set_zd_ext()</code>                                          |
| Implicitly or Explicitly Open  | Close Zone                                            | Closed                   | <code>zns_close_zone()</code>                                          |
| Implicitly Open                | another zone needs an open resource at the open limit | Closed                   | <code>zns_auto_transition_zone()</code>                                |
| Closed                         | write                                                 | Implicitly Open          | <code>zns_advance_zone_wp()</code>                                     |
| Empty, Open, Closed            | Finish Zone                                           | Full                     | <code>zns_finish_zone()</code>                                         |
| Empty, Open, Closed            | write reaches the zone capacity                       | Full                     | <code>zns_finalize_zoned_write()</code>                                |
| Open, Closed, Full             | Reset Zone                                            | Empty                    | <code>zns_reset_zone()</code>                                          |
| any but Read Only              | injected write fault on the zone                      | Read Only                | <code>zns_nvme_rw()</code>                                             |
| Read Only                      | Offline Zone                                          | Offline                  | <code>zns_offline_zone()</code>                                        |

Offline is final, and Read Only leaves only through Offline. Open, Close, Finish and Reset refuse both with Invalid Zone State Transition. Reads work in every state except Offline. Writes are refused in Full (Zone Is Full), Read Only (Zone Is Read Only) and Offline (Zone Is Offline).

Reset and Offline also deallocate the zone's slice of the DRAM backend (`zns_deallocate_zone()`), so a read of a reset zone returns zeros, as the specification requires for Empty and Offline zones.

Zone state lives only in QEMU memory. It survives a controller reset and a guest reboot, and is lost when QEMU exits. `zns_ns_shutdown()`, which moves open and closed zones to Closed, or to Empty when they hold no data and no descriptor extension, runs only when the device is removed.

### Open and active resources

`zns_max_open` and `zns_max_active` set the limits; 0 means no limit, and then FEMU does not count. An active zone is one that is open or closed; an open zone is implicitly or explicitly open.

```

write to an Empty zone, or Open of an Empty zone:
  1. need an active resource    -> else Too Many Active Zones, nothing changes
  2. at the open limit? close the oldest Implicitly Open zone
  3. need an open resource     -> else Too Many Open Zones
write to a Closed zone: steps 2 and 3 only

```

Only implicitly open zones are closed to make room, oldest first. If every open zone is explicitly open, the command fails with Too Many Open Zones. Open with Select All opens every closed zone or none: it fails up front if the closed zones would not fit under the open limit.

### 4.8.6 Commands

`zns_io_cmd()` handles Read, Write, Zone Append, Zone Management Send and Zone Management Receive. Other I/O opcodes go through the common path in `nvme-io.c`, with these zoned rules:

- Compare is checked like a Read (`zns_check_compare()`).
- Dataset Management, Write Zeroes, Copy and Write Uncorrectable fail with Invalid Opcode on a zoned namespace: they would change blocks without going through the state machine. The command effects

log for CSI 2 leaves them out, and Identify Namespace reports no Copy limits (MSSRL, MCL and MSRC are zero) even when oncs turns Copy on.

- Verify runs the common path without zone state or boundary checks.
- Flush and the I/O Management commands behave as on any namespace.

## Read and Write

A Write must start exactly at `w_ptr` and end at or below the zone capacity, or it fails with Zone Invalid Write or Zone Boundary Error. A Read must stay inside its zone unless cross-zone reads are on; see below (§4.8.7).

## Zone Append

Zone Append names the zone by its ZSLBA. Under `zone_lock` the poller takes `w_ptr` as the append's LBA and advances the pointer in one step, so appends on different queues never overlap; the data is copied afterwards, outside the lock. The completion carries the LBA where the data landed. An append fails with Invalid Field when it does not name a zone start or exceeds the Zone Append Size Limit, and with Zone Boundary Error when it would cross the zone capacity.

The limit comes from `zns_zasl_bs` in bytes. At controller enable, `zns_start_ctrl()` converts it to ZASL, a power of two in units of the 4 KiB controller page, which Identify Controller for CSI 2 reports. With `zns_zasl_bs=0` Identify reports ZASL 0, and appends are held to MDTs instead. Realize refuses a value that is not a power-of-two multiple of 4 KiB, rather than round it down.

## Zone Management Send

| Action                              | Select All acts on          | Notes                                                                                          |
|-------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------|
| Close (01h)                         | open zones                  |                                                                                                |
| Finish (02h)                        | open and closed zones       | moves <code>w_ptr</code> and <code>wp</code> to the capacity; does not program the write cache |
| Open (03h)                          | closed zones, all or none   | bit 9 of CDW13 also allocates a ZRWA (single zone only)                                        |
| Reset (04h)                         | open, closed and full zones | descriptor changes on the poller; the erase runs on the FTL thread                             |
| Offline (05h)                       | read-only zones             |                                                                                                |
| Set Zone Descriptor Extension (10h) | not allowed                 | data is staged first and copied only if the zone is Empty                                      |
| ZRWA Flush (11h)                    | not allowed                 | see ZRWA (§4.8.7)                                                                              |

A single-zone action must name the zone's start LBA, except ZRWA Flush, which names a boundary inside the zone. Every action runs under the namespace's `zone_lock`.

## Zone Management Receive

Report Zones (00h) and Extended Report Zones (01h, only with `zns_zd_ext_size` set) return a 64-byte header with the matching zone count, then one 64-byte descriptor per zone, followed in the extended report by that zone's extension. The filter field takes the states 0 to 7, plus 9 for zones with the Finished by Controller, Finish Recommended or Reset Recommended attribute; FEMU never sets those attributes, so filter 9 always reports none. The Partial Report bit limits the count to the descriptors that fit. The write pointer field is all ones for Full, Read Only and Offline zones and for conventional zones. The report is built under `zone_lock`, so it is a consistent snapshot.

## 4.8.7 Optional zone features

### Zone Random Write Area (ZRWA)

A ZRWA lets the host write anywhere in a window above the write pointer and commit the window later. `zns_zrwa_size` (ZRWA) and `zns_zrwaflush_size` (ZRWAFLUSH, the flush granularity) are in logical blocks; `zns_zrwa_num` is how many zones may hold a ZRWA at once (NUMZRWA).

```

w_ptr          w_ptr + ZRWAS          w_ptr + 2*ZRWAS
|<----- ZRWA window ----->|<-- implicit flush area -->|
zone: [...committed...|.....|.....|.....]
      ^ a write may start anywhere in here -----^
a write that ends past the window moves w_ptr up by whole ZRWAFG units
ZRWA Flush to LBA x (inside the window, x - w_ptr + 1 a multiple of
ZRWAFLUSH) sets w_ptr = x + 1
```

- A ZRWA is allocated by Open Zone, or by Set Zone Descriptor Extension, with CDW13 bit 9, on an Empty zone while `zrwa_avail` is non-zero; otherwise the command fails with Invalid Zone Operation or No ZRWA Resources.
- Writes must start in `[w_ptr, w_ptr + 2 * ZRWAS)` and stay below the zone capacity. Zone Append to a ZRWA zone fails with Invalid Zone Operation.
- When `w_ptr` reaches the capacity the zone becomes Full and the resource goes back. Finish, Reset and Offline also return it.
- Identify Namespace for CSI 2 sets OZCS bit 1 (ZRWASUP), NUMZRWA (zero-based), ZRWAS, ZRWAFG and ZRWACAP bit 0 (explicit flush).

Realize refuses a ZRWA configuration where ZRWAS or ZRWAFG exceed 65535, ZRWAS is not a multiple of ZRWAFG, the zone capacity is not a multiple of ZRWAFG, `zns_zrwa_num` is 0, or the granularity or count is set without a window.

### Conventional zones

`zns_num_conv_zones=N` makes the first N zones conventional (zone type 1). A conventional zone accepts writes anywhere below `ZSLBA + ZCAP` and keeps no write pointer. Its state changes only through an injected write fault, which can take it to Read Only like any zone. It rejects Zone Append (Invalid Field), single-zone Open, Close, Finish, Reset, Offline and Set Zone Descriptor Extension (Invalid Zone State Transition), and ZRWA Flush (Invalid Zone Operation). Zone reports describe it as usual. N above the zone count is lowered to the zone count. The ZNS specification defines only sequential-write-required zones, so Linux refuses a namespace that reports a conventional zone; the mode guide (§4.7.3) explains the consequence.

### Reads across zone boundaries

`zns_cross_zone_read=on` sets OZCS bit 0 (Read Across Zone Boundaries). A read that runs past its zone is then accepted if every zone it touches is readable (not Offline). Without it, such a read fails with Zone Boundary Error.

### Zone descriptor extensions

`zns_zd_ext_size` sets a per-zone extension in bytes, a multiple of 64 and at most 255 units of 64. Identify reports it in `LBAFE.ZDES`. Setting an extension takes an Empty zone to Closed with the Zone Descriptor Extension Valid attribute, and needs an active resource.

## Changed Zone List and Zone Descriptor Changed notices

The Changed Zone List log page (BFh) lists zones whose descriptor changed for a reason the host did not cause. The ZNS specification excludes every change that follows a Zone Management Send, a write that opens or fills a zone, and the controller closing a zone to free a resource. In FEMU the only change left is the injected write fault: with `err_write_fail_ppm` set, one write in `1000000 / err_write_fail_ppm` completes with Write Fault and its zone becomes Read Only. That zone's ZSLBA is added to the list by `zns_record_changed_zone()`.

```
failed write --> zone Read Only --> changed_zones[] (dedup, up to 511)
|
AEC bit 27 set? ----> Asynchronous Event Notice,
(Set Features 0Bh)   Zone Descriptor Changed,
                    log page BFh, this NSID

Get Log Page BFh, NSID = the zoned namespace:
8-byte count, then 8-byte ZSLBAs (count FFFFh and no entries on overflow)
read without RAE: list and pending event cleared
```

- The list is per namespace. The command must name the zoned namespace's NSID: 0, FFFFFFFFh or a non-zoned namespace fail. The offset must be a multiple of 8.
- Identify Controller advertises OAES bit 27 (ZDCN) whenever a zoned namespace exists. The notice itself is sent only when the host has set bit 27 of Asynchronous Event Configuration; the list is kept either way. The mode guide (§4.7.4) notes that Linux leaves the bit clear, which is why `scripts/zone-aen-probe.c` sets it itself.
- The supported log pages list (log 00h) shows BFh for CSI 2 only.

### 4.8.8 Threads, locks and where latency is charged

| Work                                                                 | Thread                         | Lock                                             |
|----------------------------------------------------------------------|--------------------------------|--------------------------------------------------|
| Zone checks, auto open, <code>w_ptr</code> advance, append placement | poller of the submission queue | <code>zns-&gt;zone_lock</code>                   |
| Data copy between guest memory and the DRAM backend                  | poller                         | none (the LBA range was reserved under the lock) |
| d.wp update, Full transition, injected fault                         | poller                         | <code>zone_lock</code>                           |
| Zone Management Send and Receive, reset state change                 | poller                         | <code>zone_lock</code>                           |
| Changed Zone List read                                               | admin path                     | <code>zone_lock</code>                           |
| Write cache, L2P table, block page pointers, erase, NAND timing      | the controller's FTL thread    | single owner, no lock                            |

One FTL thread serves every ZNS, bbssd and CSD namespace of a controller (`femu_ftl_thread()` in `femu.c`). It returns a latency for each request, which is added to `expire_time`; the poller posts the completion once the clock passes it. See timing model (§6.2.1).

What each command is charged, in `zftl.c`:



| Command                              | Cost                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Write, Zone Append                   | the larger of two figures: 1 us for each 4 KiB logical page put into the zone's write cache, summed over the pages added since the request's last cache flush, and the program time of any cache this write caused to be programmed: an evicted cache, or this zone's own cache when a page of this write found it full. The write that fills a cache is not charged for programming it; the next write to the zone is |
| Read                                 | the largest NAND read time over the 4 KiB pages it touches that have been programmed; pages still in a write cache, or never written, cost nothing                                                                                                                                                                                                                                                                     |
| Zone Reset                           | the largest erase time over the blocks of each reset zone; zones reset by one command share planes, so the command ends with its last erase                                                                                                                                                                                                                                                                            |
| Open, Close, Finish, Report, Offline | nothing                                                                                                                                                                                                                                                                                                                                                                                                                |

## Write cache

A write cache holds the logical page numbers of one zone's partial stripe; its capacity is one stripe unit (stripe unit / 4 KiB pages). There are `zns_num_wc` caches, or `zns_max_open` when that is 0, or 3 when both are 0.

```

write to zone z
  cache bound to z?  yes -> append LPNs
                     no  -> take an empty cache, else evict the fullest
                           cache (program it now, cost charged to this
                           write), bind it to z
  for each page:    cache already full? program it first: one program
                     unit per slot, each plane of the slot's LUN one NAND
                     program op, L2P updated; then add the page
Zone Reset of z: drop z's cache without programming it

```

Fewer caches than zones written at once makes writes evict each other's caches and pay a program each time. That is why the default follows `zns_max_open`. The timing model (§6.2.6) puts this in context with the other modes.

## NAND timing

`zns_nand_media_init()` hands the timing to the shared NAND media layer (see NAND timing (§3.5)) with:

- one latency per cell type for read, program and erase, from the `zns_flash_type` row of the built-in table in `zns/zns.h` and `nand/nand.h`, replaced by `zns_pg_rd_lat`, `zns_pg_wr_lat` and `zns_blk_er_lat` when set; MLC and PLC have no built-in row and need all three;
- the array gated per plane only (`NAND_GATE_PLANE_ONLY`): operations on different planes of one LUN overlap, operations on one plane queue;
- a shared channel bus only if `zns_cmd_addr_lat`, `zns_pg_xfer_lat` or `zns_status_lat` is non-zero;
- program and erase suspend for reads if `zns_pe_suspend` is non-zero, with `zns_tsusp_ns` charged per preempting read.

## 4.8.9 Parameters

All ZNS properties are on the femu device and are listed with types and defaults in the ZNS section of the property reference (§A.1.1). The geometry and limits are controller-wide: every zoned namespace of a controller uses the same `zns_` values with its own size.

| Group                   | Properties                                                                                                                                                                                                                                                                    | Interactions                                                                                                                                                                                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Geometry                | <code>zns_num_ch</code> , <code>zns_num_lun</code> , <code>zns_num_plane</code> ,<br><code>zns_num_blk</code> , <code>zns_chnls_per_zone</code>                                                                                                                               | Set zone size and count with the namespace size. <code>zns_num_ch</code> 1 to 128 and <code>zns_num_plane</code> 1 to 8 (PPA field widths). Pages per block must come out in 1 to 65536. <code>zns_chnls_per_zone</code> must divide <code>zns_num_ch</code> . |
| Zone shape              | <code>zns_zone_cap</code> , <code>zns_num_conv_zones</code> , <code>zns_zd_ext_size</code>                                                                                                                                                                                    | <code>zns_zone_cap</code> between one logical block and the zone size.                                                                                                                                                                                         |
| Limits                  | <code>zns_max_open</code> , <code>zns_max_active</code>                                                                                                                                                                                                                       | Each at most the zone count; open at most active when both are set. <code>zns_max_open</code> also sets the default cache count.                                                                                                                               |
| Write cache             | <code>zns_num_wc</code>                                                                                                                                                                                                                                                       | At most the zone count.                                                                                                                                                                                                                                        |
| ZRWA                    | <code>zns_zrwa_size</code> , <code>zns_zrwafg_size</code> , <code>zns_zrwa_num</code>                                                                                                                                                                                         | All three together, or none.                                                                                                                                                                                                                                   |
| Reads, appends          | <code>zns_cross_zone_read</code> , <code>zns_zasl_bs</code>                                                                                                                                                                                                                   | <code>zns_zasl_bs</code> 0 follows <code>mdts</code> .                                                                                                                                                                                                         |
| Timing                  | <code>zns_flash_type</code> , <code>zns_pg_rd_lat</code> , <code>zns_pg_wr_lat</code> , <code>zns_blk_er_lat</code> , <code>zns_cmd_addr_lat</code> ,<br><code>zns_pg_xfer_lat</code> , <code>zns_status_lat</code> , <code>zns_pe_suspend</code> , <code>zns_tsusp_ns</code> | Cell type also multiplies the program unit. All latencies are in ns and must not be negative.                                                                                                                                                                  |
| Shared with other modes | <code>devsz_mb</code> , <code>namespaces</code> , <code>namespace_sizes</code> ,<br><code>namespace_modes</code> , <code>lba_index</code> , <code>nlbaf</code> , <code>mdts</code> ,<br><code>err_write_fail_ppm</code>                                                       | <code>lba_index</code> must select a block of 4 KiB or less.                                                                                                                                                                                                   |

The BlackBox geometry and timing properties (`nchs`, `pg_rd_lat` and so on) do not apply to a zoned namespace. The controller also refuses to become ready unless the host selects a 4 KiB memory page (CC.MPS).

A device with 64 MiB zones of half width on an 8-channel geometry:

```
| -device femu,devsz_mb=2048,femu_mode=3,zns_num_ch=8,zns_num_lun=4,zns_num_plane=2,zns_num_blk=32,zns_chnls_per_zone=4,zns_max_open=8,zns_max_active=16
```

#### 4.8.10 Statistics and outputs

| Output                                | What it shows                                                                                                                            | Source                                                   |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Zone report (Zone Management Receive) | state, write pointer, capacity and attributes of each zone                                                                               | <code>zns_zone_mgmt_recv()</code>                        |
| Changed Zone List (BFh)               | zones taken read only by injected write faults                                                                                           | <code>zns_changed_zone_list()</code>                     |
| SMART / Health (02h)                  | host data units and commands read and written, counted on the poller for every mode; Media Errors includes the injected ZNS write faults | <code>nvme-io.c</code> , <code>zns_media_errors()</code> |
| Vendor log C0h                        | not filled for zoned namespaces: ZNS runs no garbage collection, so there is no write amplification to report                            | log pages and counters (§B.1)                            |
| QEMU log                              | the ZNS geometry, program unit, stripe unit and cache count at realize ([FEMU] Log: lines)                                               | <code>zns_init_params()</code>                           |

#### 4.8.11 Validation

| Check                                          | What it covers                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| qtest cases in hw/femu/tests/qtest/femu-test.c | zone-reset, zone-open-limits, zone-active-limit, zone-append-parallel, zoned-append-limit, zoned-append-mdts0, zone-bad-dptr, zone-report-length, mdts0-zone-report, zoned-format-index, zoned-compare, zone-change-notice, log-contents-zoned, sgl-zoned, dma-error-zoned, ns-mgmt-unavailable-zoned, zrwa-reopen, zrwa-write-bounds, zrwa-odd-granule, zrwa-zd-ext, io-fuzz-zoned |
| Documentation examples                         | each tagged ZNS example starts under qtest, identifies the controller and writes and reads one block                                                                                                                                                                                                                                                                                |
| hw/femu/scripts/zone-aen-probe.c               | in-guest, manual: Zone Descriptor Changed notice end to end with err_write_fail_ppm                                                                                                                                                                                                                                                                                                 |
| Guest tools                                    | blkzone, nvme_zns, fio --zonemode=zbd on a Linux guest, as in the mode guide (§4.7.6)                                                                                                                                                                                                                                                                                               |

The NAND timing goes through the same media layer as bbssd, whose timing math has unit tests in hw/femu/tests/unit/test-nand-media.c.

#### 4.8.12 Limits

- No zone attribute is ever set by the controller: no Finish Recommended, Reset Recommended or Finished by Controller, and ZOC is 0 (no variable zone capacity, no zone active excursions).
- The only zone change the host does not cause is the injected write fault. Zones never go read only or offline because of wear.
- No wear, retention or read disturb model, and no garbage collection.
- Finish does not program the zone's write cache, and Close does not either. A read of data that is still in a write cache costs nothing.
- One latency per cell type: no page-type (LSB/MSB) difference and no program/erase cycle effect.
- The zone geometry properties are shared by all zoned namespaces of a controller.
- Zone state and data exist only in QEMU memory and are lost when QEMU exits. A large namespace needs that much free host memory.
- Namespace Management cannot create zoned namespaces, and metadata and Streams are refused on them (see namespaces (§4.16)).

#### 4.8.13 How to extend it

- A new zone management action: add a case to zns\_zone\_mgmt\_send() and an op\_handler\_t that zns\_do\_zone\_op() can apply to one zone or a list. Return resources through the same ladder the existing handlers use (zns\_aor\_dec\_open(), zns\_aor\_dec\_active(), zns\_zrwa\_release()).
- A new reason for a zone to change on its own: change the state, then call zns\_record\_changed\_zone(). Never call it from a host-driven transition; the specification excludes those from the log.
- New timing: extend zns\_nand\_media\_init() and zns\_advance\_status() in zftl.c; the media layer does the plane and channel bookkeeping. Work that touches the L2P table, block page pointers or timelines must run on the FTL thread, as Zone Reset does through req->zone\_resets.
- A new property: declare it in femu.c next to the other zns\_ ones, add the field to ZNSCtrlParams, copy it per namespace in nvme\_init\_namespaces() if the zoned code reads it from the namespace, validate it in zns\_check\_params() or zns\_init\_zone\_geometry(), give it a description and topic so gen-property-docs.py documents it, and add a qtest case.

#### 4.8.14 Source map

| File                 | What is there                                                                                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hw/femu/zns/zns.h    | zone and zoned Identify structures, struct zns_ssd, write cache, built-in cell timings, open and active counters                                                       |
| hw/femu/zns/zns.c    | geometry checks, zone state machine, Read/Write/Append, Zone Management Send and Receive, ZRWA, Changed Zone List, Identify, mode registration (nvme_register_znssd()) |
| hw/femu/zns/zftl.c   | zone-to-NAND placement, write cache, L2P table, zone reset and erase, media layer setup, zns_ftl_process_req()                                                         |
| hw/femu/zns/zftl.h   | zns_zone_reset(), debug and assert macros                                                                                                                              |
| hw/femu/femu.c       | zns_ property definitions, per-namespace copies of the zone limits, FTL thread dispatch                                                                                |
| hw/femu/nvme-io.c    | opcode filtering for zoned namespaces, host counters                                                                                                                   |
| hw/femu/nvme-admin.c | Identify Controller CSI 2 (ZASL), OAES.ZDCN, supported log pages, SMART media errors                                                                                   |
| hw/femu/nand/        | shared NAND media timing                                                                                                                                               |

#### 4.8.15 Related pages

- ZNS mode guide (§4.7)
- Namespaces and subsystems (§4.16)
- Timing model (§6.2)
- Property reference: ZNS (§A.1.1)

### 4.9 Open-Channel SSD (OCSSD)

OCSSD mode (femu\_mode=0) emulates an Open-Channel SSD, also called a "white-box" SSD. The device has no FTL. The host sees the NAND geometry (channels or groups, LUNs or parallel units, blocks or chunks, pages) and addresses physical pages directly with vector read, write and erase commands. Mapping, garbage collection and wear levelling run in the host. lver=2 (the default) selects Open-Channel 2.0, lver=1 selects 1.2.

Use it for host-side FTL research: LightNVM and pblk in older Linux kernels, or SPDK.

**The guest kernel must be older than 5.15.** Linux removed LightNVM, its Open-Channel driver, in 5.15. A newer guest kernel, including the Linux 6.8 image from make-guest-image.sh, sees the controller but has no driver for the Open-Channel interface. On a newer kernel, drive the device from user space with SPDK.

#### 4.9.1 Requirements

- Guest kernel: older than 5.15, with LightNVM (CONFIG\_NVM) and pblk (CONFIG\_NVM\_PBLK) built. The minimum version for each Open-Channel release is in the mode table (§4.1.4) and requirements.md (§2.1.2). guest-image.md (§2.3.3) says how to get such a guest.
- Guest tools: nvme-cli 1.x for nvme lnvm. nvme-cli 2.x no longer has the lnvm commands.
- Host: nothing beyond the common requirements. run-whitebox.sh uses a 4 GiB device and a 4 GiB guest.

#### 4.9.2 Launch

From build-femu/:

```
| ./run-whitebox.sh
```

The FEMU device in that script, for Open-Channel 2.0, is:

```
| -device femu,devsz_mb=4096,namespaces=1,lver=2,lmetasize=16,nlbaf=5,lba_index=3,mdts=10,lnum_ch=2,lnum_lun=4, |
| lnum_pln=2,lsec_size=4096,lsecs_per_pg=4,lpgs_per_blk=512,femu_mode=0
```

For Open-Channel 1.2, set OCVER=1 at the top of run-whitebox.sh, which changes lver:

```
| -device femu,devsz_mb=4096,namespaces=1,lver=1,lmetasize=16,nlbaf=5,lba_index=3,mdts=10,lnum_ch=2,lnum_lun=4, |
| lnum_pln=2,lsec_size=4096,lsecs_per_pg=4,lpgs_per_blk=512,femu_mode=0
```

run-whitebox.sh lets you change `ssd_size`, `num_channels` and `num_chips_per_channel`. Keep the other values unless you have a reason to change them.

The same Open-Channel 2.0 device is the preset `hw/femu/scripts/configs/ocssd.conf` (set `lver = 1` in it for 1.2). `ssd-config.sh` (§5.11) expands it to:

```
| -device femu,id=nvme0,lver=2,devsz_mb=4096,namespaces=1,lnum_ch=2,lnum_lun=4,lnum_pln=2,lpgs_per_blk=512, |
| lsecs_per_pg=4,lsec_size=4096,lmetasize=16,nlbaf=5,lba_index=3,mdts=10,femu_mode=0
```

### 4.9.3 Configuration

Properties: OCSSD (§A.1.1). The BlackBox geometry and timing properties do not apply, except `ch_xfer_lat` with `oc12_channel_timing`.

#### Geometry

- `lnum_ch`: channels (groups in 2.0), 1 to 32.
- `lnum_lun`: LUNs (parallel units in 2.0) per channel. `lnum_ch * lnum_lun` is at most 128.
- `lnum_pln`: planes per LUN. Open-Channel 1.2 accepts 1, 2 or 4.
- `lpgs_per_blk`: pages per block, at most 512 for 1.2. A 2.0 chunk is one block on every plane of a parallel unit.
- `lsecs_per_pg`: sectors per page.
- `lsec_size` and `lmetasize`: sector and out-of-band metadata size for 1.2. 2.0 always uses 4096-byte sectors with 16 bytes of metadata.
- `devsz_mb` sets the capacity, and with it the number of blocks per LUN.

#### Timing

Read, program and erase times come from the `flash_type` table: 1 SLC, 2 MLC (the default), 3 TLC or 4 QLC. Open-Channel 1.2 also charges channel transfer time when `oc12_channel_timing=on`, using `ch_xfer_lat` per page or the table value when it is 0. Open-Channel 2.0 charges no channel time. Vendor admin command 0xEE changes these times at run time (NAND timing (§3.5.15)). See the timing model (§6.2.7).

#### Other

- `early_reset` (2.0): report that the host may reset a chunk it has not filled.
- OCSSD supports one namespace, does not take `namespace_modes` other than `ocssd`, and ignores `sgl`.

### 4.9.4 Use it from the guest

These commands assume a guest kernel with LightNVMe and `nvme-cli` 1.x.

Check the device:

```

sudo nvme list
sudo nvme id-ctrl /dev/nvme0 | grep -E '^(mn|sn) '
sudo nvme lnvm list

```

The model is FEMU OpenChannel-SSD Controller (v2.0) (or (v1.2)), and the serial number starts with VOCSSD.

Open-Channel 1.2 accepts only its vector commands on /dev/nvme0n1; Open-Channel 2.0 also accepts plain Read and Write. Either way the host has to follow the NAND rules below. To get a normal block device, create a pblk target over a range of LUNs. With run-whitebox.sh there are 2 x 4 = 8 LUNs, numbered 0 to 7:

```

sudo nvme lnvm create --device-name=nvme0n1 --target-name=mydev \
--target-type=pblk --lun-begin=0 --lun-end=7

```

/dev/mydev is then a block device with pblk as its host FTL. Use it like any disk:

```

sudo fio --name=oc --filename=/dev/mydev --direct=1 --ioengine=libaio \
--rw=randwrite --bs=4k --iodepth=16 --size=1G
sudo mkfs.ext4 /dev/mydev

```

With SPDK, bind the controller to SPDK's user-space driver and use its Open-Channel support. FEMU's Open-Channel 2.0 controller also accepts the plain NVMe Read and Write commands that SPDK sends.

The 2.0 chunk information log page (CAh) reports each chunk's state, write pointer and wear, 32 bytes per chunk. The Supported Log Pages log (00h) lists it for the NVM command set. It needs an explicit namespace, and --log-len must not exceed 32 times the chunk count. This reads the first 128 chunks:

```

sudo nvme get-log /dev/nvme0 --log-id=0xca --log-len=4096 --namespace-id=1

```

The vendor log page C0h stays zero in OCSSD mode: there is no device FTL to count.

## 4.9.5 Limits and refusals

| Message                                                                                                         | Cause and fix                                                          |
|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| lver must be 1 (Open-Channel 1.2) or 2 (Open-Channel 2.0)                                                       | Another lver.                                                          |
| flash_type must be 1 (SLC), 2 (MLC), 3 (TLC) or 4 (QLC)                                                         | A cell type without a timing table.                                    |
| ocssd supports a single namespace                                                                               | namespaces above 1.                                                    |
| ocssd is a controller mode: femu_mode and namespace_ modes must both select it, or neither                      | namespace_modes=ocssd on a controller of another mode, or the reverse. |
| FEMU ocssd: lnum_ch must not exceed 32 and lnum_ch * lnum_lun must not exceed 128, got 64 and 8                 | The geometry is too large.                                             |
| FEMU ocssd: lnum_ch, lnum_lun, lnum_pln, lsecs_per_pg, lpgs_per_blk and lsec_size must all be greater than zero | A geometry property is 0.                                              |
| FEMU ocssd: lnum_pln must be 1, 2 or 4, got 3                                                                   | Open-Channel 1.2 plane count.                                          |
| OC 1.2 requires SLC, MLC, TLC or QLC and at most 512 pages per block                                            | lpgs_per_blk above 512 with lver=1.                                    |

At run time, Open-Channel 2.0 enforces the chunk rules: a write must start at the chunk's write pointer, so a chunk must be reset before it is written again. Open-Channel 1.2 does not check; a rewrite of a written page silently replaces its data, which real flash would not allow. A write outside the geometry fails in both.

## 4.9.6 Verify

1. `sudo nvme list` shows the model FEMU OpenChannel-SSD Controller.
2. On a LightNVM guest, `sudo nvme lnvm list` lists `nvme0n1` with the geometry you set, and `nvme lnvm create` gives a `/dev/mydev` that `fio` can write and read.

## 4.9.7 Troubleshooting

- **The device shows up but nothing can use it.** The guest kernel is 5.15 or newer, so there is no LightNVM. Boot an older kernel, or use SPDK.
- **`nvme lnvm: invalid sub-command`.** `nvme-cli 2.x` dropped it. Build `nvme-cli 1.x` in the guest.
- **Writes fail after rewriting an address (Open-Channel 2.0).** A write must start at the chunk's write pointer; reset the chunk first. Run normal workloads on a `pblk` target, not on `/dev/nvme0n1`.
- **`dmesg` shows corrupted read LBA from `pblk`.** Users reported this warning on reads of pages that were never written; the maintainers found the data intact.
- **`mkfs` hangs on a `pblk` target.** Users reported that keeping `lnum_pln=2`, changing only `lnum_ch`, `lnum_lun` and the size, and using a power-of-two size avoided it.
- **SPDK cannot write the device.** Older FEMU accepted only vector commands on Open-Channel 2.0. Current FEMU also accepts NVMe Read and Write.

Related issues: #3, #4, #48, #49, #118.

## 4.9.8 Related pages

- Timing model: OCSSD (§6.2.7)
- Requirements: kernel per mode (§2.1.2)
- Device property reference: OCSSD (§A.1.1)

## 4.10 OCSSD: the Open-Channel extension

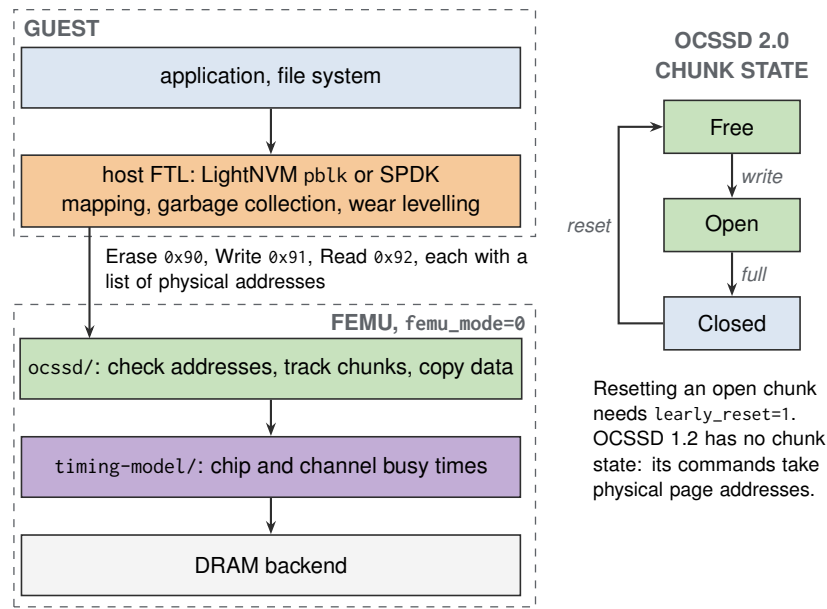
This chapter describes how FEMU emulates an Open-Channel SSD (`femu_mode=0`). It covers the internals: the two protocol versions, their address formats, the per-chunk state that 2.0 keeps, and how each command is timed. To run the mode, see the OCSSD guide (§4.9).

### 4.10.1 Purpose

An Open-Channel SSD has no device FTL. The host sees the NAND geometry and addresses physical sectors directly. Mapping, garbage collection, wear levelling and bad-block handling all run in the host: LightNVM and `pblk` in Linux before 5.15, or SPDK in user space. FEMU's job in this mode is to check the addresses the host sends, store the data, and charge NAND time per chip and per channel.

FEMU implements two versions, selected by `lver` (§A.1.1):

|               | Open-Channel 1.2 ( <code>lver=1</code> )                              | Open-Channel 2.0 ( <code>lver=2</code> , the default)      |
|---------------|-----------------------------------------------------------------------|------------------------------------------------------------|
| Address       | physical page address (PPA): channel, LUN, block, page, plane, sector | logical block address: group, parallel unit, chunk, sector |
| Unit of erase | a block on one plane, named by a PPA                                  | a chunk: one block on every plane of a parallel unit       |
| Write rule    | none enforced                                                         | sequential within a chunk, at the chunk's write pointer    |



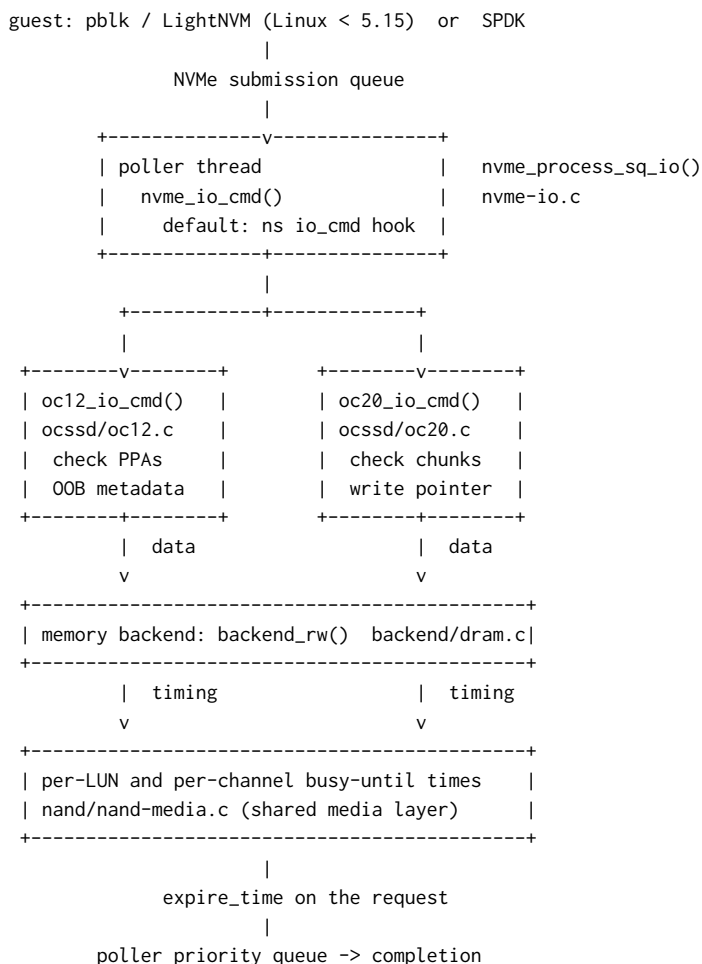
**Figure 18:** Open-Channel SSD: the FTL runs in the guest and sends commands with physical addresses; FEMU checks them, tracks chunk state (2.0), stores the data and charges chip and channel time.

|                           | Open-Channel 1.2 (lver=1)              | Open-Channel 2.0 (lver=2, the default)                                                |
|---------------------------|----------------------------------------|---------------------------------------------------------------------------------------|
| Per-unit state            | sector state and out-of-band bytes     | chunk descriptor: state, type, wear index, write pointer<br>accepted (SPDK uses them) |
| Plain NVMe Read and Write | refused                                |                                                                                       |
| Channel transfer time     | optional (oc12_channel_timing)         | none                                                                                  |
| Page-type timing          | yes (lower, center, upper page)        | no (every access is charged as a lower page)                                          |
| Log page                  | none                                   | chunk information, CAh                                                                |
| Model string              | FEMU OpenChannel-SSD Controller (v1.2) | FEMU OpenChannel-SSD Controller (v2.0)                                                |

**Guest requirement.** Linux removed LightNVM in 5.15, so a guest that uses the kernel's Open-Channel stack must run an older kernel. Requirements: kernel per mode (§2.1.2) gives the range for each version. On a newer kernel the controller still appears, and SPDK can drive it from user space.



#### 4.10.2 Place in the hierarchy



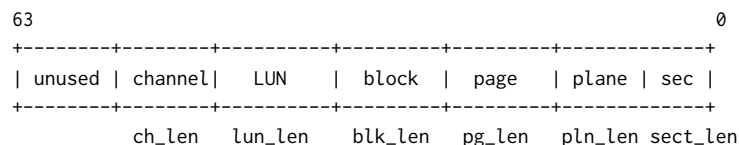
There is no FTL thread in this mode. The whole command, data copy and timing included, runs on the poller thread inside `nvme_io_cmd()`. The request then goes to the poller's own priority queue and is completed when the host clock passes its `expire_time`. The architecture page (§3.1.3) describes that completion step.

Registration happens in `nvme_register_extensions()` in `hw/femu/femu.c`: with `femu_mode=0`, `lver=1` installs `nvme_register_ocssd12()` and `lver=2` installs `nvme_register_ocssd20()`. Each fills the controller's `FemuExtCtrlOps` table: `init`, `exit`, `io_cmd` and `admin_cmd`, plus `get_log` for 2.0.

### 4.10.3 Open-Channel 1.2

## Address format

A 1.2 address is a 64-bit PPA. Each field gets just enough bits for its count, rounded up to a power of two, packed from the sector upward (`oc12_init_id_ctrl()`):



field width =  $\text{ceil}(\log_2(\text{count}))$ ; for example lsecs\_per\_pg=4  $\rightarrow$  2 bits

oc12\_ppa\_in\_geometry() checks every field against the real count. Because the fields are rounded up to powers of two, a value can fit in its field and still name a unit the device does not have; such an address is refused.

## Data structures

Oc12Ctrl (hw/femu/ocssd/oc12.h) holds the controller state:

- `id_ctrl`: the 1.2 identity structure the host reads with admin command 0xE2. One configuration group: channel, LUN, plane, block, page and sector counts, sector size (`lsec_size`), out-of-band size (`lmetasize`), the multi-plane mask (`mpos`) for 1, 2 or 4 planes, fixed timing hints, and capability bit 0 (bad block table management).
- `params`: the geometry from the properties plus derived strides (`sec_per_pl`, `sec_per_blk`, `pl_units`, `pg_units`, ...).
- `ppaf`: the masks and offsets for each address field.
- `meta_buf`: one entry per sector,  $4 + \text{lmetasize}$  bytes. The first 4 bytes are a sector state (`OC12_SEC_WRITTEN`, `OC12_SEC_ERASED`); the rest are the out-of-band bytes the host passes through the command's metadata pointer. `ppa2secidx()` flattens a PPA to the entry index.

Per namespace (NvmeNamespace):

- `bbtbl`: one bad block table per LUN,  $\text{blocks} * \text{planes}$  bytes, one byte per plane in the order block 0 plane 0, block 0 plane 1, and so on. The host reads it with admin command 0xF2 and marks blocks with 0xF1, which sets the entry for the block and plane in each address and keeps the header counts (factory bad, grown bad, device reserved, host reserved) in step.
- `tbl`: a table the host can read with admin command 0xEA (Get L2P Table). FEMU fills it with "unmapped" at start-up and never updates it.

## Commands

| Opcode     | Kind  | Handler                         | What it does                                                        |
|------------|-------|---------------------------------|---------------------------------------------------------------------|
| 0x92       | I/O   | <code>oc12_read()</code>        | vector read of up to <code>lmax_sec_per_rq</code> sectors           |
| 0x91       | I/O   | <code>oc12_write()</code>       | vector write; at least <code>lsecs_per_pg * lnum_pln</code> sectors |
| 0x90       | I/O   | <code>oc12_erase_async()</code> | vector erase of the listed blocks                                   |
| 0x01, 0x02 | I/O   |                                 | plain Write and Read are refused with Invalid Opcode                |
| 0xE2       | admin | <code>oc12_identity()</code>    | identity structure                                                  |
| 0xEA       | admin | <code>oc12_get_l2p_tbl()</code> | the <code>tbl</code> described above                                |
| 0xF2       | admin | <code>oc12_bbt_get()</code>     | bad block table of one LUN                                          |
| 0xF1       | admin | <code>oc12_bbt_set()</code>     | mark one or more blocks                                             |

The Commands Supported and Effects log (05h) lists these commands, the timing command 0xEE, and for 2.0 its own vector commands and admin 0xC1; for 1.2 it leaves plain Read and Write out.

A vector command carries either one PPA in the command or a list of PPAs in host memory. Read and write accept only PRP data pointers, check the list against the geometry and MDTS, move the out-of-band bytes, and then copy the data. Each PPA must pair with one sector of the data transfer; a data pointer that does not split into exactly one segment per address is refused with Invalid Field.

The backend stores 1.2 data at the PPA value itself, used as a sector number: `start_block + (ppa << lbads)`. Because the PPA fields are padded to powers of two, this address space can be larger than the device. An

address that lands past the end of the memory backend fails with LBA Out of Range.

Other refusals: a PPA value above the namespace size fails with LBA Out of Range before the backend is reached; an LBA format with metadata (meta) makes read and write fail with Invalid Field; and an address outside the geometry on erase or on a bad block command fails with Invalid Field.

## Bad blocks

Write and erase check every address against bbtbl. An entry with bit 0 (factory bad), bit 1 (grown bad) or bit 2 (device reserved) set makes the block unusable. Bit 3 (host reserved) and bit 4 (media manager reserved) record the host's own allocation, so the device does not refuse them. A single-plane access (control bits 1:0 zero) checks the plane in the address; a dual- or quad-plane access checks the block on every plane.

A command that names an unusable block fails as a whole with Write Fault (status type 2h, code 80h, the 1.2 error code for data that cannot be committed) and Do Not Retry. Nothing in it is written or erased, so every bit of the per-address status in completion dwords 0 and 1 is set. Reads are not checked. Clearing an entry with 0xF1 makes the block usable again. The table starts empty; nand\_bad\_blocks does not seed it.

In both versions the generic NVM commands (Flush, Dataset Management, Compare, Write Zeroes, Copy, Verify, Write Uncorrectable) are handled by nvme\_io\_cmd() on raw LBAs before the Open-Channel handler is consulted, when oncs turns them on. They do not follow the Open-Channel rules.

## What 1.2 does not enforce

1.2 keeps the sector state but does not act on it:

- A write to a sector that was already written succeeds and replaces the data. Real NAND would refuse it.
- Erase does not reset the sector states: the state update is disabled in oc12\_meta\_blk\_set\_erased(), which returns at its first line.
- Reads do not check the state, so an unwritten sector returns whatever the backend holds.
- Blocks marked bad in bbtbl can still be read; write and erase are refused (see Bad blocks (§4.10.3)).

Write the host FTL as if these rules applied, because real devices apply them.

## 4.10.4 Open-Channel 2.0

### Address format and geometry

A 2.0 address is a logical block address split into four fields (femu\_oc20\_init\_id\_ctrl(), macros in hw/femu/ocssd/oc20.h):

```

63                                     0
+-----+-----+-----+-----+-----+
| unused | group | parallel unit | chunk | sector |
+-----+-----+-----+-----+-----+
          grp_len  lun_len          chk_len sec_len

group      = channel              (NUM_GRP = lnum_ch)
parallel unit = LUN                (NUM_PU  = lnum_lun)
chunk      = one block on every plane of the LUN
sector     = index inside the chunk (CLBA = lsecs_per_pg
                                   * lpgs_per_blk * lnum_pln)
NUM_CHK = namespace blocks / (lsecs_per_pg * lpgs_per_blk
                              * lnum_ch * lnum_lun * lnum_pln)

```

NUM\_CHK is computed from the namespace's block count in the LBA format selected by lba\_index, before 2.0 switches the namespace to 4096-byte sectors. With the default lba\_index=0 (512-byte blocks), the geometry therefore describes 8 times the memory that devsz\_mb provides, and valid addresses past the first eighth fail with LBA Out of Range. Set nlbaf=5, lba\_index=3 (4096-byte blocks), as run-whitebox.sh does, so the two agree.

The namespace reports its size as  $2^{(grp\_len + lun\_len + chk\_len + sec\_len)}$  sectors, the whole address space including the padding. The sector size is always 4096 bytes. Unlike 1.2, the data path turns the address into a dense sector index (oc20\_lba\_to\_sector\_index()) before it touches the backend, so the padding costs no memory.

The geometry structure returned by admin command 0xE2 also carries fixed values: minimum write size 4 sectors, optimal write size 8, cache minimum write size units (mw\_cunits) 24, no limit on open chunks, and typical and maximum read, write and reset times. Those times are hints for the host; the timing model below does not use them.

## Data structures

Oc20Namespace (ns->state) holds:

- id\_ctrl: the 4096-byte geometry structure.
- lbaf: masks and offsets for the four address fields.
- chunk\_info: one 32-byte descriptor (Oc20CS) per chunk, the same layout the chunk information log page returns:

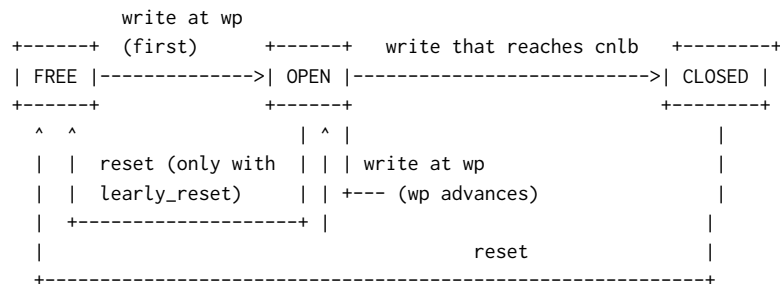
Oc20CS (32 bytes)

|       |      |            |         |      |      |     |
|-------|------|------------|---------|------|------|-----|
| state | type | wear_index | rsvd[5] | slba | cnlb | wp  |
| u8    | u8   | u8         |         | u64  | u64  | u64 |

state: FREE 0x1, CLOSED 0x2, OPEN 0x4, OFFLINE 0x8  
type : SEQ 0x1 (all chunks at start-up), RAN 0x2

Oc20Ctrl (n->ext\_ops.state) holds a header with the sector size (4096) and metadata size (16) the namespace reports.

## Chunk state machine



reset of a FREE chunk : refused (multiple resets are not offered)  
write to an OFFLINE one: Write Fault  
reset of an OFFLINE one: Offline Chunk (0x2C0)  
read of an OFFLINE one : succeeds and returns zeros  
OFFLINE is only set through Set Log Page  
each reset : wear\_index + 1, wp = 0

The rules, from oc20\_rw\_check\_chunk\_write() and oc20\_chunk\_set\_free():

- A write is split into runs, one per chunk it touches. In each run the sectors must be consecutive and the first must equal the chunk's write pointer. Otherwise the command fails with Out of Order Write (0x2F2).
- A write to a closed or offline chunk, or one that would pass the chunk's capacity, fails with Write Fault.
- After the data is stored, `oc20_advance_wp_all()` moves each chunk's write pointer by its run. A free chunk becomes open; a chunk whose write pointer reaches `cnlb` becomes closed.
- Vector erase (0x90) resets each listed chunk. Resetting a closed chunk is always allowed. Resetting an open chunk needs `learly_reset` (§A.1.1). Resetting a free chunk is refused, because FEMU does not advertise multiple resets. When the command has a metadata pointer, FEMU writes the new descriptor of each chunk there.
- A read of a sector that is not yet readable succeeds and returns zeros. A sector is readable when it is below the write pointer of a closed chunk, or more than `mw_cunits` (24) sectors below the write pointer of an open one. The 24-sector window models data still held in the device's write cache.
- Random chunks (type RAN) skip the write pointer: a write run may start at any sector of an open random chunk, though its sectors must still be consecutive, and it moves neither the write pointer nor the state. Any sector of an open random chunk can be read. Chunks only become random through Set Log Page.

The minimum write size (4 sectors) is reported but not checked, so that SPDK works.

## Commands

| Opcode                       | Kind  | Handler                            | What it does                                                               |
|------------------------------|-------|------------------------------------|----------------------------------------------------------------------------|
| 0x92                         | I/O   | <code>oc20_rw(vector=true)</code>  | vector read, list of up to 64 addresses                                    |
| 0x91                         | I/O   | <code>oc20_rw(vector=true)</code>  | vector write                                                               |
| 0x90                         | I/O   | <code>oc20_erase()</code>          | vector chunk reset                                                         |
| 0x02, 0x01                   | I/O   | <code>oc20_rw(vector=false)</code> | plain Read and Write over consecutive addresses, with the same chunk rules |
| 0xE2                         | admin | <code>oc20_identify()</code>       | geometry                                                                   |
| 0x02 (Get Log Page), LID CAh | admin | <code>oc20_get_log()</code>        | chunk descriptors                                                          |
| 0xC1                         | admin | <code>oc20_set_log()</code>        | write chunk descriptors (FEMU extension)                                   |

Read and write accept only PRP data pointers. A command with more than 64 addresses fails with Invalid Field.

Set Log Page (0xC1) with LID CAh lets the host put chunks into a given state, type, wear index and write pointer, for example to start a test from a partly used device. FEMU checks every descriptor before it changes any: `slba` and `cnlb` must match the controller's, the type must be SEQ or RAN, and the write pointer must suit the state (0 for FREE, below `cnlb` for OPEN, at most `cnlb` for CLOSED). It pauses the pollers while it copies the descriptors in.

The 2.0 data path does not transfer the metadata buffer, even though the namespace reports 16 metadata bytes per sector.

### 4.10.5 Timing

Both versions use the shared media layer (`nand_media_op()` in `hw/femu/nand/nand-media.c`), each with its own NandMedia and the LUN gate. Open-Channel 2.0 charges no channel time and uses flat lower-page times,

through `oc20_chip_op()`. Open-Channel 1.2 uses per-page-type tables, through `oc12_media_op()`; with `oc12_channel_timing` on the bus is staged, otherwise it is off. Vendor command `0xEE` refreshes the times of either version (`oc20_refresh_timing()`, `oc12_refresh_timing()`). With 1.2 channel timing on, one channel lock covers a page's transfer and its chip time together, so two commands that reach the same channel and chip at once complete in an order the host would also see from one queue; before, their phases could interleave. A single command stream gets the same times as before. The controller keeps one time per chip (LUN) and, for 1.2, one per channel:

- `lun_avail[ch * num_lun + lun]` in `Oc12Ctrl` or `Oc20Ctrl`, at most 128 chips.
- `Oc12Ctrl.ch_avail[ch]`, at most 32 channels, and the media layer's list of booked read windows per channel. For 1.2 the list has no cap (`policy.bus_res_unbounded`).

An operation starts at the larger of "now" and the chip's busy-until time, adds the operation's latency, and the end becomes the new busy-until time. Latencies come from the built-in table for `flash_type` (§A.1.1) (1 SLC, 2 MLC, 3 TLC, 4 QLC) in `hw/femu/nand/nand.h`, by operation and page type. The NAND timing properties of the black-box mode (`pg_rd_lat` and the others) do not apply.

A read or write is first grouped into buckets, one per NAND page touched, by `parse_ppa_list()` (1.2) or `oc20_parse_lba_list()` (2.0). Each bucket is then charged on its chip and channel:

```
write bucket: [ channel transfer ][ program on chip ..... ]
               ^ the channel's first free time at or after now,
               fitted around reserved read transfers      ^ done

read bucket:  [ read on chip ..... ][ channel transfer ]
               ^ now                                     ^ done
               (the transfer is fitted into a free gap of the
                channel's reservation list)

command done = now + max over buckets of (done - now)
erase        = one block erase on the chip of each listed address;
               command done = the latest of them
```

Version differences:

- **1.2.** A bucket is one page across all planes of a LUN. Its page type (lower, center and upper; QLC has two center types) comes from the page number, so MLC, TLC and QLC pages cost different times. With `oc12_channel_timing` (§A.1.1) on, each bucket also pays channel time: `ch_xfer_lat` per page, or the `flash_type` table's transfer time when `ch_xfer_lat` is 0, scaled by the sectors it carries out of `lsecs_per_pg`.
- **2.0.** A bucket is one run of addresses inside one chunk, so a command that writes 16 consecutive sectors of a chunk is charged one program. Every operation is charged as page type 0 (the lower page). No channel time is charged.

The time is computed on the poller when the command arrives, and stored as the request's `expire_time`. For 1.2 with channel timing, the channel's spin lock (`chnl_locks[]`) covers the bus and its LUNs; otherwise a chip's time is updated with a compare-and-swap. So several pollers can time commands at once.

#### 4.10.6 Parameters

All OCSSD properties are listed in the OCSSD section of the property reference (§A.1.1). How they interact:

| Property                         | 1.2                          | 2.0                                | Notes                                                                                          |
|----------------------------------|------------------------------|------------------------------------|------------------------------------------------------------------------------------------------|
| lver                             | 1                            | 2                                  | Any other value fails realize.                                                                 |
| lnum_ch                          | channels                     | groups                             | lnum_ch * lnum_lun at most 128, lnum_ch at most 32: the chip and channel arrays are that size. |
| lnum_lun                         | LUNs per channel             | parallel units per group           |                                                                                                |
| lnum_pln                         | 1, 2 or 4                    | any value above 0                  | 2.0 folds the planes into the chunk size.                                                      |
| lpgs_per_blk                     | at most 512                  | any value above 0                  | 1.2 looks up the page type in a 512-entry table.                                               |
| lsecs_per_pg                     | sectors per page             | sectors per page                   |                                                                                                |
| lsec_size                        | reported sector size         | must be above 0, otherwise ignored |                                                                                                |
| lmetasize                        | out-of-band bytes per sector | ignored (16)                       |                                                                                                |
| lmax_sec_per_rq                  | addresses per command        | ignored (64)                       |                                                                                                |
| flash_type                       | timing table                 | timing table                       | 1 to 4; others fail realize.                                                                   |
| oc12_channel_timing, ch_xfer_lat | channel time                 | ignored                            | ch_xfer_lat must not be negative when channel timing is on.                                    |
| learly_reset                     | ignored                      | early reset capability             |                                                                                                |
| devsz_mb                         | sets blocks per LUN          | sets chunks per parallel unit      | 1.2 needs 1 to 65535 blocks per LUN.                                                           |

ms\_max has no effect; a value other than the default prints a warning. An Open-Channel controller has exactly one namespace, and namespace\_modes, if given, must say ocssd.

A minimal 2.0 device with the geometry of run-whitebox.sh:

```
| -device femu,devsz_mb=4096,namespaces=1,lver=2,nlbaf=5,lba_index=3,lnum_ch=2,lnum_lun=4,lnum_pln=2,lsecs_per_pg=4,lpgs_per_blk=512,femu_mode=0
```

## 4.10.7 Counters

OCSSD has no device FTL, so the vendor log page C0h stays zero. What the mode does report:

- The SMART log's host read and write counts. The poller counts only the plain NVMe Read and Write opcodes (0x02 and 0x01), so vector commands are not counted: on 1.2 the counts stay at zero, and on 2.0 they move only for plain Read and Write.
- In 2.0, the chunk information log (CAh): each chunk's state, write pointer and wear index. The wear index counts resets.

## 4.10.8 Validation status

- The qtest suite in hw/femu/tests/qtest/femu-test.c, run in CI, has command-level cases for both versions. For 1.2: oc12-opcodes, oc12-capabilities, oc12-small-sectors, oc12-ppa-timing, oc12-flash-type, oc12-page-count, oc12-transfer-cost and the oc12-channel-\* cases for channel time. For 2.0: oc20-vector-io, oc20-set-chunks, oc20-log-length, oc20-sgl-refused and oc20-fuzz, a fuzzer over the command fields.
- oc12-trace-on and oc12-trace-off pin the time of each 1.2 command exactly, with and without channel time, through the qtest-only x-oc12-trace. The workload uses TLC pages of each type, part

pages, two planes, erases of several blocks, 40 queued reads on one channel and a 0xEE change of the NAND times. A change to the 1.2 timing model must not move these times.

- The documentation check starts each OCSSD example in this guide and in the mode guide, and sends Identify; it does not move data in this mode.
- No guest test runs in CI: LightNVM needs a guest kernel older than 5.15, and the guest image FEMU builds runs Linux 6.8.

#### 4.10.9 Limits

- One namespace per controller.
- 1.2 does not enforce erase-before-write and does not reset sector state on erase (see above). Its bad block table is enforced on write and erase only.
- 2.0 does not model page types or channel transfer, and charges one page time per chunk run regardless of how many pages the run covers.
- 2.0 does not store per-sector metadata.
- The vendor admin command 0xEE can change the read, program and erase times of the lower and upper pages and the channel time at run time, but not those of the centre pages of TLC and QLC (0xEE (§3.5.15)).
- The Get L2P Table command of 1.2 always returns unmapped entries.
- Data lives only in host memory and is lost when QEMU exits.

Refusals at realize are listed in the mode guide (§4.9.5).

#### 4.10.10 Extending the mode

- **A new cell type.** Add its latencies and page-type table in `hw/femu/nand/nand.h` and `hw/femu/nand/nand.c`, then widen the `flash_type` checks in `nvme_check_constraints()` (`hw/femu/femu.c`) and `oc12_init()`.
- **Page types or channel time in 2.0.** `oc20_advance_status()` passes page type 0 and a transfer time of 0. Look up the page from the sector index and pass a transfer time, as `oc12_advance_status()` does.
- **Enforcing 1.2 write-once.** `oc12_meta_state_set_written()` already reads the sector state; refuse a write when it is `OC12_SEC_WRITTEN`, and make `oc12_meta_blk_set_erased()` reset the states instead of returning early.
- **Another log page.** Add a case to `oc20_get_log()`; the generic Get Log Page handler offers unknown log IDs to the namespace's `get_log` hook.

#### 4.10.11 Source map

| File                                                                                          | Contents                                                                                 |
|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <code>hw/femu/ocssd/oc12.c</code> <sup>32</sup> , <code>oc12.h</code> <sup>33</sup>           | 1.2 commands, PPA format, sector metadata, bad block tables, init and exit               |
| <code>hw/femu/ocssd/oc20.c</code> <sup>34</sup> , <code>oc20.h</code> <sup>35</sup>           | 2.0 commands, chunk descriptors, write pointer rules, geometry, log page                 |
| <code>hw/femu/ocssd/oc-timing.c</code> <sup>36</sup> , <code>oc-timing.h</code> <sup>37</sup> | <code>flash_type</code> times ( <code>set_latency()</code> ), geometry bound check, 0xEE |

<sup>32</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/ocssd/oc12.c>

<sup>33</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/ocssd/oc12.h>

<sup>34</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/ocssd/oc20.c>

<sup>35</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/ocssd/oc20.h>

<sup>36</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/ocssd/oc-timing.c>

<sup>37</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/ocssd/oc-timing.h>



| File                                                     | Contents                                            |
|----------------------------------------------------------|-----------------------------------------------------|
| hw/femu/nand/nand-media.c <sup>38</sup>                  | chip and channel busy-until times for both versions |
| hw/femu/nand/nand.h <sup>39</sup> , nand.c <sup>40</sup> | per-cell-type latency tables and page-type tables   |
| hw/femu/femu.c <sup>41</sup>                             | nvme_register_extensions(), realize-time checks     |
| hw/femu/nvme-io.c <sup>42</sup>                          | nvme_io_cmd() dispatch, completion queue            |

#### 4.10.12 Related pages

- OCSSD guide (§4.9): launching, guest tools, troubleshooting
- Timing model: OCSSD (§6.2.7)
- Property reference: OCSSD (§A.1.1)

### 4.11 Key-value SSD (KV)

KV mode (femu\_mode=5) emulates an SSD with the NVMe Key Value command set. The namespace stores values under keys instead of blocks at addresses: the host sends Store, Retrieve, Delete, Exist and List commands with a key of up to 16 bytes. Values live on emulated NAND laid out with the BlackBox geometry, and each command is charged NAND read and program time.

Use it to develop or measure key-value stores and host software written for key-value devices, without key-value hardware.

#### 4.11.1 Requirements

- Host and guest: see the mode table (§4.1.4) and requirements.md (§2.1.2).
- Linux has no key-value command set driver. The namespace gets no block device (/dev/nvme0n1 does not exist for it). You drive it with NVMe passthrough commands, from nvme-cli or your own program. On a controller with one namespace, send them to the controller node /dev/nvme0. Linux refuses I/O passthrough on the controller node when the controller has more than one namespace; then use the namespace's generic node /dev/ngXnY.
- Guest kernel: 6.0 or newer. Linux 6.0 started to attach namespaces of a command set it has no driver for, with only the generic node (commit eb867ee995bd, "nvme: enable generic interface (/dev/ngXnY) for unknown command sets"). Linux 5.10 to 5.19 log unknown csi 1 for nsid N (5.9: unknown csi:1 ns:N) and skip the namespace, so passthrough on /dev/nvme0 fails too.
- Guest tools: nvme-cli (io-passthru), and a C compiler to build kv-probe.c. The image from make-guest-image.sh has no compiler; add one with ./make-guest-image.sh --packages build-essential, or run `sudo apt install build-essential` in the guest.

#### 4.11.2 Launch

From build-femu/:

```
| ./run-kvssd.sh
```

<sup>38</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nand/nand-media.c>

<sup>39</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nand/nand.h>

<sup>40</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nand/nand.c>

<sup>41</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/femu.c>

<sup>42</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nvme-io.c>

The FEMU device in that script is:

```
-device femu,devsz_mb=4096,namespaces=1,femu_mode=5,secsz=512,secs_per_pg=8,pgs_per_blk=256,blks_per_pl=256,pls_per_lun=1,luns_per_ch=8,nchs=8,pg_rd_lat=40000,pg_wr_lat=200000,blk_er_lat=2000000,gc_thres_pcent=75
```

The geometry and timing above are the defaults, so the shortest KV device is:

```
-device femu,devsz_mb=4096,namespaces=1,femu_mode=5
```

The preset `hw/femu/scripts/configs/kvssd.conf` describes the launcher's device; `ssd-config.sh` (§5.11) expands it to:

```
-device femu,id=nvme0,devsz_mb=4096,namespaces=1,secsz=512,secs_per_pg=8,pgs_per_blk=256,blks_per_pl=256,pls_per_lun=1,luns_per_ch=8,nchs=8,pg_rd_lat=40000,pg_wr_lat=200000,blk_er_lat=2000000,gc_thres_pcent=75,femu_mode=5
```

### 4.11.3 Configuration

KV uses only part of the BlackBox properties:

- NAND geometry (§A.1.1) (`nchs`, `luns_per_ch`, `pls_per_lun`, `blks_per_pl`, `pgs_per_blk`, `secs_per_pg`, `secsz`) lays out the value store.
- NAND timing (§A.1.1) (`pg_rd_lat`, `pg_wr_lat`, `blk_er_lat`, `nand_cell_type` and the channel bus properties) sets the cost of each command.
- `gc_thres_pcent` is the share of the NAND usable for values. The usable value space is the namespace size or that share of the NAND, whichever is smaller.

The FTL, mapping, cache and write buffer properties do not apply. Each command pays one page read for the index lookup, overlapped with the NAND reads or programs of its value; reclaiming space charges erases to the command that triggers it (timing model (§6.2.8)).

Limits built into the mode:

- Keys are 1 to 16 bytes.
- Values are at most 2 MiB, and one command also cannot move more than `mdts` allows (4 MiB by default).
- Each KV namespace has its own key space. A controller can have several KV namespaces, or KV next to block namespaces (multi-namespace guide (§4.15)); see the requirements above for which device node to use then.
- KV checks the BlackBox geometry properties, but not the GC capacity rule BlackBox applies: a namespace larger than the NAND allows still starts, and its usable value space is clamped to the NAND.

### 4.11.4 Use it from the guest

#### Command format

| Command  | Opcode |
|----------|--------|
| Store    | 0x01   |
| Retrieve | 0x02   |
| List     | 0x06   |
| Delete   | 0x10   |
| Exist    | 0x14   |

- Key: the low 8 bytes in CDW2 and CDW3, the high 8 bytes in CDW14 and CDW15, little-endian.
- CDW11 bits 7:0: key length in bytes.
- CDW11 bits 15:8, Store only: bit 8 stores only if the key exists, bit 9 stores only if it does not.
- CDW10: the value size for Store, the host buffer size for Retrieve and List.
- The value moves through the normal data pointer.
- The namespace ID must be given. nvme-cli sends 0 by default, which the Linux driver refuses with Invalid argument before the command reaches the device. FEMU answers the broadcast ID (0xffffffff) with Invalid Field.

## With nvme-cli

Check the controller:

```
| sudo nvme id-ctrl /dev/nvme0 | grep -E '^(mn|sn) '
```

The model is FEMU KV-SSD Controller and the serial number starts with vKVSSD.

Store a 64-byte value under the 4-byte key BBBB (0x42424242), read it back, check that it exists, then delete it:

```
| head -c 64 /dev/urandom > value.bin
| sudo nvme io-passthru /dev/nvme0 -O 0x01 -n 1 --cdw10=64 --cdw11=4 \
|   --cdw2=0x42424242 -l 64 -w -i value.bin
| sudo nvme io-passthru /dev/nvme0 -O 0x02 -n 1 --cdw10=64 --cdw11=4 \
|   --cdw2=0x42424242 -l 64 -r -b > out.bin
| cmp value.bin out.bin
| sudo nvme io-passthru /dev/nvme0 -O 0x14 -n 1 --cdw11=4 --cdw2=0x42424242
| sudo nvme io-passthru /dev/nvme0 -O 0x10 -n 1 --cdw11=4 --cdw2=0x42424242
```

Retrieve reports the full value size in the completion's result (Dword 0). If the host buffer is smaller, the device returns the first bytes and the host can retry with a larger buffer.

List up to 4096 bytes of keys from the start of the key space:

```
| sudo nvme io-passthru /dev/nvme0 -O 0x06 -n 1 --cdw10=4096 --cdw11=0 \
|   -l 4096 -r
```

The returned buffer starts with a 4-byte key count, followed by one entry per key: a 2-byte key length and the key, each entry padded to 4 bytes.

## With kv-probe

hw/femu/scripts/kv-probe.c runs the whole lifecycle: store, exist, retrieve (full and short), the conditional stores, delete, the retrieve that must then miss, and a key longer than 16 bytes. It checks status and data. It takes the node to use as its argument (default /dev/nvme0) and always addresses namespace 1. From build-femu/ on the host, copy it to the guest and run it:

```
| scp -P 8080 -i ~/images/femu-guest-key ../femu-scripts/kv-probe.c femu@localhost:
| ./run-guest-ssh.sh gcc -O2 -o kv-probe kv-probe.c
| ./run-guest-ssh.sh sudo ./kv-probe /dev/nvme0
```

## Status codes

| Status | Meaning                                                                                                             |
|--------|---------------------------------------------------------------------------------------------------------------------|
| 0x85   | Invalid Value Size: the value is larger than 2 MiB.                                                                 |
| 0x86   | Invalid Key Size: a key length of 0 on Store, Retrieve, Delete or Exist. For List, length 0 means "from the start". |

| Status            | Meaning                                                                                                                                                                                              |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x87              | Key Does Not Exist: Retrieve or Exist of a missing key, a Store with bit 8 (only if the key exists) for a missing key, or Delete of a missing key when EDNEK is set. It comes with Do Not Retry set. |
| 0x89              | Key Exists: a Store with bit 9 (only if the key is absent) found the key.                                                                                                                            |
| Capacity Exceeded | The value space or the key index is full.                                                                                                                                                            |
| Invalid Field     | A key longer than 16 bytes, a List buffer smaller than 4 bytes, or the broadcast namespace ID.                                                                                                       |

By default, Delete of a missing key succeeds. Setting the EDNEK bit (bit 0) of the Key Value Configuration feature (20h) makes it fail with 0x87:

```
| sudo nvme set-feature /dev/nvme0 -n 1 -f 0x20 -V 1
```

## Counters

The vendor log page C0h sums its counters over the controller's BlackBox, CSD and KV namespaces. This reads the host pages written, pages moved and pages programmed (log pages and counters (§B.1)):

```
| sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b | od -An -t u8 -j 8 -N 24 -w24
```

### 4.11.5 Limits and refusals

| Message                                                                                                                                              | Cause and fix                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| the key-value command set and FDP cannot share a controller: placement owns every reclaim unit and the key-value store is left with no write pointer | A KV namespace on a controller in an FDP subsystem.                                              |
| FEMU bsssd: nchs must be greater than 0, got 0                                                                                                       | KV checks the BlackBox geometry properties with the same messages, but not the GC capacity rule. |
| meta: namespace 1 runs a mode without metadata support (block or no-SSD only)                                                                        | meta with a KV namespace.                                                                        |

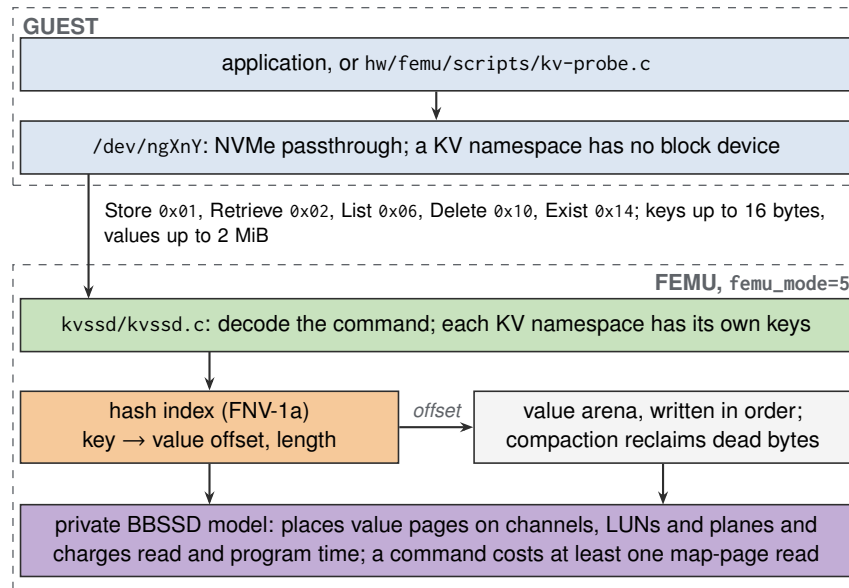
### 4.11.6 Verify

1. `sudo nvme id-ctrl /dev/nvme0` reports the model FEMU KV-SSD Controller.
2. The store, retrieve and cmp above succeed, and Exist after Delete returns 0x87.
3. `kv-probe` prints no failures.

### 4.11.7 Troubleshooting

- **There is no /dev/nvme0n1.** Expected: Linux attaches no block driver to a key-value namespace. Send commands to /dev/nvme0, or to /dev/ngXnY when the controller has several namespaces.
- **nvme-cli says Invalid argument.** Add `-n 1` (or the namespace you mean). `nvme-cli` sends namespace ID 0 by default, and the Linux driver refuses it. On a controller with several namespaces, the controller node refuses I/O passthrough altogether; use the generic node.
- **Retrieve returns 0x87 right after Store.** Check that the key bytes and the key length (CDW11) match exactly; BBBB with length 4 is a different key from BBBB with length 8.
- **FEMU\_KV\_SELFTEST:** set this environment variable to run the KV FTL self-test once at realize and log the result (environment variables (§A.1.5)).

Related issues: #68.



**Figure 19:** Key-value SSD: each KV namespace keeps a hash index and a value arena, and a private BBSSD model places the value pages and charges NAND time.

#### 4.11.8 Related pages

- Timing model: KV and CSD (§6.2.8)
- Architecture: other FTLs (§3.1.5)
- Multiple namespaces (§4.15)

## 4.12 KV: the key-value extension

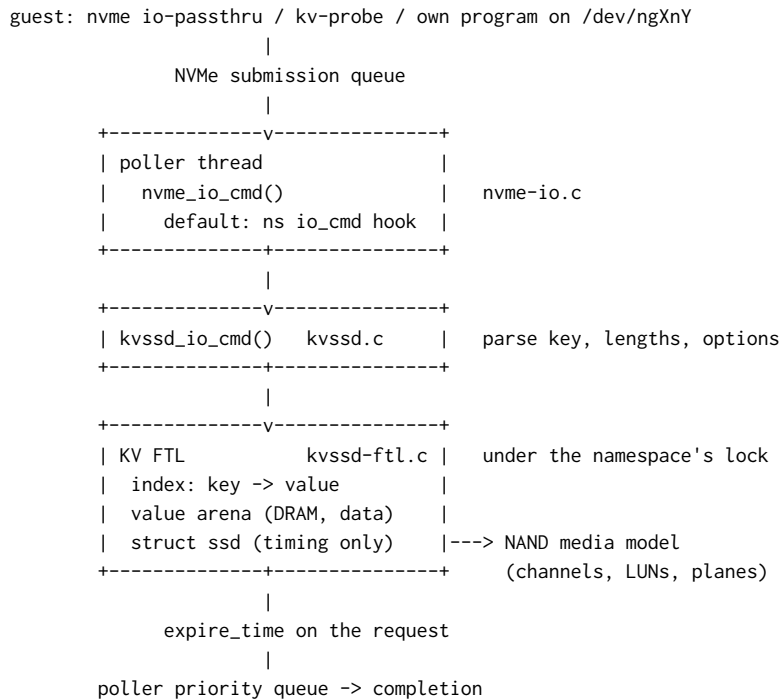
This chapter describes how FEMU emulates a key-value SSD (`femu_mode=5`): the NVMe Key Value command set as implemented, the device-side index and value store, and what each command is charged. To run the mode, see the KV guide (§4.11).

### 4.12.1 Purpose

A key-value SSD stores values under keys instead of blocks at addresses. The device owns the mapping from a key to where its value lives, so the host needs no file system or block map to find it. FEMU implements the NVM Express Key Value Command Set, revision 1.3, on top of the same NAND model the black-box mode uses. Values are placed on emulated channels, LUNs and planes, so key-value latency follows flash parallelism and space reclaim rather than a fixed number.

**Guest requirement.** Linux has no key-value driver. The guest reaches a KV namespace through NVMe passthrough on the generic character device `/dev/ngXnY`, which Linux creates for command sets it does not know since 6.0. Older kernels skip the namespace. See Requirements: kernel per mode (§2.1.2).

### 4.12.2 Place in the hierarchy



KV needs no FTL thread. The command, the data copy and the timing all run on the poller thread inside `nvme_io_cmd()`. If another namespace of the same controller starts the FTL thread (a black-box, ZNS or CSD namespace), KV requests pass through it without further charge.

Each KV namespace owns its own state: key space, value store and NAND model. `kvssd_init()` sets the namespace's command set identifier to KV (`NVME_CSI_KV`, 1h) and allocates the state with `kvssd_ftl_alloc()`. The first KV namespace also answers controller-wide admin paths that name no namespace. A controller can mix KV namespaces with block namespaces (multi-namespace guide (§4.15)), but not with FDP.

### 4.12.3 Data structures

`FemuKvssdState` (`hw/femu/kvssd/kvssd.h`), one per KV namespace:

```

FemuKvssdState
+-----+
| table[hash_slots]    open-addressing index (DRAM)                |
|   FemuKvssdMappingEntry:                                         |
|   +-----+-----+-----+-----+-----+-----+-----+-----+ |
|   | key[16] | key_len | value_off | length | ppas[] | nr_ppas | |
|   +-----+-----+-----+-----+-----+-----+-----+-----+ |
|   key_len == 0 marks an empty slot                                |
+-----+
| values[value_capacity] byte arena holding the value bytes        |
|   [ live | dead | live | live | dead | ... | free ..... ]      |
|                                     ^ value_next                  |
|   value_used, key_used, value_reclaimable (dead bytes)          |
+-----+
| ssd      a black-box struct ssd built from the NAND properties: |
|           lines, write pointer, page states, LUN busy times.    |
|           Used for placement and timing only; it holds no data. |
+-----+
| ednek    Key Value Configuration feature (20h), bit 0          |
| lock     one mutex for all of the above                         |
+-----+

```

- **Index.** A hash table with linear probing on a 32-bit FNV-1a hash of the key bytes. Deletion shifts following entries back so probe runs stay intact. It has  $\min(2^{22}, 2 * (\text{value\_capacity} / 4096 + 1))$  slots, at least 1024, and holds at most `hash_slots - 1` keys; that limit is what the namespace reports as its maximum number of keys. The index lives in host memory and costs no flash reads.
- **Value arena.** The value bytes the host stores, appended at `value_next`. An overwrite or delete leaves the old bytes behind as dead space (`value_reclaimable`).
- **Value pages.** Each entry also records the NAND pages that hold its value (`ppas`),  $\text{ceil}(\text{length} / \text{page size})$  of them, where the page size is `secsz * secs_per_pg`. Those pages exist only in the timing model.

The two views are kept in step: the arena decides where the bytes are, and the NAND model decides what storing and reading them costs.

#### 4.12.4 Capacity

`kvssd_ftl_alloc()` sizes the store from two limits and takes the smaller:

```
value_capacity = min( namespace_size,
                      NAND_pages * page_size * gc_thres_pcent / 100 )
```

The share of NAND set aside by `gc_thres_pcent` (§A.1.1) is free space for reclaim, the over-provisioning a real KV-SSD keeps. The NAND size comes from the geometry properties (§A.1.1). Identify reports the result as the namespace size (NSZE) in bytes and the live key and value bytes as NUSE.

A store is refused with Capacity Exceeded when live keys and values would pass `value_capacity`, when the index is full, or when the NAND model has too few free pages for the value.

### 4.12.5 Commands

Commands use the common command format. `kvssd.c` reads these fields:

| Field           | Meaning                                                                            |
|-----------------|------------------------------------------------------------------------------------|
| CDW2, CDW3      | key bytes 0 to 7                                                                   |
| CDW14, CDW15    | key bytes 8 to 15                                                                  |
| CDW11 bits 7:0  | key length, 1 to 16 (List also accepts 0)                                          |
| CDW11 bits 15:8 | Store options: bit 8 store only if the key exists, bit 9 store only if it does not |
| CDW10           | value size (Store) or host buffer size (Retrieve, List)                            |
| data pointer    | the value or the list, PRP or SGL                                                  |

| Opcode | Command  | FTL function         | Result                                                                                     |
|--------|----------|----------------------|--------------------------------------------------------------------------------------------|
| 01h    | Store    | kvssd_ftl_store()    | appends the value, programs its pages, updates the index                                   |
| 02h    | Retrieve | kvssd_ftl_retrieve() | returns min(host buffer, value) bytes; Dword 0 of the completion holds the full value size |
| 10h    | Delete   | kvssd_ftl_delete()   | removes the key; a missing key succeeds unless EDNEK is set                                |
| 14h    | Exist    | kvssd_ftl_exist()    | success or Key Does Not Exist                                                              |
| 06h    | List     | kvssd_ftl_list()     | key count, then one entry per key: 2-byte length and key, padded to 4 bytes                |

Store and Retrieve share their opcodes with Write and Read, so the generic code counts them in the SMART host command totals, using the bytes really moved.

Limits on the command fields:

- Key length above 16: Invalid Field. Key length 0: Invalid Key Size (86h), except for List, where 0 means "from the start".
- Value above 2 MiB: Invalid Value Size (85h).
- A value, retrieved span or list larger than MDTs allows: Invalid Field. Store checks the value before the FTL runs; Retrieve checks the span inside the FTL, and only when the key exists; List checks its buffer size inside the FTL.
- A List buffer smaller than 4 bytes: Invalid Field.
- The broadcast namespace ID: Invalid Field.

List walks the index in slot order, starting at the start key's slot when the key exists and at slot 0 otherwise. The order is stable while the namespace does not change, which is what the specification asks.

Admin side (hw/femu/kvssd/kvssd-admin.c): Identify with CSI 01h answers the I/O command set specific Namespace and Controller structures and the format query (CNS 0Ah); there is one KV format with a 16-byte key, a 2 MiB value and the key limit above. Identify CNS 00h belongs to the NVM Command Set, so for a KV namespace it fails with Invalid I/O Command Set. A host reads the common attributes and readiness from CNS 08h, as Linux does, and the size and format from the KV structure above, which reports the same NMIC and ENDGID. Set and Get Features 20h (Key Value Configuration) read and write EDNEK.



### 4.12.6 Store, step by step

```
kvssd_ftl_store(key, vsize)           lock held throughout
1. find key in index
2. option or capacity check fails? -> charge base cost, return error
3. kv_value_alloc(vsize)
   frontier fits?                     -> take [value_next, value_next + vsize)
   else dead bytes exist?             -> kv_compact(), then retry
   else                               -> Capacity Exceeded
4. copy the value from the host into the arena
5. kv_program_ppas(): for each value page
   kv_ensure_write_pointer() (may erase empty lines first)
   get_new_page(); mark valid; charge a NAND program
   kv_advance_write_pointer() (channel, LUN, plane, then page)
6. index upsert; invalidate the old value's pages, count its bytes dead
7. latency: whichever ends last of the compaction in step 3, the base
   cost, the page programs and the reclaim erases
```

The write pointer walks a line the way the black-box FTL does: across channels, then LUNs, then planes, then down the pages of the block. When a line fills, it goes to the full list if all its pages are still valid, or to the victim queue if some were invalidated.

**Reclaim.** `kv_reclaim_empty_lines()` takes lines from the victim queue whose valid page count is zero and erases them, one multi-plane erase per LUN. It runs when a line closes, when the write pointer needs a new line, and when the FTL checks how much space is left. KV never moves valid pages out of a partly valid line.

**Compaction.** Compaction runs only when the value arena's frontier cannot fit a new value and there are dead bytes. `kv_compact()` then rewrites every live value to the front of the arena, in offset order, and programs new NAND pages for each as garbage collection writes. This charges the whole live set to the store that triggered it. If the NAND model has too few free pages for the whole live set, compaction is not attempted and the Store fails with Capacity Exceeded. A Store that finds too few free NAND pages for its own value also fails with Capacity Exceeded; that case does not try compaction either.

### 4.12.7 Timing

KV charges NAND operations through the black-box media model (`ssd_advance_status()`), so every operation queues on its LUN and channel like a black-box I/O. The NAND timing properties (`pg_rd_lat`, `pg_wr_lat`, `blk_er_lat`, the channel properties (§A.1.1)) set the costs.

Every command that reaches the index pays a **base cost**: one NAND page read, charged as a mapping read on the LUN that holds the key's part of the index (`kv_index_ppa()`). The index is spread over every LUN by the key's FNV-1a hash, LUN hash % (`nchs * luns_per_ch`), so lookups of different keys run in parallel; a List uses its start key. The cost stands for the controller's index work, so that commands that touch no value are not free. The index itself adds no reads. Commands refused for their fields, and most error paths after the index lookup, are charged nothing.

| Command      | Charged                                                                                                    |
|--------------|------------------------------------------------------------------------------------------------------------|
| Store        | whichever ends last of any compaction it triggers, the base cost, its page programs and any reclaim erases |
| Retrieve     | the reads of the value pages that are transferred, and the base cost; the longest of them                  |
| Exist        | the base cost                                                                                              |
| Delete, List | the longer of the base cost and the index cost, which is 0 for the hash index                              |

| Command                                                                                    | Charged       |
|--------------------------------------------------------------------------------------------|---------------|
| a missing key, a conditional Store that does not apply, a full index or a full value space | the base cost |

All costs are measured from the command's arrival time on the LUN timelines, so a LUN that is already busy makes the command wait, and a stage that queues behind an earlier stage of the same command already contains that wait. The command therefore ends with its last stage, and costs never add up: a Store whose own programs follow its compaction on a LUN completes when they do. The result is added to the request's `expire_time` (`kv_apply_lat()`). The host-link and firmware models (host link and controller firmware (§A.1.1)) also charge Store and Retrieve as they charge Write and Read. The timing model page (§6.2.8) gives the same rules in short.

#### 4.12.8 Parameters

KV reads only part of the black-box properties:

| Group                                   | Properties                                                                                                                                                                    | Use in KV                                         |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| NAND geometry (§A.1.1)                  | <code>nchs</code> , <code>luns_per_ch</code> , <code>pls_per_lun</code> , <code>blks_per_pl</code> , <code>pgs_per_blk</code> , <code>secs_per_pg</code> , <code>secsz</code> | layout of the value pages; NAND size              |
| NAND timing (§A.1.1)                    | <code>pg_rd_lat</code> , <code>pg_wr_lat</code> , <code>blk_er_lat</code> , cell type and channel properties                                                                  | cost of each operation                            |
| Garbage collection (§A.1.1)             | <code>gc_thres_pcent</code>                                                                                                                                                   | usable share of the NAND                          |
| Mode, capacity and namespaces (§A.1.1)  | <code>devsz_mb</code> , <code>namespaces</code> , <code>namespace_sizes</code> , <code>namespace_modes</code>                                                                 | namespace size, KV next to other modes            |
| Queues, pollers and interrupts (§A.1.1) | <code>mdts</code>                                                                                                                                                             | largest value, retrieve or list one command moves |

The mapping, cache, write buffer and background GC properties do not apply. KV runs the black-box geometry check, but not its capacity check: a namespace larger than the NAND still starts, and its value space is clamped as above. meta and FDP are refused for a KV namespace.

A KV namespace with a block namespace beside it on one controller (on the QEMU command line, the comma inside the value is written `, ,`):

```
| -device femu,devsz_mb=2048,femu_mode=5,namespaces=2,namespace_modes=kvssd,,bbssd
```

#### 4.12.9 Counters

KV keeps its NAND page counts on its struct `ssd`, split the way the black-box mode splits them, and the vendor log page C0h sums them over the controller's black-box, CSD and KV namespaces (log pages and counters (§B.1)):

- pages the host asked to program: value pages written by Store;
- pages relocated by garbage collection: value pages rewritten by compaction;
- pages programmed: the same as the first, since KV has no write buffer;
- reads of the most-read block: value page reads count against their block.

The SMART log counts Store and Retrieve as host writes and reads, with the bytes actually moved.

#### 4.12.10 Validation status

- qtest cases in `hw/femu/tests/qtest/femu-test.c`, run in CI: `kv-discovery` (the command set list and its command effects log), `kv-namespaces` (separate key spaces), `kv-accounting` (SMART bytes and the most-read block figure), `kv-base-spread` and `kv-compaction-time` (timing), `namespace-kv-byte-capacity`, `kv-mdts` and `kv-list-mdts0` (transfer limits), `kv-identify-reserved` (Identify layout), `sgl-kv`, and `kv-fuzz`, a fuzzer over the command fields.
- The documentation check stores and retrieves one value in each example whose namespace 1 is a KV namespace.
- `FEMU_KV_SELFTEST`, set in the environment, runs a self-test of the FTL at realize and logs the result (environment variables (§A.1.5)).
- In a guest, `hw/femu/scripts/kv-probe.c` and the KV checks of `femu-test.sh` run the full command set; they need Linux 6.0 or newer and are not run in CI.

#### 4.12.11 Limits

- One key-value format: keys up to 16 bytes, values up to 2 MiB.
- Values are held in host memory as well as in the timing model's page states, so a KV namespace costs host memory equal to its value capacity.
- Reclaim only erases lines with no valid pages, and compaction rewrites the whole live set at once, only when the value arena is full. A workload that leaves every line partly valid pays compaction in one command instead of steady background GC, or gets Capacity Exceeded when the NAND runs out first.
- The base cost of a key always falls on the same LUN, so a workload that hammers one key queues on that LUN.
- List order follows the hash table, not key order.
- Not with FDP or meta. A controller with a KV namespace offers no namespace management: it needs a NoSSD or black-box controller whose namespaces all share its mode.

Refusal messages are listed in the KV guide (§4.11.5).

#### 4.12.12 Extending the mode

- **Another index.** `FemuKvIndexOps` in `kvssd.h` is a small table of `find`, `upsert`, `remove` and `probe_reads`. An index kept on flash, such as an LSM tree, returns the number of page reads one lookup costs from `probe_reads`, and `kv_charge_index()` charges them as mapping reads on the key's index LUN, like the base cost; an index with real page addresses should charge those instead. Set `s->index` to the new table in `kvssd_ftl_alloc()`.
- **Background reclaim.** Valid-page relocation from partly valid lines would replace whole-arena compaction; the black-box GC in `hw/femu/bbssd/` shows the line and victim bookkeeping it would reuse.
- **More KV formats.** `kvssd_fill_id_ns()` reports one format; additional formats need a Format NVM path that changes `kv_key_max` and `kv_value_max`.

#### 4.12.13 Source map

| File                                             | Contents                                     |
|--------------------------------------------------|----------------------------------------------|
| <code>hw/femu/kvssd/kvssd.c</code> <sup>43</sup> | command parsing, registration, init and exit |

<sup>43</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/kvssd/kvssd.c>

| File                                                   | Contents                                                                   |
|--------------------------------------------------------|----------------------------------------------------------------------------|
| <code>hw/femu/kvssd/kvssd-ftl.c</code> <sup>44</sup>   | index, value arena, NAND placement, reclaim, compaction, timing, self-test |
| <code>hw/femu/kvssd/kvssd-admin.c</code> <sup>45</sup> | Identify structures, Key Value Configuration feature                       |
| <code>hw/femu/kvssd/kvssd.h</code> <sup>46</sup>       | FemuKvssdState, FemuKvssdMappingEntry, FemuKvIndexOps                      |
| <code>hw/femu/bbssd/ftl-media.c</code> <sup>47</sup>   | <code>ssd_advance_status()</code> , the media model KV charges through     |
| <code>hw/femu/nvme-admin.c</code> <sup>48</sup>        | Identify and feature routing to the KV handlers                            |

#### 4.12.14 Related pages

- KV guide (§4.11): command examples, status codes, troubleshooting
- Timing model: KV and CSD (§6.2.8)
- Multiple namespaces (§4.15)

### 4.13 Computational storage (CSD)

CSD mode (`femu_mode=4`) emulates a computational storage drive. On top of a normal NVMe namespace, the device has its own memory (function data memory, FDM) and compute units that run programs next to the data. The guest allocates device memory, copies namespace data into it, loads and runs a program, and reads the result back, all with NVMe commands. Ordinary reads and writes go through the BlackBox (§4.4) FTL, so they have SSD timing.

The mode is a port of CEMU<sup>49</sup>; if you use it, cite CEMU as well as FEMU (citation (§4.13.9)). It does not need CEMU's modified guest kernel, FDMFS or a fixed VM image. CEMU's VM freezing and virtual clock changes are left out of this port.

Use it to study offloading filters, scans, compression or other kernels to the drive, and what that does to latency and host CPU use.

#### 4.13.1 Requirements

- Host and guest: see the mode table (§4.1.4) and `requirements.md` (§2.1.2). Any guest kernel with the NVMe driver works.
- Guest tools: `hw/femu/tests/csd`, built in the guest with a C compiler. The image from `make-guest-image.sh` has none; add one with `./make-guest-image.sh --packages build-essential`, or run `sudo apt install build-essential` in the guest.
- Programs: shared-library programs are built on the host and placed in the directory named by `csd_program_dir`. eBPF programs also need a FEMU build with uBPF (`./femu-compile.sh --enable-csd-ubpf`, see `build.md` (§2.2.4)) and clang with the BPF target on the host.

<sup>44</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/kvssd/kvssd-ftl.c>

<sup>45</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/kvssd/kvssd-admin.c>

<sup>46</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/kvssd/kvssd.h>

<sup>47</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/bbssd/ftl-media.c>

<sup>48</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nvme-admin.c>

<sup>49</sup><https://github.com/cs-qy Zhang/CEMU>

### 4.13.2 Security: programs run inside QEMU on the host

A program load names a file. QEMU, on the host, opens that file and runs code from it inside the QEMU process. The `run-*.sh` launchers start QEMU with `sudo`, so that code runs as root on the host.

FEMU limits what the guest can name:

- With `csd_program_dir` unset, only the built-in phantom program type loads. Shared-library and eBPF loads fail.
- With it set, the guest may name only a file in that directory: no `/`, no `...`, and the path must still resolve inside the directory after symbolic links are followed.

That protects the rest of the host file system. It does not make the programs in the directory safe: they run on QEMU's compute unit threads with all of QEMU's privileges and memory. Use CSD only with guests you trust, put only programs you built and trust in the directory, and make sure no one else can write to it.

### 4.13.3 Launch

From `build-femu/`:

```
| ./run-csd.sh
```

The FEMU device in that script is:

```
| -device femu,devsz_mb=4096,namespaces=1,femu_mode=4,secsz=512,secs_per_pg=8,pgs_per_blk=256,blks_per_pl=256,pls_per_lun=1,luns_per_ch=8,nchs=8,pg_rd_lat=40000,pg_wr_lat=200000,blk_er_lat=2000000,ch_xfer_lat=0,gc_thres_pcent=75,gc_thres_pcent_high=95,fdm_size=64,nr_cu=4,nr_thread=4,time_slice=200000,context_switch_time=200,csf_runtime_scale=3
```

The preset `hw/femu/scripts/configs/csd.conf` describes the same device without the three scheduler properties that have no effect. `ssd-config.sh` (§5.11) expands it to:

```
| -device femu,id=nvme0,devsz_mb=4096,namespaces=1,secsz=512,secs_per_pg=8,pgs_per_blk=256,blks_per_pl=256,pls_per_lun=1,luns_per_ch=8,nchs=8,pg_rd_lat=40000,pg_wr_lat=200000,blk_er_lat=2000000,gc_thres_pcent=75,gc_thres_pcent_high=95,fdm_size=64,nr_cu=4,csf_runtime_scale=3,femu_mode=4
```

`run-csd.sh` does not set `csd_program_dir`, so it runs phantom programs only. To load your own programs, add the directory after the other `FEMU_OPTIONS` lines in `run-csd.sh`:

```
| FEMU_OPTIONS=${FEMU_OPTIONS}" ,csd_program_dir=$HOME/csd-programs"
```

The shortest device line with programs enabled is:

```
| -device femu,devsz_mb=4096,femu_mode=4,fdm_size=64,csd_program_dir=/home/you/csd-programs
```

FEMU does not check the directory at start-up. A directory that does not exist makes every program load fail.

### 4.13.4 Configuration

Properties: CSD (§A.1.1). The BlackBox geometry, timing, GC and FTL properties also apply to the namespace; see the BlackBox guide (§4.4.3).

- `fdm_size`: device memory in MiB. Required.
- `nr_cu`: compute units, 1 to 64. Each is a host thread, `femu-csd-cu`, that runs programs. A program waits for the first free unit.

- `csf_runtime_scale`: a program that declares no run time and no scale of its own holds its unit for its measured host run time times this value (default 3). A completion carries the program's result, so it never arrives before the host has run the program: a run time, declared or scaled, shorter than the host's own is not reached. QEMU warns once when that happens.
- `csd_program_dir`: the host directory programs load from.
- `nr_thread`, `time_slice` and `context_switch_time` are accepted so CEMU configurations still start. They have no effect, and a value other than the default prints a warning at realize.

A copy from the namespace into device memory costs one `pg_rd_lat`, whatever its size, when any page of the range has been written, and nothing otherwise. See the timing model (§6.2.8).

## Program types

| Type           | What runs                                                                                                                                  | Needs                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Phantom        | Built in: copies the input memory range to the output range. Useful to test the command flow and timing.                                   | nothing                                                                      |
| Shared library | A function with the signature <code>int64_t fn(struct femu_csd_args *args)</code> from a <code>.so</code> file, named in the load command. | <code>csd_program_dir</code>                                                 |
| uBPF           | An eBPF ELF object, interpreted or JIT-compiled.                                                                                           | <code>csd_program_dir</code> and a build with <code>--enable-csd-ubpf</code> |

## How programs run

An Execute command is checked on the poller and then handed to one of the `nr_cu` compute unit threads, which runs the program and posts the completion. While a program runs, I/O on every queue, other CSD commands and admin commands go on as usual; only that compute unit is busy.

- At most `nr_cu` programs run at once. Runs of one program take turns, so a shared library needs no locking of its own; different programs run side by side, and a run waiting for its own program leaves the compute unit free for another.
- A program that never returns keeps its compute unit for good, and QEMU waits for it when the device is removed. FEMU cannot stop native code it has called.
- Device memory a program is using stays allocated, and counted against `fdm_size`, until the program returns, even after Free device memory.
- Deleting the I/O queue an Execute came from, or resetting the controller, while the program runs drops its result; the program still runs to the end.

## Commands

I/O commands on the namespace:

| Opcode           | Command                                |
|------------------|----------------------------------------|
| 0xb0             | Allocate device memory                 |
| 0xc0             | Free device memory                     |
| 0xd0             | Copy namespace data into device memory |
| 0xe1             | Execute a program                      |
| 0xf2             | Read device memory                     |
| 0xf5             | Write device memory                    |
| 0xf6, 0xf7, 0xf8 | Create a group, set its QoS, delete it |

Admin commands: 0x21 memory range set management, 0x22 program load and unload, 0x23 program activate and deactivate, 0x25 load program data. The command layouts follow CEMU; `hw/femu/tests/csd/csd-passthru.c` builds each one.

### 4.13.5 Use it from the guest

Check the device:

```
sudo nvme list
sudo nvme id-ctrl /dev/nvme0 | grep -E '^(mn|sn) '
```

The model is FEMU Computational Storage Controller and the serial number starts with vCSD.

#### Build the guest tool

From `build-femu/` on the host, copy the tools into the guest and build the passthrough helper:

```
scp -P 8080 -i ~/images/femu-guest-key -r ../hw/femu/tests/csd femu@localhost:
./run-guest-ssh.sh make -C csd csd-passthru
```

#### Run the phantom smoke test

This needs no program directory. It allocates, writes and reads device memory, loads and runs a phantom program, and frees everything:

```
./run-guest-ssh.sh sudo ./csd/csd-passthru /dev/nvme0n1 smoke
```

#### Run a shared-library program

On the host, build the example programs and put them in the program directory (`csd-original-kernels.so` needs the lz4 development package):

```
cd hw/femu/tests/csd # in the FEMU source tree
make csd-vadd.so csd-original-kernels.so
mkdir -p ~/csd-programs
cp csd-vadd.so csd-original-kernels.so ~/csd-programs/
```

Start FEMU with `csd_program_dir` set to that directory, then name the file alone in the guest:

```
sudo ./csd/csd-passthru /dev/nvme0n1 smoke-so csd-vadd.so
sudo ./csd/csd-passthru /dev/nvme0n1 smoke-so-all csd-original-kernels.so
sudo ./csd/csd-passthru /dev/nvme0n1 bench 4096 32
```

`bench` reports the average latency of device memory writes, reads and namespace-to-memory copies. `hw/femu/tests/csd/README.md`<sup>50</sup> lists every subcommand, the eBPF steps and the program ABI.

The vendor log page C0h counts the namespace's NAND traffic as for BlackBox (log pages and counters (§B.1)).

### 4.13.6 Limits and refusals

| Message                                                | Cause and fix                    |
|--------------------------------------------------------|----------------------------------|
| CSD mode requires <code>fdm_size</code> to be non-zero | Set <code>fdm_size</code> .      |
| CSD <code>nr_cu</code> must be in range [1, 64]        | <code>nr_cu</code> out of range. |
| CSD <code>nr_thread</code> must be non-zero            | <code>nr_thread</code> =0.       |

<sup>50</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/tests/csd/README.md>

| Message                                                                                     | Cause and fix                                                                                                                                                         |
|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CSD csf_runtime_scale must be non-zero<br>csd supports at most one namespace per controller | csf_runtime_scale=0.<br>Two CSD namespaces, from namespace_modes or from femu_mode=4 with namespaces above 1. Other namespaces of the controller may use other modes. |
| FEMU bbssd: namespace 1 exposes ...                                                         | The namespace does not fit the NAND geometry; see the BlackBox limits (§4.4.5).                                                                                       |
| FEMU csd: buffer_size has no effect under FDP                                               | A knob the FDP write path ignores, on a controller in an FDP subsystem; the same list as for BlackBox (FDP (§4.5)).                                                   |

A program load that fails returns Invalid Field to the guest, or Capacity Exceeded when the program table is full. For a missing or bad program file, QEMU prints the reason on its console after [FEMU] Err:, for example:

- CSD: loading a program needs csd\_program\_dir to be set
- CSD: a program name must be a file in csd\_program\_dir, got "..."
- CSD: <path> does not resolve inside csd\_program\_dir
- CSD: failed to load shared library <path>: <reason>

An eBPF load on a build without uBPF, a malformed load descriptor, or an unknown program type fails with Invalid Field and prints nothing.

#### 4.13.7 Verify

1. `sudo nvme id-ctrl /dev/nvme0` reports FEMU Computational Storage Controller.
2. `csd-passthru /dev/nvme0n1 smoke` completes without errors.
3. With `csd_program_dir` set, `smoke-so csd-vadd.so` completes; without it, the same command fails and the QEMU console says why.

#### 4.13.8 Troubleshooting

- **A program load fails with Invalid Field.** Read the QEMU console (or `build-femu/log`). Usually `csd_program_dir` is unset, the name contains a path, or the file is not in the directory. No message means an eBPF load on a build without uBPF, or a malformed load command.
- **An eBPF load fails.** The FEMU build has no uBPF support. Rebuild with `./femu-compile.sh --enable-csd-ubpf`.
- **make in the guest fails on csd-original-kernels.so.** It needs the lz4 development package and is meant to be built on the host. Build only `csd-passthru` in the guest.

Related issues: #60, #143, #188.

#### 4.13.9 Citation

CSD mode is derived from CEMU<sup>51</sup>. We thank the CEMU authors, Qiuyang Zhang, Jiapin Wang, You Zhou, Peng Xu, Kai Lu, Jiguang Wan, Fei Wu and Tao Lu, and Emilio (@Emilio597<sup>52</sup>), who ported it to FEMU in #188<sup>53</sup>. If you use the CSD mode, please also cite:

<sup>51</sup><https://github.com/cs-qy Zhang/CEMU>

<sup>52</sup><https://github.com/Emilio597>

<sup>53</sup><https://github.com/MoatLab/FEMU/pull/188>



```
@inproceedings{Zhang+26-CEMU,
  author    = {Qiuyang Zhang and Jiapin Wang and You Zhou and Peng Xu and
              Kai Lu and Jiguang Wan and Fei Wu and Tao Lu},
  title     = {{CEMU: Enabling Full-System Emulation of Computational Storage
              Beyond Hardware Limits}},
  booktitle = {Proceedings of the 31st ACM International Conference on
              Architectural Support for Programming Languages and Operating
              Systems (ASPLOS '26), Volume 2},
  pages     = {323--341},
  year      = {2026},
  doi       = {10.1145/3779212.3790137},
}
```

#### 4.13.10 Related pages

- [hw/femu/tests/csd/README.md](#)<sup>54</sup>: the guest tool and program ABI
- BlackBox SSD (§4.4): the FTL under the namespace
- Timing model: KV and CSD (§6.2.8)

### 4.14 CSD: the computational storage extension

This chapter describes how FEMU emulates a computational storage drive (`femu_mode=4`): the device memory, how programs are loaded and run, how compute time is charged, and what the guest is trusted with. To run the mode, see the CSD guide (§4.13).

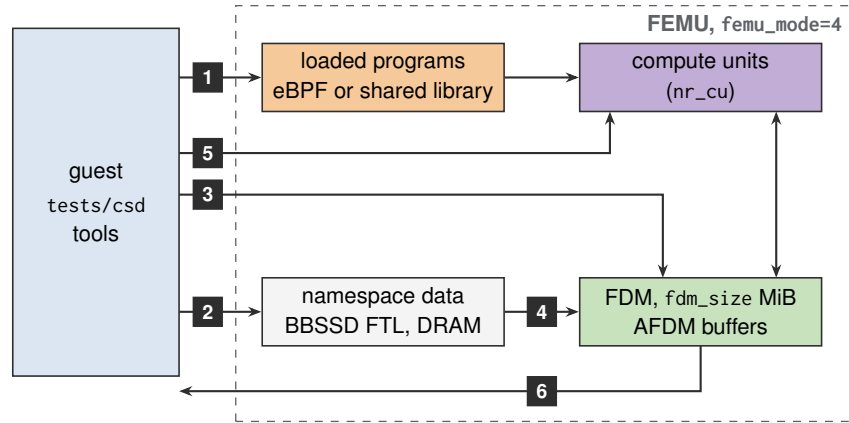
#### 4.14.1 Origin and credit

CSD mode is derived from CEMU, by Qiuyang Zhang, Jiapin Wang, You Zhou, Peng Xu, Kai Lu, Jiguang Wan, Fei Wu and Tao Lu (ASPLOS '26), and was ported to FEMU by Emilio (@Emilio597<sup>55</sup>) in #188<sup>56</sup>. The command layouts follow CEMU's. The port keeps CEMU's device-side model and leaves out its modified guest kernel, its FDMFS file system, and its VM freezing and virtual clock changes. If you use this mode, cite CEMU as well as FEMU; the BibTeX entry is in the CSD guide's citation section (§4.13.9).

#### 4.14.2 Purpose

A computational storage drive runs programs next to the data. The host allocates memory on the device, copies namespace data into it without a trip through host memory, runs a program on that memory, and reads back only the result. FEMU emulates the control path of such a device with NVMe commands, and runs each program inside the QEMU process on the host. The program's time on the device is modelled with a small set of compute units.

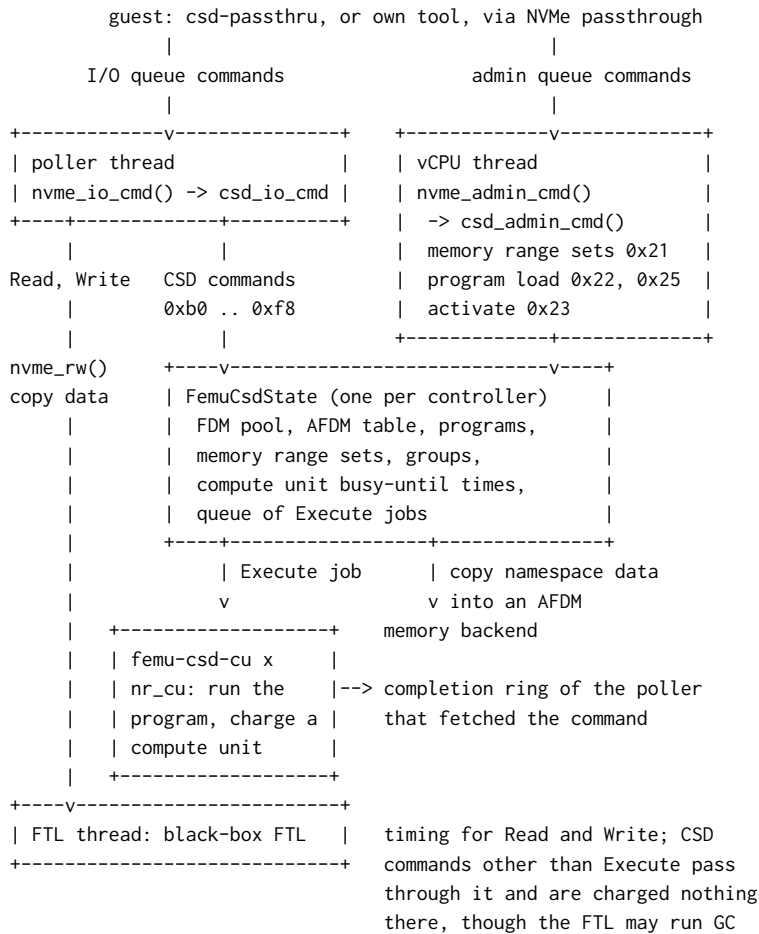
The namespace under the device memory is an ordinary block namespace on the black-box FTL, so plain reads and writes have SSD timing and garbage collection.



**1** Compute Load `0x22`, Compute Activate `0x23` (admin): load a program; shared libraries come from `csd_program_dir`. **2** Write the input with ordinary NVMe writes. **3** Alloc FDM `0xb0`: reserve a buffer (AFDM) in device memory. **4** NVM to AFDM `0xd0`: copy namespace data into the buffer, charging a page read when the pages are mapped. **5** Exec `0xe1`: run the program on the buffer on the first free compute unit; it is charged its own runtime, or the host time it took times `csf_runtime_scale`. **6** Read AFDM `0xf2`: copy the result to the guest; Dealloc AFDM `0xc0` frees the buffer.

**Figure 20:** Computational storage: load a program, stage namespace data in a device memory buffer, execute it on a compute unit, and read the result back.

### 4.14.3 Place in the hierarchy



csd\_init() runs the black-box geometry and capacity checks, builds a full black-box FTL for the namespace (ns->ssd), and allocates FemuCsdState. The state is controller-wide (n->csd\_ctrl\_state), which is why a controller may have only one CSD namespace. Its other namespaces may run other modes, but the CSD admin commands (0x21, 0x22, 0x23, 0x25) reach csd\_admin\_cmd() only through the controller's own handler table. On a controller whose femu\_mode is not 4, for example femu\_mode=2 with namespace\_modes=nossd, , csd, those commands fail with Invalid Opcode, so programs cannot be loaded. Set femu\_mode=4 on a controller that should run programs. A CSD namespace makes the controller start its FTL thread.

### 4.14.4 Data structures

FemuCsdState in hw/femu/csd/csd.c, guarded by one mutex. No program runs under it:

<sup>54</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/tests/csd/README.md>

<sup>55</sup><https://github.com/Emilio597>

<sup>56</sup><https://github.com/MoatLab/FEMU/pull/188>

```

FemuCsdState
+-----+
| fdm_capacity = fdm_size MiB      fdm_used      |
| afdms      id  -> FemuCsdAfdm { id, size, data } |
| programs   pind -> FemuCsdProgram { type, active, loading, |
|              runtime, runtime_scale, size, data, |
|              module + function, or uBPF VM + JIT } |
| mrs        rsid -> FemuCsdMrs { numr, ranges[] } |
| groups     id  -> FemuCsdGroup { prio, qos_flags, bandwidth, |
|              deadline } |
| cu_next_avail[nr_cu]  when each compute unit is free again |
| prog_used             bytes held by loaded programs |
| workers[nr_cu]        femu-csd-cu threads |
| pending, running      FemuCsdJob { req, poller, program, |
|              AFDMs, args, runtime } |
+-----+

```

AFDMs and programs are reference counted. The table holds one reference, and each Execute that is queued or running holds one on its program and on each AFDM it names. Deallocating an AFDM or unloading or replacing a program takes it out of its table at once, but its memory, and its share of the `fdm_size` quota or of the program limits, is given back only when the last Execute using it has finished. A program whose last reference goes is closed by a compute unit thread without the CSD lock, since closing a library runs its destructors.

- **FDM and AFDM.** Function data memory (FDM) is the device memory pool, `fdm_size` MiB. An allocation from it (AFDM) is a buffer in host memory with an id. `fdm_size` is a quota: FEMU allocates each AFDM when the guest asks for it, and refuses an allocation that would pass the quota with Capacity Exceeded. Only the host-visible type (0) is supported.
- **Memory range.** A 32-byte descriptor {nsid, len, sb}. FEMU accepts only nsid 0, meaning device memory, where sb is an AFDM id and len is the bytes to use (0 for the whole AFDM).
- **Memory range set.** A stored list of up to 255 memory ranges with an id (rsid), so a program can be run many times without sending its ranges again. The ranges are checked only when a program runs.
- **Program.** Indexed by pind (1 to 65535). One program may be up to 16 MiB and all loaded programs together up to 64 MiB.
- **Group.** A priority (1 to 9, 0 means 5), QoS flags, bandwidth and deadline, kept as CEMU defines them. An execute command may name a group, which must exist, but groups do not change scheduling. The group field of Execute is 8 bits wide, so only groups 1 to 255 can be named.

#### 4.14.5 Commands

I/O commands, on the CSD namespace (`csd_io_cmd()`):

| Opcode     | Command         | Effect                                                                                                        | Time charged                                                                        |
|------------|-----------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 0x01, 0x02 | Write, Read     | <code>nvme_rw()</code> then the black-box FTL                                                                 | black-box FTL timing                                                                |
| 0xb0       | Allocate FDM    | new AFDM of size bytes; Dword 0 = id                                                                          | none                                                                                |
| 0xc0       | Deallocate AFDM | frees it, returns its bytes to the quota                                                                      | none                                                                                |
| 0xd0       | NVM to AFDM     | copy <code>nlb + 1</code> blocks ( <code>nlb</code> is 0-based) from <code>slba</code> into an AFDM at offset | one <code>pg_rd_lat</code> if any page of the range is mapped in the FTL, else none |
| 0xe1       | Execute         | run a program on memory ranges; Dword 0 = its return value                                                    | compute unit time, below                                                            |

| Opcode              | Command                                | Effect                  | Time charged |
|---------------------|----------------------------------------|-------------------------|--------------|
| 0xf2                | Read AFDM                              | AFDM bytes to the host  | none         |
| 0xf5                | Write AFDM                             | host bytes into an AFDM | none         |
| 0xf6, 0xf7,<br>0xf8 | Create group, Set QoS,<br>Delete group | group bookkeeping       | none         |

Admin commands (csd\_admin\_cmd()):

| Opcode     | Command                                                                                                                  |
|------------|--------------------------------------------------------------------------------------------------------------------------|
| 0x21       | Memory range set management: sel 0 creates a set from numr ranges in the data buffer (Dword 0 = rsid), sel 1 deletes one |
| 0x22, 0x25 | Load program, or load more of its bytes; sel 1 unloads                                                                   |
| 0x23       | Activate (sel 1) or deactivate (sel 0) a program                                                                         |

hw/femu/tests/csd/csd-passthru.c builds every one of these commands and is the reference for their layouts; hw/femu/csd/csd.h declares them.

NVM to AFDM reads the namespace's slice of the memory backend directly and checks the black-box mapping table only to decide whether to charge the read. It does not go through the FTL's LUN queues. Pages still held in the write buffer and deallocated pages are not mapped, so a range of only those costs nothing.

When oncs turns them on, the generic NVM commands (Dataset Management, Compare, Write Zeroes and the others) are handled by the common layer in nvme\_io\_cmd() before csd\_io\_cmd() is consulted.

#### 4.14.6 Loading a program

```
(no program at pind)
|
| load, loff = 0: size psize, type, runtime, runtime_scale
v
+-----+ load, loff > 0: more bytes (same size, type, pid)
| LOADING |<--+
+-----+
| all psize bytes received: resolve and open by type
v
+-----+ activate sel=1 +-----+
| INACTIVE |----->| ACTIVE |
|          |<-----|          |
+-----+ activate sel=0 +-----+
| load sel=1 (only when inactive)
v
(no program at pind)
```

The diagram simplifies a few edges. A load with loff = 0 replaces whatever the index held, even an active program; an Execute already queued or running keeps the old one. A further chunk (loff > 0) is accepted only while the program is still loading, so a loaded program never changes. A load with sel=1 also unloads a program that is still loading. A further chunk must repeat the size and type, and the pid only when the command's pit field is 1. Every chunk adds its byte count toward the size, so a chunk sent twice counts twice. When the byte count reaches the size, csd\_load\_program\_data() prepares the program by type, without the CSD lock held: opening a library runs its constructors, and the pollers must not wait for them. The program is still loading then, so nothing can run it, and admin commands, the only ones that change programs, run one

at a time under QEMU's global lock:

| Type           | Value | Payload        | Prepared by                                                                                                       |
|----------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------|
| Phantom        | 0     | none needed    | nothing; runs built-in code                                                                                       |
| eBPF           | 1     | name\0symbol\0 | read the ELF file, load the section into a uBPF VM, JIT-compile it when the command's <code>jit</code> bit is set |
| Shared library | 3     | name\0symbol\0 | <code>g_module_open()</code> the file and look up the symbol                                                      |

Bitstream programs (type 2) and any other type fail with Invalid Field, but only when the last byte arrives. If preparing fails, the load command fails and the program stays at that index in the loading state until a new load replaces it.

### Program files and the trust model

`csd_program_path()` turns the guest's name into a host path:

- With `csd_program_dir` (§A.1.1) unset, every shared-library and eBPF load fails; only phantom programs load.
- The name must be a plain file name: not empty, no `/`, not `.` or `...`
- The joined path is resolved with `realpath()`, symbolic links included, and must still lie inside the resolved directory.

This keeps the guest from naming any other file on the host. It does not make the programs safe. A shared library runs as native code in the QEMU process, with QEMU's privileges and access to all of its memory, and the launchers start QEMU with `sudo`. A uBPF program runs in uBPF's interpreter, or as native code when JIT-compiled, also inside QEMU; FEMU adds no checks of its own to what uBPF does. So the directory must hold only programs you built and trust, and must not be writable by anyone else. FEMU does not check the directory at realize; a missing one makes every load fail.

eBPF needs a build with uBPF (`./femu-compile.sh --enable-csd-ubpf`, meson option `femu_csd_ubpf`). Without it, an eBPF load fails with Invalid Field.

### 4.14.7 Running a program

Execute (0xel), on the poller thread (csd\_exec):

1. read numr inline ranges (+ optional extra data, dlen <= 1 MiB) from the data buffer, or take the stored set named by rsid
2. lock; check the program exists and is active, and the group exists
3. map each range to its AFDM buffer (nsid 0, sb = AFDM id), and take a reference on each AFDM and on the program
4. runtime = command runtime, else program runtime; if there is one, take the compute unit now (see below); unlock
5. hand the request to the CSD as a job (req->defer); the poller moves on without sending it to the FTL thread

on one of nr\_cu femu-csd-cu threads (csd\_worker):

6. take the oldest job whose program is not running already, mark the program running, and call it without the CSD lock, timing it with the host clock:
  - phantom copy range 1 into range 0
  - shared lib fn(&args)
  - eBPF JIT function or ubpf\_exec(&args)
7. lock; Dword 0 = return value (above 2<sup>32</sup> - 1 clamped to that; a negative value keeps its low 32 bits)
8. with no runtime from step 4: runtime = measured host time x scale, and take the compute unit now
9. put the request on the completion ring of the poller that fetched it, retrying while the ring is full; drop the references; unlock

Taking the compute unit means: pick the unit that is free first, hold it for the runtime from the command's arrival, and set expire\_time to the end of that hold. A known runtime takes its unit on the poller, in arrival order, so the order in which the host happens to finish programs does not change it; a failed run still holds it. A measured runtime is known only after the run, so those take their unit in the order runs end.

The poller then posts the completion once expire\_time has passed, as for any other command. A long program therefore holds one femu-csd-cu thread and nothing else: the pollers keep serving every queue, CSD admin commands and other CSD I/O commands take the CSD lock only briefly, and the vCPU that sends an admin command never waits for a program.

When a submission queue is deleted, or the controller is reset, while one of its Execute commands is queued or running, nvme\_drain\_sq() first calls femu\_csd\_drain\_sq(), which frees its queued jobs and detaches the request from a running one. A running program still runs to the end, but its result is dropped, as the specification allows for commands on a deleted queue, and nothing touches the freed request.

The program sees one argument, struct femu\_csd\_args (hw/femu/tests/csd/femu-csd-kernel.h):

|           |                  |                                                                                |
|-----------|------------------|--------------------------------------------------------------------------------|
| int       | numr             | number of memory ranges                                                        |
| void      | **mr_addr        | one pointer per range; by convention<br>mr_addr[0] is output, mr_addr[1] input |
| long long | *mr_len          | bytes per range                                                                |
| long long | cparam1, cparam2 | from the command                                                               |
| void      | *data_buffer     | extra bytes after the ranges, or NULL                                          |
| long long | buffer_len       |                                                                                |

### Compute units and runtime scaling

The program runs to completion on the host as soon as a femu-csd-cu thread is free; the compute units' busy-until times then decide when the guest sees the completion. This is the same "compute, then hold" approach the rest of FEMU uses (timing model (§6.2.1)).

```
runtime (ns), first that applies:
1. the runtime field of the Execute command
2. the runtime given when the program was loaded
3. measured host time x program runtime_scale / 10
4. measured host time x csf_runtime_scale

compute units (nr_cu = 3):
cu0  |==== A ====|
cu1  |== B ==|    |===== D =====|
cu2  |===== C =====|
      ^ D arrives: cu1 is already free, D starts at once

cu  = the unit with the smallest cu_next_avail
start = max(arrival, cu_next_avail[cu])
cu_next_avail[cu] = start + runtime
expire_time = cu_next_avail[cu]
```

nr\_cu (§A.1.1) bounds how many programs run at once in device time; csf\_runtime\_scale (§A.1.1) stands for how much slower the device's cores are than the host's.

The completion carries the program's return value, so the poller cannot post it before the worker has finished the run. A runtime shorter than the host's own run time, declared or scaled below 1, is therefore not reached: the guest sees the host's time. csd\_finish\_locked() warns once when a run's runtime is shorter than its measured host time.

#### 4.14.8 Parameters

Properties: CSD (§A.1.1). The NAND geometry (§A.1.1), NAND timing (§A.1.1) and garbage collection (§A.1.1) properties apply to the namespace as in the black-box mode.

| Property                                   | Role                                                                                                                  |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| fdm_size                                   | device memory quota in MiB; required                                                                                  |
| nr_cu                                      | compute units, 1 to 64                                                                                                |
| csf_runtime_scale                          | host-to-device time factor for programs that give no runtime; not 0                                                   |
| csd_program_dir                            | where shared-library and eBPF programs load from                                                                      |
| pg_rd_lat                                  | the cost of an NVM to AFDM copy                                                                                       |
| nr_thread, time_slice, context_switch_time | accepted for CEMU configurations; no effect, and a non-default value warns at realize (nr_thread must still not be 0) |

The smallest device line that loads programs:

```
| -device femu,devsz_mb=4096,femu_mode=4,fdm_size=64,nr_cu=2,csd_program_dir=/home/you/csd-programs
```

#### 4.14.9 Counters

- The vendor log page C0h counts the namespace's NAND traffic as for the black-box mode (log pages and counters (§B.1)). NVM to AFDM copies do not appear there, because they do not go through the FTL.
- The SMART log counts plain Read and Write. CSD commands are not counted.
- FEMU keeps no counter of programs run or compute unit use. Time the Execute commands from the guest; csd-passthru benchmark-kernels times whole program runs, and csd-passthru bench times the device memory and NVM to AFDM transfers.



#### 4.14.10 Validation status

- qtest cases in CI: `csd-program-dir`, run with the property set, checks that a missing file, a symbolic link out of the directory and a file that is not a library are refused; `csd-fuzz` fuzzes the fields of the CSD I/O commands; `csd-exec-concurrency` builds a shared-library program that sleeps (it skips when the host has no C compiler), loads and runs it, and checks that a read on another queue, an admin command and another program complete while it runs, that a loaded program refuses a further load piece, and that deleting the program's queue or disabling the controller under it leaves the controller working. No qtest covers the case where `csd_program_dir` is unset, or the `/`, `.` and `..` name rules. Phantom and eBPF programs and compute unit timing are tested only in a guest.
- The documentation check starts each CSD example and writes and reads one block of the namespace.
- Shared-library and eBPF programs run only in a guest, with `hw/femu/tests/csd/csd-passthru` and the example programs in `hw/femu/tests/csd/`. When `femu-test.sh` finds a CSD namespace, it runs a smoke check that allocates device memory, copies into it and reads it back; it runs no program. CI does not build uBPF, so the eBPF path is not tested there.

#### 4.14.11 Limits

- Programs run on `nr_cu` host threads, one per compute unit, so at most `nr_cu` run at once on the host; further Execute commands queue for a thread. Two runs of the same program never overlap; different programs do. A program that never returns keeps its thread and, at device removal (`device_del`), keeps QEMU waiting for it; FEMU cannot stop native code it has called.
- A program runs on the AFDMs it was given while other commands may read or write the same AFDMs; FEMU does not order them. Device memory freed while a program uses it counts against `fdm_size` until the program returns.
- The guest's program runs in the QEMU process. Treat `csd_program_dir` as code you are choosing to run on the host.
- NVM to AFDM charges one page read whatever its size, and does not queue on LUNs or channels.
- Groups and their QoS settings are stored but do not affect scheduling.
- Device memory transfers (Read and Write AFDM) take no device time.
- One CSD namespace per controller. No meta and no namespace management.

Refusal messages are listed in the CSD guide (§4.13.6).

#### 4.14.12 Extending the mode

- **A new program type.** Add a case to `csd_load_program_data()` to prepare it, and one to the switch in `csd_run_program()` to run it.
- **Group scheduling.** `csd_exec()` already looks up the group. Choosing the compute unit by group priority or deadline would happen in `csd_finish_locked()`, where it picks the unit with the smallest `cu_next_avail`, and the order jobs leave the queue in `csd_worker()`.
- **Copy cost from the FTL.** `csd_nvm_to_afdm()` has the mapped page count from `csd_check_nvm_ftl_range()`; charging each mapped page through the media model (`ssd_advance_status()`) instead of one `pg_rd_lat` would add LUN and channel contention.
- **Preemption.** A compute unit thread runs a program to the end. A time slice, which CEMU's `time_slice` and `context_switch_time` describe, would need programs that yield.

#### 4.14.13 Source map

| File                                          | Contents                                                                   |
|-----------------------------------------------|----------------------------------------------------------------------------|
| <code>hw/femu/csd/csd.c</code> <sup>57</sup>  | state, every CSD command, program loading, compute unit threads, execution |
| <code>hw/femu/csd/csd.h</code> <sup>58</sup>  | command layouts, opcodes, program types                                    |
| <code>hw/femu/tests/csd/</code> <sup>59</sup> | guest tool, example programs, program ABI                                  |
| <code>hw/femu/bbssd/</code> <sup>60</sup>     | the black-box FTL under the namespace                                      |
| <code>hw/femu/femu.c</code> <sup>61</sup>     | the one-CSD-namespace check, FTL thread start                              |

#### 4.14.14 Related pages

- CSD guide (§4.13): launching, guest tool, citation
- Timing model: KV and CSD (§6.2.8)
- BlackBox SSD guide (§4.4): the FTL under the namespace

### 4.15 Several namespaces and devices

One FEMU controller can expose several namespaces, and each namespace can run its own mode. A controller can, for example, offer a BlackBox namespace for random writes next to a ZNS namespace for zoned data. You can also start several FEMU controllers in one guest.

Namespaces created this way exist from boot. To create and delete namespaces from the guest while it runs, see namespace management (§4.17).

#### 4.15.1 Requirements

- Host and guest: see the mode table (§4.1.4) and requirements.md (§2.1.2). Each namespace needs what its own mode needs: a ZNS namespace needs a guest kernel with ZNS support, and a KV namespace gets no block device.
- Host memory: `devsz_mb` is split between the namespaces, so the total does not grow with the namespace count. Each extra controller adds its own `devsz_mb`.

#### 4.15.2 Launch

There is no launcher for this. Take any `run-*.sh` script and change its `-device femu` line.

Two BlackBox namespaces that split 4 GiB evenly:

```
| -device femu,devsz_mb=4096,femu_mode=1,namespaces=2
```

The same with explicit sizes of 3 GiB and 1 GiB. A comma inside a property value is written twice on the QEMU command line:

```
| -device femu,devsz_mb=4096,femu_mode=1,namespaces=2,namespace_sizes=3G,,1G
```

Four namespaces, each in a different mode:

```
| -device femu,devsz_mb=4096,femu_mode=1,namespaces=4,namespace_modes=nossd,,bbssd,,znssd,,kvssd
```

<sup>57</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/csd/csd.c>

<sup>58</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/csd/csd.h>

<sup>59</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/tests/csd/README.md>

<sup>60</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/bbssd/ftl.c>

<sup>61</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/femu.c>

Two controllers, one BlackBox and one ZNS, each with its own memory:

```
| -device femu,id=nvme0,devsz_mb=1024,femu_mode=1 -device femu,id=nvme1,devsz_mb=1024,femu_mode=3
```

### 4.15.3 Configuration

Properties: mode, capacity and namespaces (§A.1.1).

- namespaces: 1 to 256 namespaces.
- namespace\_sizes: the size of each namespace, in bytes or with a QEMU size suffix (4G, 512M). One entry per namespace, each at least one 512-byte sector; the sum may be all of devsz\_mb. Unset, devsz\_mb is split evenly, each share rounded down to 512 bytes. Each size, given or split, is then rounded down to whole logical blocks ( $512 \ll \text{lba\_index}$  bytes; 512 bytes for a KV namespace), and a size under one block makes an empty namespace. Capacity that no namespace takes, because the sizes sum to less than devsz\_mb or because of this rounding, stays unused and is not reported in tnvmcap.
- namespace\_modes: the mode of each namespace, one of nossd, bssd, znssd, ocssd, csd and kvssd, one entry per namespace. Unset, every namespace runs femu\_mode. The controller's model number and serial still come from femu\_mode.

The namespaces are packed one after another in the controller's memory, so none can overwrite another.

The mode properties apply to each namespace of that mode:

- Each BlackBox or CSD namespace has its own FTL with the full NAND geometry (nchs, blks\_per\_pl and so on). Each namespace must fit in that geometry on its own, with room for GC.
- Each ZNS namespace builds its zones from the zns\_ properties and its own size, so a smaller ZNS namespace gets smaller zones (zone size (§4.7.3)). Keep each ZNS namespace's size a power of two, or Linux may refuse its zone size.
- Each KV namespace has its own key space.

### 4.15.4 Use it from the guest

List the namespaces and look at one:

```
| sudo nvme list  
| sudo nvme list-ns /dev/nvme0  
| sudo nvme id-ns /dev/nvme0n2
```

Block namespaces appear as /dev/nvme0n1, /dev/nvme0n2 and so on, in namespace order; nvme list shows which ID each one has. A ZNS namespace is a zoned block device. In the four-mode example it is the third namespace:

```
| cat /sys/block/nvme0n3/queue/zoned
```

A KV namespace has no block device. Address it by namespace ID through the controller, as in the KV guide (§4.11.4), with -n set to its ID.

The SMART log and the vendor log page C0h report the controller as a whole: C0h sums its counters over every BlackBox, CSD and KV namespace, and reports the largest block read count among them. The second command below prints the controller's WAF times 1000 (log pages and counters (§B.1)).

```
| sudo nvme smart-log /dev/nvme0  
| sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b | od -An -t u4 -N 4
```

A second controller is /dev/nvme1 with its own namespaces.

#### 4.15.5 Limits and refusals

| Message                                                                           | Cause and fix                                                                                                      |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| namespaces must be in [1, 256]                                                    | namespaces out of range.                                                                                           |
| namespace_sizes count must equal namespaces=2                                     | Give one size per namespace. Check that every comma inside the value is doubled.                                   |
| namespace_sizes: invalid size '3X'                                                | A size QEMU cannot parse.                                                                                          |
| namespace_sizes sum (2147483648 B) exceeds the backend capacity (1073741824 B)    | Raise devsz_mb or lower the sizes.                                                                                 |
| backend capacity N B is too small for M namespaces                                | devsz_mb=0.                                                                                                        |
| namespace_sizes: '100' is smaller than a sector                                   | Each size must be at least 512 bytes.                                                                              |
| namespace_modes count must equal namespaces=2                                     | Give one mode per namespace.                                                                                       |
| namespace_modes: unknown mode 'foo'                                               | Use nossd, bbssd, znssd, ocssd, csd or kvssd.                                                                      |
| ocssd supports a single namespace                                                 | OCSSD cannot share a controller.                                                                                   |
| csd supports at most one namespace per controller                                 | Only one CSD namespace. Its neighbours may use other modes.                                                        |
| meta: namespace 2 runs a mode without metadata support (block or no-SSD only)     | meta needs every namespace to be NoSSD or BlackBox.                                                                |
| streams requires NVM namespaces                                                   | Streams needs every namespace to be NoSSD or BlackBox.                                                             |
| FDP supports a single namespace; set namespaces=1 or disable FDP on the subsystem | FDP takes one namespace.                                                                                           |
| FEMU bbssd: namespace 1 exposes 4096 MiB of the 2048 MiB this geometry has, ...   | A BlackBox namespace is larger than the NAND geometry allows. Each namespace is checked against the full geometry. |

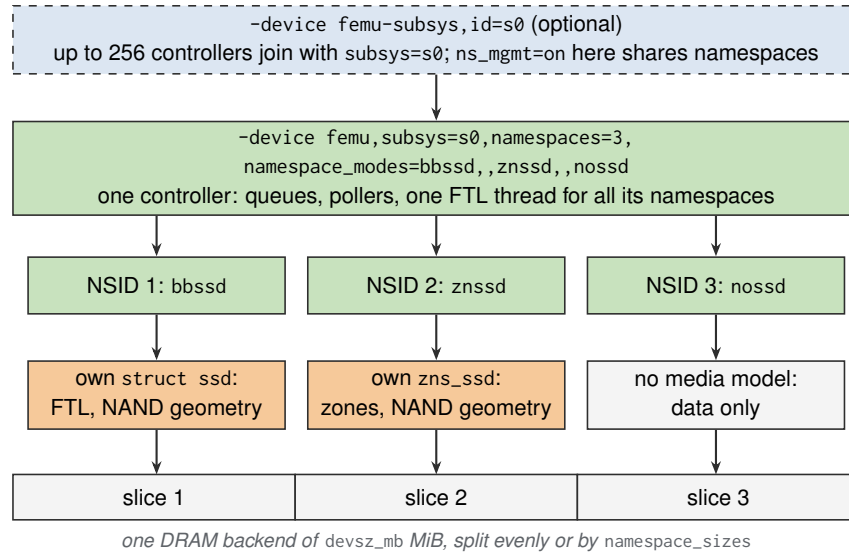
#### 4.15.6 Verify

1. `sudo nvme list-ns /dev/nvme0` lists IDs 1 to namespaces.
2. `sudo nvme list` shows one block device per non-KV namespace, with the sizes you set.
3. Data written to one namespace does not appear in another.

#### 4.15.7 Troubleshooting

- **Only one namespace appears.** namespaces defaults to 1. Set it, and check that the commas in namespace\_sizes and namespace\_modes are doubled. A single comma ends the property and QEMU reads the next item as a new property.
- **I want random-write and zoned capacity on one ZNS device.** Linux rejects ZNS conventional zones, so use a BlackBox namespace next to a ZNS namespace instead.
- **I want several separate SSDs.** Add one -device femu per SSD, each with its own id and devsz\_mb.
- **Data is gone after QEMU exits.** Every namespace lives in host memory only. A guest reboot keeps the data.

Related issues: #26, #52, #121.



**Figure 21:** One controller with three namespaces of different modes: each BBSSD or ZNS namespace builds its own FTL and NAND model, and all of them share slices of one DRAM backend. Sharing a namespace between controllers is limited to NoSSD or BBSSD without namespace\_modes.

#### 4.15.8 Related pages

- Namespace management, metadata and protection information (§4.17)
- Choosing a mode: which features combine (§4.1.3)
- Architecture: mode backends (§3.1.4)

### 4.16 Subsystems, controllers and namespaces

This chapter describes how FEMU arranges NVMe subsystems, controllers and namespaces: how a controller's memory backend is split between namespaces, how each namespace runs its own mode, how Namespace Management creates and deletes namespaces at run time, how a subsystem shares namespaces between controllers, and where metadata and protection information are kept. For usage, see several namespaces (§4.15) and namespace management, metadata and PI (§4.17).

#### 4.16.1 Purpose

An NVMe device is a subsystem of one or more controllers; each controller exposes namespaces, and each namespace is a range of logical blocks with its own command set. FEMU models this so you can:

- give one controller several namespaces, each in a different mode, such as a BlackBox namespace next to a ZNS namespace;
- create, delete, attach and detach namespaces from the guest;
- attach one namespace to several controllers;
- store per-block metadata and protection information.

### 4.16.2 Place in the hierarchy

```
femu-subsys device id=S [ns_mgmt=on] [fdp=on]      optional; not on any bus
| CNTLID table (up to 256), SUBNQN, endurance group (FDP),
| shared storage object (ns_mgmt=on only)
|
+-- femu device, subsys=S      controller CNTLID 0   (PCIe endpoint)
|   queues, pollers, one FTL thread
|   namespace table: 256 slots, NSID = slot + 1
|   NSID 1  mode A  -> backend bytes [0, size1)
|   NSID 2  mode B  -> backend bytes [size1, size1 + size2)
|   ...
|   DRAM backend (devsz_mb)
|
+-- femu device, subsys=S      controller CNTLID 1
    its own namespaces and backend, unless the subsystem has ns_mgmt=on

femu device, no subsys=      a controller without a subsystem: the same,
                             with CNTLID 0 and no subsystem features
```

There is no namespace -device. Namespaces live inside the controller (or the subsystem's storage object) and are built from controller properties. femu-subsys creates an nvme-bus, but no FEMU device plugs into it.

### 4.16.3 Data structures

| Structure     | File   | Fields that matter                                                                                                                                                                                                                                                                                  |
|---------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NvmeSubsystem | nvme.h | ctrls[256] (CNTLID slots), subnqn, endgrp (FDP), ns_mgmt, storage (the shared storage object), ns_lock                                                                                                                                                                                              |
| FemuCtrl      | nvme.h | namespaces (256 slots), num_namespaces (boot count), namespace_limit (slots in use: the boot count, or 256 with Namespace Management), namespace_pool_size, mbe (DRAM backend), attached_ns (per-controller attachment bitmap for shared namespaces), cntlid, ext_ops (the controller's mode table) |
| NvmeNamespace | nvme.h | id, size (bytes the host sees), extent_size (bytes reserved in the backend), backend_offset, start_block, femu_mode, csi, ext_ops (this namespace's mode table), allocated, attached, util and uncorrectable bitmaps, mdata, the mode state (ssd for bbssd and CSD, zns for ZNS), endgrp and fdp    |

### 4.16.4 Namespace sizes and backend partitioning

Every controller has one DRAM backend of devsz\_mb MiB, created in femu\_realize(). For bbssd with op\_pcent the backend is the full NAND capacity of the geometry instead, and the exposed capacity is that NAND capacity divided by  $1 + \text{op\_pcent} / 100$ , so the spare space is op\_pcent percent of the exposed size (op\_pcent=25 exposes 80 % of the NAND).

nvme\_resolve\_ns\_sizes() then sizes the boot namespaces:

- namespace\_sizes unset: the exposed capacity is split evenly, each share rounded down to 512 bytes.
- namespace\_sizes set: one size per namespace, each at least one 512-byte sector; the sum must fit in the exposed capacity (devsz\_mb, or the over-provisioned NAND capacity with op\_pcent), all of it, whatever the namespace count.

Either way each size is then rounded down to whole logical blocks of the format lba\_index selects ( $512 \ll \text{lba\_index}$  bytes), so no slice ends in part of a block the host cannot address and every slice starts on a block

boundary. A KV namespace is sized in bytes and is rounded to 512 bytes only. A size smaller than one logical block leaves an empty namespace.

`nvme_init_namespaces()` packs the slices in NSID order. Each namespace records where its slice starts in `backend_offset`, and every data path adds it to the byte offset of an LBA, so no namespace can reach another's data:

```
devsz_mb=4096, namespaces=4, namespace_modes=nossd,,bbssd,,znssd,,kvssd
```

```
backend  0          1 GiB      2 GiB      3 GiB      4 GiB
+-----+-----+-----+-----+
| NSID 1 | NSID 2 | NSID 3 | NSID 4 |
| nossd  | bbssd  | znssd  | kvssd  |
+-----+-----+-----+-----+
backend_offset: 0      1 GiB      2 GiB      3 GiB
```

```
byte of LBA x in NSID n = backend_offset(n) + x * logical block size
```

The namespace's NSZE, NCAP and NUSE start as its slice size divided by the logical block size of its format. A zoned namespace then lowers NSZE to its whole zones (see ZNS (§4.8.4)); a KV namespace measures its capacity in bytes.

The pool reported to the host is the sum of the boot slices: TNVMCAP is the pool size and UNVMCAP the part no namespace holds. Backend space past the last boot slice, including what the rounding above leaves, is not in the pool and no namespace uses it.

#### 4.16.5 Per-namespace modes

`namespace_modes` gives each namespace its own mode, from the tokens `nossd`, `bbssd`, `znssd`, `ocssd`, `csd` and `kvssd`. Unset, every namespace runs `femu_mode`. `nvme_resolve_ns_modes()` parses the list.

```
femu_realize()
nvme_init_namespaces()      sizes, slices, modes, per-namespace checks
nvme_register_extensions()  controller ext_ops = table of femu_mode
                             (used by the admin paths)

for each namespace:
    nvme_register_extensions_ns()  ns->ext_ops = table of the ns's mode
    ns->ext_ops.init(n, ns)        bb_init, zns_init, kvssd_init, ...
start one FTL thread if any namespace is bbssd, ZNS or CSD
```

At run time the mode is chosen per command, not per controller:

- I/O commands that the common path in `nvme-io.c` does not handle itself go to `ns->ext_ops.io_cmd` of the namespace the command names.
- The controller's single FTL thread routes each request by `req->ns`: ZNS namespaces to `zns_ftl_process_req()`, `bbssd` and CSD namespaces to `bb_ftl_process_req()`; NoSSD and KV cost nothing there.
- Each namespace reports its own command set (CSI 2 for ZNS, 1 for KV, 0 otherwise). The controller always advertises NVM, Zoned and KV in its I/O Command Set list; only the Changed Zone List entry in the supported log pages depends on a zoned namespace being present.

The controller's model number and serial come from its own `femu_mode`. Every namespace still advances its mode's serial counter as it starts, so serials keep their values, but only a namespace of the controller's mode names the controller. When none does (a KV controller whose KV namespace is not namespace 1, for example), the controller is named from `femu_mode` once its namespaces are up (`nvme_set_ctrl_name()`).

Each namespace keeps its own mode state. Every bbssd or CSD namespace has a complete FTL with the full NAND geometry, and must fit in that geometry on its own with room for GC. Every ZNS namespace builds its own zones from the shared zns\_ properties and its own size. Every KV namespace has its own key space.

| Rule                                                                 | Why                                                            | Where                  |
|----------------------------------------------------------------------|----------------------------------------------------------------|------------------------|
| ocssd only as the controller's own mode, and only with one namespace | Open-Channel tables are controller-wide                        | nvme_init_namespaces() |
| at most one csd namespace per controller                             | CSD keeps controller-wide state                                | nvme_init_namespaces() |
| no kvssd with FDP                                                    | placement takes every line, leaving the KV store none          | nvme_init_namespaces() |
| meta and streams need every namespace to be nossd or bbssd           | only the block data path carries them                          | nvme_init_namespaces() |
| one namespace with FDP                                               | the FDP write path addresses the FTL by command LBA, not slice | nvme_init_namespaces() |

#### 4.16.6 Namespace management

With ns\_mgmt=on on a controller, and when the controller and every boot namespace run the same mode, NoSSD or bbssd, with no dps and no FDP (nvme\_ns\_mgmt\_supported()), the controller:

- reports NN = 256 and opens all 256 slots (namespace\_limit);
- sets OACS bit 3 (Namespace Management) and OAES Namespace Attribute notices;
- reports each namespace's NVMCAP.

In any other configuration ns\_mgmt=on adds no Namespace Management. Realize refuses it outright with a subsystem that lacks ns\_mgmt, and with power\_loss or cxl\_ssd.

#### Create, attach, detach, delete

```
pool = sum of boot slices          (all of it in use at boot)

| NSID 1 (2 GiB) | NSID 2 (2 GiB) | Delete NSID 2
| NSID 1 (2 GiB) |   free 2 GiB   | Create 1 GiB
| NSID 1 (2 GiB) | NSID 2 (1 GiB) | free | detached until Attach
```

create: lowest free NSID; extent = first gap that fits (first fit)

Namespace Management Create (nvme\_ns\_mgmt() in nvme-admin.c, then nvme\_ns\_create() in femu.c):

- accepts only the NVM command set, with NCAP equal to NSZE (a smaller NCAP fails with Thin Provisioning Not Supported), an LBA format the controller offers, and a PI setting only if pi=on;
- rounds the size up to a NAND page for bbssd (secs\_per\_pg \* secsz) and to 4 KiB for NoSSD;
- takes the lowest free NSID and the first gap in the pool that fits (nvme\_ns\_find\_extent()); no gap fails with Namespace Insufficient Capacity;
- for bbssd, refuses a new namespace once bbssd\_ns\_limit namespaces are allocated, and refuses one that would leave its FTL no room for GC;
- zeroes the slice, builds the mode state, and leaves the namespace detached.

The controller pauses its pollers and FTL thread around create and delete, so no request runs against a half-built or half-freed namespace. Attach and detach flip attached (or a bit of attached\_ns for shared namespaces) and send Attached Namespace Attribute Changed notices to the controllers they affect. Delete releases the slice; the gap is reused by the next create.



### 4.16.7 Subsystems and shared namespaces

A controller joins a subsystem with `subsys=<id>`; the subsystem must come first on the command line. `nvme_init_subsys()` gives it the lowest free CNTLID. What else is shared depends on the subsystem's properties:

| Subsystem  | Controllers                           | What they share                                                                                                                                                   |
|------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| plain      | any number                            | only the CNTLID space; each keeps its own namespaces and backend, and none can use Namespace Management, so the controller lists (CNS 12h, 13h) are not available |
| fdp=on     | one                                   | the endurance group and its FDP state (FDP (§4.6))                                                                                                                |
| ns_mgmt=on | any number, NoSSD or bbssd, all alike | one namespace table, backend and set of FTLs, and the SUBNQN                                                                                                      |

A controller with `streams=on` also cannot share a subsystem.

#### Shared namespaces

```
femu-subsys,ns_mgmt=on
storage object (an unrealized femu, a copy of the first controller's
properties): namespace table, DRAM backend, one FTL per bbssd namespace
      ^               ^               ^
      | attached_ns bitmap | attached_ns bitmap |
controller A (CNTLID 0) controller B (CNTLID 1) ...
own queues, pollers, FTL thread (bbssd only), namespace-change log
all I/O decode and FTL work of every controller under subsys->ns_lock
```

- The subsystem must be realized before its controllers. The first controller builds the boot namespaces; `nvme_subsys_take_storage()` then moves them, with the backend, to a storage object that copies the first controller's properties. That controller starts with every boot namespace attached; later controllers start with none.
- Later controllers must match the first in mode, meta, pi, mc, dpc, nlbaf, vwc and oncs. Their own capacity and geometry properties do not create storage.
- Identify Controller reports CMIC bit 1 (multiple controllers) and the subsystem's NQN; each namespace reports NMIC from its creation, and a private namespace cannot be attached to two controllers at once.
- `subsys->ns_lock` serializes command decode on the pollers and request processing on the FTL threads of all controllers, so two controllers never run the same FTL at once.
- The namespaces and backend outlive any one controller. They are released when the subsystem is removed and its last controller has left.

`femu-subsys` with both `ns_mgmt=on` and `fdp=on` is refused.

### 4.16.8 Metadata and protection information

```
Identify Namespace LBA formats with meta=8, nlbaf=5:
index 0..4   512 B, 1 KiB, 2 KiB, 4 KiB, 8 KiB      no metadata
index 5..9   the same sizes                          8 bytes metadata
boot format = nlbaf + lba_index (5 with lba_index=0)
```

where the bytes live:

```
data      DRAM backend, at backend_offset + LBA * block size
metadata  ns->mdata, a separate host buffer of (blocks * meta) bytes,
          one entry per LBA, under ns->mdata_lock
transfer  separate buffer (MPTR, mc bit 1), or interleaved after each
          block in the data buffer (extended LBA, mc bit 0)
```

- meta sets the metadata bytes per block and doubles the format list. nlbaf is then at most 8.
- With pi=on and meta of 8 or more, Identify reports DPC 1Fh: PI types 1 to 3, in the first or last 8 bytes of the metadata. The host picks a type with Format NVM or at Namespace Management create. FEMU generates the PI when PRACT is set and checks the guard (CRC-16 T10-DIF), application tag and reference tag as the PRCHK bits ask (nvme-pi.c).
- dpc and dps describe PI without these paths. A non-zero dps cannot pass the realize checks, so use pi.
- Metadata and PI work only on NoSSD and bbssd namespaces and not with FDP.

#### 4.16.9 Threads and locks

| Lock                                | Protects                                                     | Taken by                                                       |
|-------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------|
| subsys->ns_lock                     | the shared namespace table and FTLs (shared namespaces only) | pollers around command decode, FTL threads around each request |
| ns->mdata_lock                      | the metadata buffer of one namespace                         | every metadata read and write                                  |
| poller pause (nvme_pause_pollers()) | namespace lifecycle, Format, Sanitize                        | admin path; pauses every controller of a shared subsystem      |

#### 4.16.10 Parameters

Properties are in the property reference under mode, capacity and namespaces (§A.1.1), namespace management, streams and power loss (§A.1.1), LBA formats, metadata and protection (§A.1.1) and shared namespaces (§A.1.2).

| Property                                           | On                | Effect and interactions                                                                                             |
|----------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------|
| devsz_mb                                           | femu              | Backend size, split between the boot namespaces.                                                                    |
| namespaces                                         | femu              | Boot namespaces, 1 to 256.                                                                                          |
| namespace_sizes                                    | femu              | Per-namespace sizes; commas doubled on the QEMU command line.                                                       |
| namespace_modes                                    | femu              | Per-namespace modes; see the rules above. Refused with a shared subsystem.                                          |
| op_pcent                                           | femu              | bbssd: backend = NAND capacity; exposed capacity = NAND capacity / (1 + op_pcent / 100).                            |
| ns_mgmt                                            | femu, femu-subsys | Namespace Management; on the subsystem, shared namespaces. A controller with it cannot join a subsystem without it. |
| bbssd_ns_limit                                     | femu              | bbssd namespaces allocated at once, 1 to 256 and at least namespaces. Each has a full FTL in host memory.           |
| subsys                                             | femu              | Joins a femu-subsys.                                                                                                |
| nqn                                                | femu-subsys       | SUBNQN suffix; reaches Identify only on a shared subsystem.                                                         |
| meta, mc, extended, pi, nlbaf, lba_index, dpc, dps | femu              | Formats, metadata and PI, above.                                                                                    |

A NoSSD namespace and a ZNS namespace of different sizes on one controller:

```
| -device femu,devsz_mb=2048,femu_mode=2,namespaces=2,namespace_sizes=1G,,1G,namespace_modes=nossd,,znssd
```

#### 4.16.11 Statistics and outputs

| Output                                                          | Scope                                                                                                        |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Identify Controller TNVMCAP, UNVMCAP                            | the controller's namespace pool                                                                              |
| Identify Namespace NSZE, NCAP, NUSE, NVMCAP                     | one namespace (NVMCAP with Namespace Management)                                                             |
| Namespace lists (CNS 02h, 10h), controller lists (CNS 12h, 13h) | 12h and 13h answer only with Namespace Management; 02h and 10h on any controller                             |
| Changed Namespace List (04h)                                    | namespaces attached, detached, deleted or reformatted, per controller; filled only with Namespace Management |
| SMART / Health (02h)                                            | the controller, summed over its namespaces                                                                   |
| Vendor log C0h                                                  | summed over the bbssd, CSD and KV namespaces (log pages and counters (§B.1))                                 |

#### 4.16.12 Validation

qtest cases in `hw/femu/tests/qtest/femu-test.c`:

- layout and modes: namespace-capacity, namespace-identity, namespace-mixed-identity and its -bbssd and -kv variants, namespace-sizes-rounding, namespace-large, namespace-empty-slice, namespace-kv-byte-capacity, kv-namespaces, namespace-sparse;
- management: the ns-mgmt-\* cases (lifecycle, capacity, bbssd\_ns\_limit, notices, validation, unsupported modes and FDP), namespace-lifecycle, namespace-allocated, ns-retire-\*;
- shared namespaces: the ns-shared-\* cases and ns-mgmt-subsys;
- metadata and PI: metadata, metadata-bbssd, metadata-extended, and the pi-\* cases.

Every tagged documentation example with several namespaces or controllers is also started under qtest and moves one block on namespace 1.

#### 4.16.13 Limits

- Namespace Management creates only NoSSD or bbssd namespaces of the controller's own mode, and only on controllers whose namespaces all run that mode. It does not create zoned, KV, CSD or Open-Channel namespaces.
- Each bbssd namespace has a full-size FTL; there is no scaled geometry for small namespaces, which is why bbssd\_ns\_limit exists.
- Allocated Namespace Attribute notices are not advertised.
- Shared namespaces need homogeneous NoSSD or bbssd controllers without Streams, dps or namespace\_modes.
- All namespaces and metadata live in host memory and are lost when QEMU exits.

#### 4.16.14 How to extend it

- A new mode token: add it to `nvme_mode_from_token()`, register its table in `nvme_register_extensions()`, add any per-namespace restriction to `nvme_init_namespaces()`, and keep `hw/femu/docs/modes.py` in step.
- A mode that keeps controller-wide state must refuse a second namespace of that mode, as CSD does, or move the state into the namespace.
- Namespace Management for another mode: `nvme_ns_mgmt_supported()`, `nvme_ns_create()` and the capacity checks in `nvme_ns_mgmt()` are the places that limit it to NoSSD and bbssd.
- Any new data path must add `ns->backend_offset` to its byte offsets.

#### 4.16.15 Source map

| File                                 | What is there                                                                                                                                                                                                              |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hw/femu/femu.c                       | femu-subsys device, CNTLID registration, namespace sizing, packing and per-namespace mode setup (nvme_init_namespaces()), nvme_ns_create(), extent search, shared storage (nvme_subsys_take_storage()), capacity reporting |
| hw/femu/nvme.h                       | NvmeSubsystem, NvmeNamespace, nvme_ns() and attachment helpers                                                                                                                                                             |
| hw/femu/nvme-admin.c                 | Namespace Management and Attachment, Identify lists, Format NVM                                                                                                                                                            |
| hw/femu/nvme-io.c                    | per-namespace dispatch, backend offsets, metadata transfer                                                                                                                                                                 |
| hw/femu/nvme-pi.c, hw/femu/nvme-pi.h | protection information generation and checks                                                                                                                                                                               |

#### 4.16.16 Related pages

- Several namespaces and devices (§4.15)
- Namespace management, metadata and PI (§4.17)
- ZNS (§4.8) and FDP (§4.6)
- Architecture (§3.1)

### 4.17 Namespace management, metadata and protection information

This page covers two optional NVMe features of NoSSD and BlackBox (§4.4) controllers:

- Namespace management (§4.17.2): the guest creates, deletes, attaches and detaches namespaces while it runs.
- Metadata and protection information (§4.17.3): extra bytes stored with every logical block, and end-to-end protection information (PI) types 1 to 3 in them.

Both are off by default. Use them to test host software that manages namespaces, or that writes and checks per-block metadata and PI, without hardware that supports them.

#### 4.17.1 Requirements

- Host and guest: see the mode table (§4.1.4) and requirements.md (§2.1.2).
- Namespace management: any guest kernel with the NVMe driver, and nvme-cli (create-ns, attach-ns).
- Metadata through the block layer: a guest kernel with CONFIG\_BLK\_DEV\_INTEGRITY=y. Without it, use passthrough commands.

#### 4.17.2 Namespace management

##### What it does

With ns\_mgmt=on, the controller advertises Namespace Management and Namespace Attachment and reports 256 as its namespace count. The namespaces from namespaces and namespace\_sizes are created at boot, and together they define the capacity pool: the pool is the sum of their sizes, and at boot all of it is in use (unvmcap is 0). To create a namespace, the guest first deletes one, then allocates from the space it freed. A namespace the guest creates is detached until the guest attaches it.

It takes effect only when the controller and every boot namespace run the same mode, NoSSD or BlackBox. With another mode it is accepted and stays off.

## Launch

A BlackBox controller with two 2 GiB boot namespaces, a 4 GiB pool:

```
| -device femu,devsz_mb=4096,femu_mode=1,ns_mgmt=on,namespaces=2
```

A NoSSD controller:

```
| -device femu,devsz_mb=4096,femu_mode=2,ns_mgmt=on,namespaces=2
```

To share namespaces between controllers, set `ns_mgmt=on` on a `femu-subsys` and join it with `subsys=`. The subsystem comes first:

```
| -device femu-subsys,id=shared,ns_mgmt=on -device femu,id=ctrl-a,subsys=shared,femu_mode=2,devsz_mb=64 -device femu,id=ctrl-b,subsys=shared,femu_mode=2
```

The first controller sets up the pool and the boot namespaces; later ones join with nothing attached. Shared namespaces in detail (§4.17.2) describes the shared model in full.

## Configuration

Properties: namespace management, streams and power loss (§A.1.1), shared namespaces (§A.1.2).

- `ns_mgmt`: on a standalone controller, or on `femu-subsys` for shared namespaces. A controller with `ns_mgmt=on` cannot join a subsystem that does not have it.
- `bbssd_ns_limit` (BlackBox only, default 4): the most namespaces that may be allocated at once, detached ones included. Each BlackBox namespace has its own FTL with the full NAND geometry, so this bounds host memory.
- `namespaces` and `namespace_sizes`: the boot namespaces, whose sizes add up to the pool. Space that no boot namespace covers is not in the pool.

A shared subsystem takes NoSSD or BlackBox controllers that all have the same mode, meta, mc, pi, dpc, nlbaf, vwc and oncs, and no Streams, dps or `namespace_modes`.

## Use it from the guest

Check that the controller offers it, and find its controller ID:

```
| sudo nvme id-ctrl /dev/nvme0 | grep -E '^(oacs|nn|cntlid|tnvmcap|unvmcap) '
```

`oacs` has bit 3 (0x8) set, `nn` is 256, and `unvmcap` is the free capacity in bytes.

With the BlackBox example above, delete boot namespace 2 to free 2 GiB, create a 1 GiB namespace of 512-byte blocks (LBA format 0) in it, attach it to this controller, use it, then remove it:

```
| sudo nvme delete-ns /dev/nvme0 --namespace-id=2
| sudo nvme create-ns /dev/nvme0 --nsze=2097152 --ncap=2097152 --flbas=0
| sudo nvme attach-ns /dev/nvme0 --namespace-id=2 --controllers=0
| sudo nvme ns-rescan /dev/nvme0
| sudo nvme list
| sudo nvme detach-ns /dev/nvme0 --namespace-id=2 --controllers=0
| sudo nvme delete-ns /dev/nvme0 --namespace-id=2
```

`create-ns` prints the new namespace ID; use it in the commands after it. `--controllers` takes the `cntlid` from `id-ctrl`. `nsze` and `ncap` are in logical blocks and must be equal: an `ncap` below `nsze` fails with Thin

Provisioning Not Supported, and a zero or larger one with Invalid Field. A BlackBox namespace is rounded up to whole NAND pages.

## Limits

| Message or status                                                                          | Cause and fix                                                                                                     |
|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| bbssd_ns_limit must be between 1 and 256 and cover the boot namespaces                     | namespaces above bbssd_ns_limit, or the limit out of range.                                                       |
| ns_mgmt=on does not support subsys; use a standalone controller                            | ns_mgmt=on on a controller whose subsystem lacks it. Set it on the femu-subsys instead.                           |
| shared namespaces require homogeneous NoSSD or bbssd without Streams or default protection | A ZNS, KV, CSD or OCSSD controller, or Streams, dps or namespace_modes, in a shared subsystem.                    |
| shared namespace mode and capabilities must match                                          | Controllers in one shared subsystem differ in mode or format properties.                                          |
| namespace management does not support FDP                                                  | ns_mgmt=on and fdp=on on one subsystem.                                                                           |
| Namespace Insufficient Capacity                                                            | The pool has no free space (delete a namespace first), or a BlackBox namespace would not leave GC its free lines. |
| Namespace Identifier Unavailable                                                           | All 256 IDs are used, or bbssd_ns_limit is reached.                                                               |
| Thin Provisioning Not Supported                                                            | ncap below nsze.                                                                                                  |
| I/O Command Set Not Supported                                                              | A command set other than NVM; the guest can create only block namespaces.                                         |

## Shared namespaces in detail

With femu-subsys, ns\_mgmt=on, the subsystem owns one namespace table, backend capacity pool and full-geometry FTL per bbssd namespace. This prevents controllers allocating the same NSID independently or deleting only a private copy. Realize the subsystem before its controllers. The first controller establishes the pool and boot namespaces; later controllers join that pool with no initial attachments. Namespace Attachment selects their active namespaces.

The subsystem copies the first controller's properties into an unrealized configuration object as its storage context, independently of PCI transport lifetime. This context has no queues or workers. Its namespace table and backend survive removal of that controller, including removal of every controller. They are released at subsystem teardown after its last controller has left. Namespace identity and the creation sequence belong to this context. Controllers must agree on the storage mode and format capabilities. Runtime construction uses the original storage configuration, including capacity, geometry, media options and the bbssd namespace cap; those properties on later controllers do not create additional storage.

Each controller keeps its own queues, pollers, FTL thread, attachment bitmap and namespace-change log. Pollers reach the common table through their controller; active lookup additionally checks that controller's attachment bitmap. A subsystem mutex serializes data and FTL processing. Existing pause/resume interfaces pause all controllers in a managed subsystem, so Format, Sanitize and lifecycle changes cannot race another controller's workers. Delete retires namespace references in every controller's request containers before releasing storage, without waiting for guest consumption of a full CQ. Detach retires only the selected controllers. Reset preserves attachments. Transport removal stops its workers and drops its attachments without releasing the subsystem's namespaces.

CNS 10h/11h describe the common allocated set, and CNS 1Fh gives the command-set-independent structure (CNS 08h) of an allocated namespace, attached or not. CNS 02h describes the issuing controller's attachments. CNS 12h lists attached controllers and CNS 13h lists eligible controllers, in ascending order with inclusive

CNTID filtering. Attach/detach records changes and sends enabled Attached Namespace Attribute Changed notices on each affected controller. Delete does the same except that its issuer receives no notice. Allocated Namespace Attribute notices are not advertised. Create produces a detached namespace and no attached-set notice.

The model is opt-in on the subsystem. Existing standalone management and unmanaged subsystems retain their behavior. Only homogeneous NoSSD or bbssd is supported; FDP and Streams remain excluded. Full-geometry bbssd allocation and its namespace-count cap are retained. Scaled FTL geometry, allocated-namespace notices, persistent storage and other namespace modes are not supported.

With the shared example under Launch (§4.17.2), only ctrl-a initially has the boot namespace attached. Controller lists report the assigned CNTLIDs; use Namespace Attachment to expose it through ctrl-b too. Create honors NMIC: private namespaces can move between controllers but cannot be attached to two simultaneously. Shared namespaces permit both attachments.

### 4.17.3 Metadata and protection information

#### What it does

meta=<bytes> adds that many metadata bytes to every logical block. The controller then offers each LBA format twice: formats 0 to nlbaf - 1 without metadata, and formats nlbaf to 2 \* nlbaf - 1 with it. The namespace boots on the metadata version of lba\_index. With the default nlbaf=5 and lba\_index=0, that is format 5: 512-byte blocks with metadata.

The metadata travels either in a separate buffer (mc bit 1, 0x2) or interleaved with the data as extended LBAs (mc bit 0, 0x1). extended=1 boots on the interleaved layout.

With pi=on and meta of 8 or more, the controller also offers protection information types 1, 2 and 3, in the first or the last 8 bytes of the metadata. The guest selects a type with Format NVM, or with --dps when it creates a namespace. The controller generates and checks PI as the command asks (PRACT and PRCHK bits).

#### Launch

8 bytes of separate metadata with PI available:

```
| -device femu,devsz_mb=1024,femu_mode=1,meta=8,mc=2,pi=on
```

8 bytes of interleaved metadata, booting on the extended layout:

```
| -device femu,devsz_mb=1024,femu_mode=1,meta=8,mc=1,extended=1
```

#### Configuration

Properties: LBA formats, metadata and protection (§A.1.1).

- meta: metadata bytes per block. NoSSD and BlackBox namespaces only.
- mc: which layouts the controller supports, separate (0x2), interleaved (0x1) or both (0x3). Required with meta.
- extended: boot on the interleaved layout. Needs bit 0 of mc.
- pi: offer PI types 1 to 3. Needs meta of 8 or more; with less it is accepted and offers nothing.
- nlbaf: at most 8 with meta, since each format is offered twice.
- dpc and dps describe PI without metadata support; leave them at 0 and use pi.

## Use it from the guest

Look at the formats and the protection capabilities:

```
| sudo nvme id-ns /dev/nvme0n1 -H | grep -E 'LBA Format|dpc|dps|mc'
```

Reformat to 512-byte blocks with 8 bytes of separate metadata and PI type 1 in the last 8 bytes (--pil=0), then check that the namespace changed:

```
| sudo nvme format /dev/nvme0n1 --lbaf=5 --pi=1 --pil=0 --ms=0 --force  
| sudo nvme id-ns /dev/nvme0n1 | grep -E '^(flbas|dps)'
```

Write and read one block and let the controller generate and check the PI (--prinfo=0xf: PRACT plus the guard, application tag and reference tag checks; the reference tag of LBA 0 is 0):

```
| head -c 512 /dev/urandom > blk.bin  
| sudo nvme write /dev/nvme0n1 -s 0 -c 0 -z 512 -d blk.bin --prinfo=0xf --ref-tag=0  
| sudo nvme read /dev/nvme0n1 -s 0 -c 0 -z 512 -d out.bin --prinfo=0xf --ref-tag=0  
| cmp blk.bin out.bin
```

With CONFIG\_BLK\_DEV\_INTEGRITY=y, Linux generates and verifies PI for ordinary block I/O on such a namespace by itself.

## Limits

| Message                                                                                        | Cause and fix                                            |
|------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| meta/extended need a matching metadata capability (mc)                                         | mc must have bit 0 when extended=1, and bit 1 otherwise. |
| meta: at most 8 block sizes (nlbaf), each is also offered with metadata                        | Lower nlbaf.                                             |
| meta: namespace N runs a mode without metadata support (block or no-SSD only)                  | Every namespace must be NoSSD or BlackBox.               |
| meta: not supported with placement (fdp)                                                       | Metadata and FDP do not combine.                         |
| meta: protection information (dpc, dps) is not supported                                       | dpc or dps with meta; use pi.                            |
| dps needs 8 bytes of metadata and a matching dpc                                               | Use pi=on instead of dps.                                |
| power_loss requires bbssd, buffer_size > 0, and no metadata, namespace management or subsystem | meta or pi with power_loss.                              |

At run time, a Format NVM that asks for a PI type when pi is off, a PI type with less than 8 bytes of metadata, or a layout mc does not allow fails with Invalid Format. On a type 1 namespace, a command whose reference tag is not the low 32 bits of its LBA fails with Invalid Protection Information. A check against stored PI that fails returns the matching End-to-end Guard, Application Tag or Reference Tag Check Error.

### 4.17.4 Verify

1. Namespace management: oacs has bit 3 set, and a created and attached namespace appears in nvme list with the size you asked for.
2. Metadata: nvme id-ns /dev/nvme0n1 shows a non-zero ms for the current format, and dpc is 0x1f with pi=on.
3. PI: the write and read above succeed, and a read with --ref-tag=1 for LBA 0 fails with Invalid Protection Information.



### 4.17.5 Troubleshooting

- **ns\_mgmt=on but oacs has no Namespace Management bit.** The controller or a boot namespace is not NoSSD or BlackBox, or the modes differ. The property is silently off in that case.
- **create-ns fails with Insufficient Capacity.** The pool is the boot namespaces, and they use all of it at boot. Delete a namespace first.
- **A new namespace does not appear.** It is detached after creation. Attach it with the controller ID from `id-ctrl`, then run `nvme ns-rescan`.
- **The namespace shows a size of 0 after formatting with metadata.** The guest kernel cannot use that format through the block layer, usually because it lacks `CONFIG_BLK_DEV_INTEGRITY` or the format interleaves the metadata. Use passthrough commands on `/dev/ng0n1`, or a separate-buffer format on a kernel with integrity support.

Related issues: #121.

### 4.17.6 Related pages

- Several namespaces (§4.15)
- Changelog (§D)
- Choosing a mode: which features combine (§4.1.3)

## 4.18 CXL SSD

`femu-cxl-ssd` is a CXL Type-3 memory device whose capacity is backed by emulated NAND flash. The guest sees ordinary CXL memory: it creates a region, then uses it as a DAX device or onlines it as system RAM on its own NUMA node. Behind the memory interface, FEMU keeps a DRAM page cache in front of the same BBSSD FTL and NAND timing model that `femu_mode=1` uses. A cache hit costs no media time; a miss costs a NAND read, and evicting a dirty page costs a NAND program.

This page shows how to run it. The design note<sup>62</sup> explains how it works inside, and the property reference (§A.1.3) lists every property. The device comes from Cylon (FAST '26); if you use it, cite Cylon as well as FEMU (citation (§4.18.11)).

Related pages:

- CXL caching API (§4.20): let the guest pin, drop and uncache pages through BAR5.
- CXL NVMe link (§4.21): serve the same medium as an NVMe namespace too.

### 4.18.1 What it emulates

| Part      | What FEMU does                                                                                                                                                                                      |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interface | A CXL Type-3 volatile memory device, a subclass of QEMU's <code>cxl-type3</code> . The guest CXL driver enumerates it as a <code>memdev</code> ( <code>mem0</code> ).                               |
| Capacity  | The memory backend named by <code>volatile-memdev</code> : 256 MiB to 120 GiB, in steps of 256 MiB. The data lives there.                                                                           |
| Cache     | <code>cache-pages</code> pages of 4 KiB (default 1024, so 4 MiB), <code>cache-ways</code> ways per set, policy <code>fifo</code> , <code>lifo</code> , <code>clock</code> or <code>s3-fifo</code> . |

<sup>62</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md>

| Part           | What FEMU does                                                                                          |
|----------------|---------------------------------------------------------------------------------------------------------|
| Media          | The BBSSD FTL: page mapping, garbage collection, NAND read, program and erase times, channels and LUNs. |
| Direct mapping | Optional (der). Cached pages are mapped into the guest so hits run at DRAM speed with no exit to QEMU.  |
| Persistence    | None. It is volatile memory. Data is lost when QEMU exits.                                              |
| Migration      | Not supported. migrate fails with "State blocked by non-migratable device '...'".                       |

Every guest access that is not direct-mapped is an MMIO exit to QEMU. The vCPU that made the access waits out the modelled media time before the load or store completes. The timing model (§6.2.10) lists what each kind of access costs.

### 4.18.2 Host requirements

- An x86-64 Linux host with KVM. QEMU must be built for x86\_64-softmmu; femu-cxl-ssd is built by default (see build (§2.2)).
- Host RAM for the media size, the guest RAM, and the FTL mapping tables.
- Hugepages are optional for der=off and der=memslot. der=cylons needs a shared, preallocated hugetlbfs backend (see der=cylons (§4.18.7)).
- der=cylons also needs a modified host kernel. The other modes run on a stock kernel.

To reserve 2 MiB hugepages for a 256 MiB device, for example:

```
echo 128 | sudo tee /sys/kernel/mm/hugepages/hugepages-2048kB/nr_hugepages
grep HugePages_Free /proc/meminfo
mount | grep hugetlbfs          # usually /dev/hugepages
```

Give the pages back with `echo 0 | sudo tee ...` after QEMU exits.

### 4.18.3 Guest requirements

The guest kernel needs the CXL stack with region support and the DAX drivers:

| Option                                                             | Why                                                                                                                                                               |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONFIG_CXL_BUS, CXL_PCI, CXL_ACPI, CXL_MEM, CXL_PORT               | Enumerate the host bridge, root port and memdev                                                                                                                   |
| CONFIG_CXL_REGION                                                  | Create a region over the device                                                                                                                                   |
| CONFIG_CXL_REGION_INVALIDATION_TEST                                | Needed inside a VM. A guest cannot invalidate CPU caches by address range, and without this option Linux refuses to commit the region. The option skips that step |
| CONFIG_DEV_DAX, CONFIG_DEV_DAX_CXL                                 | Expose the region as /dev/daxX.Y                                                                                                                                  |
| CONFIG_DEV_DAX_KMEM                                                | Online the region as system RAM. kmem may claim a new RAM region on its own; daxctl reconfigure-device switches it back                                           |
| CONFIG_MEMORY_HOTPLUG, CONFIG_MEMORY_HOTREMOVE, CONFIG_ZONE_DEVICE | Memory hotplug support that kmem and DAX need                                                                                                                     |
| CONFIG_FS_DAX                                                      | Not needed for RAM regions; the validated guests had it on                                                                                                        |

The guest runs validated in FEMU's own testing used Linux 6.12 with these options. Check a guest kernel's configuration with:

```
grep -E 'CONFIG_(CXL_REGION|CXL_REGION_INVALIDATION_TEST|DEV_DAX|DEV_DAX_CXL|DEV_DAX_KMEM|FS_DAX)=' \
/boot/config-$(uname -r)
```

Guest tools: `cxl` (`cxl-cli`), `daxctl` and `ndctl`. They come from the `ndctl` project; `./make-guest-image.sh --cxl` installs them (see guest image (§2.3)). `numactl` helps when you use the region as system RAM.

#### 4.18.4 Launching

##### With `run-cxlssd.sh`

`hw/femu/scripts/run-cxlssd.sh` builds a command line with one `pxb-cxl` host bridge, one `cxl-rp` root port, the `femu-cxl-ssd` and a single-target CXL window. Settings come from environment variables; arguments after the script name go to QEMU unchanged. It does not add a guest disk or network, so pass them yourself. From `build-femu/`:

```
../femu-scripts/run-cxlssd.sh \
-drive file=$HOME/images/u20s.qcow2,if=none,id=hd0 \
-device virtio-blk-pci,drive=hd0,bus=pcie.0 \
-netdev user,id=net0,hostfwd=tcp::8080-:22 \
-device virtio-net-pci,netdev=net0,bus=pcie.0
```

The script looks for QEMU at `./qemu-system-x86_64`; set QEMU to use another binary. `DRY_RUN=1` prints the command instead of running it. It never runs `sudo` and changes no host settings.

Its defaults follow the published Cylon experiments and differ from the device's own defaults:

| Variable                   | Script default                                        | Device default | Property                    |
|----------------------------|-------------------------------------------------------|----------------|-----------------------------|
| CXL_SIZE                   | 256M                                                  | (required)     | backend size, M or G suffix |
| CACHE_PAGES                | size in MiB / 20 * 256 (a cache of 1/20 of the media) | 1024           | cache-pages                 |
| CACHE_WAYS                 | 1 (direct mapped); full means CACHE_PAGES             | 16             | cache-ways                  |
| CACHE_POLICY               | fifo                                                  | fifo           | cache-policy                |
| CHANNELS, LUNS_PER_CHANNEL | 8, 8                                                  | 4, 4           | channels, luns-per-channel  |
| LSA_CONTROL                | on                                                    | off            | lsa-control                 |
| BLOCKS_PER_PLANE           | 768 for 48G, 1536 for 96G, else 0                     | 0              | blocks-per-plane            |

The other variables (`DER`, `CYLON_KERNEL_ACK`, `PREFETCH_DEGREE`, `PREFETCH_STRIDE`, `PAGES_PER_BLOCK`, `READ_NS`, `PROGRAM_NS`, `ERASE_NS`, `CHANNEL_NS`, `GC_THRESHOLD`, `GC_THRESHOLD_HIGH`, `FTL`, `CYLON_FIRST_TOUCH_PROGRAM`, `CYLON_FREE_WRITEBACK`, `LOG_DIR`, `LOG_LIMIT`, `TRACEFS_DIR`, `CXL_BACKEND`, `ACCEL`, `CPU`, `CPUS`, `RAM`) keep the device defaults; the comment at the top of the script lists them all. The 48G and 96G presets leave no spare NAND blocks, as in Cylon. Without spare blocks the NAND fills up and accesses complete uncached (see Counters (§4.18.9), `media-full`).

A property the script has no variable for can be set with `-global`, which applies to every `femu-cxl-ssd`:

```
../femu-scripts/run-cxlssd.sh -global femu-cxl-ssd.concurrent-misses=on
```

## The equivalent command line

This is equivalent to the script with its defaults. The script also passes the remaining properties at their default values:

```
./qemu-system-x86_64 -machine q35,cxl=on,smm=off -accel kvm \
  -cpu host -smp 4 -m 4G \
  -object memory-backend-ram,id=cxlmem,size=256M \
  -device pxb-cxl,id=cxl.0,bus=pcie.0,bus_nr=52 \
  -device cxl-rp,id=cxl-rp0,bus=cxl.0,chassis=0,slot=0 \
  -device femu-cxl-ssd,id=cxlssd,bus=cxl-rp0,volatile-memdev=cxlmem,cache-pages=3072,cache-ways=1,cache-policy=fifo,der=off,channels=8,luns-per-channel=8,lsa-control=on \
  -M cxl-fmw.0.targets.0=cxl.0,cxl-fmw.0.size=256M \
  -drive file=$HOME/images/u20s.qcow2,if=none,id=hd0 \
  -device virtio-blk-pci,drive=hd0,bus=pcie.0 \
  -netdev user,id=net0,hostfwd=tcp::8080-:22 \
  -device virtio-net-pci,netdev=net0,bus=pcie.0 \
  -nographic
```

The pieces:

- `-machine q35,cxl=on` turns on CXL support in the machine.
- `pxb-cxl` is a CXL host bridge, `cxl-rp` a root port below it, and the `femu-cxl-ssd` sits below the root port.
- `cxl-fmw.0` is the fixed memory window, the host physical address range where the guest maps the region. Its size must be at least the media size.
- `volatile-memdev` names the backend that holds the data.
- `smm=off` is required by `der=cyl` and harmless otherwise.

To give the guest its own kernel, add `-kernel bzImage -append "root=/dev/vda1 console=ttyS0"` with the root partition of your image.

### 4.18.5 Creating the region in the guest

Log in (`ssh -i ~/images/femu-guest-key -p 8080 femu@localhost` with the image from `make-guest-image.sh`) and work as root. Find the memdev and the root decoder:

```
cxl list -M          # the memdev, normally mem0, with its ram_size
cxl list -D -d root  # the root decoder, normally decoder0.0
```

Create a RAM region over the whole device:

```
cxl create-region -d decoder0.0 -t ram -w 1 -m mem0
daxctl list
```

`daxctl list` shows one device, for example `dax0.0`. If `kmem` already claimed it as system RAM (`daxctl list` shows `"mode":"system-ram"`), switch it back with `daxctl reconfigure-device --mode=devdax --force dax0.0`. If `cxl create-region` fails and `dmesg` reports that the CPU cache could not be synchronized, the kernel lacks `CONFIG_CXL_REGION_INVALIDATION_TEST`.

#### As a DAX device

In `devdax` mode, programs `mmap()` `/dev/dax0.0` and load and store directly. Mapping offsets and lengths must be multiples of the device's alignment, which `daxctl list` shows as `align` (normally 2 MiB). Every access

reaches the device; nothing in the guest page cache sits in front of it. The caching API (§4.20) tools expect this mode.

### As system RAM on a NUMA node

daxctl hands the region to the kmem driver, which onlines it as memory:

```
daxctl reconfigure-device --mode=system-ram --force dax0.0
numactl -H                # a new node with memory and no CPUs
numactl --membind=1 ./my-workload
```

The region becomes its own NUMA node only when the guest has an ACPI SRAT, which QEMU generates when the guest RAM is described with `-numa`. Without it the guest has no NUMA information and the new memory gets no node of its own. Pass the guest RAM as a NUMA node; the memory backend size must equal `-m` (4G with the script's default RAM):

```
../femu-scripts/run-cxlssd.sh \
  -object memory-backend-ram,id=ram0,size=4G \
  -numa node,nodeid=0,cpus=0-3,memdev=ram0 \
  -drive file=$HOME/images/u20s.qcow2,if=none,id=hd0 \
  -device virtio-blk-pci,drive=hd0,bus=pcie.0 \
  -netdev user,id=net0,hostfwd=tcp::8080-:22 \
  -device virtio-net-pci,netdev=net0,bus=pcie.0
```

With this the CXL window becomes node 1. FEMU's guest runs wrote and verified data on node 1 in all three der modes; long workloads on it have not been checked. With `der=off` each access to the node exits to QEMU, so expect a workload there to run orders of magnitude slower than on node 0; use `der=mems1ot` or `der=cylon` to serve cache hits at memory speed.

### 4.18.6 Cache

| Property     | Default | Meaning                                                                                                                                                                   |
|--------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cache-pages  | 1024    | Pages of 4 KiB the cache holds. 0 disables the cache: every read is a NAND read and every write a NAND program. At most the media page count, and divisible by cache-ways |
| cache-ways   | 16      | Entries per set. 1 is direct mapped; equal to cache-pages is fully associative                                                                                            |
| cache-policy | fifo    | Replacement within a set: fifo drops the oldest entry, lifo the newest, clock the first without a recent reference, s3-fifo uses small, main and ghost queues             |

What an access costs:

- A hit costs no media time.
- A read miss reads the page from NAND and inserts it. A page the FTL has never mapped costs no media time, though it still counts in media-reads; its contents come from the backend, which reads as zeros for a fresh memory-backend-ram.
- A write miss reads the page as a read miss does, then the write makes it dirty.
- Evicting a dirty page programs it to NAND, and the access that caused the eviction pays for it. A clean eviction is free.

cache-ways, prefetch-degree and prefetch-stride can be changed while the guest runs with `qom-set`. Changing cache-ways writes back dirty pages and rebuilds the cache. `flush-cache` writes back every dirty

page and drops the cache, waiting for the modelled media time; it is how you start a measurement from a cold cache:

```
{ "execute": "qom-set", "arguments": { "path": "/machine/peripheral/cxlssd", "property": "flush-cache", "value": true } }  
{ "execute": "qom-set", "arguments": { "path": "/machine/peripheral/cxlssd", "property": "cache-ways", "value": 4 } }
```

The path ends with the id= of the device; run-cxlssd.sh uses cxlssd.

## Fast load

fast-load=true speeds up warmup and data loading phases that are not measured. While it is on, an access does not wait for its modelled media time. Everything else still runs: the FTL request, cache inserts and evictions, prefetch, direct mapping and every counter. The NAND timelines still advance, so the skipped time builds up as a backlog on the LUNs.

fast-load=false is a barrier. It waits for the accesses in progress, then waits until the modelled NAND is idle, and only then returns. It does not flush the cache. fast-load-drain-ns gives the time that this wait took. After it returns, accesses pay the full media time again.

```
{ "execute": "qom-set", "arguments": { "path": "/machine/peripheral/cxlssd", "property": "fast-load", "value": true } }  
{ "execute": "qom-set", "arguments": { "path": "/machine/peripheral/cxlssd", "property": "fast-load", "value": false } }  
{ "execute": "qom-get", "arguments": { "path": "/machine/peripheral/cxlssd", "property": "fast-load-drain-ns" } }
```

The contract is narrow:

- Only the wait at the end of an access is skipped. Flush, cache-ways changes, caching API commands and a linked NVMe controller keep their timing.
- A serial access sequence gives the same cache and media counters with fast-load on or off. media-time-ns can differ, because accesses that arrive sooner queue behind each other on a LUN.
- With concurrent accesses, the final state is not guaranteed to be the same. Shorter page holds change which evictions succeed, and a guest that runs faster can issue its accesses in a different order.
- Use it for warmup and loading only. Compare measured phases against a normal run if the starting state matters.

## Prefetch

On a miss, after inserting the missed page, the device inserts up to prefetch-degree more pages starting prefetch-stride pages after it: the pages [page + stride, page + stride + degree). Pages already cached or past the end of the media are skipped. Prefetched pages are inserted with no NAND read and no media time; a dirty page they evict is still programmed. The degree is capped at cache-pages. The default degree is 0, which turns prefetch off.

```
CACHE_PAGES=4096 CACHE_WAYS=16 CACHE_POLICY=s3-fifo PREFETCH_DEGREE=4 \  
../femu-scripts/run-cxlssd.sh
```

## NAND geometry and timing

| Property                      | Default                | Meaning                                                                                                |
|-------------------------------|------------------------|--------------------------------------------------------------------------------------------------------|
| channels                      | 4                      | NAND channels, 1 to 4096                                                                               |
| luns-per-channel              | 4                      | LUNs per channel, 1 to 128, one plane each                                                             |
| pages-per-block               | 256                    | 4 KiB pages per block                                                                                  |
| blocks-per-plane              | 0                      | 0 sizes the NAND to 5/4 of the media plus 4 blocks per plane, which leaves room for garbage collection |
| read-ns, program-ns, erase-ns | 40000, 200000, 2000000 | Page read, page program and block erase times, at most one second                                      |

| Property                        | Default | Meaning                                                                                    |
|---------------------------------|---------|--------------------------------------------------------------------------------------------|
| channel-ns                      | 0       | Channel transfer time per page                                                             |
| gc-threshold, gc-threshold-high | 75, 95  | Percent of lines in use at which garbage collection starts and is forced                   |
| ftl                             | on      | off charges no media time at all; the device behaves as plain memory with a cache in front |

Sectors are 512 bytes, eight per 4 KiB page. Misses on different LUNs overlap in the NAND model. Two switches reproduce the media model of the published Cylon experiments and are off by default: `cylon-first-touch-program=on` charges a program instead of a free read the first time an unmapped page is read, and `cylon-free-writeback=on` makes dirty write-backs free.

```
CXL_SIZE=1G CHANNELS=8 LUNS_PER_CHANNEL=4 READ_NS=50000 PROGRAM_NS=500000 \
  ../femu-scripts/run-cxlssd.sh
```

#### 4.18.7 Direct mapping: der

With `der=off` every load and store traps to QEMU, even a cache hit. That gives exact counts and exact timing, but a hit costs microseconds. The other two modes map cached pages straight into the guest, so hits run at DRAM speed without an exit.

| Mode          | How hits are served                                                    | Needs                                                                            | Limits                                               |
|---------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------|
| off (default) | Every access traps to QEMU                                             | Nothing                                                                          | Slow hits; every hit is counted and timed            |
| memslot       | Cached pages become KVM memory slot aliases                            | KVM. Refused under TCG                                                           | At most 1024 mapped pages (4 MiB) across all devices |
| cylon         | Cached pages are written into KVM's page tables by a Cylon host kernel | The fixed Cylon host kernel, a hugetlb backend, <code>cylon-kernel-ack=on</code> | Only on that kernel; falls back to MMIO elsewhere    |

As a reference point, one host (Xeon Gold 6548Y+, 256 MiB device, 1024-page cache) measured a median cached load of about 3.0 us with off and 105 ns with memslot and with cylon.

What changes in the direct modes:

- A direct hit never reaches QEMU, so it is not counted in cache-hits, does not update the clock or s3-fifo reference state, and has no media time.
- Direct mapping works only in the topology `run-cxlssd.sh` builds: one endpoint directly below the only root port of a host bridge, in a window with one target, not interleaved. Elsewhere the device still works, all accesses use MMIO, and `der-fallbacks` counts the refused mappings.
- Mappings are revoked before an eviction, a flush, a PCI configuration or decoder change, a CXL mailbox command (except a control Get LSA, below), a reset and unplug. The page maps again on its next access.
- `der-ratio` maps a fixed fraction of all pages, whether cached or not, to reproduce Cylon's direct ratio experiments. See the design note.

#### der=memslot

```
DER=memslot ../femu-scripts/run-cxlssd.sh
```

Each mapped page is a one-page alias in the guest address space. All femu-cxl-ssd devices share a budget of 1024 aliases (4 MiB), fewer when KVM runs short of free memory slots. A cached page that finds the budget full is served by MMIO at `der=off` cost and counts in `der-fallbacks`. When the hot set moves, a cached page that takes 256 trapped hits while the budget is full replaces this device's oldest alias that does not hold a pinned page, at most `der-replace-rate` times per second (default 64; 0 disables replacement). Every mapped page is treated as dirty, because writes through an alias are invisible to QEMU, so its eviction costs a program.

Realize refuses `der=memslot` under TCG ("`der=memslot` is not supported with TCG; use KVM, or `der=off`"): the alias changes would race with other vCPUs' TLBs. Use `memslot` for hot sets of up to 4 MiB on a stock kernel.

### **der=cylon**

`cylon` writes direct entries into KVM's EPT page tables through two ioctls of the Cylon host kernel. It has no per-page alias cost, samples EPT dirty bits so clean pages are not programmed on eviction, and suits hot sets and caches larger than 4 MiB.

It needs all of these on the host; FEMU checks the ones it can:

- A Cylon kernel with the dual-slot fixes on MoatLab/Cylon master. The published CylonLinux 6.4.6 corrupts host memory with this interface. FEMU cannot tell the two apart, so realize fails unless you set `cylon-kernel-ack=on` to state that the fixed kernel is running.
- KVM on Intel with EPT, EPT A/D bits, the TDP MMU and MMIO caching enabled, 4 KiB host base pages and no dirty ring.
- A hugetlbfs memory backend with `share=on` and `prealloc=on`. The device locks it with `mlock()` and reads `/proc/self/pagemap`, so run QEMU as root.
- `-machine smm=off` and the same CPU model on every vCPU (`run-cxlssd.sh` does both).

```
sudo DER=cylon CYLON_KERNEL_ACK=on \  
    CXL_BACKEND=memory-backend-file,mem-path=/dev/hugepages,share=on,prealloc=on \  
    ../femu-scripts/run-cxlssd.sh
```

When a check fails the device prints one warning naming the first missing requirement, keeps working on MMIO as with `der=off`, and `der-active` reads false. On a stock kernel:

```
DER=cylon CYLON_KERNEL_ACK=on ../femu-scripts/run-cxlssd.sh  
  
qemu-system-x86_64: -device femu-cxl-ssd,...: warning: FEMU CXL DER unavailable: Cylon GET_LINEAR_SPT/SET_SPT_FLAG require a Cylon KVM host; using MMIO
```

While a `cylon` VM runs, do not migrate, offline or soft-offline the hugetlb pages, and do not run idle page tracking (`/sys/kernel/mm/page_idle`) or DAMON on the QEMU process. The design note's host and guest restrictions<sup>63</sup> explain why.

### **Concurrent misses**

`concurrent-misses` decides whether misses to different pages wait for the media together or one at a time. `auto` (default) overlaps them only while a direct mode is active; `on` and `off` force it. With `der=off`, a guest's lock-prefixed read-modify-write reaches the device as a separate read and write and is never atomic. Overlapping misses make lost updates common (about half of `lock` add increments with four vCPUs in FEMU's tests); one miss at a time keeps them rare but not absent. Do not rely on atomic operations to `der=off` memory, and keep `concurrent-misses` at `auto` or `off` there.

---

<sup>63</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md#host-and-guest-restrictions>



### 4.18.8 Control channel through the label area

`lsa-control=on` lets the guest send experiment commands with the CXL Get LSA mailbox command, in the form Cylon's scripts use: the read size is the command and the read offset its argument.

```
cxl read-labels mem0 -s 1 -O 42 -o /tmp/status.bin # command 1, argument 42
od -An -tu1 -N1 /tmp/status.bin                  # 0 ok, 1 error, 2 queued
```

| Command | Argument and effect                                                                                                                                                                                                                                                                                                                  |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Append a statistics snapshot tagged with the argument to <code>cxlssd-stats.log</code> , then reset the event counters as <code>stats-reset</code> does                                                                                                                                                                              |
| 2       | Write dirty pages back and drop unpinned pages, as <code>flush-cache</code>                                                                                                                                                                                                                                                          |
| 3       | Set the ways: 0 to 4 mean 1, 2, 4, 8, 16 ways; 5 means fully associative                                                                                                                                                                                                                                                             |
| 5       | Set <code>prefetch-degree</code>                                                                                                                                                                                                                                                                                                     |
| 7       | Set <code>prefetch-stride</code>                                                                                                                                                                                                                                                                                                     |
| 9, 11   | Flush and clear the cache, as 2                                                                                                                                                                                                                                                                                                      |
| 13      | Open the next per-access log, <code>cxlssd-io-N.log</code> (64 names, reused in turn)                                                                                                                                                                                                                                                |
| 15      | Close the per-access log                                                                                                                                                                                                                                                                                                             |
| 17      | Dump the current direct mappings to <code>cxlssd-spt.log</code>                                                                                                                                                                                                                                                                      |
| 90, 80  | Set a direct ratio / remove it. The ratio is one of 50, 75, 90, 95, 97, 98, 99, 995 (99.5%), 999 (99.9%) or 100 percent, and 0 means 100. Setting one needs <code>der=memslot</code> or <code>der=cylon</code> ; <code>memslot</code> refuses a ratio needing more aliases than are free, which on a 256 MiB device leaves 99 and up |
| 91, 81  | Empty the host trace buffer and start tracing / stop tracing, in <code>tracefs-dir</code>                                                                                                                                                                                                                                            |

Byte zero of the data returned is the status: 0 success, 1 error, 2 queued. Commands 1, 5, 7, 13, 15, 17, 80, 81, 90 and 91 run at once unless the device is busy, and report 0 or 1. Flushes and way changes (2, 3, 9, 11), unknown commands and anything that finds the device busy are queued and run after the read returns, from QEMU's main loop; their byte zero is 2, and an error shows only in `control-status` over QMP, which reads 2 until the queue drains and then 0 or 1 for the last command. A read larger than the mailbox payload also returns 1. Commands 2, 3, 9 and 11 also clear the cache event counters.

`lsa-control` is off by default on the device and on in `run-cxlssd.sh`. With it on, the device serves a 128 MiB internal label area and refuses an `lsa` backend. Turn it off for guests you do not trust: commands 1, 13 and 17 let the guest create and write files on the host in `log-dir` (the working directory by default), and commands 91 and 81 let it write the host's `tracing_on` and `trace` files when `tracefs-dir` is set. Each file is capped at `log-limit` (64 MiB by default), statistics appends are rate limited, and `log-dropped` counts what was refused.

```
LSA_CONTROL=off LOG_DIR=/tmp/cxl-logs ../femu-scripts/run-cxlssd.sh
```

The same commands are available from the host whatever `lsa-control` says: set `control-argument`, then `control-command`, with `qom-set`. The command runs before `qom-set` returns, and an invalid argument is reported as a QMP error.

### 4.18.9 Counters

The device's counters are QOM properties. Start QEMU with a QMP socket by adding `-qmp unix:/tmp/qmp.sock,server=on,wait=off` after the script name, then read them with QEMU's `scripts/qmp/qom-get`, from the QEMU source tree:

```
export QMP_SOCKET=/tmp/qmp.sock
scripts/qmp/qom-get /machine/peripheral/cxlssd.cache-misses
```

```
scripts/qmp/qom-get /machine/peripheral/cxlssd.media-time-ns
```

or send raw QMP, for example with `socat - UNIX-CONNECT:/tmp/qmp.sock:`

```
{"execute": "qmp_capabilities"}
{"execute": "qom-get", "arguments": {"path": "/machine/peripheral/cxlssd", "property": "media-reads"}}
```

The ones you need most:

| Counter                                                                    | Meaning                                                                                                                                                                                             |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cache-hits, cache-misses, read-hits, read-misses, write-hits, write-misses | Trapped lookups; direct hits are not counted                                                                                                                                                        |
| cache-entries, cache-evictions, prefetch-inserts                           | Cache occupancy and churn                                                                                                                                                                           |
| media-reads, media-writes, media-time-ns                                   | NAND page reads, page programs and total modelled media time                                                                                                                                        |
| media-full                                                                 | Accesses whose NAND program found no free page. That program is not timed and the access completes uncached; a fill read already issued is still charged. A measurement is valid only while it is 0 |
| der-active, der-mapped, der-fallbacks                                      | Whether direct mapping is on, how many pages are mapped now, and refused mappings                                                                                                                   |

`qom-set ... stats-reset true` copies the counters to the `last-*` properties and clears the cache, prefetch and caching API event counters. The media counters, `media-full` and the `der-*` counters are never cleared; measure them as differences between two reads. The full list is in runtime properties (§A.2.2).

#### 4.18.10 Limits

- No live migration and no snapshots of device state.
- Volatile only: no persistent memory or dynamic capacity (`memdev`, `persistent-memdev`, `volatile-dc-memdev` and `num-dc-regions` are refused). An `lsa` label backend works only with `lsa-control=off`.
- Direct mapping needs the single-endpoint topology above.
- With `der=off`, every access costs an exit to QEMU; large workloads take hours. Use a small device and cache when you only need correct behaviour.

#### 4.18.11 Citation

`femu-cxl-ssd` comes from Cylon, described in Cylon: Fast and Accurate Full-System Emulation of CXL-SSDs<sup>64</sup> (FAST '26). If you use the CXL SSD mode, please also cite:

```
@inproceedings{Yoon+26-Cylon,
  author    = {Dongha Yoon and Hansen Idden and Jinshu Liu and Berkay Inceisci and
              Sam H. Noh and Huaicheng Li},
  title     = {{Cylon: Fast and Accurate Full-System Emulation of CXL-SSDs}},
  booktitle = {24th USENIX Conference on File and Storage Technologies (FAST 26)},
  pages     = {313--327},
  year      = {2026},
}
```

<sup>64</sup><https://www.usenix.org/conference/fast26/presentation/yoon>

### 4.18.12 Related pages

- CXL SSD design note<sup>65</sup>
- CXL caching API (§4.20)
- CXL NVMe link (§4.21)
- Timing model (§6.2.10)
- Device properties (§A.1.3)

## 4.19 CXL SSD (`femu-cxl-ssd`)

The device comes from Cylon (FAST '26); the user guide has its citation (§4.18.11).

This chapter describes the CXL SSD as a component: where it sits, what state it keeps, how an access moves through it, which thread charges which cost, and which parameters and counters belong to it. It ties together three existing pages:

- The design note<sup>66</sup> is the detailed reference. It records every locking rule, every KVM corner case and every validation result. This chapter links to it for depth instead of repeating it.
- The CXL SSD guide (§4.18) shows how to run the device.
- The caching API guide (§4.20) and the CXL NVMe link guide (§4.21) cover the two optional features.

### 4.19.1 Purpose

`femu-cxl-ssd` is a CXL Type-3 memory device whose capacity is backed by emulated NAND flash. The guest sees ordinary volatile CXL memory: it creates a region and uses it as a DAX device or as system RAM. Behind the memory interface the device keeps a DRAM page cache in front of a private instance of the BBSSD FTL and NAND timing model. A cache hit costs no media time; a miss costs a NAND page read; evicting a dirty page costs a NAND page program.

The device is a subclass of QEMU's `cxl-type3`. It keeps the parent's decoder translation, capacity reporting, CDAT and media-disable behaviour, and adds the cache, the media timing, optional direct mapping of cached pages into the guest, a guest-controlled caching API and an optional NVMe front end over the same medium. It changes no other FEMU mode.

Three properties set it apart from the NVMe modes:

1. The data path is a load or store, not a command. Each access that is not direct-mapped is an MMIO exit on a vCPU thread, and that vCPU waits out the modelled media time before the load or store completes.
2. The payload is a QEMU host memory backend (`volatile-memdev`), not FEMU's own DRAM backend. The cache holds metadata only: page numbers, dirty state and replacement state.
3. The FTL never touches payload bytes. It receives page numbers, opcodes and timestamps, and returns a latency.

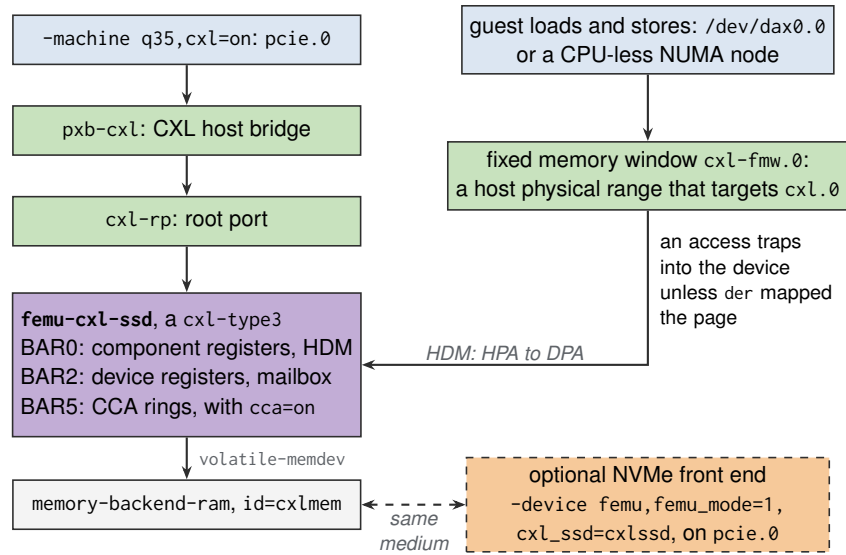
### 4.19.2 Place in the CXL topology

The guest reaches the device through a CXL fixed memory window (CFMW), a host physical address range that the machine routes to one or more CXL host bridges. Below a host bridge sit root ports, optionally a

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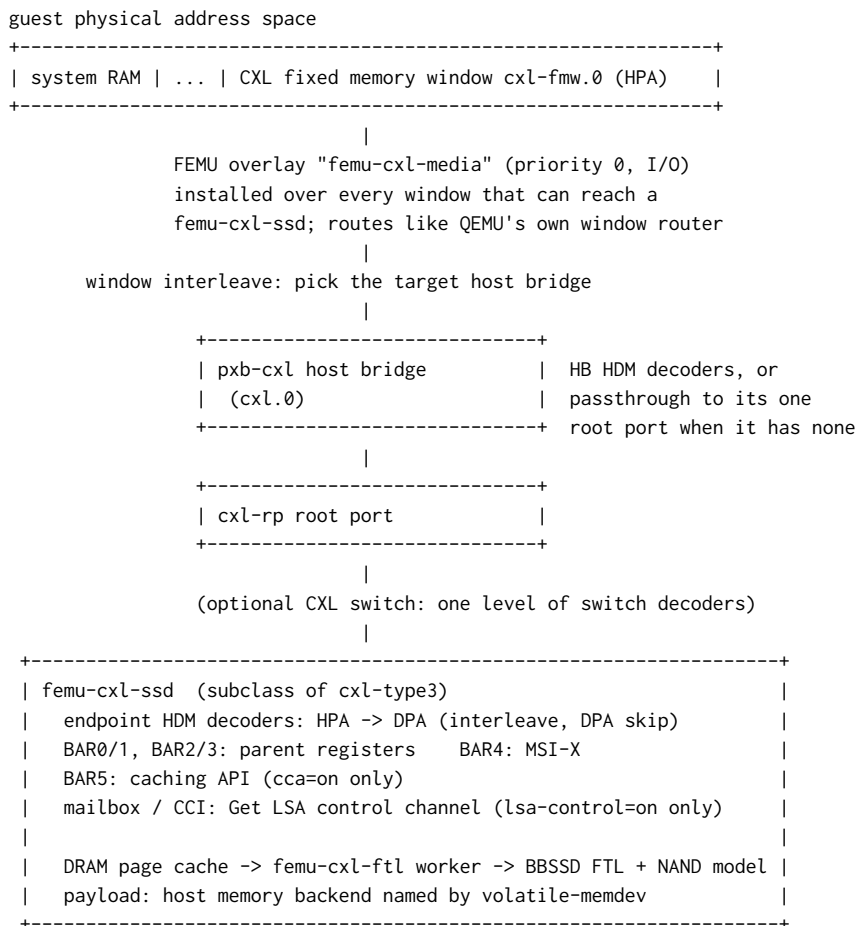
<sup>65</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md>

<sup>66</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md>



**Figure 22:** Where femu-cxl-ssd sits: a CXL type-3 device under pxb-cxl and cxl-rp on a q35 machine, backed by a RAM memory backend; guest loads and stores to the fixed memory window trap into the device unless der has mapped the page. An optional NVMe controller can serve the same medium.

CXL switch, and the Type-3 endpoint. Each level has HDM decoders that the guest's CXL driver programs when it creates a region.



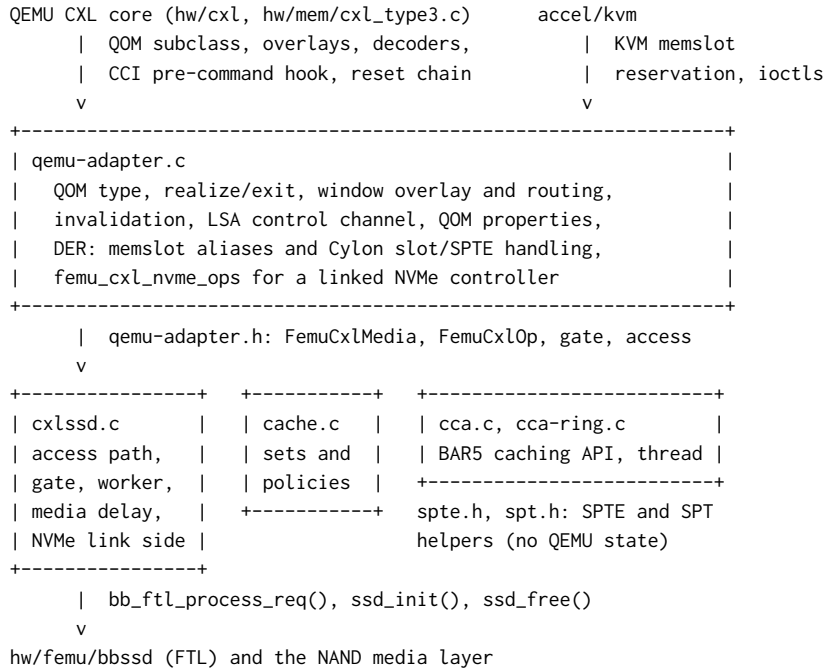
run-cxlssd.sh builds the simplest form: one pxb-cxl, one cxl-rp, the device, and a single-target window. The device realizes in any topology a plain cxl-type3 accepts: interleaved windows, host bridges with HDM decoders and several root ports, and below a CXL switch. Address routing follows the window interleave, the host bridge decoders (or its single root port when it has none), one level of switch decoders, then endpoint translation, the same order QEMU's own window router uses. A selected endpoint that is not a femu-cxl-ssd is forwarded to the window's original dispatch, so a plain cxl-type3 in the same window keeps working. An address that no decoder claims reads as zero with a transaction error, and a write to it is dropped.

Only the direct mapping modes are tied to one topology: a single endpoint directly below the only root port of a host bridge without HDM decoders, in a single-target window, with no committed interleaved endpoint decoder (der\_windows\_scan()). Elsewhere the device works, accesses stay on MMIO and refused mappings count in der-fallbacks.

Where it lives: adapter\_route(), adapter\_translate(), adapter\_access() and adapter\_machine\_done() in hw/femu/cxlssd/qemu-adapter.c.

### 4.19.3 The adapter boundary

All dependencies on QEMU's CXL device layout, fixed windows, decoders, CCI state and KVM ioctls are in one file, hw/femu/cxlssd/qemu-adapter.c. The rest of the component talks to it through hw/femu/cxlssd/qemu-adapter.h.



Two generic hooks live outside hw/femu/, and both are inert unless a femu-cxl-ssd uses them:

| Hook                     | Where                                                                                                                               | What it does                                                                                                                                                   |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KVM memslot reservation  | kvm_reserve_memslot(), kvm_reserved_memslot_id() and kvm_release_memslot() in accel/kvm/kvm-all.c, declared in include/system/kvm.h | Reserves a slot ID that the KVM memory listener never allocates and that does not count as free capacity. der=cylon registers its dual-mode slot under that ID |
| CCI pre-command callback | pre_command and pre_command_opaque in CXLCI (include/hw/cxl/cxl_device.h), called from hw/cxl/cxl-mailbox-utils.c                   | Runs before every CCI command on every transport, without replacing any command handler. FEMU uses it to revoke direct mappings                                |

Everything else is done by chaining parent methods: PCI configuration writes call the inherited method after FEMU revokes, component register writes go through a FEMU overlay on the component register block, reset chains the parent's hold phase, and the LSA class methods are always overridden but pass through to the parent's unless lsa-control=on.

The NVMe side never includes CXL headers. hw/femu/femu.c, and hw/femu/bbssd/bb.c for flips, call the medium through femu\_cxl\_nvme\_ops, a table of five functions (prepare, attach, detach, ftl, flip) that qemu-adapter.c registers. The CXL files are built only when both CONFIG\_FEMU\_PCI and CONFIG\_CXL\_MEM\_DEVICE are on (hw/femu/meson.build); without them the table stays empty and cxl-ssd= is refused.

#### 4.19.4 Data structures

| Structure  | File           | Fields that matter                                                                          |
|------------|----------------|---------------------------------------------------------------------------------------------|
| FemuCxlSsd | qemu-adapter.c | parent_obj (the CXLType3Dev), media, component_overlay, lsa_queue (queued control commands) |

| Structure                              | File           | Fields that matter                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FemuCxlWindow                          | qemu-adapter.c | One per fixed window that can reach a FEMU endpoint: the window and its io overlay region. Shared by all FEMU devices; removed when the last one leaves                                                                                                                                                                                                                     |
| FemuCxlMedia                           | qemu-adapter.h | backend (payload pointer and size), cache, direct (DER state), cca; the gate (busy, accesses, exclusive_waiters, idle condition variable); pages (pages held by accesses in progress); invalidations (generation); lock, worker_cond, work, worker (the FTL worker); ns.ssd (the private FTL); NVMe link fields nvme, nvme_ranges, nvme_taken, nvme_done, nvme_bh; counters |
| FemuCxlOp                              | qemu-adapter.h | One access, flush or eviction chain: the media time ns it has accumulated and the pages it holds itself                                                                                                                                                                                                                                                                     |
| FemuCxlWork                            | qemu-adapter.h | One FTL request on the worker queue: an NvmeRequest, the returned latency, done, and done_cond, the waiter's condition variable                                                                                                                                                                                                                                             |
| FemuCxlCache, FemuCxlSet, FemuCxlEntry | cache.h        | nsets, ways, policy, entries and ghosts hash tables; per set the small, main, ghost and pinned queues; per entry lpn, dirty, freq, queue, der_hits, der_displaced                                                                                                                                                                                                           |
| FemuCxlDer                             | der.h          | maps (mapped pages), available, cylon, fast (Cylon state), ratio, installed (memslot aliases, oldest first), replacement state, DER counters                                                                                                                                                                                                                                |
| FemuCxlMap                             | qemu-adapter.c | One memslot alias: first page, page count, alias region, queue link                                                                                                                                                                                                                                                                                                         |
| FemuCylon, CylonPage                   | qemu-adapter.c | Slot reservation, memory listener, SPT chunk areas, resolved huge page frames; per mapped page the SPTE pointer and the saved MMIO SPTE                                                                                                                                                                                                                                     |
| FemuCxlCca                             | cca.h          | BAR5 regions, host ring cursors, status, epoch, uncached_map (one bit per media page), thread, kick, stop, counters                                                                                                                                                                                                                                                         |

Keys in the cache and DER tables are full 64-bit page numbers ( $LPN = DPA / 4096$ ). The cache library (cache.c) has no QEMU dependency and builds standalone for its unit test.

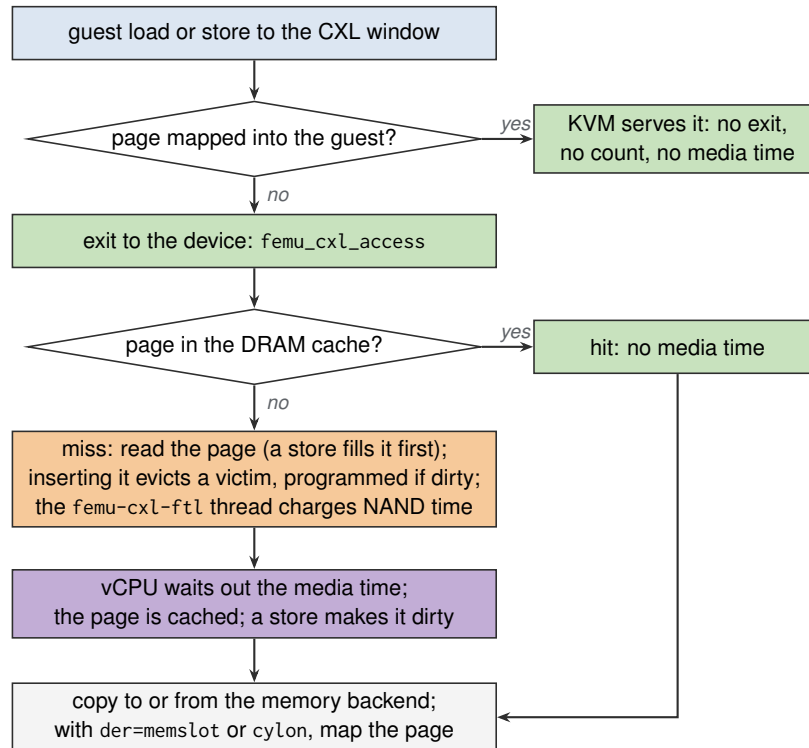
## 4.19.5 The media path

### From a guest access to the cache

The overlay callback `adapter_access()` runs on the vCPU thread that took the MMIO exit, holding the BQL. It routes and translates the address, takes the device's operation gate, routes again (decoders may have changed while it waited), and calls `femu_cxl_access()` in `cxlssd.c`. If media is disabled (the parent's media-disable state), reads return random bytes and writes are discarded, as the parent Type-3 device does.

`femu_cxl_access()` handles an access of 1 to 8 bytes, which may cross one page boundary:

1. Hold each page it touches, in ascending order, in `s->pages`. A second access to a held page waits. This keeps accesses to one page ordered and makes a second miss wait for the first fill instead of repeating it.
2. Look up each page in the cache. On a miss, ask the FTL worker for a page read (or a program, when the page is not cacheable and the access is a write), then insert the page. Insertion may evict a victim; a dirty victim costs a program. A write marks the entry dirty.
3. On a demand miss, prefetch (see Prefetch (§4.19.6)).
4. Wait out the accumulated media time with the BQL released.



der=off (default): nothing is mapped, so every access exits and is counted. der=memslot: a cached page becomes a one-page KVM memory slot, up to 1024 for all devices; without KVM realize fails, and a page that finds no free slot stays on MMIO. der=cylon: a Cylon host kernel writes the page into KVM's EPT; realize fails without cylon-kernel-ack=on, and when the host lacks Cylon support the device warns once and runs as der=off. Evicting a page unmaps it first.

**Figure 23:** One guest access to CXL SSD memory: a page mapped by der=memslot or der=cylon never exits; otherwise the device looks the page up in its DRAM cache, and a miss evicts a victim, reads the page through the FTL and makes the vCPU wait out the NAND time.

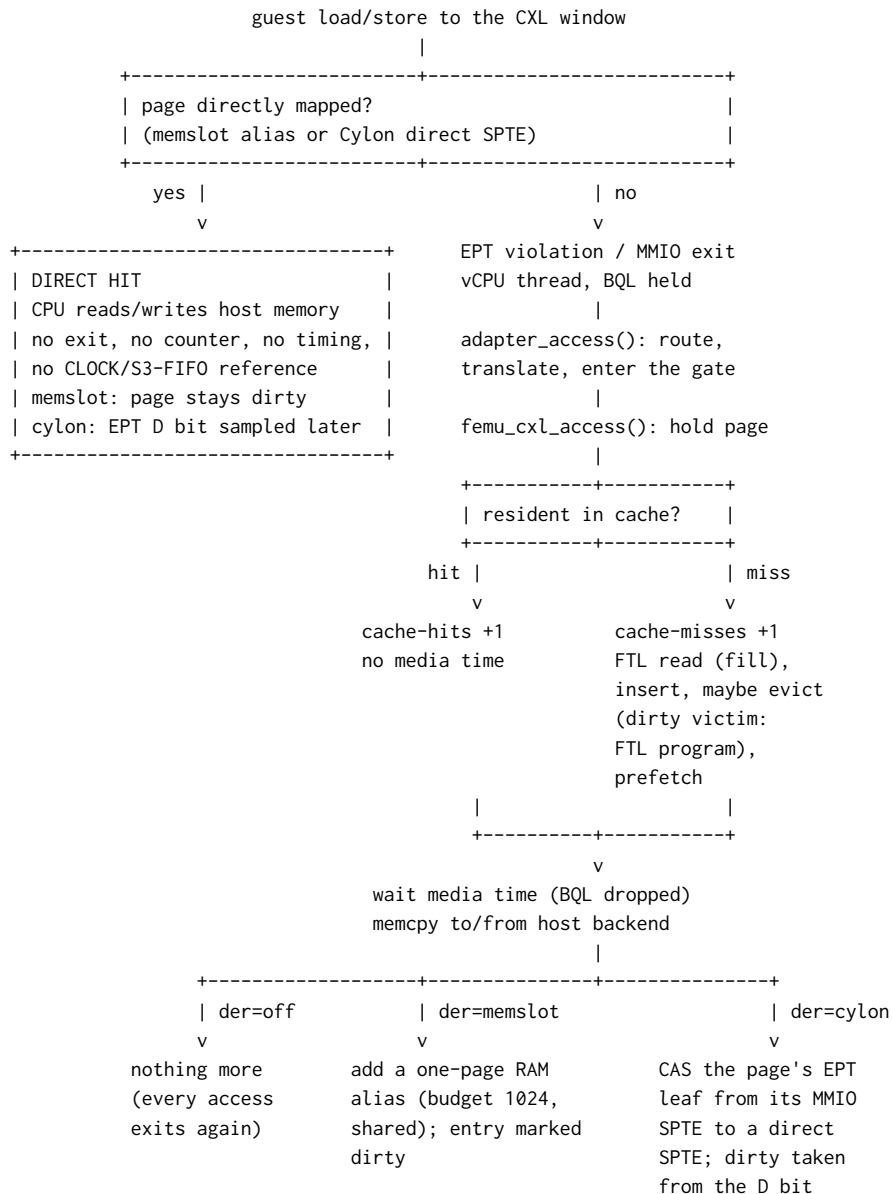


5. Copy the bytes between the guest and the host memory backend.
6. For a single-page access to a cached (or ratio-selected) page, try to map the page directly into the guest (see Direct mapping (§4.19.8)).

A page is not cacheable when the cache is off (`cache-pages=0`), when the caching API marked it uncached, or when every way of its set is pinned. Such an access goes to the media every time: a read is a NAND read, a write a NAND program. An access also goes uncached when the only victim its insert could evict is held by another access in progress.

The write miss is write-allocate with fill: a store to a page that is not resident first reads the page from NAND, then dirties it. An unmapped page (one the FTL has never programmed) costs no media time on a read, though it still counts in media-reads; its bytes come from the backend, which reads as zeros for a fresh memory-backend-ram.

### Access path under each der mode



With `der=off` the same page exists on every access, so every hit is counted and timed. In the direct modes the first access after a miss completes in QEMU and installs a mapping; the guest's later accesses to that page are direct hits until the page is evicted or an invalidation revokes it.

### The operation gate

The gate (`femu_cxl_enter()`, `femu_cxl_enter_access()` and their leave pairs in `cxlssd.c`) is a reader/writer style lock built on the BQL and the `idle` condition variable:

| Holder                                                                                                                                                                                   | Mode                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Guest access, <code>concurrent-misses</code> in effect                                                                                                                                   | Shared ( <code>femu_cxl_enter_access()</code> ) |
| Guest access otherwise                                                                                                                                                                   | Exclusive ( <code>femu_cxl_enter()</code> )     |
| <code>flush-cache</code> , <code>stats-reset</code> , <code>fast-load</code> , <code>cache-ways</code> , <code>prefetch</code> and <code>der-ratio</code> changes, every control command | Exclusive                                       |
| Each caching API chunk                                                                                                                                                                   | Exclusive                                       |
| The NVMe link bottom half that drops cache entries                                                                                                                                       | Exclusive                                       |

An exclusive waiter blocks new shared holders, so a flush is not starved by a stream of accesses. Waiters release the BQL while they wait. Invalidation and teardown never wait for the gate (see Invalidation (§4.19.9)).

`concurrent-misses` decides whether accesses share the gate. With `auto`, they share it only while a direct mode is available (`femu_cxl_concurrent()`). Sharing lets misses to different pages wait for the media together, so their NAND reads overlap where they reach different LUNs.

**Atomicity caveat.** With `der=off`, a guest's lock-prefixed read-modify-write reaches the device as a read and a separate write. It is never atomic. One access at a time keeps other vCPUs out from between the two most of the time; overlapping misses make lost updates common. FEMU's tests with four vCPUs lost 0.6% to 0.9% of `lock_add` increments with serialized accesses and 51% to 52.5% with overlapping misses. In the direct modes the write lands on the page the read mapped, where KVM's emulator exchanges and retries, and no increments were lost. Do not rely on atomic operations to `der=off` memory, and keep `concurrent-misses` at `auto` or `off` there.

### 4.19.6 The DRAM page cache

The cache (`hw/femu/cxlssd/cache.c`) is set-associative over LPNs. It holds cache-pages entries in `nsets = cache-pages / cache-ways` sets; page `lpn` belongs to set `lpn % nsets`. One way is direct mapped; cache-ways equal to cache-pages is fully associative.

```

FemuCxlCache
+-----+
| entries: hash LPN -> FemuCxlEntry  ghosts: hash LPN -> node |
| nsets = cache-pages / cache-ways  policy                    |
+-----+
      set index = lpn % nsets
      |
      v
sets[0] ... sets[i] ... sets[nsets-1]
      |
      v
      FemuCxlSet (ways = cache-ways)
+-----+
| small : [head: next victim] e e e e [tail: newest]           |
| main  : [head] e e e [tail]                                (S3-FIFO only) |
| ghost : LPNs recently evicted from small, at most `ways`   |
| pinned: e e          (caching API; never evicted)           |
+-----+
      small + main + pinned <= ways

FemuCxlEntry: lpn | dirty | freq (0..3) | queue | der_hits | link

```

## Replacement policies

cache-policy selects one policy for all sets. Every policy evicts only from `small` and `main`; pinned entries have left those queues.

| Policy  | Insert                                                                                        | Victim                                                                                                                                                                                                                                                                                                                                             |
|---------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fifo    | Tail of <code>small</code>                                                                    | Head of <code>small</code> (oldest)                                                                                                                                                                                                                                                                                                                |
| lifo    | Tail of <code>small</code>                                                                    | Tail of <code>small</code> (newest)                                                                                                                                                                                                                                                                                                                |
| clock   | Tail of <code>small</code> with its reference set                                             | Rotate from the head, clearing references, until an entry has none                                                                                                                                                                                                                                                                                 |
| s3-fifo | Tail of <code>small</code> , or of <code>main</code> when the LPN is in the set's ghost queue | Evict from <code>small</code> while it holds at least a tenth of the unpinned ways (minimum one) or <code>main</code> is empty. An entry leaving <code>small</code> with more than one hit moves to <code>main</code> ; otherwise it is evicted and its LPN joins the ghost queue. In <code>main</code> , an entry with hits loses one and rotates |

Hits increment `freq` up to 3. With one way, S3-FIFO uses only `small`. Removal of a known entry is constant time because each entry keeps its own queue node. The design note's media and cache<sup>67</sup> section has the remaining details.

## Prefetch

Only a demand miss triggers prefetch, after the demanded page is inserted. The device inserts pages [`lpn` + `prefetch-stride`, `lpn` + `prefetch-stride` + `degree`), where the `degree` is `prefetch-degree` capped at `cache-pages`. Resident, out-of-range and uncached pages, and pages whose set is fully pinned, are skipped. A prefetch performs no NAND read and adds no media time, but a dirty page it evicts is still programmed. If an insert cannot evict, prefetching stops for that access without failing it. Prefetched pages may be mapped directly. A small cache may evict the demanded page itself while prefetching, following insertion order.

<sup>67</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md#media-and-cache>

## Run-time changes

cache-ways, prefetch-degree and prefetch-stride can change while the guest runs (qom-set, or control commands 3, 5 and 7). A cache-ways change takes the gate exclusively, refuses before touching anything if the pinned pages do not fit the new geometry, then revokes mappings, writes dirty pages back, rebuilds the cache with the pinned pages still resident, and maps a direct ratio again.

### 4.19.7 Dirty data and write-back to the FTL

The cache is write-back. A dirty page is programmed to NAND when:

| Event                                       | Who pays                                                                     |
|---------------------------------------------|------------------------------------------------------------------------------|
| Eviction of a dirty victim by an insert     | The access that caused the insert, or the femu-cxl-cca thread for a PIN fill |
| flush-cache=true, control commands 2, 9, 11 | The caller (the main loop, waiting with the BQL dropped)                     |
| cache-ways change                           | The caller                                                                   |
| Caching API INVALIDATE and CACHE_DISABLE    | The femu-cxl-cca thread, per chunk                                           |

cylone-free-writeback=on skips the program on eviction and flush, as the published Cylon experiments do.

What counts as dirty depends on the der mode, because a direct store never reaches QEMU:

- der=off: a store through MMIO sets dirty.
- der=memslot: QEMU cannot see stores through an alias, so every page that gets an alias is marked dirty when it is mapped. Its eviction always costs a program.
- der=cylone: revocation samples the EPT dirty bit. A page read but never written through the direct entry is evicted clean, with no program.

The FTL worker (cxl\_worker() in cxlssd.c, thread femu-cxl-ftl) is the only thread of the device itself that changes FTL and NAND state; with a linked NVMe controller, that controller's FTL thread is the other one, and both run under s->lock. Callers queue a FemuCxlWork on s->work under s->lock, wake the worker on s->worker\_cond and wait for done on the request's own done\_cond, which lives on the caller's stack. The worker signals only that condition, so each request is woken individually and no waiter wakes for another request. The worker calls bb\_ftl\_process\_req() with an 8-sector request (one 4 KiB page of 512-byte sectors) and returns its latency. Requests are served in arrival order, and the NAND model overlaps them where they reach different LUNs. cylone-first-touch-program=on turns a read of an unmapped page into a program.

The FTL is a private BBSSD instance built in femu\_cxl\_start(): 512-byte sectors, eight per page, one plane per LUN, and the geometry, timing and GC thresholds of the device properties. It has no PCI queues and none of the optional NVMe features. With ftl=off there is no FTL and no worker: accesses have no media time and write-back is metadata only. GC, mapping and NAND timing are those of the BBSSD FTL, which the timing model (§6.2) and the architecture page's BBSSD section (§3.1.5) describe.

When NAND fills up (a geometry with no spare blocks), a write-back has nowhere to go. The access still completes, uncached, and media-full counts it; the refused program adds no media time, though a fill read already issued is charged. A measurement is valid only while media-full is 0.

#### 4.19.8 Direct mapping

der selects how cached pages may be served without an exit. The design note's Direct Endpoint Remapping<sup>68</sup> section is the full reference.

| Mode    | Mechanism                                                                                                            | Host needs                                                                                                                                                                                                                                                                         | Mapping limit                                                                                      |
|---------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| off     | None: every access is MMIO                                                                                           | Nothing                                                                                                                                                                                                                                                                            | n/a                                                                                                |
| memslot | One-page <code>memory_region_init_alias()</code> RAM aliases over the window; KVM makes each one an ordinary memslot | KVM; refused under TCG at realize                                                                                                                                                                                                                                                  | 1024 aliases shared by all devices, and never more than KVM's free slots minus 8                   |
| cylon   | A dual-mode KVM slot over the window, and direct leaves written into KVM's EPT through the Cylon kernel's linear SPT | The fixed Cylon host kernel, Intel EPT with A/D bits, TDP MMU, MMIO caching, a shared preallocated hugetlbfs backend, root (for <code>mlock()</code> and <code>/proc/self/pagemap</code> ), <code>cylon-kernel-ack=on</code> , <code>smm=off</code> , identical CPUID on all vCPUs | No per-page alias cost; the window is at most 120 GiB (60 SPT chunks of 4 MiB at 8 bytes per page) |

Every mapping's HPA is checked against the endpoint decoders before it is installed (`femu_cxl_der_map()`), so a prefetched page whose derived HPA decodes to another DPA, or to none, is not mapped. Pages in caching API uncached ranges are never mapped. `der-ratio` maps a fixed fraction of all pages, cached or not, to reproduce Cylon's direct ratio experiments; see direct ratios<sup>69</sup>.

##### **der=memslot: the alias budget and replacement**

Each alias, and the MMIO gap beside it, is a separate section of the system address space, and QEMU aborts once an address space needs 4096 sections. So all `femu-cxl-ssd` devices share one budget of 1024 one-page aliases (`FEMU_CXL_DER_ALIASES`), further capped at KVM's free slots minus 8 (`der_alias_budget()`). Ratio runs use the same budget. A cached page that finds the budget full stays on MMIO and counts in `der-fallbacks`.

When the hot set moves, the aliases may hold cold pages. Replacement (`der_replace_due()`, `der_replace_victim()`, `der_displace()`):

<sup>68</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md#direct-endpoint-remapping>

<sup>69</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md#direct-ratios>

```

MMIO access to a cached page, budget full
|
v
der-replace-rate == 0 ? ---- yes -----> stay on MMIO
| no                                     (no counting)
v
entry->der_hits += 1 ---- < 256 (FEMU_CXL_DER_HOT) ----> stay on MMIO
| >= 256          (the miss that inserted the page counts too)
v
first replacement, or time since the last one
>= (1 s / rate) << backoff ? ----- no -----> stay on MMIO
| yes
v
victim = oldest alias of this device whose page is not pinned
| none -----> stay on MMIO
v
one memory transaction: remove victim alias, add this page's alias
der-replacements +1, der-remaps +1, der-revocations +1
|
v
backoff: this page was displaced before -> backoff + 1 (max 8, so the
         interval grows up to 256x); 8 consecutive promotions of pages
         not displaced before -> backoff - 1

```

Accesses through an alias never reach QEMU, so installation order is the only recency the device can see. Pinned pages keep their aliases.

### **der=cylon: dual-mode slot and SPTE protocol**

At realize, femu\_cylon\_prepare() probes the two Cylon ioctls (KVM\_GET\_LINEAR\_SPT, VM ioctl 0xde; KVM\_SET\_SPTE\_FLAG, vCPU ioctl 0xdd), checks the KVM module parameters, the hugetlbfs backend, mlock() and the pagemap frames. Any failure prints one warning, FEMU CXL DER unavailable: <reason>; using MMIO, and the device continues as with der=off. cylon-kernel-ack=on is required even then, because the device cannot tell the fixed kernel from the published one.

The first eligible decoded access schedules cylon\_install\_bh(), which pauses all vCPUs, reserves a slot ID through the generic reservation hook, and registers a slot over the window with flag KVM\_CYLON\_DUAL\_MODE (bit 17). Its host address is the payload backend itself. The kernel then hands back the slot's linear SPT in at most 60 chunks of 4 MiB. Installation needs at least 8 free memslots.

Per page, the SPTE moves through these states:

```

      KVM's first fault on the page
(empty) -----> MMIO SPTE (kernel-created,
    |                                     generation bits saved by FEMU)
    | ratio only: CAS empty -> direct    |
    v                                     | access completes in QEMU:
DIRECT SPTE <-----+ CAS MMIO -> direct
RWX, WB, A=0, D=0                                     (der-remaps +1)
    |
    | revoke (eviction, invalidation, teardown)
    v
A bit clear? -- yes --> CAS direct -> saved MMIO, no TLB flush
    |                                     (der-quiet-revocations +1, der-revocations +1,
    |                                     page clean)
    | no (or the CAS lost to the CPU setting A)
    v
CAS clear W and MMU-writable; flush this GFN
read D; CAS -> saved MMIO; flush this GFN
D set -> cache entry dirty      (der-revocations +1)

KVM replaced the entry itself (zero, MMIO, frozen): treated as a KVM
revocation: entry made dirty, record dropped after a flush.
Unexpected entry or ioctl failure: restore saved MMIO entries, delete
the slot, disable Cylon until the next device reset.

```

An entry whose accessed bit is still clear has not been used by any page walk since it was installed, so no TLB holds it: that is the quiet path. The full path costs two single-GFN flushes per page. `der-revocations` counts every revocation, quiet or not. A whole-slot invalidation still samples each page (one flush per accessed page) but skips the final flush, because the slot deletion that follows flushes everything; later accesses reinstall the slot and pages map again lazily. Global dirty logging (migration) revokes everything and disables Cylon. A failed slot deletion aborts QEMU, because continuing could leave the kernel using a slot whose backing FEMU has released.

**The Cylon host kernel.** The published CylonLinux 6.4.6 corrupts host memory with this interface. The fixes are on MoatLab/Cylon master; the design note's host kernel<sup>70</sup> section lists them and the requirements the kernel still does not check (SMM off, identical CPUID, no move or flag change of the dual slot). Its host and guest restrictions<sup>71</sup> section explains why `hugetlb` migration, memory offline, `page_idle` and DAMON must not touch the QEMU process during a run.

**Known kernel-side limitation: first touch per 2 MiB under a ratio.** A direct ratio installs direct leaves into empty SPT entries, including pages in 2 MiB guest regions the guest has never touched. For such a region the level above the leaf is still empty. The first guest access in the region faults, KVM links the page table, and in doing so replaces FEMU's direct leaf with an MMIO SPTE. FEMU sees a KVM revocation (`der-revocations +1`), serves that access by MMIO and maps the page again (`der-remaps +1`); the rest of the region stays direct. Expect one exit per untouched 2 MiB region under a `der=cylon` ratio. This is kernel behaviour, not a device bug; the fix belongs in the kernel (keep a present direct leaf when a fault only links its parent table). Ordinary cache mappings are not affected, because they are installed only over MMIO SPTEs KVM has already created.

<sup>70</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md#host-kernel>

<sup>71</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md#host-and-guest-restrictions>

### 4.19.9 Invalidation

Direct mappings are revoked whenever the guest could change what an address means, and whenever the device needs to observe a page again. `cxl_invalidate()` revokes all mappings and bumps the invalidations generation; an access in flight that sees the generation move during its media delay does not install the mapping it translated before.

| Trigger                                          | Where                                                                         | Effect                                                                                                                                                                                        |
|--------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PCI configuration write                          | <code>adapter_config_write()</code>                                           | Revoke all, then the parent's method                                                                                                                                                          |
| Component register write (HDM decoders)          | <code>adapter_component_write()</code> through <code>component_overlay</code> | Revoke all, then the parent's registers                                                                                                                                                       |
| Any CCI command on any transport                 | <code>adapter_pre_command()</code>                                            | Revoke all; except Get LSA on the primary mailbox with <code>lsa-control=on</code> , which carries a control command                                                                          |
| Device reset                                     | <code>adapter_reset_hold()</code>                                             | Revoke all, let a failed Cylon retry, caching API reset (unpin all, end uncached ranges), free CCI background state, then the parent's hold phase                                             |
| Unplug                                           | <code>cxl_exit()</code>                                                       | Tell the caching API thread to stop, mark the device closing, disable DER, and free the media now or leave that to the last gate holder                                                       |
| Eviction                                         | <code>femu_cxl_evict()</code>                                                 | Revoke that page only (a ratio-selected page keeps its mapping; its dirty bit is sampled)                                                                                                     |
| Flush, way change, commands 2, 3, 9, 11          | <code>cxl_flush()</code> , <code>cxl_runtime_set()</code>                     | Revoke all, write dirty pages back, drop unpinned entries, map a ratio again                                                                                                                  |
| Caching API INVALIDATE, CACHE_DISABLE            | <code>cca.c</code>                                                            | Revoke the chunk's pages, or all mappings once for more than 64 candidates                                                                                                                    |
| Linked NVMe write, zeroes, copy, deallocate      | <code>femu_cxl_nvme_bh()</code>                                               | Revoke and drop the pages without write-back; pinned pages stay, made clean; ratio-selected pages keep their mapping; a range over 64 pages revokes all mappings at once when no ratio is set |
| Global dirty logging (Cylon)                     | <code>cylon_log_start()</code>                                                | Revoke all, disable Cylon                                                                                                                                                                     |
| Memory map change overlapping the window (Cylon) | <code>cylon_region_change()</code>                                            | Delete the dual slot; re-admit after fresh checks                                                                                                                                             |

Invalidation never waits for the gate. Configuration writes, component writes and CCI commands run inside another owner's re-entrancy guard, and waiting there would refuse other vCPUs' accesses. Revocation therefore touches only DER state, which is safe under the BQL alone. Warm reset keeps the volatile payload and the cache and FTL contents; only mappings, pins and uncached ranges go. Pages map again on their next access.

### 4.19.10 The LSA control channel

With `lsa-control=on` the device serves an internal, zero-filled 128 MiB label storage area (`FEMU_CXL_LSA_SIZE`) instead of an `lsa` backend, and a guest Get LSA mailbox command (opcode 0x4102) carries an experiment command: the read length is the command and the read offset its argument. This is the convention of Cylon's scripts: `cxl read-labels mem0 -s COMMAND -O ARGUMENT`. Byte 0 of the returned data is the status: 0 success, 1 error, 2 queued.



```

guest: cxl read-labels mem0 -s CMD -O ARG
      |
      v mailbox Get LSA (0x4102), vCPU thread, device re-entrancy guard held
cxl_get_lsa()
      |
      +--- CMD in {1,5,7,13,15,17,80,81,90,91}, gate free, queue empty
      |       -> run inline (cxl_command()), byte 0 = 0 or 1
      |
      +--- otherwise (2, 3, 9, 11, unknown, or device busy)
              -> append to lsa_queue, byte 0 = 2
                  main-loop BH cxl_lsa_bh() runs the queue in order;
                  control-status reads 2 until it drains

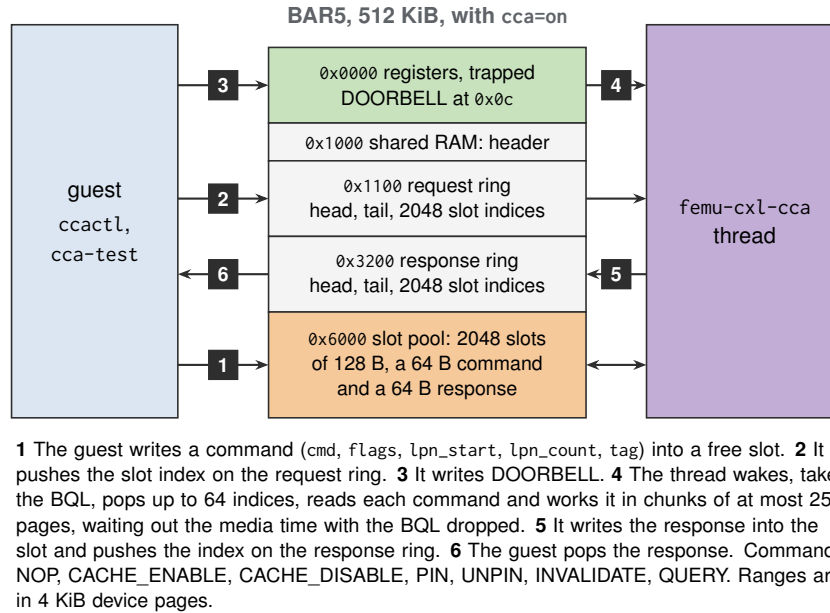
```

The command table is in the guide (§4.18.8) and the design note<sup>72</sup>. The same commands are available from the host through control-argument and control-command, whatever lsa-control says.

**Trust.** The channel lets the guest act on the host. Commands 1, 3, 5, 7, 13 and 17 create or write files (cxlssd-stats.log, cxlssd-io-N.log, cxlssd-spt.log) in log-dir, QEMU's working directory by default. When tracefs-dir is set, command 91 truncates the host's tracefs trace file and writes tracing\_on, and 81 writes tracing\_on. The device bounds this: each file stops at log-limit, statistics appends pass a token bucket (a burst of 64, then 100 per second), I/O log names cycle through 64 files, and log-dropped counts what was refused. It does not authenticate the guest. Keep lsa-control off for guests you do not trust; it is off by default on the device and on in run-cxlssd.sh. The security page (§5.13.1) says more.

#### 4.19.11 The caching API (CCA)

cca=on gives the guest control of the cache through PCI BAR5: pin and unpin pages, write back and drop ranges, mark ranges uncached, and query residency. The device side is hw/femu/cxlssd/cca.c and cca-ring.c; the ABI is one header, hw/femu/cxlssd/cca-abi.h, compiled by both the device and the guest library in hw/femu/tools/cca/.



**Figure 24:** The CXL caching API in BAR5: the guest puts a command in a slot, queues the slot on the request ring and rings the doorbell; the femu-cxl-cca thread runs it and queues the response.

## BAR5 layout

BAR5, 512 KiB, 32-bit memory BAR

offset

```
0x00000 +-----+
| registers, 4 KiB, trapped (4- and 8-byte access) |
| 0x00 MAGIC 0x43434131 "CCA1" 0x04 VERSION 2 |
| 0x08 STATUS READY|FATAL|CACHE|BUSY |
| 0x0c DOORBELL (any write) 0x10 RESET (1, 2) |
| 0x14 FATAL_REASON 0x18 MEDIA_PAGES (64-bit) |
| 0x20 CACHE_PAGES 0x24 CACHE_WAYS |
| 0x28 PIN_LIMIT 0x30 COMPLETED (64) 0x38 EPOCH |
0x01000 +-----+
| RAM "femu-cxl-cca-shm" (offsets below are BAR5) |
| 0x01000 header (magic, version, ring offsets; |
| data ring fields stay 0) |
| 0x01100 request ring: head | tail | 2048 u32 |
| 0x03200 response ring: head | tail | 2048 u32 |
| 0x06000 slot pool: 2048 slots x 128 B |
| slot = 64 B command + 64 B response |
0x46000 | (unused to the end of the BAR) |
0x80000 +-----+
```

command (64 B): cmd | flags | lpn\_start | lpn\_count | tag | rsvd[4] = 0

response (64 B): status (0 or -errno) | lpn\_start | lpn\_count (pages acted on) | tag | resident | dirty | pinned | uncached

The rings are single producer, single consumer, with free-running 32-bit head and tail; their size is a constant, never read from shared memory. The device keeps its own cursors, reads each shared index once, and copies a command out of its slot before validating it. A head too far ahead, a slot index out of range, or a response ring the guest never drains sets STATUS.FATAL with a reason and stops the device side until RESET.

<sup>72</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md#experiment-controls>

## Commands

| Command       | Effect                                                                                                                                                                                                                                                                                       |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NOP           | Status 0                                                                                                                                                                                                                                                                                     |
| PIN           | Room is checked for the whole range first (-ENOSPC). Resident pages are pinned; others are filled by a NAND read, then pinned. A later fill failure (-EIO) or a way change between chunks (-EAGAIN) leaves the pages already pinned. -EBUSY on an uncached page, -EOPNOTSUPP without a cache |
| UNPIN         | Returns pinned pages to main under S3-FIFO with more than one way, else to the tail of small; dirty state is kept                                                                                                                                                                            |
| INVALIDATE    | Revoke mappings, write dirty pages back, drop them; pinned pages need CCA_F_FORCE                                                                                                                                                                                                            |
| CACHE_DISABLE | Mark pages uncached, then drop resident ones as INVALIDATE does                                                                                                                                                                                                                              |
| CACHE_ENABLE  | End the uncached marking                                                                                                                                                                                                                                                                     |
| QUERY         | Count resident, dirty, pinned and uncached pages                                                                                                                                                                                                                                             |

Ranges are 4 KiB device pages; CCA\_F\_ALL selects the whole medium. Every way of a set may be pinned; a miss to such a set is served uncached and counted in cca-pinned-set-misses. der-ratio and uncached ranges exclude each other.

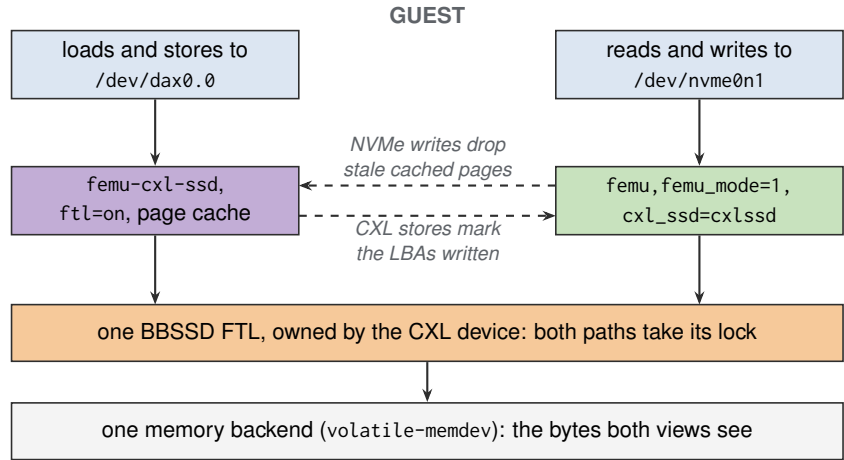
### Thread and chunking

A doorbell write signals the detached femu-cxl-cca thread and never waits. The thread takes the BQL, runs up to 64 commands in submission order, then gives the BQL up and wakes itself again, so a guest that keeps the ring full cannot hold the BQL. Each command runs in chunks of at most 256 pages of work and 4096 lookups. A chunk holds the gate exclusively, then waits out its media time with the BQL released; before each chunk the thread lets one waiting access take the gate first. Guest accesses therefore run between chunks, and the worst stall a chunk adds to an access is about 256 programs. Device reset reformats the rings, bumps EPOCH and drops a command in flight without a completion. The caching API section<sup>73</sup> of the design note covers the reset and unplug races.

### 4.19.12 The NVMe front end

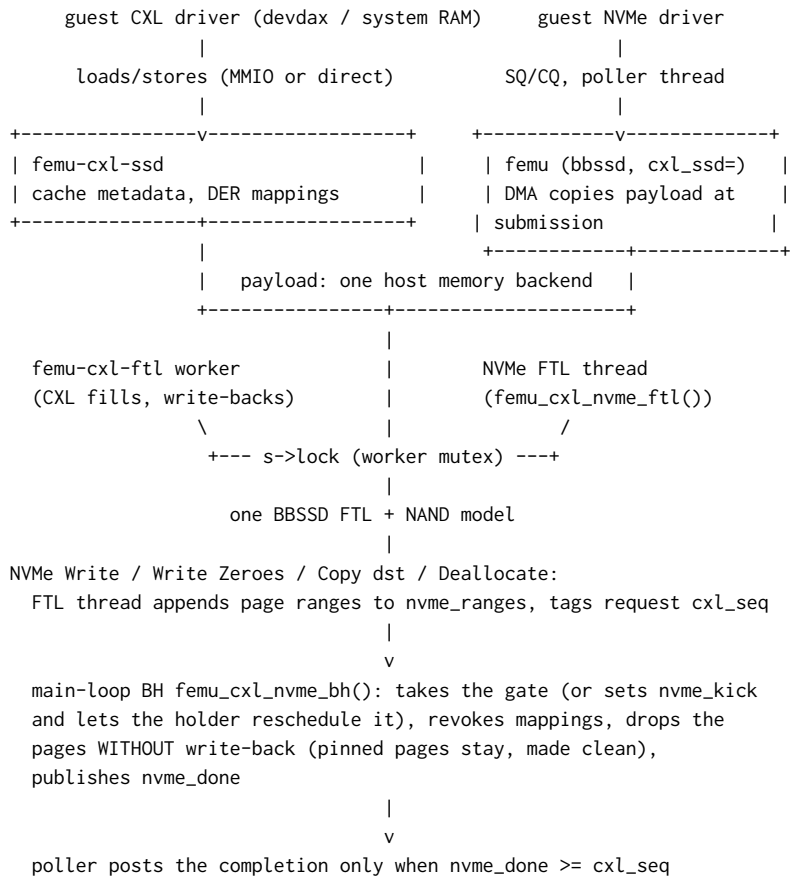
-device femu,femu\_mode=1,cxl\_ssd=<id> puts a BBSSD NVMe controller in front of the same medium. Its one namespace uses the device's memory backend as payload and the device's FTL as its own, so both front ends see one set of bytes and one mapping table.

<sup>73</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md#caching-api-cca>



The femu-cxl-ssd comes first on the command line; the femu controller sits on pcie.0 or a PCIe root port, not under pxb-cxl, has one namespace and uses the CXL device's geometry and timing. Avoid mounting a file system on the namespace while the CXL region is in use.

**Figure 25:** One medium, two views: a linked BBSSD controller serves the CXL SSD's medium as an NVMe namespace, sharing its FTL and memory backend, and each path keeps the other's view consistent.



Consistency rules (the full list is in the design note's coherence<sup>74</sup> section):

<sup>74</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md#coherence>

1. Payload needs no code: CXL accesses, direct mappings and NVMe DMA all touch the same host memory. An NVMe read returns the latest CXL store whether the page is clean, dirty in the cache or direct-mapped.
2. Write, Write Zeroes, Copy (destination) and Deallocate make each page they cover non-resident before the command completes. The entry is dropped without write-back, because the command already programmed or unmapped the page.
3. NVMe reads leave the cache alone and charge a NAND read even for resident pages.
4. A CXL store, and every direct mapping, marks the namespace's written-block bitmap, so DULBE and Get LBA Status see CXL writes.
5. NVMe Flush does not flush the cache; the medium is volatile memory.
6. Flips 1 to 4 change the shared timings and so affect CXL misses too; flip 3 restores the device's own read-ns, program-ns, erase-ns and channel-ns.

Lock order is BQL, gate, worker mutex. The NVMe FTL thread never takes the BQL. The controller must be BBSSD with one namespace and none of namespace management, a subsystem, streams, power\_loss, buffer\_size, op\_pcent, metadata or protection information (femu\_cxl\_link\_check() in femu.c). Format NVM is not advertised and Sanitize is refused, since both would rewrite the medium behind the cache. While linked, device\_del of the medium fails with an unplug blocker; the NVMe link guide (§4.21) covers the slot power-off case and the realize messages.

#### 4.19.13 Threads, locks and where latency is charged

| Thread                              | Runs                                                                                                   | Takes                                                                               |
|-------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| vCPU                                | MMIO overlay callback, the whole access path, Get LSA                                                  | BQL, gate, page holds; drops the BQL to wait for the worker and for the media delay |
| femu-cxl-ftl                        | cxl_worker(): every FTL request of the medium                                                          | s->lock only; never the BQL, never guest memory                                     |
| femu-cxl-cca                        | Caching API commands                                                                                   | BQL, gate per chunk, cca->lock for the doorbell flag                                |
| QEMU main loop                      | QMP qom-set (flush, way change, control commands), queued LSA commands, NVMe drop BH, Cylon install BH | BQL; the gate, except the Cylon install BH, which pauses all vCPUs instead          |
| NVMe FTL thread (linked controller) | NVMe I/O on the shared FTL (femu_cxl_nvme_ftl())                                                       | s->lock; never the BQL                                                              |
| NVMe pollers (linked controller)    | Hold a completion until its cache drop is published                                                    | Read nvme_done only                                                                 |

Latency model (femu\_cxl\_media(), femu\_cxl\_delay() in cxlssd.c):

- Each FTL request is stamped now + op.ns, so the media operations of one access (a fill, then the write-back of the victim its insert evicts) are serialized in modelled time, and the returned latencies accumulate in op.ns and in media-time-ns.
- After its media work, the access waits op.ns minus the time already spent, with the BQL released. The wait sleeps until 100 us before the deadline and spins on the realtime clock for the last 100 us (FEMU\_CXL\_SPIN\_NS), which absorbs timer slack without holding a host CPU for long waits such as a flush.
- The stall is charged to the vCPU inside the MMIO exit: the guest's load or store does not complete until the media time has passed.

- Direct hits never enter QEMU and have no modelled time. `flush-cache` and `way` changes wait on the main loop. Caching API chunks wait on the CCA thread.

The timing model (§6.2.10) lists what each kind of access costs, and the architecture page's CXL load walkthrough (§3.1.11) follows one load end to end.

#### 4.19.14 Parameters

The property reference (§A.1.3) lists every property with its type, default and range; it is generated from the binary. This table explains how they interact.

| Group                                          | Properties                                                                                                                                                                                                                                                                                              | Interactions                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Backend (§A.1.3)                               | <code>volatile-memdev</code>                                                                                                                                                                                                                                                                            | Required and the only backend accepted: <code>memdev</code> , <code>persistent-memdev</code> , <code>volatile-dc-memdev</code> and <code>num-dc-regions</code> are refused, and <code>lsa</code> only with <code>lsa-control=off</code> . Size a nonzero multiple of 256 MiB, at most 120 GiB. <code>der=cylon</code> needs it <code>hugetlbfs</code> , <code>share=on</code> , <code>prealloc=on</code>           |
| Cache (§A.1.3)                                 | <code>cache-pages</code> , <code>cache-policy</code>                                                                                                                                                                                                                                                    | <code>cache-pages=0</code> disables the cache. Otherwise at most the media page count and divisible by <code>cache-ways</code>                                                                                                                                                                                                                                                                                     |
| Cache tunables (§A.2.2)                        | <code>cache-ways</code> , <code>prefetch-degree</code> , <code>prefetch-stride</code>                                                                                                                                                                                                                   | Accepted on <code>-device</code> and changeable with <code>qom-set</code> . Ways must be nonzero and divide <code>cache-pages</code> . Prefetch values are bounded by the media page count; the effective degree is capped at <code>cache-pages</code>                                                                                                                                                             |
| NAND geometry and timing (§A.1.3)              | <code>ftl</code> , <code>channels</code> , <code>luns-per-channel</code> , <code>pages-per-block</code> , <code>blocks-per-plane</code> , <code>gc-threshold</code> , <code>gc-threshold-high</code> , <code>read-ns</code> , <code>program-ns</code> , <code>erase-ns</code> , <code>channel-ns</code> | Feed the private BBSSD FTL. <code>blocks-per-plane=0</code> sizes the NAND to 5/4 of the media plus 4 blocks per plane; an explicit value must cover the media. With no spare blocks, <code>media-full</code> rises. <code>ftl=off</code> drops all media timing and forbids an NVMe link. When linked, these also apply to the NVMe namespace and the controller's own geometry and timing properties are ignored |
| Cylon media switches (same table)              | <code>cylon-first-touch-program</code> , <code>cylon-free-writeback</code>                                                                                                                                                                                                                              | Change the media model to match published Cylon experiments; off for normal use                                                                                                                                                                                                                                                                                                                                    |
| Direct mapping (§A.1.3)                        | <code>der</code> , <code>der-replace-rate</code> , <code>cylon-kernel-ack</code> , <code>concurrent-misses</code>                                                                                                                                                                                       | <code>der=memslot</code> is refused under TCG. <code>der=cylon</code> without <code>cylon-kernel-ack=on</code> is refused. <code>der-replace-rate</code> matters only for <code>memslot</code> . <code>concurrent-misses=auto</code> follows whether DER is available                                                                                                                                              |
| Caching API, control channel and logs (§A.1.3) | <code>cca</code> , <code>lsa-control</code> , <code>log-dir</code> , <code>tracefs-dir</code> , <code>log-limit</code>                                                                                                                                                                                  | <code>cca=off</code> registers no BAR5. <code>lsa-control=on</code> refuses an <code>lsa</code> backend. <code>log-limit=0</code> opens no I/O log and takes no statistics appends. <code>tracefs-dir</code> unset makes commands 91 and 81 no-ops on the host                                                                                                                                                     |
| Actions and control (§A.2.2)                   | <code>der-ratio</code> , <code>control-command</code> , <code>control-argument</code> , <code>control-status</code> , <code>flush-cache</code> , <code>stats-reset</code> , <code>fast-load</code> , <code>fast-load-drain-ns</code>                                                                    | Run time only, except <code>fast-load</code> , which <code>-device</code> also accepts. <code>der-ratio</code> needs a direct mode and no uncached ranges; <code>memslot</code> refuses a ratio that needs more aliases than are free. <code>fast-load=false</code> holds the gate alone while it waits for the NAND timelines, with the BQL dropped and lock released                                             |

`run-cxlssd.sh` uses defaults that follow Cylon's launch script and differ from the device's (a cache of 1/20 of the media, direct mapped, 8 by 8 channels and LUNs, `lsa-control=on`, and no spare NAND blocks for the 48G and 96G presets); the guide (§4.18.4) has the table.

#### 4.19.15 Counters

All counters are read-only QOM properties on the device (/machine/peripheral/<id>). The runtime property reference (§A.2.2) describes each one; this table says where each family comes from.

| Family                                                                                                                                           | Counted where                                 | Notes                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Cache (§A.2.2): cache-hits, cache-misses, read-*, write-*, cache-inserts, cache-evictions, cache-entries, prefetch-inserts                       | femu_cxl_access() and the cache library       | Trapped lookups only; direct hits are invisible                                                                   |
| Snapshots (§A.2.2): last-*                                                                                                                       | stats-reset, control command 1                | Copies taken before the event counters are cleared                                                                |
| Media (§A.2.2): media-reads, media-writes, media-time-ns, media-full                                                                             | femu_cxl_media()                              | Never cleared by stats-reset; take differences. media-writes comes from the FTL and includes linked NVMe programs |
| Direct mapping (§A.2.2): der-active, der-probes, der-mapped, der-remaps, der-revocations, der-quiet-revocations, der-replacements, der-fallbacks | DER code in qemu-adapter.c                    | der-mapped is a gauge and the evidence that mapping is active                                                     |
| Caching API (§A.2.2): cca-*                                                                                                                      | cca.c                                         | cca-pinned and cca-uncached are gauges                                                                            |
| Other (§A.2.2): invalidations, nvme-drops, log-dropped                                                                                           | Invalidation sites, NVMe drop BH, log writers | invalidations is a generation, not an error count                                                                 |

Host-side files, all bounded by log-limit and written in log-dir: cxlssd-stats.log (command 1 snapshots, ways and prefetch changes), cxlssd-io-N.log (one line per trapped access: realtime start, R/W, DPA, length, modelled ns) and cxlssd-spt.log (command 17 dump of tracked direct mappings and Cylon SPTE values).

#### 4.19.16 Validation

| Layer                                    | What runs                                                                              | What it covers                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unit tests (hw/femu/tests/unit/)         | test-cxl-cache.c, test-cxl-spte.c, test-cxl-ccaring.c, built by hw/femu/tests/Makefile | Policy ordering, S3-FIFO promotion and ghost admission, full-width keys, dirty accounting at 1 to 32 ways and a large fully associative cache; SPTE encodings, MMIO generation range and huge-page address arithmetic; the guest ring library against the device ring consumer, with every guest-writable index fuzzed                                                                                                                            |
| qtests (hw/femu/tests/qtest/femu-test.c) | About a hundred cxl-* cases under qos-test                                             | Real PCI bridges and HDM registers programmed, then accesses through the window: all policies, readback, dirty programming, media timing, invalid configuration, ftl=off, mixed-window forwarding, interleaved, switched and multi-root-port topologies, memslot mappings, budget and replacement, ratios, invalidation, unplug during a wait, concurrency stress, LSA and QOM control, every caching API command, and the NVMe link (cxl-nvme-*) |
| Doc examples                             | hw/femu/scripts/check-doc-examples.py                                                  | Each tagged command line in the guides starts under qtest; der=cylon is checked to fall back with its warning                                                                                                                                                                                                                                                                                                                                     |

| Layer                    | What runs                                                                                    | What it covers                                                                                                                                                                                                                                                  |
|--------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Guest runs on real hosts | hw/femu/tools/cca/run-guest-tests.sh in the guest, plus the runs recorded in the design note | A bare-metal Intel host on the fixed Cylon kernel, and a nested run under KASAN, passed write/readback across eviction cycles in all three der modes; caching API self-test; fio on a linked NVMe namespace next to devdax traffic with checksums on both paths |

Under QTest Cylon is deliberately inert (no KVM), so QTest cannot validate the custom kernel: dual-slot registration, SPT mapping, SPT revocation and the quiet path are covered only by guest runs on a Cylon host. Long workloads, sustained traffic past the GC threshold in a guest, hugetlb migration during a run, and guests that online the range as system RAM for long periods are not covered. The design note's scope and validation limits<sup>75</sup> has the exact list.

#### 4.19.17 Limits

- No migration or snapshot: the device's vmstate is unmigratable, because the FTL and cache have no migration state. migrate fails with "State blocked by non-migratable device".
- Volatile only: no persistent memory, no dynamic capacity. Data is lost when QEMU exits.
- TCG: der=memslot is refused at realize (other vCPUs could keep TLB entries for a revoked alias). der=cylon needs KVM and falls back to MMIO under TCG and QTest. der=off works everywhere.
- Direct mapping only in the single-endpoint topology described above.
- der=off costs an exit per access: a hit takes microseconds, so large workloads are slow.
- With der=off, lock-prefixed operations to the window are not atomic.
- Direct hits are untimed and uncounted, and do not update CLOCK or S3-FIFO reference state.
- der=memslot holds at most 1024 aliases at once across all devices: one page each for cache mappings, while a ratio run counts as one alias.
- der=cylon needs the fixed host kernel and has the host restrictions and the 2 MiB first-touch effect described above.
- The caching API data rings are not implemented; their header fields are 0.

#### 4.19.18 Extending the component

| To add            | Change                                                                                                                                                                                                                                                                                                                      | Check                                                                                                              |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| A cache policy    | FemuCxlPolicy in cache.h, the name table in femu_cxl_policy(), insert and victim rules in femu_cxl_cache_insert() and cache_evict() and the requeue rule in femu_cxl_cache_unpin() in cache.c, the name table in cxl_policy_name() and the realize error message in qemu-adapter.c, the cache-policy description in props.c | test-cxl-cache.c builds cache.c without QEMU; add ordering cases there first                                       |
| A counter         | A QOM property in cxl_init() in qemu-adapter.c and its description in cxl_runtime_descs in props.c                                                                                                                                                                                                                          | gen-property-docs.py --check fails until the description exists and reference/runtime-properties.md is regenerated |
| A control command | cxl_command() in qemu-adapter.c; add it to cxl_lsa_inline() only if it never drops the BQL (no media wait, flush or way change)                                                                                                                                                                                             | qtests cxl-control-qom and cxl-control-lsa run the same table both ways                                            |

<sup>75</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md#scope-and-validation-limits>



| To add                | Change                                                                                                                                                                                                                                                                                                    | Check                                                                                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| A caching API command | The command enum in <code>cca-abi.h</code> (raise <code>CCA_LAYOUT_VERSION</code> if the layout changes), <code>cca_validate()</code> , <code>cca_prepare()</code> and the per-page handler table in <code>cca_exec()</code> in <code>cca.c</code> , the guest library in <code>hw/femu/tools/cca/</code> | <code>test-cxl-cca-ring.c</code> and the <code>cxl-cca-* qtests</code>                                                                            |
| A direct mapping mode | The der check in <code>cxl_realize()</code> , <code>femu_cxl_der_init()</code> , and the map, remove, sample and clear entry points in <code>qemu-adapter.c</code>                                                                                                                                        | Every invalidation trigger in the table above must reach it                                                                                       |
| Another media model   | <code>cxl_worker()</code> calls <code>bb_ftl_process_req()</code> ; replace the call and the FTL construction in <code>femu_cxl_start()</code>                                                                                                                                                            | Keep the worker free of guest memory: payload copies stay on the vCPU thread, and any future worker-side copy must use <code>femu_dma_rw()</code> |

Keep new QEMU or KVM dependencies in `qemu-adapter.c`, so `cache.c`, `cca-ring.c`, `spte.h` and `spt.h` stay testable without QEMU and `cxlssd.c` stays free of CXL and KVM details.

#### 4.19.19 Source map

| File                                                                                      | Contents                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hw/femu/cxlssd/qemu-adapter.c</code>                                                | QOM type <code>femu-cxl-ssd</code> , realize and exit, window overlay and routing, invalidation hooks, reset chain, LSA control channel and logs, QOM properties, DER (memslot aliases, budget, replacement, ratios; Cylon prepare, install, SPTE map and revoke, listener), NVMe link ops |
| <code>hw/femu/cxlssd/qemu-adapter.h</code>                                                | <code>FemuCxlMedia</code> , <code>FemuCxlOp</code> , <code>FemuCxlWork</code> , gate and access API                                                                                                                                                                                        |
| <code>hw/femu/cxlssd/cxlssd.c</code>                                                      | Gate, media delay, FTL worker, <code>femu_cxl_access()</code> , eviction callback, geometry checks, FTL start and stop, NVMe link side ( <code>femu_cxl_nvme_ftl()</code> , drop BH, flips)                                                                                                |
| <code>hw/femu/cxlssd/cache.c</code> , <code>cache.h</code>                                | Set-associative cache and the four policies, pinning                                                                                                                                                                                                                                       |
| <code>hw/femu/cxlssd/der.h</code>                                                         | <code>FemuCxlDer</code> and the ratio selection helpers                                                                                                                                                                                                                                    |
| <code>hw/femu/cxlssd/spte.h</code> , <code>spt.h</code>                                   | Cylon EPT SPTE encodings and SPT chunk limits                                                                                                                                                                                                                                              |
| <code>hw/femu/cxlssd/cca.c</code> , <code>cca.h</code>                                    | Caching API device side: BAR5, thread, commands                                                                                                                                                                                                                                            |
| <code>hw/femu/cxlssd/cca-ring.c</code> , <code>cca-ring.h</code> , <code>cca-abi.h</code> | Ring consumer and the shared ABI                                                                                                                                                                                                                                                           |
| <code>hw/femu/cxlssd/props.c</code>                                                       | Property help text used by the generated reference                                                                                                                                                                                                                                         |
| <code>hw/femu/tools/cca/</code>                                                           | Guest library, <code>ccactl</code> , <code>cca-test</code> , <code>run-guest-tests.sh</code>                                                                                                                                                                                               |
| <code>hw/femu/scripts/run-cxlssd.sh</code>                                                | One-device launcher                                                                                                                                                                                                                                                                        |
| <code>hw/femu/femu.c</code>                                                               | <code>cxl_ssd</code> link property, <code>femu_cxl_link_check()</code> , calls through <code>femu_cxl_nvme_ops</code>                                                                                                                                                                      |
| <code>accel/kvm/kvm-all.c</code> , <code>include/system/kvm.h</code>                      | Generic memslot reservation hook                                                                                                                                                                                                                                                           |
| <code>hw/cxl/cxl-mailbox-utils.c</code> , <code>include/hw/cxl/cxl_device.h</code>        | Generic CCI pre-command hook                                                                                                                                                                                                                                                               |

#### 4.19.20 Related pages

- CXL SSD design note<sup>76</sup>
- CXL SSD guide (§4.18)

<sup>76</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md>

- CXL caching API (§4.20)
- CXL NVMe link (§4.21)
- Timing model (§6.2.10)
- Architecture (§3.1)
- Security and limits (§5.13)

## 4.20 CXL caching API (CCA)

The caching API lets software in the guest control the DRAM cache of a `femu-cxl-ssd` (§4.18): pin pages so they stay cached, write back and drop ranges, mark ranges uncached so every access goes to the media, and ask what is resident. Ranges are in 4 KiB device pages. It follows the caching API of the Cylon CXL SSD emulator.

The guest talks to the device through PCI BAR5 of the CXL function. FEMU ships a small C library, the `ccactl` command-line tool and a self-test in `hw/femu/tools/cca/`. The design note's caching API section<sup>77</sup> describes the register layout and the rings in full.

### 4.20.1 Turning it on

`cca=on` on the device (default off). With it off there is no BAR5 and the device behaves exactly as without the feature. `run-cxlssd.sh` has no variable for it, so pass a `-global`:

```
../femu-scripts/run-cxlssd.sh -global femu-cxl-ssd.cca=on \
    -drive file=$HOME/images/u20s.qcow2,if=none,id=hd0 \
    -device virtio-blk-pci,drive=hd0,bus=pcie.0 \
    -netdev user,id=net0,hostfwd=tcp::8080-:22 \
    -device virtio-net-pci,netdev=net0,bus=pcie.0
```

On a full command line, add `cca=on` to the device:

```
./qemu-system-x86_64 -machine q35,cxl=on,smm=off -accel kvm -cpu host -smp 4 -m 4G \
    -object memory-backend-ram,id=cxlmem,size=256M \
    -device pxb-cxl,id=cxl.0,bus=pcie.0,bus_nr=52 \
    -device cxl-rp,id=rp0,bus=cxl.0,chassis=0,slot=0 \
    -device
        femu-cxl-ssd,id=cxlssd,bus=rp0,volatile-memdev=cxlmem,cache-pages=1024,cache-ways=16,cca=on
    \
    -M cxl-fmw.0.targets.0=cxl.0,cxl-fmw.0.size=256M
```

The device starts a `femu-cxl-cca` thread that runs the commands. PIN and UNPIN need a cache (cache-pages above 0).

### 4.20.2 Guest setup

The tools map `/sys/bus/pci/devices/<function>/resource5`. That needs root and a guest kernel without PCI lockdown; no kernel module or UIO driver is needed. Check that firmware assigned BAR5 under the CXL host bridge:

```
fn=$(basename "$(readlink -f /sys/bus/cxl/devices/mem0/..)") # PCI function of mem0
lspci -vv -s "$fn" | grep Region # a Region 5 line must be present
ls -l /sys/bus/pci/devices/$fn/resource5
```

<sup>77</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md# caching-api-cca>

Build the tools inside the guest from a copy of the FEMU source tree:

```
sudo apt install build-essential
make -C hw/femu/tools/cca
```

This builds `libcca.a`, `ccactl` and `cca-test`. Commands work without a region. The address helpers of the library and most self-test cases need a devdax region on the memdev; create one as in creating the region (§4.18.5) and keep it in devdax mode.

### 4.20.3 ccactl

```
ccactl [-d DEV] [-f] [-t MS] COMMAND [LPN COUNT | all]
```

| Option    | Meaning                                                                                                                                                   |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| -d DEV    | A memdev name ( <code>mem0</code> ), a PCI address or a resource5 path. Without it, the only CCA device is used                                           |
| -f        | FORCE: lets invalidate and disable act on pinned pages, unpinning them                                                                                    |
| -t MS     | Command timeout in milliseconds, default 10000; negative waits forever                                                                                    |
| LPN COUNT | First device page (DPA / 4096) and number of pages, decimal, <code>0x</code> hexadecimal or, with a leading 0, octal; <code>all</code> is the whole media |

| Command              | What it does                                                                                                |
|----------------------|-------------------------------------------------------------------------------------------------------------|
| info                 | Print the layout version, media pages, cache pages, cache ways, pins allowed per set and commands completed |
| nop                  | Round trip with no effect                                                                                   |
| pin LPN COUNT        | Keep the pages cached until unpinned                                                                        |
| unpin LPN COUNT      | Let the pages be evicted again                                                                              |
| invalidate LPN COUNT | Write dirty pages back to NAND and drop them from the cache                                                 |
| disable LPN COUNT    | Mark the pages uncached and drop them                                                                       |
| enable LPN COUNT     | End the uncached marking                                                                                    |
| query LPN COUNT      | Count resident, dirty, pinned and uncached pages                                                            |
| reset                | Reset the rings only                                                                                        |
| reset all            | Reset the rings, unpin every page and end every uncached range                                              |

```
sudo ./ccactl -d mem0 info
sudo ./ccactl -d mem0 pin 0 16          # pin device pages 0..15
sudo ./ccactl -d mem0 query all
sudo ./ccactl -d mem0 -f invalidate all  # unpin, write back and drop all
sudo ./ccactl -d mem0 disable 100 8     # pages 100..107 uncached
sudo ./ccactl -d mem0 enable all
```

Range commands print status `N` (text) and pages `N`, the number of pages acted on; query adds resident, dirty, pinned and uncached. The exit status is 0 when the device returns status 0 (always for info), 1 on an error status or when the device cannot be opened, and 2 on a usage error. `ccactl` opens the device before it checks the command, so with no device a usage error also exits 1.

### 4.20.4 What each command does

**PIN** : Refused as a whole with `-ENOSPC` when the pins cannot fit. Resident pages are pinned. Other pages are read from NAND as a miss would read them (counted in `media-reads` and `cca-pin-fills`, not as guest misses), possibly evicting an unpinned page, then pinned. Pinned pages leave the eviction queues, so eviction

and prefetch never touch them. Every way of a set may be pinned; a miss to a set whose ways are all pinned is served from NAND without caching and counted in `cca-pinned-set-misses`. On `-EIO` or `-EAGAIN` partway through, the pages pinned before the failure stay pinned; pages says how many.

**UNPIN** : Returns pinned pages to the replacement queue a newly inserted page would join. Dirty pages stay dirty.

**INVALIDATE** : Writes dirty pages back to NAND and drops them from the cache. Pinned pages need **FORCE**; without it nothing changes and the status is `-EBUSY`. Direct mappings of the pages are revoked first.

**CACHE\_DISABLE** (`disable`) : Marks the pages uncached, then drops resident ones as **INVALIDATE** does. An uncached page is never inserted, prefetched or direct-mapped: every read is a NAND read and every write a NAND program. A failed command clears the mark from the pages it could not drop, so an uncached page is never resident.

**CACHE\_ENABLE** (`enable`) : Ends the uncached marking; the next access caches the page again.

**QUERY** : Counts resident, dirty, pinned and uncached pages. Dirty comes from the cache metadata; with `der=cyl` on a page written through a direct mapping may not show as dirty until its EPT dirty bit is sampled.

Commands run one at a time in submission order, in chunks of at most 256 pages, and each chunk waits out its media time. Guest accesses run between chunks. A page is changed atomically within its chunk, but a range is not: if a range must stay non-resident, stop accessing it while the command runs.

Pins and uncached ranges survive a cache flush (`flush-cache`, control commands 2, 9 and 11) and a cache-ways change; a flush writes pinned dirty pages back and keeps them resident. A way change that cannot fit the pinned pages is refused. A device reset, including a guest reboot, unpins everything and ends every uncached range. A direct ratio (`der-ratio`) and uncached ranges exclude each other.

## 4.20.5 Errors

Status values from the device:

| Status                   | Meaning                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-EINVAL</code>     | Unknown command or flag, nonzero reserved field, or an empty range                                                                                                    |
| <code>-ERANGE</code>     | The range goes past the end of the media                                                                                                                              |
| <code>-ENOSPC</code>     | <b>PIN</b> : a set has no way left to pin                                                                                                                             |
| <code>-EBUSY</code>      | <b>INVALIDATE</b> or <b>CACHE_DISABLE</b> over pinned pages without <b>FORCE</b> ; <b>PIN</b> over an uncached page; <b>CACHE_DISABLE</b> while a direct ratio is set |
| <code>-EOPNOTSUPP</code> | <b>PIN</b> or <b>UNPIN</b> without a cache                                                                                                                            |
| <code>-ENODEV</code>     | The CXL media is disabled; every command but <b>NOP</b> fails. Nothing a guest does through the CXL mailbox in this QEMU disables it                                  |
| <code>-EIO</code>        | NAND refused a read or write. <b>INVALIDATE</b> leaves that page and the rest of the range resident                                                                   |
| <code>-EAGAIN</code>     | A cache-ways change during a <b>PIN</b> left no room for the rest                                                                                                     |

The library adds its own:

| Status                  | Meaning                                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------------------------------|
| <code>-ETIMEDOUT</code> | No answer within the timeout                                                                                       |
| <code>-ECANCELED</code> | A device reset cancelled the command while it waited                                                               |
| <code>-EPROTO</code>    | The rings hit a fatal error; every call fails until <code>cca_reset()</code> , or the next <code>cca_open()</code> |

| Status                                 | Meaning                          |
|----------------------------------------|----------------------------------|
| -EBUSY (from <code>cca_open()</code> ) | Another process holds the device |

## 4.20.6 Library

`hw/femu/tools/cca/cca.h` documents every call. Link with `libcca.a` and `-pthread`. A short example:

```
#include "cca.h"

struct cca_dev *d;
struct cca_result r;

if (cca_open("mem0", &d) == 0) {
    cca_pin(d, 0, 16, &r);          /* returns r.status */
    cca_query(d, 0, 16, &r);        /* r.resident, r.pinned, ... */
    cca_unpin(d, 0, 16, &r);
    cca_close(d);
}
```

- Synchronous calls (`cca_pin`, `cca_unpin`, `cca_invalidate`, `cca_cache_disable`, `cca_cache_enable`, `cca_query`) return the device status and fill struct `cca_result`; pages is the number of pages acted on. `cca_nop` returns the status only. Pass `CCA_FLAG_FORCE` to `cca_invalidate` and `cca_cache_disable`, and `CCA_WHOLE` as the count with a start of 0 for the whole media.
- `cca_submit()` and `cca_reap()` keep up to 2048 commands in flight; `CCA_SUBMIT_DEFER` batches doorbells, and `cca_kick()` rings once.
- `cca_attach_dax()` takes a `devdax` mapping of a non-interleaved region on the `memdev`; afterwards `cca_pin_addr()` and the other `*_addr` calls take virtual addresses in the mapping and widen them to whole pages.
- All calls are thread safe. The library locks `resource5` with `flock()`, so one process uses a device at a time. A process that dies holding it leaves its ring state behind; the next `cca_open()` resets the rings. Its pins and uncached ranges stay; `ccactl reset all` clears them.
- `cca_set_timeout()` changes the 10 second default; negative waits forever.

## 4.20.7 Guest tests

`run-guest-tests.sh` builds the tools, records the kernel and whether BAR5 was assigned, finds the `devdax` device of the `memdev` and runs `cca-test`. Run it as root inside the guest:

```
cd hw/femu/tools/cca
sudo ./run-guest-tests.sh -d mem0
```

`-x dax0.0` names the `devdax` device and `-o LOG` the log file (default `cca-guest-<date>-<time>.log`). Cases can be named after the options. Each case prints PASS, FAIL or SKIP, and the script exits 1 if any failed. The query, thrash, invalidate and disable cases map the `devdax` device and are skipped without one of at least 2 MiB. The tools README<sup>78</sup> lists every case.

The thrash and disable cases compare access times and expect a factor of 5. With `der=off` a cache hit already costs microseconds of emulation, so `disable` can fail that ratio without anything being wrong; with `der=memslot` and `der=cylon` all cases passed in FEMU's guest runs. `invalidate` brackets its step with CCA-MARK lines; between them the host's media-writes counter must rise by 64.

<sup>78</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/tools/cca/README.md#guest-tests>

## 4.20.8 Counters

Read with `qom-get` as described in counters (§4.18.9):

| Counter                                                | Meaning                                                                                                       |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <code>cca-commands</code> , <code>cca-errors</code>    | Commands completed, and those with a nonzero status                                                           |
| <code>cca-pinned</code> , <code>cca-uncached</code>    | Pages pinned now, pages in uncached ranges now                                                                |
| <code>cca-pin-fills</code>                             | Pages PIN read from NAND                                                                                      |
| <code>cca-writebacks</code> , <code>cca-dropped</code> | Dirty pages programmed, and resident pages dropped, by <code>INVALIDATE</code> and <code>CACHE_DISABLE</code> |
| <code>cca-pinned-set-misses</code>                     | Misses served from NAND because every way of the set is pinned                                                |

`stats-reset` clears the event counters and keeps the two gauges.

## 4.20.9 Related pages

- CXL SSD (§4.18)
- CCA guest tools README<sup>79</sup>
- CXL SSD design note<sup>80</sup>

## 4.21 CXL NVMe link

A BBSSD NVMe controller can serve the medium of a `femu-cxl-ssd` (§4.18) as its namespace. The guest then sees the same bytes twice: as CXL memory and as an NVMe block device. Both front ends share one memory backend and one FTL, so one mapping table, one garbage collector and one set of NAND counters cover both.

Use it to study software that moves data between a memory path and a block path on one device. For most other uses, give the guest one view at a time.

### 4.21.1 Turning it on

Add a `femu` controller with `femu_mode=1` and `cxl_ssd=<id>` after the `femu-cxl-ssd` on the command line. With `run-cxlssd.sh`, whose device id is `cxlssd`:

```
../femu-scripts/run-cxlssd.sh \  
-device femu,id=nvme0,bus=pcie.0,femu_mode=1,cxl_ssd=cxlssd \  
-drive file=$HOME/images/u20s.qcow2,if=none,id=hd0 \  
-device virtio-blk-pci,drive=hd0,bus=pcie.0 \  
-netdev user,id=net0,hostfwd=tcp::8080-:22 \  
-device virtio-net-pci,netdev=net0,bus=pcie.0
```

On a full command line:

```
./qemu-system-x86_64 -machine q35,cxl=on,smm=off -accel kvm -cpu host -smp 4 -m 4G \  
-object memory-backend-ram,id=cxlmem,size=256M \  
-device pxb-cxl,id=cxl.0,bus=pcie.0,bus_nr=52 \  
-device cxl-rp,id=rp0,bus=cxl.0,chassis=0,slot=0 \  
-device femu-cxl-ssd,id=cxlssd,bus=rp0,volatile-memdev=cxlmem \  

```

<sup>79</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/tools/cca/README.md>

<sup>80</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md#caching-api-cca>

```
-device femu,id=nvme0,bus=pcie.0,femu_mode=1,cxl_ssd=cxlssd \
-M cxl-fmw.0.targets.0=cxl.0,cxl-fmw.0.size=256M
```

Give the controller a bus outside the CXL hierarchy. Without `bus=`, QEMU picks the `pxb-cxl` bus and refuses with "Only PCI/PCIe bridges can be plugged into pxb-cxl". `bus=pcie.0` is the simplest choice; use a `pcie-root-port` instead if you want to unplug the controller at run time (see Unplugging (§4.21.6)).

In the guest the namespace appears as `/dev/nvme0n1` (with the NVMe driver), next to the CXL `memdev`. The namespace size is the medium's size.

### 4.21.2 Rules at realize

Realize fails with a message naming the rule when one is broken:

| Rule                                                                                                  | Message                                                                                                                             |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| The femu-cxl-ssd exists and comes first on the command line                                           | Device 'cxlssd' not found                                                                                                           |
| <code>cxl_ssd</code> names a femu-cxl-ssd                                                             | Invalid parameter type for 'cxl_ssd', expected: femu-cxl-ssd                                                                        |
| The femu-cxl-ssd is realized and not being removed                                                    | <code>cxl_ssd</code> must name a realized femu-cxl-ssd                                                                              |
| The femu-cxl-ssd has <code>ftl=on</code> (the default)                                                | <code>cxl_ssd</code> requires the femu-cxl-ssd to have <code>ftl=on</code>                                                          |
| One controller per medium                                                                             | the femu-cxl-ssd already serves an NVMe controller                                                                                  |
| <code>femu_mode=1</code>                                                                              | <code>cxl_ssd</code> requires <code>femu_mode=1</code>                                                                              |
| One namespace; no <code>namespace_modes</code> or <code>namespace_sizes</code>                        | <code>cxl_ssd</code> requires a single <code>bbssd</code> namespace                                                                 |
| No <code>ns_mgmt</code> and no <code>subsys</code>                                                    | <code>cxl_ssd</code> requires no namespace management or subsystem                                                                  |
| No <code>streams</code> , <code>power_loss</code> , <code>buffer_size</code> or <code>op_pcent</code> | <code>cxl_ssd</code> requires no <code>streams</code> , <code>power_loss</code> , <code>buffer_size</code> or <code>op_pcent</code> |
| No <code>meta</code> , <code>pi</code> or <code>dps</code>                                            | <code>cxl_ssd</code> requires no metadata or protection information                                                                 |
| <code>devsz_mb</code> unset, 1024 or the medium's size in MiB                                         | <code>devsz_mb</code> must be unset or N, the size of the femu-cxl-ssd                                                              |

The medium's geometry and NAND timings apply. The controller's own geometry and timing properties are ignored.

### 4.21.3 What the controller offers

- Read, Write, Flush, Dataset Management (Deallocate), and the optional commands you enable with `oncs` (for example `oncs=0x10c` adds Write Zeroes and Copy). Write Zeroes is off by default, as on any FEMU controller.
- Format NVM is not advertised and Sanitize is refused. Both would rewrite the whole medium behind the CXL cache.
- SMART, vendor log page C0h and the write amplification factor report the shared FTL. CXL traffic does not count as NVMe host I/O.
- The timing admin command 0xEF works. Codes 1 and 2 (GC delay on and off) and 3 and 4 (NAND times) act on the shared FTL and so affect CXL misses too; code 3 restores the medium's read-ns, program-ns, erase-ns and channel-ns. Codes 5 to 7 affect the NVMe side only.

#### 4.21.4 Consistency between the two views

The data is consistent without any action from you:

- An NVMe read returns the most recent CXL store, whether the page is clean, dirty in the cache or direct-mapped.
- A CXL access after an NVMe write completes sees the NVMe data.
- Accesses that overlap in time with no synchronization in the guest have no defined order, as on real hardware.

What happens to the cache and the counters:

1. Write, Write Zeroes, Copy (its destination) and Deallocate drop every page they touch from the CXL cache before the command completes, without writing it back: the command already programmed or unmapped the page. `nvme-drops` counts these pages, and the pinned pages of rule 2.
2. A page pinned through the caching API (§4.20) stays pinned and resident but becomes clean.
3. NVMe reads leave the cache alone and are charged a NAND read even when the page is cached.
4. CXL stores mark the blocks they cover as written, so Deallocated or Unwritten Logical Block errors (DULBE) and Get LBA Status see them. A trapped CXL load marks nothing, but with `der=memslot` or `der=cylon` any access that maps a page, a load or a prefetch included, marks the whole page, since stores through the mapping are invisible. Linking to a medium that was already accessed marks every block. A Deallocate does not clear pages that a direct ratio maps; they stay marked as written.
5. NVMe Flush does not flush the CXL cache. The medium is volatile memory.
6. A long cache operation (a flush, a cache-ways change, a caching API chunk) delays NVMe write completions until it ends. Reads do not wait.

Do not mount a filesystem on the namespace while the guest also uses the CXL region. The filesystem would overwrite CXL-resident data, and kernel memory when the region is onlined as system RAM. Use one view at a time unless your workload coordinates the two.

`media-writes` on the `femu-cxl-ssd` includes NVMe writes; it is updated before each NVMe write, Write Zeroes, Copy or Deallocate completes.

#### 4.21.5 Concurrent misses and atomicity

`concurrent-misses` on the `femu-cxl-ssd` controls whether CXL misses to different pages wait for the media together:

| Value          | Behaviour                                                                                                              |
|----------------|------------------------------------------------------------------------------------------------------------------------|
| auto (default) | Together while a direct mode ( <code>der=memslot</code> or <code>der=cylon</code> ) is active, one at a time otherwise |
| on             | Always together                                                                                                        |
| off            | Always one at a time                                                                                                   |

With `der=off`, a guest lock-prefixed read-modify-write on CXL memory reaches the device as a separate read and write, so it is never atomic. With misses one at a time, other vCPUs rarely get between the two (under 1% of lock add increments were lost with four vCPUs in FEMU's tests). With `concurrent-misses=on` about half were lost. Do not combine `der=off` with `concurrent-misses=on` for anything that uses atomic operations on the region. In the direct modes, on a page that is direct-mapped the write lands on the page the read mapped, and none were lost either way. A `memslot` page left on MMIO because the alias budget is full behaves as with `der=off`.



```
DER=memslot ../femu-scripts/run-cxlssd.sh \
    -global femu-cxl-ssd.concurrent-misses=on \
    -device femu,id=nvme0,bus=pcie.0,femu_mode=1,cxl_ssd=cxlssd
```

#### 4.21.6 Unplugging

While the link exists, `device_del` of the `femu-cxl-ssd` fails with "femu-cxl-ssd is in use by NVMe controller `nvme0`". Remove the controller first, then the medium. A controller on `pcie.0` cannot be unplugged ("Bus '`pcie.0`' does not support hotplugging"), so for this put it on a root port:

```
../femu-scripts/run-cxlssd.sh \
    -device pcie-root-port,id=hp0,bus=pcie.0,chassis=5,slot=5 \
    -device femu,id=nvme0,bus=hp0,femu_mode=1,cxl_ssd=cxlssd
```

Then, over QMP, with the guest running so it can acknowledge each removal:

```
{"execute": "device_del", "arguments": {"id": "nvme0"}}
{"execute": "device_del", "arguments": {"id": "cxlssd"}}
```

The guest can still remove the medium by powering off its root port slot, which QEMU allows without asking. The controller then keeps working as a plain BBSSD over the same bytes until you remove it; the CXL cache goes away with the device. A CXL reset or disabled CXL media leaves the NVMe path working.

#### 4.21.7 Checking it in a guest

FEMU's guest runs checked, in all three der modes, that data written through `devdax` reads back through NVMe and the reverse, that `Deallocate` reports DULBE errors and reads zeros through `devdax`, and that `fio` random writes on one half of the namespace run alongside a `devdax` writer on the other half with both halves verified through both paths. Long runs past the garbage collection threshold have not been checked in a guest.

The guest needs the NVMe driver (`CONFIG_BLK_DEV_NVME`) in addition to the CXL options in guest requirements (§4.18.3). Its `nvme-cli` reads the logs as on any BBSSD:

```
sudo nvme list
sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b | xxd | head
```

#### 4.21.8 Related pages

- CXL SSD (§4.18)
- CXL caching API (§4.20)
- CXL SSD design note<sup>81</sup>
- Log pages and counters (§B)

<sup>81</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md#nvme-front-end>

## 5 Running and configuring

### 5.1 Parameter manual

This manual explains how to configure an emulated device: the ways a parameter reaches FEMU, then every group of parameters by component, with what each one means, its unit, the values it accepts, and how it interacts with the others. It ends with worked configurations.

The property reference (§A.1) is generated from the binary and is the authority on names, types and defaults. This manual never states a default; each group links to its table there. Run-time properties and counters are in `runtime-properties.md` (§A.2).

#### 5.1.1 Contents

1. How parameters reach FEMU (§5.1.2)
2. Devices and how they combine (§5.1.3)
3. Controller (§5.1.4)
4. Namespaces and subsystems (§5.1.5)
5. NAND geometry (§5.1.6)
6. NAND timing (§5.1.7)
7. FTL: mapping, caches and write buffer (§5.1.8)
8. Garbage collection (§5.1.9)
9. Reliability, wear and fault insertion (§5.1.10)
10. Host link and controller firmware (§5.1.11)
11. ZNS (§5.1.12)
12. FDP (§5.1.13)
13. OCSSD, CSD and KV (§5.1.14)
14. CXL SSD (§5.1.15)
15. Accepted for compatibility, no effect (§5.1.16)
16. Worked configurations (§5.1.17)

#### 5.1.2 How parameters reach FEMU

##### 1.1 Device options on the QEMU command line

Every FEMU parameter is a property of one of three QEMU devices, given on the command line when QEMU starts:

|                                                         |                                         |
|---------------------------------------------------------|-----------------------------------------|
| <code>-device femu-subsys,id=SUB,prop=value,...</code>  | NVMe subsystem (FDP, shared namespaces) |
| <code>-device femu-cxl-ssd,id=CXL,prop=value,...</code> | CXL Type-3 SSD                          |
| <code>-device femu,id=CTRL,prop=value,...</code>        | NVMe controller and its namespaces      |

The rules of QEMU's option syntax apply:

- Options are separated by commas. A comma inside a value is written twice: `namespace_modes=bbssd,,znsd` is the list `bbssd,znsd`.
- Booleans take on or off (QEMU also accepts yes/no and true/false).
- Properties of type size take a number of bytes or a QEMU size with a suffix (64M, 4G). Integer properties take plain numbers, decimal or 0x hexadecimal; their unit is in the property name or in this manual (`_mb` MiB, `_ns` and `_lat` nanoseconds, `_pcent` percent).

- A property that links to another device (subsys, cxl\_ssd) names the other device's id, and that device must come earlier on the command line.
- A misspelled name stops QEMU with Property 'femu.NAME' not found.

Static properties are read once, when QEMU creates (realizes) the device. Realize checks the values and their combinations and stops QEMU with a message naming the rule that failed; the "Limits and refusals" section of each mode page (§4.4.5) lists them. `./qemu-system-x86_64 -device femu,help` prints every property with its type, default and description; `femu-subsys,help` and `femu-cxl-ssd,help` do the same for the other two devices.

## 1.2 Configuration files

`hw/femu/scripts/ssd-config.sh` expands a file of `key = value` lines into the same `-device` options, so a configuration can be read, commented and kept in version control. Keys are property names, `mode = bbssd` stands for `femu_mode`, and a `[subsys]` section becomes a `femu-subsys` device that the controller joins. The script checks names against `-device femu,help` (and `femu-subsys,help` for `[subsys]` keys) of the binary in `FEMU_BIN`, or of one it finds in the build directories; without a binary it only warns. It also rejects an unknown mode and malformed lines. QEMU still checks the values. Tutorial 09 (§5.11) walks through it, and scripts and tools (§C.4) describes the shipped files.

## 1.3 Run-time properties over QMP

A few properties can be read or changed while the guest runs, with QMP `qom-get` and `qom-set` on `/machine/peripheral/<id>`:

- `femu: simulate-power-loss`, a write-only trigger for the power-cut model (power loss (§5.1.8)).
- `femu-cxl-ssd`: the cache tunables `cache-ways`, `prefetch-degree` and `prefetch-stride` (also accepted on `-device`), the actions `flush-cache`, `stats-reset`, `der-ratio` and `control-command`, and every counter.

The full list is `runtime-properties.md` (§A.2). `femu` and `femu-subsys` have no counter properties; their counters are NVMe log pages (log pages and counters (§B)).

## 1.4 Run-time switches from the guest

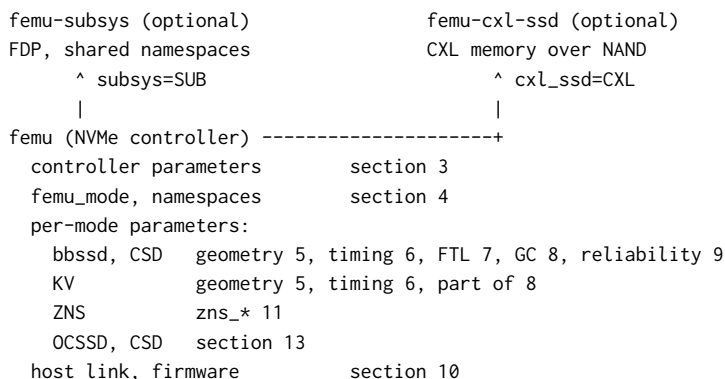
These are not parameters, but they change the model while the guest runs:

- Vendor admin command `0xEF` on a BlackBox controller turns GC time and the flat NAND times off and on, and counts completions posted after their due time. Code 3 sets the built-in flat times, not the ones on your command line; code 4 sets them to 0 (timing model (§6.2.11)).
- Set Features `06h` (Volatile Write Cache) turns the write buffer off and on when `vwc=1`; Set Features `20h` (Key Value Configuration) sets `EDNEK` on a KV namespace; Set Features `04h` and `0Bh` set the temperature threshold and the asynchronous events the host wants.

## 1.5 Environment variables

A few debugging and host-placement aids are environment variables of the QEMU process, not properties: environment variables (§A.1.5).

### 5.1.3 Devices and how they combine



A femu controller runs one mode per namespace. `femu_mode` picks it for every namespace, and `namespace_modes` per namespace. Only the parameters of the modes in use have an effect; the description of each property in `properties.md` (§A.1) names the modes it applies to. A property of a mode that no namespace runs has no effect. The properties of section 15 (§5.1.16) have no effect in any mode and warn when set.

### 5.1.4 Controller

These shape the NVMe controller the guest driver sees, whatever the mode.

### 3.1 Queues and pollers

Reference: queues, pollers and interrupts (§A.1.1).

| Parameter                    | Unit              | Valid values              | Meaning                                                                                                                                    |
|------------------------------|-------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| queues                       | count             | 1 to 2047                 | I/O submission and completion queue pairs the controller offers; MSI-X vectors are <code>queues + 1</code>                                 |
| entries                      | count, 0's based  | 1 to 65534                | CAP.MQES; queues may have up to <code>entries + 1</code> entries                                                                           |
| multipoller_enabled          | 0 or 1            | 0, 1                      | 0: one poller thread for every I/O queue; 1: several, see <code>poller_ratio</code>                                                        |
| poller_ratio                 | queues per poller | 0 or more (0 counts as 1) | with <code>multipoller_enabled=1</code> , <code>ceil(queues / poller_ratio)</code> pollers, each serving a round-robin share of the queues |
| stride                       | power of two      | 0 to 12                   | doorbell stride CAP.DSTRD: doorbells are $4 \ll \text{stride}$ bytes apart                                                                 |
| max_sqes, max_cqes           | power of two      | 6 and 4 only              | Identify SQES and CQES; the only accepted values are the 64-byte and 16-byte entries                                                       |
| aerl                         | count, 0's based  | 0 to 255                  | outstanding Asynchronous Event Requests the controller holds                                                                               |
| elpe                         | count, 0's based  | 0 to 255                  | Error Information log entries kept                                                                                                         |
| hiops_inline                 | bool              | on, off                   | NoSSD only: performance option, on by default; set off only when debugging                                                                 |
| intc, intc_thresh, intc_time | as named          | intc 0 or 1               | initial values of the Interrupt Coalescing and Interrupt Vector Configuration features; reported only, interrupts are not coalesced        |

Interactions:

- Every poller spins on a host core while the controller is enabled. The number of pollers follows queues, not the number of queues the guest creates, so set queues to the guest's vCPU count when you turn on more pollers (performance tuning (§5.12.2)).

### 3.2 Identity, capabilities and transfers

Reference: controller identity and capabilities (§A.1.1).

| Parameter      | Unit         | Valid values                                                                                                              | Meaning                                                                                                |
|----------------|--------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| vid, did       | PCI IDs      | 16-bit                                                                                                                    | PCI vendor and device ID; vid is also Identify VID                                                     |
| mdts           | power of two | 0 to 255; 0 = no limit                                                                                                    | largest transfer, $2^{(12 + \text{mpsmin} + \text{mdts})}$ bytes                                       |
| mpsmin, mpsmax | power of two | $\text{mpsmin} \leq \text{mpsmax} \leq 15$                                                                                | host memory page sizes the controller accepts, $2^{(12 + n)}$ bytes                                    |
| cqr            | 0 or 1       | 0, 1                                                                                                                      | 1 requires physically contiguous queues                                                                |
| vwc            | 0 or 1       | 0, 1                                                                                                                      | advertise a volatile write cache and make feature 06h usable; Flush drains the write buffer either way |
| oacs           | bit mask     | only bit 1 (0x2)                                                                                                          | Format NVM support; clearing it refuses Format NVM                                                     |
| oncs           | bit mask     | 0x1 Compare, 0x2 Write Uncorrectable, 0x4 Dataset Management, 0x8 Write Zeroes, 0x10 Save/Select, 0x80 Verify, 0x100 Copy | optional NVM commands; Timestamp is always added                                                       |
| sgl            | bool         | on, off                                                                                                                   | accept scatter gather lists; OCSSD ignores it                                                          |
| cmbasz, cmbloc | registers    | cmbasz size x unit a power of two; cmbloc BAR field 2                                                                     | Controller Memory Buffer; 0 means none                                                                 |
| temperature    | kelvin       | 16-bit                                                                                                                    | composite temperature in the SMART log, compared with the threshold feature                            |
| pel_file       | host path    | a writable path                                                                                                           | keeps the Persistent Event log and power cycle count across QEMU runs (retention (§B.3))               |

Interactions:

- Write Zeroes, Compare, Copy and Verify are unreachable until you set their oncs bit. A test of those commands on a default controller tests nothing. Flush is always accepted and drains the bbssd write buffer even with vwc=0, though Linux sends no Flush to a controller that advertises no cache.
- With ZNS, mdts also caps Zone Append when zns\_zasl\_bs=0.

### 3.3 LBA formats, metadata and protection

Reference: LBA formats, metadata and protection (§A.1.1).

| Parameter | Unit            | Valid values                      | Meaning                                                                    |
|-----------|-----------------|-----------------------------------|----------------------------------------------------------------------------|
| nlbaf     | count           | 1 to 16, at most 8 with meta      | LBA formats offered: 512 bytes, doubling with each                         |
| lba_index | index           | below nlbaf                       | format the namespaces boot with; ZNS needs 4 KiB or less                   |
| meta      | bytes per block | NoSSD and bbssd only              | metadata per logical block; each format is then also offered with metadata |
| mc        | bit mask        | bit 0 interleaved, bit 1 separate | Metadata Capabilities; required with meta                                  |
| extended  | 0 or 1          | needs mc bit 0                    | boot with metadata interleaved (extended LBAs)                             |
| pi        | bool            | takes effect only with meta >= 8  | offer protection information types 1 to 3 to Format and Create             |
| dpc, dps  | bit masks       | dpc 0 with meta; leave dps 0      | Identify fields; use pi instead                                            |

Interactions: meta is refused with FDP, with dpc or dps, and with any namespace that is not NoSSD or bbssd; pi is refused with power\_loss and cxl\_ssd (namespace management, metadata and PI (§4.17)).

### 5.1.5 Namespaces and subsystems

Reference: mode, capacity and namespaces (§A.1.1), namespace management, streams and power loss (§A.1.1), shared namespaces (§A.1.2). Design: subsystems, controllers and namespaces (§4.16).

| Parameter             | Unit              | Valid values                                               | Meaning                                                                     |
|-----------------------|-------------------|------------------------------------------------------------|-----------------------------------------------------------------------------|
| femu_mode             | mode number       | 0 OCSSD, 1 bbssd, 2 NoSSD, 3 ZNS, 4 CSD, 5 KV              | mode of every namespace unless namespace_modes is set                       |
| devsz_mb              | MiB               | 32-bit                                                     | host memory backend, split across the namespaces; ignored with op_pcent     |
| namespaces            | count             | 1 to 256; OCSSD and FDP 1                                  | namespaces present from boot                                                |
| namespace_sizes       | bytes, QEMU sizes | one entry per namespace, each >= 512 bytes, sum <= backend | size of each namespace; unset splits the backend evenly                     |
| namespace_modes       | list              | nosssd, bbssd, znssd, ocssd, csd, kvssd, one per namespace | mode of each namespace                                                      |
| op_pcent              | percent           | bbssd; not with cxl_ssd                                    | back the device with the full NAND and expose NAND / (1 + op_pcent/100)     |
| ns_mgmt (femu)        | bool              | standalone NoSSD or bbssd controller                       | Namespace Management and Attachment from the guest                          |
| bbssd_ns_limit        | count             | 1 to 256, >= namespaces                                    | most bbssd namespaces ns_mgmt may allocate, each with its own FTL           |
| streams, streams.max  | bool, count       | streams.max 1 to 32                                        | the Streams directive; bbssd separates streams per FTL page                 |
| power_loss            | bool              | see interactions                                           | roll back writes still in the write buffer on a simulated power cut         |
| subsys                | device id         | a femu-subsys listed earlier                               | join a subsystem                                                            |
| ns_mgmt (femu-subsys) | bool              | NoSSD and bbssd controllers, not with fdp                  | one namespace table and backend shared by every controller of the subsystem |
| nqn (femu-subsys)     | string            | any                                                        | subsystem NQN suffix                                                        |

Interactions:

- Namespaces are packed one after another in the backend. Each bbssd or CSD namespace has its own FTL built from the whole NAND geometry, so each must fit that geometry on its own with room for GC (the reserve (§3.4.12)). Each ZNS namespace builds its zones from its own size.
- `op_pcent` overrides `devsz_mb`: the backend becomes the raw NAND, and each namespace gets its share of NAND  $\times 100 / (100 + \text{op\_pcent})$ .
- A controller may hold at most one CSD namespace; OCSSD takes the whole controller.
- `ns_mgmt` on femu stays off without an error unless every namespace runs `femu_mode` with `dps 0`; with a shared subsystem use the subsystem's `ns_mgmt`.
- `streams` needs `bbssd` with mapping page or `dftl` (or NoSSD, where it has no placement effect). It is refused with FDP, with shared namespaces (`ns_mgmt` on the subsystem), and when a second controller joins the subsystem. `bbssd` reserves `streams.max + 1` lines for it.
- `power_loss` needs `bbssd` (`femu_mode=1`), `buffer_size > 0` and page-aligned namespaces. Without `vwc=1` it is accepted but the write buffer is off, so there is nothing to roll back. It is refused with `meta`, `pi`, `ns_mgmt`, `subsys`, `namespace_modes` and `cxl_ssd`.

### 5.1.6 NAND geometry

Applies to `bbssd`, CSD and KV. Reference: NAND geometry (§A.1.1). Design: geometry (§3.5.3).

| Parameter                | Unit     | Valid values                                            | Meaning                                                                          |
|--------------------------|----------|---------------------------------------------------------|----------------------------------------------------------------------------------|
| <code>secsz</code>       | bytes    | > 0                                                     | sector size                                                                      |
| <code>secs_per_pg</code> | sectors  | 1 to 256                                                | sectors per NAND page; page size = <code>secs_per_pg</code> x <code>secsz</code> |
| <code>pgs_per_blk</code> | pages    | 1 to 65536; $\leq 512$ with <code>nand_cell_type</code> | pages per block                                                                  |
| <code>blks_per_pl</code> | blocks   | 1 to 65536                                              | blocks per plane; also the number of lines                                       |
| <code>pls_per_lun</code> | planes   | 1 to 16                                                 | planes per LUN                                                                   |
| <code>luns_per_ch</code> | LUNs     | 1 to 128                                                | LUNs (dies) per channel                                                          |
| <code>nchs</code>        | channels | 1 to 4096                                               | channels                                                                         |

Derived quantities:

```

page           = secs_per_pg x secsz bytes
NAND capacity  = nchs x luns_per_ch x pls_per_lun x blks_per_pl x pgs_per_blk x page
lines          = blks_per_pl           (a line is one block on every plane of every LUN)
line size      = nchs x luns_per_ch x pls_per_lun x pgs_per_blk x page
parallel units = nchs x luns_per_ch    (LUNs work in parallel; planes of a LUN together)

```

Interactions:

- The total sector count must fit in a signed 32-bit integer.
- Without `op_pcent`, the namespace is `devsz_mb` and must leave GC its reserve of lines; the reserve grows with `gc_thres_pcent_high`, `hot_cold_sep`, log-block mapping and Streams (the reserve (§3.4.12)).
- Fewer, larger lines make GC coarser; more LUNs and channels raise throughput, not single-request latency.

### 5.1.7 NAND timing

Applies to `bbssd`, CSD and KV; ZNS has its own in section 11 (§5.1.12). Reference: NAND timing (§A.1.1). Design: NAND media and timing (§3.5).

| Parameter                        | Unit                | Valid values                                              | Meaning                                                                              |
|----------------------------------|---------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------|
| pg_rd_lat, pg_wr_lat, blk_er_lat | ns                  | 32-bit                                                    | flat page read, page program and block erase times, used when nand_cell_type is 0    |
| nand_cell_type                   | type                | 0 flat, 1 SLC, 2 MLC, 3 TLC, 4 QLC; others fall back to 0 | built-in per-page-type tables instead of the flat times                              |
| pgtype_lat, cell_pages           | flag, bits per cell | cell_pages 0 to 5                                         | with nand_cell_type 0, scale the program time by the page's position in its wordline |
| cmd_addr_lat                     | ns                  | >= 0                                                      | channel bus command and address phase                                                |
| pg_xfer_lat                      | ns                  | >= 0                                                      | channel bus data phase per page; 0 uses ch_xfer_lat                                  |
| ch_xfer_lat                      | ns                  | >= 0                                                      | data phase when pg_xfer_lat is 0; also the OCSSD 1.2 transfer time                   |
| status_lat                       | ns                  | >= 0                                                      | channel bus status phase                                                             |
| tplebsy                          | ns                  | >= 0                                                      | busy time between the planes of a multi-plane erase (GC with pls_per_lun > 1)        |
| pe_suspend, tsusp_ns             | flag, ns            | tsusp_ns >= 0                                             | let a read suspend a program or erase on its LUN, at a cost of tsusp_ns              |
| trim_lat_ns                      | ns per range        | >= 0; bbssd and CSD; refused with FDP                     | time charged per Dataset Management deallocate range                                 |

#### Interactions:

- The channel bus is modelled only when cmd\_addr\_lat, pg\_xfer\_lat (or ch\_xfer\_lat) or status\_lat is non-zero; phases on one channel then run one at a time.
- Vendor command 0xEF codes 3 and 4 change only the flat times: code 3 sets the built-in values, not the configured ones, and code 4 sets them to 0. Neither has an effect while nand\_cell\_type is set.
- Read time also grows with ecc\_step\_ns (section 9 (§5.1.10)).
- Tutorial 05 (§5.7) measures each of these.

### 5.1.8 FTL: mapping, caches and write buffer

Applies to bbssd and CSD. Reference: garbage collection, mapping and caches (§A.1.1). Design: the BlackBox FTL (§3.4).

| Parameter        | Unit | Valid values                                                                                          | Meaning                                                                                            |
|------------------|------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| mapping          | name | page, dftl, hybrid, fast                                                                              | logical-to-physical scheme: a full page table, a cached page table, BAST or FAST log-block mapping |
| mapping_cache_mb | MiB  | 32-bit; 0 picks the built-in size                                                                     | with mapping=dftl, the cached part of the table; a miss costs a NAND read                          |
| read_cache_mb    | MiB  | 0 disables                                                                                            | DRAM read cache; a hit costs DRAM time instead of a NAND read                                      |
| cache_evict      | name | clock, random, lru, arc                                                                               | read cache eviction policy                                                                         |
| hot_cold_sep     | bool | acts with page or dftl; ignored under hybrid and fast, which still reserve its line; refused with FDP | write overwrites of mapped pages to separate hot lines                                             |



| Parameter          | Unit                  | Valid values                  | Meaning                                                   |
|--------------------|-----------------------|-------------------------------|-----------------------------------------------------------|
| buffer_size        | NAND pages, not bytes | >= 0; refused with FDP        | DRAM write buffer; 0 programs every write directly        |
| buffer_thres_pcent | percent               | 1 to 100 when buffer_size > 0 | fill level at which buffered pages are written to NAND    |
| debug_ftl          | bool                  |                               | print FTL invariant violations and log-block merge counts |

Interactions:

- The write buffer models timing; the data lives in the backend. With vwc=1 the guest sees a volatile write cache and feature 06h turns it off. Flush drains the buffer and FUA writes skip it with either setting. With power\_loss=on the buffer also holds data, which simulate-power-loss drops.
- hybrid and fast reserve one more line and count merges in log page C0h; hot\_cold\_sep reserves one more line.
- mapping\_cache\_mb has no effect except under dftl.
- Unknown names for mapping, gc\_policy and cache\_evict are refused.

### 5.1.9 Garbage collection

Applies to bbssd and CSD; KV uses gc\_thres\_pcent only (but see KV (§5.1.14)). Reference: garbage collection, mapping and caches (§A.1.1). Design: garbage collection (§3.4.6).

| Parameter           | Unit                    | Valid values                                               | Meaning                                                            |
|---------------------|-------------------------|------------------------------------------------------------|--------------------------------------------------------------------|
| gc_thres_pcent      | percent of lines in use | 1 to 100                                                   | background GC starts; KV: share of the NAND usable for values      |
| gc_thres_pcent_high | percent of lines in use | gc_thres_pcent to 100                                      | forced GC inside writes; also sizes the reserve                    |
| gc_policy           | name                    | greedy, random, cost-benefit, fifo, d-choice; without FDP  | victim line policy                                                 |
| gc_strategy         | number                  | 0 greedy, 1 cost-benefit, 2 random, 4 per-handle; FDP only | victim reclaim unit policy                                         |
| fdp_trim_erase_all  | flag                    | FDP only                                                   | a deallocate resets every reclaim unit instead of the given ranges |

Interactions:

- Background GC takes only a line with at least 1/8 of its pages invalid; forced GC takes the best line whatever it holds.
- Set gc\_thres\_pcent above the share of the NAND the namespace fills. Below it, background GC never stops and the WAF settles near 8 (tutorial 02 (§5.4.4)).
- The reserve is  $(1 - \text{gc\_thres\_pcent\_high} / 100) \times \text{blks\_per\_pl}$  lines, rounded down, plus the write pointers; the namespace must fit in the rest.

### 5.1.10 Reliability, wear and fault insertion

Applies to bbssd, CSD and KV unless noted. Reference: reliability and wear (§A.1.1). Design: wear, read reclaim and retention refresh (§3.4.10).

| Parameter           | Unit        | Valid values                                | Meaning                                                                                                         |
|---------------------|-------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| ecc_step_ns         | ns per tier | 0 disables                                  | extra read time per ECC tier: one per 750 erases of the block, one per ecc_retention_sec of data age, at most 4 |
| ecc_retention_sec   | seconds     | 0 counts wear only; refused with FDP        | data age that adds one tier                                                                                     |
| pe_cycles Rated     | P/E cycles  | 0 takes the rating of nand_cell_type        | denominator of SMART Percentage Used                                                                            |
| nand_bad_blocks     | blocks      | capped at the block count                   | blocks bad from the start; lowers SMART Available Spare                                                         |
| err_read_unc_ppm    | per million | 0 disables; bbssd and CSD                   | reads that fail as Unrecovered Read Error, at a fixed period                                                    |
| err_write_fail_ppm  | per million | 0 disables; bbssd, CSD and ZNS              | writes that fail; a ZNS zone then becomes read only                                                             |
| read_reclaim_limit  | reads       | 0 disables; bbssd and CSD; refused with FDP | a block read this often since its erase gets its line rewritten on a following write                            |
| retention_limit_sec | seconds     | 0 disables; bbssd and CSD; refused with FDP | a read that hits a line filled at least this long ago queues the line, which is rewritten on a following write  |

Interactions: faults come at a fixed period, so a run repeats exactly. Read reclaim and retention refresh act only when the host reads and then writes; their cost appears as write amplification.

### 5.1.11 Host link and controller firmware

Applies to every NVMe mode. Reference: host link and controller firmware (§A.1.1). Design: host link and firmware CPU (§3.5.13).

| Parameter           | Unit                         | Valid values | Meaning                                                                                                              |
|---------------------|------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------|
| pcie_bandwidth_mbps | MB/s (10 <sup>6</sup> bytes) | 0 disables   | each Read and Write is charged its size at this rate, on one queue per direction                                     |
| pcie_prop_delay_ns  | ns                           | 0 disables   | fixed delay added to each Read and Write after its transfer                                                          |
| fw_cpu_ns           | ns per command               | 0 disables   | firmware time per Read, Write and Zone Append on one modelled core; caps the rate at about one command per fw_cpu_ns |

### 5.1.12 ZNS

Reference: ZNS (§A.1.1). Design: ZNS (§4.8).

| Parameter          | Unit     | Valid values                | Meaning                                        |
|--------------------|----------|-----------------------------|------------------------------------------------|
| zns_num_ch         | channels | 1 to 128                    | channels                                       |
| zns_num_lun        | LUNs     | >= 1                        | LUNs per channel                               |
| zns_num_plane      | planes   | 1 to 8                      | planes per LUN; the program unit grows with it |
| zns_num_blk        | blocks   | >= 1                        | blocks per plane                               |
| zns_chnls_per_zone | channels | divides zns_num_ch; 0 = all | zone width                                     |

| Parameter                                         | Unit                  | Valid values                                  | Meaning                                                        |
|---------------------------------------------------|-----------------------|-----------------------------------------------|----------------------------------------------------------------|
| zns_zone_cap                                      | bytes                 | one block to the zone size; 0 = zone size     | writable part of each zone                                     |
| zns_flash_type                                    | type                  | 1 SLC, 2 MLC, 3 TLC, 4 QLC, 5 PLC             | cell type; MLC and PLC need the three times below              |
| zns_pg_rd_lat, zns_pg_wr_lat, zns_blk_er_lat      | ns                    | >= 0; 0 = built-in                            | override the cell type's times                                 |
| zns_cmd_addr_lat, zns_pg_xfer_lat, zns_status_lat | ns                    | >= 0                                          | channel bus phases                                             |
| zns_pe_suspend, zns_tsusp_ns                      | flag, ns              |                                               | reads suspend a program or erase on their plane                |
| zns_max_open, zns_max_active                      | zones                 | <= zone count, open <= active; 0 = no limit   | Maximum Open and Active Resources                              |
| zns_num_wc                                        | write caches          | <= zone count; 0 picks from zns_max_open      | zone write caches                                              |
| zns_zasl_bs                                       | bytes                 | power-of-two multiple of 4 KiB; 0 follows mdt | Zone Append size limit                                         |
| zns_zd_ext_size                                   | bytes                 | multiple of 64, up to 16320                   | zone descriptor extension size                                 |
| zns_num_conv_zones                                | zones                 | capped at the zone count                      | leading conventional zones; Linux rejects them                 |
| zns_zrwa_size, zns_zrwaflush_size, zns_zrwa_num   | blocks, blocks, zones | all three set or all 0                        | Zone Random Write Area window, flush granularity and resources |
| zns_cross_zone_read                               | bool                  |                                               | allow reads across zone boundaries                             |

Derived quantities (when the divisions are exact):

```

pages per block = namespace size / 16 KiB / (zns_num_ch x zns_num_lun x zns_num_blk)
zone width      = zns_chnls_per_zone, or zns_num_ch when 0
zone size       = zone width x zns_num_lun x zns_num_plane x pages per block x 16 KiB
zone count      = zns_num_ch x zns_num_blk / (zone width x zns_num_plane)

```

Interactions:

- The zone count does not depend on the namespace size; the zone size grows with it. Linux uses a zoned namespace only when the zone size is a power of two, so keep the size and the geometry powers of two.
- lba\_index must select a block of 4 KiB or less.
- zns\_zrwa\_size must be a multiple of zns\_zrwaflush\_size, and the zone capacity a multiple of zns\_zrwaflush\_size.
- The BlackBox geometry and timing properties do not apply to ZNS.

### 5.1.13 FDP

Set on femu-subsys, which a bbssd controller joins with subsys=. Reference: Flexible Data Placement (§A.1.2). Design: FDP (§4.6).

| Parameter | Unit | Valid values | Meaning                                             |
|-----------|------|--------------|-----------------------------------------------------|
| fdp       | bool |              | FDP in endurance group 1 for the joining controller |

| Parameter          | Unit          | Valid values                       | Meaning                                                                                           |
|--------------------|---------------|------------------------------------|---------------------------------------------------------------------------------------------------|
| fdp.nruh           | handles       | 1 to fdp.nru; required with fdp=on | reclaim unit handles, the placement identifiers the host can name                                 |
| fdp.nru            | reclaim units | fdp.nruh to 65536                  | reclaim units per group; bbssd uses at most one per line and needs $2 \times \text{fdp.nruh} + 1$ |
| fdp.nrg            | groups        | 1 only                             | reclaim groups                                                                                    |
| fdp.runs           | bytes         | 0, or one line's size with bbssd   | reclaim unit size; 0 lets FEMU choose                                                             |
| fdp.isolation_mode | number        | 0, or any other value              | 0: every handle Persistently Isolated; otherwise the last one Initially Isolated                  |

Interactions:

- The subsystem takes one controller with one namespace, and must come first on the command line.
- The controller's `gc_strategy` and `fdp_trim_erase_all` apply; `gc_policy` other than greedy, mapping other than page, `buffer_size`, `hot_cold_sep`, `read_reclaim_limit`, `retention_limit_sec`, `ecc_retention_sec`, `trim_lat_ns`, `meta`, `streams` and `ns_mgmt` are refused.
- KV is refused under FDP; NoSSD, ZNS and OCSSD report FDP but ignore it for placement.

### 5.1.14 OCSSD, CSD and KV

#### OCSSD

Reference: OCSSD (§A.1.1).

| Parameter                             | Unit                  | Valid values                               | Meaning                                                                                         |
|---------------------------------------|-----------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------|
| lver                                  | version               | 1 (1.2) or 2 (2.0)                         | Open-Channel version                                                                            |
| flash_type                            | type                  | 1 SLC, 2 MLC, 3 TLC, 4 QLC                 | cell type for the built-in timing tables                                                        |
| lnum_ch, lnum_lun                     | count                 | lnum_ch 1 to 32, lnum_ch x lnum_lun <= 128 | channels (2.0 groups) and LUNs (parallel units)                                                 |
| lnum_pln                              | planes                | > 0; 1.2: 1, 2 or 4                        | planes per LUN                                                                                  |
| lpgs_per_blk, lsecs_per_pg            | count                 | > 0; 1.2: pages <= 512                     | pages per block, sectors per page                                                               |
| lsec_size, lmetasize, lmax_sec_per_rq | bytes, bytes, sectors | 1.2 only                                   | sector size, out-of-band bytes, most sectors per vector command                                 |
| oc12_channel_timing                   | bool                  | 1.2 only                                   | charge channel transfer per page, <code>ch_xfer_lat</code> or the <code>flash_type</code> value |
| learly_reset                          | flag                  | 2.0 only                                   | report the early reset capability                                                               |

OCSSD takes the whole controller: one namespace, no namespace\_modes neighbours.

#### CSD

Reference: CSD (§A.1.1). CSD builds the bbssd FTL, so sections 5 to 9 apply.

| Parameter | Unit          | Valid values  | Meaning                             |
|-----------|---------------|---------------|-------------------------------------|
| fdm_size  | MiB           | > 0, required | functional data memory              |
| nr_cu     | compute units | 1 to 64       | programs run on the first free unit |

| Parameter         | Unit           | Valid values | Meaning                                                                                      |
|-------------------|----------------|--------------|----------------------------------------------------------------------------------------------|
| csf_runtime_scale | multiplier     | > 0          | host run time multiplier for programs that declare no run time                               |
| csd_program_dir   | host directory |              | where shared-object and uBPF programs load from; unset allows only the built-in program type |

## KV

KV uses the geometry (section 5), the NAND timing (section 6), and `gc_thres_pcent` as the share of the NAND usable for values. The FTL, mapping, cache, write buffer and other GC properties have no effect on KV, but KV runs the same validation as `bbssd`, so an out-of-range or unknown value is still refused. The GC capacity check of `bbssd` is not made: a namespace larger than the NAND starts, with its value space clamped. Design: KV (§4.12).

### 5.1.15 CXL SSD

`femu-cxl-ssd` is a separate device below a CXL root port. Reference: `femu-cxl-ssd` (§A.1.3) and its runtime properties (§A.2.2). Design: CXL SSD (§4.19).

| Parameter                                 | Unit              | Valid values                                                 | Meaning                                                                 |
|-------------------------------------------|-------------------|--------------------------------------------------------------|-------------------------------------------------------------------------|
| volatile-memdev                           | memory backend id | required; a non-zero multiple of 256 MiB, at most 120 GiB    | the backend that holds the data; its size is the device capacity        |
| cache-pages                               | 4 KiB pages       | 0, or up to the media page count and divisible by cache-ways | DRAM page cache; 0 sends every access to the media                      |
| cache-ways                                | ways              | non-zero, divides cache-pages                                | set associativity; 1 is direct mapped; changeable at run time           |
| cache-policy                              | name              | fifo, lifo, clock, s3-fifo                                   | replacement within a set                                                |
| prefetch-degree, prefetch-stride          | pages             | up to the media page count                                   | pages inserted after a miss, and their distance; changeable at run time |
| ftl                                       | bool              |                                                              | off charges no media time and cannot be linked to an NVMe controller    |
| channels, luns-per-channel                | count             | 1 to 4096, 1 to 128                                          | NAND channels and LUNs (one plane each)                                 |
| pages-per-block, blocks-per-plane         | count             | 1 to 65536; blocks 2 to 65536, or 0 to size it               | NAND blocks; 0 leaves spare room for GC                                 |
| read-ns, program-ns, erase-ns, channel-ns | ns                | at most one second                                           | NAND times                                                              |
| gc-threshold, gc-threshold-high           | percent           | 1 to 100, high $\geq$ low                                    | GC watermarks                                                           |
| der                                       | name              | off, memslot, cylon                                          | direct mapping of cached pages into the guest                           |
| der-replace-rate                          | per second        | 0 disables                                                   | memslot alias replacements a hot page may cause                         |
| cylon-kernel-ack                          | bool              | must be on with der=cylon                                    | states that the host runs a fixed Cylon kernel                          |
| concurrent-misses                         | on, off, auto     |                                                              | misses to different pages wait for the media together                   |

| Parameter                                       | Unit              | Valid values                 | Meaning                                                  |
|-------------------------------------------------|-------------------|------------------------------|----------------------------------------------------------|
| cylon-first-touch-program, cylon-free-writeback | bool              |                              | media rules of the published Cylon experiments           |
| cca                                             | bool              |                              | the caching API on BAR 5                                 |
| lsa-control                                     | bool              | not with an lsa backend      | experiment commands through Get LSA; trusted guests only |
| log-dir, tracefs-dir, log-limit                 | path, path, bytes | log-limit 0 opens no I/O log | where the control channel writes, and how much           |

#### Interactions:

- The machine needs `cxl=on`, a `pxb-cxl` host bridge, a `cxl-rp` root port and a `cxl-fmw` window at least as large as the backend; `run-cxlssd.sh` builds them.
- `der=memslot` needs KVM. `der=cylon` needs a Cylon host kernel and a shared, preallocated `hugetlbfs` backend; without them, or when the memory cannot be locked, it falls back to MMIO with a warning. It also needs `-machine smm=off`, which the device does not check (`der=cylon` (§4.18.7)).
- A femu controller with `cxl_ssd=<id>` serves the same medium as an NVMe namespace. It needs `femu_mode=1`, one namespace, `ftl=on` on the CXL device, `devsz_mb` unset or equal to the medium's size, and none of `ns_mgmt`, `subsys`, `streams`, `power_loss`, `buffer_size`, `op_pcent`, `meta`, `pi` or `dps`. The medium's geometry and timing apply, not the controller's (CXL NVMe link (§4.21)).

### 5.1.16 Accepted for compatibility, no effect

These properties exist so that old command lines still start. Setting one to anything but its default prints a warning at realize.

| Device | Properties                                                                                        |
|--------|---------------------------------------------------------------------------------------------------|
| femu   | serial, ms, ms_max, dlfeat, tplpbsy, tplrbsy, trcb sy, nr_thread, time_slice, context_switch_time |

`nr_thread` is still refused at 0 by CSD. Identify Controller reports a serial number FEMU generates, whatever `serial` says.

### 5.1.17 Worked configurations

Each configuration below is started by the documentation checks, which also write and read one block through it.

#### A TLC drive with a channel bus and read suspend

A 4 GiB namespace over the default geometry, TLC page-type timing, a 10 us page transfer and 2 us command and status phases on each channel, and reads that suspend a program or erase for 15 us of overhead:

```
| -device femu,femu_mode=1,devsz_mb=4096,nand_cell_type=3,cmd_addr_lat=2000,pg_xfer_lat=10000,status_lat=2000,pe_suspend=1,tsusp_ns=15000
```

#### DFTL with caches and a write buffer

A DFTL table with 8 MiB cached, a 64 MiB read cache with LRU eviction, and a write buffer of 4096 NAND pages that the guest sees as a volatile write cache. `oncs=0x1c` enables Write Zeroes (0x8) together with Dataset Management (0x4) and Save/Select (0x10):

```
| -device femu,femu_mode=1,devsz_mb=2048,mapping=dftl,mapping_cache_mb=8,read_cache_mb=64,cache_evict=lru,buffer_size=4096,buffer_thres_pcent=75,vwc=1,oncs=0x1c
```

## A GC study drive

512 MiB of NAND exposed as NAND / 1.25, background and forced GC both starting at 95% of lines in use, cost-benefit victims and hot/cold separation, as in tutorial 02 (§5.4):

```
| -device femu,femu_mode=1,nchs=2,luns_per_ch=4,blks_per_pl=64,op_pcent=25,gc_thres_pcent=95,gc_thres_pcent_high=95,gc_policy=cost-benefit,hot_cold_sep=on
```

## A ZNS drive with 4 KiB blocks, limits and ZRWA

64 zones of 64 MiB, 4 KiB logical blocks, 8 open and 16 active zones, and a ZRWA of 64 blocks flushed in units of 8 on up to 4 zones at once:

```
| -device femu,femu_mode=3,devsz_mb=4096,zns_num_ch=8,zns_num_lun=4,zns_num_plane=2,zns_num_blk=128,lba_index=3,zns_max_open=8,zns_max_active=16,zns_zrwa_size=64,zns_zrwafg_size=8,zns_zrwa_num=4
```

## FDP with eight handles

Eight placement handles over 256 lines (blks\_per\_pl=256), with cost-benefit reclaim unit selection:

```
| -device femu-subsys,id=fdp0,fdp=on,fdp_nruh=8 -device femu,femu_mode=1,devsz_mb=2048,blks_per_pl=256,subsys=fdp0,gc_strategy=1
```

## Mixed namespaces on a modelled link

A 3 GiB BlackBox namespace and a 1 GiB ZNS namespace on one controller, with a 3500 MB/s host link, 1 us of propagation delay and 2 us of firmware time per command:

```
| -device femu,femu_mode=1,devsz_mb=4096,namespaces=2,namespace_sizes=3G,,1G,namespace_modes=bbssd,znssd,pcie_bandwidth_mbps=3500,pcie_prop_delay_ns=1000,fw_cpu_ns=2000
```

## Wear and faults

A drive rated for 3000 program/erase cycles, with 100 bad blocks, an ECC step of 5 us, one read in 100,000 uncorrectable and one write in 1,000,000 failed:

```
| -device femu,femu_mode=1,devsz_mb=1024,pe_cycles_rated=3000,nand_bad_blocks=100,ecc_step_ns=5000,err_read_unc_ppm=10,err_write_fail_ppm=1
```

## Namespace management

A controller that starts with one bbssd namespace and lets the guest allocate up to four bbssd namespaces in all, the boot one included, so three more:

```
| -device femu,femu_mode=1,devsz_mb=2048,ns_mgmt=on,bbssd_ns_limit=4
```

## A CXL SSD and an NVMe view of it

A 256 MiB femu-cxl-ssd with a 1024-page, 4-way s3-fifo cache below a CXL host bridge, and a femu controller that serves the same medium as an NVMe namespace:

```
| ./qemu-system-x86_64 -machine q35,cxl=on \  
|   -object memory-backend-ram,id=cxlmem,size=256M \  
|   -device pxb-cxl,id=cxl.0,bus=pcie.0,bus_nr=52 \  
|   -device cxl-rp,id=cxl-rp0,bus=cxl.0,chassis=0,slot=0 \  
|   -device femu-cxl-ssd,id=cxlssd,bus=cxl-rp0,volatile-memdev=cxlmem,cache-pages=1024,cache-ways=4,cache-policy=s3-fifo \  
|   \  
|   -device femu,id=nvme0,bus=pcie.0,femu_mode=1,cxl_ssd=cxlssd \  
|   -M cxl-fmw.0.targets.0=cxl.0,cxl-fmw.0.size=256M
```

### 5.1.18 Related pages

- Device property reference (§A.1) and runtime properties (§A.2), generated from the binary
- Tutorials (§5.2)
- Design (§3.2)
- Log pages and counters (§B)

## 5.2 Tutorials

Hands-on walkthroughs. Each one starts a guest with an emulated device, runs a workload inside it, and reads what the device reports. Every tutorial stands on its own, but they build on each other in this order.

| Tutorial                             | You learn to                                                                                                                                                      | Device               |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 01 Your first SSD (§5.3)             | boot a BlackBox SSD from your own QEMU command line, measure read and write latency, read the write amplification factor (WAF)                                    | BBSSD, 2 GiB of NAND |
| 02 Garbage collection and WAF (§5.4) | provoke garbage collection (GC), measure the WAF of one run, and see how over-provisioning, the GC threshold, the victim policy and hot/cold separation change it | BBSSD                |
| 03 Zoned namespaces (§5.5)           | read the zone report, move zones through their states, use Zone Append, fio's zoned mode and zonefs                                                               | ZNS                  |
| 04 Flexible Data Placement (§5.6)    | write through placement identifiers, read the FDP log pages, and measure what placement does to the WAF                                                           | BBSSD with FDP       |
| 05 Latency tuning (§5.7)             | change NAND times, the cell type, the channel bus, the host link and the firmware cost, and measure each with fio                                                 | BBSSD                |
| 06 Several namespaces (§5.8)         | run a BlackBox, a ZNS, a NoSSD and a KV namespace on one controller                                                                                               | mixed                |
| 07 Key-value SSD (§5.9)              | store, retrieve, list and delete values with nvme-cli                                                                                                             | KV                   |
| 08 CXL SSD as memory (§5.10)         | start a femu-cxl-ssd, find it in the guest, use it as a DAX device or a NUMA node, and read its counters                                                          | CXL Type-3 SSD       |
| 09 Configuration files (§5.11)       | describe a device in a file and expand it with <code>ssd-config.sh</code> instead of writing a long <code>-device</code> line                                     | any                  |

### 5.2.1 Before you start

You need a built FEMU and a guest image. The quick start (§2.4) does both in about five minutes of work: `femu-compile.sh` in `build-femu/` and `make-guest-image.sh` for the image.

The tutorials use three shell variables on the host. Set them once in the terminal you start QEMU from and in the one you use to reach the guest:

```
cd build-femu                # the directory femu-compile.sh built in
export IMGDIR=$HOME/images    # where make-guest-image.sh put the image
export OSIMGF=$IMGDIR/u20s.qcow2 # the guest image
export SSH_PORT=8080          # host port forwarded to the guest's SSH
```

`run-guest-ssh.sh` reads `IMGDIR` and `SSH_PORT`, so with these set, `./run-guest-ssh.sh` opens a shell in the guest and `./run-guest-ssh.sh CMD` runs one command there.



## 5.2.2 How to read the tutorials

- Blocks marked as a FEMU example hold a QEMU command line or -device options. The documentation checks start each of them under QEMU's QTest accelerator, so they are known to be accepted by the current code. The checks do not boot a guest.
- sh blocks run inside the guest unless the text says "on the host". Run them in the shell `./run-guest-ssh.sh` gives you.
- text blocks show output. Unless a block says it is illustrative, it was captured from a real run: FEMU at commit 1a03ded0f, the Ubuntu 24.04 guest from `make-guest-image.sh` (Linux 6.8, nvme-cli 2.8, fio 3.36), on a 20-core host. Latencies and throughput depend on the host. Counter values such as the WAF depend only on the workload, and fio's random offsets change between runs, so expect them within a few percent of the values shown.
- The QEMU command lines run without `sudo`. That works when your user is in the `kvm` group; otherwise put `sudo` in front, as the `run-*.sh` launchers do. Without root, FEMU usually cannot lock the device memory (`ulimit -l` is too small) and says so at start; the device works, but host paging can add jitter to latency, so use `sudo` or raise the limit for latency measurements.
- Each emulated device lives in host memory: plan for the device size plus the guest's `-m` in free host RAM.

## 5.2.3 Where to go next

- Measuring (§6.1) and performance tuning (§5.12) explain how to get numbers that repeat.
- The parameter manual (§5.1) explains every group of device parameters and how they interact.
- The design pages (§3.2) explain what happens inside the device.

## 5.3 Tutorial 01: your first SSD

You start a guest with a small BlackBox SSD (BBSSD) from a QEMU command line you write yourself, look at the device the guest sees, measure read and write latency with `fio`, and read the write amplification factor (WAF) from the device. It takes about ten minutes.

The quick start (§2.4) does something similar with `run-blackbox.sh`, which builds a 12 GiB device. This tutorial uses a 2 GiB device so that the whole drive can be filled in seconds, and it explains each option.

You need: a built FEMU and guest image, the variables from Before you start (§5.2.1), and about 7 GiB of free host memory (2 GiB for the device, 4 GiB for the guest).

### 5.3.1 Choose the geometry

A BBSSD is NAND flash organised as channels, LUNs (dies) per channel, planes per LUN, blocks per plane and pages per block. This tutorial uses 4 channels of 4 LUNs, one plane, 128 blocks per plane and the default 256 pages of 4 KiB per block:

|      |                                                                          |
|------|--------------------------------------------------------------------------|
| NAND | = 4 channels x 4 LUNs x 1 plane x 128 blocks x 256 pages x 4 KiB = 2 GiB |
| line | = one block on every LUN = 16 blocks = 16 MiB; there are 128 lines       |

A line (a superblock) is the unit garbage collection (GC) reclaims. The guest must not see all of the NAND: GC needs spare room to move data into. `op_pcent=25` exposes NAND / 1.25, about 1.6 GiB, and keeps the rest as spare. With `op_pcent` set, `devsz_mb` is ignored and the device is sized from the geometry.

### 5.3.2 Start the guest (on the host)

From build-femu/:

```
./qemu-system-x86_64 -name femu-tut01,debug-threads=on \
  -enable-kvm -cpu host -smp 4 -m 4G \
  -device virtio-scsi-pci,id=scsi0 -device scsi-hd,drive=hd0 \
  -drive file=$OSIMGF,if=none,cache=none,format=qcow2,id=hd0 \
  -net user,hostfwd=tcp::$SSH_PORT-:22 -net nic,model=virtio \
  -device femu,femu_mode=1,nchs=4,luns_per_ch=4,blks_per_pl=128,op_pcent=25 \
  -nographic
```

Everything except the last `-device` line is an ordinary guest: KVM, 4 vCPUs, 4 GiB of RAM, the guest image on a virtio-scsi disk, and user networking that forwards `SSH_PORT` to the guest's SSH port. `debug-threads=on` names FEMU's threads, which performance tuning (§5.12.1) needs. The FEMU device:

| Option                                            | Meaning                                                                         |
|---------------------------------------------------|---------------------------------------------------------------------------------|
| <code>femu_mode=1</code>                          | BlackBox mode: the device runs its own FTL, GC and NAND timing                  |
| <code>nchs=4,luns_per_ch=4,blks_per_pl=128</code> | the geometry from step 1; the page size and pages per block keep their defaults |
| <code>op_pcent=25</code>                          | 25% over-provisioning                                                           |

Wait for the `femu-guest` login: prompt, about 30 seconds. Leave this terminal running. To stop QEMU, press `Ctrl-a` then `x`.

### 5.3.3 Look at the device

In a second terminal, open a shell in the guest with `./run-guest-ssh.sh`, then:

```
sudo nvme list
```

| Node         | Generic    | SN    | Model                        | Namespace | Usage             | Format      | FW Rev |
|--------------|------------|-------|------------------------------|-----------|-------------------|-------------|--------|
| /dev/nvme0n1 | /dev/ng0n1 | vSSD0 | FEMU BlackBox-SSD Controller | 0x1       | 1.72 GB / 1.72 GB | 512 B + 0 B | 1.0    |

1.72 GB is 2 GiB / 1.25. The guest's own disk is `/dev/sda`. The namespace uses 512-byte blocks; `cat /sys/block/nvme0n1/size` prints 3355443, the size in 512-byte sectors.

### 5.3.4 Read before writing: no NAND time

Measure 4 KiB random reads at queue depth 1 on the empty drive:

```
sudo fio --name=rr0 --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
  --rw=randread --bs=4k --iodepth=1 --runtime=10 --time_based
```

The median completion latency (`clat 50.00th`) is about 8 us. A page that was never written has no mapping, so the FTL charges it no NAND read: what you see is the guest's and FEMU's own overhead. A read benchmark on a fresh device measures nothing, so always write the range first.

### 5.3.5 Fill the drive

Write the whole namespace once, sequentially:

```
sudo fio --name=fill --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
  --rw=write --bs=128k --iodepth=16
```

```
| write: IOPS=2499, BW=312MiB/s (328MB/s)(1638MiB/5244msec); 0 zone resets
```

### 5.3.6 Read the WAF

The vendor log page C0h holds FEMU's media counters. The first 4 bytes are the WAF times 1000. The three 8-byte counters from byte 8 are the pages the host wrote, the pages GC moved, and the pages programmed for the host (log pages and counters (§B.1)):

```
| sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b | od -An -t u4 -N 4
| sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b | od -An -t u8 -j 8 -N 24 -w24

|          1000
|          419424          0          419424
```

The WAF is (pages programmed + pages GC moved) x 1000 / pages the host wrote. 1000 means 1.000: each 4 KiB page the host wrote was programmed once, and GC has moved nothing. 419424 pages are the 13107 whole 128 KiB blocks fio wrote.

### 5.3.7 Measure written data

Now read the written data at queue depth 1:

```
| sudo fio --name=rr --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
| --rw=randread --bs=4k --iodepth=1 --runtime=15 --time_based

| clat percentiles (usec):
| 1.00th=[ 44], 5.00th=[ 45], 10.00th=[ 45], 20.00th=[ 45],
| 30.00th=[ 45], 40.00th=[ 45], 50.00th=[ 46], 60.00th=[ 47],
| 70.00th=[ 48], 80.00th=[ 49], 90.00th=[ 49], 95.00th=[ 50],
| 99.00th=[ 58], 99.50th=[ 66], 99.90th=[ 116], 99.95th=[ 141],
```

The median is 46 us: the default NAND page read time `pg_rd_lat` of 40 us plus about 6 us of guest and poller overhead. Then random writes:

```
| sudo fio --name=rw --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
| --rw=randwrite --bs=4k --iodepth=1 --runtime=15 --time_based

| clat percentiles (usec):
| 1.00th=[ 204], 5.00th=[ 206], 10.00th=[ 206], 20.00th=[ 208],
| 30.00th=[ 208], 40.00th=[ 208], 50.00th=[ 208], 60.00th=[ 210],
| 70.00th=[ 210], 80.00th=[ 212], 90.00th=[ 215], 95.00th=[ 219],
| 99.00th=[ 269], 99.50th=[ 297], 99.90th=[54789], 99.95th=[55313],
```

The median is 208 us, the program time `pg_wr_lat` of 200 us plus overhead. The 99.9th percentile is 55 ms. After the fill, more than 75% of the lines are in use, so GC is armed. Every overwrite leaves an invalid page behind, and once a line has 1/8 of its pages invalid, GC reads its valid pages, programs them elsewhere and erases it. This run gets there towards its end, and a write that needs a LUN GC is busy with waits for it. That is why GC shows in the slowest 0.1% and not in the median. Read the counters again:

```
|          1477
|          472826          225792          472826
```

GC has moved 225792 pages for 472826 host pages: WAF 1.477.

### 5.3.8 Shut down

```
| sudo poweroff
```

QEMU exits in the first terminal. The data on the emulated SSD is gone: FEMU keeps it only in host memory. A guest reboot keeps it.

### 5.3.9 What you learned

- A FEMU SSD is one `-device femu` option on an ordinary QEMU command line.
- `op_pcent` sets the spare area directly; the `geometry` sets the NAND size.
- Unwritten pages cost no NAND time; write before you read.
- Queue depth 1 latency is the NAND time plus a few microseconds of overhead; GC shows up in the tail.
- Log page C0h gives the WAF and the page counters since the device started.

### 5.3.10 Next

- Tutorial 02 (§5.4) makes GC run on purpose and changes what it does.
- The BlackBox FTL (§3.4) explains lines, write pointers and GC; NAND media and timing (§3.5) explains where the 40 and 200 us come from.
- BlackBox SSD (§4.4) lists every option of the mode.

## 5.4 Tutorial 02: garbage collection and write amplification

You make garbage collection (GC) run on a BlackBox SSD, measure the write amplification factor (WAF) of one run from the counters in log page C0h, and then change four things and watch the WAF move: the GC threshold, the over-provisioning, the victim policy, and hot/cold separation. Each measurement takes under a minute.

You need: tutorial 01 (§5.3) done once, the variables from Before you start (§5.2.1), and about 7 GiB of free host memory.

### 5.4.1 Background

The FTL writes every host page to a fresh NAND page and marks the old copy invalid. A line (one block on every LUN) fills up with a mix of valid and invalid pages. GC picks a victim line, copies its valid pages elsewhere and erases it. Every copy is a NAND program the host did not ask for, so

| 
$$\text{WAF} = (\text{pages programmed for the host} + \text{pages GC copied}) / \text{pages the host wrote}$$

A WAF of 1 means GC copied nothing. The C0h counters hold the three page counts since the device started. The BlackBox FTL (§3.4.6) describes GC in full; this tutorial needs two facts from it:

- **Background GC** runs after a request once the share of lines in use reaches `gc_thres_pcent`. It collects one line at a time, and only a line with at least 1/8 of its pages invalid.
- **Forced GC** runs inside a write once the share reaches `gc_thres_pcent_high`. It takes the policy's victim however few invalid pages it holds, and repeats until enough lines are free.

### 5.4.2 The device

Every run in this tutorial uses the drive from tutorial 01, 2 GiB of NAND in 128 lines of 16 MiB, and changes one or two options:

| `-device femu,femu_mode=1,nchs=4,luns_per_ch=4,blks_per_pl=128,op_pcent=25`

Start QEMU with the command line of tutorial 01, step 2 (§5.3.2), with this `-device` line in place of its own. You restart QEMU for every configuration: the counters, the mapping and the NAND state start from zero only when QEMU starts.

### 5.4.3 The measurement script

In the guest, save this as `waf-run.sh`:

```
cat > waf-run.sh <<'EOF'
# Usage: bash waf-run.sh [extra fio options]
c0() {
    sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b |
        od -An -t u8 -j 8 -N 24 -w24
}
# Optional: GC and NAND operations take no time, so the runs finish in
# seconds. The page counts do not depend on time.
sudo nvme admin-passthru /dev/nvme0 --opcode=0xef --cdw10=2 >/dev/null
sudo nvme admin-passthru /dev/nvme0 --opcode=0xef --cdw10=4 >/dev/null
# 1. Fill the whole drive once.
sudo fio --name=fill --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
    --rw=write --bs=128k --iodepth=16 >/dev/null
# 2. Age it: random overwrites until GC is in a steady state.
sudo fio --name=age --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
    --rw=randwrite --bs=4k --iodepth=32 --io_size=4G --randrepeat=0 "$@" >/dev/null
# 3. Measure one run as the difference of two counter reads.
c0 > before.txt
sudo fio --name=run --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
    --rw=randwrite --bs=4k --iodepth=32 --io_size=4G --randrepeat=0 "$@" >/dev/null
c0 > after.txt
paste before.txt after.txt | awk '{ h = $4 - $1; g = $5 - $2; n = $6 - $3;
    printf "host %d gc %d nand %d WAF %.3f\n", h, g, n, (n + g) / h }'
EOF
```

Three things in it matter:

- **Steady state.** The first pass over a fresh drive copies nothing, so its WAF is 1. The script fills the drive, then ages it with 4 GiB of random writes (about 2.5 times its size) before it measures.
- **Differences, not totals.** The counters count since QEMU started. The script reads them before and after the measured run and divides the differences.
- **Time switched off.** Vendor command 0xEF with code 2 makes GC take no time and code 4 sets the flat NAND times to 0. GC still copies the same pages, so the WAF is unchanged, and a run takes seconds instead of minutes. Code 3 would restore the built-in times, not the ones on your command line, so restart QEMU before you measure latency again (run-time controls (§3.4.13)). Leave the two lines out if you want the latency as well.

Run it:

```
| bash waf-run.sh
| host 1048576 gc 7357952 nand 1048576 WAF 8.017
```

The measured run wrote 1048576 pages (4 GiB) and GC copied 7.36 million: eight NAND programs for each host write. That is far more than an SSD with 25% spare should need. The next step shows why.

### 5.4.4 The GC threshold

`gc_thres_pcent` defaults to 75. This drive exposes  $1 / 1.25 = 80\%$  of its NAND, so once it is full, more than 75% of its lines always hold data and background GC never stops. It collects any line that is at least 1/8 invalid, so it takes lines that are still 7/8 valid: it copies 7 valid pages for each invalid page it frees. Seven GC copies plus the host's own program per page written is the WAF of 8.

Raise the threshold above the fill level. Restart QEMU for each line, then run `bash waf-run.sh`:

```
-device femu,femu_mode=1,nchs=4,luns_per_ch=4,blks_per_pl=128,op_pcent=25,gc_thres_pcent=90
-device femu,femu_mode=1,nchs=4,luns_per_ch=4,blks_per_pl=128,op_pcent=25,gc_thres_pcent=95
```

| <code>gc_thres_pcent</code> | WAF   |
|-----------------------------|-------|
| 75 (default)                | 8.017 |
| 90                          | 4.590 |
| 95                          | 3.273 |

At 95, `gc_thres_pcent` equals `gc_thres_pcent_high` in this configuration, so background GC starts only where forced GC does. By then the lines hold more invalid pages, and each collection frees more space per copy. Set `gc_thres_pcent` above the share of the NAND your namespace fills, or the background GC rate is what you measure.

### 5.4.5 Over-provisioning

Keep `gc_thres_pcent=90` and change `op_pcent`:

```
-device femu,femu_mode=1,nchs=4,luns_per_ch=4,blks_per_pl=128,op_pcent=7,gc_thres_pcent=90
-device femu,femu_mode=1,nchs=4,luns_per_ch=4,blks_per_pl=128,op_pcent=15,gc_thres_pcent=90
-device femu,femu_mode=1,nchs=4,luns_per_ch=4,blks_per_pl=128,op_pcent=50,gc_thres_pcent=90
-device femu,femu_mode=1,nchs=4,luns_per_ch=4,blks_per_pl=128,op_pcent=100,gc_thres_pcent=90
```

| <code>op_pcent</code> | Share of NAND exposed | WAF    |
|-----------------------|-----------------------|--------|
| 7                     | 93%                   | 43.945 |
| 15                    | 87%                   | 8.000  |
| 25                    | 80%                   | 4.590  |
| 50                    | 67%                   | 1.970  |
| 100                   | 50%                   | 1.028  |

Read the table this way:

- At 15% the namespace fills 87% of the NAND. With the open lines and the invalid pages on top of that, the share of lines in use stays at the 90% threshold, background GC keeps running, and it is the case of step 3 again: WAF 8.
- At 7% almost no line is free, forced GC runs inside most writes, and it takes victims that are nearly all valid. Forced GC has no 1/8 filter, so the WAF has no ceiling: 44 programs per host write.
- From 25% up, more spare means emptier victims and a WAF that falls towards 1.

FEMU refuses a namespace that leaves GC too few lines at start-up (the reserve (§3.4.12)); 7% is close to the smallest this geometry accepts.

### 5.4.6 The victim policy

Go back to `op_pcent=25` with `gc_thres_pcent=95`, and try each `gc_policy`:

```
-device femu,femu_mode=1,nchs=4,luns_per_ch=4,blks_per_pl=128,op_pcent=25,gc_thres_pcent=95,gc_policy=greedy
-device femu,femu_mode=1,nchs=4,luns_per_ch=4,blks_per_pl=128,op_pcent=25,gc_thres_pcent=95,gc_policy=fifo
-device femu,femu_mode=1,nchs=4,luns_per_ch=4,blks_per_pl=128,op_pcent=25,gc_thres_pcent=95,gc_policy=random
-device femu,femu_mode=1,nchs=4,luns_per_ch=4,blks_per_pl=128,op_pcent=25,gc_thres_pcent=95,gc_policy=cost-benefit
-device femu,femu_mode=1,nchs=4,luns_per_ch=4,blks_per_pl=128,op_pcent=25,gc_thres_pcent=95,gc_policy=d-choice
```

Run the uniform workload (`bash waf-run.sh`) and a skewed one, where a few pages take most of the writes:

```
bash waf-run.sh --random_distribution=zipf:1.2
```

| gc_policy    | WAF, uniform | WAF, zipf 1.2 |
|--------------|--------------|---------------|
| greedy       | 3.277        | 5.961         |
| fifo         | 3.277        | 6.052         |
| random       | 5.087        | 6.122         |
| cost-benefit | 3.269        | 5.798         |
| d-choice     | 3.660        | 5.898         |

With uniform writes, every line ages the same way, so the line with the fewest valid pages is also the oldest, and greedy, fifo and cost-benefit choose alike. random and d-choice (best of 4 random lines) pick worse victims. With skewed writes the WAF is higher for every policy: hot and cold pages share lines, and each collection copies cold pages that will not change again. The policy cannot fix that; separating the data can.

### 5.4.7 Hot/cold separation

`hot_cold_sep=on` writes overwrites (pages that already had a mapping) to their own lines, so hot pages are invalidated together and cold pages stay together (hot/cold separation (§3.4.5)):

```
-device femu,femu_mode=1,nchs=4,luns_per_ch=4,blks_per_pl=128,op_pcent=25,gc_thres_pcent=95,hot_cold_sep=on
-device femu,femu_mode=1,nchs=4,luns_per_ch=4,blks_per_pl=128,op_pcent=25,gc_thres_pcent=95,gc_policy=cost-benefit,hot_cold_sep=on
```

| Configuration, zipf 1.2       | WAF   |
|-------------------------------|-------|
| greedy                        | 5.961 |
| greedy, hot_cold_sep=on       | 2.394 |
| cost-benefit, hot_cold_sep=on | 1.646 |

Separation more than halves the WAF. With hot and cold data in different lines, the age term of cost-benefit now pays off: it leaves young hot lines alone until more of their pages are invalid.

### 5.4.8 What you learned

- Measure the WAF of a run as the ratio of counter differences, after the drive is full and aged.

- Background GC with a threshold below the fill level runs all the time and holds the WAF near 8; set `gc_thres_pcent` above the share of NAND you expose.
- Spare capacity is the strongest lever. Below a few percent, forced GC has no limit.
- Victim policies differ little under uniform writes; under skew, separate hot from cold data first.

### 5.4.9 Next

- Tutorial 04 (§5.6) lets the host do the separating, with Flexible Data Placement.
- Tutorial 05 (§5.7) measures latency instead of page counts.
- Measuring (§6.1.2) covers the counters and repeatable runs in more depth.

## 5.5 Tutorial 03: zoned namespaces

You start a guest with a Zoned Namespace (ZNS) SSD, read its zone report, move one zone through every state by hand, append data and let the device pick the address, hit the open-zone limit on purpose, then write with `fio`'s zoned mode and mount `zonefs`. It takes about fifteen minutes.

You need: the variables from Before you start (§5.2.1) and about 5 GiB of free host memory. The guest image from `make-guest-image.sh` has a kernel with ZNS support (Linux 6.8), `nvme-cli`, `blkzone` and `fio`. The `zonefs` step installs two more packages, so the guest needs network access.

### 5.5.1 Background

A zoned namespace is divided into zones. Each zone has a write pointer: the host writes the zone only at its write pointer (or with Zone Append, where the device picks the address), and must reset the zone before it writes it again. The device runs no garbage collection; reclaiming space is the host's job. Each zone is in one state, and these are the moves you make in this tutorial:

| From                               | Event                                                       | To                |
|------------------------------------|-------------------------------------------------------------|-------------------|
| Empty or Closed                    | a write or an append                                        | Implicitly Opened |
| Empty, Implicitly Opened or Closed | Open Zone                                                   | Explicitly Opened |
| Implicitly or Explicitly Opened    | Close Zone                                                  | Closed            |
| Empty, Opened or Closed            | Finish Zone, or the write pointer reaches the zone capacity | Full              |
| Opened, Closed or Full             | Reset Zone                                                  | Empty             |

Opened zones count against Maximum Open Resources, and opened plus closed zones against Maximum Active Resources. The ZNS design page (§4.8.5) has the full state machine.

### 5.5.2 Start the guest (on the host)

```
./qemu-system-x86_64 -name femu-tut03,debug-threads=on \
  -enable-kvm -cpu host -smp 4 -m 4G \
  -device virtio-scsi-pci,id=scsi0 -device scsi-hd,drive=hd0 \
  -drive file=$OSIMGF,if=none,cache=none,format=qcow2,id=hd0 \
  -net user,hostfwd=tcp::SSH_PORT-:22 -net nic,model=virtio \
  -device femu,femu_mode=3,devsz_mb=1024,zns_max_open=4,zns_max_active=6 \
  -nographic
```



| Option                           | Meaning                                                                        |
|----------------------------------|--------------------------------------------------------------------------------|
| femu_mode=3                      | ZNS                                                                            |
| devsz_mb=1024                    | a 1 GiB namespace                                                              |
| zns_max_open=4, zns_max_active=6 | at most 4 open zones and 6 open or closed ones; 0, the default, means no limit |

With the default ZNS geometry (2 channels, 4 LUNs, 2 planes, 32 blocks) the namespace has 16 zones; their size grows with the namespace, so 1 GiB gives 64 MiB zones (zone size (§4.7.3)). The default cell type is QLC.

### 5.5.3 Check that Linux sees a zoned device

Open a shell in the guest with `./run-guest-ssh.sh`, then:

```

| sudo nvme list
| cd /sys/block/nvme0n1/queue
| cat zoned nr_zones chunk_sectors max_open_zones max_active_zones
| cd
|
/dev/nvme0n1      /dev/ng0n1      vZNSSD0      FEMU ZMS-SSD Controller [by Misao]      0x1      1.07 GB / 1.07 GB      512 B + 0 B 1.0
host-managed
16
131072
4
6

```

`chunk_sectors` is the zone size in 512-byte sectors:  $131072 \times 512 \text{ bytes} = 64 \text{ MiB}$ . The ZNS Identify Namespace data reports the limits 0's based:

```

| sudo nvme zns id-ns /dev/nvme0n1 | grep -E '^(mar|mor) '
|
mar      : 0x5
mor      : 0x3

```

### 5.5.4 Read the zone report

```

| sudo nvme zns report-zones /dev/nvme0n1 -d 3
| sudo blkzone report -c 3 /dev/nvme0n1
|
nr_zones: 16
SLBA: 0      WP: 0      Cap: 0x20000 State: 0x10 Type: 0x2 Attrs: 0 AttrsInfo: 0
SLBA: 0x20000 WP: 0x20000 Cap: 0x20000 State: 0x10 Type: 0x2 Attrs: 0 AttrsInfo: 0
SLBA: 0x40000 WP: 0x40000 Cap: 0x20000 State: 0x10 Type: 0x2 Attrs: 0 AttrsInfo: 0
start: 0x00000000, len 0x020000, cap 0x020000, wptr 0x000000 reset:0 non-seq:0, zcond: 1(em) [type: 2(SEQ_WRITE_REQUIRED)]
start: 0x000020000, len 0x020000, cap 0x020000, wptr 0x000000 reset:0 non-seq:0, zcond: 1(em) [type: 2(SEQ_WRITE_REQUIRED)]
start: 0x000040000, len 0x020000, cap 0x020000, wptr 0x000000 reset:0 non-seq:0, zcond: 1(em) [type: 2(SEQ_WRITE_REQUIRED)]

```

`nvme-cli` prints the write pointer as an absolute LBA, `blkzone` relative to the zone start. The two tools name the states differently:

| State             | nvme-cli State | blkzone zcond |
|-------------------|----------------|---------------|
| Empty             | 0x10           | 1 (em)        |
| Implicitly Opened | 0x20           | 2 (oi)        |
| Explicitly Opened | 0x30           | 3 (oe)        |
| Closed            | 0x40           | 4 (cl)        |
| Read Only         | 0xd0           | 13 (ro)       |
| Full              | 0xe0           | 14 (fu)       |

Every zone is sequential-write-required (type 2) and starts Empty.

### 5.5.5 Append, then move a zone through its states

Append 4 KiB to zone 0 twice. The device writes at the write pointer and reports where the data went:

```
head -c 4096 /dev/urandom > data.bin
sudo nvme zns zone-append /dev/nvme0n1 -s 0 -z 4096 -d data.bin
sudo nvme zns zone-append /dev/nvme0n1 -s 0 -z 4096 -d data.bin
sudo nvme zns report-zones /dev/nvme0n1 -d 1

Success appended data to LBA 0
Success appended data to LBA 8
nr_zones: 16
SLBA: 0          WP: 0x10          Cap: 0x20000    State: 0x20 Type: 0x2  Attrs: 0    AttrsInfo: 0
```

The write pointer moved 16 blocks (2 x 4 KiB), and the write opened the zone implicitly (0x20). Read the data back and compare:

```
sudo nvme read /dev/nvme0n1 -s 0 -c 7 -z 4096 -d out.bin
cmp data.bin out.bin && echo same
```

Close it, then explicitly open four other zones, and try a fifth:

```
sudo nvme zns close-zone /dev/nvme0n1 -s 0
for z in 1 2 3 4; do
    sudo nvme zns open-zone /dev/nvme0n1 -s $((z * 0x20000))
done
sudo nvme zns open-zone /dev/nvme0n1 -s 0xa0000

zns-close-zone: Success, action:1 zone:0 all:0 zcapc:0 nsid:1
zns-open-zone: Success zone slba:20000 nsid:1
zns-open-zone: Success zone slba:40000 nsid:1
zns-open-zone: Success zone slba:60000 nsid:1
zns-open-zone: Success zone slba:80000 nsid:1
NVMe status: Too Many Open Zones: The controller does not allow additional open zones(0x41be)
```

Four zones are open, the limit `zns_max_open=4` allows no fifth. Zone 0 is Closed (0x40): it no longer counts as open, but it still counts as active. Finish one of the open zones, which makes it Full without writing it:

```
sudo nvme zns finish-zone /dev/nvme0n1 -s 0x20000
sudo nvme zns report-zones /dev/nvme0n1 -d 2 | tail -1

zns-finish-zone: Success, action:2 zone:20000 all:0 zcapc:0 nsid:1
SLBA: 0x20000    WP: 0xffffffffffffffff Cap: 0x20000    State: 0xe0 Type: 0x2  Attrs: 0    AttrsInfo: 0
```

A Full zone has no write pointer; it reads as all ones. Reset every zone to start over:

```
sudo nvme zns reset-zone /dev/nvme0n1 -a
```

### 5.5.6 Write with fio's zoned mode

`--zonemode=zbd` makes fio write each zone at its write pointer and respect the open-zone limit:

```
sudo fio --name=zv --filename=/dev/nvme0n1 --direct=1 --ioengine=psync \
--zonemode=zbd --rw=write --bs=128k --size=256M
sudo blkzone report -c 5 /dev/nvme0n1
```

```
write: IOPS=650, BW=81.3MiB/s (85.3MB/s)(256MiB/3148msec); 0 zone resets
start: 0x00000000, len 0x020000, cap 0x020000, wptr 0x020000 reset:0 non-seq:0, zcond:14(fu) [type: 2(SEQ_WRITE_REQUIRED)]
start: 0x00002000, len 0x020000, cap 0x020000, wptr 0x020000 reset:0 non-seq:0, zcond:14(fu) [type: 2(SEQ_WRITE_REQUIRED)]
start: 0x00004000, len 0x020000, cap 0x020000, wptr 0x020000 reset:0 non-seq:0, zcond:14(fu) [type: 2(SEQ_WRITE_REQUIRED)]
start: 0x00006000, len 0x020000, cap 0x020000, wptr 0x020000 reset:0 non-seq:0, zcond:14(fu) [type: 2(SEQ_WRITE_REQUIRED)]
start: 0x00008000, len 0x020000, cap 0x020000, wptr 0x000000 reset:0 non-seq:0, zcond: 1(em) [type: 2(SEQ_WRITE_REQUIRED)]
```

256 MiB filled four 64 MiB zones, which are now Full. Read them back at random:

```
sudo fio --name=zr --filename=/dev/nvme0n1 --direct=1 --ioengine=psync \
--zonemode=zbd --rw=randread --bs=4k --size=256M --runtime=8 --time_based
```

The average completion latency is about 89 us: the built-in QLC page read time of 85 us plus overhead. A ZNS write goes to a zone write cache at 1 us per 4 KiB page. The cached pages are programmed when a write finds the cache full, or when another zone takes the cache over, and that write pays for the program (write cache (§4.8.8)). Reset the zones before the next step:

```
sudo blkzone reset /dev/nvme0n1
```

## 5.5.7 Mount zonefs

zonefs exposes each zone as a file. The cloud kernel ships the module in linux-modules-extra:

```
sudo apt-get install -y zonefs-tools linux-modules-extra-$(uname -r)
sudo mkzonefs -f /dev/nvme0n1
sudo modprobe zonefs
sudo mkdir -p /mnt/zonefs
sudo mount -t zonefs /dev/nvme0n1 /mnt/zonefs
ls /mnt/zonefs; ls /mnt/zonefs/seq | wc -l
```

```
seq
15
```

zonefs keeps its super block in zone 0, and the other 15 zones are the files seq/0 to seq/14. A file can only grow by appending, with direct I/O:

```
sudo dd if=/dev/urandom of=/mnt/zonefs/seq/0 bs=1M count=8 oflag=direct,append conv=notrunc
ls -l /mnt/zonefs/seq/0
sudo blkzone report -c 2 /dev/nvme0n1
```

```
8388608 bytes (8.4 MB, 8.0 MiB) copied, 0.109279 s, 76.8 MB/s
-rw-r----- 1 root root 8388608 Oct 1 16:28 /mnt/zonefs/seq/0
start: 0x00000000, len 0x020000, cap 0x020000, wptr 0x020000 reset:0 non-seq:0, zcond:14(fu) [type: 2(SEQ_WRITE_REQUIRED)]
start: 0x00002000, len 0x020000, cap 0x020000, wptr 0x004000 reset:0 non-seq:0, zcond: 2(oi) [type: 2(SEQ_WRITE_REQUIRED)]
```

The file size is the zone's write pointer (0x4000 sectors = 8 MiB), and the zone is implicitly open. Truncating the file to 0 resets the zone:

```
sudo truncate -s 0 /mnt/zonefs/seq/0
sudo blkzone report -c 2 /dev/nvme0n1 | tail -1
sudo umount /mnt/zonefs
```

```
start: 0x00002000, len 0x020000, cap 0x020000, wptr 0x000000 reset:0 non-seq:0, zcond: 1(em) [type: 2(SEQ_WRITE_REQUIRED)]
```

Do not run mkfs.ext4 on a zoned namespace: it needs random writes. A zoned file system such as f2fs also needs a randomly writable device for its metadata; tutorial 06 (§5.8) shows how to put a BlackBox namespace next to the zoned one. That combination was not run for this page.

## 5.5.8 What you learned

- A ZNS namespace is a host-managed zoned block device; the zone count follows from the `zns_` geometry, and the zone size from the geometry and the namespace size.
- Writes and appends open zones implicitly; open, close, finish and reset move them by hand; the device enforces the open and active limits.
- `fio` (`--zonemode=zbd`) and `zonefs` work on FEMU's ZNS as on real hardware.

## 5.5.9 Next

- ZNS mode (§4.7) covers ZRWA, zone capacity, conventional zones and the other options.
- The ZNS design page (§4.8) explains the zone layout over NAND and where time is charged.
- Tutorial 04 (§5.6) shows the other way to let the host place data: Flexible Data Placement.

## 5.6 Tutorial 04: Flexible Data Placement

You start a BlackBox SSD with Flexible Data Placement (FDP), read its FDP configuration and status with `nvme-cli`, write through a placement identifier and watch the reclaim unit fill, and then run a workload with data of two lifetimes, with and without placement, to see what placement does to the write amplification factor (WAF). It takes about fifteen minutes.

You need: tutorial 02 (§5.4) is helpful background, the variables from Before you start (§5.2.1), and about 5 GiB of free host memory. The guest image has `nvme-cli` 2.8, which has the `fdp` commands, and `fio` 3.36, which can place writes through its `io_uring_cmd` engine.

### 5.6.1 Background

FDP lets the host say which writes belong together. The device groups its NAND into reclaim units and offers a few reclaim unit handles. A write may carry a placement identifier that names a handle; all writes of one handle go into that handle's open reclaim unit. If data that dies together shares a reclaim unit, garbage collection (GC) finds it mostly invalid and copies little.

In FEMU, FDP is not a mode. It is a property of an NVMe subsystem (`femu-subsys`) that a BlackBox controller joins, and a reclaim unit is one line (superblock) of the BlackBox FTL. See the FDP design page (§4.6.4).

### 5.6.2 Start the guest (on the host)

```
./qemu-system-x86_64 -name femu-tut04,debug-threads=on \  
-enable-kvm -cpu host -smp 4 -m 4G \  
-device virtio-scsi-pci,id=scsi0 -device scsi-hd,drive=hd0 \  
-drive file=$OSIMGF,if=none,cache=none,format=qcow2,id=hd0 \  
-net user,hostfwd=tcp::$SSH_PORT-:22 -net nic,model=virtio \  
-device femu-subsys,id=fdp0,fdp=on,fdp.nruh=4 \  
-device femu,femu_mode=1,devsz_mb=384,nchs=2,luns_per_ch=4,blks_per_pl=64,subsys=fdp0 \  
-nographic
```

| Option                                  | Meaning                                                                                     |
|-----------------------------------------|---------------------------------------------------------------------------------------------|
| <code>femu-subsys,id=fdp0,fdp=on</code> | a subsystem with FDP in endurance group 1; it must come before the controller that names it |
| <code>fdp.nruh=4</code>                 | four reclaim unit handles, so placement identifiers 0 to 3                                  |

| Option                                | Meaning                                                             |
|---------------------------------------|---------------------------------------------------------------------|
| nchs=2, luns_per_ch=4, blks_per_pl=64 | 512 MiB of NAND in 64 lines of 8 MiB; each line is one reclaim unit |
| devsz_mb=384                          | a 384 MiB namespace, 75% of the NAND                                |
| subsys=fdp0                           | the controller joins the subsystem                                  |

fdp.nru keeps its default of 128 reclaim units, but FEMU uses at most one per line, so this device has 64.

### 5.6.3 Check FDP from the guest

In the guest (./run-guest-ssh.sh):

```

sudo nvme id-ctrl /dev/nvme0 | grep ctratt
sudo nvme fdp configs /dev/nvme0 -e 1

ctratt      : 0x80010
FDP Attributes: 0x80
Vendor Specific Size: 0
Number of Reclaim Groups: 1
Number of Reclaim Unit Handles: 4
Number of Namespaces Supported: 256
Reclaim Unit Nominal Size: 8388608
Estimated Reclaim Unit Time Limit: 0
Reclaim Unit Handle List:
  [0]: Persistently Isolated
  [1]: Persistently Isolated
  [2]: Persistently Isolated
  [3]: Persistently Isolated

```

Bit 19 (0x80000) of CTRATT says the controller supports FDP. The reclaim unit is 8 MiB, one line of this geometry. The namespace's placement identifiers and the room left in each one's reclaim unit:

```

sudo nvme fdp status /dev/nvme0n1

Placement Identifier 0; Reclaim Unit Handle Identifier 0
  Estimated Active Reclaim Unit Time Remaining (EARUTR): 0
  Reclaim Unit Available Media Writes (RUAMW): 16384
...

```

RUAMW counts logical blocks: 16384 x 512 bytes = 8 MiB, an empty reclaim unit. The output repeats for identifiers 1 to 3.

### 5.6.4 Write through a placement identifier

A write with directive type 2 (data placement) carries its placement identifier in the directive specific field (-s). Write 4 KiB three times through identifier 1:

```

head -c 4096 /dev/urandom > d.bin
for i in 1 2 3; do
  sudo nvme write /dev/nvme0n1 -s $((i * 8)) -c 7 -z 4096 -d d.bin -T 2 -S 1
done
sudo nvme fdp status /dev/nvme0n1 | grep RUAMW

write: Success
write: Success

```

```

write: Success
Reclaim Unit Available Media Writes (RUAMW): 16384
Reclaim Unit Available Media Writes (RUAMW): 16360
Reclaim Unit Available Media Writes (RUAMW): 16384
Reclaim Unit Available Media Writes (RUAMW): 16384

```

Only identifier 1's reclaim unit lost room: 3 writes of 8 blocks. A write without a directive, or with an identifier the namespace does not have, goes to handle 0; an identifier the namespace does not have also logs an FDP event (`sudo nvme fdp events /dev/nvme0 -e 1 -E` lists them). The endurance group statistics count bytes:

```

sudo nvme fdp stats /dev/nvme0 -e 1

Host Bytes with Metadata Written (HBMW): 12288
Media Bytes with Metadata Written (MBMW): 12288
Media Bytes Erased (MBE): 0

```

On a device that has seen only these three writes, HBMW is the 12 KiB the host wrote, and no reclaim unit has been erased yet. HBMW counts what the host wrote; MBMW also counts what GC copied, so MBMW / HBMW is the WAF over the endurance group.

### 5.6.5 Measure what placement does

The workload mixes two lifetimes: a hot 64 MiB region that takes random overwrites, and a cold 320 MiB region rewritten sequentially. Without placement both go into the same reclaim units, and GC copies cold pages whenever it collects a unit full of dead hot pages. With placement, hot writes use identifier 0 and cold writes identifier 1.

Save this in the guest as `fdp-run.sh`. It follows the method of tutorial 02 (§5.4.3): switch time off, precondition, then measure the counter differences of one run.

```

cat > fdp-run.sh <<'EOF'
# Usage: FDP=0 bash fdp-run.sh, or FDP=1 bash fdp-run.sh
c0() {
    sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b |
        od -An -t u8 -j 8 -N 24 -w24
}
sudo nvme admin-passthru /dev/nvme0 --opcode=0xef --cdw10=2 >/dev/null
sudo nvme admin-passthru /dev/nvme0 --opcode=0xef --cdw10=4 >/dev/null
P= H= C=
if [ "$FDP" = 1 ]; then P=--fdp=1 H=--fdp_pli=0 C=--fdp_pli=1; fi
J="--filename=/dev/ng0n1 --ioengine=io_uring_cmd --cmd_type=nvme --bs=4k --iodepth=8 $P"
sudo fio --name=prefill $J $C --rw=write --size=384M >/dev/null
c0 > before.txt
sudo fio $J --randrepeat=0 \
    --name=hot $H --rw=randwrite --offset=0 --size=64M --io_size=1G \
    --name=cold $C --rw=write --offset=64M --size=320M --io_size=1G >/dev/null
c0 > after.txt
paste before.txt after.txt | awk '{ h = $4 - $1; g = $5 - $2; n = $6 - $3;
    printf "host %d gc %d nand %d WAF %.3f\n", h, g, n, (n + g) / h }'
sudo nvme fdp stats /dev/nvme0 -e 1
EOF

```

fio's `io_uring_cmd` engine sends NVMe commands through the generic node `/dev/ng0n1`. With `--fdp=1` it reads the namespace's placement identifiers and uses the ones whose indexes `--fdp_pli` lists, one per job here.

Restart QEMU (step 1) before each run, so that both start from an empty device, and run once each way:

```
FDP=0 bash fdp-run.sh

host 524288 gc 3524608 nand 524288 WAF 7.723
Host Bytes with Metadata Written (HBMW): 2550136832
Media Bytes with Metadata Written (MBMW): 16986931200
Media Bytes Erased (MBE): 16567500800

FDP=1 bash fdp-run.sh

host 524288 gc 1977855 nand 524288 WAF 4.772
Host Bytes with Metadata Written (HBMW): 2550136832
Media Bytes with Metadata Written (MBMW): 10651430912
Media Bytes Erased (MBE): 10234101760
```

The host wrote the same 2.55 GB (384 MiB of prefill plus 2 GiB) in both runs. Placement cut the pages GC copied by 44% and the WAF of the measured run from 7.72 to 4.77. The FDP statistics count since the device started, prefill included: MBMW / HBMW falls from 6.66 to 4.18.

The WAF with placement is still well above 1 because the hot region is written at random: its own reclaim units still hold a mix of live and dead pages when GC takes them. Try `gc_strategy=1` (cost-benefit) on the controller, or more handles and a third lifetime.

### 5.6.6 What you learned

- FDP lives on `femu-subsys`; the controller joins it with `subsys=`.
- `nvme fdp configs, status and stats` show the handles, the room left in each reclaim unit, and host against media bytes.
- Placement only helps when the writes carry it: `nvme write -T 2 -S N`, or `fio` with `--ioengine=io_uring_cmd --fdp=1` on the generic node.
- Separating lifetimes by placement cuts GC copies.

### 5.6.7 Next

- Flexible Data Placement (§4.5) lists every option, refusal and the other log pages.
- The FDP design page (§4.6) explains how reclaim units map to lines and how GC chooses one.
- Tutorial 05 (§5.7) turns from page counts to latency.

## 5.7 Tutorial 05: latency tuning

You change the parts of FEMU's timing model one at a time: the flat NAND times, the cell type, the channel bus, the host link and the controller firmware cost. For each one you predict the effect, measure it with `fio`, and compare. It takes about twenty minutes.

You need: tutorial 01 (§5.3), the variables from Before you start (§5.2.1), and about 5 GiB of free host memory. Read performance tuning (§5.12) first if you want numbers that repeat to the microsecond: a poller or FTL thread that has to share a core adds delay the model did not ask for.

### 5.7.1 Background

FEMU copies the data of a command at once, computes when the command would have finished on the emulated device, and posts the completion no earlier than that (the timing model (§6.2.1)). A queue depth 1

latency is therefore the model's time plus a fixed overhead of the guest and of FEMU itself. Each step below changes one term of the model; NAND media and timing (§3.5) has the formulas.

## 5.7.2 The device and the measurement

Every run uses a small BlackBox SSD, 512 MiB of NAND in 2 channels of 4 LUNs, and adds one option to it:

```
-device femu,femu_mode=1,nchs=2,luns_per_ch=4,blks_per_pl=64,op_pcent=25
```

Start QEMU with the command line of tutorial 01, step 2 (§5.3.2) and this `-device` line, restarting QEMU for each configuration. In the guest, fill the first 256 MiB (a read of a page never written costs no NAND time), then run four measurements:

```
sudo fio --name=fill --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
--rw=write --bs=128k --iodepth=16 --size=256M
F="--filename=/dev/nvme0n1 --direct=1 --ioengine=libaio --size=256M --runtime=8 --time_based"
sudo fio --name=rd4k $F --rw=randread --bs=4k --iodepth=1
sudo fio --name=rd128k $F --rw=randread --bs=128k --iodepth=1
sudo fio --name=qd32 $F --rw=randread --bs=4k --iodepth=32
sudo fio --name=wr4k $F --rw=randwrite --bs=4k --iodepth=1
```

Read the median (c<sub>lat</sub> 50.00th percentile) of the QD1 tests and the IOPS of the QD32 test. Keep the write test last. Random overwrites move pages to new places, so a 128 KiB read after them finds its 32 pages spread unevenly over the LUNs and takes longer than the model of a freshly filled range (in a first run with the write test second, it took 289 us instead of 165 us). On this small drive, background garbage collection may also start in the last second of the write test; that moves its tail, not its median.

## 5.7.3 The baseline

With the default timing (40 us read, 200 us program, no channel bus):

| Test              | Result       |
|-------------------|--------------|
| 4 KiB read, QD1   | 44.3 us      |
| 4 KiB write, QD1  | 203.8 us     |
| 128 KiB read, QD1 | 164.9 us     |
| 4 KiB read, QD32  | 176,880 IOPS |

A 4 KiB read is `pg_rd_lat` plus about 4 us. A 128 KiB read is 32 pages. The FTL spreads consecutive pages over the channels and then the LUNs, so the 32 pages sit on all 8 LUNs, 4 each. The LUNs work in parallel and each reads its 4 pages one after another: 4 x 40 us = 160 us. Copying the 128 KiB takes about 30 us (measured with NAND times set to 0), but FEMU copies at once and holds only the completion, so the copy overlaps the modelled time instead of adding to it. At queue depth 32, 8 LUNs reading in parallel allow at most 8 / 40 us = 200,000 reads per second; FEMU delivers 88% of that.

## 5.7.4 Flat NAND times

```
-device femu,femu_mode=1,nchs=2,luns_per_ch=4,blks_per_pl=64,op_pcent=25,pg_rd_lat=80000,pg_wr_lat=400000
```

Prediction: reads and writes take 40 us and 200 us longer, and QD32 throughput halves.



| Test             | Baseline     | 80 us / 400 us |
|------------------|--------------|----------------|
| 4 KiB read, QD1  | 44.3 us      | 84.5 us        |
| 4 KiB write, QD1 | 203.8 us     | 423.9 us       |
| 4 KiB read, QD32 | 176,880 IOPS | 89,317 IOPS    |

### 5.7.5 Cell type

nand\_cell\_type replaces the flat times with built-in tables that give each page type in a wordline its own time (cell types (§3.5.6)). TLC (3) has lower, center and upper pages read in 56.5, 77.5 and 106 us:

```
| -device femu,femu_mode=1,nchs=2,luns_per_ch=4,blks_per_pl=64,op_pcent=25,nand_cell_type=3
```

| Test                             | Baseline     | TLC         |
|----------------------------------|--------------|-------------|
| 4 KiB read, QD1                  | 44.3 us      | 81.4 us     |
| 4 KiB read, QD1, 99th percentile | 52.0 us      | 114.2 us    |
| 4 KiB write, QD1                 | 203.8 us     | 2310.1 us   |
| 4 KiB read, QD32                 | 176,880 IOPS | 89,449 IOPS |

The pages of a block are split about equally over the lower, center and upper types, so the read median is the center page's 77.5 us plus overhead, and the 99th percentile is near the upper page. TLC programs take milliseconds. pg\_rd\_lat, pg\_wr\_lat and blk\_er\_lat have no effect while nand\_cell\_type is set.

### 5.7.6 The channel bus

By default, moving a page between controller and NAND takes no time. Any non-zero bus phase turns the channel bus on; pg\_xfer\_lat is the data transfer per page, and transfers on one channel run one at a time:

```
| -device femu,femu_mode=1,nchs=2,luns_per_ch=4,blks_per_pl=64,op_pcent=25,pg_xfer_lat=20000
```

| Test              | Baseline     | 20 us transfer |
|-------------------|--------------|----------------|
| 4 KiB read, QD1   | 44.3 us      | 63.7 us        |
| 128 KiB read, QD1 | 164.9 us     | 383.0 us       |
| 4 KiB read, QD32  | 176,880 IOPS | 98,056 IOPS    |

A single page read gains exactly one transfer. A 128 KiB read moves 16 pages over each channel, one at a time: 320 us of transfers, which now dominate the read (with NAND times set to 0 it still takes 342 us). Under load the two channels become the limit: each moves one page per 20 us, 100,000 pages per second for both, which is where QD32 lands.

### 5.7.7 The host link

pcie\_bandwidth\_mbps charges each Read and Write its size divided by the link bandwidth, on one queue per direction (host link (§3.5.13)):

```
| -device femu,femu_mode=1,nchs=2,luns_per_ch=4,blks_per_pl=64,op_pcent=25,pcie_bandwidth_mbps=1000
```

| Test            | Baseline | 1000 MB/s link |
|-----------------|----------|----------------|
| 4 KiB read, QD1 | 44.3 us  | 47.9 us        |

| Test              | Baseline | 1000 MB/s link |
|-------------------|----------|----------------|
| 128 KiB read, QD1 | 164.9 us | 305.2 us       |

128 KiB at 1000 MB/s is 131 us, and the 128 KiB read grew by 140 us. A 4 KiB read pays 4 us. `pcie_prop_delay_ns` adds a fixed delay on top.

### 5.7.8 Controller firmware

`fw_cpu_ns` charges a fixed cost per Read, Write and Zone Append on one modelled controller core, so commands queue behind each other there:

```
| -device femu,femu_mode=1,nchs=2,luns_per_ch=4,blks_per_pl=64,op_pcent=25,fw_cpu_ns=20000
```

| Test             | Baseline     | 20 us firmware |
|------------------|--------------|----------------|
| 4 KiB read, QD1  | 44.3 us      | 63.7 us        |
| 4 KiB write, QD1 | 203.8 us     | 230.4 us       |
| 4 KiB read, QD32 | 176,880 IOPS | 49,534 IOPS    |

At queue depth 1 the cost simply adds. Under load one core finishes one command per 20 us, 50,000 per second, and that becomes the limit no matter how many LUNs the NAND has.

### 5.7.9 What you learned

- Queue depth 1 latency is the modelled time plus a few microseconds; each timing option moves it by what you configured.
- Throughput is set by whichever resource saturates first: LUNs, channels with the bus on, or the firmware core.
- `nand_cell_type` gives a spread of read times rather than one value.

The numbers above come from one run per configuration on a 20-core host where other work was running, so expect yours to differ by a few microseconds. The relations between them should hold.

### 5.7.10 Next

- NAND media and timing (§3.5) has the rules these measurements follow, including program suspend and ECC read time.
- Measuring (§6.1.5) and performance tuning (§5.12) explain how to pin FEMU's threads and get repeatable numbers.
- Tutorial 06 (§5.8) puts several modes on one controller.

## 5.8 Tutorial 06: several namespaces, several modes

You start one FEMU controller with four namespaces, each in a different mode: a BlackBox SSD, a zoned namespace, a NoSSD namespace with no media timing, and a key-value namespace. You find each one in the guest, use it, and see which counters belong to which. It takes about ten minutes.

You need: the variables from Before you start (§5.2.1) and about 6 GiB of free host memory. Tutorials 01 (§5.3), 03 (§5.5) and 07 (§5.9) explain each mode on its own.

### 5.8.1 Background

A controller's `femu_mode` sets its default mode. `namespaces` sets how many namespaces exist from boot, `namespace_sizes` how big each is (the default splits `devsz_mb` evenly), and `namespace_modes` the mode of each. The namespaces are packed one after another in the controller's memory. Each BlackBox namespace gets its own FTL built from the whole NAND geometry; each ZNS namespace builds its own zones; each KV namespace has its own key space (namespaces design page (§4.16.5)).

### 5.8.2 Start the guest (on the host)

```
./qemu-system-x86_64 -name femu-tut06,debug-threads=on \
    -enable-kvm -cpu host -smp 4 -m 4G \
    -device virtio-scsi-pci,id=scsi0 -device scsi-hd,drive=hd0 \
    -drive file=$OSIMGF,if=none,cache=none,format=qcow2,id=hd0 \
    -net user,hostfwd=tcp:::$SSH_PORT-:22 -net nic,model=virtio \
    -device femu,femu_mode=1,devsz_mb=2048,namespaces=4,namespace_modes=bbssd,,znssd,,nosssd,,kvssd \
    -nographic
```

A comma inside a property value is written twice on the QEMU command line, so `namespace_modes=bbssd,,znssd,,nosssd,,kvssd` is the list `bbssd,znssd,nosssd,kvssd`. A single comma would end the property, and QEMU would read `znssd` as the name of the next one. The 2 GiB backend is split into four 512 MiB namespaces.

### 5.8.3 Find the namespaces

In the guest (`./run-guest-ssh.sh`):

```
sudo nvme list
sudo nvme list-ns /dev/nvme0
```

| Node         | Generic    | SN    | Model                        | Namespace | Usage                 | Format      | FW Rev |
|--------------|------------|-------|------------------------------|-----------|-----------------------|-------------|--------|
| /dev/nvme0n1 | /dev/ng0n1 | vSSD0 | FEMU BlackBox-SSD Controller | 0x1       | 536.87 MB / 536.87 MB | 512 B + 0 B | 1.0    |
| /dev/nvme0n2 | /dev/ng0n2 | vSSD0 | FEMU BlackBox-SSD Controller | 0x2       | 536.87 MB / 536.87 MB | 512 B + 0 B | 1.0    |
| /dev/nvme0n3 | /dev/ng0n3 | vSSD0 | FEMU BlackBox-SSD Controller | 0x3       | 536.87 MB / 536.87 MB | 512 B + 0 B | 1.0    |
| /dev/nvme0n4 | /dev/ng0n4 | vSSD0 | FEMU BlackBox-SSD Controller | 0x4       | 536.87 MB / 0.00 B    | 512 B + 0 B | 1.0    |
| [ 0]:0x1     |            |       |                              |           |                       |             |        |
| [ 1]:0x2     |            |       |                              |           |                       |             |        |
| [ 2]:0x3     |            |       |                              |           |                       |             |        |
| [ 3]:0x4     |            |       |                              |           |                       |             |        |

Namespaces 1 to 3 have block devices. Namespace 4, the KV one, has only the generic character device `/dev/ng0n4`: Linux has no block driver for the key-value command set.

The model and serial number are the controller's, and they come from its own `femu_mode` (BlackBox here), not from the namespaces' modes. Tell the namespaces apart by what they report:

```
for n in 1 2 3; do
    echo nvme0n$n $(cat /sys/block/nvme0n$n/queue/zoned) $(cat /sys/block/nvme0n$n/size)
done

nvme0n1 none 1048576
nvme0n2 host-managed 1048576
nvme0n3 none 1048576
```

Namespace 2 is the zoned one. Each is 1048576 sectors of 512 bytes, 512 MiB.

### 5.8.4 Use each one

The BlackBox namespace pays NAND program time, the NoSSD namespace pays none:

```
sudo fio --name=n1 --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
--rw=randwrite --bs=4k --iodepth=1 --size=64M --runtime=5 --time_based
sudo fio --name=n3 --filename=/dev/nvme0n3 --direct=1 --ioengine=libaio \
--rw=randwrite --bs=4k --iodepth=1 --size=64M --runtime=5 --time_based
```

The average completion latency is about 206 us on nvme0n1 (the 200 us program) and about 5.5 us on nvme0n3.

The zoned namespace builds its zones from its own size: 16 zones of 32 MiB, half the size of the 1 GiB namespace in tutorial 03 (§5.5):

```
sudo blkzone report -c 2 /dev/nvme0n2

start: 0x00000000, len 0x010000, cap 0x010000, wptr 0x000000 reset:0 non-seq:0, zcond: 1(em) [type: 2(SEQ_WRITE_REQUIRED)]
start: 0x000010000, len 0x010000, cap 0x010000, wptr 0x000000 reset:0 non-seq:0, zcond: 1(em) [type: 2(SEQ_WRITE_REQUIRED)]
```

Store and retrieve a value in the KV namespace. With several namespaces, Linux refuses I/O passthrough on the controller node /dev/nvme0, so use the generic node, and name the namespace with -n 4:

```
head -c 64 /dev/urandom > value.bin
sudo nvme io-passthru /dev/ng0n4 -O 0x01 -n 4 --cdw10=64 --cdw11=4 \
--cdw2=0x42424242 -l 64 -w -i value.bin
sudo nvme io-passthru /dev/ng0n4 -O 0x02 -n 4 --cdw10=64 --cdw11=4 \
--cdw2=0x42424242 -l 64 -r -b > out.bin
cmp value.bin out.bin && echo match

IO Command Write is Success and result: 0x00000040
IO Command Read is Success and result: 0x00000040
match
```

The same Retrieve sent to /dev/nvme0 fails with passthru: Invalid argument. Tutorial 07 (§5.9) explains the command fields.

### 5.8.5 Whose counters

Log page C0h is per controller. It sums the BlackBox, CSD and KV namespaces, and ignores NoSSD and ZNS (counters (§B.1)). After the two fio runs above, before the KV store:

```
sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b | od -An -t u8 -j 8 -N 24 -w24

24046          0          24046
```

24046 pages are the BlackBox run alone (5 seconds at about 4,800 writes per second); the 2.7 GiB written to the NoSSD namespace in the same time is not counted. The SMART log (sudo nvme smart-log /dev/nvme0) counts the successful Read, Write and Zone Append commands of every namespace, NoSSD and ZNS included; KV Store and Retrieve share those opcodes and are counted too.

### 5.8.6 What you learned

- namespaces, namespace\_sizes and namespace\_modes build several namespaces on one controller; commas inside a value are doubled.
- Each mode keeps its own behaviour: zone geometry, NAND timing, key space.
- A KV namespace has only a generic node, and on a controller with several namespaces passthrough needs that node.

- C0h counts only BlackBox, CSD and KV namespaces; NoSSD and ZNS do not add to it.

### 5.8.7 Next

- Several namespaces and devices (§4.15) lists the limits and refusals, and how to start several controllers.
- Namespace management (§4.17) creates and deletes namespaces from the guest while it runs.
- Subsystems, controllers and namespaces (§4.16) explains how the backend is partitioned.

## 5.9 Tutorial 07: key-value SSD

You start an SSD with the NVMe Key Value command set, then store, retrieve, check, list and delete values with `nvme-cli`'s passthrough command, and read the status codes the device returns. It takes about ten minutes.

You need: the variables from Before you start (§5.2.1), about 5 GiB of free host memory, and a guest kernel of Linux 6.0 or newer. Older kernels skip a namespace whose command set they do not know, so the commands below fail. The Ubuntu 24.04 image from `make-guest-image.sh` runs Linux 6.8.

### 5.9.1 Background

A KV namespace stores values under keys instead of blocks at addresses. Linux has no driver for the key-value command set, so the namespace gets no block device, only the generic node `/dev/ngXnY`, and you send each command as a passthrough command. The fields of a command:

| Field           | Holds                                                                      |
|-----------------|----------------------------------------------------------------------------|
| opcode          | 0x01 Store, 0x02 Retrieve, 0x06 List, 0x10 Delete, 0x14 Exist              |
| CDW2, CDW3      | bytes 0 to 7 of the key, little-endian                                     |
| CDW14, CDW15    | bytes 8 to 15 of the key                                                   |
| CDW11 bits 7:0  | key length, 1 to 16 bytes                                                  |
| CDW11 bits 15:8 | Store only: bit 8 stores only if the key exists, bit 9 only if it does not |
| CDW10           | the value size for Store, the buffer size for Retrieve and List            |
| NSID            | the namespace; it must be given                                            |

Values are up to 2 MiB and live on emulated NAND with the BlackBox geometry and timing (the KV design page (§4.12.5)).

### 5.9.2 Start the guest (on the host)

```
./qemu-system-x86_64 -name femu-tut07,debug-threads=on \
  -enable-kvm -cpu host -smp 4 -m 4G \
  -device virtio-scsi-pci,id=scsi0 -device scsi-hd,drive=hd0 \
  -drive file=$OSIMGF,if=none,cache=none,format=qcow2,id=hd0 \
  -net user,hostfwd=tcp:::$SSH_PORT-:22 -net nic,model=virtio \
  -device femu,femu_mode=5,devsz_mb=1024 \
  -nographic
```

`femu_mode=5` is KV; the namespace is 1 GiB.

### 5.9.3 Find the namespace

In the guest (`./run-guest-ssh.sh`):

```

| sudo nvme list
| ls /dev/ng0n1 /dev/nvme0n1
nvme0n1          /dev/ng0n1          vKVSSD0          FEMU KV-SSD Controller          0x1          1.07 GB / 0.00 B          512 B + 0 B          1.0
/dev/ng0n1
ls: cannot access '/dev/nvme0n1': No such file or directory

```

There is a generic node and no block device. On a controller with one namespace you may also send passthrough commands to the controller node `/dev/nvme0`, which this tutorial does. With several namespaces use `/dev/ngXnY` (tutorial 06 (§5.8.4)).

### 5.9.4 Store and retrieve

Store a 64-byte value under the 4-byte key BBBB (0x42424242), then read it back:

```

| head -c 64 /dev/urandom > value.bin
| sudo nvme io-passthru /dev/nvme0 -O 0x01 -n 1 --cdw10=64 --cdw11=4 \
|   --cdw2=0x42424242 -l 64 -w -i value.bin
| sudo nvme io-passthru /dev/nvme0 -O 0x02 -n 1 --cdw10=64 --cdw11=4 \
|   --cdw2=0x42424242 -l 64 -r -b > out.bin
| cmp value.bin out.bin && echo match
| IO Command Write is Success and result: 0x00000040
| IO Command Read is Success and result: 0x00000040
| match

```

`nvme-cli` names opcodes 0x01 and 0x02 after the NVM command set's Write and Read; the device treats them as Store and Retrieve. The result (completion Dword 0) of a Retrieve is the full size of the value, 0x40. `-n 1` is required: `nvme-cli` sends namespace 0 by default, which Linux refuses with Invalid argument.

A buffer smaller than the value gets the first bytes, and the result still reports the full size, so the host can retry with a larger buffer:

```

| sudo nvme io-passthru /dev/nvme0 -O 0x02 -n 1 --cdw10=16 --cdw11=4 \
|   --cdw2=0x42424242 -l 16 -r -b | od -An -tx1

```

The command prints result: 0x00000040 and 16 bytes of the value.

### 5.9.5 Conditional stores, Exist and List

Check that the key exists, then try to store it again with bit 9 set ("only if the key does not exist"), which CDW11 0x204 encodes with the key length 4:

```

| sudo nvme io-passthru /dev/nvme0 -O 0x14 -n 1 --cdw11=4 --cdw2=0x42424242
| printf hello > v2.bin
| sudo nvme io-passthru /dev/nvme0 -O 0x01 -n 1 --cdw10=5 --cdw11=0x204 \
|   --cdw2=0x42424242 -l 5 -w -i v2.bin
| IO Command Vendor Specific is Success and result: 0x00000000
| NVMe status: unrecognized(0x4089)

```

`nvme-cli` does not know the key-value status codes. 0x4089 is status 0x89, Key Exists, with the Do Not Retry bit (0x4000). Store a second key, CCCC, and list the keys. List walks the device's key index from its first slot, so keys come back in index order, not sorted:

```

| sudo nvme io-passthru /dev/nvme0 -O 0x01 -n 1 --cdw10=5 --cdw11=4 \
|   --cdw2=0x43434343 -l 5 -w -i v2.bin

```

```

| sudo nvme io-passthru /dev/nvme0 -O 0x06 -n 1 --cdw10=4096 --cdw11=0 \
|     -l 4096 -r -b | od -An -tx1 -N 24
|
| 02 00 00 00 04 00 43 43 43 43 00 00 04 00 42 42
| 42 42 00 00 00 00 00 00

```

The buffer starts with a 4-byte key count (2), then one entry per key: a 2-byte key length (4), the key, and padding to a multiple of 4 bytes.

## 5.9.6 Delete

```

| sudo nvme io-passthru /dev/nvme0 -O 0x10 -n 1 --cdw11=4 --cdw2=0x42424242
| sudo nvme io-passthru /dev/nvme0 -O 0x14 -n 1 --cdw11=4 --cdw2=0x42424242
|
| IO Command Vendor Specific is Success and result: 0x00000000
| NVMe status: unrecognized(0x4087)

```

0x87 is Key Does Not Exist. Deleting a key that does not exist succeeds unless the host sets the EDNEK bit of the Key Value Configuration feature (20h).

| Status | Meaning                                                                 |
|--------|-------------------------------------------------------------------------|
| 0x85   | Invalid Value Size: the value is larger than 2 MiB                      |
| 0x86   | Invalid Key Size: a key length of 0 on Store, Retrieve, Delete or Exist |
| 0x87   | Key Does Not Exist                                                      |
| 0x89   | Key Exists, from a Store with bit 9 set                                 |

## 5.9.7 Counters

The KV namespace adds to log page C0h like a BlackBox one:

```

| sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b | od -An -t u8 -j 8 -N 24 -w24
|
|           2           0           2

```

Two Stores of small values, two pages programmed. The key index lives in device DRAM, but each command is charged a base cost of one NAND page read time, which overlaps the media time of its value (KV timing (§4.12.7)).

## 5.9.8 What you learned

- A KV namespace has a generic node and no block device; Linux 6.0 or newer is needed to see it.
- `nvme io-passthru` with the opcode, the key in CDW2 and CDW3 (and CDW14 and CDW15), the key length in CDW11 and an explicit namespace drives every command.
- `nvme-cli` prints KV status codes as unrecognized; read the low byte.

## 5.9.9 Next

- KV mode (§4.11) has `kv-probe.c`, a C program that runs the whole command cycle and checks the results.
- The KV design page (§4.12) explains the key index and the value store.

## 5.10 Tutorial 08: a CXL SSD as memory

You start a guest with `femu-cxl-ssd`, a CXL Type-3 memory device whose capacity is backed by emulated NAND behind a DRAM page cache. You find the device in the guest, turn it into a memory region, use the region as a DAX device and as a NUMA node, and read the device's counters from the host over QMP. It takes about twenty minutes.

You need: the variables from Before you start (§5.2.1), about 3 GiB of free host memory, and `socat` on the host. The guest needs the CXL tools (`cxl`, `daxctl`) and a kernel that can commit a CXL region inside a virtual machine; step 4 explains what that takes.

### 5.10.1 Background

The guest sees ordinary CXL memory. Behind it, FEMU keeps a cache of 4 KiB pages in front of the BlackBox FTL and NAND model. A cache hit costs no media time, a miss costs a NAND page read (nothing for a page never written), and evicting a dirty page costs a NAND program. With the default `der=off`, every guest load or store to the device exits to QEMU, and the vCPU waits out the media time before the access completes (CXL SSD design (§4.19.5)).

The device needs a CXL topology: a CXL host bridge, a root port below it, the device below the root port, and a fixed memory window on the machine. `run-cxlssd.sh` builds all of it.

### 5.10.2 Start the guest (on the host)

`run-cxlssd.sh` adds no guest disk and no network; you pass them after the script name. Put them on the main PCIe bus with `bus=pcie.0`: without it, QEMU may place them on the CXL host bridge, which accepts only bridges, and stops with `Only PCI/PCIe bridges can be plugged into pxb-cxl`. From `build-femu/`:

```
RAM=2G ../femu-scripts/run-cxlssd.sh \
-object memory-backend-ram,id=ram0,size=2G \
-numa node,nodeid=0,cpus=0-3,memdev=ram0 \
-drive file=$OSIMGF,if=none,id=hd0 \
-device virtio-blk-pci,drive=hd0,bus=pcie.0 \
-netdev user,id=net0,hostfwd=tcp::SSH_PORT-:22 \
-device virtio-net-pci,netdev=net0,bus=pcie.0 \
-qmp unix:$PWD/qmp-cxl.sock,server=on,wait=off
```

| Part                                              | Why                                                                                                                                                           |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RAM=2G                                            | the guest's RAM; the script's default is 4G                                                                                                                   |
| -object memory-backend-ram,... and -numa node,... | describe the guest RAM as NUMA node 0, so QEMU builds an ACPI SRAT and the CXL memory can become a node of its own; the backend size must equal the guest RAM |
| -drive,-device virtio-blk-pci                     | the guest disk                                                                                                                                                |
| -netdev,-device virtio-net-pci                    | SSH to the guest on SSH_PORT                                                                                                                                  |
| -qmp unix:...                                     | the socket you read the counters through in step 6                                                                                                            |

The script's defaults give a 256 MiB device with a 3072-page (12 MiB) direct-mapped cache, 8 channels of 8 LUNs, and `lsa-control=on`, which lets the guest send experiment commands that write files on the host (`run-cxlssd.sh` (§C.3) lists every variable). `DRY_RUN=1` in front prints the QEMU command instead of running it.



### 5.10.3 Install the guest tools

The image from `make-guest-image.sh --cxl` already has them. Otherwise, in the guest (`./run-guest-ssh.sh`):

```
sudo apt-get install -y cxl daxctl ndctl linux-modules-extra-$(uname -r)
sudo modprobe cxl_acpi
sudo modprobe cxl_pci
```

### 5.10.4 Find the device

```
sudo cxl list -M
sudo cxl list -D -d root

[
  {
    "memdev": "mem0",
    "ram_size": 268435456,
    "serial": 0,
    "host": "0000:35:00.0"
  }
]
[
  {
    "decoder": "decoder0.0",
    "resource": 4563402752,
    "size": 268435456,
    "interleave_ways": 1,
    "max_available_extent": 268435456,
    "pmem_capable": true,
    "volatile_capable": true,
    "acclmem_capable": true,
    "nr_targets": 1
  }
]
```

`mem0` is the `femu-cxl-ssd` with 256 MiB of volatile capacity, and `decoder0.0` is the root decoder of the fixed memory window.

### 5.10.5 Create a region

```
sudo cxl create-region -d decoder0.0 -t ram -w 1 -m mem0
```

With the stock Ubuntu 24.04 kernel this fails:

```
cxl region: create_region: region0: failed to commit decode: No such device or address
cxl region: cmd_create_region: created 0 regions
```

and `dmesg` says `cxl region0: Failed to synchronize CPU cache state. Linux invalidates the CPU caches over the new range before it commits a region, and a virtual machine cannot do that. A guest kernel built with CONFIG_CXL_REGION_INVALIDATION_TEST=y skips the step. Check yours with:`

```
grep -E 'CONFIG_(CXL_REGION|CXL_REGION_INVALIDATION_TEST|DEV_DAX_CXL|DEV_DAX_KMEM)=' \
/boot/config-$(uname -r)
```

CXL SSD guest requirements (§4.18.3) lists every option the guest kernel needs. The rest of this tutorial assumes such a kernel. Its output is illustrative: it was not captured for this page. FEMU's own guest runs used Linux 6.12 with these options and wrote and verified data on the region as a NUMA node.

On such a kernel the region is created and appears as a DAX device:

```
sudo daxctl list

[
  {
    "chardev": "dax0.0",
    "size": 268435456,
    "target_node": 1,
    "align": 2097152,
    "mode": "devdax"
  }
]
```

If mode is system-ram, the kmem driver claimed the region on its own; `sudo daxctl reconfigure-device --mode=devdax --force dax0.0` gives it back.

### 5.10.6 Use the region

#### As a DAX device

A program maps `/dev/dax0.0` and loads and stores directly. Offsets and lengths must be multiples of align (2 MiB). This writes and reads back one page:

```
sudo python3 -c '
import mmap, os
fd = os.open("/dev/dax0.0", os.O_RDWR)
m = mmap.mmap(fd, 2 << 20)
m[0:4096] = b"femu" * 1024
print(m[0:8])
'
```

Every access traps to QEMU (der=off), so each costs microseconds even on a cache hit; keep test working sets small.

#### As a NUMA node

```
sudo daxctl reconfigure-device --mode=system-ram --force dax0.0
numactl -H
numactl --membind=1 dd if=/dev/zero of=/dev/null bs=1M count=64
```

`numactl -H` shows a node 1 with memory and no CPUs: the CXL SSD. Memory a program allocates with `--membind=1` lives on the emulated NAND behind the cache. Expect such a program to run orders of magnitude slower than on node 0 with `der=off`; `der=memslot` maps cached pages into the guest so hits run at DRAM speed (direct mapping (§4.18.7)).

### 5.10.7 Read the counters (on the host)

The counters are QOM properties of the device, whose id is `cxlssd`. QEMU's `scripts/qmp/qom-get` reads one; run it from the FEMU source tree:

```
export QMP_SOCKET=build-femu/qmp-cxl.sock
for p in cache-pages cache-ways cache-hits cache-misses media-reads media-writes media-time-ns
    media-full der-active; do
    echo "$p: $(python3 scripts/qmp/qom-get /machine/peripheral/cxlssd.$p)"
done
```

On the guest of this tutorial, which could not create a region, every access counter is still 0:

```
cache-pages: 3072
cache-ways: 1
cache-hits: 0
cache-misses: 0
media-reads: 0
media-writes: 0
media-time-ns: 0
media-full: 0
der-active: False
```

After a workload on the region, cache-misses, media-reads and media-time-ns grow, and cache-hits counts the accesses the cache served. media-full must stay 0: a non-zero value means some NAND programs found no free page and were not timed, and the run is not valid.

To change a setting or start from a cold cache, send raw QMP, for example with socat. flush-cache writes dirty pages back and drops every page that is not pinned; cache-ways can change while the guest runs:

```
printf '%s\n' '{"execute": "qmp_capabilities"}' \
    '{"execute": "qom-set", "arguments": {"path": "/machine/peripheral/cxlssd", "property": "flush-cache",
    "value": true}}' \
    \
    '{"execute": "qom-set", "arguments": {"path": "/machine/peripheral/cxlssd", "property": "cache-ways",
    "value": 4}}' \
    \
    '{"execute": "qom-get", "arguments": {"path": "/machine/peripheral/cxlssd", "property": "cache-ways"}}'
|
socat -t 2 - UNIX-CONNECT:build-femu/qmp-cxl.sock

{"QMP": {"version": {"qemu": {"micro": 0, "minor": 1, "major": 10}, "package": ""}, "capabilities": ["oob"]}}
{"return": {}}
{"return": {}}
{"return": {}}
{"return": 4}
```

stats-reset (also a qom-set to true) clears the cache event counters but not the media counters, so measure those as the difference of two reads (runtime properties (§A.2.2)).

### 5.10.8 What you learned

- run-cxlssd.sh builds the CXL topology; guest devices you add go on bus=pcie.0.
- The guest sees mem0 and a root decoder; committing a region inside a VM needs CONFIG\_CXL\_REGION\_INVALIDATION\_TEST.
- The region works as a DAX device or, with an SRAT from -numa, as a CPU-less NUMA node.
- The counters are QOM properties read over QMP; media-full must stay 0.

### 5.10.9 Next

- CXL SSD (§4.18) covers the cache options, prefetch, direct mapping and the control channel.

- The CXL SSD design page (§4.19) explains the cache, the write-back path and direct mapping.
- The caching API (§4.20) lets the guest pin and drop pages itself.

## 5.11 Tutorial 09: configuration files

You describe a device in a short INI-style file instead of a long `-device femu,...` line, expand it with `ssd-config.sh`, let the script catch a misspelled property, and boot the result. It takes about ten minutes.

You need: a built FEMU and the variables from Before you start (§5.2.1). Steps 1 to 4 run on the host and start no guest.

### 5.11.1 Background

`hw/femu/scripts/ssd-config.sh` turns a file of `key = value` lines into QEMU `-device` arguments. Keys are the femu device's property names and mean exactly what the property reference (§A.1) says. Two additions:

- `mode = bbssd` (or `znssd`, `nosssd`, `ocssd`, `csd`, `kvssd`) stands for `femu_mode`.
- Keys under a `[subsys]` section become a separate `-device femu-subsys,...`, which the femu device is joined to. FDP lives there.

Other section headers are labels for the reader. `#` and `;` start a comment, and a key with an empty value is ignored, so a file can list a property and leave it at its default. The script checks every key against `-device femu,help` (`femu-subsys,help` for `[subsys]` keys) of the binary named by `FEMU_BIN`, or of one it finds in `build-femu/`, `build/` or `build-official/` of the source tree.

### 5.11.2 Write a configuration

Save the drive of tutorial 02, step 6 (§5.4.7) as `gc-study.conf` in `build-femu/`:

```
# Tutorial 02's skewed-workload drive: 2 GiB of NAND, 25% spare,
# GC from 95% of lines in use, cost-benefit victims, hot and cold separated.

[device]
mode      = bbssd
op_pcent  = 25

[geometry]
nchs      = 4
luns_per_ch = 4
blks_per_pl = 128

[gc]
gc_thres_pcent = 95
gc_policy      = cost-benefit
hot_cold_sep   = on
```

### 5.11.3 Expand it

From `build-femu/`, on the host:

```
export FEMU_BIN=$PWD/qemu-system-x86_64
../femu-scripts/ssd-config.sh gc-study.conf
```

```
| -device femu,id=nvme0,op_pcent=25,nchs=4,luns_per_ch=4,blks_per_pl=128,gc_thres_pcent=95,gc_policy=cost-benefit,hot_cold_sep=on,femu_mode=1
```

That is the line you would otherwise type, and the documentation checks start it under QEMU.

The script adds `id=nvme0` unless the file sets an id. `--device-only` prints the arguments without the `-device` words, for a launcher that adds them itself. Set `FEMU_BIN`: run through the `../femu-scripts` link, the script cannot find the binary on its own and then skips the key check with a warning.

#### 5.11.4 Let it catch mistakes

Misspell a key:

```
| printf '[device]\nmode = bbssd\ngc_policy = greedy\n' > bad.conf
| ../femu-scripts/ssd-config.sh bad.conf; echo "exit $?"

ssd-config: unknown property 'gc_policy' -- not one FEMU accepts
ssd-config: config rejected; see the warnings above
exit 1
```

Put a subsystem property on the controller:

```
| printf '[device]\nmode = bbssd\nfdp = on\n' > bad.conf
| ../femu-scripts/ssd-config.sh bad.conf

ssd-config: 'fdp' is a subsystem property; move it under [subsys]
ssd-config: config rejected; see the warnings above
```

`--check` validates a file without printing the arguments: it prints the same warnings for a bad file, nothing for a good one, and its exit status says which. Without a binary to check against, it only warns and exits 0. Besides names, the script rejects an unknown mode, mode together with `femu_mode`, and lines that are not `key = value`. Values, ranges and combinations are checked when QEMU creates the device, which stops with a message; the "Limits and refusals" section of each mode page lists them, for example BlackBox (§4.4.5).

#### 5.11.5 Lists and subsystems

The shipped files in `hw/femu/scripts/configs/` show the two conveniences. A list value is written with plain commas, and the script doubles them for QEMU:

```
| ../femu-scripts/ssd-config.sh ../femu-scripts/configs/heterogeneous.conf

-device femu,id=nvme0,devsz_mb=6144,namespaces=3,namespace_modes=bbssd,,znssd,,nosssd,secsz=512,secs_per_pg=8,
pgs_per_blk=256,blks_per_pl=256,pls_per_lun=1,luns_per_ch=8,nchs=8,femu_mode=1
```

A `[subsys]` section becomes its own device, placed first, with the controller joined to it:

```
| ../femu-scripts/ssd-config.sh ../femu-scripts/configs/fdp.conf

-device femu-subsys,id=femu-subsys-0,nqn=subsys0,fdp=on,fdp.nruh=4,fdp.nrg=1,fdp.nru=256 -device
femu,id=nvme0,devsz_mb=4096,namespaces=1,secsz=512,secs_per_pg=8,pgs_per_blk=256,blks_per_pl=256,pls_
per_lun=1,luns_per_ch=8,nchs=8,femu_mode=1,subsys=femu-subsys-0
```

Every mode has a file to start from: `bbssd.conf`, `znssd.conf`, `nosssd.conf`, `ocssd.conf`, `csd.conf` and `kvssd.conf`. The scripts reference (§C.4) lists them all.

#### 5.11.6 Boot it

The launchers take no arguments, so put the expansion on your own QEMU command line, the one from tutorial 01, step 2 (§5.3.2), in place of its `-device femu` option. In bash:

```

QEMU_ARGS=$(../femu-scripts/ssd-config.sh gc-study.conf)
./qemu-system-x86_64 -name femu-tut09,debug-threads=on \
    -enable-kvm -cpu host -smp 4 -m 4G \
    -device virtio-scsi-pci,id=scsi0 -device scsi-hd,drive=hd0 \
    -drive file=$OSIMGF,if=none,cache=none,format=qcow2,id=hd0 \
    -net user,hostfwd=tcp:::$SSH_PORT-:22 -net nic,model=virtio \
    $QEMU_ARGS -nographic

```

`$QEMU_ARGS` is left unquoted on purpose: `bash` splits it into the `-device` word and its options. `zsh` does not split an unquoted variable, and `QEMU` then stops with invalid option; in `zsh` write `${=QEMU_ARGS}`, or run the command under `bash`. In the guest, `sudo nvme list` shows the 1.72 GB BlackBox drive of tutorial 01.

### 5.11.7 What you learned

- A configuration file names the same properties as `-device femu`, with `mode` for `femu_mode` and a `[subsys]` section for FDP.
- `ssd-config.sh` checks names against the binary in `FEMU_BIN`; `QEMU` checks values when it starts.
- List values need no doubled commas in the file.

### 5.11.8 Next

- Scripts and tools (§C.4) describes `ssd-config.sh` and the shipped files.
- The parameter manual (§5.1) explains what to put in a configuration, group by group.

## 5.12 Performance tuning

FEMU's timing model computes when each command should complete, and a poller thread posts the completion once the host clock passes that time (timing model (§6.2.1)). The model gives a lower bound: if a poller does not get a host CPU when a completion is due, the guest waits longer than the model says. Tuning is mostly about giving FEMU's threads the CPUs and memory they need, so that the latency you measure is the model's and not the host's.

### 5.12.1 Threads and cores

These threads run inside `QEMU`:

| Thread name     | How many                                                                               | Started                                     | Uses a core                                      |
|-----------------|----------------------------------------------------------------------------------------|---------------------------------------------|--------------------------------------------------|
| CPU N/KVM       | one per guest vCPU ( <code>-smp</code> )                                               | at start                                    | while the vCPU runs                              |
| femu-poller     | 1, or <code>ceil(queues / poller_ratio)</code> with <code>multipoller_enabled=1</code> | when the guest first enables the controller | always: it spins while the controller is enabled |
| FEMU-FTL-Thread | one per controller with a BlackBox, ZNS or CSD namespace                               | at start                                    | always: it spins while the controller is enabled |
| femu-cxl-ftl    | one per <code>femu-cxl-ssd</code> with <code>ftl=on</code> (the default)               | at start                                    | only while it serves a miss                      |
| femu-cxl-cca    | one per <code>femu-cxl-ssd</code> with <code>cca=on</code>                             | at start                                    | only while it serves a command                   |
| femu-csd-cu     | <code>nr_cu</code> per controller with a CSD namespace                                 | at start                                    | only while it runs a program                     |

Linux shows these names only when `QEMU` runs with `-name NAME,debug-threads=on`. The `run-*.sh` launchers pass it, for example `-name "FEMU-BBSSD-VM",debug-threads=on` in `run-blackbox.sh`. Without it,

every thread is called `qemu-system-x86`, so add it to any QEMU command line of your own.

Plan one host core for every poller and FTL thread, on top of one per vCPU. `run-blackbox.sh` (4 vCPUs, one poller, one FTL thread) needs at least 6 cores to itself; 8 is comfortable. When cores are short, the first sign is latency above the configured NAND time and run-to-run variation.

### 5.12.2 Pollers and queues

Properties: queues, pollers and interrupts (§A.1.1).

| Setting                                            | Pollers                                                  | Use it when                                                           |
|----------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------|
| <code>multipoller_enabled=0</code> (default)       | 1, serving every I/O queue                               | You measure latency at low queue depth, or have few spare cores       |
| <code>multipoller_enabled=1, poller_ratio=1</code> | one per I/O queue                                        | You need throughput from several guest jobs and have a core per queue |
| <code>multipoller_enabled=1, poller_ratio=R</code> | $\text{ceil}(\text{queues} / R)$ , each serving R queues | You need more than one poller but have fewer spare cores than queues  |

Other values of `multipoller_enabled` are refused at start. `poller_ratio=0` counts as 1.

The number of pollers comes from the `queues` property (default 8), not from the number of queues the guest creates. Linux creates about one I/O queue pair per guest CPU, up to `queues`. A poller whose queues the guest never created still spins. So set `queues` to the guest's vCPU count when you turn on more pollers. With 4 vCPUs and a poller per queue:

```
| -device femu,devsz_mb=4096,femu_mode=1,queues=4,multipoller_enabled=1
```

What you trade:

- One poller walks every queue on each pass. It needs only one core, but its pass gets longer with each busy queue, and all completions share it.
- A poller per queue keeps each queue's path short and scales with guest jobs, at the cost of one spinning core per queue.
- `poller_ratio` sits in between. Each poller serves its queues round-robin: poller *i* serves queues *i*, *i* + *P*, *i* + 2*P* and so on, where *P* is the number of pollers.

### 5.12.3 Pin the threads

Unpinned, the scheduler moves vCPUs and pollers between cores and lets them share a core with other work. Pin them to separate cores. With `debug-threads=on` on the command line (see above), boot the guest, then on the host:

```
| pid=$(pgrep -x qemu-system-x86)
| ps -T -p "$pid" -o tid=,comm=
```

Pin the vCPUs with the QMP helper that `femu-copy-scripts.sh` copies to `build-femu/ftk/`. This puts vCPU 0 to 3 on host CPUs 0 to 3:

```
| sudo ./ftk/qmp-vcpu-pin -s ./qmp-sock 0 1 2 3
```

Then give each poller and the FTL thread a core of its own, here starting at host CPU 4. The pollers exist only after the guest has enabled the controller, so run this after the guest has booted:

```

cpu=4
for tid in $(ps -T -p "$pid" -o tid=,comm= |
    awk '$2 == "femu-poller" || $2 == "FEMU-FTL-Thread" {print $1}'); do
    sudo taskset -pc "$cpu" "$tid"
    cpu=$((cpu + 1))
done

```

Before you pin anything, move every existing QEMU thread off the cores you reserve for FEMU, for example with `sudo taskset -apc 8-15 "$pid"`, then pin the vCPUs and FEMU's threads as above. A thread created later inherits the affinity of the thread that creates it. Choose cores that are not hardware-thread siblings of each other (`lscpu -e` shows the core of each CPU), so that two spinning threads do not share one physical core.

`pin.sh` does all of this in one step. From `build-femu/`, after the guest has booted:

```

| ./pin.sh 4      # vCPUs, then pollers and FTL threads, from host CPU 4

```

It gives each vCPU, `femu-poller`, `FEMU-FTL-Thread`, `femu-cxl-ftl` and `femu-cxl-cca` a CPU of its own, starting at the CPU you name (default 0), and moves the other QEMU threads, `femu-csd-cu` included, to the CPUs after those. It finds the threads by name, so it stops with an error when QEMU runs without `debug-threads=on`. Set `QEMU_PID` when more than one QEMU runs (scripts reference (§C.6)).

#### 5.12.4 Hugepages

FEMU's device memory is ordinary anonymous memory; FEMU does not request hugepages for it (a host with transparent hugepages set to `always` may still use them). You can back the guest's RAM with hugepages, which reduces TLB misses in the guest. Reserve them on the host (2048 pages of 2 MiB for a 4 GiB guest):

```

| echo 2048 | sudo tee /proc/sys/vm/nr_hugepages
| grep HugePages_Free /proc/meminfo

```

Then, in the launcher, replace `-m 4G` with these options:

```

| -m 4G -object memory-backend-memfd,id=ram0,size=4G,hugetlb=on,prealloc=on \
| -machine memory-backend=ram0

```

If the host has too few free hugepages, QEMU stops at start with `unable to map backing store for guest RAM: Cannot allocate memory`. CI does not test this setup.

`femu-cxl-ssd` with `der=cylon` needs its CXL memory on a shared, preallocated hugetlb backend; see the CXL SSD guide (§4.18.7).

#### 5.12.5 NUMA

On a host with more than one NUMA node, a thread that reads memory on the other node pays the inter-socket latency on every access. Each NVMe command copies its data between the guest's RAM and FEMU's device memory, so keep the vCPUs, the pollers, the guest RAM and the device memory on one node, unless you mean to study the other case.

To keep the whole QEMU process on node 0, start the launcher under `numactl`:

```

| sudo numactl --cpunodebind=0 --membind=0 ./run-blackbox.sh

```

To place only the device memory, set `FEMU_MBE_INTERLEAVE` in QEMU's environment: 0 or 1 binds it to that node, on interleaves it across nodes 0 and 1 (environment variables (§A.1.5)). The launchers other than `run-`



cxlssd.sh start QEMU with sudo, which drops your environment, so put the variable on the launcher's sudo line:

```
| sudo FEMU_MBE_INTERLEAVE=1 ./qemu-system-x86_64 \
```

FEMU prints backend: N MB bound via FEMU\_MBE\_INTERLEAVE=1 when the binding worked. One layout that keeps the guest and the emulator from competing for memory bandwidth puts the vCPUs and guest RAM on one node, and the pollers, the FTL thread and the device memory on the other.

### 5.12.6 Host settings

- **CPU frequency.** A core that changes frequency changes how long FEMU's own work takes. Set every core to the performance policy: `sudo cpupower frequency-set -g performance`, or `sudo ../femu-scripts/set_cpu_perf_mode.sh`.
- **Locked device memory.** Under sudo, FEMU locks its device memory so that page faults do not add latency. As a normal user it needs `ulimit -l unlimited` or a matching `/etc/security/limits.conf` entry; without it FEMU prints cannot pin the N MiB memory backend and runs with less precise latency.
- **Other work.** Keep other busy processes off the cores you gave FEMU. The `isolcpus=` kernel parameter keeps the scheduler from placing other tasks there.
- **Physical host.** Nested virtualization and WSL add their own delays (requirements (§2.1.1)).
- **Priority.** `sudo nice -n -10 ./run-blackbox.sh` starts QEMU, and every thread it creates, at a higher scheduling priority.

Inside the guest, services that wake up on their own add noise to a measurement, and a guest that suspends stops it. For example:

```
| sudo systemctl disable cups bluetooth
| sudo systemctl mask sleep.target suspend.target
| cat /sys/block/nvme0n1/queue/scheduler          # the guest's I/O scheduler
| echo mq-deadline | sudo tee /sys/block/nvme0n1/queue/scheduler
```

The guest's I/O scheduler queues requests before they reach FEMU, so state which one a measurement used.

### 5.12.7 What each knob trades

| Knob                                          | Gains                                                      | Costs                                                                                  |
|-----------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------|
| multipoller_enabled=1<br>poller_ratio above 1 | throughput that scales with guest jobs<br>fewer cores used | one spinning core per poller<br>longer passes, more completion delay per poller        |
| queues                                        | one queue per guest CPU, no queue sharing in the guest     | with multipoller_enabled=1 and poller_ratio=1, each extra queue adds a spinning poller |
| Pinning                                       | stable latency, no migrations                              | cores reserved for FEMU                                                                |
| Guest RAM on hugepages                        | fewer TLB misses in the guest                              | memory reserved up front                                                               |
| One NUMA node                                 | no cross-socket copies                                     | that node's cores and memory bandwidth only                                            |
| Performance CPU frequency                     | stable service times                                       | power and heat                                                                         |
| Locked device memory                          | no page faults in the I/O path                             | memory resident from start-up                                                          |

### 5.12.8 Related pages

- Measuring (§6.1): repeatable numbers once the host is tuned
- Timing model (§6.2)
- Security and limits: host sizing (§5.13.4)
- Requirements (§2.1)

Related issues: #7, #69, #77, #93, #101.

## 5.13 Security and limits

What a guest can do to the host through FEMU, what QEMU features FEMU does not support, how device properties change between versions, and how much host memory a device takes. To report a vulnerability, follow SECURITY.md<sup>82</sup>.

### 5.13.1 Security model

FEMU runs inside the QEMU process on the host. Everything a FEMU device does, it does with QEMU's privileges. The NVMe `run-*.sh` launchers start QEMU with `sudo`, so by default that means root; `run-cxlssd.sh` runs QEMU as you.

Treat the guest as trusted unless you have turned off every feature below. For a guest you do not trust:

- Run QEMU as a normal user in the `kvm` group, not with `sudo`. FEMU then cannot pin the device memory unless you raise the memory lock limit (requirements (§2.1.1)). It starts anyway and prints `[FEMU] Err: cannot pin the N MiB memory backend, which is a warning, not a failure.`
- Do not use CSD mode with `csd_program_dir`.
- Keep `lsa-control` off on `femu-cxl-ssd`.

### CSD programs

In CSD mode (`femu_mode=4`), the guest can ask the device to load a program by file name. For a shared-library program, QEMU opens that file with `g_module_open()` and runs its code inside the QEMU process. An eBPF program is read from the same directory and run by the uBPF library, when FEMU is built with it.

FEMU limits which files the guest can name:

- With `csd_program_dir` unset (the default, and what `run-csd.sh` does), only the built-in program type loads.
- With it set, a name must be a file in that directory. FEMU refuses an empty name, a name containing `/`, the names `.` and `..`, and a name whose path leaves the directory after symbolic links are resolved.

This keeps the guest away from the rest of the host file system. It does not make the programs in the directory safe: any shared object there runs as QEMU, which is root under the launchers. Put only programs you built in the directory, and make sure no other user can write to it. See the CSD guide (§4.13.2).

### CXL SSD control channel

`femu-cxl-ssd,lsa-control=on` lets the guest send experiment commands through the CXL Get LSA mailbox command (CXL SSD guide (§4.18.8)). With it on, the guest can:

---

<sup>82</sup><https://github.com/MoatLab/FEMU/blob/master/SECURITY.md>

- create and write `cxlssd-stats.log`, `cxlssd-io-N.log` and `cxlssd-spt.log` in `log-dir`, which is QEMU's working directory when unset;
- write the host's trace and `tracing_on` files in `tracefs-dir`, which clears the host trace buffer and turns host tracing on or off, when `tracefs-dir` is set;
- flush and reconfigure the device cache, reset the statistics, and set or remove a direct-mapping ratio.

Each file is capped at `log-limit` (64 MiB by default) and FEMU reuses a fixed set of 64 I/O log names. With the statistics and mapping dump files, a guest can therefore make FEMU write about 66 times `log-limit` (about 4.1 GiB with the default) in `log-dir`, and no more.

`lsa-control` is off by default on the device. `run-cxlssd.sh` turns it on, because Cylon's scripts use it. Turn it off for a guest you do not trust:

```
| LSA_CONTROL=off ../femu-scripts/run-cxlssd.sh
```

The same commands stay available from the host through QMP (`control-command`), whatever `lsa-control` says.

### Other host-side files and output

These are chosen on the host, not by the guest:

- `pel_file` keeps the Persistent Event Log in a host file that QEMU creates if it is missing.
- `FEMU_DUMP_LPN` (BlackBox) hex-dumps the contents of one logical page to QEMU's standard error on every read, which `run-blackbox.sh` copies into its log file. `FEMU_EXP_LOG` traces only the addresses of pages that contain the `FEMU_SECRET` marker. Leave `FEMU_DUMP_LPN` unset when the guest's data is private.

## 5.13.2 Migration and snapshots

FEMU devices cannot be migrated or saved. Both `femu` and `femu-cxl-ssd` mark their state unmigratable, so QEMU refuses these operations and the guest keeps running:

| Operation                                         | What QEMU reports                                                                                                     |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| QMP migrate                                       | State blocked by non-migratable device ' <code>0000:00:04.0/femu</code> '                                             |
| HMP savevm NAME                                   | Error: State blocked by non-migratable device ' <code>0000:00:04.0/femu</code> '                                      |
| Either, with a CXL SSD                            | State blocked by non-migratable device ' <code>&lt;path&gt;/femu-cxl-ssd</code> '                                     |
| <code>--only-migratable</code> with a FEMU device | Device <code>femu</code> is not migratable, but <code>--only-migratable</code> was specified, and QEMU does not start |

The part before `/femu` is the device's PCI address, so it changes with your command line.

Device data lives only in host memory:

- A guest reboot or a controller reset keeps the data.
- QEMU exiting, for any reason, loses it. FEMU never writes an NVMe namespace to a file, and there is no option to do so. (`femu-cxl-ssd` keeps its data in the memory backend you give it; a file-backed `memory-backend-file` leaves the bytes in that file.)
- A snapshot of the guest's boot disk (for example with `qemu-img`) does not include the FEMU device.

To start every run from the same state, write the data again after boot, or run a fixed preconditioning workload. Related issue: #52.

### 5.13.3 Property compatibility

FEMU's device properties are its configuration interface, but they are not versioned: FEMU has no machine-type compatibility settings for them, so a command line means what the current code says it means.

#### Values refused at realize

Recent versions check more property values when the device is created and stop QEMU with a message, where older versions accepted the value and then misbehaved. A command line that used to start may therefore stop. The most recent additions:

| Property            | Now refused                             |
|---------------------|-----------------------------------------|
| femu_mode           | above 5                                 |
| multipoller_enabled | other than 0 or 1                       |
| lver (OCSSD)        | other than 1 or 2                       |
| flash_type (OCSSD)  | outside 1 to 4                          |
| zns_chnls_per_zone  | a value that does not divide zns_num_ch |

The message names the property, for example:

```
qemu-system-x86_64: -device femu,devsz_mb=64,femu_mode=7: femu_mode must be 0 (OpenChannel), 1 (black-box), 2 (no-SSD), 3 (zoned), 4 (computational storage) or 5 (key-value)
```

The changelog (§D.1.3) lists these and other refusals, and the changes that move numbers without stopping a run. Most mode guides list their own refusals under "Limits and refusals"; the CXL SSD guide has them under "Limits".

#### Properties that are accepted but do nothing

These are kept so that old command lines still start. Nothing reads them, so setting them changes nothing the guest can observe. A value other than the default prints one warning at realize, for example warning: femu: dlfeat has no effect and is accepted only for compatibility:

| Property                                         | Why                                                                                         |
|--------------------------------------------------|---------------------------------------------------------------------------------------------|
| serial                                           | Identify Controller reports a serial number FEMU generates                                  |
| ms                                               | the metadata size comes from meta (the warning says so)                                     |
| dlfeat                                           | Identify Namespace always reports 0x9                                                       |
| ms_max                                           | OCSSD 2.0 reports a single LBA format (NLBAF 0), and no other mode reads it                 |
| tplpbsy, tplrbsy, trcbisy                        | programs and reads are issued one plane at a time, and no mode enables the cache read model |
| nr_thread, time_slice, context_switch_time (CSD) | accepted for CEMU configurations; CSD still refuses nr_thread=0                             |

intc, intc\_thresh and intc\_time do not warn, because Get Features returns them: intc\_thresh and intc\_time in Interrupt Coalescing (feature 08h) and intc (0 or 1) in Interrupt Vector Configuration (09h). Interrupts are still not coalesced.

The property reference (§A.1) says this in each description, and `-device femu,help` prints the same text.

### 5.13.4 Host sizing

#### Memory

A FEMU device keeps its whole capacity in host DRAM. For each device, plan for:

- **The backend:** `devsz_mb` MiB, or the raw NAND capacity with `op_pcent`. Under `sudo`, FEMU locks it in memory at start-up, so all of it is resident from the first second. Without the lock (a normal user with the default limit), pages become resident as the guest first touches them.
- **FTL tables:** allocated and filled at start-up. With the default BlackBox geometry (16 GiB of NAND in 4 KiB pages), QEMU's resident size was 116 MiB against 35 MiB for a NoSSD device of the same size, so the FTL took about 80 MiB. It grows with the number of NAND pages.
- **The guest:** its `-m` size.
- **QEMU itself:** the requirements page budgets about 1 GiB.

The requirements page (§2.1.1) gives the totals for each launcher. If the allocation is more than the kernel will commit, QEMU aborts at start with `failed to allocate N bytes`.

In BlackBox, CSD and KV mode the NAND geometry as a whole (every namespace plus spare space) may hold at most  $2^{31} - 1$  sectors of `secsz` bytes, just under 1 TiB with the default 512-byte sectors.

#### Hugepages

FEMU allocates its backend from ordinary anonymous memory, not `hugetlbfs`, so the NVMe modes need no hugepages. You can back the guest's RAM with hugepages to reduce TLB misses in the guest; see performance tuning (§5.12.4). `femu-cxl-ssd` with `der=cylon` is the exception: it needs a shared, preallocated `hugetlb` memory backend (CXL SSD guide (§4.18.7)).

#### CPU

Each `femu-poller` thread, and the `FEMU-FTL-Thread` of a controller with a BlackBox, ZNS or CSD namespace, spin on a host core while the guest has the controller enabled. Plan one core for each of them on top of the guest's vCPUs (performance tuning (§5.12.1)).

### 5.13.5 Related pages

- Architecture (§3.1)
- Performance tuning (§5.12)
- Troubleshooting (§7.1)
- Changelog (§D)

## 6 Measuring and validation

### 6.1 Measuring

How to read FEMU's counters, measure latency and throughput from the guest, and get numbers that repeat from run to run. Commands in sh blocks run inside the guest unless they say otherwise. From the host, prefix a single command with `./run-guest-ssh.sh` as in the quick start (§2.4); run multi-line blocks in a shell inside the guest (`./run-guest-ssh.sh` with no arguments).

Before you measure anything, tune the host as in performance tuning (§5.12). Most surprising numbers come from a FEMU thread that did not get a core.

#### 6.1.1 What FEMU can tell you

| Source              | Read with                               | Modes                                   | What it holds                                                                         |
|---------------------|-----------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------|
| Vendor log page C0h | <code>nvme get-log --log-id=0xc0</code> | BlackBox (with or without FDP), CSD, KV | WAF, host, GC and NAND page counts, write buffer hits, read reclaim, log-block merges |
| SMART log           | <code>nvme smart-log</code>             | every NVMe mode                         | data read and written, command counts, Percentage Used, Available Spare, media errors |
| FDP statistics      | <code>nvme fdp stats</code>             | BlackBox with FDP                       | host and media bytes written                                                          |
| QOM counters        | <code>QMP qom-get</code> on the host    | <code>femu-cxl-ssd</code>               | cache hits and misses, media reads, programs and time, direct mapping                 |
| fio                 | in the guest                            | every block mode                        | latency and throughput as the guest sees them                                         |

ZNS, NoSSD and OCSSD leave the C0h counters at zero. ZNS has no device garbage collection: the host resets zones.

The guest's own tools show I/O from the guest's side: `iostat -x 1` per device, `iotop` per process, `dstat -cdn` for CPU, disk and network together.

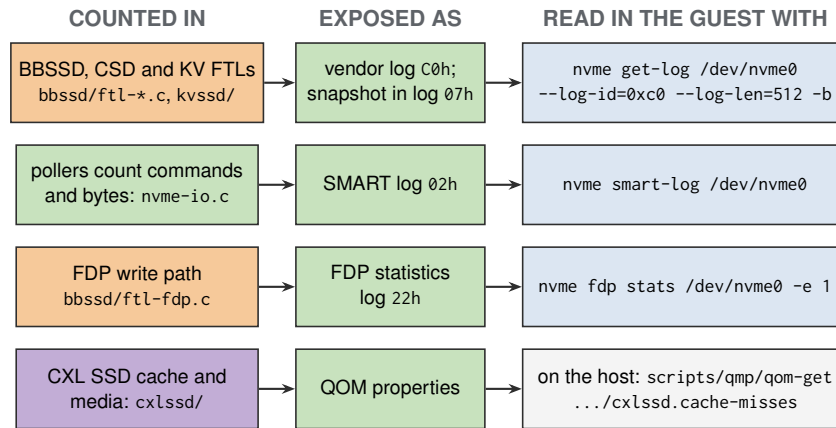
#### 6.1.2 Write amplification and media counters (C0h)

The first 4 bytes of log page C0h are the write amplification factor (WAF) times 1000. The three 8-byte counters from byte 8 are pages the host wrote, pages garbage collection moved, and pages programmed for the host:

```
sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b | od -An -t u4 -N 4
sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b | od -An -t u8 -j 8 -N 24 -w24
```

FEMU computes the WAF as (pages programmed + pages GC moved) x 1000 / pages the host wrote. The counters are summed over every BlackBox, CSD and KV namespace of the controller, and they count from the moment QEMU started the device (or from when namespace management created the namespace). The WAF in the log is therefore the WAF since start, not the WAF of your last run. To measure one run, read the counters before and after it and use the differences:

```
c0() {
    sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b |
        od -An -t u8 -j 8 -N 24 -w24
}
```



C0h, little endian: bytes 0 to 3 hold the WAF times 1000, (NAND pages + GC pages) × 1000 / host pages; 8-byte host, GC and NAND page counts follow at bytes 8, 16 and 24, then the most reads of any block since its erase, read reclaims, retention refreshes, write buffer and merge counts. It stays zero in ZNS, NoSSD and OCSSD modes. fio in the guest gives the latency and bandwidth the guest sees.

**Figure 26:** Where FEMU’s counters come from and how you read them: FTL media counters in vendor log C0h, command and byte counts in the SMART log, FDP statistics, and CXL SSD counters as QOM properties on the host.

```
c0 > before.txt
sudo fio --name=run --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
--rw=randwrite --bs=4k --iodepth=32 --size=4G
c0 > after.txt
paste before.txt after.txt | awk '{ h = $4 - $1; g = $5 - $2; n = $6 - $3;
printf "host %d gc %d nand %d WAF %.3f\n", h, g, n, (n + g) / h }'
```

Pages the write buffer absorbs (buffer\_size) count as host pages but not as programmed pages until they leave the buffer, so with a buffer the WAF of a short run can be below 1. The other fields (read reclaim, write buffer hits, log-block merges) are listed in log pages and counters (§B.1).

GC starts when gc\_thres\_pcent (75% by default) of the lines are in use. A WAF of 1.000 after a run means GC has not run yet, or found no valid pages to move. To make GC run, write more than the free space, as in the BlackBox guide (§4.4.4).

### 6.1.3 SMART and FDP statistics

```
| sudo nvme smart-log /dev/nvme0
```

FEMU fills these SMART fields:

- data\_units\_read, data\_units\_written: thousands of 512-byte units the host moved, rounded up.
- host\_read\_commands, host\_write\_commands.
- percentage\_used: wear of the most worn namespace, from erase counts and pe\_cycles\_rated (or the rating of nand\_cell\_type); 0 when neither is set.
- available\_spare: the worst namespace, from nand\_bad\_blocks.
- media\_errors, num\_err\_log\_entries, temperature (the temperature property), power-on hours and unsafe shutdowns.

With FDP, the endurance group statistics give host and media bytes written. Their ratio is the WAF over the endurance group, every FDP namespace of the subsystem:

```
| sudo nvme fdp stats /dev/nvme0 -e 1
```

### 6.1.4 CXL SSD counters

The femu-cxl-ssd counters are QOM properties, read on the host through QMP. Add a QMP socket to the run-cxlssd.sh command line (-qmp unix:/tmp/qmp.sock,server=on,wait=off), then on the host, from the QEMU source tree:

```
| export QMP_SOCKET=/tmp/qmp.sock
| scripts/qmp/qom-get /machine/peripheral/cxlssd.cache-misses
| scripts/qmp/qom-get /machine/peripheral/cxlssd.media-time-ns
| scripts/qmp/qom-get /machine/peripheral/cxlssd.media-full
```

cxlssd is the id= that run-cxlssd.sh gives the device.

- stats-reset clears the cache, prefetch and caching API (cca-\*) event counters. It copies the cache and prefetch counters to the last-\* properties first. It does not clear the media counters (media-reads, media-writes, media-time-ns) or the der-\* counters, so measure those as differences between two reads.
- media-full must stay 0. A non-zero value means some NAND programs found no free page and were not timed, and the run is not valid.
- Accesses served through a direct mapping (der=memslot or cylon) never reach QEMU and are not counted as hits.

The full list is in runtime properties (§A.2.2).

### 6.1.5 Latency and throughput with fio

Use --direct=1 so the guest page cache does not answer for the device. Report the completion latency (clat) percentiles, not only the mean.

#### NoSSD

NoSSD has no media time, so it measures FEMU's own path. Unwritten blocks read as fast as written ones.

```
| sudo fio --name=lat --filename=/dev/nvme0n1 --direct=1 --ioengine=io_uring \
|   --rw=randread --bs=4k --iodepth=1 --runtime=30 --time_based
| sudo fio --name=tput --filename=/dev/nvme0n1 --direct=1 --ioengine=io_uring \
|   --rw=randread --bs=4k --iodepth=64 --numjobs=4 --group_reporting \
|   --runtime=30 --time_based
```

For throughput beyond one poller, see pollers and queues (§5.12.2).

#### BlackBox and CSD

A read of a page that was never written costs no NAND time. Fill the range first, then read it:

```
| sudo fio --name=fill --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
|   --rw=write --bs=128k --iodepth=16 --size=4G
| sudo fio --name=rr --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
|   --rw=randread --bs=4k --iodepth=1 --size=4G --runtime=30 --time_based
```

On an idle device at queue depth 1, a 4 KiB read takes about pg\_rd\_lat (40 us by default) plus the guest's and the poller's overhead, and a 4 KiB write about pg\_wr\_lat (200 us) when there is no write buffer. Measure the overhead on a NoSSD device with the same guest and subtract it.



To see GC in the latency, age the device first: random writes over the whole namespace, several times, until the C0h WAF stops rising. Then measure. The tail percentiles (c<sub>lat</sub> 99th and above) show GC; the median mostly does not.

## ZNS

Write sequentially inside zones, and reset the zones between runs:

```
sudo fio --name=zns --filename=/dev/nvme0n1 --direct=1 --ioengine=psync \
--zonemode=zbd --rw=write --bs=128k --size=1G
sudo fio --name=zr --filename=/dev/nvme0n1 --direct=1 --ioengine=psync \
--zonemode=zbd --rw=randread --bs=4k --size=1G --runtime=30 --time_based
sudo blkzone reset /dev/nvme0n1
```

A write costs 1 us per 4 KiB page while it fits in the zone's write cache; the write that fills the cache pays for programming it (timing model (§6.2.6)).

## FDP

fio places data with its `io_uring_cmd` engine on the generic node. Run the same job with and without `--fdp=1` and compare the WAF:

```
sudo fio --name=fdp --filename=/dev/ng0n1 --ioengine=io_uring_cmd --cmd_type=nvme \
--fdp=1 --fdp_pli=0,1,2,3 --rw=randwrite --bs=4k --iodepth=16 --size=4G
```

## KV, OCSSD and CXL

- KV has no block device, and fio has no key-value engine. Time the passthrough commands in your own program; `kv-probe.c` is a starting point (KV guide (§4.11.4)).
- OCSSD needs a guest kernel older than 5.15 with `pblk`, or `SPDK` (OCSSD guide (§4.9)).
- For the CXL SSD, run your workload on the DAX device or NUMA node of the region (CXL SSD guide (§4.18.5)), and read the QOM counters above before and after.

### 6.1.6 Repeatable numbers

1. **Fix the host.** Pin vCPUs, pollers and the FTL thread to their own cores, set the CPU frequency policy to performance, and keep other work off those cores (performance tuning (§5.12)).
2. **Record the setup.** The FEMU commit, the `-device` line (`run-blackbox.sh`, `run-blackbox-fdp.sh` and `run-csd.sh` print it; `DRY_RUN=1 ../femu-scripts/run-cxlssd.sh` prints the CXL command), the guest kernel and the fio job file. A number without these cannot be compared.
3. **Start from a known state.** Restart QEMU between configurations. The device memory and every counter start from zero when QEMU starts; a guest reboot or controller reset keeps both. A namespace created with namespace management starts its counters at zero, and deleting one removes its counts from C0h. Then precondition: fill the range you will read, and for steady-state write numbers, overwrite at random until the WAF stops changing.
4. **Measure intervals, not totals.** Read C0h, SMART or the QOM counters before and after each run and use the differences.
5. **Run long enough, more than once.** Use `--time_based` with a `--ramp_time`, run each point at least three times, and report the median and the spread.

6. **Check FEMU kept up.** On a BlackBox controller, vendor command 0xEF with CDW10 5 prints to QEMU's console how many completions were posted 20 us or more after they were due, out of all completions, and resets both counts. `run-blackbox.sh` copies the console to `build-femu/log`. Its standard output goes through a pipe there, so the line can appear only later; put `stdbuf -oL` in front of `./qemu-system-x86_64` on the launcher's `sudo` line to see it at once. Reset the counts before a run, read them after:

```
| sudo nvme admin-passthru /dev/nvme0 --opcode=0xef --cdw10=5
```

The line ends with the two counts as A/B. If A is more than a small share of B, the pollers did not get enough CPU and the latencies are longer than the model.

7. **Keep the timing switches in mind.** 0xEF codes 2 and 4 turn GC time and NAND time off until you turn them back on with 1 and 3, and code 3 restores the built-in times, not the ones on your command line (timing model (§6.2.11)). Restart QEMU after using them.

### 6.1.7 Related pages

- Timing model (§6.2)
- Log pages and counters (§B)
- Runtime properties (§A.2)
- Performance tuning (§5.12)

Related issues: #7, #15, #92, #130, #137, #151.

## 6.2 Timing model

FEMU makes an emulated SSD take as long as a real one would. This page explains how it computes that time, how it makes the guest wait for it, which properties change it, and how to check what it did. The architecture page (§3.1) shows where each step runs.

### 6.2.1 Compute, then hold

FEMU does not slow down the data copy. An NVMe command's data is copied between guest memory and the memory backend as soon as the command is parsed. Separately, a model computes when the command would have finished on the emulated device, and the poller does not post the completion before that time:

1. The poller stamps the command with the current host time, `stime`, read from `QEMU_CLOCK_REALTIME`. Its completion time starts equal to it.
2. The model's model adds the command's latency to the completion time.
3. The poller keeps completed commands in a priority queue ordered by completion time and posts each one once the host clock passes it (`nvme_process_cq_cpl()` in `hw/femu/nvme-io.c`).

Two things follow from this:

- Time in the model is host wall-clock time, and the guest sees the same clock. A 200 us program makes the guest wait 200 us.
- The model gives a lower bound. A completion is posted on the first poller sweep after it is due, so a poller that does not get a host CPU (too few cores, no pinning) adds delay on top. Give each `femu-poller` and the `FEMU-FTL-Thread` its own core when you measure latency.

The CXL SSD works differently: the vCPU thread that made the access waits out the media time itself before the access completes (see CXL SSD (§6.2.10)).

## 6.2.2 Which mode charges what

| Mode  | Time charged for                                                                                                                    | Computed in                                                                |
|-------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| NoSSD | nothing                                                                                                                             | n/a                                                                        |
| BBSSD | NAND reads, programs and erases; write buffer and read cache hits; DFTL mapping misses; GC; deallocate ( <code>trim_lat_ns</code> ) | FEMU-FTL-Thread                                                            |
| CSD   | as BBSSD, plus program run time on a compute unit and copies into device memory                                                     | FTL thread; compute units on the femu-csd-cu threads; copies in the poller |
| ZNS   | NAND reads, programs and erases; write cache accesses                                                                               | FEMU-FTL-Thread                                                            |
| KV    | NAND reads and programs for values, erases on reclaim, and a fixed per-command cost of one page read                                | poller                                                                     |
| OCSSD | NAND reads, programs and erases, optional channel transfer                                                                          | poller                                                                     |

Optional host-link and firmware-CPU models apply on top of any NVMe mode (see Host link and controller firmware (§6.2.9)).

## 6.2.3 NAND operations

`nand_media_op()` in `hw/femu/nand/nand-media.c` computes one read, program or erase. BBSSD, CSD, KV, ZNS and both Open-Channel versions use it.

### Operation times

The array time of an operation comes from one of two sources:

- **Flat times** (BBSSD, CSD, KV with `nand_cell_type=0`, the default): every read takes `pg_rd_lat`, every program `pg_wr_lat`, every erase `blk_er_lat`, in nanoseconds. Defaults are 40 us, 200 us and 2 ms. `pgtype_lat` with `cell_pages` scales the program time by the page's position in its wordline (lower to upper).
- **Cell-type tables**: `nand_cell_type` 1 to 4 (SLC, MLC, TLC, QLC) uses the built-in per-page-type read and program times and erase time in `hw/femu/nand/nand.h`, and ignores the flat times.
- **ZNS** uses one built-in value per cell type for `zns_flash_type` (`hw/femu/zns/zns.h`; SLC, TLC and QLC have one), unless `zns_pg_rd_lat`, `zns_pg_wr_lat` or `zns_blk_er_lat` overrides it.

Properties: NAND timing (§A.1.1), ZNS (§A.1.1).

### Parallelism

The media layer keeps a busy-until time for each unit that can work on its own:

- **BBSSD, CSD, KV**: one per LUN. An operation starts at the later of its own start time and the LUN's busy-until time, and the LUN is busy until the operation ends. All planes of a LUN are busy together. Erasing a whole line (GC, FDP GC and KV reclaim) with `pls_per_lun > 1` erases the same block on every plane of a LUN as one operation, which takes one erase time instead of one per plane.
- **ZNS**: one per plane, so the planes of a LUN work in parallel.

Different LUNs (or ZNS planes) never wait for each other. A command that covers several pages issues them all at its start time, and its latency is that of the slowest page. Because the BBSSD write pointer moves across

channels first and then LUNs, consecutive pages of a large write fall on different LUNs and are programmed in parallel.

Properties: NAND geometry (§A.1.1).

### Channel bus

The channel bus is off by default: transfers between controller and NAND take no time and channels never contend. It is turned on when any bus phase has a non-zero time:

| Phase               | BBSSD, CSD, KV                                    | ZNS              |
|---------------------|---------------------------------------------------|------------------|
| Command and address | cmd_addr_lat                                      | zns_cmd_addr_lat |
| Page data transfer  | pg_xfer_lat, or ch_xfer_lat when pg_xfer_lat is 0 | zns_pg_xfer_lat  |
| Status read         | status_lat                                        | zns_status_lat   |

With the bus on, phases on one channel run one at a time. A program sends its command and data over the bus before the array starts. A read sends its command, waits for the array, then moves its data out; the data-out is booked for when it will happen, so other LUNs can use the bus in the meantime.

### Program and erase suspend, ECC

- **pe\_suspend** (zns\_pe\_suspend for ZNS) lets a read go ahead of a program or erase that is running on its LUN (plane for ZNS). The first such read pays tsusp\_ns (zns\_tsusp\_ns); the suspended operation then ends later by tsusp\_ns plus the time of every read that went ahead of it.
- **ecc\_step\_ns** adds read time for worn or old blocks: one step per 750 erases of the block, plus one per ecc\_retention\_sec seconds since its line was filled, at most four steps.

Properties: NAND timing (§A.1.1), Reliability and wear (§A.1.1).

## 6.2.4 BBSSD FTL costs

On top of the NAND operations, the BBSSD FTL (also used by CSD) charges:

- **Write buffer** (buffer\_size > 0, in pages): a write the buffer accepts costs one DRAM access, pg\_rd\_lat / 16 (2.5 us with the defaults). When the buffer passes buffer\_thres\_pcent, the write that pushed it there also pays for programming a batch of buffered pages. A read of a buffered page costs the same DRAM access and does not reach NAND. Flush programs everything buffered and pays for it. FUA writes and stream writes skip the buffer.
- **Read cache** (read\_cache\_mb > 0): a hit costs a DRAM access instead of a NAND read.
- **Mapping cache** (mapping=dftl): a miss in the cached mapping table of mapping\_cache\_mb adds a NAND read of the translation page, after a NAND program if the entry it replaces is dirty.
- **Deallocate**: trim\_lat\_ns per Dataset Management range.

Properties: Garbage collection, mapping and caches (§A.1.1).

## 6.2.5 Garbage collection

GC moves valid pages out of a victim line and erases it. Its NAND reads, programs and erases go through the same per-LUN timelines as host commands, so a host command that needs a LUN that GC is using waits for it. That is how GC shows up in host latency.

- **Background GC** runs one line after a request, on the FTL thread, when the share of lines in use has reached `gc_thres_pcent` (default 75).
- **Foreground GC** runs inside a write when the share has reached `gc_thres_pcent_high` (default 95), and repeats until it drops below. The write waits for the LUNs it needs, which GC has just made busy.
- `gc_policy` picks the victim line. Write amplification depends on it and on how full the device is.

GC time can be switched off at run time with the vendor command below; GC still moves the pages, but its NAND operations then take no time.

ZNS has no device GC: the host resets zones, and a reset charges the erase of the zone's blocks.

### 6.2.6 ZNS write cache

A ZNS write goes into a per-zone SRAM write cache and costs 1 us per 4 KiB page. When the cache is full, or another zone needs it, the cached pages are programmed across the zone's planes and the write that triggered it pays for the program. `zns_num_wc` sets the number of caches (default: `zns_max_open`, or 3 when that is 0).

### 6.2.7 OCSSD

Both Open-Channel versions keep their busy-until times in the shared media layer above, with the LUN gate. Open-Channel 1.2 keeps one per chip (LUN) and one per channel; Open-Channel 2.0 keeps one per LUN, with the bus off. A write first moves its data over the channel, then programs the chip; a read occupies the chip, then moves its data out. Open-Channel 1.2 charges channel transfer only with `oc12_channel_timing=on`, using `ch_xfer_lat` per page or the `flash_type` table value when that is 0. Open-Channel 2.0 charges no channel time. Read, program and erase times come from the `flash_type` table (SLC, MLC, TLC, QLC) and can be changed at run time with vendor admin command 0xEE. `hw/femu/ocssd/oc-timing.c` copies the table times at init and handles 0xEE.

Properties: OCSSD (§A.1.1).

### 6.2.8 KV and CSD

KV computes its latency in the poller with the BBSSD NAND model: value pages are read and programmed, reclaim erases are charged to the command that triggers them, and every command pays one page read for the index lookup, on a LUN chosen by the key's hash. A command completes when its last NAND operation does.

CSD runs a program on the compute unit that frees up first. A program holds its unit for its declared run time, or for its measured host run time multiplied by the program's own scale, or by `csf_runtime_scale` when it gives none. `nr_cu` sets how many units run at once. A copy from the namespace into device memory costs `pg_rd_lat`.

Properties: CSD (§A.1.1).

### 6.2.9 Host link and controller firmware

Three optional models apply to every NVMe mode. They are added in the poller after the media time and before the completion is queued:

- `pcie_bandwidth_mbps`: each Read and Write is charged its size divided by this bandwidth, on one queue per direction, so transfers in the same direction wait for each other.
- `pcie_prop_delay_ns`: a fixed delay added to each Read and Write after its transfer.

- `fw_cpu_ns`: a fixed cost per Read, Write and Zone Append on one modelled controller core, which caps the command rate at about one per `fw_cpu_ns`.

All three are 0 (off) by default. With any of them on, NoSSD also completes through the priority queue instead of inside the sweep.

Properties: Host link and controller firmware (§A.1.1).

## 6.2.10 CXL SSD

A `femu-cxl-ssd` access is timed on the vCPU thread that made it:

| Access                                                                             | Media time                                                                                                                              |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Cache hit                                                                          | none                                                                                                                                    |
| Read miss                                                                          | one NAND page read; nothing if the FTL has never mapped the page                                                                        |
| Write miss                                                                         | one NAND page read to fill the page, which then becomes dirty                                                                           |
| Eviction of a dirty page                                                           | one NAND page program, charged to the access that caused it                                                                             |
| Prefetched page                                                                    | none for the page itself; a dirty page it evicts is written back at the access's cost                                                   |
| Any access with <code>cache-pages=0</code>                                         | a NAND read or program for every access                                                                                                 |
| Access through a direct mapping ( <code>der=memslot</code> or <code>cylon</code> ) | none, and it is not counted; it never reaches QEMU. A <code>memslot</code> mapped page counts as dirty, so its eviction costs a program |

The access hands the page to the `femu-cxl-ftl` worker, which runs the same BBSSD FTL and NAND model as above and returns the latency. The vCPU then waits out what is left of it with the BQL released: it sleeps until 100 us before the deadline and spins for the rest. Other properties:

- Geometry and NAND times: `channels`, `luns-per-channel`, `pages-per-block`, `blocks-per-plane`, `read-ns`, `program-ns`, `erase-ns`, `channel-ns`, and GC thresholds `gc-threshold`, `gc-threshold-high`.
- `ftl=off` charges no media time at all.
- `cylon-first-touch-program=on` charges a program instead of a free read on the first access to an unmapped page; `cylon-free-writeback=on` makes dirty write-backs free. Both reproduce published Cylon experiments and are off by default.
- `concurrent-misses` decides whether misses to different pages wait for the media together or one after another.

`flush-cache` (QMP) and cache control commands on BAR5 also wait for the media time of the write-backs they cause, on the thread that runs them.

`fast-load=true` removes only the wait at the end of an access; the FTL and the NAND timelines still advance. Setting it back to false waits until the latest LUN and channel busy-until time has passed, so the next access sees an idle model. See Fast load (§4.18.6).

Properties: `femu-cxl-ssd` cache (§A.1.3), NAND geometry and timing (§A.1.3).

## 6.2.11 Changing timing at run time

A BBSSD controller (`femu_mode=1`) accepts the vendor admin command 0xEF, with the action in CDW10. It applies to every namespace of the controller that has a BBSSD FTL. From the guest:

```
| sudo nvme admin-passthru /dev/nvme0 --opcode=0xef --cdw10=2
```

| CDW10 | Effect                                                                                                                                                  |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | GC NAND operations take time (the default)                                                                                                              |
| 2     | GC NAND operations take no time                                                                                                                         |
| 3     | Set the flat read, program and erase times to the built-in 40 us, 200 us and 2 ms                                                                       |
| 4     | Set the flat read, program and erase times to 0                                                                                                         |
| 5     | Print the number of completions posted 20 us or more after they were due and the total number of completions to QEMU's standard output, then reset both |
| 6, 7  | Turn per-command debug logging on or off; it prints only in a build with FEMU_DEBUG_NVME defined                                                        |

Code 3 restores the built-in values, not the ones given on the command line. Codes 3 and 4 change only the flat times: channel bus phases, the `nand_cell_type` tables, `ecc_step_ns` and `trim_lat_ns` stay in force, and write buffer and read cache hits still cost 1 us when `pg_rd_lat` is 0. For a controller linked to a CXL SSD, it restores the medium's read-ns, program-ns, erase-ns and channel-ns instead.

### 6.2.12 Measuring

- **Latency seen by the guest:** run `fio` with a direct I/O engine and read the completion latency (`clat`). With an idle device and queue depth 1, a 4 KiB random read on BBSSD should take about `pg_rd_lat` plus guest and poller overhead. Write the range first: a read of a page that was never written is not charged any NAND time.
- **Write amplification and media counters:** the vendor log page C0h holds the write amplification factor, host and NAND page counts, GC copies, write buffer hits and more. Read it with `sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b`. The layout is in log pages and counters (§B.1).
- **Health:** `sudo nvme smart-log /dev/nvme0` reports Percentage Used and Available Spare from the wear model, and host read and write totals.
- **CXL SSD:** the device's QOM counters, read with `qom-get` through QMP, include `media-time-ns` (total media time charged), `media-reads`, `media-writes`, `cache-hits`, `cache-misses` and the direct mapping counters. `media-full` must stay 0 for a valid measurement. See runtime properties (§A.2.2).

### 6.2.13 Related pages

- Architecture (§3.1)
- Choosing a mode (§4.1)
- Device property reference (§A.1)
- CXL SSD design note<sup>83</sup>

<sup>83</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/cxlssd.md>

## 7 Troubleshooting

### 7.1 Troubleshooting and FAQ

Answers to the questions that come up most often in FEMU's issue tracker. Each one gives a short answer, the fix or workaround, and the issues where it came up. For build errors, see `build.md` (§2.2.6). For crashes and debug output, see `debugging` (§7.2).

1. Which device and mode do I want? (§7.1.1)
2. Which guest kernel does each mode need? (§7.1.2)
3. Where do I get a guest image? (§7.1.3)
4. The build fails (§7.1.4)
5. How large an SSD can I emulate? (§7.1.5)
6. How do I change the geometry or latency without recompiling? (§7.1.6)
7. How do I set the ZNS zone size, zone count and limits? (§7.1.7)
8. How do I choose the NAND cell type? (§7.1.8)
9. Why are reads impossibly fast? (§7.1.9)
10. How do I see garbage collection and write amplification? (§7.1.10)
11. What does FEMU model, and what not? (§7.1.11)
12. How do I run several SSDs or namespaces? (§7.1.12)
13. How do I run FDP? (§7.1.13)
14. Can I still use OCSSD after LightNVM was removed? (§7.1.14)
15. How do I get debug output? (§7.1.15)
16. How do I tune for performance? (§7.1.16)
17. Where is the code for X, and how do I add a command? (§7.1.17)
18. Does data survive a reboot? (§7.1.18)

#### 7.1.1 Which device and mode do I want?

`-device femu` is an NVMe SSD; `femu_mode` picks what is inside it. With no `femu_mode`, you get NoSSD (`femu_mode=2`): an NVMe drive with no media timing. For a conventional SSD with an FTL, garbage collection and write amplification, use BlackBox (`femu_mode=1`, `run-blackbox.sh`). ZNS is `femu_mode=3`, Open-Channel 0, computational storage 4, key-value 5. FDP is not a mode: it is BlackBox with a `femu-subsys,fdp=on`. A CXL memory-semantic SSD is a different device, `femu-cxl-ssd`.

**Fix:** pick from the decision table in choosing a mode (§4.1), then follow that mode's guide.

Related issues: #21, #36, #60, #68, #153.

#### 7.1.2 Which guest kernel does each mode need?

| Mode                      | Guest kernel                                                             |
|---------------------------|--------------------------------------------------------------------------|
| NoSSD, BlackBox, CSD, FDP | any with the NVMe driver                                                 |
| ZNS                       | 5.9 or newer, with <code>CONFIG_BLK_DEV_ZONED=y</code>                   |
| KV                        | 6.0 or newer; the namespace appears only as <code>/dev/ngXnY</code>      |
| OCSSD                     | older than 5.15, with LightNVM; 1.2 needs 4.16, 2.0 needs 4.17           |
| CXL SSD                   | CXL region support ( <code>CONFIG_CXL_REGION</code> and related options) |



The image from `make-guest-image.sh` runs Linux 6.8 and covers every mode except OCSSD. Build it with `--cxl` for the CXL SSD tools.

**Fix:** if `nvme list` shows the controller but no usable namespace, check `uname -r` in the guest against this table and `dmesg | grep nvme`. `requirements.md` (§2.1.2) has the details.

Related issues: #20, #57, #64, #112, #115, #161.

### 7.1.3 Where do I get a guest image?

Build one. `make-guest-image.sh` makes an Ubuntu 24.04 image from the official cloud image, with a femu user, an SSH key, `nvme-cli` and `fio`, and writes it where the launchers look for it, `$HOME/images/u20s.qcow2`:

```
cd build-femu
./make-guest-image.sh
```

#### Fixes for older instructions:

- `qemu-system-x86_64: -localtime: invalid option: QEMU removed -localtime in version 3.1. Use -rtc base=localtime.`
- A launcher says VM disk image couldn't be found: set `OSIMGF` to your image, or `IMGDIR` to its directory.
- WSL is not supported, and in WSL the guest may use only one or two cores. Use a Linux host.

See `guest-image.md` (§2.3).

Related issues: #1, #36, #88, #89, #93.

### 7.1.4 The build fails

FEMU is based on QEMU 10.1, which needs Python 3.9 and GLib 2.66 or newer, so Ubuntu 20.04 and older cannot build it with their own packages.

#### Fixes for the common errors:

- `-Werror` stops the build on a new compiler warning: add `--disable-werror` to the configure line, and report the warning.
- Errors around `nfs_pread_async`: older FEMU trees failed against `libnfs 6`, whose API changed. The current QEMU base ignores `libnfs 6`, and `femu-compile.sh` passes `--disable-libnfs` as well.
- Cannot find Ninja or python venv creation failed: install `ninja-build` and `python3-venv`.
- Errors such as `memfd_create` declared twice, or syntax errors in the ZNS code, were reported against older FEMU versions. Build current master.

The full table is in `build.md` (§2.2.6).

Related issues: #2, #56, #136, #150, #168.

### 7.1.5 How large an SSD can I emulate?

As large as the host's free memory. FEMU keeps the whole device in host DRAM, so a 256 GiB device needs 256 GiB of free host RAM, plus the guest's RAM and the FTL tables. In BlackBox, CSD and KV mode the NAND geometry as a whole (every namespace plus spare space) may hold at most  $2^{31} - 1$  sectors of `secsz` bytes, just under 1 TiB with the default 512-byte sectors; a larger geometry is refused at start.

FEMU allocates the whole device at start-up and tries to lock it in memory. Under sudo the lock normally succeeds, so all of it must be free when QEMU starts. If the allocation itself is more than the kernel will commit, QEMU aborts with failed to allocate N bytes. If only the lock fails, FEMU prints [FEMU] Err: cannot pin the N MiB memory backend and runs anyway.

**Fix:** for NoSSD and ZNS, set devsz\_mb. For BlackBox, CSD and KV, also size the geometry (nchs, luns\_per\_ch, blks\_per\_pl and so on) to hold the namespace plus spare space; with op\_pcent, the host memory used is the full NAND capacity. Host sizing (§5.13.4) explains the memory a device takes.

Related issues: #19, #33, #52, #73, #144; discussion #119.

### 7.1.6 How do I change the geometry or latency without recompiling?

Every setting is a device property on the QEMU command line. Nothing needs a code change, and the old vssd1.conf file is gone.

**Fix:**

- Edit the variables at the top of the launcher (run-blackbox.sh and others), or write the -device femu, . . . line yourself.
- Or describe the device in a config file and expand it with ssd-config.sh (scripts reference (§C.4)).
- BlackBox: pg\_rd\_lat, pg\_wr\_lat, blk\_er\_lat (ns) and the geometry properties. ZNS: zns\_pg\_rd\_lat, zns\_pg\_wr\_lat, zns\_blk\_er\_lat and zns\_num\_ch, zns\_num\_lun, zns\_num\_plane, zns\_num\_blk.
- qemu-system-x86\_64 -device femu,help lists every property, and the property reference (§A.1) explains each one.

This example sets a 4 GiB BlackBox device with 50 us reads:

```
| -device femu,devsz_mb=4096,femu_mode=1,pg_rd_lat=50000,pg_wr_lat=500000,blk_er_lat=3000000
```

Related issues: #19, #21, #73, #193.

### 7.1.7 How do I set the ZNS zone size, zone count and limits?

There is no zone size property. The zone size follows from the namespace size and zns\_num\_ch, zns\_num\_lun, zns\_num\_plane, zns\_num\_blk and zns\_chnls\_per\_zone. Raising zns\_num\_blk gives more, smaller zones. Linux needs a power-of-two zone size.

**Fix:**

- Use the formula and table in the ZNS guide (§4.7.3).
- For 4 KiB logical blocks, set lba\_index=3; no code change is needed.
- zns\_max\_open and zns\_max\_active set the open and active zone limits (0 means no limit).

```
| -device femu,devsz_mb=4096,femu_mode=3,lba_index=3,zns_num_ch=8,zns_num_lun=4,zns_num_plane=2,zns_num_blk=128,zns_max_active=16,zns_max_open=8
```

Related issues: #76, #126, #144, #177, #182.

### 7.1.8 How do I choose the NAND cell type?

Each mode has its own numeric property. Names such as tlc are not accepted:

| Mode              | Property       | Values                                             |
|-------------------|----------------|----------------------------------------------------|
| BlackBox, CSD, KV | nand_cell_type | 0 flat times (default), 1 SLC, 2 MLC, 3 TLC, 4 QLC |

| Mode  | Property                    | Values                                                                                                                                          |
|-------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| ZNS   | <code>zns_flash_type</code> | 1 SLC, 3 TLC, 4 QLC; 2 MLC and 5 PLC only with explicit <code>zns_pg_rd_lat</code> , <code>zns_pg_wr_lat</code> and <code>zns_blk_er_lat</code> |
| OCSSD | <code>flash_type</code>     | 1 SLC, 2 MLC, 3 TLC, 4 QLC                                                                                                                      |

Property `'femu.cell_type'` not found means the name is wrong: use `nand_cell_type`. FEMU has no mode that mixes cell types, such as an SLC cache in front of TLC.

```
| -device femu,devsz_mb=1024,femu_mode=1,nand_cell_type=4
```

Related issues: #50, #113, #132.

### 7.1.9 Why are reads impossibly fast?

In BlackBox and CSD, a read of a page that was never written has no mapping, so it costs no NAND time and returns zeroes. A read benchmark on a fresh device therefore measures only FEMU's overhead. Old versions printed `ppn[-1]` not mapped for such reads; current versions print nothing.

**Fix:** write the range before you read it, for example with a sequential fio write job, as in measuring (§6.1.5).

Related issues: #16, #92.

### 7.1.10 How do I see garbage collection and write amplification?

Read the vendor log page C0h in the guest. Its first 4 bytes are the write amplification factor times 1000, followed by the host, GC and NAND page counts:

```
| sudo nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b | od -An -t u4 -N 4
```

The old SMART vendor bytes no longer hold these counters. GC starts when 75% of the lines are in use (`gc_thres_pcent`), so on a fresh device you must write more than the free space before the WAF rises above 1.000. Random overwrites leave valid pages in every line, which GC must move; sequential overwrites free whole lines, so the WAF stays near 1.

**Fix:** see measuring (§6.1.2) for per-run WAF and the BlackBox guide (§4.4.4) for a workload that makes GC run.

Related issues: #130, #137.

### 7.1.11 What does FEMU model, and what not?

FEMU models:

- NAND read, program and erase times per LUN (per plane for ZNS), with optional channel bus, program and erase suspend, and ECC retry time;
- the BlackBox FTL: page or cached mapping, GC, write buffer, read cache, wear and errors;
- optional PCIe bandwidth, propagation delay and controller firmware time (`pcie_bandwidth_mbps`, `pcie_prop_delay_ns`, `fw_cpu_ns`).

It does not model:

- the time to copy data between guest memory and the device; the copy happens at once, and the model only delays the completion;

- persistence: the device lives in host memory;
- power, heat or temperature changes (the reported temperature is the temperature property);
- interrupt coalescing (the features are reported but not applied).

The latency is a lower bound. A completion is posted on the first poller pass after it is due, so a poller without a host core adds delay. The guest's own driver and block layer add time that the model does not include.

**Fix:** read the timing model (§6.2), and measure FEMU's own overhead on a NoSSD device before you compare against the model.

Related issues: #7, #15, #69, #151.

### 7.1.12 How do I run several SSDs or namespaces?

For several SSDs, add one `-device femu` per SSD, each with its own `id` and `devsz_mb`. For several namespaces on one controller, set `namespace_sizes` and `namespace_modes` give each namespace its own size and mode; double the commas inside them on the QEMU command line:

```
| -device femu,id=nvme0,devsz_mb=4096,femu_mode=1,namespaces=2,namespace_sizes=3G,,1G -device femu,id=nvme1,devsz_mb=1024,femu_mode=3
```

The `serial` property has no effect, and setting it prints a warning at realize. Each controller reports a serial number made of a mode prefix (vSSD, vZNSSD, vNoSSD and so on) and that mode's own counter, which advances once for each namespace the mode sets up. In the example above the serials are vSSD1 and vZNSSD0. Names under `/dev/disk/by-id` therefore depend on the order of the `-device` options.

**Fix:** see several namespaces and devices (§4.15). OCSSD and FDP controllers have one namespace each.

Related issues: #26, #121.

### 7.1.13 How do I run FDP?

FDP is configured on an NVMe subsystem, not on the controller. Create a `femu-subsys` with `fdp=on` first, then a `BlackBox` controller that joins it. `run-blackbox-fdp.sh` does both.

```
| -device femu-subsys,id=subsys0,nqn=subsys0,fdp=on,fdp.nruh=4 -device femu,devsz_mb=4096,femu_mode=1,subsys=subsys0
```

In the guest, `sudo nvme fdp configs /dev/nvme0 -e 1` shows the configuration. See the FDP guide (§4.5).

Related issues: #153.

### 7.1.14 Can I still use OCSSD after LightNVM was removed?

Linux removed LightNVM, its Open-Channel driver, in 5.15. On a newer guest kernel the OCSSD controller appears, but nothing in the kernel can use it.

**Fix:** use a guest kernel older than 5.15 with `CONFIG_NVM` and `pblk`, or drive the device from user space with `SPDK`. Open-Channel 2.0 in current FEMU also accepts plain NVMe Read and Write, which `SPDK` uses. `nvme-cli 2.x` dropped `nvme lnvm`; build `nvme-cli 1.x` in the guest for it. See the OCSSD guide (§4.9).

Related issues: #4, #48, #139.

### 7.1.15 How do I get debug output?

FEMU prints lines that start with `[FEMU]` (`Log:`, `Err:`, `FTL-Log:`, `FTL-Err:`) and, for the ZNS FTL, `[Misao]` `ZFTL-` on QEMU's console. `run-blackbox.sh`, `run-zns.sh` and `run-csd.sh` save the console in `build-femu/`

log; run-blackbox-fdp.sh writes /tmp/femu-fdp.log. There is no FEMU\_DEBUG environment variable, and FEMU defines no QEMU trace events.

**Fix:** compile the debug messages in with `--extra-cflags="-DFEMU_DEBUG_NVME -DFEMU_DEBUG_FTL"`, or use the run-time variables such as `FEMU_FDP_DEBUG`. See debugging (§7.2).

Related issues: #18, #105; discussion #133.

### 7.1.16 How do I tune for performance?

Give every FEMU thread a core. The femu-poller threads and, with a BlackBox, ZNS or CSD namespace, the FEMU-FTL-Thread spin while the guest has the controller enabled, so each needs a host core of its own, on top of the vCPUs. One poller serves every queue by default; `multipoller_enabled=1` starts one per queue, or one per `poller_ratio` queues.

**Fix:** follow performance tuning (§5.12): pin the vCPUs, the pollers and the FTL thread, set the CPU frequency policy to performance, and keep the threads on one NUMA node. `pin.sh` pins the vCPUs, the pollers and the FTL threads (CXL ones included) in one step.

Related issues: #69, #77, #101.

### 7.1.17 Where is the code for X, and how do I add a command?

All FEMU code is under `hw/femu/`. Architecture (§3.1) maps each layer to its files: `femu.c` creates the device, `nvme-admin.c` and `nvme-io.c` handle NVMe commands, and each mode has its own directory (`bbssd/`, `zns/`, `ocssd/`, `nosssd/`, `kvssd/`, `csd/`, `cxlssd/`).

#### To add a command:

- An I/O command for every mode: add a case to `nvme_io_cmd()` in `nvme-io.c`. Opcodes it does not handle go to the namespace's mode, for example `bb_io_cmd()` in `bbssd/bb.c`.
- An admin command: add it to `nvme_admin_cmd()` in `nvme-admin.c`, or to the mode's `admin_cmd` hook, as `bb_admin_cmd()` does for the vendor command `0xEF`.
- To fail a command, return an NVMe status from the handler, for example `return NVME_INVALID_FIELD | NVME_DNR;`. A status found later, in the FTL, goes in `req->status`.

Add a qtest for it (testing (§8.2.6)).

Related issues: #72, #114, #162.

### 7.1.18 Does data survive a reboot?

A guest reboot keeps the data, unless you trigger `simulate-power-loss` on a `power_loss=on` device, which drops the write buffer. Stopping QEMU loses it. FEMU keeps every NVMe namespace in host memory and never writes it to a file; there is no option to make it persistent. Snapshots and migration are refused (security and limits (§5.13.2)).

**Workaround:** keep data you need on the guest's boot disk or copy it out before you stop QEMU, and recreate the device contents after each start.

Related issues: #52.

## 7.2 Debugging

Where FEMU reports problems, how to get more output, and how to run it under a debugger. For answers to common problems, see troubleshooting (§7.1).

### 7.2.1 Where messages go

FEMU prints its messages on QEMU's console:

| Prefix                                            | Stream         | Meaning                                                                 |
|---------------------------------------------------|----------------|-------------------------------------------------------------------------|
| [FEMU] Log:                                       | stdout         | information, such as a vendor command taking effect                     |
| [FEMU] Err:                                       | stderr         | an error FEMU recovered from, such as a CSD program that failed to load |
| [FEMU] FTL-Log:, [FEMU] FTL-Err:                  | stdout, stderr | the BlackBox FTL                                                        |
| [Misao] ZFTL-Log:, [Misao] ZFTL-Err:              | stdout, stderr | the ZNS FTL                                                             |
| [FEMU] FDP-Log:, [FEMU] FDP-Trace:                | stderr         | FDP setup; placement and reclaim traces with FEMU_FDP_DEBUG             |
| qemu-system-x86_64: -device femu,.<br>... MESSAGE | stderr         | the device refused its options and QEMU did not start                   |

run-blackbox.sh, run-zns.sh and run-csd.sh copy the console to build-femu/log, and run-blackbox-fdp.sh to /tmp/femu-fdp.log. run-nossd.sh, run-whitebox.sh and run-cxlssd.sh print only to the terminal. Through the tee pipe, standard output is buffered, so stdout lines can reach the log later than stderr lines. Put `stdbuf -oL` in front of `./qemu-system-x86_64` on the launcher's sudo line to write them at once.

Check the guest's side too. A guest that gives up on the device logs it in dmesg:

```
| sudo dmesg | grep -i nvme
```

FEMU defines no QEMU trace events and does not use QEMU's `-d` log categories, so `-d` and `-trace` show nothing from FEMU itself. They still help with the QEMU code around it. `-d guest_errors,unimp -D qemu.log` logs bad register accesses that QEMU's MSI-X and CXL code detect. Add them to the QEMU command line in the launcher:

```
| -d guest_errors,unimp -D qemu.log
```

### 7.2.2 Run FEMU under gdb

Do not use `gdb-run.sh`; it starts QEMU's stock nvme device, not FEMU. Instead, run a launcher's own command line under gdb. From `build-femu/`, make a copy of the launcher that starts QEMU through gdb, and run it:

```
| sed -e 's|\. /qemu-system-x86_64|gdb -ex "handle SIGUSR1 nostop noprint pass" --args  
./qemu-system-x86_64|' \  
-e 's/ 2>&1 | tee .*$// ' run-blackbox.sh > gdb-blackbox.sh  
bash gdb-blackbox.sh
```

At the gdb prompt, set breakpoints and start QEMU:

```
| (gdb) break femu_realize  
(gdb) run
```

KVM uses SIGUSR1 to kick vCPU threads, which is why gdb is told to pass it on. The guest's serial console shares the terminal with gdb; press Ctrl-C to get back to the gdb prompt.

To attach to a QEMU that is already running instead, on the host:

```
| sudo gdb -p "$(pgrep -x qemu-system-x86)" -ex "handle SIGUSR1 nostop noprint pass"
```

When QEMU crashes, thread apply all bt in gdb prints every thread's stack. The launchers pass -name NAME, debug-threads=on, so info threads shows FEMU's thread names (femu-poller, FEMU-FTL-Thread, femu-csd-cu, femu-cxl-ftl, femu-cxl-cca); add it to a QEMU command line of your own (performance tuning (§5.12.1)).

### 7.2.3 Debug builds and compile-time switches

femu-compile.sh builds with -O2 -g. gdb works on that build, but many variables are optimized out. To single-step, configure an unoptimized build yourself from build-femu/:

```
| ../configure --enable-kvm --target-list=x86_64-softmmu --enable-slirp \  
|     --disable-libnfs --disable-libiscsi --disable-curl \  
|     --enable-debug --enable-debug-info  
| make -j"$(nproc)"
```

The debug messages (Dbg:.) have no run-time switch. These macros, passed with --extra-cflags, compile them in:

| Macro           | Effect                                                                                                                        |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------|
| FEMU_DEBUG_NVME | controller debug messages ([FEMU] Dbg:), and the per-command log that vendor command 0xEF with CDW10 6 and 7 turns on and off |
| FEMU_DEBUG_FTL  | BlackBox FTL debug messages ([FEMU] FTL-Dbg:), and the FTL invariant checks of BlackBox and ZNS                               |
| FEMU_DEBUG_ZFTL | ZNS FTL debug messages ([Misao] ZFTL-Dbg:)                                                                                    |
| FEMU_FTL_ASSERT | the FTL invariant checks only, without the messages                                                                           |

For example:

```
| ../configure --enable-kvm --target-list=x86_64-softmmu --enable-slirp \  
|     --disable-libnfs --disable-libiscsi --disable-curl \  
|     --extra-cflags="-DFEMU_DEBUG_FTL -DFEMU_DEBUG_NVME"  
| make -j"$(nproc)"
```

The invariant checks (ftl\_assert) are compiled out of a normal build. A mapping or bound error there corrupts FTL state quietly instead of stopping QEMU. When a BlackBox or ZNS run produces wrong data or impossible counters, rebuild with -DFEMU\_FTL\_ASSERT first. The debug messages print on the I/O path, so use them with small workloads.

The sanitizer build (§8.2.3) adds AddressSanitizer and UndefinedBehaviorSanitizer on top of the FTL checks. Run the failing workload on it to find memory errors.

### 7.2.4 Run-time switches

FEMU reads a few environment variables (environment variables (§A.1.5)):

| Variable                             | Prints to stderr                                                                         |
|--------------------------------------|------------------------------------------------------------------------------------------|
| FEMU_FDP_DEBUG (any value)           | FDP placement and reclaim traces                                                         |
| FEMU_EXP_LOG=1 with FEMU_SECRET=TEXT | writes, overwrites, deallocations, GC moves and erases of pages whose data contains TEXT |
| FEMU_DUMP_LPN=N                      | a hex dump of logical page N on every read not served from the write buffer              |
| FEMU_KV_SELFTEST (any value)         | the result of a KV FTL self-test run once at start                                       |

The launchers other than `run-cxlssd.sh` start QEMU with `sudo`, which drops your environment. `run-blackbox.sh` passes `FEMU_EXP_LOG`, `FEMU_SECRET` and `FEMU_DUMP_LPN` through, so this works:

```
| FEMU_EXP_LOG=1 FEMU_SECRET=MARKER ./run-blackbox.sh
```

For the others, add the variable to the launcher's `sudo` line, for example `sudo FEMU_FDP_DEBUG=1 ./qemu-system-x86_64 \` in `run-blackbox-fdp.sh`.

On a running BlackBox controller, the vendor admin command `0xEF` switches GC time and NAND time on and off and prints poller counts (timing model (§6.2.11)).

## 7.2.5 Inspect a running device

QEMU's monitor shows what QEMU built. With the QMP socket the launchers other than `run-cxlssd.sh` create (`build-femu/qmp-sock`), on the host:

```
| printf '%s\n' '{"execute":"qmp_capabilities"}' '{"execute":"query-pci"}' |
| sudo socat - UNIX-CONNECT:./qmp-sock
```

`qom-list` and `qom-get` on `/machine/peripheral/<id>` read a device's properties and counters (runtime properties (§A.2)). `run-nossd.sh` names its controller `nvme0` and `run-cxlssd.sh` names its device `cxlssd`. The other launchers give the controller no `id=`, so add one to address it by name.

## 7.2.6 Common crash reports

| Symptom                                                                                         | What it means                                                                         | What to do                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>qemu-system-x86_64: -device femu,...: MESSAGE and QEMU exits</code>                       | A property value was refused at start                                                 | Read the message; the mode guide's "Limits and refusals" table explains it                                                                                                       |
| <code>[FEMU] Err: cannot pin the N MiB memory backend</code>                                    | Not a crash. QEMU runs without <code>sudo</code> and the memory lock limit is too low | Ignore it, or raise <code>ulimit -l</code> (requirements (§2.1.1))                                                                                                               |
| <code>failed to allocate N bytes, then QEMU stops with a trap or abort and may dump core</code> | The host has less free memory than the device needs                                   | Lower <code>devsz_mb</code> or free memory (host sizing (§5.13.4))                                                                                                               |
| <code>Guest dmesg shows nvme nvme0: I/O ... timeout, then a controller reset</code>             | The guest's NVMe driver waited too long for a completion and reset the controller     | Check the QEMU console for an error at the same time. If there is none, check that the pollers have cores (performance tuning (§5.12)). If it repeats on an idle host, report it |
| The device is missing in the guest, and the console shows <code>[FEMU] Err:</code> lines        | FEMU refused a command or a configuration the guest used                              | Read the console; for ZNS, see the ZNS troubleshooting (§4.7.7)                                                                                                                  |



| Symptom                                              | What it means | What to do                                                                  |
|------------------------------------------------------|---------------|-----------------------------------------------------------------------------|
| QEMU exits with a segmentation fault or an assertion | A FEMU bug    | Run it under gdb or the sanitizer build, and report it with the steps below |

Older reports of crashes in FDP garbage collection (#186, #189, #191) and of controllers disabled under load (#22, #30) were made against older FEMU versions. Reproduce on current master before you report one.

### 7.2.7 Reporting a bug

Open an issue with:

- the FEMU commit (`git log -1 --oneline`) and how you built it;
- the full QEMU command line, or the launcher and every change you made to it;
- the guest kernel (`uname -r`) and the commands you ran in the guest;
- the QEMU console output and the guest's `dmesg`;
- for a crash, the thread `apply_all` bt output from gdb, or the sanitizer report.

Report a security problem privately, as SECURITY.md<sup>84</sup> says.

### 7.2.8 Related pages

- Testing (§8.2)
- Troubleshooting (§7.1)
- Architecture (§3.1)

Related issues: #18, #22, #30, #105; discussion #133.

<sup>84</sup><https://github.com/MoatLab/FEMU/blob/master/SECURITY.md>

## 8 Contributing

### 8.1 Code structure

All FEMU code, scripts, tests and documentation live under `hw/femu/`, so the rest of the tree stays QEMU. A top-level `femu-scripts` link points to `hw/femu/scripts/`, which keeps the `cd build-femu && ../femu-scripts/...` workflow working. How the pieces fit together at run time, layer by layer, is in architecture (§3.1).

#### 8.1.1 Directory map

| Path                                      | What is there                                                                                                                                                                                   |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>femu.c</code> <sup>85</sup>         | QOM types <code>femu</code> and <code>femu-subsys</code> , property definitions, <code>realize</code> and <code>exit</code> , the FTL thread, mode registration                                 |
| <code>femu-props.c</code> <sup>86</sup>   | Help text for every <code>femu</code> and <code>femu-subsys</code> property ( <code>what -device femu,help</code> prints)                                                                       |
| <code>nvme.h</code> <sup>87</sup>         | NVMe structures, the <code>femu_mode</code> enum and the controller state <code>FemuCtrl</code>                                                                                                 |
| <code>nvme-admin.c</code> <sup>88</sup>   | Admin commands, starting the pollers, namespace management, asynchronous events                                                                                                                 |
| <code>nvme-caps.c</code> <sup>89</sup>    | What the controller advertises: the Commands Supported and Effects and Supported Log Pages entries, and the Identify bits derived from them                                                     |
| <code>nvme-io.c</code> <sup>90</sup>      | I/O commands and the poller loop that fetches submissions and posts completions                                                                                                                 |
| <code>nvme-util.c</code> <sup>91</sup>    | Deallocation state per LBA (TRIM, Write Zeroes with deallocate, DULBE), queue head and tail and completion posting helpers, poller pause and resume, the Timestamp feature                      |
| <code>nvme-pel.c</code> <sup>92</sup>     | Persistent Event log and its <code>pel_file</code>                                                                                                                                              |
| <code>nvme-pi.c</code> <sup>93</sup>      | Metadata and protection information                                                                                                                                                             |
| <code>nvme-streams.c</code> <sup>94</sup> | Streams directive                                                                                                                                                                               |
| <code>dma.c</code> <sup>95</sup>          | PRP and SGL mapping, copies between guest memory and the device                                                                                                                                 |
| <code>intr.c</code> <sup>96</sup>         | MSI-X, MSI and pin interrupts                                                                                                                                                                   |
| <code>bbssd</code> <sup>97</sup>          | BlackBox mode ( <code>bb.c</code> ) and its FTL: geometry, data path, mapping schemes, read cache, GC and lines, FDP, the bridge to the NAND media layer                                        |
| <code>zns</code> <sup>98</sup>            | ZNS mode ( <code>zns.c</code> ) and its zone FTL ( <code>zftl.c</code> )                                                                                                                        |
| <code>ocssd</code> <sup>99</sup>          | Open-Channel 1.2 ( <code>oc12.c</code> ) and 2.0 ( <code>oc20.c</code> ); their <code>flash_type</code> times, geometry check and vendor command <code>0xEE</code> ( <code>oc-timing.c</code> ) |
| <code>nosssd</code> <sup>100</sup>        | NoSSD mode ( <code>nop.c</code> )                                                                                                                                                               |

<sup>85</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/femu.c>

<sup>86</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/femu-props.c>

<sup>87</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nvme.h>

<sup>88</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nvme-admin.c>

<sup>89</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nvme-caps.c>

<sup>90</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nvme-io.c>

<sup>91</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nvme-util.c>

<sup>92</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nvme-pel.c>

<sup>93</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nvme-pi.c>

<sup>94</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/nvme-streams.c>

<sup>95</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/dma.c>

<sup>96</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/intr.c>

<sup>97</sup><https://github.com/MoatLab/FEMU/tree/master/hw/femu/bbssd>

<sup>98</sup><https://github.com/MoatLab/FEMU/tree/master/hw/femu/zns>

<sup>99</sup><https://github.com/MoatLab/FEMU/tree/master/hw/femu/ocssd>

<sup>100</sup><https://github.com/MoatLab/FEMU/tree/master/hw/femu/nosssd>

| Path                                      | What is there                                                                                  |
|-------------------------------------------|------------------------------------------------------------------------------------------------|
| kvssd/ <sup>101</sup>                     | Key-value mode: commands, its FTL, Identify and features                                       |
| csd/ <sup>102</sup>                       | Computational storage mode and its private commands                                            |
| cxlssd/ <sup>103</sup>                    | femu-cxl-ssd: QOM glue, the page cache, the DER modes, the caching API                         |
| nand/ <sup>104</sup>                      | NAND media layer: per-cell-type timing tables and the timing of each operation                 |
| backend/ <sup>105</sup>                   | The DRAM backend that holds the emulated medium                                                |
| lib/ <sup>106</sup> , inc/ <sup>107</sup> | Lock-free rings and the priority queue, and their headers                                      |
| scripts/ <sup>108</sup>                   | Build and launch scripts, configs, guest tools, documentation tooling (scripts reference (§C)) |
| tools/ <sup>109</sup>                     | Guest tools for the CXL caching API                                                            |
| tests/ <sup>110</sup>                     | Unit tests, the qtest file, CSD guest tests (testing (§8.2))                                   |
| docs/ <sup>111</sup>                      | This documentation (doc map <sup>112</sup> )                                                   |

### 8.1.2 Making a change

CONTRIBUTING.md<sup>113</sup> has the process: style (checkpatch.pl), tests, sign-off and pull requests. For a new property, add its help text in femu-props.c and regenerate the property reference (keeping the documentation correct (§8.3)). For a new mode, add it to the femu\_mode enum in nvme.h, register its handlers the way the existing modes do in femu.c, and add an entry to docs/modes.py.

## 8.2 Testing

FEMU has five kinds of tests. The first four run without a guest and run in CI; the last runs inside a booted guest.

| Tests                | Where                               | What they cover                                                                                                                                      | Run with                                                                            |
|----------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Unit tests           | hw/femu/tests/unit/                 | the NAND timing math, the completion priority queue, the hybrid mapping, the CXL cache, page table entries and caching API ring                      | make -C hw/femu/tests check and meson test in a build; each runs a different subset |
| Device tests (qtest) | hw/femu/tests/qtest/<br>femu-test.c | about 400 cases that drive the controller through its registers with no guest: every mode, admin and I/O commands, error paths, fuzzers, the CXL SSD | qos-test                                                                            |

<sup>101</sup><https://github.com/MoatLab/FEMU/tree/master/hw/femu/kvssd>

<sup>102</sup><https://github.com/MoatLab/FEMU/tree/master/hw/femu/csd>

<sup>103</sup><https://github.com/MoatLab/FEMU/tree/master/hw/femu/cxlssd>

<sup>104</sup><https://github.com/MoatLab/FEMU/tree/master/hw/femu/nand>

<sup>105</sup><https://github.com/MoatLab/FEMU/tree/master/hw/femu/backend>

<sup>106</sup><https://github.com/MoatLab/FEMU/tree/master/hw/femu/lib>

<sup>107</sup><https://github.com/MoatLab/FEMU/tree/master/hw/femu/inc>

<sup>108</sup><https://github.com/MoatLab/FEMU/tree/master/hw/femu/scripts>

<sup>109</sup><https://github.com/MoatLab/FEMU/tree/master/hw/femu/tools>

<sup>110</sup><https://github.com/MoatLab/FEMU/tree/master/hw/femu/tests>

<sup>111</sup><https://github.com/MoatLab/FEMU/tree/master/hw/femu/docs>

<sup>112</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/README.md>

<sup>113</sup><https://github.com/MoatLab/FEMU/blob/master/CONTRIBUTING.md>

| Tests                | Where                                                    | What they cover                                                          | Run with                         |
|----------------------|----------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------|
| Documentation checks | hw/femu/scripts/                                         | the property reference, mode tables, links and every example in the docs | make -C hw/femu/tests check-docs |
| Configuration files  | hw/femu/scripts/ssd-config-test.sh                       | every file in scripts/configs/ starts QEMU                               | ssd-config-test.sh<br>BINARY     |
| Guest-side tests     | hw/femu/scripts/, hw/femu/tests/csd/, hw/femu/tools/cca/ | the device as Linux sees it                                              | inside the guest                 |

CONTRIBUTING.md asks for a test that fails without your change and passes with it.

### 8.2.1 Build for testing

The qtests and the documentation checks need QEMU and qos-test. A debug build is the usual choice:

```
mkdir build && cd build
../configure --target-list=x86_64-softmmu --disable-docs --enable-debug --enable-slirp
ninja qemu-system-x86_64 tests/qtest/qos-test
```

### 8.2.2 Unit tests

Without a QEMU build, from the source tree:

```
make -C hw/femu/tests check
make -C hw/femu/tests clean
```

This compiles the NAND media, hybrid mapping, CXL cache, page table entry and caching API ring tests against a stub `qemu/osdep.h` and runs them, in about a second. The CXL cache test needs GLib development files and is skipped without them. In a configured build directory, meson runs the NAND media, priority queue and hybrid mapping tests against QEMU's real headers. CI runs both:

```
cd build
./pyvenv/bin/meson test test-femu-nand-media test-femu-pqueue test-femu-hybrid-oracle \
--print-errorlogs
```

### 8.2.3 Device tests (qtest)

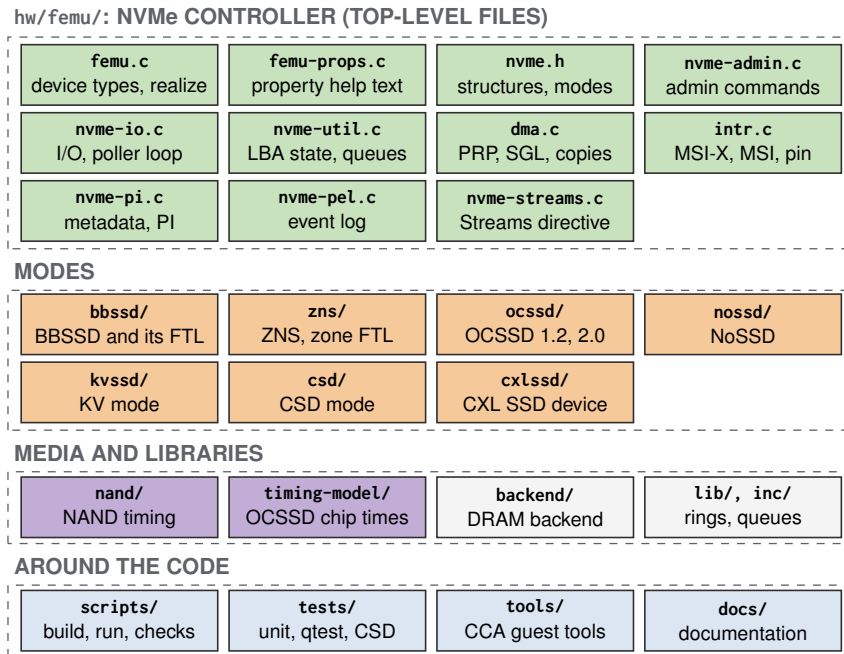
From the build directory, run every FEMU qtest:

```
QTEST_QEMU_BINARY=./qemu-system-x86_64 ./tests/qtest/qos-test -m quick \
-p /x86_64/pc/i440FX-pcihost/pci-bus-pc/pci-bus/femu/femu-tests
```

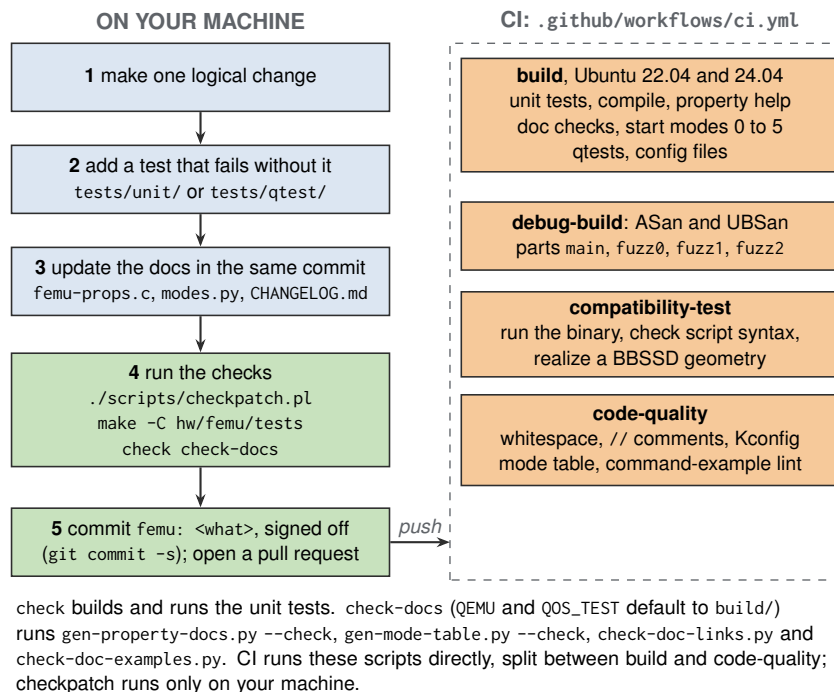
List them, or run one by its full path:

```
QTEST_QEMU_BINARY=./qemu-system-x86_64 ./tests/qtest/qos-test -l |
sed -n 's|^# \(.*femu-tests/.*)|\1|p'
QTEST_QEMU_BINARY=./qemu-system-x86_64 ./tests/qtest/qos-test -m quick \
-p /x86_64/pc/i440FX-pcihost/pci-bus-pc/pci-bus/femu/femu-tests/cc-states
```

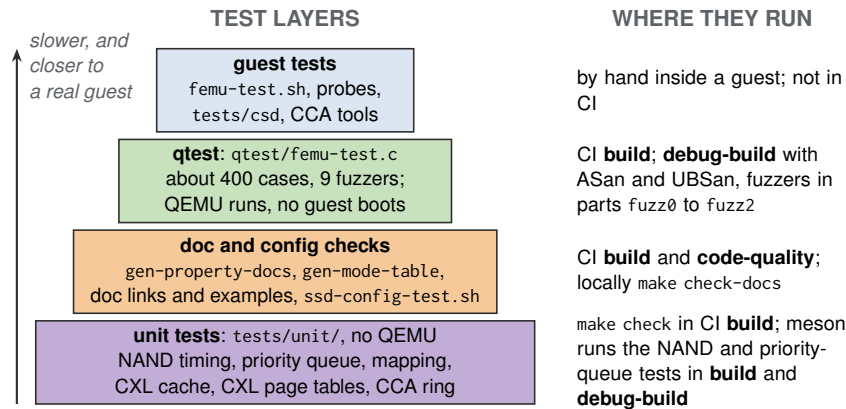
Each case starts its own QEMU under the qtest accelerator, so no KVM and no guest are needed. The default device is a 64 MiB NoSSD controller at PCI address 04.0; a case can add its own device options. The CXL cases start their own machine with a `femu-cxl-ssd`, which the `x86_64-softmmu` build includes by default.



**Figure 27:** The FEMU source tree under hw/femu/: the NVMe controller files at the top level, one directory per mode, the shared media and library code, and the scripts, tests, tools and docs around them.



**Figure 28:** Contributing a change: one logical change with a failing test and its docs, checked locally with checkpatch, make check and make check-docs, then the four CI jobs run on the pull request.



**Figure 29:** FEMU’s test layers, fastest at the bottom: unit tests without QEMU, doc and config checks, qtests that run QEMU without a guest, and guest tests run by hand.

## Sanitizer build

CI also runs the unit tests and qtests in a build with AddressSanitizer, UndefinedBehaviorSanitizer and the FTL invariant checks on. A bound violation that the normal build silently survives aborts there:

```
mkdir build-debug && cd build-debug
../configure --enable-kvm --target-list=x86_64-softmmu --enable-slirp \
  --disable-libnfs --disable-libiscsi --disable-curl \
  --enable-asan --enable-ubsan --extra-cflags=-DFEMU_FTL_ASSERT
make -j"$(nproc)"
```

Run the qtests in that build with the sanitizer options CI uses. Without `abort_on_error=1`, a sanitizer report does not fail the test:

```
export ASAN_OPTIONS=detect_leaks=0:abort_on_error=1
export UBSAN_OPTIONS=print_stacktrace=1:halt_on_error=1
QTEST_QEMU_BINARY=./qemu-system-x86_64 ./tests/qtest/qos-test -m quick \
  -p /x86_64/pc/i440FX-pcihost/pci-bus-pc/pci-bus/femu/femu-tests
```

The fuzz cases (names containing fuzz) take most of the sanitized run time, and CI splits them over three jobs.

### 8.2.4 Documentation checks

From the source tree, with the build above in `build/`:

```
make -C hw/femu/tests check-docs
```

It regenerates the property reference and mode tables and compares them, checks every relative link, and runs every example tagged `femu-example` under `qtest`. Set `QEMU=` and `QOS_TEST=` for a build elsewhere. `docs-maintenance.md` (§8.3) says how to tag a new example.

### 8.2.5 Guest-side tests

These need a booted guest; CI does not run them. Copy the files into the guest first, for example with `scp -P 8080` and the key from `make-guest-image.sh`.

`femu-test.sh` checks one namespace end to end: data written comes back, the counters move, deallocate works, and zones or key-value commands answer when the namespace has them. **It overwrites the whole**

## namespace:

```
| sudo bash femu-test.sh --yes /dev/nvme0n1
```

It prints PASS, FAIL and SKIP lines and a summary, and exits non-zero when a check failed, so it can gate a script. A read that fails counts as a failure, the same as a bad checksum. It refuses a mounted device. A key-value namespace has no block node, so it is driven by passthrough through its generic node, /dev/ngXnY; name it as /dev/nvmeXnY or /dev/ngXnY. It makes no timing claims. A BlackBox run prints lines like these (N stands for the counts; ... for lines left out here). max\_block\_reads is the read count of the most-read block since its erase:

```
== device ==
PASS controller answers Identify
== data survives the FTL ==
PASS random write then verify (crc32c)
== deallocate ==
PASS deallocate accepted
...
== counters ==
PASS SMART log readable
data_units_written=N host_write_commands=N
...
waf_x1000=N host_pages=N nand_pages=N max_block_reads=N read_reclaims=N
PASS host writes counted
PASS write amplification reported

FEMU_TEST pass=N fail=0 skip=N
```

Smaller probes, built in the guest with `gcc -O2 -o NAME NAME.c`:

| Probe            | Run                                                        | Checks                                                                       |
|------------------|------------------------------------------------------------|------------------------------------------------------------------------------|
| kv-probe.c       | <code>sudo ./kv-probe /dev/nvme0</code>                    | key-value Store, Exist, Retrieve and Delete                                  |
| aer-probe.c      | <code>sudo ./aer-probe /dev/nvme0</code>                   | a temperature event completes an Asynchronous Event Request                  |
| zone-aen-probe.c | <code>sudo ./zone-aen-probe /dev/nvme0 /dev/nvme0n1</code> | Zone Descriptor Changed notices, on a ZNS device with err_write_fail_ppm set |

## Mode-specific suites:

- FDP: `fdp-test-nvme-admin.sh` checks the FDP admin commands against the device `run-blackbox-fdp.sh` creates.
- CSD: the programs and guest tool in `hw/femu/tests/csd/` (CSD guide (§4.13.5)).
- CXL caching API: `hw/femu/tools/cca/run-guest-tests.sh` (CCA guide (§4.20.7)).

All of them are listed in the scripts reference (§C.5).

## 8.2.6 Adding a test

### A qtest

1. Write a function in `hw/femu/tests/qtest/femu-test.c` with the signature `static void femu_test_NAME(void *obj, void *data, QGuestAllocator *alloc)`. `obj` is the QFemu device; the existing helpers enable the controller, create queues and submit commands. Read a similar case first.

2. Register it in `femu_register_nodes()`. For the default device:

```
| qos_add_test("my-case", "femu", femu_test_my_case, NULL);
```

For other device options:

```
| qos_add_test("my-bbssd-case", "femu", femu_test_my_case,
|             &(QOSGraphTestOptions) {
|                 .edge.extra_device_opts =
|                 "serial=my-bbssd-case,devsz_mb=4,femu_mode=1,secs_per_pg=8,"
|                 "pgs_per_blk=4,blks_per_pl=128,pls_per_lun=1,luns_per_ch=2,nchs=2"
|             });
```

3. Rebuild `tests/qtest/qos-test` and run the case by its path.
4. Show that it can fail: revert your fix, or plant the bug it is meant to catch, and check that the case goes red.

### A unit test

Put it in `hw/femu/tests/unit/` and add it to `UNIT_TESTS`, a build rule and the `clean` rule in `hw/femu/tests/Makefile`. If it should also build against QEMU's headers, add it to `femu_unit_tests` in `hw/femu/tests/meson.build`.

### A documentation example

Put the command line in a code block with a `femu-example` tag. The check starts it, enables each controller, and writes and reads one block. See `docs-maintenance.md` (§8.3.2).

## 8.2.7 Related pages

- Debugging (§7.2)
- Scripts reference (§C)
- CONTRIBUTING.md<sup>114</sup>

## 8.3 Keeping the documentation correct

Four checks keep the documentation in step with the code. CI runs all of them, and so does

```
| make -C hw/femu/tests check-docs QEMU=$PWD/build/qemu-system-x86_64 \
|     QOS_TEST=$PWD/build/tests/qtest/qos-test
```

| Check              | Script                                                    | Fails when                                                                                                                                     |
|--------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Property reference | <code>hw/femu/scripts/gen-property-docs.py --check</code> | a property has no description, or <code>reference/properties.md</code> or <code>reference/runtime-properties.md</code> differs from the binary |
| Mode table         | <code>hw/femu/scripts/gen-mode-table.py --check</code>    | a mode table differs from <code>modes.py</code> , or <code>modes.py</code> disagrees with the tree                                             |
| Links              | <code>hw/femu/scripts/check-doc-links.py</code>           | a relative link or heading anchor does not exist                                                                                               |
| Examples           | <code>hw/femu/scripts/check-doc-examples.py</code>        | a code block is not tagged, or a tagged example does not work                                                                                  |

<sup>114</sup><https://github.com/MoatLab/FEMU/blob/master/CONTRIBUTING.md>



Only two reference pages are generated: `reference/properties.md` and `reference/runtime-properties.md`, from the built binary plus `reference/property-topics.py` and `reference/environment.inc.md`. The others, `reference/log-pages-and-counters.md` and `reference/scripts.md`, are written by hand and no check compares them with the code, so update them in the same commit as the change they describe. A user-visible change also needs an entry in `CHANGELOG.md` (§D).

### 8.3.1 Per-mode facts: `modes.py`

`hw/femu/docs/modes.py`<sup>115</sup> holds one entry per mode or feature: how to turn it on, a minimal example, the guest kernel and tools it needs, host requirements, its launcher and the page that documents it. Change a fact there, never in a table, then run

```
| python3 hw/femu/scripts/gen-mode-table.py
```

The script rewrites the text between `<!-- modes-table:start -->` and `<!-- modes-table:end -->` in `README.md` and in any page under `hw/femu/docs/` that has the two markers. To put the table on a new page, add the markers and run the script. `README.md` must keep its markers.

`--check` also compares `modes.py` with the code. Each symbol must have the stated `femu_mode` value in `hw/femu/nvme.h`, every mode in that enum must have an entry, each launcher must exist in `hw/femu/scripts/`, and each guide link must resolve. Each entry's example is run by the example check below, so the "Checked" column states what CI really does with it.

### 8.3.2 Code blocks in the documentation

Every fenced code block in `README.md` and `hw/femu/docs/**` falls into one of these classes:

| Block                                                                                        | Meaning                                                                                                                         | Checked                                   |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| preceded by <code>&lt;!-- femu-example: NAME --&gt;</code>                                   | a QEMU command line, <code>-device</code> options or a <code>run-*.sh</code> launcher                                           | run under <code>qtest</code>              |
| preceded by <code>&lt;!-- femu-untested: REASON --&gt;</code>                                | a command that cannot run in CI                                                                                                 | the reason must have at least three words |
| <code>```sh</code>                                                                           | shell commands that do not start FEMU: <code>git</code> , <code>apt</code> , <code>configure</code> , commands inside the guest | must not start FEMU                       |
| <code>```bash</code> , <code>```shell</code> , <code>```console</code> , <code>```zsh</code> | not allowed untagged                                                                                                            | fails                                     |
| any other info string, or none                                                               | output, configuration files, code                                                                                               | must not start FEMU                       |

A block "starts FEMU" when, outside shell comments, it runs `qemu-system-*` as a command, passes `-device femu...`, or names a `run-*.sh` launcher other than `run-guest-ssh.sh`. The tag comment goes on the line right before the fence. GitHub does not show it and still highlights the block.

#### Writing an example

```
| <!-- femu-example: zns-small -->
| -device femu,devsz_mb=1024,femu_mode=3
```

- One example per block, or several separated by blank lines; each is named `NAME`, `NAME.2`, and so on in the report.

<sup>115</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/docs/modes.py>

- ... stands for options left out and is dropped, so `-device femu,...,femu_mode=1,read_reclaim_limit=100000` is tested as `-device femu,femu_mode=1,read_reclaim_limit=100000`.
- A launcher such as `./run-zns.sh` (with optional `VAR=value` settings in front) is run with stand-ins for `sudo` and `QEMU` that record the command line it builds, so the launcher's own options are what gets tested. `OSIMGF` is replaced by an empty file, and lines that run `run-guest-ssh.sh` are skipped.
- A list of `key=value` lines can be tested as one device: `<!-- femu-example: zns-params; device: femu,femu_mode=3 -->`.
- `io: rw, io: kv, io: identify` or `io: none` overrides what the I/O stage does; by default it follows the mode, from `modes.py`.
- `allow-warning: TEXT` allows one expected warning, for example `allow-warning: FEMU CXL DER unavailable for der=cylon`, which falls back to MMIO without a Cylon host.

### What the example check does

1. QEMU starts with every `-device` of the example, in its order, plus the memory backends and `-machine` options (q35 when none is given), under `-accel qtest -S`. Guest disks and NICs are realized too, so one the machine cannot plug where it lands fails here: on a `cxl=on` machine a `virtio-net-pci` or `virtio-blk-pci` without `bus=pcie.0` fails with "Only PCI/PCIe bridges can be plugged into pxb-cxl". Their backends are replaced by stand-ins that touch nothing on the host: each `-drive` by a `null-co` block device with the same `id` and `if`, each `-netdev` and `-net user` by user networking with no forwarded ports (so the QEMU under test needs `--enable-slirp`). The other options are dropped, among them `-enable-kvm`, `-cpu`, `-smp`, `-m`, `-numa`, `-name` and `-qmp`. QMP must show every `femu`, `femu-subsys` and `femu-cxl-ssd` created, and `query-pci` must list each `femu` as an NVMe controller. Anything on `stderr` fails the example, except FEMU's `[FEMU] Log:` lines, the notice that the memory backend could not be pinned (the check lowers `RLIMIT_MEMLOCK` so that it never pins), and allowed warnings. Test-only properties (`x-...`) are refused.
2. For an example with an NVMe controller, the `doc-examples` case in `hw/femu/tests/qtest/femu-test.c` starts the FEMU devices, the CXL topology and the memory backends from step 1, without the guest's disks and NICs, enables each controller, sends `Identify`, and writes and reads back one block of namespace 1 (stores and retrieves one value in KV mode). Open-Channel controllers stop after `Identify`. A controller whose namespace 1 is not attached stops after `Identify`, but at least one controller in the example must move data.

The check does not boot a guest. Guest kernel versions, guest tools and anything done inside the guest are outside what it can see.

`--self-test` runs planted mistakes (an untagged block, a misspelled property, `femu_mode=9`, a warning-only `der=cylon`, an I/O-only failure, a test-only property, a `virtio-net-pci` without `bus=` on a `cxl=on` machine, a guest disk naming a drive that does not exist) and requires each to fail with the expected message, and two good examples, one with guest devices on a CXL machine, to pass, so a check that stopped catching them fails rather than passing everything.

```
python3 hw/femu/scripts/check-doc-examples.py --lint          # tags only
python3 hw/femu/scripts/check-doc-examples.py --list          # what would run
python3 hw/femu/scripts/check-doc-examples.py \
    --qemu build/qemu-system-x86_64 --qos-test build/tests/qtest/qos-test \
    --only quick-start-bbssd
```

## A Property reference

### A.1 Device property reference

Every property of the FEMU devices, as the binary reports it. `./qemu-system-x86_64 -device femu,help` prints the same descriptions at the terminal. Properties are set on the command line, for example `-device femu,femu_mode=1,devsz_mb=4096`, and only before the device is realized; the QOM properties that can be read or changed while the guest runs are in `runtime-properties.md` (§A.2).

Mode names follow `femu_mode`: OCSSD (0), bbssd (1, black box), NoSSD (2), ZNS (3), CSD (4) and KV (5). CSD builds the bbssd FTL, so the bbssd geometry, timing and FTL properties apply to it; KV uses only the bbssd geometry and NAND timing. Each description names the modes it applies to.

Devices:

- `femu`: NVMe controller (§A.1.1)
- `femu-subsys`: NVMe subsystem (§A.1.2)
- `femu-cxl-ssd`: CXL Type-3 SSD (§A.1.3)
- Test-only properties (§A.1.4)
- Environment variables (§A.1.5)

#### A.1.1 `femu`: NVMe controller

`-device femu` has 146 properties of its own and 1 QOM property listed in `runtime-properties.md` (§A.2).

##### Mode, capacity and namespaces

| Property                     | Type                                  | Default | Description                                                                                                                                                                                                                                           |
|------------------------------|---------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>femu_mode</code>       | <code>uint8</code>                    | 2       | Emulated SSD type: 0 OCSSD (Open-Channel), 1 bbssd (black-box FTL), 2 NoSSD (no media timing), 3 ZNS, 4 CSD (computational), 5 KV; other values fail realize                                                                                          |
| <code>devsz_mb</code>        | <code>uint32</code>                   | 1024    | Size of the host memory backend in MiB, split across namespaces unless <code>namespace_sizes</code> is set; bbssd with <code>op_pcent</code> sizes from the NAND geometry instead                                                                     |
| <code>namespaces</code>      | <code>uint32</code>                   | 1       | Number of namespaces created at boot, 1 to 256; OCSSD and FDP support only 1                                                                                                                                                                          |
| <code>namespace_sizes</code> | <code>str</code>                      | unset   | Comma-separated size in bytes of each namespace (QEMU size syntax such as 4G,2G), one non-empty entry per namespace, summing to at most the backend; unset splits the backend evenly                                                                  |
| <code>namespace_modes</code> | <code>str</code>                      | unset   | Comma-separated mode of each namespace from <code>nossd</code> , <code>bbssd</code> , <code>znssd</code> , <code>ocssd</code> , <code>csd</code> and <code>kvssd</code> , one entry per namespace; unset gives every namespace <code>femu_mode</code> |
| <code>op_pcent</code>        | <code>uint32</code>                   | 0       | Over-provisioning in percent for bbssd: back the device with the full NAND capacity and expose <code>capacity/(1 + op_pcent/100)</code> ; 0 keeps <code>devsz_mb</code> sizing, not with <code>cxl-ssd</code>                                         |
| <code>subsys</code>          | <code>link&lt;femu-subsys&gt;</code>  | unset   | ID of a <code>femu-subsys</code> device to join, created before this controller; needed for FDP, shared namespaces and the Endurance Group log                                                                                                        |
| <code>cxl-ssd</code>         | <code>link&lt;femu-cxl-ssd&gt;</code> | unset   | ID of a <code>femu-cxl-ssd</code> , listed before this controller, whose memory and FTL this bbssd controller serves as its one namespace                                                                                                             |

| Property | Type | Default | Description                                                                                                                                       |
|----------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| pel_file | str  | unset   | Host file that keeps the Persistent Event Log and power cycle count across runs, created if missing; a corrupt or incompatible file fails realize |
| serial   | str  | unset   | No effect, kept for compatibility; Identify Controller reports a serial number FEMU generates. Setting it warns at realize                        |

### Queues, pollers and interrupts

| Property            | Type   | Default | Description                                                                                                                                                       |
|---------------------|--------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| queues              | uint32 | 8       | Number of I/O submission and completion queue pairs, 1 to 2047; MSI-X vectors are queues + 1                                                                      |
| entries             | uint32 | 2047    | Value reported as CAP.MQES (0's based), 1 to 65534; the controller accepts queues of up to entries + 1 entries                                                    |
| max_sqes            | uint8  | 6       | Submission queue entry size as a power of two (Identify SQES); must be 6, 64-byte entries                                                                         |
| max_cqes            | uint8  | 4       | Completion queue entry size as a power of two (Identify CQES); must be 4, 16-byte entries                                                                         |
| stride              | uint8  | 0       | Doorbell stride (CAP.DSTRD): doorbells are 4 « stride bytes apart, 0 to 12                                                                                        |
| multipoller_enabled | uint8  | 0       | 0 runs one poller thread for all I/O queues; 1 runs ceil(queues / poller_ratio) pollers, each owning a round-robin share of the queues; other values fail realize |
| poller_ratio        | uint32 | 1       | I/O queues per poller thread when multipoller_enabled=1; 0 is treated as 1, one poller per queue                                                                  |
| hiops_inline        | bool   | on      | NoSSD only: performance option, on by default; set off only when debugging                                                                                        |
| aerl                | uint8  | 3       | Asynchronous Event Request Limit (Identify AERL, 0's based): the controller holds up to aerl + 1 outstanding requests                                             |
| elpe                | uint8  | 3       | Error Log Page Entries (Identify ELPE, 0's based): the Error Information log keeps the newest elpe + 1 entries                                                    |
| mdts                | uint8  | 10      | Maximum Data Transfer Size as a power of two of the minimum memory page size (2 <sup>12</sup> + mpsmin + mdts) bytes; 0 means no limit                            |
| intc                | uint8  | 0       | Initial Coalescing Disable bit (0 or 1) of the Interrupt Vector Configuration feature; the value is reported, interrupts are not coalesced                        |
| intc_thresh         | uint8  | 0       | Initial aggregation threshold of the Interrupt Coalescing feature; the value is reported, interrupts are not coalesced                                            |
| intc_time           | uint8  | 0       | Initial aggregation time, in 100 microsecond units, of the Interrupt Coalescing feature; the value is reported, interrupts are not coalesced                      |

### Controller identity and capabilities

| Property | Type   | Default | Description                                                                                                                    |
|----------|--------|---------|--------------------------------------------------------------------------------------------------------------------------------|
| vid      | uint16 | 7453    | PCI vendor ID, also reported as the Identify Controller PCI Vendor ID                                                          |
| did      | uint16 | 7967    | PCI device ID of the controller function                                                                                       |
| acl      | uint8  | 3       | Abort Command Limit reported in Identify Controller (0's based); an Abort run with more than acl others queued behind it fails |

| Property    | Type   | Default | Description                                                                                                                                                                                                         |
|-------------|--------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cqr         | uint8  | 1       | CAP.CQR: 1 requires physically contiguous queues, 0 allows PRP-list queues                                                                                                                                          |
| vwc         | uint8  | 0       | 1 advertises a volatile write cache, which the host can turn off with feature 06h to stop bbssd buffering writes; 0 advertises none and refuses feature 06h. Flush drains the bbssd write buffer either way; 0 or 1 |
| temperature | uint16 | 323     | Composite temperature in kelvin reported by the SMART log and compared with the temperature threshold feature; default 323 (50 C)                                                                                   |
| mpsmin      | uint8  | 0       | CAP.MPSMIN: smallest host memory page size as $2^{(12 + mpsmin)}$ bytes; must not exceed mpsmax                                                                                                                     |
| mpsmax      | uint8  | 0       | CAP.MPSMAX: largest host memory page size as $2^{(12 + mpsmax)}$ bytes, from mpsmin to 15                                                                                                                           |
| oacs        | uint16 | 2       | Optional Admin Command Support; only bit 1 (Format NVM, 0x2) may be set, and clearing it refuses Format NVM                                                                                                         |
| oncs        | uint16 | 20      | Optional NVM Command Support bit mask: 0x1 Compare, 0x2 Write Uncorrectable, 0x4 Dataset Management, 0x8 Write Zeroes, 0x10 Save/Select, 0x80 Verify, 0x100 Copy; Timestamp is always added                         |
| sgl         | bool   | off     | Advertise and accept address scatter gather lists for data transfer; OCSSD ignores it                                                                                                                               |
| cmbasz      | uint32 | 0       | Controller Memory Buffer size register; 0 means no buffer, otherwise the size field times the unit must be a non-zero power of two                                                                                  |
| cmbloc      | uint32 | 0       | Controller Memory Buffer location register; its BAR field must be 2 when cmbasz is set                                                                                                                              |

### LBA formats, metadata and protection

| Property  | Type  | Default | Description                                                                                                                                                   |
|-----------|-------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| nlbaf     | uint8 | 5       | Number of LBA formats, 1 to 16 (at most 8 with meta): 512-byte blocks doubling with each format                                                               |
| lba_index | uint8 | 0       | LBA format the namespaces boot with, below nlbaf; ZNS needs a block size of 4 KiB or less                                                                     |
| extended  | uint8 | 0       | 1 boots with metadata interleaved with the data (extended LBAs); needs bit 0 of mc; 0 or 1                                                                    |
| meta      | uint8 | 0       | Metadata bytes per logical block, NoSSD and bbssd only, not with FDP, dpc or dps; each format is then also offered with metadata                              |
| mc        | uint8 | 0       | Metadata Capabilities bit mask: bit 0 interleaved (extended LBAs), bit 1 separate buffer; required when meta is set                                           |
| pi        | bool  | off     | Offer end-to-end protection information types 1 to 3 when meta is at least 8 bytes and allow Format and Create to select them; not with power_loss or cxl_ssd |
| dpc       | uint8 | 0       | Data Protection Capabilities reported in Identify Namespace when pi is off; must be 0 with meta                                                               |
| dps       | uint8 | 0       | Data Protection Type Settings reported in Identify Namespace; a non-zero value needs 8 bytes of metadata, which meta refuses, so leave it 0 and use pi        |
| ms        | uint8 | 16      | No effect, kept for compatibility; meta sets the metadata size. A value other than the default warns at realize                                               |
| ms_max    | uint8 | 64      | No effect, kept for compatibility; OCSSD 2.0 reports a single LBA format. A value other than the default warns at realize                                     |

| Property | Type  | Default | Description                                                                                                                                                          |
|----------|-------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dlfeat   | uint8 | 1       | No effect, kept for compatibility; Identify Namespace always reports DLFEAT 0x9 (deallocated blocks read as zeroes). A value other than the default warns at realize |

### Namespace management, streams and power loss

| Property       | Type   | Default | Description                                                                                                                                                                                                                    |
|----------------|--------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ns_mgmt        | bool   | off     | Enable Namespace Management and Attachment on a standalone NoSSD or bbssd controller whose namespaces all run femu_mode with dps 0, otherwise it stays off without an error; with a shared subsystem use femu-subsys ns_mgmt   |
| bbssd_ns_limit | uint32 | 4       | Most bbssd namespaces that may be allocated with ns_mgmt, including detached ones, 1 to 256 and at least namespaces; each has its own FTL                                                                                      |
| streams        | bool   | off     | Enable the Streams directive; bbssd separates streams per FTL page (SWS) and needs page or dftl mapping, NoSSD has no placement effect; not with FDP or a shared subsystem                                                     |
| streams.max    | uint16 | 8       | Number of stream slots with streams=on, 1 to 32; bbssd reserves streams.max + 1 lines for them                                                                                                                                 |
| power_loss     | bool   | off     | bbssd: roll back writes still in the write buffer on a simulated power cut and enable simulate-power-loss; needs buffer_size > 0, vwc=1, page-aligned namespaces, and no meta, pi, ns_mgmt, subsys, namespace_modes or cxl_ssd |

### NAND geometry (bbssd, CSD, KV)

| Property    | Type  | Default | Description                                                                                               |
|-------------|-------|---------|-----------------------------------------------------------------------------------------------------------|
| secsz       | int32 | 512     | bbssd, CSD, KV: sector size in bytes, greater than 0                                                      |
| secs_per_pg | int32 | 8       | bbssd, CSD, KV: sectors per NAND page, 1 to 256                                                           |
| pgs_per_blk | int32 | 256     | bbssd, CSD, KV: pages per NAND block, 1 to 65536, at most 512 with nand_cell_type                         |
| blks_per_pl | int32 | 256     | bbssd, CSD, KV: blocks per plane, 1 to 65536; also the number of lines (superblocks)                      |
| pls_per_lun | int32 | 1       | bbssd, CSD, KV: planes per LUN, 1 to 16                                                                   |
| luns_per_ch | int32 | 8       | bbssd, CSD, KV: LUNs (dies) per channel, 1 to 128                                                         |
| nchs        | int32 | 8       | bbssd, CSD, KV: number of channels, 1 to 4096; the total sector count must fit in a signed 32-bit integer |

### NAND timing (bbssd, CSD, KV)

| Property    | Type  | Default | Description                                                                                                                                                |
|-------------|-------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pg_rd_lat   | int32 | 40000   | bbssd, CSD, KV: NAND page read time in ns when nand_cell_type is 0                                                                                         |
| pg_wr_lat   | int32 | 200000  | bbssd, CSD, KV: NAND page program time in ns when nand_cell_type is 0                                                                                      |
| blk_er_lat  | int32 | 2000000 | bbssd, CSD, KV: NAND block erase time in ns when nand_cell_type is 0                                                                                       |
| ch_xfer_lat | int32 | 0       | Channel transfer time per page in ns: the data phase for bbssd, CSD and KV when pg_xfer_lat is 0, and the OCSSD 1.2 transfer time with oc12_channel_timing |

| Property       | Type  | Default | Description                                                                                                                                                        |
|----------------|-------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cmd_addr_lat   | int32 | 0       | bbssd, CSD, KV: command and address phase on the channel bus in ns; the bus is modelled only when this, pg_xfer_lat (or ch_xfer_lat) or status_lat is non-zero     |
| pg_xfer_lat    | int32 | 0       | bbssd, CSD, KV: page data transfer phase on the channel bus in ns; 0 uses ch_xfer_lat                                                                              |
| status_lat     | int32 | 0       | bbssd, CSD, KV: status read phase on the channel bus in ns                                                                                                         |
| tplpbsy        | int32 | 0       | No effect, kept for compatibility; programs are issued one plane at a time. A value other than the default warns at realize                                        |
| tplrbsy        | int32 | 0       | No effect, kept for compatibility; reads are issued one plane at a time. A value other than the default warns at realize                                           |
| tplbsy         | int32 | 0       | bbssd, CSD, KV: busy time in ns between the planes of a multi-plane erase, which garbage collection issues when pls_per_lun > 1                                    |
| trcb sy        | int32 | 0       | No effect, kept for compatibility; no mode enables the cache read model. A value other than the default warns at realize                                           |
| trim_lat_ns    | int32 | 0       | bbssd, CSD: time in ns charged per Dataset Management deallocate range; refused with FDP                                                                           |
| pe_suspend     | int32 | 0       | bbssd, CSD, KV: non-zero lets a read suspend a program or erase on its LUN instead of waiting for it to finish                                                     |
| tsusp_ns       | int32 | 0       | bbssd, CSD, KV: overhead in ns added to a read that suspends a program or erase, 0 or more                                                                         |
| nand_cell_type | uint8 | 0       | bbssd, CSD, KV: 0 uses the flat pg_rd_lat, pg_wr_lat and blk_er_lat; 1 SLC, 2 MLC, 3 TLC or 4 QLC uses built-in per-page-type timing (other values fall back to 0) |
| cell_pages     | int32 | 0       | bbssd, CSD, KV: bits per cell (pages per wordline) for the pgtype_lat model, 0 to 5; 0 with pgtype_lat means 3                                                     |
| pgtype_lat     | int32 | 0       | bbssd, CSD, KV: non-zero scales the program time by page type (lower to upper) using cell_pages, when nand_cell_type is 0                                          |

## Reliability and wear

| Property           | Type   | Default | Description                                                                                                                                                                                      |
|--------------------|--------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ecc_step_ns        | int32  | 0       | bbssd, CSD, KV: extra read time in ns per ECC tier, one tier per 750 erases of the block plus one per ecc_retention_sec of data age, at most 4 tiers; 0 turns the model off                      |
| ecc_retention_sec  | int32  | 0       | bbssd, CSD, KV: data age in seconds that adds one ECC tier, with ecc_step_ns; 0 counts wear only; refused with FDP                                                                               |
| pe_cycles_rated    | uint32 | 0       | bbssd, CSD, KV: rated program/erase cycles used for SMART Percentage Used; 0 takes the rating of nand_cell_type, or reports none                                                                 |
| nand_bad_blocks    | uint32 | 0       | bbssd, CSD, KV: blocks marked bad at start, capped at the block count, which lowers SMART Available Spare                                                                                        |
| err_read_unc_ppm   | uint32 | 0       | bbssd, CSD: reads per million that fail as Unrecovered Read Error, injected at a fixed period; 0 disables                                                                                        |
| err_write_fail_ppm | uint32 | 0       | bbssd, CSD and ZNS: writes per million that fail, injected at a fixed period (a ZNS zone then goes read-only); 0 disables                                                                        |
| read_reclaim_limit | int32  | 0       | bbssd, CSD: when a host read finds its block has taken this many reads since its erase, that line is queued and rewritten on a following write, one line at a time; 0 disables, refused with FDP |

| Property            | Type  | Default | Description                                                                                                                                                             |
|---------------------|-------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| retention_limit_sec | int32 | 0       | bbssd, CSD: when a host read hits a line filled at least this many seconds earlier, the line is queued and rewritten on a following write; 0 disables, refused with FDP |

### Garbage collection, mapping and caches

| Property            | Type   | Default | Description                                                                                                                                                                       |
|---------------------|--------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gc_thres_pcent      | int32  | 75      | bbssd, CSD: percent of lines in use at which background garbage collection starts, 1 to 100; KV uses it only as the fraction of NAND usable for values                            |
| gc_thres_pcent_high | int32  | 95      | bbssd, CSD: percent of lines in use at which garbage collection is forced, from gc_thres_pcent to 100; under FDP it keeps at least one reclaim unit free unless it is 100         |
| gc_policy           | str    | unset   | bbssd, CSD without FDP: line victim policy, one of greedy, random, cost-benefit, fifo or d-choice; unset is greedy                                                                |
| gc_seed             | uint64 | 0       | bbssd, CSD: seed for the victims the random and d-choice gc_policy and the FDP random gc_strategy pick; the same seed and workload pick the same victims, so vary it to vary runs |
| gc_strategy         | int32  | 0       | bbssd with FDP: reclaim unit victim strategy, 0 greedy, 1 cost-benefit, 2 random or 4 per-handle                                                                                  |
| mapping             | str    | unset   | bbssd, CSD: logical-to-physical mapping scheme, one of page, dftl, hybrid or fast; unset is page, and FDP supports only page                                                      |
| mapping_cache_mb    | uint32 | 0       | bbssd, CSD with mapping=dftl: size of the cached mapping table in MiB; 0 means 4                                                                                                  |
| read_cache_mb       | uint32 | 0       | bbssd, CSD: size of the DRAM read cache in MiB; 0 disables it                                                                                                                     |
| cache_evict         | str    | unset   | bbssd, CSD: read cache eviction policy, one of clock, random, lru or arc; unset is clock                                                                                          |
| hot_cold_sep        | bool   | off     | bbssd, CSD with mapping page or dftl: write overwrites of mapped pages to separate hot lines; refused with FDP                                                                    |
| buffer_size         | int32  | 0       | bbssd, CSD: DRAM write buffer capacity in NAND pages, not bytes; 0 programs every write directly                                                                                  |
| buffer_thres_pcent  | int32  | 90      | bbssd, CSD: buffer fill level in percent at which buffered pages are written to NAND, 1 to 100 when buffer_size > 0                                                               |
| fdp_trim_erase_all  | int32  | 0       | bbssd with FDP: non-zero makes a deallocate reset every reclaim unit instead of the given ranges                                                                                  |
| debug_ftl           | bool   | off     | bbssd, CSD, KV: print a message when a page is programmed while not free or invalidated while not valid, and print merge counts for hybrid and fast mapping                       |

### Host link and controller firmware

| Property            | Type   | Default | Description                                                                                                                                                         |
|---------------------|--------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pcie_bandwidth_mbps | uint32 | 0       | Host link bandwidth in MB/s (10 <sup>6</sup> bytes); each Read or Write is charged its transfer time on a per-direction link queue; 0 disables the bandwidth charge |
| pcie_prop_delay_ns  | uint32 | 0       | Host link propagation delay in ns added to each Read or Write after its link transfer; 0 disables it                                                                |



| Property  | Type   | Default | Description                                                                                                                |
|-----------|--------|---------|----------------------------------------------------------------------------------------------------------------------------|
| fw_cpu_ns | uint64 | 0       | Controller firmware time in ns charged to each Read, Write and Zone Append, serialized on one modelled core; 0 disables it |

## ZNS

| Property            | Type   | Default | Description                                                                                                                                       |
|---------------------|--------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| zns_num_ch          | uint8  | 2       | ZNS: number of channels, 1 to 128                                                                                                                 |
| zns_num_lun         | uint8  | 4       | ZNS: LUNs (dies) per channel, 1 or more                                                                                                           |
| zns_num_plane       | uint8  | 2       | ZNS: planes per LUN, 1 to 8; the program unit grows with it                                                                                       |
| zns_num_blk         | uint8  | 32      | ZNS: blocks per plane, 1 or more; the zone count is zns_num_blk times zns_num_ch divided by the zone width and zns_num_plane                      |
| zns_flash_type      | int32  | 4       | ZNS: cell type, 1 SLC, 2 MLC, 3 TLC, 4 QLC or 5 PLC; MLC and PLC have no built-in timing and need zns_pg_rd_lat, zns_pg_wr_lat and zns_blk_er_lat |
| zns_pg_rd_lat       | int64  | 0       | ZNS: page read time in ns, 0 or more; 0 uses the built-in time of zns_flash_type                                                                  |
| zns_pg_wr_lat       | int64  | 0       | ZNS: page program time in ns, 0 or more; 0 uses the built-in time of zns_flash_type                                                               |
| zns_blk_er_lat      | int64  | 0       | ZNS: block erase time in ns, 0 or more; 0 uses the built-in time of zns_flash_type                                                                |
| zns_cmd_addr_lat    | int64  | 0       | ZNS: command and address phase on the channel bus in ns, 0 or more                                                                                |
| zns_pg_xfer_lat     | int64  | 0       | ZNS: page data transfer phase on the channel bus in ns, 0 or more                                                                                 |
| zns_status_lat      | int64  | 0       | ZNS: status read phase on the channel bus in ns, 0 or more                                                                                        |
| zns_pe_suspend      | int32  | 0       | ZNS: non-zero lets a read suspend a program or erase on its plane                                                                                 |
| zns_tsusp_ns        | int64  | 0       | ZNS: overhead in ns added to a read that suspends a program or erase, 0 or more                                                                   |
| zns_max_active      | uint32 | 0       | ZNS: Maximum Active Resources (zones), at most the zone count; 0 means no limit                                                                   |
| zns_max_open        | uint32 | 0       | ZNS: Maximum Open Resources (zones), at most the zone count and zns_max_active; 0 means no limit                                                  |
| zns_num_wc          | uint32 | 0       | ZNS: number of zone write caches; 0 uses zns_max_open, or 3 when that is 0                                                                        |
| zns_zd_ext_size     | uint32 | 0       | ZNS: zone descriptor extension size in bytes, a multiple of 64 up to 16320; 0 means none                                                          |
| zns_num_conv_zones  | uint32 | 0       | ZNS: number of leading conventional zones, which take random writes; capped at the zone count                                                     |
| zns_zone_cap        | size   | 0       | ZNS: zone capacity in bytes, at least one logical block and at most the zone size; 0 means the zone size                                          |
| zns_chnls_per_zone  | uint32 | 0       | ZNS: channels one zone spans (zone width), dividing zns_num_ch; 0 means all channels                                                              |
| zns_zrwa_size       | uint64 | 0       | ZNS: Zone Random Write Area size in logical blocks, at most 65535 and a multiple of zns_zrwafg_size; 0 disables ZRWA                              |
| zns_zrwafg_size     | uint64 | 0       | ZNS: ZRWA flush granularity in logical blocks, 1 to 65535 with ZRWA and 0 without; the zone capacity must be a multiple of it                     |
| zns_zrwa_num        | uint32 | 0       | ZNS: number of zones that may hold a ZRWA at once, 1 or more with ZRWA and 0 without                                                              |
| zns_cross_zone_read | bool   | off     | ZNS: allow reads that cross zone boundaries (Read Across Zone Boundaries)                                                                         |

| Property    | Type   | Default | Description                                                                           |
|-------------|--------|---------|---------------------------------------------------------------------------------------|
| zns_zasl_bs | uint32 | 131072  | ZNS: Zone Append size limit in bytes, a power-of-two multiple of 4 KiB; 0 follows mdt |

### OCSSD (Open-Channel)

| Property            | Type   | Default | Description                                                                                                                                                                                               |
|---------------------|--------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lver                | uint8  | 2       | OCSSD: Open-Channel version, 1 for 1.2 or 2 for 2.0; other values fail realize                                                                                                                            |
| flash_type          | uint8  | 2       | OCSSD: cell type for the built-in timing tables, 1 SLC, 2 MLC, 3 TLC or 4 QLC; other values fail realize                                                                                                  |
| oc12_channel_timing | bool   | off     | OCSSD 1.2: charge channel transfer time for each page accessed, scaled by the sectors used out of lsecs_per_pg; ch_xfer_lat sets ns per page and 0 uses the flash_type value; off, transfers take no time |
| lsec_size           | uint16 | 4096    | OCSSD 1.2: sector size in bytes reported in the geometry, greater than 0 in both versions; data moves at the namespace block size, and OCSSD 2.0 always uses 4096                                         |
| lsecs_per_pg        | uint8  | 4       | OCSSD: sectors per page, greater than 0                                                                                                                                                                   |
| lpgs_per_blk        | uint16 | 512     | OCSSD: pages per block, greater than 0 and at most 512 for OCSSD 1.2                                                                                                                                      |
| lmax_sec_per_rq     | uint8  | 64      | OCSSD 1.2: most sectors in one vector command; OCSSD 2.0 uses 64                                                                                                                                          |
| lnum_ch             | uint8  | 2       | OCSSD: channels (2.0 groups), 1 to 32, with lnum_ch * lnum_lun at most 128                                                                                                                                |
| lnum_lun            | uint8  | 8       | OCSSD: LUNs (2.0 parallel units) per channel, 1 or more, with lnum_ch * lnum_lun at most 128                                                                                                              |
| lnum_pln            | uint8  | 2       | OCSSD: planes per LUN, greater than 0; OCSSD 1.2 accepts 1, 2 or 4                                                                                                                                        |
| lmetasize           | uint16 | 16      | OCSSD 1.2: out-of-band metadata bytes per sector; OCSSD 2.0 uses 16                                                                                                                                       |
| learly_reset        | uint8  | 0       | OCSSD 2.0: non-zero reports the early reset capability, so the host may reset a chunk it has not filled                                                                                                   |

### CSD (computational storage)

| Property            | Type   | Default | Description                                                                                                      |
|---------------------|--------|---------|------------------------------------------------------------------------------------------------------------------|
| fdm_size            | uint64 | 0       | CSD: functional data memory size in MiB; required, greater than 0                                                |
| nr_cu               | uint8  | 4       | CSD: number of compute units, 1 to 64; programs wait for the first free unit                                     |
| nr_thread           | uint8  | 4       | No effect, kept for CEMU compatibility; CSD still refuses 0. A value other than the default warns at realize     |
| time_slice          | uint64 | 200000  | No effect, kept for CEMU compatibility. A value other than the default warns at realize                          |
| context_switch_time | uint64 | 200     | No effect, kept for CEMU compatibility. A value other than the default warns at realize                          |
| csf_runtime_scale   | uint16 | 3       | CSD: non-zero multiplier applied to the host run time of a program that sets neither a runtime nor its own scale |

| Property        | Type | Default | Description                                                                                                                                                                                   |
|-----------------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| csd_program_dir | str  | unset   | CSD: host directory that shared-object and uBPF programs are loaded from, named by a file name with no slash that must resolve inside it; unset allows only the built-in phantom program type |

### Generic PCI device properties

Common to every QEMU PCI device and documented by QEMU: acpi-index, addr, busnr, failover\_pair\_id, multifunction, rombar, romfile, romsize, sriov-pf. QEMU's internal compatibility properties (x-max-bounce-buffer-size, x-pcie-ari-nextfn-1, x-pcie-err-unc-mask, x-pcie-ext-tag, x-pcie-extcap-init, x-pcie-lnksta-dllla) are not listed.

### A.1.2 femu-subsys: NVMe subsystem

-device femu-subsys has 8 properties of its own.

#### Shared namespaces

| Property | Type | Default | Description                                                                                                                                                          |
|----------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ns_mgmt  | bool | off     | Keep one namespace table and one backend in the subsystem, shared by every controller that names it with subsys=; NoSSD and bbssd controllers only, and not with fdp |
| nqn      | str  | unset   | Subsystem name reported as nqn.2019-08.org.qemu:<nqn> by controllers that share namespaces through this subsystem; unset uses the device id                          |

#### Flexible Data Placement

| Property           | Type   | Default | Description                                                                                                                                                                                                                                                                                                                      |
|--------------------|--------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fdp                | bool   | off     | Enable Flexible Data Placement in endurance group 1 for controllers that join this subsystem; only bbssd places data by reclaim unit                                                                                                                                                                                             |
| fdp.runs           | size   | 0       | Reclaim unit size in bytes; 0 means 96 MiB, and a bbssd controller accepts only 0 or the size of one superblock, which it then uses                                                                                                                                                                                              |
| fdp.nrg            | uint32 | 1       | Number of FDP reclaim groups; must be 1, placement into other groups is not implemented                                                                                                                                                                                                                                          |
| fdp.nruh           | uint16 | 0       | Number of FDP reclaim unit handles (placement handles), from 1 to fdp.nru; must be set when fdp=on                                                                                                                                                                                                                               |
| fdp.nru            | uint64 | 128     | Number of reclaim units in each reclaim group, from fdp.nruh to 65536; bbssd uses at most one per superblock and needs at least 2 * fdp.nruh + 1 of them, and the namespace must fit in the units left once each handle has one open, each Persistently Isolated handle one to collect into, and forced collection its free ones |
| fdp.isolation_mode | uint32 | 0       | 0 makes every reclaim unit handle Persistently Isolated; any other value makes the last handle Initially Isolated                                                                                                                                                                                                                |

### A.1.3 femu-cxl-ssd: CXL Type-3 SSD

-device femu-cxl-ssd has 24 properties of its own, 10 inherited from cxl-type3 and 52 QOM properties listed in runtime-properties.md (§A.2).

## Cache

| Property     | Type   | Default | Description                                                                                                                                         |
|--------------|--------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| cache-pages  | uint32 | 1024    | Number of 4 KiB pages the device cache holds; 0 sends every access to the media, otherwise at most the media page count and divisible by cache-ways |
| cache-policy | str    | unset   | Cache replacement policy, one of fifo, lifo, clock or s3-fifo; unset is fifo                                                                        |

## NAND geometry and timing

| Property                  | Type   | Default | Description                                                                                                                                               |
|---------------------------|--------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ftl                       | bool   | on      | Charge cache misses and write-backs to the FTL and NAND model; off keeps memory behaviour with no media timing and cannot be linked to an NVMe controller |
| channels                  | uint32 | 4       | Number of NAND channels, 1 to 4096                                                                                                                        |
| luns-per-channel          | uint32 | 4       | NAND LUNs per channel, 1 to 128, with one plane per LUN                                                                                                   |
| pages-per-block           | uint32 | 256     | 4 KiB pages per NAND block, 1 to 65536                                                                                                                    |
| blocks-per-plane          | uint32 | 0       | NAND blocks per plane, 2 to 65536 and enough to cover the media; 0 sizes it to 5/4 of the media plus 4 blocks per plane                                   |
| gc-threshold              | uint32 | 75      | Percent of lines in use at which background garbage collection starts, 1 to 100                                                                           |
| gc-threshold-high         | uint32 | 95      | Percent of lines in use at which garbage collection is forced, from gc-threshold to 100                                                                   |
| read-ns                   | uint64 | 40000   | NAND page read time in ns, at most one second                                                                                                             |
| program-ns                | uint64 | 200000  | NAND page program time in ns, at most one second                                                                                                          |
| erase-ns                  | uint64 | 2000000 | NAND block erase time in ns, at most one second                                                                                                           |
| channel-ns                | uint64 | 0       | NAND channel transfer time per page in ns, at most one second                                                                                             |
| cylon-first-touch-program | bool   | off     | Charge a NAND program instead of a free read when a read reaches a page the FTL has never mapped, as the Cylon experiments do                             |
| cylon-free-writeback      | bool   | off     | Write dirty pages back on eviction and flush with no NAND program and no media time, as the Cylon experiments do                                          |

## Direct mapping (DER)

| Property          | Type      | Default | Description                                                                                                                                                                                    |
|-------------------|-----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| der               | str       | unset   | Direct mapping of cached pages into the guest: off (MMIO only, the default), memslot (KVM memory slot aliases, not under TCG) or cylon (a Cylon host kernel)                                   |
| der-replace-rate  | uint32    | 64      | With der=memslot and no free alias (1024 shared by all devices, fewer if KVM has fewer free slots), the most aliases per second a repeatedly missing page may displace; 0 disables replacement |
| cylon-kernel-ack  | bool      | off     | Must be on with der=cylon to state that the host runs a Cylon kernel with the dual-slot fixes; the device does not check it                                                                    |
| concurrent-misses | OnOffAuto | auto    | Let misses to different pages wait for the media together; auto does so only while direct mapping is active                                                                                    |

## Caching API, control channel and logs

| Property    | Type | Default           | Description                                                                                                                            |
|-------------|------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| cca         | bool | off               | Expose the caching API on BAR 5 (pin, unpin, invalidate, uncached ranges, query) and start its worker thread                           |
| lsa-control | bool | off               | Accept experiment control commands through Get LSA on an internal 128 MiB label area; trusted guests only, and not with an lsa backend |
| log-dir     | str  | unset             | Host directory for cxlssd-stats.log, cxlssd-io-N.log and cxlssd-spt.log; unset is the working directory                                |
| tracefs-dir | str  | unset             | Host tracefs directory whose tracing control commands 91 and 81 write; unset, those commands change nothing on the host                |
| log-limit   | size | 67108864 (64 MiB) | Size limit in bytes for each log file the device writes; 0 opens no I/O log and takes no statistics appends                            |

### Inherited from cxl-type3

These belong to QEMU's cxl-type3 device. The descriptions say how femu-cxl-ssd treats them.

| Property           | Type                 | Default              | Description                                                                                                                                                                                                                                      |
|--------------------|----------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cdat               | str                  | unset                | Host file holding the CDAT table returned over DOE; unset builds a table from the backends; femu-cxl-ssd leaves it unchanged                                                                                                                     |
| lsa                | link<memory-backend> | unset                | Label storage area backend served by Get and Set LSA; femu-cxl-ssd accepts it only with lsa-control=off                                                                                                                                          |
| memdev             | link<memory-backend> | unset                | Legacy persistent memory backend of cxl-type3; femu-cxl-ssd refuses it at realize                                                                                                                                                                |
| num-dc-regions     | uint8                | 0                    | Number of dynamic capacity regions; femu-cxl-ssd requires it to stay 0                                                                                                                                                                           |
| persistent-memdev  | link<memory-backend> | unset                | Persistent memory backend of cxl-type3; femu-cxl-ssd refuses it at realize                                                                                                                                                                       |
| sn                 | uint64               | 18446744073709551615 | PCIe Device Serial Number; the default ( $2^{64} - 1$ ) means unset, which gives no serial number capability; femu-cxl-ssd leaves it unchanged                                                                                                   |
| volatile-dc-memdev | link<memory-backend> | unset                | Dynamic capacity memory backend; femu-cxl-ssd refuses it at realize                                                                                                                                                                              |
| volatile-memdev    | link<memory-backend> | unset                | Required: ID of the host memory backend that holds the device data, a non-zero multiple of 256 MiB, at most 120 GiB; der=cylon needs a hugetlbfs file backend with share=on and prealloc=on, or direct mapping falls back to MMIO with a warning |
| x-speed            | PCIELinkSpeed        | 32                   | PCIe link speed the device reports; femu-cxl-ssd leaves it unchanged and its timing does not depend on it                                                                                                                                        |
| x-width            | PCIELinkWidth        | 16                   | PCIe link width the device reports; femu-cxl-ssd leaves it unchanged and its timing does not depend on it                                                                                                                                        |

## Generic PCI device properties

Common to every QEMU PCI device and documented by QEMU: `acpi-index`, `addr`, `busnr`, `failover_pair_id`, `multifunction`, `rombar`, `romfile`, `romsize`, `sriov-pf`. QEMU's internal compatibility properties (`x-max-bounce-buffer-size`, `x-pcie-ari-nextfn-1`, `x-pcie-err-unc-mask`, `x-pcie-ext-tag`, `x-pcie-extcap-init`, `x-pcie-lnksta-dllla`) are not listed.

### A.1.4 Test-only properties

These exist only when QEMU runs under `qtest` (`-accel qtest`) and serve FEMU's own tests. They are not part of the user interface.

- `femu`: `x-ftl-check`, `x-ftl-trace`, `x-ns-test`, `x-oc12-clock`, `x-oc12-trace`, `x-stream-test`
- `femu-cxl-ssd`: `test-change-dpa`, `test-media-disabled`, `test-slot-reservation`

### A.1.5 Environment variables

FEMU reads these variables from the environment of the QEMU process. They are debugging and host-placement aids, not device configuration, so they have no `-device` property. When QEMU runs under `sudo`, pass them through, for example `sudo FEMU_EXP_LOG=1 ./run-blackbox.sh` or `sudo -E`.

| Variable                         | Read by                                                            | Effect                                                                                                                                                                                                                                                        |
|----------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>FEMU_MBE_INTERLEAVE</code> | memory backend, every mode ( <code>hw/femu/backend/dram.c</code> ) | on interleaves the backend memory across NUMA nodes 0 and 1; 0 or 1 binds it to that node. Other values are ignored with a message. Unset leaves the host default policy.                                                                                     |
| <code>FEMU_FDP_DEBUG</code>      | bbssd FTL ( <code>hw/femu/bbssd/ftl.c</code> )                     | Any value, even empty, prints FDP placement and reclaim traces to <code>stderr</code> .                                                                                                                                                                       |
| <code>FEMU_EXP_LOG</code>        | bbssd FTL ( <code>hw/femu/bbssd/ftl-exp.c</code> )                 | A non-empty value prints [EXP] lines to <code>stderr</code> that trace the writes, overwrites, deallocations, garbage collection moves and erases of pages whose data contains <code>FEMU_SECRET</code> ; without <code>FEMU_SECRET</code> nothing is traced. |
| <code>FEMU_SECRET</code>         | bbssd FTL ( <code>hw/femu/bbssd/ftl-exp.c</code> )                 | A non-empty marker string that selects the pages <code>FEMU_EXP_LOG</code> traces.                                                                                                                                                                            |
| <code>FEMU_DUMP_LPN</code>       | bbssd FTL ( <code>hw/femu/bbssd/ftl-exp.c</code> )                 | A logical page number (decimal or 0x hex) whose backend page is hex-dumped to <code>stderr</code> on every read not served from the write buffer, independent of <code>FEMU_EXP_LOG</code> .                                                                  |
| <code>FEMU_KV_SELFTEST</code>    | KV FTL ( <code>hw/femu/kvssd/kvssd-ftl.c</code> )                  | Any value runs the KV FTL self-test once at realize and logs the result.                                                                                                                                                                                      |

## A.2 Runtime properties and counters

QOM properties that a running device exposes. Read and set them through QMP at `/machine/peripheral/<id>`, where `<id>` is the `id=` given on `-device`:

```
{ "execute": "qom-get", "arguments": { "path": "/machine/peripheral/cxl-ssd0", "property": "read-hits" } }
{ "execute": "qom-set", "arguments": { "path": "/machine/peripheral/cxl-ssd0", "property": "stats-reset", "value": true } }
```

or in the HMP monitor with `qom-get` and `qom-set`. Event counters are cleared by `stats-reset`; gauges and totals keep their value. Descriptions are the ones `-device <type>,help` prints.

### A.2.1 femu: NVMe controller

#### Power loss trigger

| Property            | Type | Description                                                                                                                                |
|---------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------|
| simulate-power-loss | bool | Write-only: setting true on a realized controller with power_loss=on drops the write buffer and resets the controller as a power cut would |

### A.2.2 femu-cxl-ssd: CXL Type-3 SSD

#### Cache tunables (also accepted on -device)

| Property        | Type   | Description                                                                                                                                               |
|-----------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| cache-ways      | uint32 | Entries per cache set, non-zero and dividing cache-pages (1 is direct mapped); default 16, and a qom-set at run time writes back and rebuilds the cache   |
| prefetch-degree | uint32 | Up to this many pages inserted into the cache after each demand miss, at most the media page count, with no NAND read; default 0, changeable with qom-set |
| prefetch-stride | uint32 | Distance in pages from a missed page to the first prefetched page, at most the media page count; default 1, changeable with qom-set                       |

#### Actions and control

| Property           | Type   | Description                                                                                                                                                                                                                                                      |
|--------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| der-ratio          | uint64 | Run time only: direct-map this fraction of pages, one of 0, 50, 75, 90, 95, 97, 98, 99, 995 (99.5%), 999 (99.9%) or 100 percent; needs der other than off and no uncached ranges, and der=memslot refuses a ratio whose unmapped gaps outnumber the free aliases |
| control-command    | uint64 | Run time only: writing runs that experiment control command with control-argument before qom-set returns; reading gives the last command                                                                                                                         |
| control-argument   | uint64 | Argument for the next control-command write; reading gives the last argument used, which a Get LSA control command also sets                                                                                                                                     |
| control-status     | uint64 | Read-only result of the last control command: 0 success, 1 error, 2 a Get LSA command still queued                                                                                                                                                               |
| flush-cache        | bool   | Write-only: true revokes direct mappings, writes dirty pages back, drops unpinned pages and waits for the modelled media time                                                                                                                                    |
| stats-reset        | bool   | Write-only: true copies the counters to the last-* properties, then clears the event counters                                                                                                                                                                    |
| fast-load          | bool   | Accesses skip only their wait for the modelled media time; the FTL, cache and counters still run. For warmup and loading, not for measurement. Setting false waits for the queued NAND work first; default off, changeable with qom-set                          |
| fast-load-drain-ns | uint64 | Read-only: ns the last fast-load switch to false waited for queued NAND work                                                                                                                                                                                     |

#### Cache counters

| Property      | Type   | Description                                                           |
|---------------|--------|-----------------------------------------------------------------------|
| cache-entries | uint64 | Read-only gauge: pages resident in the cache, pinned ones included    |
| cache-hits    | uint64 | Read-only event counter: MMIO page lookups that found a resident page |

| Property         | Type   | Description                                                                                                         |
|------------------|--------|---------------------------------------------------------------------------------------------------------------------|
| cache-misses     | uint64 | Read-only event counter: MMIO page lookups that missed, including accesses with no cache                            |
| read-hits        | uint64 | Read-only event counter: cache-hits caused by reads                                                                 |
| read-misses      | uint64 | Read-only event counter: cache-misses caused by reads                                                               |
| write-hits       | uint64 | Read-only event counter: cache-hits caused by writes                                                                |
| write-misses     | uint64 | Read-only event counter: cache-misses caused by writes                                                              |
| cache-inserts    | uint64 | Read-only event counter: pages admitted by demand misses, prefetch and PIN fills                                    |
| cache-evictions  | uint64 | Read-only event counter: pages the replacement policy removed, including those a flush or a cache-ways change drops |
| prefetch-inserts | uint64 | Read-only event counter: pages inserted by prefetch                                                                 |

### Snapshot counters

| Property              | Type   | Description                                                              |
|-----------------------|--------|--------------------------------------------------------------------------|
| last-read-hits        | uint64 | Read-only: read-hits at the last stats-reset or control command 1        |
| last-read-misses      | uint64 | Read-only: read-misses at the last stats-reset or control command 1      |
| last-write-hits       | uint64 | Read-only: write-hits at the last stats-reset or control command 1       |
| last-write-misses     | uint64 | Read-only: write-misses at the last stats-reset or control command 1     |
| last-inserts          | uint64 | Read-only: cache-inserts at the last stats-reset or control command 1    |
| last-evictions        | uint64 | Read-only: cache-evictions at the last stats-reset or control command 1  |
| last-entries          | uint64 | Read-only: cache-entries at the last stats-reset or control command 1    |
| last-prefetch-inserts | uint64 | Read-only: prefetch-inserts at the last stats-reset or control command 1 |

### Media counters

| Property      | Type   | Description                                                                                                                                                                                                                               |
|---------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| media-time-ns | uint64 | Read-only: total modelled media time in ns returned by FTL requests, including contention                                                                                                                                                 |
| media-reads   | uint64 | Read-only: page reads the FTL performed for cache fills, uncached reads and PIN fills, not counting reads that cylon-first-touch-program turned into programs; stays 0 with ftl=off                                                       |
| media-writes  | uint64 | Read-only: user page programs counted by the FTL, garbage collection copies excluded, refreshed at each media request of this device, so writes from a linked NVMe controller appear after the next one                                   |
| media-full    | uint64 | Read-only: accesses whose NAND program found no free page; that program is not timed and the access completes uncached, though a fill read already issued is charged; stats-reset keeps it, and a measurement is valid only while it is 0 |

### Direct mapping counters

| Property   | Type   | Description                                                     |
|------------|--------|-----------------------------------------------------------------|
| der-active | bool   | Read-only: whether direct mapping is available on this device   |
| der-probes | uint64 | Read-only: probe attempts, one per realize with der=cylon       |
| der-mapped | uint64 | Read-only gauge: pages currently mapped for direct guest access |
| der-remaps | uint64 | Read-only: direct page mappings installed                       |



| Property              | Type   | Description                                                                                    |
|-----------------------|--------|------------------------------------------------------------------------------------------------|
| der-revocations       | uint64 | Read-only: direct page mappings removed                                                        |
| der-quiet-revocations | uint64 | Read-only: Cylon revocations of entries whose accessed bit was clear, done without a TLB flush |
| der-replacements      | uint64 | Read-only: memslot aliases displaced by a hotter page                                          |
| der-fallbacks         | uint64 | Read-only: refused direct mapping attempts and device disablements                             |

### Caching API counters

| Property              | Type   | Description                                                                                  |
|-----------------------|--------|----------------------------------------------------------------------------------------------|
| cca-commands          | uint64 | Read-only event counter: caching API commands completed, whatever their status               |
| cca-errors            | uint64 | Read-only event counter: caching API commands completed with a non-zero status               |
| cca-pinned            | uint64 | Read-only gauge: pages currently pinned in the cache                                         |
| cca-uncached          | uint64 | Read-only gauge: pages currently in caching API uncached ranges                              |
| cca-pin-fills         | uint64 | Read-only event counter: pages PIN read from the media                                       |
| cca-writebacks        | uint64 | Read-only event counter: dirty pages programmed by caching API commands                      |
| cca-dropped           | uint64 | Read-only event counter: resident pages dropped by INVALIDATE and CACHE_DISABLE              |
| cca-pinned-set-misses | uint64 | Read-only event counter: misses served from the media because every way of the set is pinned |

### Other counters

| Property      | Type   | Description                                                                                                                                                                                |
|---------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| invalidations | uint64 | Read-only: generation count bumped by PCI configuration writes, component register writes, every mailbox command other than a control Get LSA, and reset; each revokes all direct mappings |
| nvme-drops    | uint64 | Read-only: resident pages a linked NVMe write, copy or deallocate dropped or cleaned                                                                                                       |
| log-dropped   | uint64 | Read-only: statistics appends refused, I/O logs closed at log-limit and SPT dumps truncated                                                                                                |

## B Log pages and counters

How to read FEMU's own counters from the guest. Device properties are in properties.md (§A.1), and the counters of femu-cxl-ssd are QOM properties listed in runtime-properties.md (§A.2).

### B.1 Vendor log page C0h

`nvme get-log /dev/nvme0 --log-id=0xc0 --log-len=512 -b` returns the emulator's own media counters, little-endian at these offsets (FemuStatsLog in hw/femu/nvme.h):

| Offset | Size | Field                                                                                                                                           |
|--------|------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | 4    | Write amplification factor, scaled by 1000: (pages programmed + pages relocated by garbage collection) x 1000 / pages the host asked to program |
| 8      | 8    | Pages the host asked to program                                                                                                                 |
| 16     | 8    | Pages relocated by garbage collection                                                                                                           |
| 24     | 8    | Pages actually programmed                                                                                                                       |
| 32     | 8    | Reads of the most-read block since its erase                                                                                                    |
| 40     | 8    | Lines rewritten because of read stress                                                                                                          |
| 48     | 8    | Lines rewritten because of retention age                                                                                                        |
| 56     | 8    | Host read pages the write buffer saw                                                                                                            |
| 64     | 8    | Of those, pages the buffer held                                                                                                                 |
| 72     | 8    | Host write pages the write buffer saw                                                                                                           |
| 80     | 8    | Of those, pages the buffer already held                                                                                                         |
| 88     | 8    | Log-block switch merges (mapping=hybrid only)                                                                                                   |
| 96     | 8    | Log-block full merges                                                                                                                           |
| 104    | 8    | Erases charged to log-block merges                                                                                                              |

Bytes 4-7 and 112-511 are reserved and read as zero. The counters are summed over the controller's bbssd, CSD and KV namespaces (the block read count is the largest of them); other modes leave them zero. The write amplification factor stays zero until the host has written a page. Bytes 8 and 24 are equal when no write buffer is configured; a buffer that absorbs repeated writes to one page makes the factor drop below 1.

They were previously written into the SMART log from byte 192, which NVMe Base 2.0 assigned to the composite temperature times, the temperature sensors and the thermal transition counts.

The same counters can be captured through the standard Telemetry Host-Initiated log (07h): `nvme telemetry-log /dev/nvme0 --output-file=telemetry.bin` takes a snapshot and saves it. Data Area 1 is one 512-byte block laid out as above, and it stays as captured until the next capture. The Controller-Initiated log (08h) never holds data, because the controller does not capture on its own.

`nvme get-log /dev/nvme0 --log-id=0 --log-len=1024 -b` lists every log page the controller answers, four bytes per identifier with bit 0 set for the ones it supports, so this page can be discovered rather than assumed.

### B.2 Asynchronous events

The controller completes an outstanding Asynchronous Event Request when one of these happens:

| Event                       | Raised when                                                                                                                                                                | Log page it names            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| SMART temperature warning   | the host has enabled it with Async Event Configuration and set a temperature threshold at or below the reported value (temperature, in Kelvin, default 323, which is 50 C) | SMART / Health (02h)         |
| Error                       | the host writes a doorbell that does not exist, or a value past the end of its queue                                                                                       | Error Information (01h)      |
| Namespace Attribute Changed | with ns_mgmt=on, a namespace is attached, detached, deleted or formatted, and the host enabled the notice                                                                  | Changed Namespace List (04h) |
| Zone Descriptor Changed     | an injected write fault (err_write_fail_ppm) made a ZNS zone read only, and the host enabled Zone Descriptor Changed notices (bit 27)                                      | Changed Zone List (BFh)      |

An event of a given type is reported once and then held back until the host reads the log page it named without Retain Asynchronous Event (RAE), so the same condition is not reported again before the host has looked. A controller reset drops anything outstanding. `hw/femu/scripts/aer-probe.c` checks the temperature path from inside the guest:

```
gcc -O2 -o aer-probe femu-scripts/aer-probe.c # inside the guest
sudo ./aer-probe /dev/nvme0
```

### B.3 Persistent Event log retention

The Persistent Event log (0Dh) lives in memory and is lost when QEMU exits. Set `pel_file=/absolute/path/events.pel` on a femu device to keep it across QEMU runs. A missing file starts an empty history. A corrupt or incompatible file refuses device creation and is left unchanged.

The file holds the encoded events, the log generation and the power cycle count. Each realize adds one power cycle, a power-on event and a SMART snapshot. The PEL header's power cycle count (PWRCC), the Controller Power Cycle and SMART Power Cycles all use the retained count. Reporting contexts, the other SMART counters, namespace data and power-on hours are not retained.

The file is written on the main loop after events and generation changes, and again at controller reset, device removal and normal process exit. Each write goes to a temporary file in the same directory, which is synced, renamed over the old file, and followed by a sync of the directory. At normal exit, event collection closes under the log mutex before the last write: events added before that point are saved, and later ones, including completions of outstanding I/O, are dropped. The exit path does not wait for the pollers, whose MMIO DMA may need the main thread's lock. Killing QEMU abruptly can lose changes still queued for the main loop. A write error is printed to stderr, and the next event or lifecycle save tries again.

Use one file per device and one QEMU writing it. The file is not a way to share a log between devices or to migrate one.

## C Scripts and tools

Every script and tool shipped under `hw/femu/scripts/` and `hw/femu/tools/`: what it does, its arguments and the environment variables it reads. The top-level `femu-scripts` link points to `hw/femu/scripts/`, so from `build-femu/` you can also run them as `../femu-scripts/NAME`.

Scripts marked **legacy** do not start FEMU. They start QEMU's stock nvme device or another disk on the old `x86_64-softmmu/` binary path, set thread affinity by hand, or prepare a specific lab machine. They are kept for reference. Do not use them.

### C.1 Build and setup

| Script                            | Run from                 | What it does                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>pkgdep.sh</code>            | anywhere, as root        | Installs the build dependencies with <code>apt-get</code> on Debian and Ubuntu. Exits with <code>pkgdep: unsupported system type</code> elsewhere. CI does not run it.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>femu-compile.sh</code>      | <code>build-femu/</code> | Runs <code>make clean</code> , then <code>../configure --enable-kvm --target-list=x86_64-softmmu --enable-slirp --disable-libnfs --disable-libiscsi --disable-curl</code> , then <code>make</code> with one job per CPU. <code>--enable-csd-ubpf</code> adds uBPF CSD programs, <code>--enable-csd-ubpf=PATH</code> uses the uBPF tree at <code>PATH</code> . Any other argument is an error. See <code>build.md</code> (§2.2).                                                                                                                                                                                                     |
| <code>femu-copy-scripts.sh</code> | <code>build-femu/</code> | Copies <code>pkgdep.sh</code> , <code>femu-compile.sh</code> , <code>make-guest-image.sh</code> , <code>run-guest-ssh.sh</code> , the <code>run-blackbox.sh</code> , <code>run-blackbox-fdp.sh</code> , <code>run-whitebox.sh</code> , <code>run-nossd.sh</code> , <code>run-zns.sh</code> , <code>run-csd.sh</code> and <code>run-kvssd.sh</code> launchers, <code>pin.sh</code> and <code>ftk/</code> into the current directory, overwriting earlier copies. It does not copy <code>run-cxlssd.sh</code> , <code>ssd-config.sh</code> , <code>configs/</code> or the guest tools; run those from <code>../femu-scripts/</code> . |

### C.2 Guest image and access

| Script                           | What it does                                                                                                                                                                                                                                                                                                                                                                                                                            | Arguments and environment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>make-guest-image.sh</code> | Builds an Ubuntu 24.04 guest image from the official cloud image. It checks the download's SHA256 sum, and its signature when <code>gpgv</code> and the Ubuntu keyring are installed, with user <code>femu</code> , an SSH key, a serial console, <code>nvme-cli</code> and <code>fio</code> . Needs no root, but needs access to <code>/dev/kvm</code> unless you pass <code>--no-kvm</code> . See <code>guest-image.md</code> (§2.3). | <code>-o</code> DIR output directory, <code>-n</code> FILE image name (default <code>u20s.qcow2</code> , the name the launchers expect), <code>-s</code> SIZE (default 32G), <code>-k</code> FILE public key, <code>-p</code> PW console password, <code>--cxl</code> adds <code>ndctl</code> , <code>daxctl</code> and <code>cxl-cli</code> , <code>--packages</code> LIST extra packages, <code>--qemu</code> PATH, <code>--qemu-img</code> PATH, <code>--no-kvm</code> , <code>--timeout</code> SEC (default 1800), <code>-f</code> replace an existing image. Reads <code>IMGDIR</code> (default output directory <code>\$HOME/images</code> ), <code>QEMU</code> and <code>QEMU_IMG</code> . |
| <code>run-guest-ssh.sh</code>    | Logs in to a running guest built by <code>make-guest-image.sh</code> , or runs one command in it: <code>./run-guest-ssh.sh sudo nvme list</code> .                                                                                                                                                                                                                                                                                      | <code>IMGDIR</code> (default <code>\$HOME/images</code> ), <code>SSH_KEY</code> (default <code>\$IMGDIR/femu-guest-key</code> ), <code>SSH_PORT</code> (default 8080), <code>GUEST_USER</code> (default <code>femu</code> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### C.3 Launchers

Run each NVMe launcher from `build-femu/` after `femu-copy-scripts.sh`. They start `./qemu-system-x86_64` with `sudo`, `KVM`, `-cpu` `host`, `-nographic`, `-name` `NAME`, `debug-threads=on` (so the host shows FEMU's thread

names), a virtio-scsi boot disk and user networking that forwards host port SSH\_PORT to the guest's port 22. run-cxlssd.sh is different: it is not copied, so run it as `../femu-scripts/run-cxlssd.sh`, and it adds no disk, no network and no sudo (see below). All except run-cxlssd.sh read:

| Variable | Default             | Meaning                                                                                               |
|----------|---------------------|-------------------------------------------------------------------------------------------------------|
| IMGDIR   | \$HOME/images       | Directory of the guest image                                                                          |
| OSIMGF   | \$IMGDIR/u20s.qcow2 | Guest image file; the script exits with status 1 if it does not exist                                 |
| SSH_PORT | 8080                | Host port forwarded to the guest's SSH port; use a different one for each VM you run at the same time |

The device geometry, size and timing are shell variables at the top of each script. Edit them there; they are not read from the environment.

| Launcher            | Device                                                                                                                                | Guest                                                      | Output files                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| run-blackbox.sh     | BlackBox SSD, 12 GiB namespace on 16 GiB of NAND (BlackBox (§4.4))                                                                    | 4 vCPUs, 4 GiB                                             | log, qmp-sock                                                                                                    |
| run-blackbox-fdp.sh | BlackBox with a femu-subsys that has FDP on, 4 handles, 1 reclaim group (FDP (§4.5))                                                  | 4 vCPUs, 4 GiB                                             | /tmp/femu-fdp.log, qmp-sock                                                                                      |
| run-nossd.sh        | NoSSD, 4 GiB (NoSSD (§4.2))                                                                                                           | 4 vCPUs, 4 GiB                                             | qmp-sock                                                                                                         |
| run-zns.sh          | ZNS, 4 GiB, QLC timing, 16 zones of 256 MiB (ZNS (§4.7))                                                                              | 4 vCPUs, 4 GiB                                             | log, qmp-sock                                                                                                    |
| run-whitebox.sh     | Open-Channel 2.0 (OCVER=2 in the script; 1 selects 1.2), 4 GiB (OCSSD (§4.9))                                                         | 4 vCPUs, 4 GiB                                             | qmp-sock                                                                                                         |
| run-csd.sh          | Computational storage, 4 GiB, 4 compute units (CSD (§4.13)). It does not set csd_program_dir, so only the built-in program type loads | 4 vCPUs, 4 GiB                                             | log, qmp-sock                                                                                                    |
| run-kvssd.sh        | Key-value SSD, 4 GiB of value space on the BlackBox geometry (KV (§4.11)); the guest needs Linux 6.0 or newer                         | 4 vCPUs, 4 GiB                                             | log, qmp-sock                                                                                                    |
| run-cxlssd.sh       | One femu-cxl-ssd below a CXL host bridge (CXL SSD (§4.18))                                                                            | 4 vCPUs, 4 GiB, no disk and no network unless you add them | cxlssd-stats.log, cxlssd-io-N.log and cxlssd-spt.log in LOG_DIR when the guest asks for them through lsa-control |

run-blackbox.sh also passes FEMU\_EXP\_LOG, FEMU\_SECRET and FEMU\_DUMP\_LPN through sudo to QEMU (environment variables (§A.1.5)). The other launchers pass no variables to QEMU.

qmp-sock is created by root in the current directory; log is written by tee as you. Two launchers started from the same directory share them, so start a second VM from another directory.

run-cxlssd.sh runs QEMU without sudo and adds its own arguments after the ones it builds, so you append a boot disk, a network and a QMP socket on its command line. Its settings are environment variables:

| Variable    | Default                | Sets                                    |
|-------------|------------------------|-----------------------------------------|
| QEMU        | ./qemu-system-x86_64   | the QEMU binary                         |
| CXL_SIZE    | 256M                   | media size, a number followed by M or G |
| CACHE_PAGES | size in MiB / 20 x 256 | cache-pages                             |

| Variable                                            | Default                              | Sets                                        |
|-----------------------------------------------------|--------------------------------------|---------------------------------------------|
| CACHE_WAYS                                          | 1                                    | cache-ways; full means CACHE_PAGES          |
| BLOCKS_PER_PLANE                                    | 768 for 48G, 1536 for 96G,<br>else 0 | blocks-per-plane                            |
| CACHE_POLICY                                        | fifo                                 | cache-policy                                |
| DER                                                 | off                                  | der                                         |
| CYLON_KERNEL_ACK                                    | off                                  | cylon-kernel-ack                            |
| PREFETCH_DEGREE, PREFETCH_STRIDE                    | 0, 1                                 | prefetch-degree, prefetch-stride            |
| CHANNELS, LUNS_PER_CHANNEL, PAGES_<br>PER_BLOCK     | 8, 8, 256                            | NAND geometry                               |
| READ_NS, PROGRAM_NS, ERASE_NS,<br>CHANNEL_NS        | 40000, 200000, 2000000, 0            | NAND timing                                 |
| GC_THRESHOLD, GC_THRESHOLD_HIGH                     | 75, 95                               | GC thresholds                               |
| FTL                                                 | on                                   | ftl                                         |
| LSA_CONTROL                                         | on                                   | lsa-control; see security (§5.13.1)         |
| CYLON_FIRST_TOUCH_PROGRAM, CYLON_<br>FREE_WRITEBACK | off                                  | the matching properties                     |
| LOG_DIR, LOG_LIMIT                                  | ., 64M                               | log-dir, log-limit                          |
| TRACEFS_DIR                                         | unset                                | tracefs-dir, only when set                  |
| CXL_BACKEND                                         | memory-backend-ram                   | memory backend type and options             |
| ACCEL, CPU, CPUS, RAM                               | kvm, host, 4, 4G                     | accelerator, CPU model, vCPUs, guest<br>RAM |
| DRY_RUN                                             | 0                                    | 1 prints the command instead of running it  |

These defaults follow Cylon's launch script and differ from the device's own defaults: one cache way instead of 16, a cache of size / 20 (3072 pages for 256 MiB) instead of 1024 pages, 8x8 channels and LUNs instead of 4x4, fixed blocks-per-plane for the 48G and 96G sizes, and lsa-control on instead of off.

## C.4 Configuration files

| Script                                             | What it does                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ssd-config.sh CONFIG [-<br>-device-only   --check] | Expands an INI-style file into -device femu,... arguments (and a -device femu-<br>subsys,... for a [subsys] section). Keys are device property names; mode = bbssd and<br>the like stand for femu_mode. With --device-only it omits the -device words; with --<br>check it only validates. It checks keys against -device femu,help of the binary in FEMU_<br>BIN, or of build-femu/, build/ or build-official/ under the source tree. |
| ssd-config-test.sh<br>[QEMU]                       | Expands every file in configs/, starts QEMU with each and requires the device to come<br>up, and checks that the parser rejects bad input. The binary is the argument, FEMU_BIN, or<br>the first one found as above. CI runs it.                                                                                                                                                                                                       |

The files in configs/:

| File                       | Device                                                                                        |
|----------------------------|-----------------------------------------------------------------------------------------------|
| bbssd.conf                 | BlackBox, 4 GiB, 8 channels of 8 LUNs                                                         |
| bbssd-overprovisioned.conf | BlackBox sized with op_pcent=10                                                               |
| csd.conf                   | Computational storage, the device of run-csd.sh without its no-effect scheduler<br>properties |

| File               | Device                                                                         |
|--------------------|--------------------------------------------------------------------------------|
| fdp.conf           | BlackBox with FDP on a [subsys] section                                        |
| heterogeneous.conf | One controller with a BlackBox, a ZNS and a NoSSD namespace                    |
| kvssd.conf         | Key-value SSD, the device of run-kvssd.sh                                      |
| nossd.conf         | NoSSD with link and firmware time from the NVMeCHA controller (NoSSD (§4.2.3)) |
| ocssd.conf         | Open-Channel 2.0, the device of run-whitebox.sh; lver = 1 selects 1.2          |
| qlc.conf           | BlackBox with QLC per-page-type timing                                         |
| write-buffer.conf  | BlackBox with a 2048-page write buffer, vwc=1 and Write Zeroes                 |
| zns.conf           | ZNS, 16 zones of 256 MiB, at most 16 active and 8 open                         |

Run it from build-femu/ with FEMU\_BIN set. Through the ../femu-scripts link the script cannot find the binary on its own, and then it skips the key check:

```
| FEMU_BIN=./qemu-system-x86_64 ../femu-scripts/ssd-config.sh ../femu-scripts/configs/zns.conf
```

The launchers take no arguments. To boot a config, copy a launcher and put the output in place of its -device femu option, or call QEMU yourself:

```
| QEMU_ARGS=$(FEMU_BIN=./qemu-system-x86_64 ../femu-scripts/ssd-config.sh my-ssd.conf)
| ./qemu-system-x86_64 -enable-kvm -cpu host -smp 4 -m 4G $QEMU_ARGS ...
```

The file format: keys are femu device properties and mean what -device femu,help says. # and ; start comments, section headers are labels for the reader (except [subsys]), and a key with an empty value is ignored. List values such as namespace\_modes = bbssd,znssd,nossd get their commas doubled for QEMU automatically.

```
| [device]
| mode      = bbssd          # friendly name for femu_mode
| devsz_mb  = 4096
|
| [geometry]
| secs_per_pg = 8            # 4 KiB pages
| luns_per_ch = 8
| nchs       = 8
|
| [timing]
| pg_rd_lat  = 40000         # ns
| pg_wr_lat  = 200000
```

A misspelled key stops the expansion when the binary is found:

```
| ssd-config: unknown property 'gc_policy' -- not one FEMU accepts
| ssd-config: config rejected; see the warnings above
```

## C.5 Guest-side test tools

Copy these into the guest and run them there. The C programs build with gcc -O2 -o NAME NAME.c and need no headers beyond libc.

| Tool                   | Arguments                                                  | What it checks                                                                                                                                                                                                                      |
|------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| femu-test.sh           | --yes [DEVICE], default /dev/nvme0n1                       | Data integrity, counters, deallocate, and zone or key-value commands, chosen by what the namespace reports. <b>It overwrites the whole namespace</b> , so --yes is required, and it refuses a mounted device. See testing (§8.2.5). |
| kv-probe.c             | [CONTROLLER], default /dev/nvme0                           | A key-value Store, Exist, Retrieve, Delete and a Retrieve of the deleted key (KV (§4.11))                                                                                                                                           |
| aer-probe.c            | [CONTROLLER], default /dev/nvme0                           | That crossing the temperature threshold completes an Asynchronous Event Request, and that the event is re-armed after the log is read                                                                                               |
| zone-aen-probe.c       | [CONTROLLER] [NAMESPACE], defaults /dev/nvme0 /dev/nvme0n1 | That a zone taken read only raises the Zone Descriptor Changed notice; needs a ZNS device with err_write_fail_ppm set (ZNS (§4.7))                                                                                                  |
| fdp-test-nvme-admin.sh | none; uses /dev/nvme0, /dev/nvme0n1 and /dev/ng0n1         | The FDP admin commands of nvme-cli against the configuration run-blackbox-fdp.sh creates (4 handles, 1 reclaim group); written for a Linux 6.12 guest                                                                               |

## C.6 Host tuning helpers

| Script                            | What it does                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ftk/qmp-vcpu-pin -s SOCKET CPU... | Pins each vCPU thread to a host CPU with taskset, using QMP query-cpus-fast on SOCKET; vCPU i goes to the i-th CPU in the list, wrapping around. It imports ftk/qmp.py. Run it with sudo when QEMU runs as root. A Unix socket path longer than about 107 bytes fails with AF_UNIX path too long.                                                                                                                                                                                                                                                                                                     |
| pin.sh [FIRST_CPU]                | Pins each vCPU thread, then each femu-poller, FEMU-FTL-Thread, femu-cxl-ftl and femu-cxl-cca, to its own host CPU, starting at FIRST_CPU (default 0), and moves the other QEMU threads, femu-csd-cu included, to the CPUs after those. It finds the threads by name, so QEMU must run with -name NAME, debug-threads=on (the launchers pass it), and it finds QEMU with pgrep -x qemu-system-x86; set QEMU_PID when several run. Run it after the guest has booted: the pollers start when the guest enables the controller. It stops if the host has too few CPUs. See performance tuning (§5.12.3). |
| set_cpu_perf_mode.sh              | Sets every CPU's cpufreq scaling policy to performance through sysfs. Run it as root.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## C.7 Documentation tooling

These run in CI. docs-maintenance.md (§8.3) explains each.

| Script                | Arguments                                                                                                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gen-property-docs.py  | --qemu BINARY regenerates reference/properties.md and runtime-properties.md from the binary; --check compares instead of writing (--qemu is still required)      |
| gen-mode-table.py     | rewrites the mode tables from docs/modes.py; --check compares and checks modes.py against the code                                                               |
| check-doc-links.py    | [--root DIR] [PATH...]; checks every relative link and anchor                                                                                                    |
| check-doc-examples.py | --lint, --list, --self-test, --qemu BINARY, --qos-test BINARY, --only NAME, --timeout SEC, [PATH...]; checks every code block's tag and runs the tagged examples |



## C.8 CXL caching API tools: hw/femu/tools/cca/

Guest code for femu-cxl-ssd,cca=on. make -C hw/femu/tools/cca in the guest builds libcca.a, the ccactl command and the cca-test self-test. run-guest-tests.sh [-d MEMDEV] [-x DAX] [-o LOG] [CASE...] builds them and runs the self-test as root, logging to cca-guest-YYYYMMDD-HHMMSS.log by default. See the caching API guide (§4.20) and the tool README<sup>116</sup>.

## C.9 Legacy scripts

None of these start FEMU. Several hard-code paths on one lab machine (/home/huaicheng/images), the old x86\_64-softmmu/qemu-system-x86\_64 binary, or a u14s.qcow2 image.

| Script                                                         | What it was for                                                                                                         |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| femu-run.sh, m.sh, s1.sh, s2.sh                                | Start a VM with QEMU's stock -device nvme                                                                               |
| f.sh, dp.sh, dp-run.sh, ide-run.sh, virtio-run.sh, null-run.sh | Start a VM with a raw, tmpfs-backed, IDE, virtio-blk or null data disk instead of FEMU, for comparison runs             |
| gdb-run.sh                                                     | Start the stock nvme device under gdb; see debugging (§7.2.2) for a version that runs FEMU                              |
| valgrind-run.sh                                                | Start the stock nvme device under valgrind, with properties that no longer exist                                        |
| aff.sh PID                                                     | Pin 20 consecutive thread IDs starting at PID to CPUs 5 to 24                                                           |
| getaff.sh                                                      | Print the CPU affinity of every thread of every qemu process                                                            |
| pre.sh                                                         | Turn off address space randomization, format /dev/nvme0n1 with ext4 and mount it for one lab user. <b>Destroys data</b> |
| pre-all.sh                                                     | Run pre.sh and network setup over SSH on three named lab hosts                                                          |
| tuning.sh                                                      | Stop a list of services on one lab machine                                                                              |

One more NoSSD launcher, and a benchmark harness in a subdirectory of hw/femu/scripts/ with its own README, are not covered here.

## C.10 Related pages

- Build (§2.2)
- Guest image (§2.3)
- Performance tuning (§5.12)
- Testing (§8.2)

<sup>116</sup><https://github.com/MoatLab/FEMU/blob/master/hw/femu/tools/cca/README.md>

## D Changelog

User-visible changes to FEMU. Each entry names the commits it comes from; `git show <hash>` has the details.

### D.1 Unreleased (since femu-v9.0.1)

655 commits after the `femu-v9.0.1` tag (2024-06-20), dated 2024-08-07 to 2026-10-01, plus the documentation and script fixes that close this list. Tests, CI-only changes, internal refactoring and wording fixes are not listed one by one.

#### D.1.1 QEMU base

- FEMU now builds on QEMU 10.1.0 instead of 9.0.1, with device registration moved to the current QEMU APIs (eafddfd0).
- Two QEMU TCG fixes are carried from upstream QEMU. Without them a vCPU could resolve a stale memory section after a device was unplugged and stop QEMU with an assertion (ef516ed58, c66409e3b).
- Two QEMU CXL fixes are carried: the background command timer is freed with its CCI, and a CXL Type 3 device releases its mapped backends when realize fails (43040a04c, 6153becf8).
- QEMU's network block drivers are left out of the FEMU build, which fixes compilation against libnfs 6 (81c4385fc).

#### D.1.2 New devices, modes and features

##### CXL SSD (new device)

- New `femu-cxl-ssd` device: a CXL memory device whose accesses go through a DRAM buffer cache and FEMU's FTL and NAND timing (bb9aea670, a8627fcf9, 7777c0d53, 6ffe34f17).
- The buffer cache gained runtime controls, scalable associativity, per-page pin and remove, and NAND geometry properties for larger capacities (f5ba3ff12, 38b314181, 76f180b9d).
- Direct mapping modes `der=memslot` and `der=cylon` map hot pages straight into the guest, with configurable direct mapping ratios and LSA mailbox commands to control them (80803505b, b3e042a84, 1f26fb2df, 8266a291e, a1e550885).
- A CXL caching API is served on BAR5, with a guest library, the `ccactl` tool and a guest self-test (924714c18, 1235e87d2, ec3af4ba4, 0a357d11e, defd999b8).
- Cache misses to different pages wait for the media together, by default only while a direct mode is active, and long media waits sleep instead of spinning (b72d14dbe, 64d424dd2, 133c553ca, 145fcfe4b).
- A `bbssd` NVMe controller can share a `femu-cxl-ssd` medium through `cxl_ssd=<id>`, with writes, deallocates and flips kept consistent between the two front ends (146a240dc, d73a5b780, a7a3c5678, 353bbe107, 287e5c885, 6783d50af).
- A runtime fast-load switch on `femu-cxl-ssd` lets warmup and data loading skip the wait for modelled media time while the FTL, cache and counters still run. Switching it off waits for the queued NAND work, reported in `fast-load-drain-ns`, so measurement starts on an idle model.
- The `femu-cxl-ssd` FTL worker wakes only the access whose request it finished, instead of every waiting access. On a `der=cylon` test with 8 and 16 vCPUs doing random reads that all miss a 256-page cache, throughput rose about 40% (65k to 91k and 70k to 98k accesses/s), because waiters no longer wake for other requests.

## New modes and data placement

- Flexible Data Placement (FDP) on a new femu-subsys device, with reclaim unit handles, FDP log pages, I/O Management commands and several GC strategies (c966d341a, 6df00de01).
- FDP placement features can be read and enabled by a Linux host, and the host sees its active reclaim unit and remaining space (8f00c7117, 9a66fe52d).
- Computational Storage Drive mode (femu\_mode=4) with program load and execute commands, routed through the bbssd FTL (eb01bb73b, 50b65ca91).
- CSD nr\_cu and csf\_runtime\_scale now take effect (202ab7afb).
- A CSD runtime shorter than the program's own run on the host cannot be reached, since the completion carries the result; QEMU now warns once when that happens (b8e8d6b45).
- Key-value SSD mode (femu\_mode=5) with store, retrieve, list, delete and exist commands (e72465909).
- Every KV namespace has its own command set and key space, the KV command set is advertised to the host, and the KV Configuration feature is served (39664d242, e62db84f2, e4949509c).

## Namespaces

- Multiple namespaces per controller via namespaces and namespace\_sizes (140b2891e, 2e25897fe).
- Namespaces on one controller can run different modes via namespace\_modes (053fd8936).
- Opt-in Namespace Management and Attachment (ns\_mgmt=on) for NoSSD and bbssd, with a bbssd namespace cap (bbssd\_ns\_limit) (b12bc8754, d9c9f2251, b48799800, e0fa92f19, 42369e0bd).
- Namespace attribute changes are reported through AER and log page 04h, and recorded in the Persistent Event log (3630d6fa0, cc9e271b4).
- Namespaces can be created with protection information (c213b120c).
- Shared namespace management across controllers of one femu-subsys, ns\_mgmt=on, with the subsystem owning namespace storage (ca80e3dc5, 23844afaa, feebe635c, 76cc58f63, 97fc6ae0a, 349f12d40).

## Metadata, protection information and Streams

- Separate LBA metadata (meta, mc) on bbssd and NoSSD namespaces (62d149dd5).
- Metadata interleaved with data (extended LBAs, extended=1 or Format with MSET) (ba36f0f68).
- Opt-in protection information formats (pi=on), with generation and checking on Read, Write, Verify, Compare, Write Zeroes and Copy (bb2f3fb06, d1d8e1593, 617f09c10, 1854c1940, f6dc00f46, 98fcaedea, 8f7382c2a, 9661e5adb).
- Opt-in Streams directive (streams, streams.max), with stream writes placed on separate bbssd write frontiers (623637942, 8e7387268, cb459a65b, e3529d48f).

## NVMe commands, features and log pages

- SGL data transfers (sgl), now honoured on every I/O command that carries data (1fa43530d, d24705851).
- DULBE and Write Zeroes with Deallocate report deallocated blocks (98247b96f).
- TRIM (Dataset Management deallocate) on bbssd (d749928e9).
- Verify command (e8c2c8933).
- Copy command, including copies between namespaces (descriptor format 2) (b16b3b3ae, 04b3da484).
- Device Self-test and its log (b3e5965d4).
- Sanitize block erase and the Sanitize Status log (4997dbdd9).
- Telemetry Host-Initiated and Controller-Initiated logs (d859ad4d6).
- Get LBA Status and the LBA Status Information log (bfc02d6a3).
- Timestamp feature (5323f363b).
- Persistent Event log, with format, sanitize, telemetry, error and Set Features events, optionally kept across runs in pel\_file (87ea8154c, 93dddf1d2, 276c93e23, 3a368e984).

- Asynchronous Event Requests are accepted and events are reported (7d5d2f84f, b53ec7122).
- Supported Log Pages (log 00h) and the I/O Command Set Profile feature are answered (1b15f38df, 481eb6df2).
- SMART reports host data units and command counts, available spare (with an optional nand\_bad\_blocks model), percentage used (from the cell type or pe\_cycles\_rated) and media errors (f07d117f1, a20d9ac94, a8a28d451, df8a3492a, e25ddf7be).
- The Endurance Group log is filled in (074b63f4f).
- Media counters (write amplification, host and relocated pages, write buffer hits, read reclaims, retention refreshes) are reported in vendor log page C0h (e57448e41, 4cb1f9ba7, e9746f6bf, e7f182b87).
- Opt-in power loss model (power\_loss) that drops data still held in the volatile write buffer and records the event in SMART and the event log (46602b0e7, 303c77aef, 7679c443d).

## **ZNS**

- Write buffer and FTL for the ZNS SSD, and zone reset charged with erase time (0326ab489, b3272c013).
- Configurable zone resource limits (zns\_max\_active, zns\_max\_open) and zone descriptor extension size (6404dd233).
- Optional conventional zones (zns\_num\_conv\_zones) (6e7395b55).
- Configurable zone width (zns\_chnls\_per\_zone) (e04e4e901).
- Zone Random Write Area support (zns\_zrwa\_\*) (b18f56b2b, 9f51854d7).
- An injected write fault (err\_write\_fail\_ppm) takes the zone read only and lists it in the Changed Zone List (ccdae5d3b).
- Changed Zone List log page (569608a41).
- Configurable reads across zone boundaries (zns\_cross\_zone\_read) and Zone Append size limit (zns\_zasl\_bs) (78caee058, bca834133).
- ZNS takes NAND timing from the shared media layer, with configurable read, program and erase times and an opt-in channel bus (3d6056020, 34bbe45fa, 2a4ebc2cb).
- Configurable write cache count (zns\_num\_wc) (66f432117).
- Optional program/erase suspend for reads (zns\_pe\_suspend, zns\_tsusp\_ns) (9ec423985).

## **BlackBox SSD (bbssd) and NAND media**

- NAND operation timing moved into a shared media layer used by bbssd and ZNS (46cb582c9, 711d537bc).
- Pluggable GC victim selection (gc\_policy: greedy, random, cost-benefit, fifo, d-choice) (7222d6744).
- Opt-in DRAM read cache (read\_cache\_mb, cache\_evict) (3b21542de, fc5d3ded0).
- Pluggable L2P mapping (mapping: page, dftl, hybrid, fast) with a modelled mapping cache (mapping\_cache\_mb) (64a19a6f3, 46e4261f9, 0f9b6aeab).
- Write amplification accounting, and a debug\_ftl switch that prints page-state violations found on the GC path (e57448e41).
- Optional cell-type NAND timing (nand\_cell\_type, cell\_pages, pgtype\_lat, ONFI phase times, ecc\_step\_ns) (e405dbcbc).
- Optional modelled TRIM time (trim\_lat\_ns) and explicit over-provisioning (op\_pcent) (1027eb240, 6a4559ee4).
- DRAM write buffer (buffer\_size, buffer\_thres\_pcent) with Flush, FUA and volatile write cache support (6a8daec7a, d4d03f127, 0ee6d211b).
- Hot/cold separation of overwritten pages (hot\_cold\_sep) (4ded2e928).
- Read reclaim (read\_reclaim\_limit) and retention refresh (retention\_limit\_sec) (7f9b4f6af, 4e09c4797, e7f182b87).

- Data age feeds the ECC read model (ecc\_retention\_sec) (f9433e9ab, 13f2b85de).
- More than one plane per LUN (pls\_per\_lun) in bbssd, FDP and KV, with a line's planes erased in one operation (0f554fb7d, 3699e980d, 9c8228d28, 7276af2d8).
- Channel bus phases (cmd\_addr\_lat, pg\_xfer\_lat, status\_lat, ch\_xfer\_lat) are added to the timing when set (c274ba7d9).
- A read can suspend an in-flight program or erase (pe\_suspend, tsusp\_ns) (9ec423985, 6a5c498a8).
- Optional debug logging to study whether deleted data remains on the device (18ba6557c, 45e61ae41).

## Timing and fault models

- Fault insertion (err\_read\_unc\_ppm, err\_write\_fail\_ppm), a host link model (pcie\_bandwidth\_mbps, pcie\_prop\_delay\_ns) and a controller CPU model (fw\_cpu\_ns), now also applied to NoSSD (7bfac4c3a, 2cdd55c52).

## Open-Channel SSD

- Opt-in OC 1.2 channel transfer timing (oc12\_channel\_timing) (d174f9551, 5cf3996c0).
- OC 2.0 gets media timings, chunk resets are charged as erases, an unwritten block reads as the predefined pattern, and a host can reset a chunk it has not filled (b652c3d87, 9d176f891, 00daad11c, 90ff859c1).

## NoSSD

- Poller threads decoupled from queues (poller\_ratio), and I/O counters kept per poller (0b192893b, 6b1592746).
- The backend memory can be bound to or interleaved across NUMA nodes with FEMU\_MBE\_INTERLEAVE (3594d27c6).

## D.1.3 Configuration changes and new refusals

A command line that ran before may stop at realize, with a message naming the property. Each of these was accepted before and then ignored, or failed a check without saying so. Nothing here changes a configuration that was already being honoured. If one of these stops a run, remove the property: it was not doing anything.

### Refused at realize (previously accepted and ignored)

| Property                                                           | Why it is refused                                                                                                                                                                                                                                    | Commits                                    |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| any violated controller constraint                                 | The check ran but returned silently, leaving QEMU up with no FEMU PCI device and no namespaces. The reason is now reported.                                                                                                                          | 9716c0a9f                                  |
| mpsmax below mpsmin or above 15                                    | The test used to be inverted, so mpsmax=1 was rejected and mpsmin=1, mpsmax=0 accepted, advertising CAP.MPSMIN above CAP.MPSMAX.                                                                                                                     | 9716c0a9f                                  |
| meta with dpc or dps, with nlbaf above 8, or without a matching mc | Metadata is implemented for NoSSD and bbssd (separate buffer or interleaved with the data), but not together with the legacy protection settings; use pi for protection information. Before metadata was implemented, any non-zero meta was refused. | d589ce2b8, 62d149dd5, ba36f0f68, bb2f3fb06 |
| cell_pages above 5                                                 | Indexed past the page-type multiplier table.                                                                                                                                                                                                         | d589ce2b8                                  |
| nand_cell_type with pgs_per_blk above 512                          | Read past the page-type latency tables.                                                                                                                                                                                                              | d589ce2b8                                  |
| gc_strategy outside {0,1,2,4}                                      | Other values silently fell back to greedy or never collected at all.                                                                                                                                                                                 | d589ce2b8                                  |

| Property                                                                                                                                                    | Why it is refused                                                                                                                                                           | Commits   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| zns_flash_type 0, 6 or above, or MLC/PLC without explicit latencies                                                                                         | 0 gives a zero-length write cache and an endless flush loop; 6+ indexes past the timing tables; MLC and PLC have no built-in figures, so every NAND operation cost nothing. | b234d27c8 |
| femu_mode above 5                                                                                                                                           | No mode registers command handlers for it (6 was a SmartSSD placeholder), so the controller came up with none.                                                              | c3b8e88ef |
| multipoller_enabled other than 0 or 1                                                                                                                       | Values above 1 started several pollers that each swept every queue, so two pollers could run and complete the same command.                                                 | de5fcd972 |
| lver other than 1 or 2 with femu_mode=0                                                                                                                     | No Open-Channel handlers were registered for it.                                                                                                                            | 498afdd7a |
| flash_type outside 1 to 4 with femu_mode=0                                                                                                                  | OCSSD 2.0 indexed the SLC to PLC timing tables with it unchecked, so 6 or above read past them; 0 and 5 have no built-in figures. OCSSD 1.2 already refused them.           | 4891d870e |
| zns_num_plane above 8, zns_num_ch above 128, or a page count above 65536                                                                                    | Wrapped and aliased onto lower indices in the PPA.                                                                                                                          | b234d27c8 |
| zns_chnls_per_zone that does not divide zns_num_ch                                                                                                          | Was silently replaced by the full channel width.                                                                                                                            | b234d27c8 |
| bbssd knobs under FDP: buffer_size, hot_cold_sep, read_reclaim_limit, retention_limit_sec, ecc_retention_sec, trim_lat_ns, non-default mapping or gc_policy | FDP keeps its own write and reclaim path; none of these reach it.                                                                                                           | d589ce2b8 |

## Other new refusals

- Invalid bbssd and zoned geometries, and a bbssd namespace with no room left for garbage collection; CSD and KV, and every mode a controller serves, now run the same checks (b4bbd93e0, 1ac931d86, ceb2b5bf7, 5ccbe3f24, bf8961b9b, 0b82aa58a).
- Invalid ZRWA configurations, page sizes ZNS cannot serve, and more write caches than zones (b89cde46d, 154775e62, fcfa0d581).
- Unknown mapping or gc\_policy names, and a negative suspend overhead (29c397f38, 2ae58aa55).
- A CSD controller with more than one namespace (b425007c2).
- FDP with more than one reclaim group, and other placement configurations the device cannot serve (f08762450, 8ed17a495).
- An FDP namespace larger than what is left once each handle has a reclaim unit open, each Persistently Isolated handle one to collect into, and forced GC its free units. Such a namespace was accepted and then failed most random writes. A partly exposed last page now counts as a whole one in this check, with or without placement (89f34939f).
- Open-Channel namespaces on a controller in another mode, and OCSSD geometries the timing model cannot index (8cd92eec5, 598498e93, f4e85634a, 707d8c8c1).
- Controller memory buffer settings that do not fit, including any CMB BAR other than 2 (1c3a5a2a2, 9d6e13cd3).
- Queue entry sizes other than 64 and 16 bytes (23b21cc38).
- A format with protection information but no metadata room for it (1c9808bcd).

- On femu-cxl-ssd, a direct ratio without DER, and der=mems1ot under TCG (34b9f8f9e, d03b0606e, f1182b72e).

### Accepted with a warning (still no effect)

- serial, ms, ms\_max, dlfeat, tplpbsy, tplrbsy, trcbsy, and the CSD nr\_thread, time\_slice and context\_switch\_time are still accepted, but nothing reads them, so a value other than the default now prints one warning at realize naming the property; for ms it points to meta (f67568880).

### Behaviour changes (still boots, numbers move)

- A CSD namespace now goes through its FTL, so reads and writes take NAND time instead of completing instantly. A pure-CSD device previously timed out on its first I/O and the kernel disabled the controller (50b65ca91).
- A namespace whose mode differs from the controller's is routed by its own mode. A bbssd namespace on a NoSSD controller no longer completes inline (50b65ca91).
- pcie\_bandwidth\_mbps, pcie\_prop\_delay\_ns and fw\_cpu\_ns now apply to NoSSD. Setting any of them takes the request off the inline completion path, which costs throughput; leaving them unset keeps the previous path (2cdd55c52).
- cmd\_addr\_lat, pg\_xfer\_lat, status\_lat and ch\_xfer\_lat now add channel bus time on bbssd. The bundled run scripts pass 0 and are unaffected (c274ba7d9).
- zns\_cmd\_addr\_lat, zns\_pg\_xfer\_lat and zns\_status\_lat add the same channel bus to ZNS. They default to 0, which leaves ZNS timing exactly as it was; a negative value is refused at realize (2a4ebc2cb).
- Temperature threshold Set Features accepts only TMPSEL 0 and Fh; other selectors are rejected rather than stored as part of the value (c7eaa6b4a).
- An aborted command completes as Command Abort Requested rather than Invalid Opcode (03a4eee3c).
- FDP RUAMW counts down in LBAs, so it drains at the documented rate and reaches zero; cost-benefit victim selection now orders by age and utilisation rather than insertion (9a66fe52d).
- The data plane starts when the host enables the controller rather than on Doorbell Buffer Config, so hosts that never send that command (SPDK, FreeBSD, Windows, older Linux) now work (04f04b86e).
- Failure to lock the backend memory is a warning instead of a fatal error (847864637).
- The media counters moved from the SMART log's temperature fields to vendor log page C0h (4cb1f9ba7).
- The register BAR is at least 16 KiB (01c79c442).
- Under FDP, forced GC keeps at least one reclaim unit free unless gc\_thres\_pcent\_high is 100. Below 20 units the watermark rounded to none, and a pass that filled its destination part way had no unit to go on with (89f34939f).
- With hot\_cold\_sep, Streams, or a hybrid or fast mapping, forced GC keeps at least one line free. Below 20 lines, where the watermark used to round to zero, such a namespace can expose one line less (5a81d02cd).
- OC 1.2 Identify no longer advertises hybrid commands, and unsupported OC 1.2 block opcodes are rejected (f5fc7d573, 6b5081821).
- The run-\*.sh launchers name QEMU's threads (-name ..., debug-threads=on), so ps -T and top -H show femu-poller, FEMU-FTL-Thread and CPU N/KVM (e29fe6ee2).

The shared namespace model behind femu-subsys,ns\_mgmt=on is described in namespace management (§4.17.2).

## D.1.4 Fixes

### Crashes, memory safety and teardown

- Fixed a memory leak and build errors from the previous release (b11ce9631, 1c2d358c1).
- Fixed ZNS plane allocation and the DSM buffer size (f863d7a8e, 805b9c8e6).
- Fixed a submission queue setup race and a poller use-after-free on controller reset or guest reboot (7100e6179, ac89117b2).
- Fixed priority queue heap repair, which affected FDP victim selection (e2d5413ff, 66067fab, a4ed1d8c9).
- I/O past the device geometry is refused instead of accessing out of bounds (8848d724d).
- Widened bbssd and ZNS physical address fields so large channel, LUN and plane counts no longer alias (2e4c7fe82, d7c5aff58).
- Fixed an out-of-bounds write in ZNS zone reset with a narrow zone width, and ZNS backend addressing within a namespace (02a11cc51, 316ecf9df, cba095fb5).
- Fixed OCSSD writes past the backing store and many OCSSD bounds checks on vector lists, chunk info, bad block tables, offsets and addresses (fca17a8c6, f2ccb06ab, e8d415a07, 11aafda03, b37e7dd58, ff6e8bb4b, 58ba9f515, 98641558d, d44f0c4c1, 761f78757, 42ae8360e, 1f135b961, d5a5b788b, 91412b74d, 3838cf6bf, 1c9f737fe, 0d6d5926f).
- The FTL thread and pollers are stopped and every mode's state released on device removal, controller reset and failed realize (da76e206a, 06c7579b2, 81fa5697a, b3608d3f4, d0998ea16, 5a4107c14, 6610ecba1, f050a8bce, 3ba1ddec4, 42ba3cf4a, 817d34363, 36d1ef4b0, 52a579830, 8e8dc4431, 8c1fb192f, 0b7c53826, 9b0dd6eb0).
- Fixed leaks of aborted requests, DSM range lists, PRP lists and dropped command lists (03a4eee3c, f3fdbe7c0, 1c498adad, 10aaf11e2, c501af95e).
- A host address is no longer written into guest memory when setting up a non-contiguous queue, and Identify transfers the structure rather than a whole memory page (037d9c9c2, 14f4224d3).
- Queues with 64 KiB pages no longer crash QEMU, and queues are mapped by their length in bytes (cbebe5e10, 49880d0a0).
- Shadow doorbell values are bounds-checked as ring positions and the buffers are one page each (e082da2c1, b95d305b5, 33584d6b8).
- Controller DMA from device threads is restricted to directly reachable memory, which avoids a VM hang on data pointers into device registers (fbfe674c2, 60e4b2d5d).
- Report and log buffers are bounded and their lengths checked before transfer (608749b8d, 40656e5a1, 062bcd02, 0ddcc33bd, 3106c1205).
- CSD program loads are bounded and confined to csd\_program\_dir (f1d5f3736).
- CSD programs run on nr\_cu compute unit threads (femu-csd-cu) instead of the poller, so a long program no longer stalls I/O on every queue or the vCPU that sends a CSD admin command; a loaded program can no longer be reloaded in place while it runs, and deleting a queue or resetting the controller under a running program drops its result safely (4596ba498).
- Running out of lines refuses the write instead of aborting, and FDP reports device full instead of asserting or following a null reclaim unit (49ada8834, e1c4e9174, 59128ab66, e7913d89c, c0ae29cb4, cbda78bf6, 37b9b84f9, f05f128a9).
- Zone state is locked, reset zones are erased on the media thread, and asynchronous events are raised from the main loop (fe8e931e3, 5001ae15c, 89e1c4b04).
- FDP event rings are serialized and the written and uncorrectable bitmaps are updated atomically across pollers (645424c86, daaa6a3db).
- The BlackBox flip command applies to every namespace of the controller (3b7d88d63).
- femu-cxl-ssd no longer advertises that unmapped pages read as zeros (3c85bfd86).



- Many femu-cxl-ssd lifecycle, decoder and mapping fixes, including teardown while callers wait on invalidation, little-endian forwarding, and keeping stores when NAND has no page for write-back (9e0223c19, bb7e28722, 51c420143, d31c48d92, ec8b1ac7d, 02a872661, 4505cd9a7, 86ba15078, 842d4ff72, b322efb8f, 2d0c1404e).

## Data correctness

- Compare reads the stored data instead of a zeroed buffer (7589ee012).
- FDP TRIM honours the requested LBA ranges instead of erasing the whole device (0cc20e897).
- Fixed FDP victim accounting, foreground GC under write pressure, the initial reclaim unit, and the FDP Events log offset (cee9670a5, 247151a16, f9fc94d4f, 438e485a1).
- bbssd maps blocks by their real size, so 4 KiB formats no longer share logical pages (bb63df5a3).
- Fixed ZNS zone append, Zone Append bounds, zone open on write, zone resource counts, multi-zone reset, and zone write caches keyed by zone (07acf491a, 27949b51f, 3fd4f34e6, 8129cfd16, 340522142, 99ce4fb2b, 9a3449362, 347a8a691).
- A ZNS write refused for its data pointer no longer moves the write pointer (6343c4d22).
- FDP Write Zeroes is placed by its own directive fields and counted in HBMW and MBMW, instead of reusing the placement of an earlier command (834abe2cb).
- Write Zeroes and deallocate address the backend per namespace, program the media where required, and drop buffered copies (f44ce6499, 9005cf952, 373de039b, 783131b7d, 22988e07b).
- The write buffer no longer resurrects deallocated pages and is bounded (d4d03f127, 0ee6d211b).
- A Format rebuilds per-block state (a79474e73, 7e7e36161).
- Writes clear the invalid status set by Write Uncorrectable (026eb0ef0).
- CSD and KV commands address the namespace they name, and each namespace gets its own mode state (a93de9d54, aa6a5a8f7, d2a770601, ce5ad913f).
- A KV command completes when its last NAND operation does: a Store that compacts no longer counts the compaction wait twice, and still pays for it when it then fails. The per-command index read is charged on a LUN chosen by the key's hash instead of always on the LUN at address 0, which had every KV command queue on one LUN (4ca11c2a0, 1e5385ae4).
- Vendor admin command 0xEE sets an Open-Channel controller's read, program, erase and channel times; it used to write fields nothing read. Other modes refuse it with Invalid Field (949d01eb9, cdbd5bb36).
- namespace\_sizes may add up to all of devsz\_mb when the namespace count does not divide it, and each size is rounded down to whole logical blocks, so TNVMCAP counts only addressable capacity (d8cdd4d22).
- A CSD controller in an FDP subsystem refuses the same knobs as a BlackBox one and reports the superblock as its reclaim unit size, instead of 96 MiB (207bb12d4).
- bbssd GC no longer erases a line it could not empty, which left mappings pointing at erased pages and made every later write fail. Forced GC runs before every page a command programs, so a large write on a device with fewer than 20 lines no longer runs it out of space (5a81d02cd).
- FDP GC moves all of a reclaim unit's pages before erasing any block, retires each moved page's old copy, and runs foreground GC per page. A pass that stopped part way used to leave the unit with erased blocks counted again later (c7b373186).
- FDP GC takes a new unit for a collection destination that was dropped when it filled with nothing free, and collects a unit with no valid pages without one. GC used to stop for good once the destination was gone, so a full device stayed full even after the host deallocated everything (22a0fc5ee).
- An FDP handle whose last unit filled with nothing free reports no room in RUH Status. It used to report the room of its retired unit, which GC could free and give to another handle (4e7a06666).

- The random and d-choice GC policies and FDP's random reclaim strategy draw victims from a generator seeded by the new `gc_seed` property instead of the wall clock and `rand()`, so the same configuration and workload give the same victims and WAF on every run. `fifo` finds its victim at the top of a queue ordered by close order instead of scanning every line, with the same victims as before (04ba1c0aa).

### Spec conformance and host compatibility

- Commands complete on the completion queue they were bound to, and submission queues that share a completion queue are served (7a3f1d9fa, dc4289f91).
- Completions are not posted into a full queue and the phase tag is written last (6f998742f, 04ce8b1a7).
- I/O interrupts are delivered without a KVM route, and pin interrupts work with shadow doorbells (8ebffbba4, 34621540e, 4ca1be945).
- Create SQ/CQ reject out-of-range queue IDs and invalid interrupt vectors, and return the specified statuses (b3bbaedb0, 7580dd341, edefcd7c5).
- Fixed controller configuration and admin queue registers, feature reset on controller reset, and the fatal status cleared on disable (330ed2aab, f2ea41361, ab63717c5).
- Get Log Page reads its identifier and offset correctly and serves every log from the requested offset (a6b0d5897, ef8b755ee, 23fc538ea).
- Feature identifier, selector and save bit are decoded, and namespace-scoped features take a namespace (2380e4ecc, 1729244c2).
- Identify reports total NVM capacity and the fields a host relies on (907ad6ff1, 8530bc287).
- Fixed the error and SMART logs, the log directories, Format secure erase decoding, data pointer and transfer size statuses, and bad doorbell reporting (325e02cfd, c52d9ff9c, adfc12e93, bdb6f6b59, 225c6bdee).
- Fixed the AER count, Zone Append limit, KV MDTs and vector masking (7b5dbbc77).
- A failed data transfer is reported as an error, PRP lists starting inside a page are followed, and SGL errors use their own status codes (29175cb05, ecbeef8416, c9344bd4c, e310fdb43).
- The Changed Zone List reports only unsolicited changes, ZNS follows the zone resource, compare and notice rules, and commands that rewrite a zone behind its back are refused (7a6f01f8a, 41a6c189f, 99d2194e0, ed38b0728).
- A reset zone holds no data, an offline zone returns its resources, and zone geometry is reported for the namespace's format (bce5e1c3e, 2c7062ed0, 901d0908f, b4d9585ca, f0f989c21).
- KV list entries are padded to 4 bytes, NVM-only commands are refused on KV namespaces, and KV Identify requires a KV namespace (240e21c43, f5a7d7bc8, 5ec9ee859).
- FDP reports the reclaim unit size the FTL uses and moves a handle to a new reclaim unit on update (7fe32c556, 78de60448).
- The FDP Reclaim Unit Handle Usage log uses 8 byte descriptors, so `nvme fdp usage` reports every handle as host specified instead of reading some as unused, and the placement logs store their fields little endian (ca844c994, a85b869fc).
- With `namespace_modes`, the controller's model number and serial come from its own `femu_mode` instead of the namespace brought up last (efdb98643, c3f968029).
- SMART wear counters are summed across namespaces (4a4f0d9bf).
- Counters fixed to move in every mode: FDP and KV write amplification and bytes, KV relocations, FDP erases (f4b3ac376, 980fb7886, 37464e4bd, ce20d8c06, 4c7d9c22a, 749e09fef, 2a16553c9, ba0c49338).
- OC 1.2 enforces its bad block table: a write or erase that names a factory bad, grown bad or device reserved block fails with Write Fault. Set Bad Block Table marks the plane it is given instead of another block's plane, and keeps the table's counts current (933addb72).
- Abort no longer rewrites the aborted command in the host's submission queue: the controller marks it

and completes it with Command Abort Requested when fetched. An Abort run with more than acl others queued behind it fails with Abort Command Limit Exceeded (7b7eaf133).

- Logs 12h (Feature Identifiers Supported and Effects) and 13h (NVMe-MI Commands Supported and Effects) are answered and listed in log 00h. A feature that needs something absent, such as Volatile Write Cache without vwc=1, is an invalid field for every selector (7b36b4c25).
- Enabling with a CC.CSS value that CAP.CSS does not offer fails the controller, and admin commands that name SGLs are refused (992dc9911, 2ff5e3925).
- Every Unrecovered Read Error and Compare Failure sets Do Not Retry (f8efff85a).
- Identify reports one read-only firmware slot, matching the firmware log (989df3cea).
- OACS, ONCS, OCFS, LPA, SANICAP and logs 00h and 05h are built from one capability registry, and a qtest checks them against what every mode answers (a1c67d97f, 933fb3f15).
- Log 05h lists each mode's own commands (BBSSD 0xEF, the Open-Channel and CSD commands, Flush on KV namespaces) and no longer lists Read and Write for Open-Channel 1.2, which refuses them (99eabe2c1).
- Log 00h lists the Open-Channel 2.0 chunk information page (CAh) (209b499b6).
- Get Log Page answers a page log 00h lists for no command set with Invalid Log Page, so the endurance group and FDP pages without a subsystem no longer answer Invalid Field (87d5f1372).
- A zoned namespace reports no Copy limits in Identify Namespace, since it refuses Copy (92b3763a3).
- The controller reports NVMe 2.1 instead of 1.4, with what that requires: CAP.CRMS and the CRT0 register, BPCAP 01b, Identify CNS 1Fh, CNS 00h refused for a Key Value namespace (Invalid I/O Command Set), CSI-specific log pages refused for an unknown command set, and CNS 07h refused for a set CC.CSS does not enable. Open-Channel stays at 1.4 (409db7bb3).
- CAP.AMS no longer claims weighted round robin, which nothing arbitrated by, and enabling with another CC.AMS fails (409db7bb3).
- A controller in a subsystem reports its endurance group (CTRATT bit 4, ENDGIDMAX, each namespace's ENDGID, the Key Value Identify structure included) whether or not FDP is on, as log 09h already did (409db7bb3).

### D.1.5 Documentation and tooling

- Documentation moved under hw/femu/docs/ with getting-started, concepts, mode and feature guides, an FAQ, security notes and a doc map (e00c70f99, 034670c4c, 62a565ae0, fb1cf3ce6, 28aa224b7, 0bcb5cae2).
- CXL SSD, caching API and NVMe link guides, and a reference for every femu-cxl-ssd property and counter (41b318ef2, 9ec16a701, 93bfbf5c9, ccaa5d190, dcea2b9a8).
- A property reference generated from the binary and checked in CI, with every device property described in its help text (a6118d051, 9489cf692, 37aaccf27, c2a12a2eb).
- The mode table is generated from modes.py, and every documented command line is tested (6a97f8755, 12361d7f7, 573da0a1b).
- Refusals and behaviour changes that affect existing command lines are listed (9e93cb94f); that list is now part of this changelog.
- The vwc description no longer claims vwc=0 makes Flush a no-op: Flush drains the bbssd write buffer either way (f48a6bc6f).
- Documentation of the SMART vendor area, vendor log page, endurance rating, mapping schemes, GC and cache options (4e1f5bce5, 77fa23981, 9f52a098d, 658e4c9e6).
- README refreshed: build hosts for the QEMU 10.1 base, binary path, guest image steps, namespace and plane limits, debugging switches, and links to the new docs (4f327242c, a967b15fa, 4dbeea46d, b5c68a96f, f2d9dee99, b2ae45483, 65332b21a, 10f30e231).

- Added CONTRIBUTING, CODE\_OF\_CONDUCT, SECURITY, CITATION.cff and ROADMAP files (599a36676, da692f5d9).
- `ssd-config.sh` expands an INI-style config file into the device arguments, checking keys against the binary (7f605016e).
- `femu-test.sh` checks a device from inside the guest for block, zoned, KV and CSD namespaces (9fee8190c, 2244a1b9c, c17a31ea9).
- `make-guest-image.sh` builds an Ubuntu 24.04 guest image, and the run scripts accept IMGDIR, OSIMGF and another SSH port (e877f2dc0, e74535012, 66bb3068e).
- The build script fails on a compile error, and the config self-test fails when FEMU does not survive (a1bf37caf, da7c1fcfe).
- The config self-test requires each device to come up, not only its property names to be accepted. That caught `zns.conf`, which asked for more active zones than it has and was refused at every size (b509a23a5).
- Config presets for OCSSD, CSD, KV and NoSSD (NoSSD with link and firmware time from the NVMeCHA controller), and a `run-kvssd.sh` launcher (04eb9b06d).
- A key-value probe tool and a corrected KV wire format description (9ab75f5dc).
- `pin.sh` no longer names a CPU past the last one, and now pins the pollers, the FTL thread and the CXL SSD's `femu-cxl-ftl` and `femu-cxl-cca` threads as well as the vCPUs (e29fe6ee2, 56e0e2f27).
- Every launcher, the legacy scripts and `make-guest-image.sh` name QEMU's threads with `debug-threads=on` (e29fe6ee2, 56e0e2f27).
- The documentation example check realizes the guest's disks and NICs too, with stand-in backends, so a device the machine cannot plug (a NIC without `bus=pcie.0` on a `cxl=on` machine) fails the check (ac5a152fe).
- `kv-probe.c` comments describe what it checks and which device nodes it can use (dfc1bd606).
- GitHub Actions CI on several Ubuntu releases, with pinned actions, a read-only token, sanitizer builds, link checks and steps that can actually fail (3b4708763, daf890b7f, c613ae5ac, 4164e551b, 8d4762608, 89ff3ed38, e3700422c).
- qtests and unit tests for FEMU now live under `hw/femu/tests/`, including a NAND media unit test run by meson and FTL checks turned on in CI (d72e30a34, 6549d6094, cc926e1e9, 8241ccc7f).
- Seeded fuzzers for admin, I/O, zoned, FDP, KV, Open-Channel 2.0 and CSD commands, run on sanitizer builds in CI (8a57a3e27, 6d6d927d0, 9798a6574, df3d9f72, 28eaf2afa, 2dbb18a82, 243fc0b66, 2f1208402, 692d9a106).

### D.1.6 Removed or legacy

- `femu_mode=6` (an unimplemented SmartSSD placeholder) is no longer accepted (c3b8e88ef).
- QEMU network block drivers are no longer built (81c4385fc).
- Three NAND media policy flags that nothing read were dropped (2e29d7efb).
- OC 1.2 no longer advertises hybrid commands it does not implement (f5fc7d573).